Figures that communicate

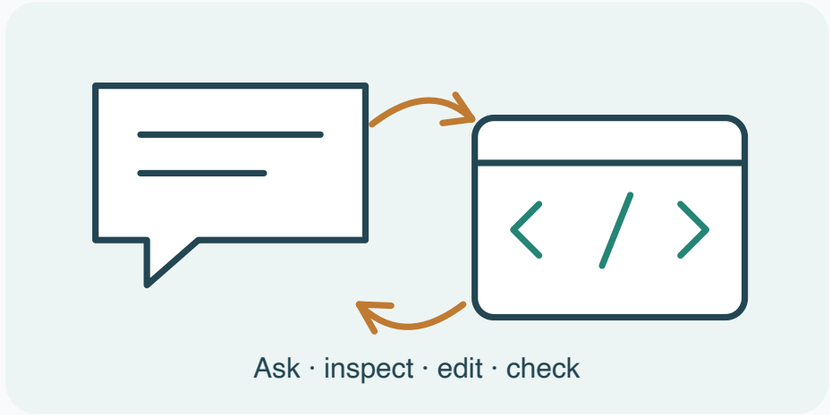
Choose a course day to explore its topics.

Day 2 · Part 2

Heatmap focus

AI tools, reading heatmaps, scaling, clustering and hands-on work in R.

Open Day 2 topics →



Day 3

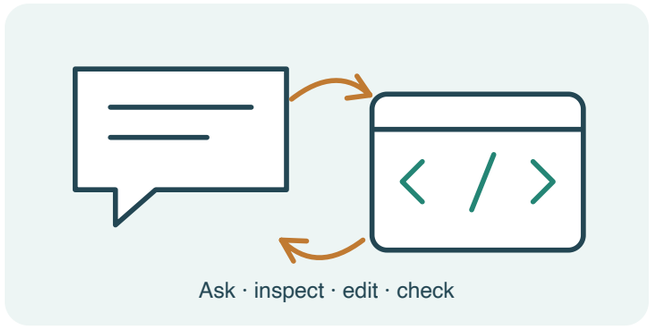
From plot to publication

Plot design, consistent panels, figure assembly, export and generative tools.

Open Day 3 topics →

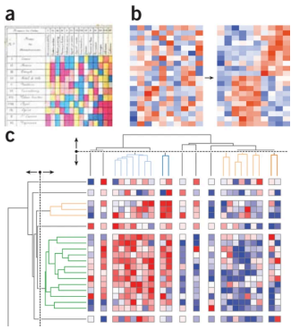
Heatmap focus

Read a cell, uncover a pattern, then test the design choices.



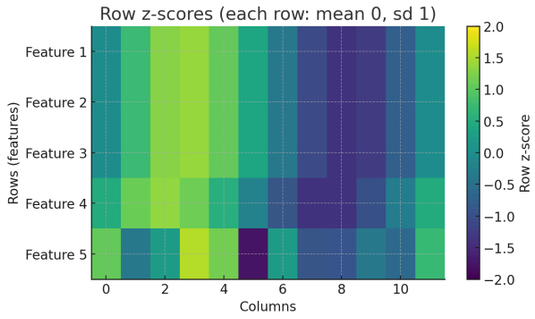
1 AI tools

Chat versus agents, data protection, shared experiments and good practices.



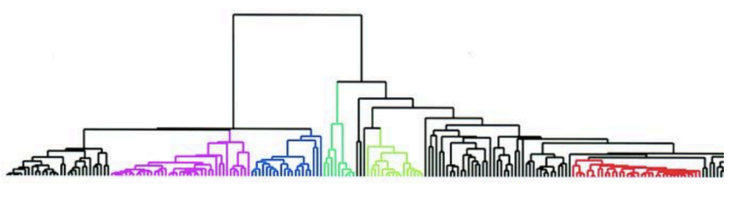
2 Reading heatmaps

Why heatmaps, why ComplexHeatmap, and how to read a matrix.



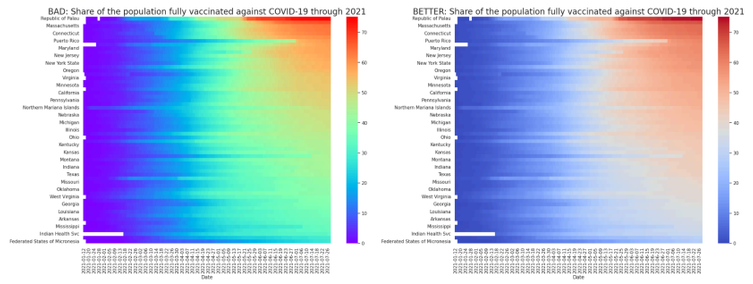
3 Scaling and z-scores

Switch between raw values and row z-scores.



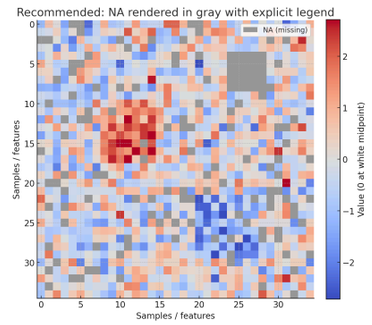
4 Clustering

See how ordering and distance change the picture.



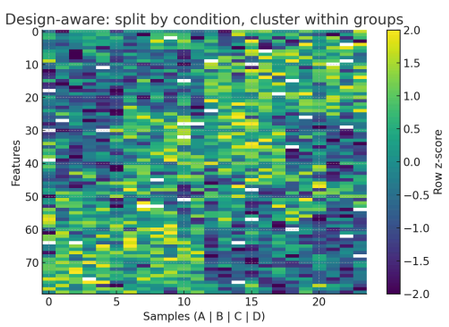
5 Colour and perception

Palettes, colour keys and visual context.



6 Pitfalls and special cases

Missing values, outliers and matrix layout.



7 Build & check

Improve one heatmap and check the result.

8 Hands-on · create a heatmap in R

Build dataset 1 together → explore dataset 2 in groups → share your heatmap.

AI tools · collaborate, check, protect

A dedicated section for using AI in the heatmap exercise.

1

Understand the workflow

Chat-based help versus agents that inspect files, edit code and run commands.

2

Check data protection

Know what leaves your computer and which files your institution permits you to share.

3

Experiment together

Compare results, spot misunderstandings and collect practices that help.

AI · chatbox versus agentic tools

Chat-based help · 1 / 5 · Give the task

Chat-based help

You choose which suggestions to apply in RStudio.

1 **You** **Ask + paste context**

Explain this heatmap error.

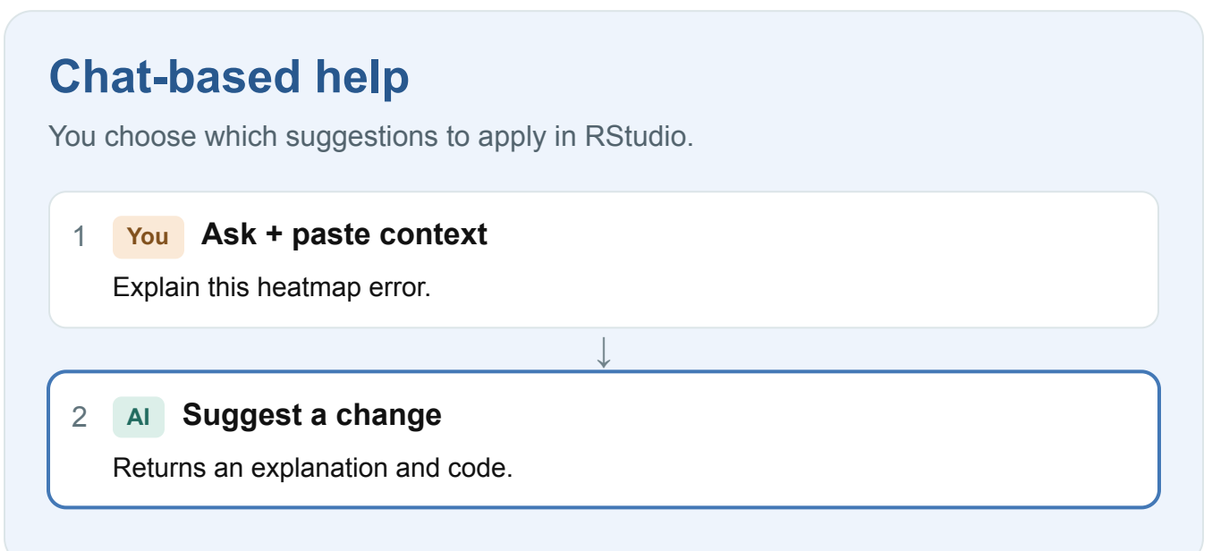
Chat-based tools are completely fine for this exercise. With a basic chat workflow, you decide what to copy, change and run, giving you direct control over each edit. Agents can make more changes for you, so set their scope and review the result.

Tomorrow · live demo: I'll use **Claude Code** to bring several plots into a consistent aesthetic for a journal submission.

Tools, permissions and examples

AI · chatbox versus agentic tools

Chat-based help · 2 / 5 · Inspect & suggest



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Returns an explanation and code.

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3 **You** **Copy & edit**

Apply the suggestion in RStudio.

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5 **You** **Check the science**
Compare data, labels and meaning.

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Agentic AI · 1 / 5 · Give the task

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4 **Agent** **Run + check + revise**
Uses tool results to try the next step.

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Writes changes within its permissions.

↓

4 **Agent** **Run + check + revise**
Uses tool results to try the next step.

↓

5 **You** **Review edits + plot**
Accept only after checking the result.

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Tomorrow · live demo: I'll use **Claude Code** to bring several plots into a consistent aesthetic for a journal submission.

The agent can run the technical loop. You still check the scientific meaning.

> **Tools, permissions and examples**

AI · data protection before you start

1 / 4 · You choose what to share

You + your AI app

Prompt

Help me improve this heatmap.

▤ Code + table + plot

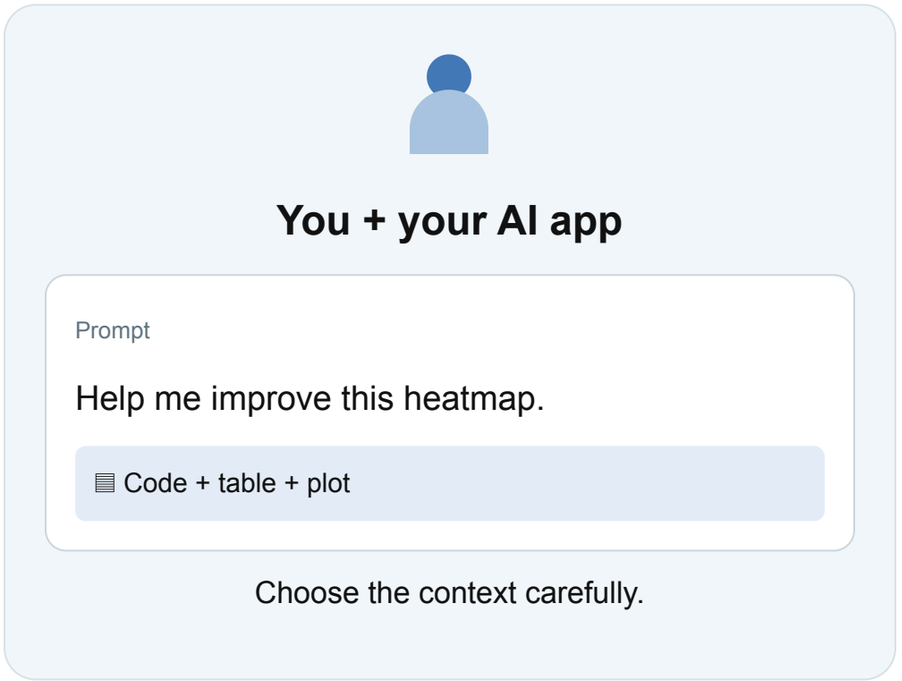
Choose the context carefully.

You choose what to share

A question can contain more than words: pasted tables, screenshots, code and uploaded files may carry research data.

AI · data protection before you start

2 / 4 · Context travels to the provider

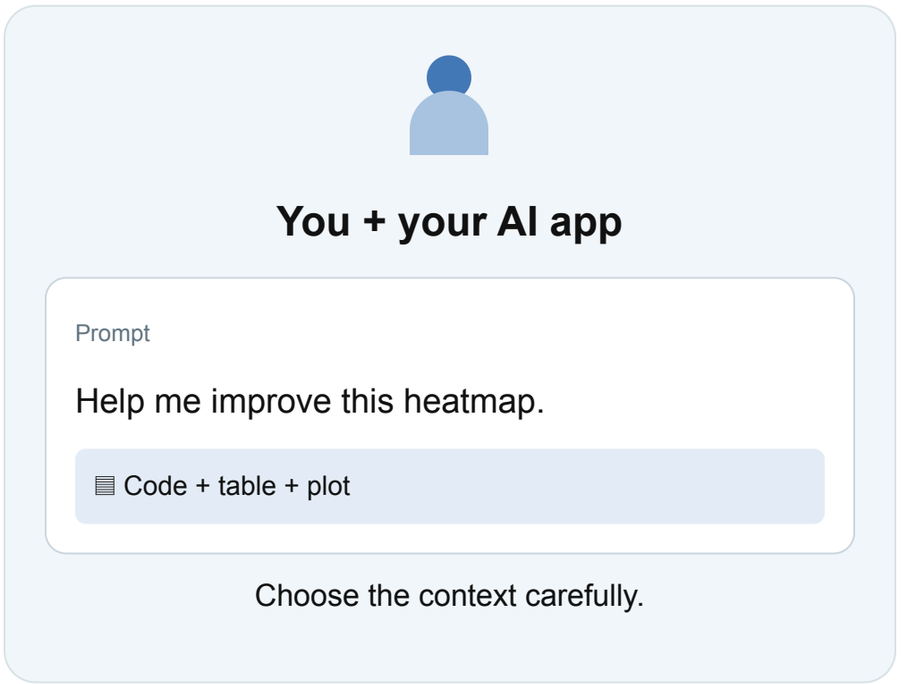


Context travels to the provider

For an online model, the selected context is sent to remote servers. A locally installed chat or agent app can still use a remote model.

AI · data protection before you start

3 / 4 · Check where it goes and who may access it



> **Check where it goes and who may access it**
>
> Processing location, retention and access rules depend on the provider, account and institutional agreement. Check them before sharing restricted files.

Do not infer location from a brand name. Check the actual service and contract: a provider may process data in the US, China, Europe or other regions. This diagram does not imply that a person reads every prompt.

AI · data protection before you start

4 / 4 · An answer returns; you stay responsible

You + your AI app

Prompt

Help me improve this heatmap.

▤ Code + table + plot

Choose the context carefully.

Selected context

→

←

Answer / tool instructions

Network connection

Provider’s servers

The remote model processes the context.

Check before sending

Which country or region?

Who may access it?

How long is it retained?

Is this account approved?

An answer returns; you stay responsible

A chat returns a suggestion. An agent may also act through tools and send results back. Use approved services and only data you are permitted to share.

“Not used for training” does not mean “never leaves your computer”. Fully local models are a different setup. If unsure, ask your institution’s IT or data-protection contact. Example data-use documentation ↗

Working with AI · explore together

Use the AI tools you already have: Claude, Codex, Gemini or another assistant.

Claude

Codex

Gemini

You are encouraged to use AI while working on your heatmap. Work with your assistant to understand the code, try design changes and investigate errors.

1

Try a real task

Start from your working plot. Ask for one improvement, an explanation of an argument, or help with an error. Run the suggested code and inspect the result.

2

Compare experiences

What worked? What failed? Did the assistant misunderstand the data, use the wrong library or change something you wanted to keep?

3

Collect good practices

Share one useful prompt, one mistake and one check that caught it. Together, turn these examples into advice for the next person.

Small experiment: try the same question with only your code, then add a plot image and the supplied dataset or a pasted table with column names. Does the answer improve? What is still misunderstood?

Record: task → context supplied → suggestion → what you tested → outcome. A suggestion that sounds convincing still needs checking.

Breakout rooms · improve one heatmap together

3–5 people per room · one shared screen · one change at a time

Discuss as a group

“Can we make the NA values grey?”

“Let’s find a clearer colour scale.”

“Did the data or sample order change?”

One person shares their screen

AI chat · e.g. ChatGPT

“Use R + ComplexHeatmap. Show missing cells in grey. Keep the data, order and 0–1 scale unchanged.”

Read the suggestion together.
Decide what to try.

RStudio · your notebook

```
Heatmap(mat,
  na_col = "grey80",
  ...
)
```

Apply the change → run the chunk → inspect the plot.

Rotate who types. Everyone helps choose, interpret and check.

Bring back: one before/after plot · one useful prompt · one mistake or surprise

Working with AI · good practices

Give context. Make one change. Check the result.

Keep the original

Save your code and plot.
Compare before and after.

Give context

Code + plot + approved data.
Name **ComplexHeatmap** and link its docs.

Keep meaning fixed

Same values, samples and NA.
Check scale and annotation matches.

Run & inspect

Try one change at a time.
Check the plot, not just the code.

Ask for ideas

“What would make this heatmap easier to interpret? Suggest three improvements before editing.”

Ask for a check

“Compare both versions. Did any data, scaling or sample annotations change unintentionally?”

▸ **Two more questions to try**

Why heatmaps? · see the data’s structure

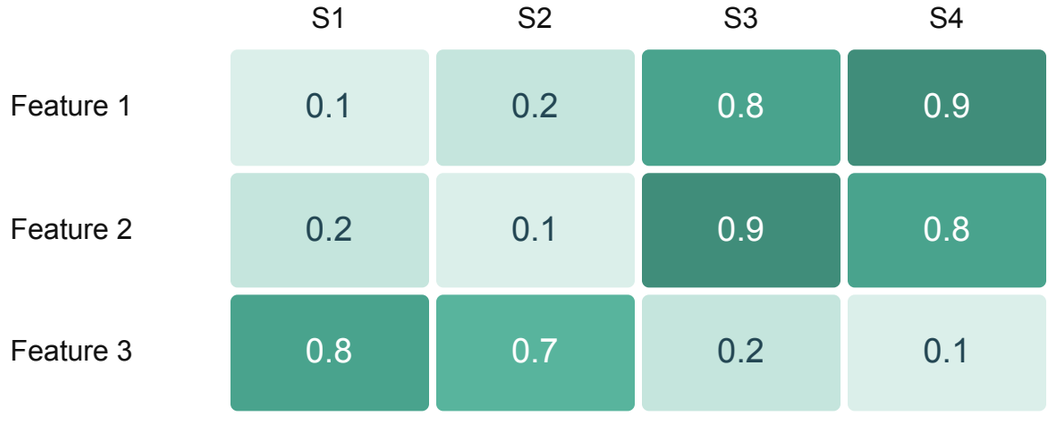
Turn a table of measurements into a visual overview, without reducing everything to one average.

A table gives you the values

	S1	S2	S3	S4
Feature 1	0.1	0.2	0.8	0.9
Feature 2	0.2	0.1	0.9	0.8
Feature 3	0.8	0.7	0.2	0.1

Exact numbers are available, but comparing many rows and columns takes effort.

Colour makes patterns easier to spot



Which samples look similar? Which features change together? Where are unusual or missing values?

Illustrative values · light = low, dark = high · scale 0–1.

Build an intuition for the data: see variation, similarities and exceptions across individual measurements.

A heatmap can show original measurements, but it may also show transformed or scaled values. Always state what the colours represent; a visible pattern is a starting point for investigation, not proof of an effect.

Our tool · ComplexHeatmap in R

Biological heatmaps can be challenging to design, but very useful when the choices are clear.

1 · Start simply

A small first step

```
ht <- Heatmap(mat)
draw(ht)
```

With the package loaded and a numeric matrix ready, a basic plot takes just a few lines.

2 · Add context

Rich annotations

Add treatment, mouse or study-day strips; label features; include bars or other summaries beside the matrix.

Annotations explain the rows and columns.

3 · Refine the figure

Room to grow

Control colours, ordering, clustering, splits and legends. Combine related heatmaps while keeping their rows or columns aligned.

We will add one feature at a time.

In biology, the effort pays off: a well-designed heatmap can reveal patterns across genes, microbes, treatments and time points. Clear scales, ordering and annotations help readers interpret those patterns correctly.

ComplexHeatmap reference ↗ · Annotation examples ↗

What a heatmap encodes

Rows and columns identify items. Colour represents one value at their intersection.

Original values · illustrative data

Condition	C	C	C	T	T	T
Feature	S1	S2	S3	S4	S5	S6
A	2	3	4	8	9	10
B	22	23	24	28	29	30
C	10	9	8	4	3	2
D	40	38	36	28	26	24

0 40 Value

Add colour

Labels and values remain visible so you can track each observation.

One cell, one measurement

Feature **A**, sample **S1**

2

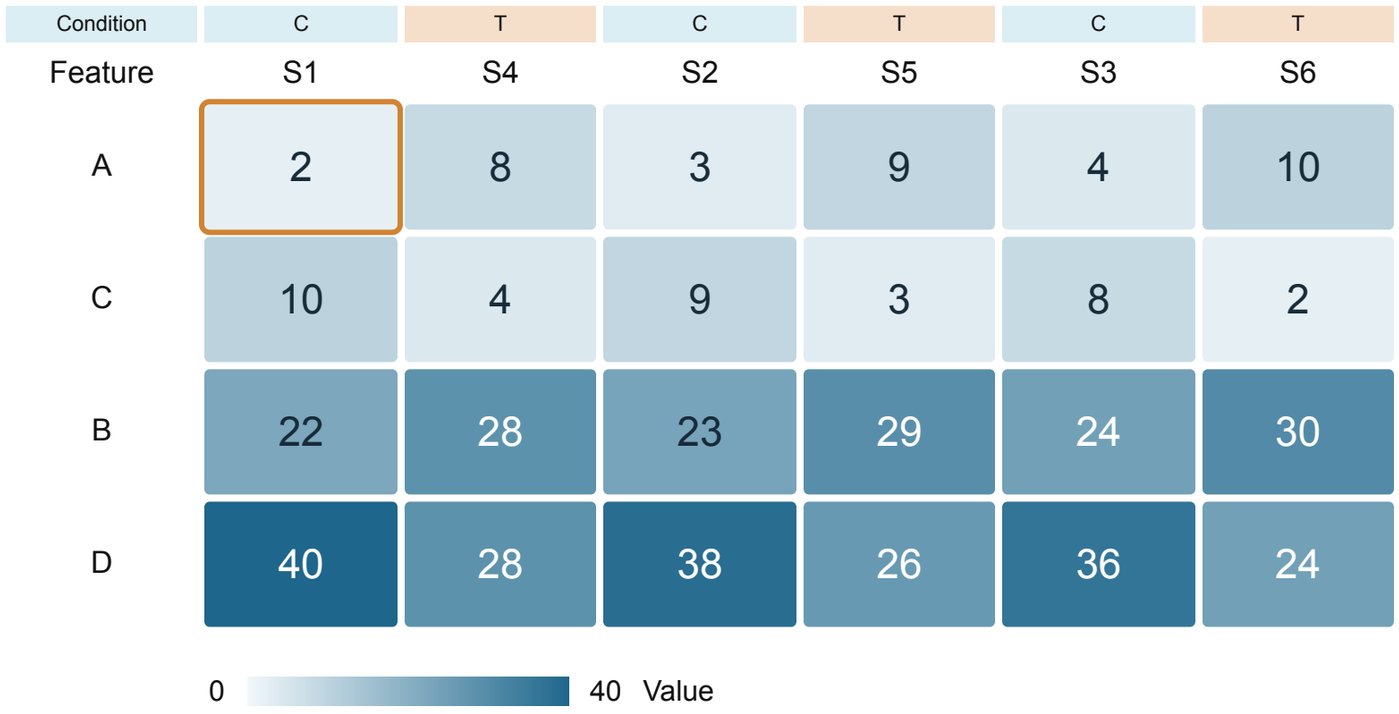
Hover or click a cell to read its value. C: control; T: treated. Then add colour to see the pattern.

▸ **Original course example**

Measurements, ordering and groups

Reordering brings similar profiles together without changing the measurements.

Original values · illustrative data



Same values, different neighbours

Follow the row and sample labels as the order changes.

Grouping similar profiles reveals two blocks. No measurements are changed.

This is a hand-arranged example. The clustering slides explain how an algorithm chooses an order.

Group similar profiles

Labels and values remain visible so you can track each observation.

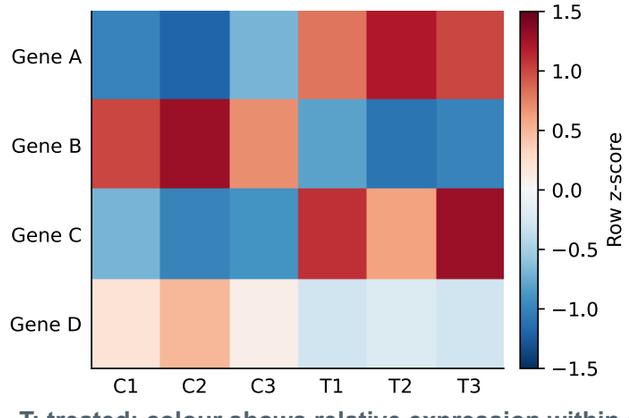
Original course example

When a heatmap helps

A matrix is useful when the question is about patterns across many items.

Gene expression

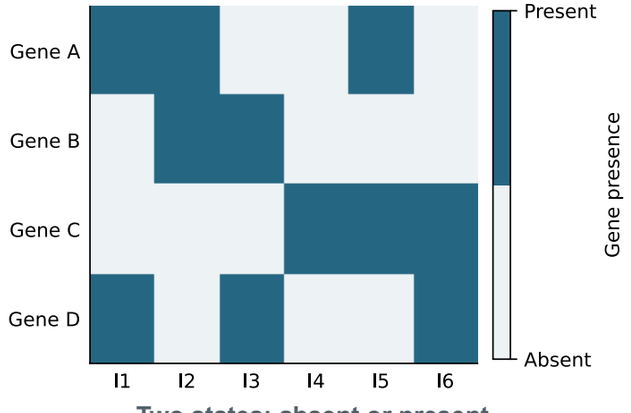
Genes × samples



C: control · T: treated; colour shows relative expression within each gene.

Resistance-gene presence

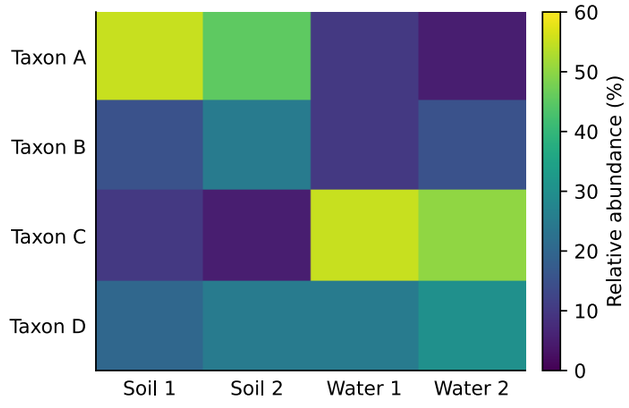
Genes × isolates



Two states: absent or present.

Taxon abundance

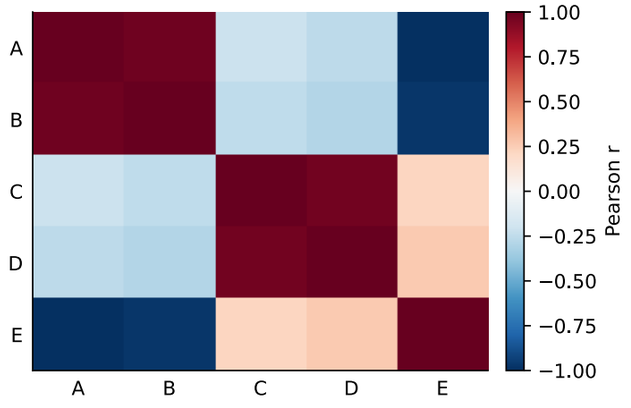
Taxa × environments



Colour shows relative abundance; columns sum to 100%.

Pairwise correlation

Variables × variables



A symmetric matrix; the diagonal is 1.

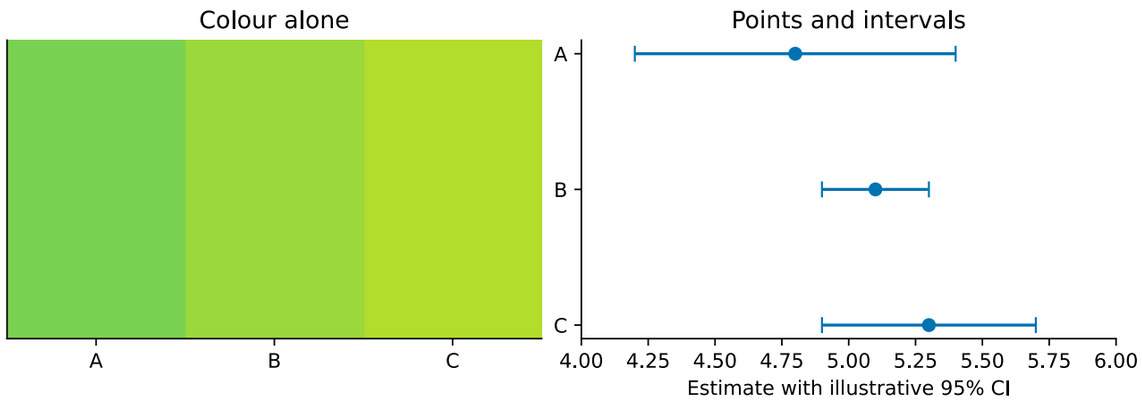
Illustrative data generated for teaching, not experimental results.

Original course example

When another display helps

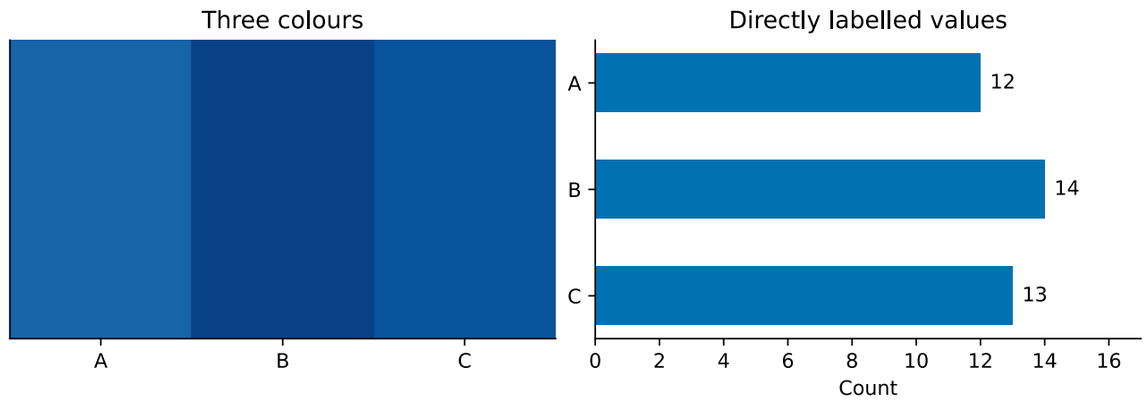
Choose the display for the comparison you want the reader to make.

When uncertainty matters



Intervals show uncertainty that a single colour cannot encode.

When a few exact values matter



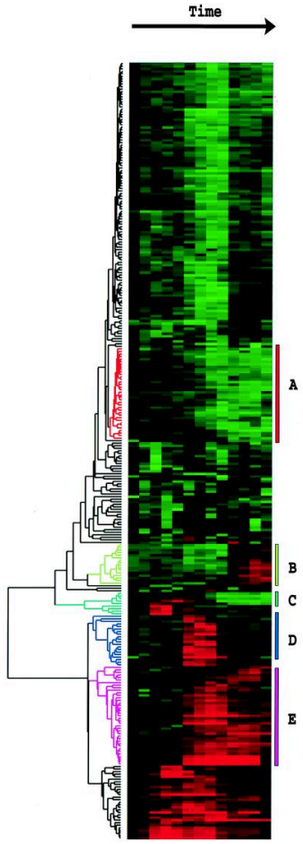
A short labelled plot can make small differences easier to compare.

Illustrative data. A heatmap remains useful for an overview; choose the display for the question.

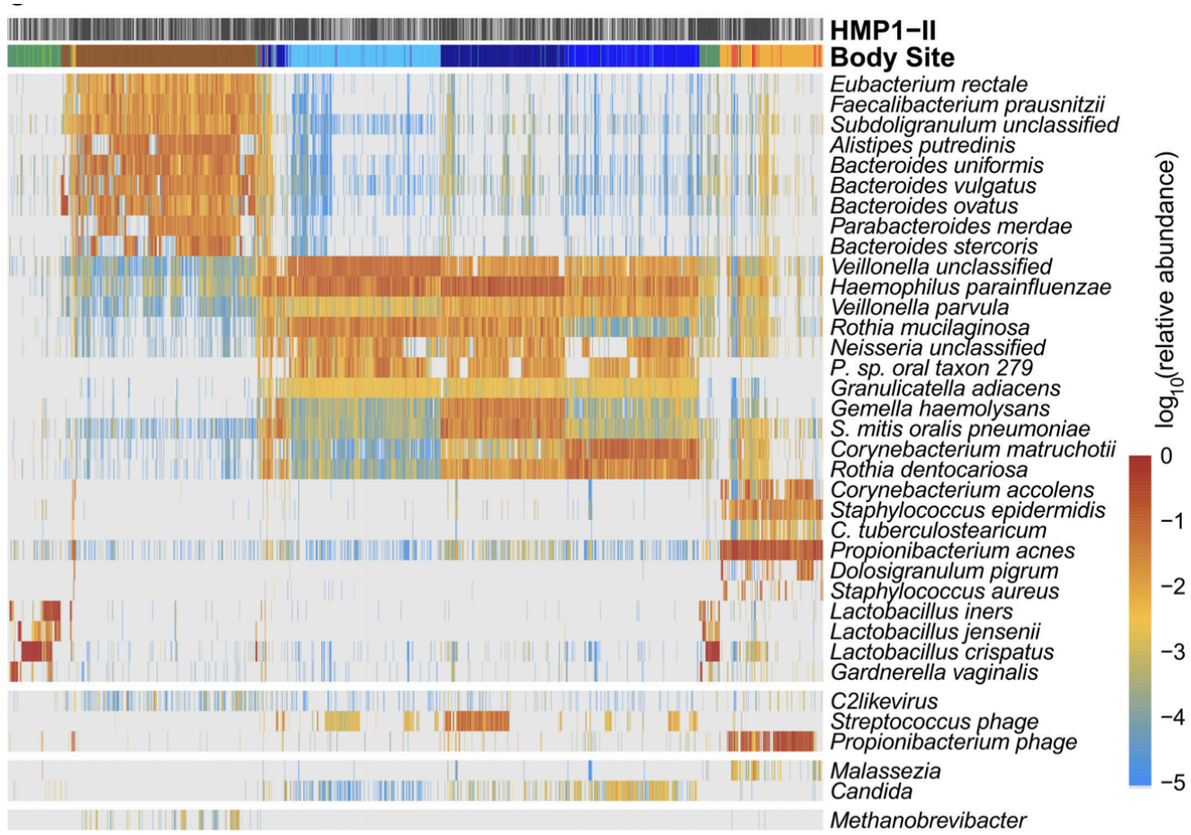
> **Original course example**

Heatmaps in biological research

Compare the row labels, column annotations and colour keys.



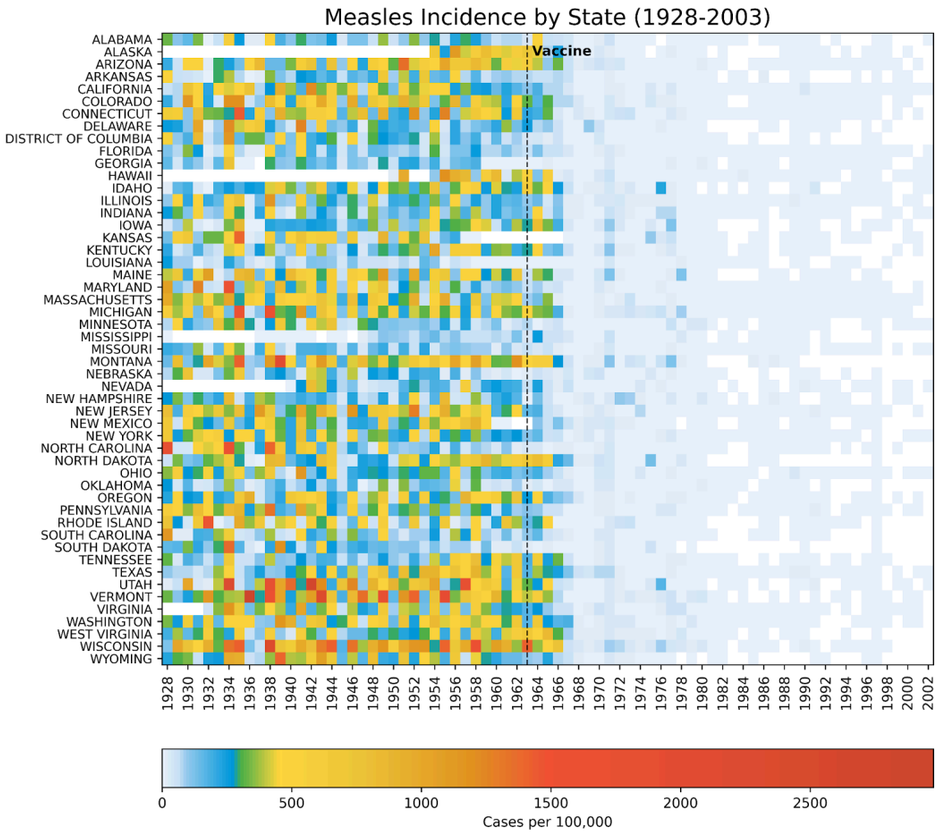
Genome-wide expression



Microbiome profiles

A heatmap across place and time

The same matrix idea works beyond gene expression.



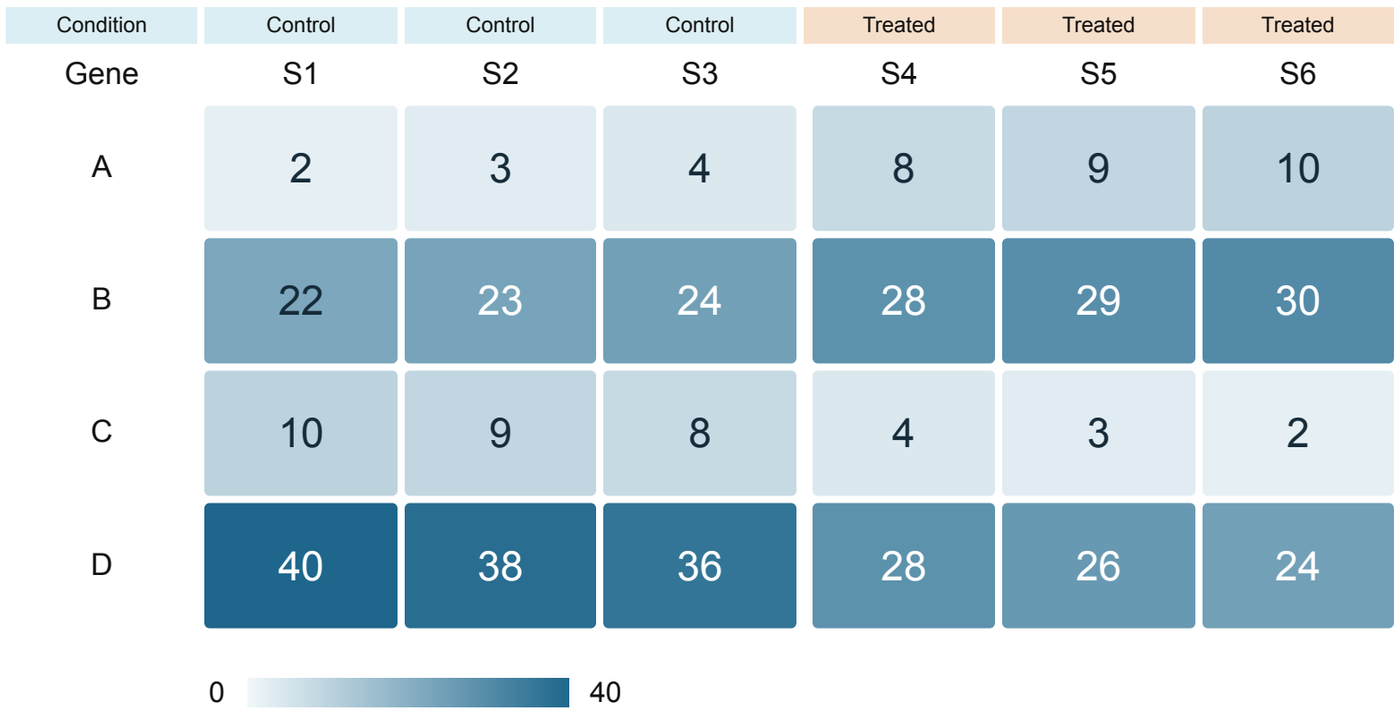
Vaccination example from the original deck

Why standardise rows?

A row z-score expresses each value relative to that row’s mean and variability.

Original | $\log_2(\text{value} + 1)$ | Row z-score

Same 4 genes × 6 samples · original values



Abundance or response pattern?

A and B rise in the same way, but B has higher values.

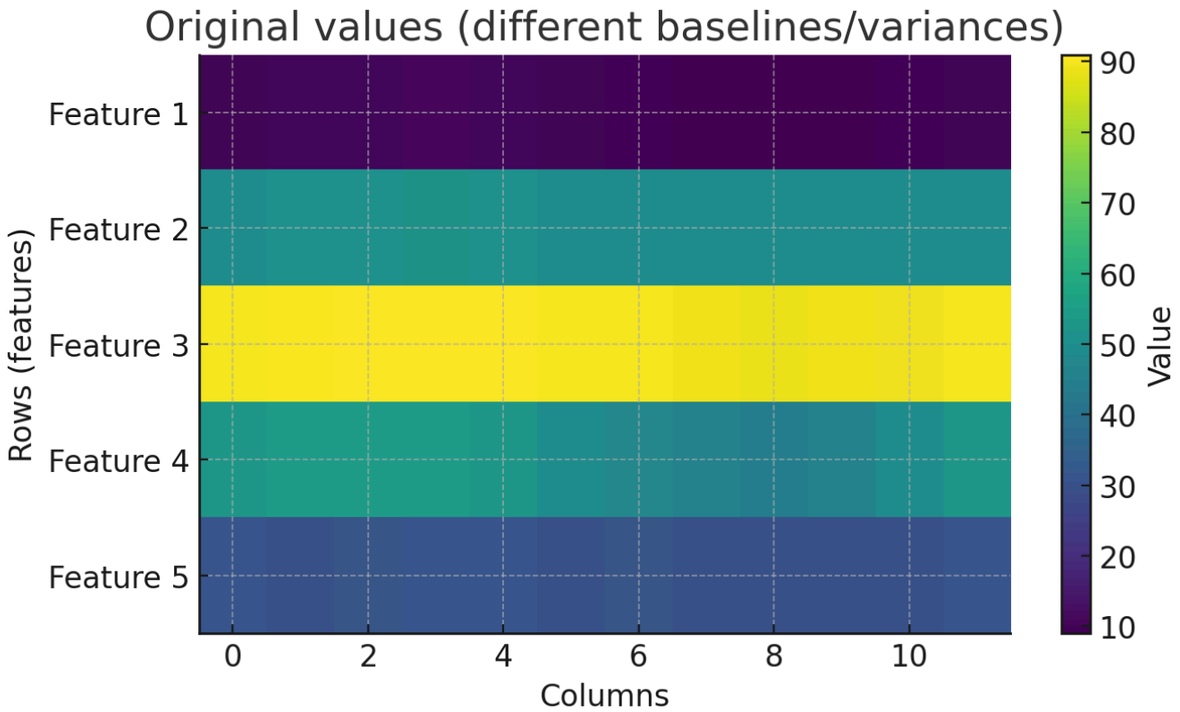
Changed: values and colour scale.

Fixed: samples, gene order and original observations.

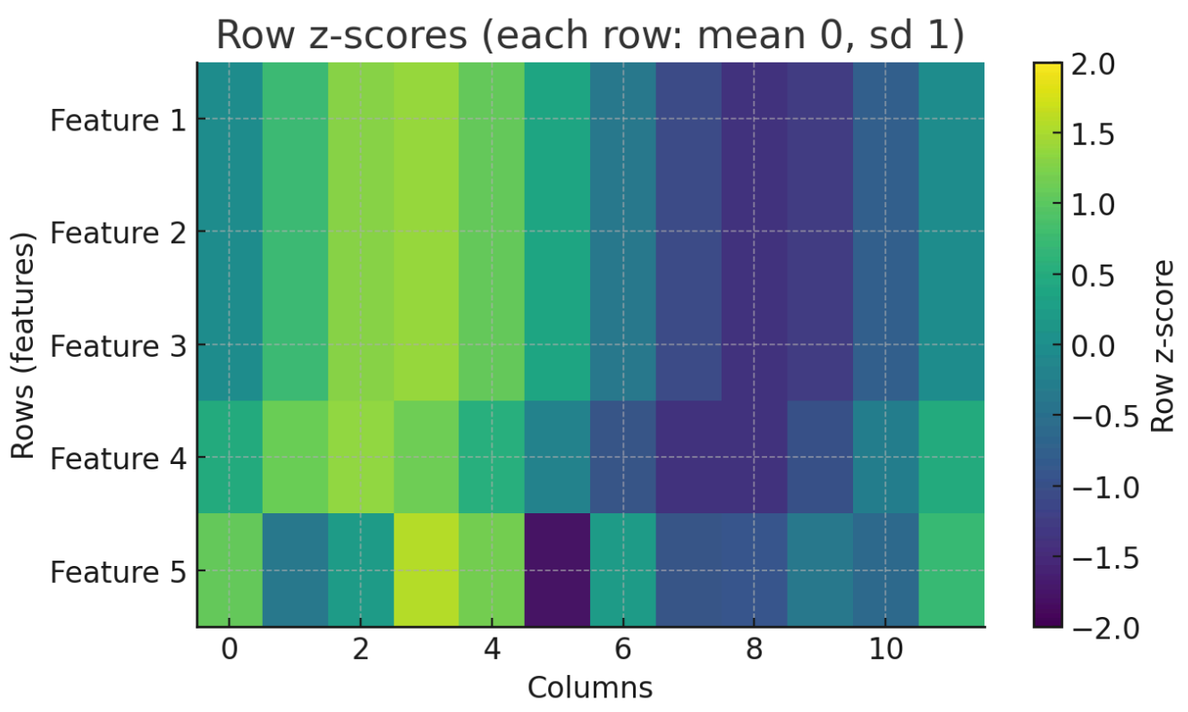
Original course example

Before and after row scaling

The same measurements answer a different question after standardisation.



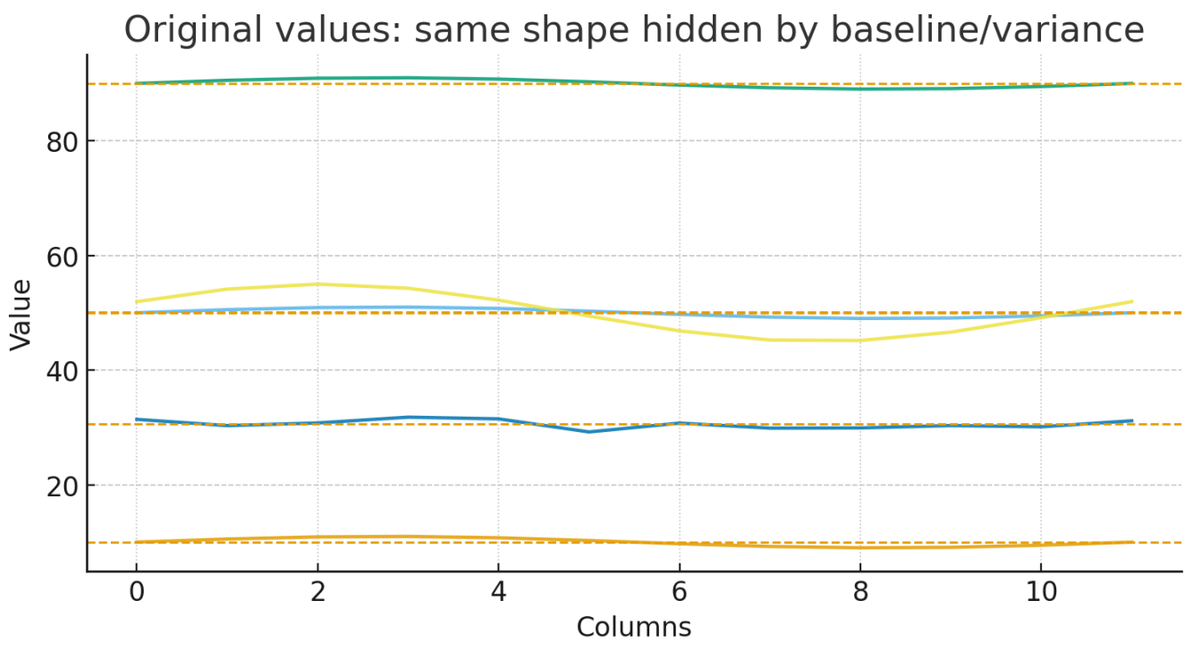
Original values



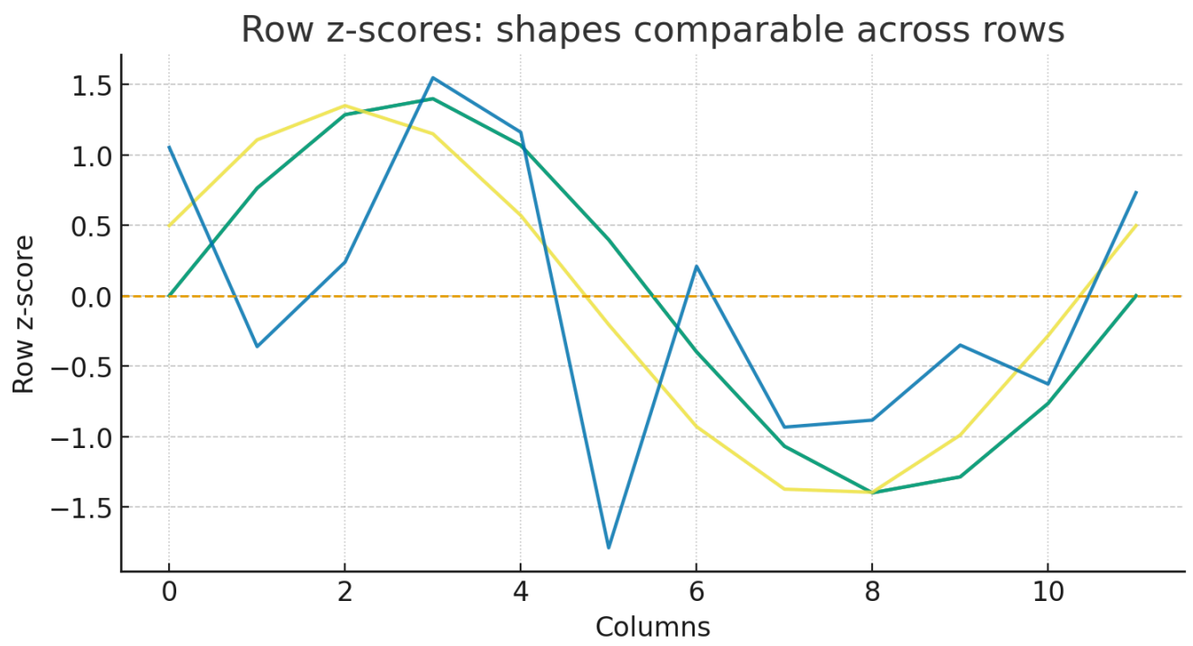
Row z-scores

Profiles before and after scaling

Baseline differences can hide shared changes across samples.



Original profiles



Standardised profiles

Calculating a row z-score

Subtract the row mean, then divide by the row standard deviation.

$$z_{ri} = (x_{ri} - \mu_r) / s_r$$

x_{ri}	μ_r	s_r
Value in row r, column i	Mean of row r	Standard deviation of row r

A constant row has zero standard deviation: its z-scores are undefined. Handle it explicitly.

What row scaling changes

Each nonconstant row has mean approximately zero and standard deviation one.

Feature	Mean before	SD before	Mean after	SD after
1	10.00	0.71	≈ 0	1
2	50.00	0.71	≈ 0	1
3	90.00	0.71	≈ 0	1
4	50.16	3.58	≈ 0	1
5	30.60	0.77	≈ 0	1

Statistics from the original example, rounded for display.

What a dendrogram shows

Hierarchical clustering records successive merges of items and groups.

Distance: 1 − Pearson r | Euclidean

Average linkage

Branch length shows merge distance; pale branches are not merged yet.

Same 4 genes × 6 samples · row z-scores

Condition		Control	Control	Control	Treated	Treated	Treated
	Gene	S1	S2	S3	S4	S5	S6
	A	-1.2	-0.9	-0.6	0.6	0.9	1.2
	C	1.2	0.9	0.6	-0.6	-0.9	-1.2
	B	-1.2	-0.9	-0.6	0.6	0.9	1.2
	D	1.2	0.9	0.6	-0.6	-0.9	-1.2

−2 2.0 Midpoint: 0

Merge step **4 clusters** | Next merge | Swap first two rows manually

Distances use original values; colours show row z-scores. Drag through the merge steps to cut the tree into 4, 3, 2 or 1 clusters. Equal-height merges are resolved in a fixed order.

> **Original course example**

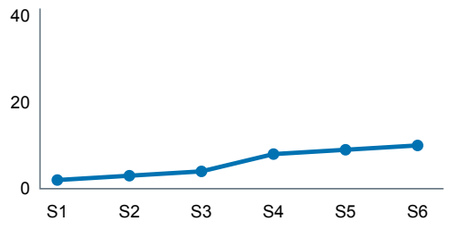
Distance defines “similar”

Choose the distance for the biological question.

Each row is one gene measured across the same six samples. The small line plot shows that row’s profile, not its mean.

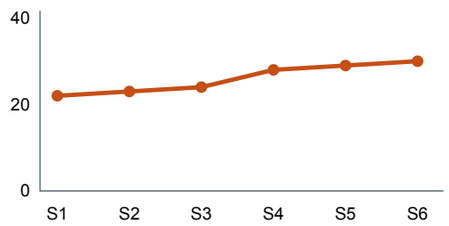
Gene A

2 · 3 · 4 · 8 · 9 · 10



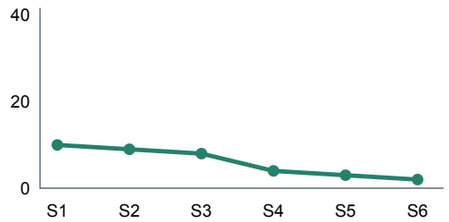
Gene B

22 · 23 · 24 · 28 · 29 · 30



Gene C

10 · 9 · 8 · 4 · 3 · 2



Identical y-axis limits (0–40) · illustrative expression values.

Which genes are most similar?

Pair	Euclidean	1 − Pearson r
A ↔ B	49.0	0.0
A ↔ C	15.2	2.0
B ↔ C	51.3	2.0

Smaller distance = more similar.

Reveal the interpretation

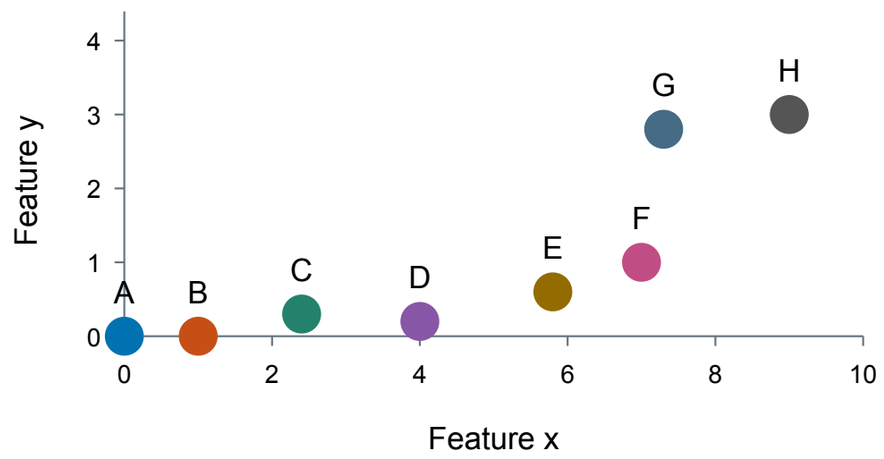
Original course example

Linkage defines the next merge

The same distances can produce different trees.

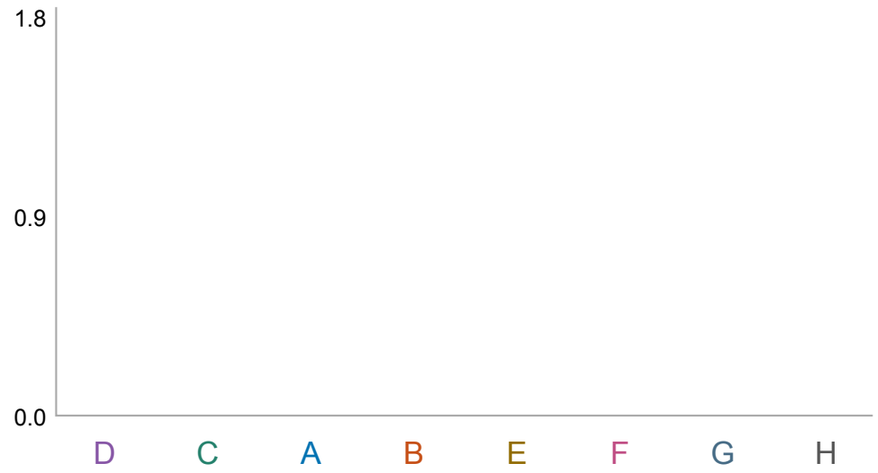
Single | Complete | Ward — Nearest pair between two groups.

Same eight points



Equal units on both axes · colour shows current groups.

Build the tree



Height = nearest between-group distance

Start with eight separate groups. Which will merge first?

8 groups remain. Switch methods at the same step to compare.

Reset | Undo merge | Next merge → | Show full tree

> **Original course example**

Match the distance to the data

The measurement type matters as much as the appearance of the heatmap.

Continuous profiles

`2, 3, 5, 4`

`12, 13, 15, 14`

Euclidean: differences in level and magnitude.

1 − Pearson correlation: differences in profile shape.

Scale features when units differ. Correlation is undefined for a constant profile.

Presence / absence

`1, 0, 1, 0`

`1, 0, 0, 0`

Jaccard compares shared presences

Abundance

`60, 30, 10`

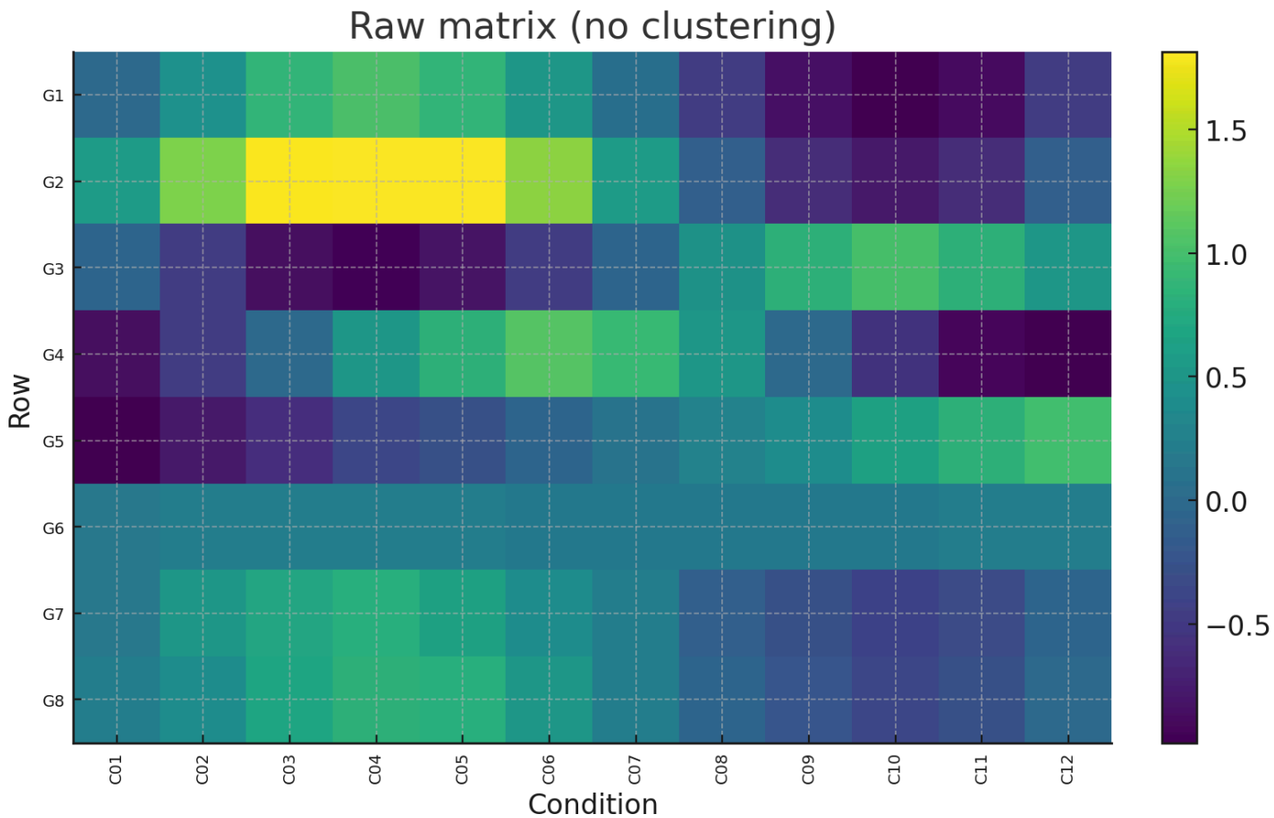
`10, 30, 60`

Bray–Curtis compares abundance profiles

- Profile shape: consider correlation or rank correlation.
- Absolute differences: consider Euclidean or Manhattan distance.
- Binary presence/absence: consider Jaccard distance.
- Nonnegative abundance: consider Bray–Curtis with appropriate preprocessing.

The original matrix order

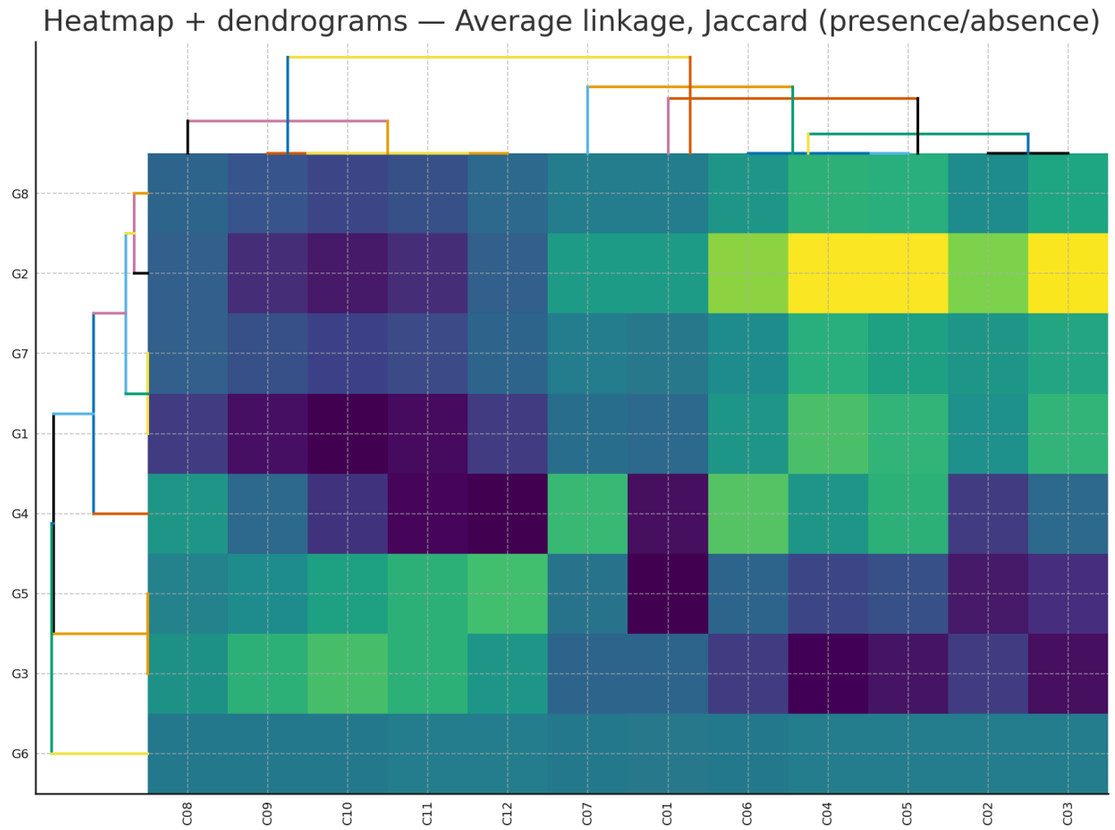
Use this as the reference before changing the clustering.



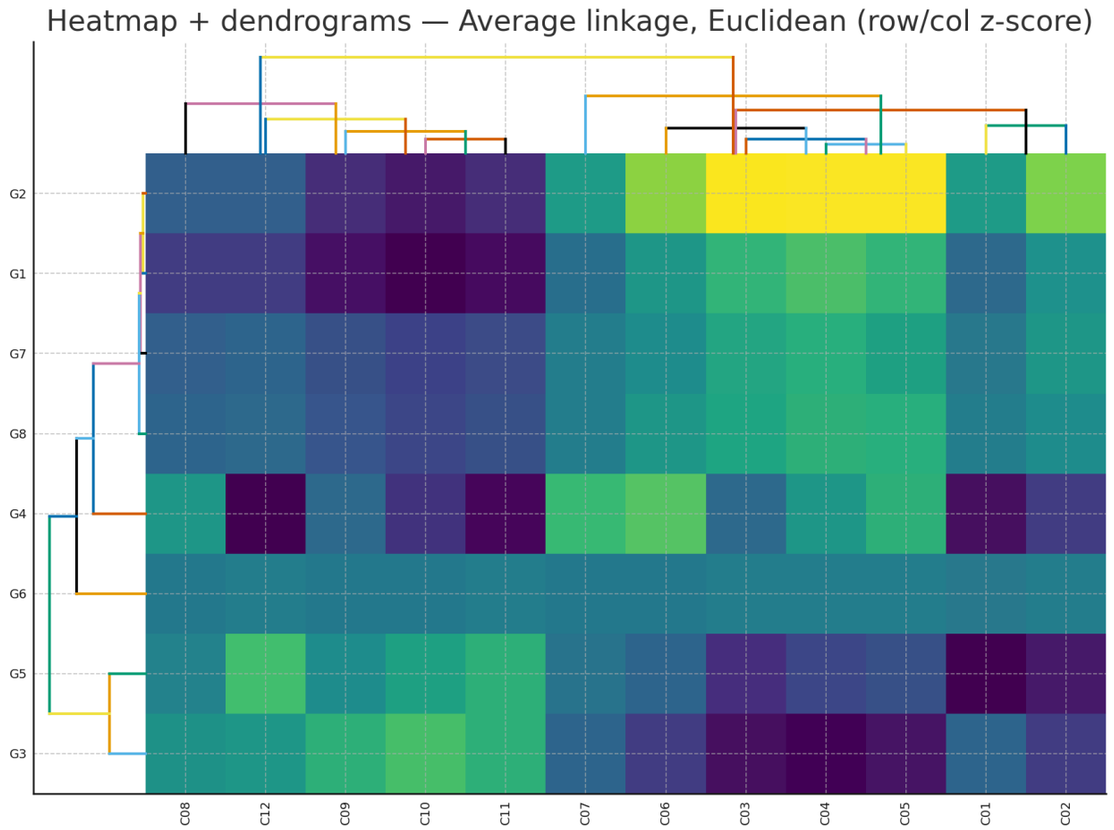
Original matrix

Correlation or Euclidean distance

The same data can yield different row orders.



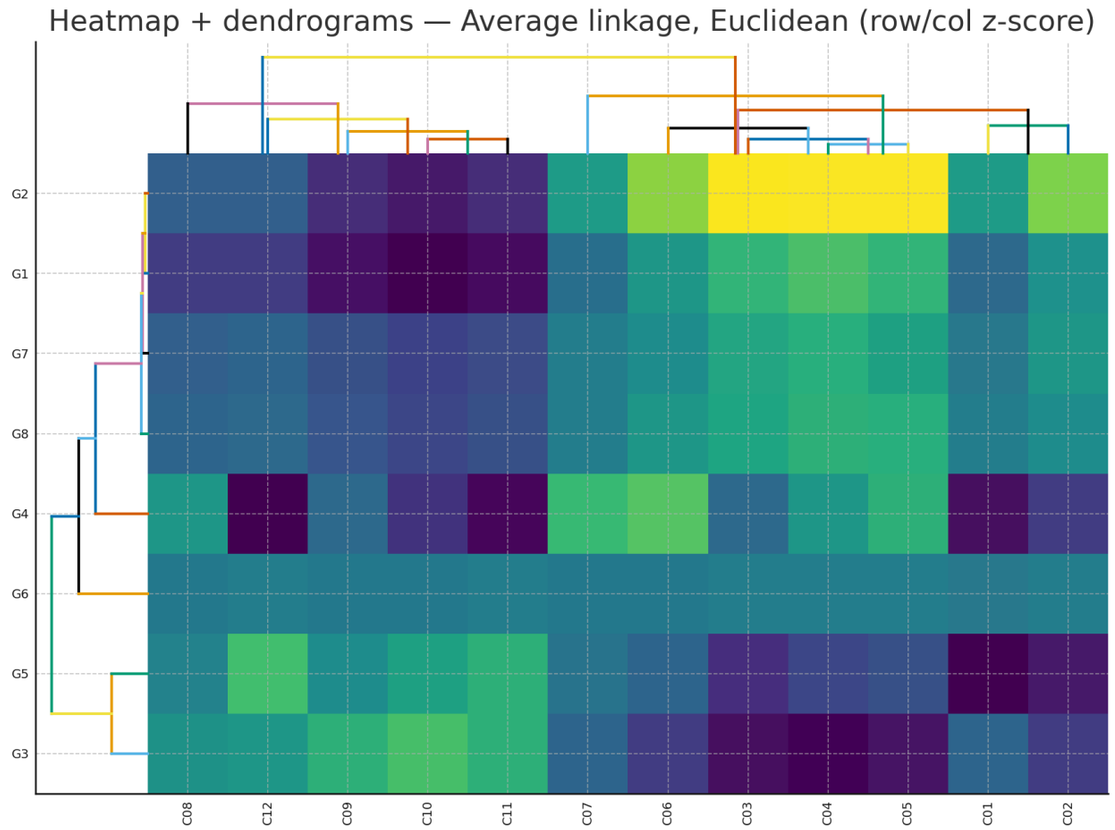
Average linkage · correlation distance



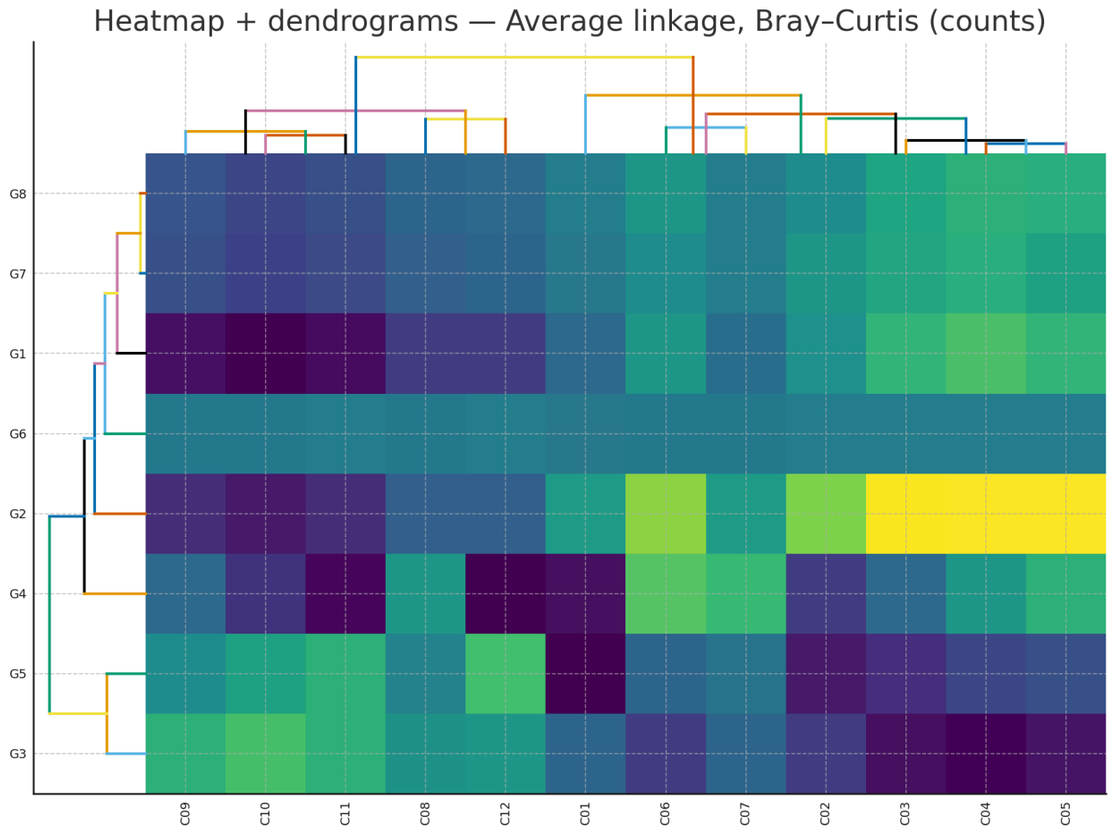
Average linkage · Euclidean distance

Scaling changes the clustering too

Changing preprocessing or distance changes what “close” means.



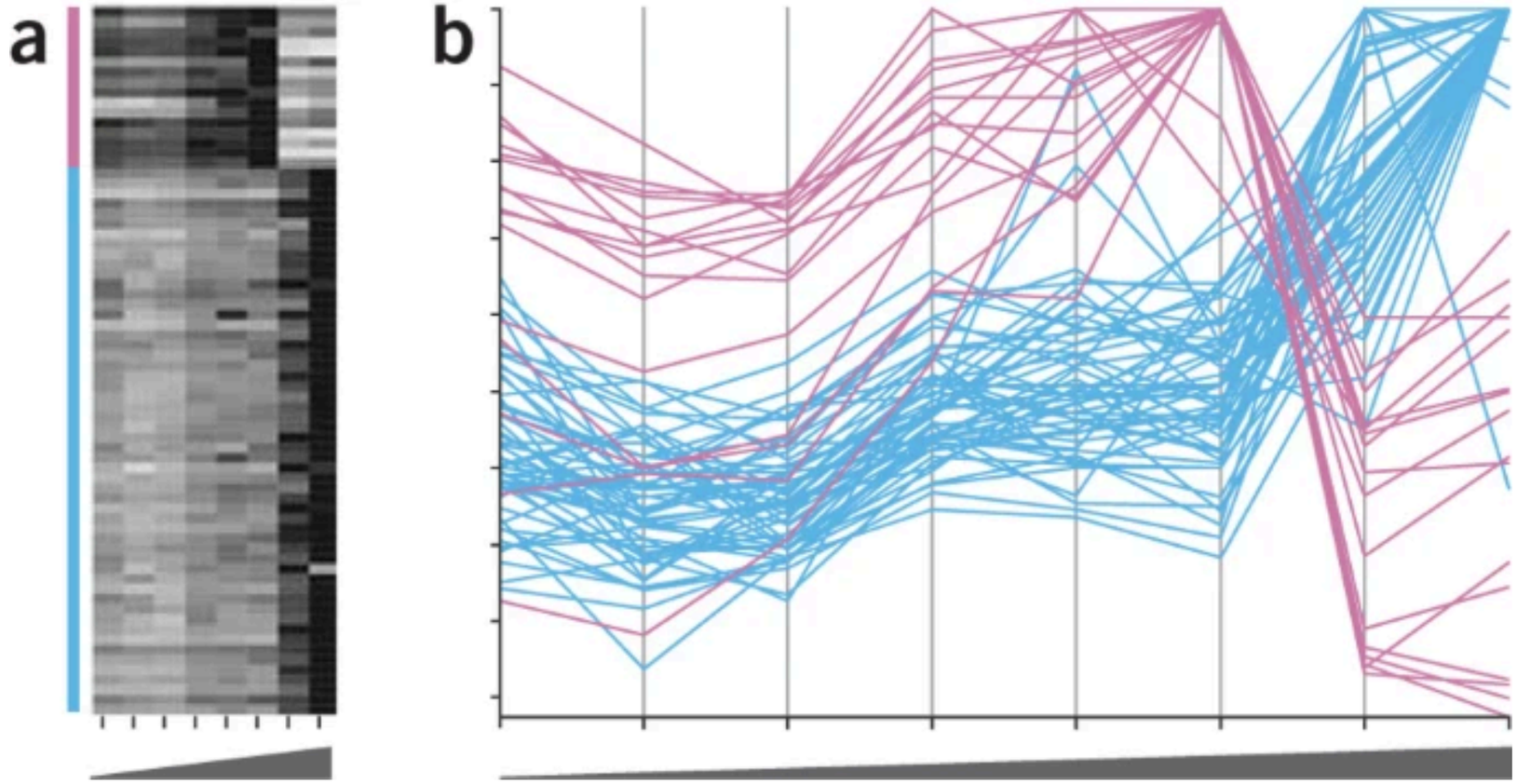
Average linkage · Euclidean on row z-scores



Average linkage · Bray–Curtis on counts

A heatmap and its profiles

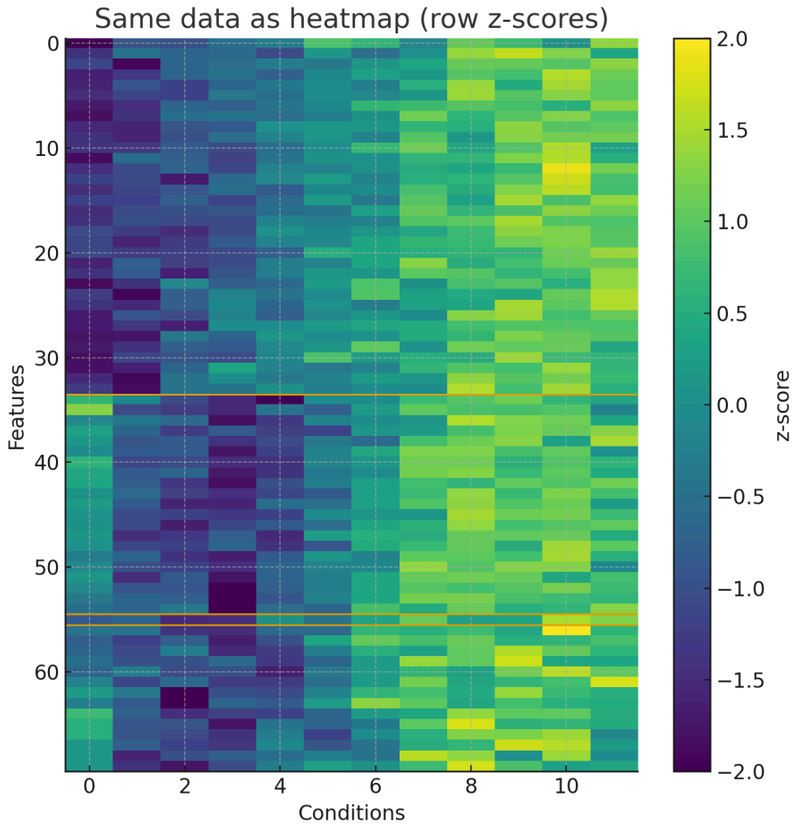
A line view can help explain the patterns visible as colour blocks.



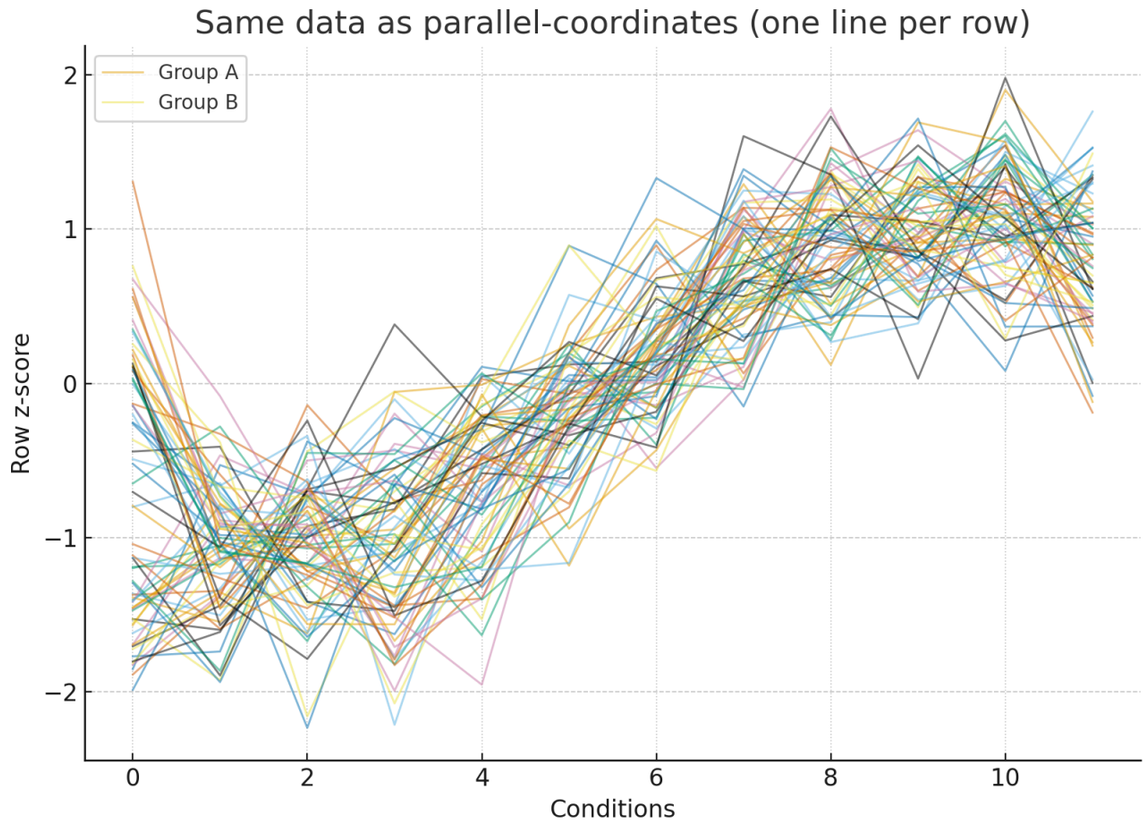
Linked heatmap and profile view

Same data, different display

Use the heatmap for an overview; use profiles to follow changes across conditions.



Heatmap



Parallel profiles

Order samples with the design in mind

Keep meaningful groups explicit; cluster within groups when appropriate.

Group samples by condition **Conditions are interleaved**

The annotation belongs to the sample and moves with it. Colour in the strip shows **condition**; colour in the cells shows **row z-score**.

> **When to cluster within groups**

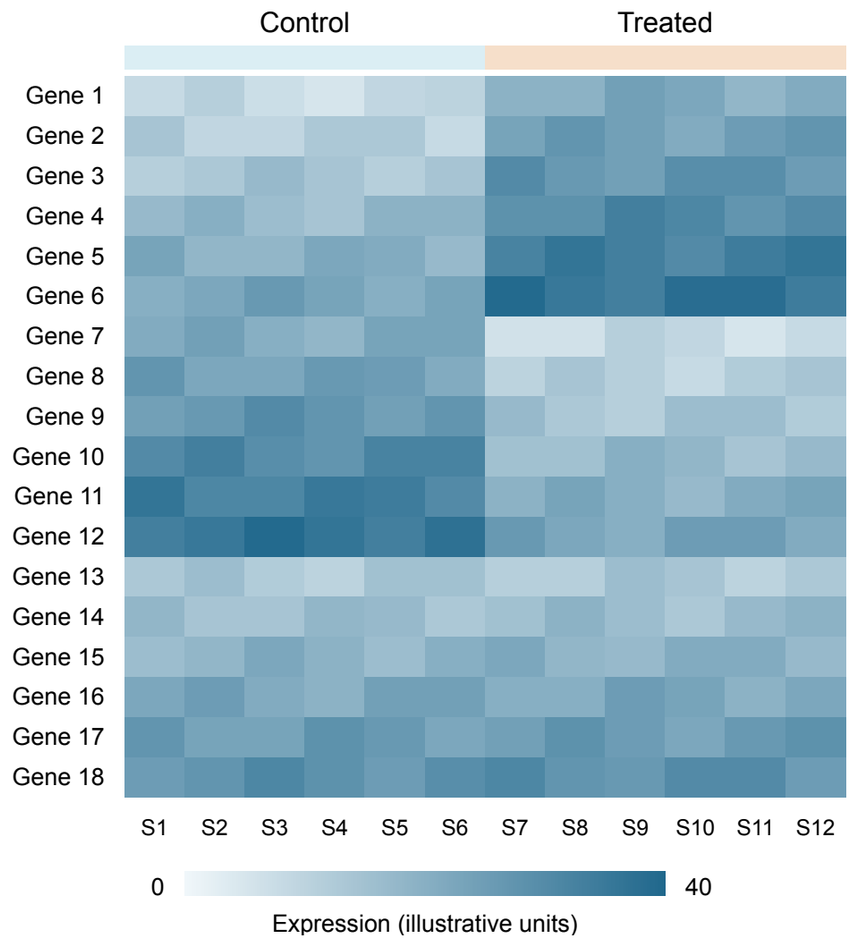
> **Original course example**

> **Reveal the design lesson**

Palette pitfalls

Rainbow and red–green scales can make patterns harder to read. Non-uniform lightness means equal changes in value do not look equally large: bright bands can stand out and suggest boundaries that are not in the data.

sequential | diverging | rainbow | Upper limit | Midpoint



Recommended for these data

Does the colour tell the right story?

Sequential: ordered magnitude.
Diverging: deviations from a meaningful reference.
Rainbow: hue changes can suggest boundaries.

18 genes × 12 samples · illustrative data. Hover a cell to read its value.

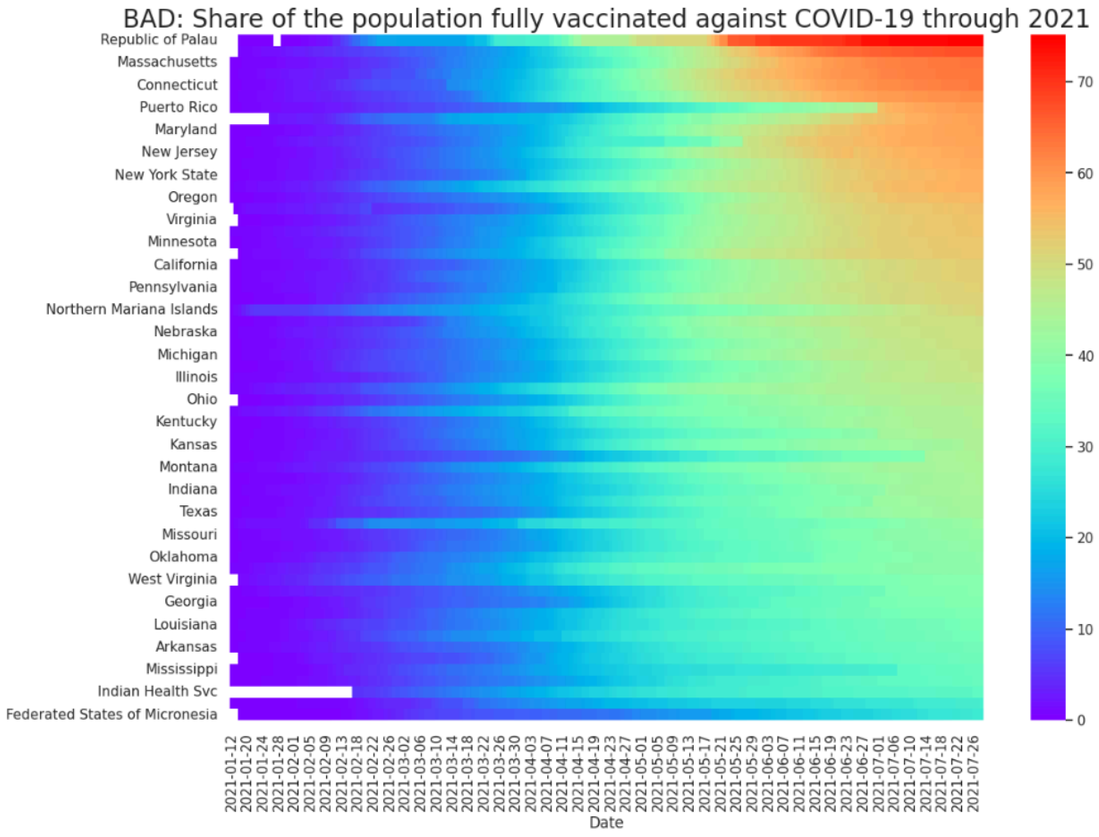
Lower the upper limit: values above it get the same end colour. For centred z-scores, choose a diverging palette instead.

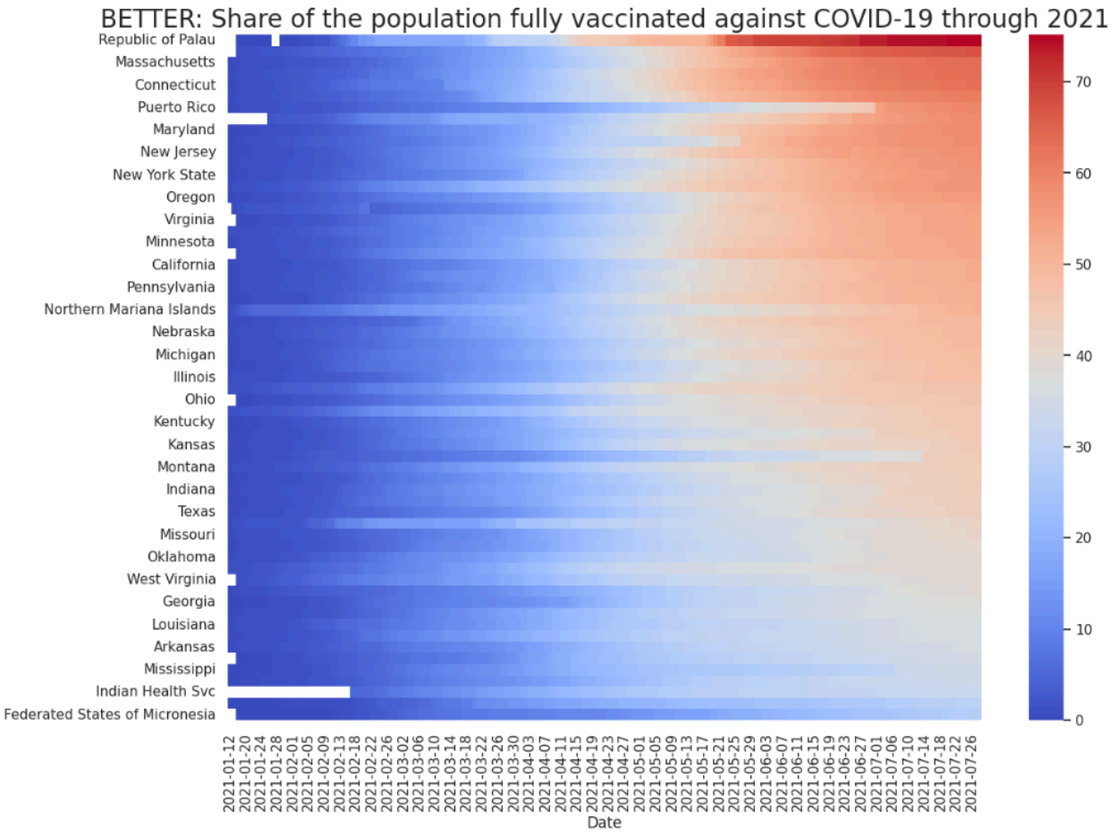
Fixed: every measurement.
Changed: only the colour mapping.

Original course example

Colour lightness affects emphasis

Bright colours can attract attention even when they do not represent the largest values.

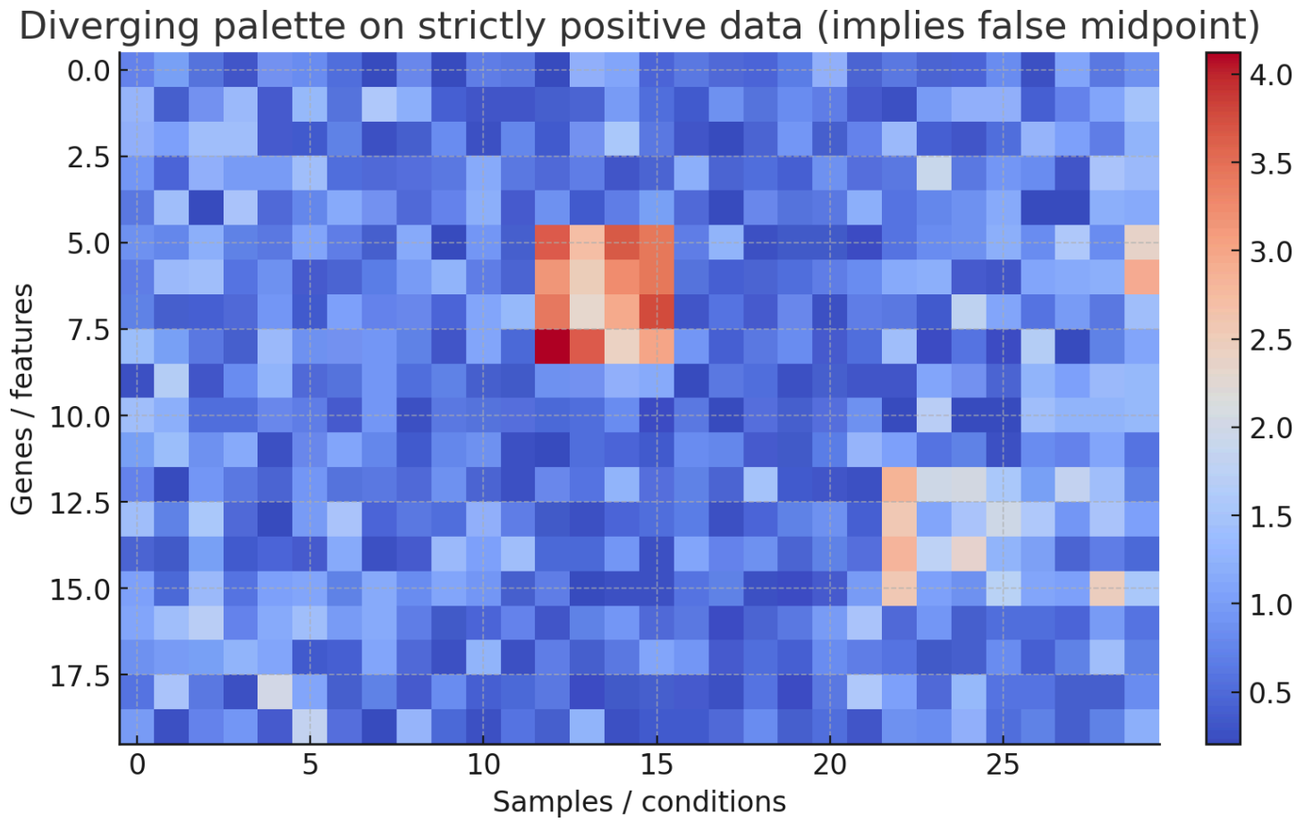




Palette comparison from the original deck

A diverging scale needs a meaningful centre

Use a sequential scale for ordered magnitudes unless there is a meaningful reference value.

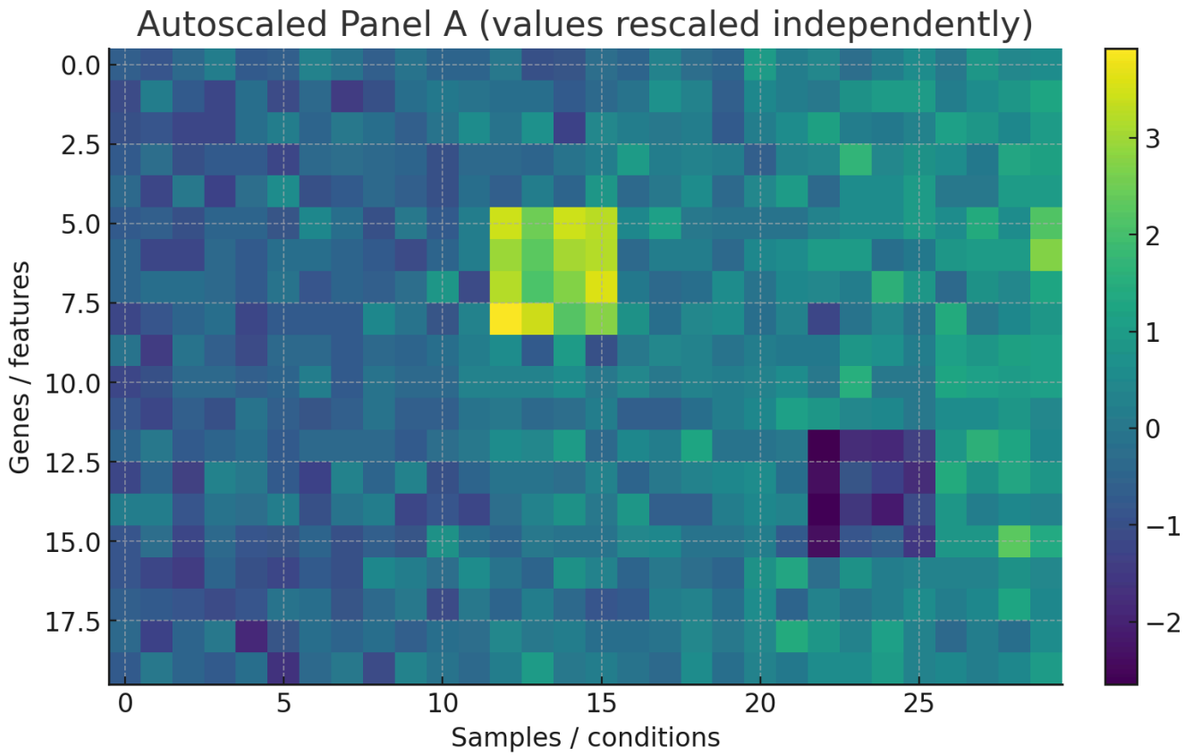


A diverging palette on positive values

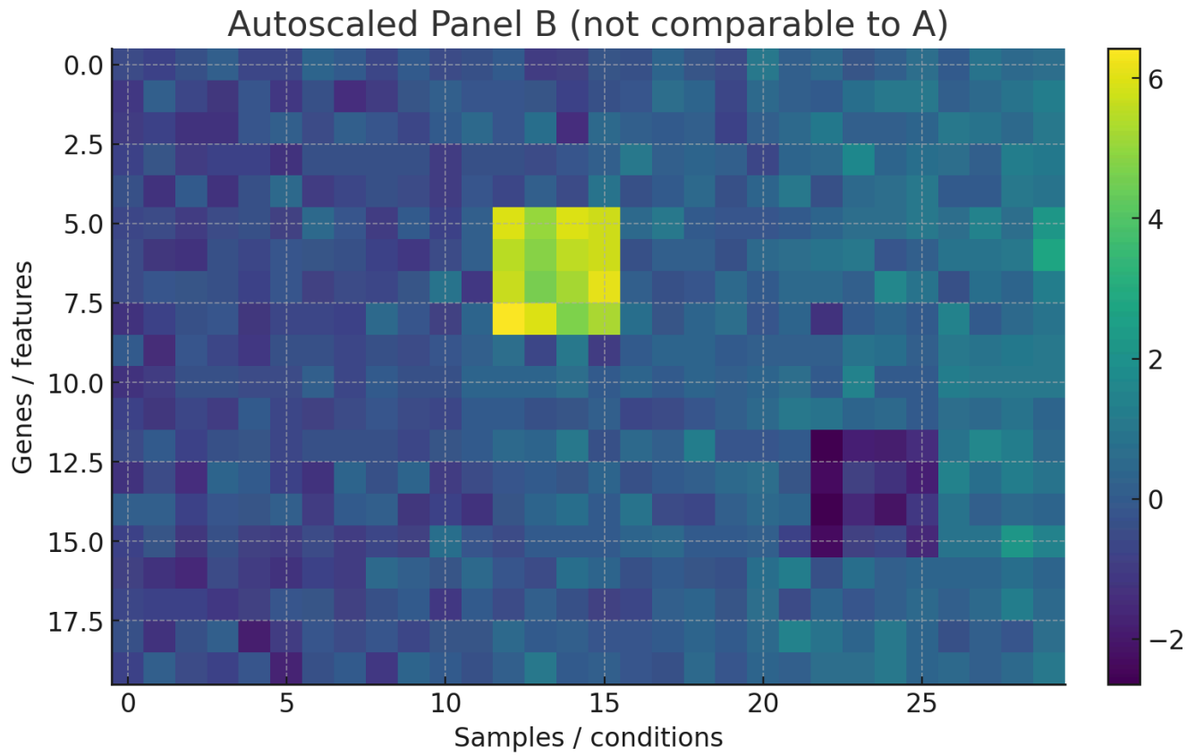
A meaningful centre can be a baseline other than zero. State it explicitly.

Comparable panels need comparable colour keys

The same colour should represent the same value when panels invite direct comparison.



Panel A · independent scale

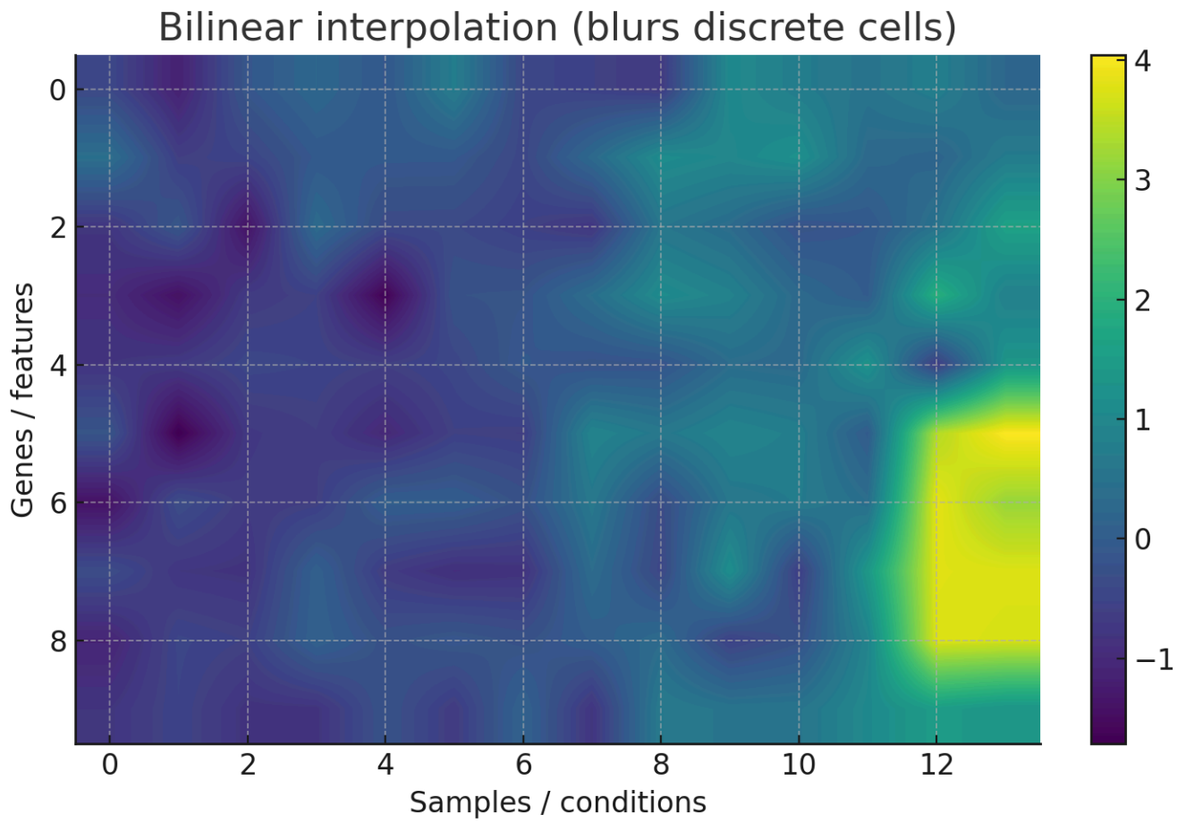


Panel B · independent scale

Reveal the design lesson

Interpolation changes the visual impression

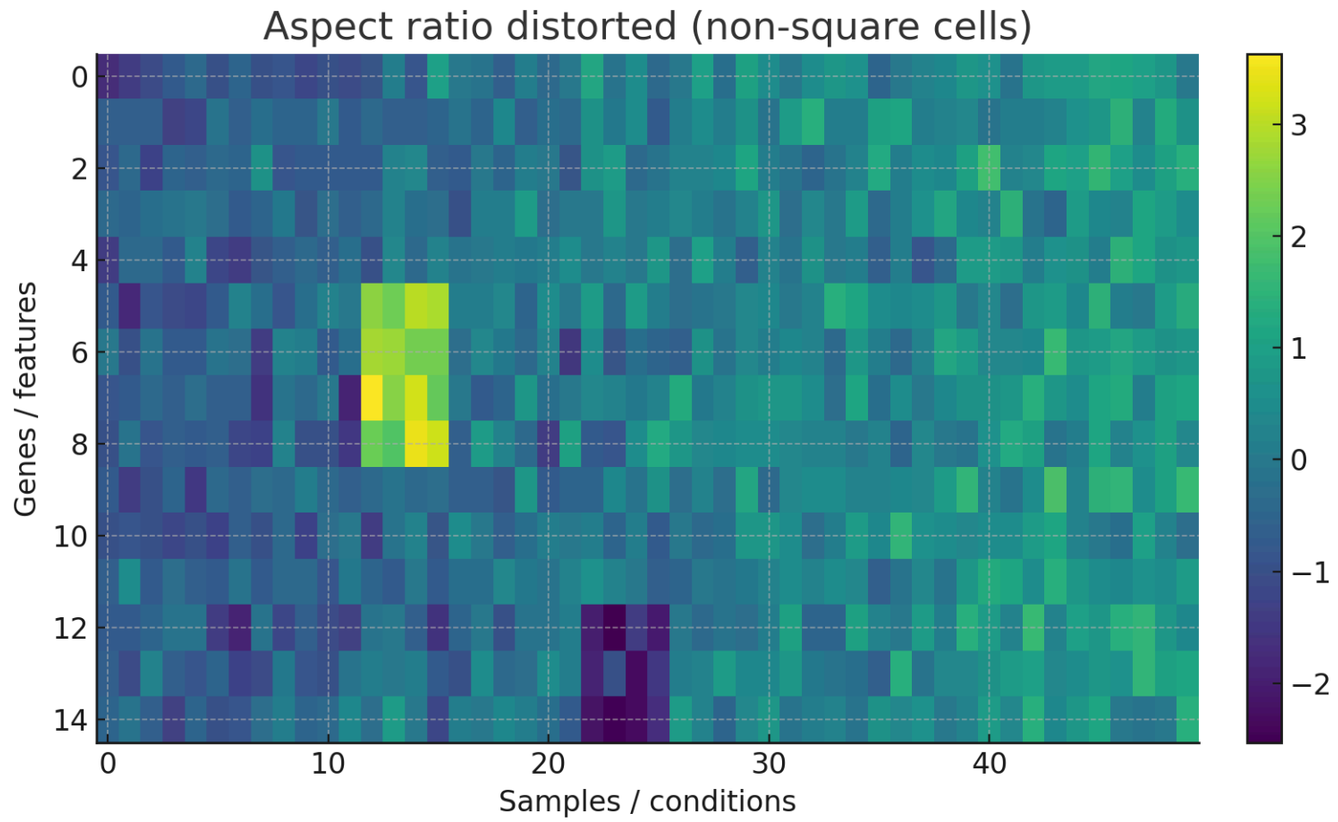
For discrete measurements, crisp cells show what was actually measured.



Bilinear interpolation suggests smooth transitions

Panel shape changes the appearance of blocks

Choose the aspect ratio deliberately and keep labels readable.

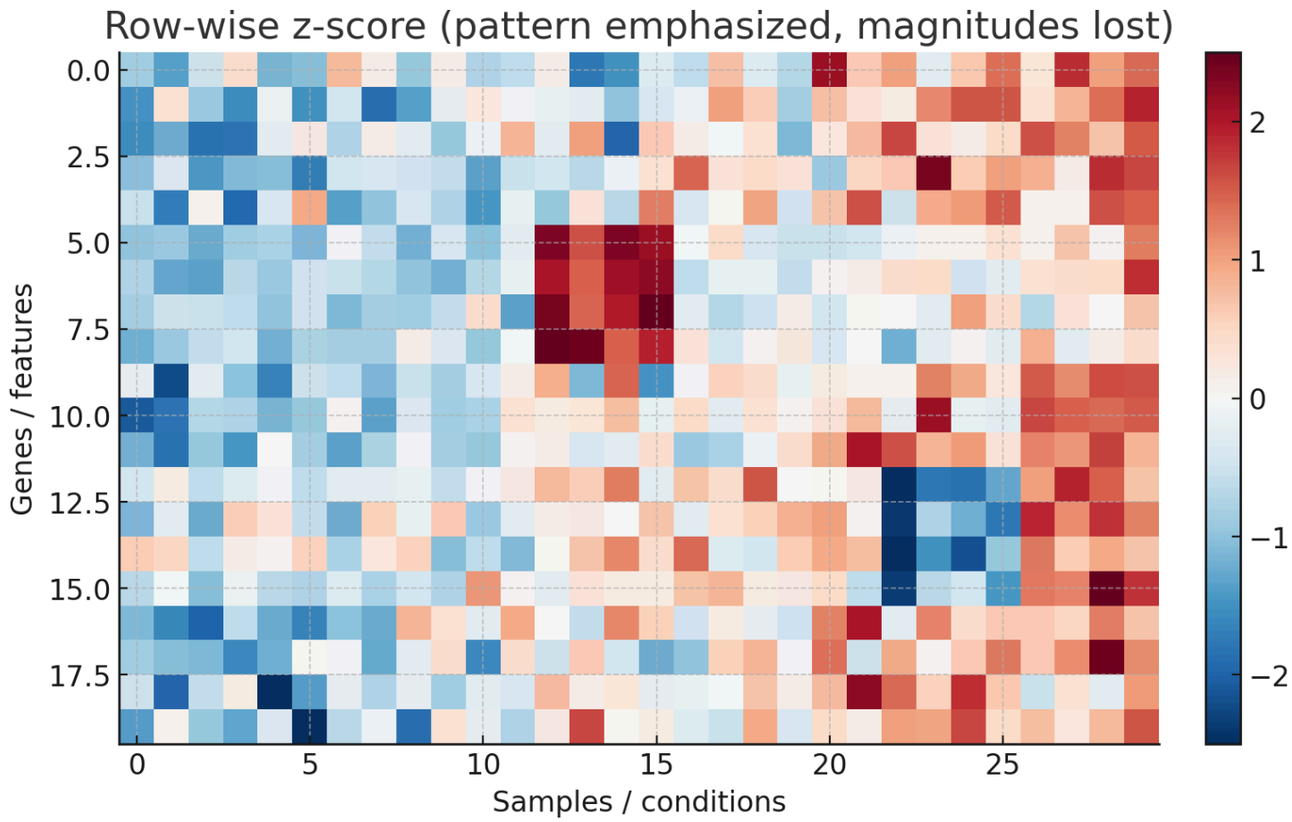


Stretched heatmap cells

Rectangular cells are not automatically wrong. Avoid inferring geometric relationships from arbitrary row and column spacing.

State what the colours now mean

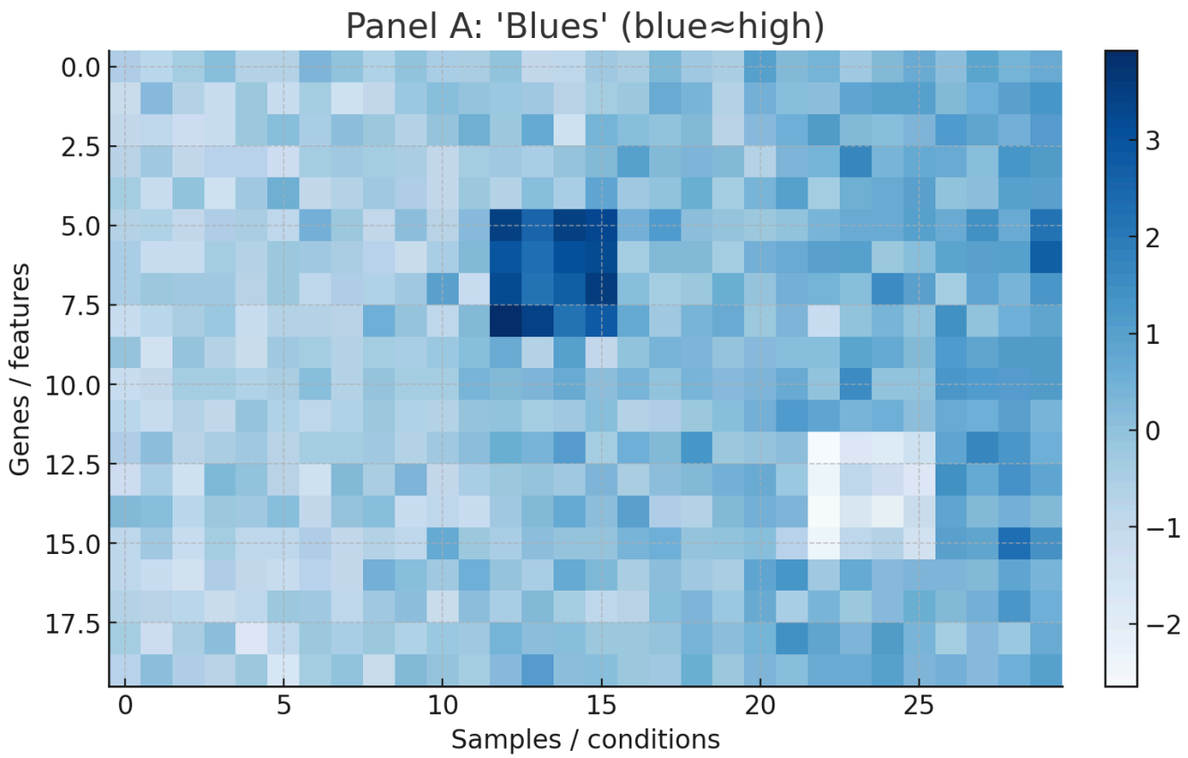
After row scaling, colours compare deviations within each row, not absolute levels between features.



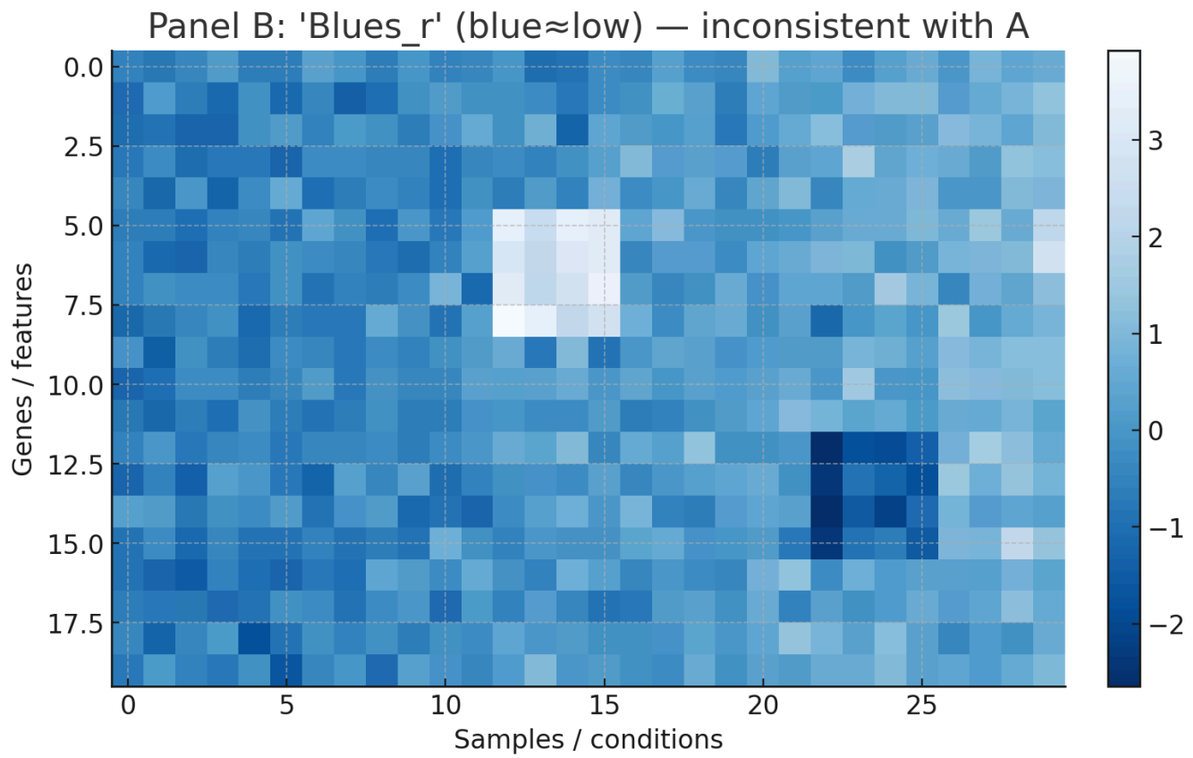
Row z-scores: pattern emphasised, absolute magnitude lost

Keep colour direction consistent

A dark cell should not mean high in one panel and low in the next without an explicit reason.



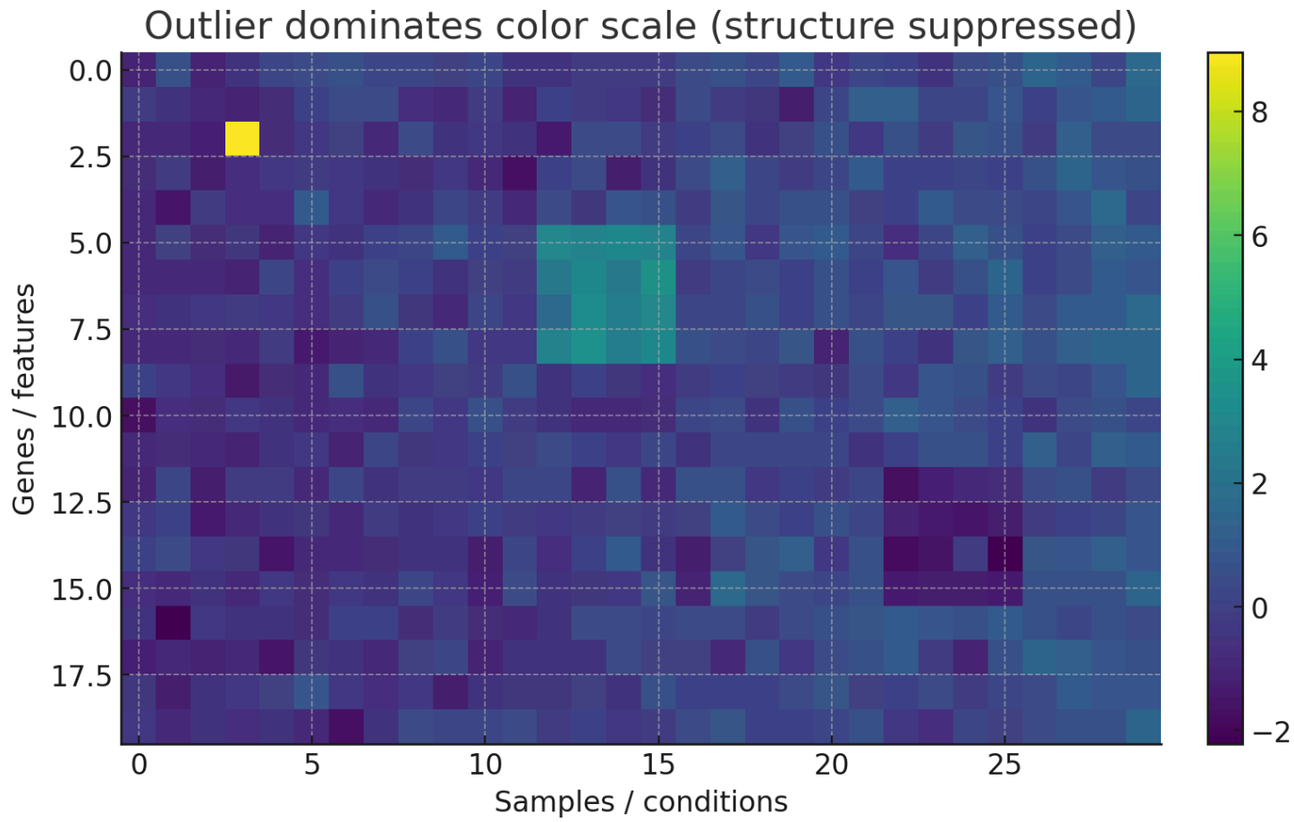
Dark blue means high



Dark blue means low

An extreme value can dominate the scale

Check whether a few extremes hide variation in the rest of the matrix.



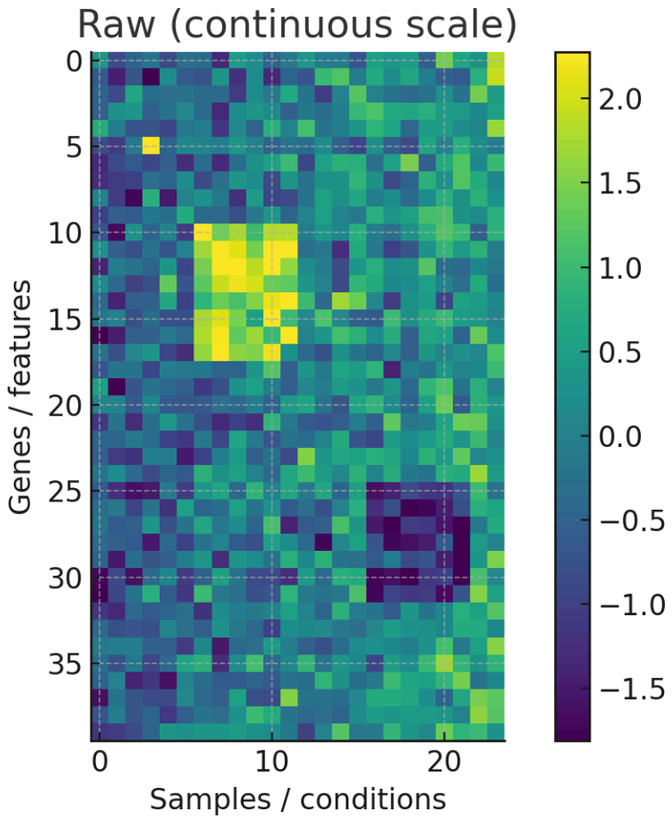
One extreme compresses the visible variation

Consider a justified transformation or disclosed colour limits. Keep the data intact and indicate values beyond the scale.

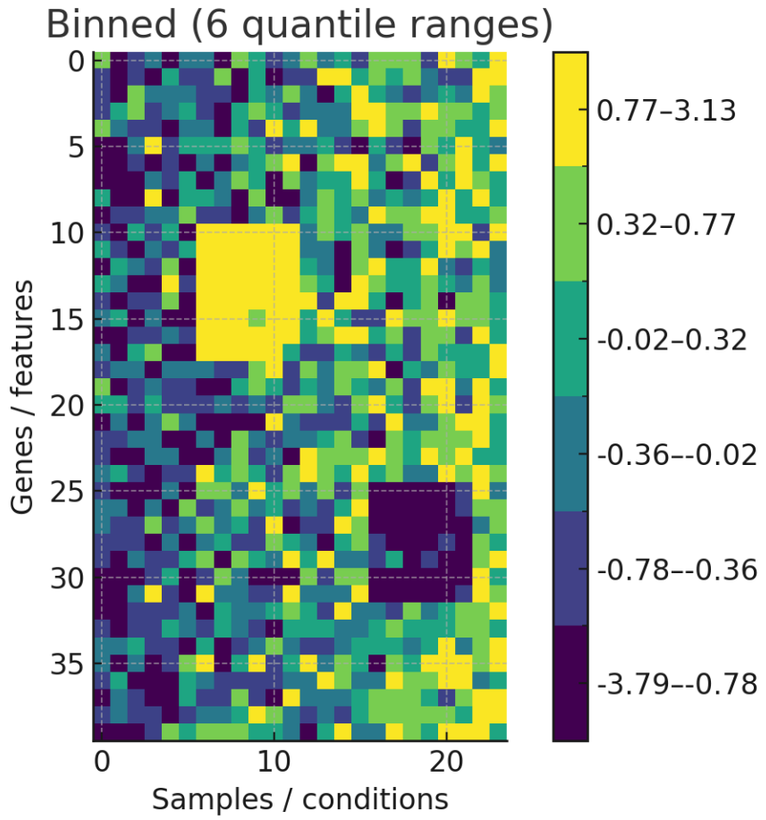
Reveal the design lesson

Continuous colours or bins?

Bins simplify the display, but values within each bin become visually indistinguishable.



Continuous colour scale

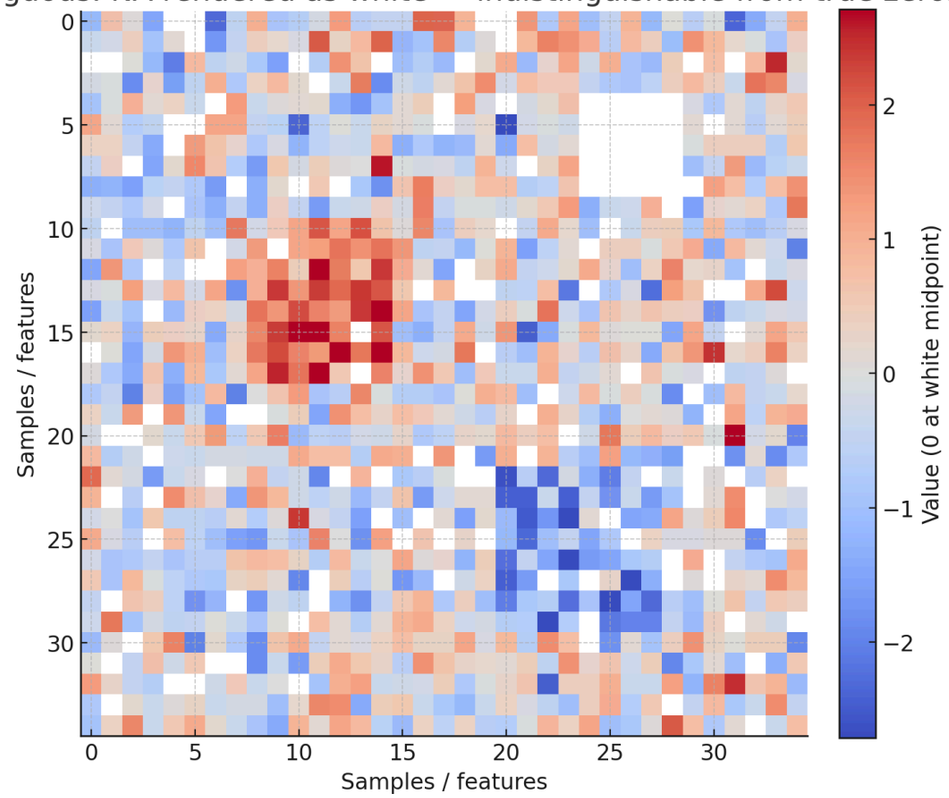


Six quantile bins

Missing values need their own appearance

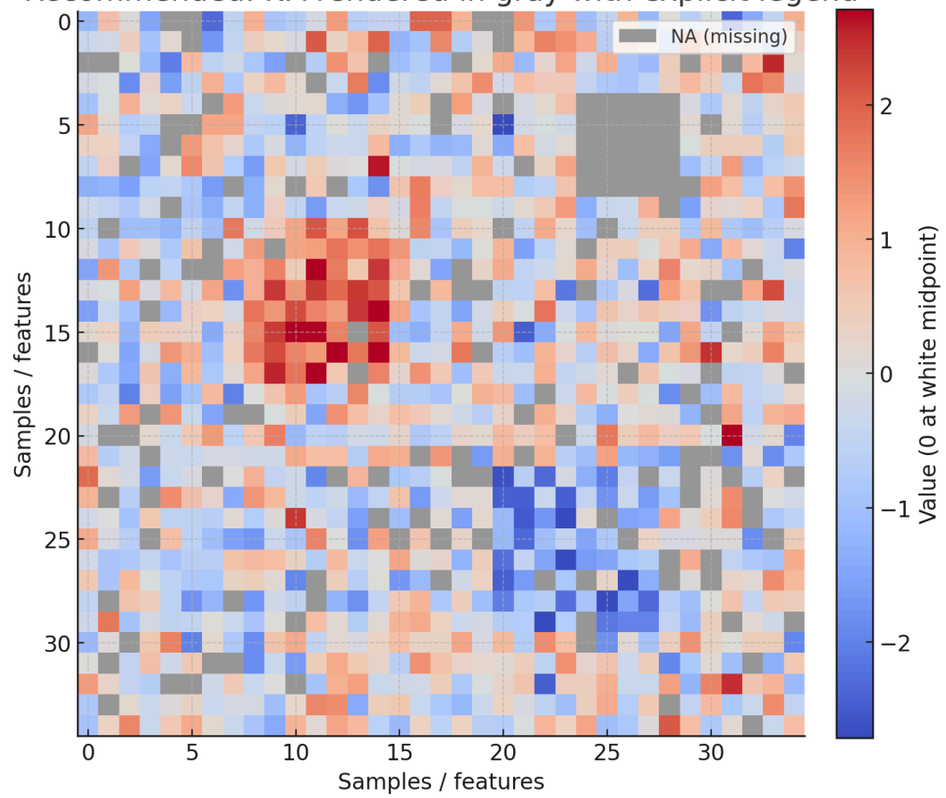
A missing measurement is not the same as zero.

Ambiguous: NA rendered as white — indistinguishable from true zeros



White is ambiguous

Recommended: NA rendered in gray with explicit legend

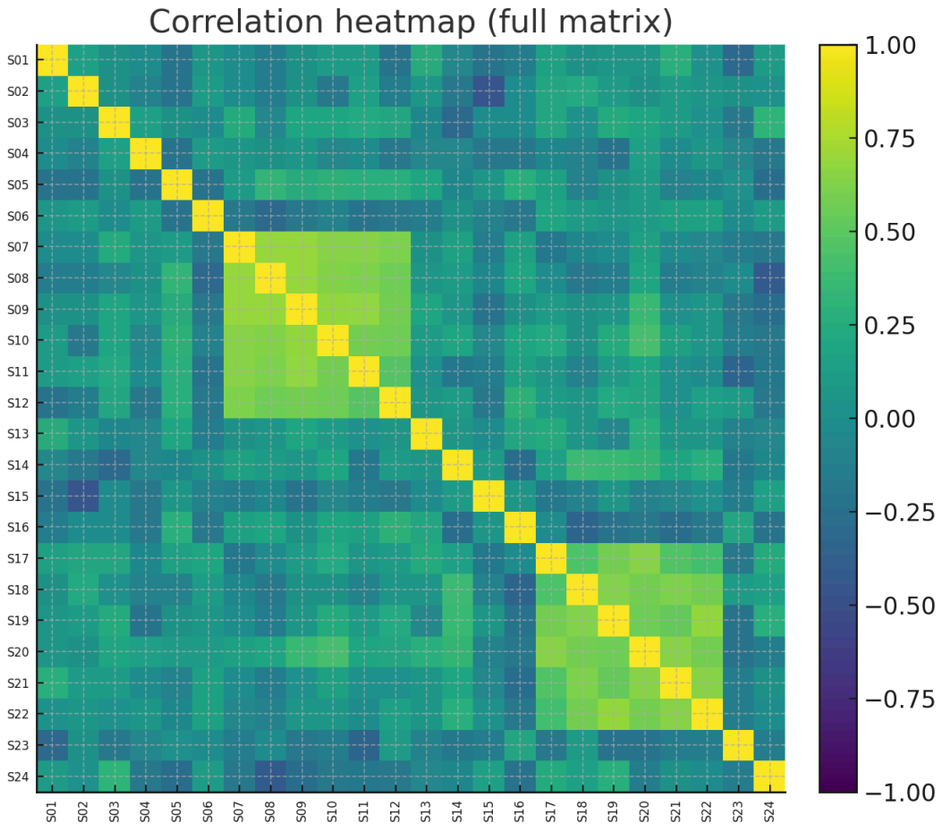


Grey identifies missing values

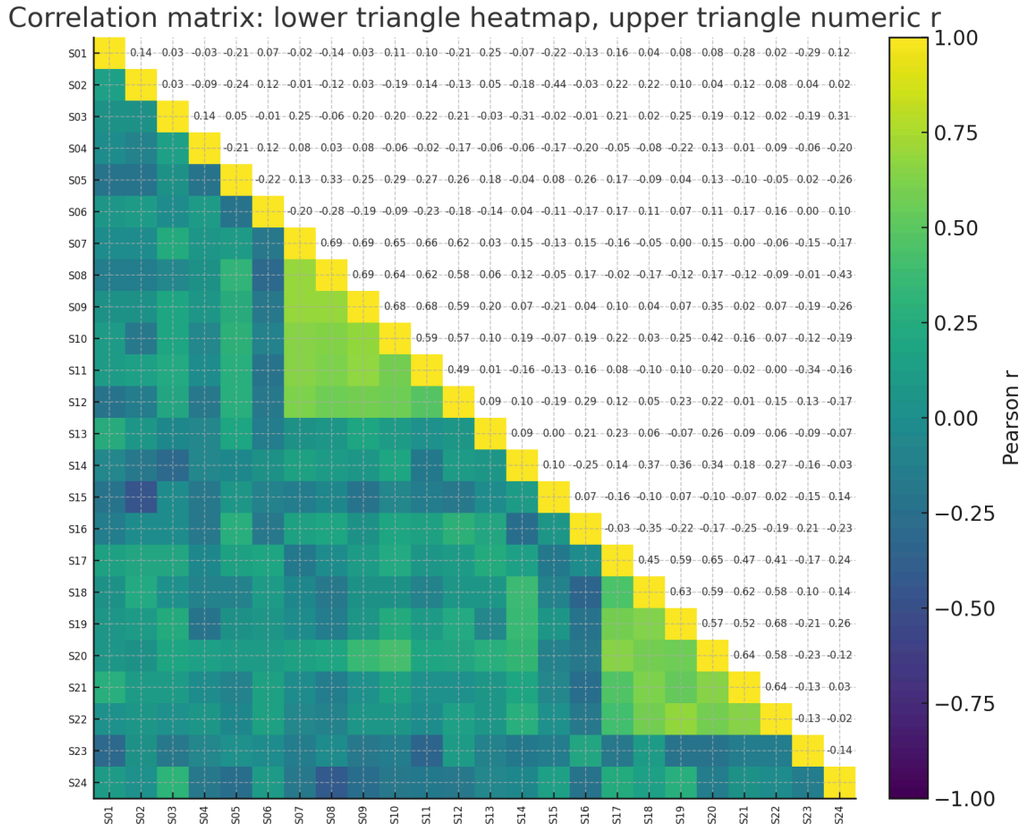
Reveal the design lesson

A symmetric matrix can show one triangle

When both halves repeat the same information, one triangle can leave room for values.



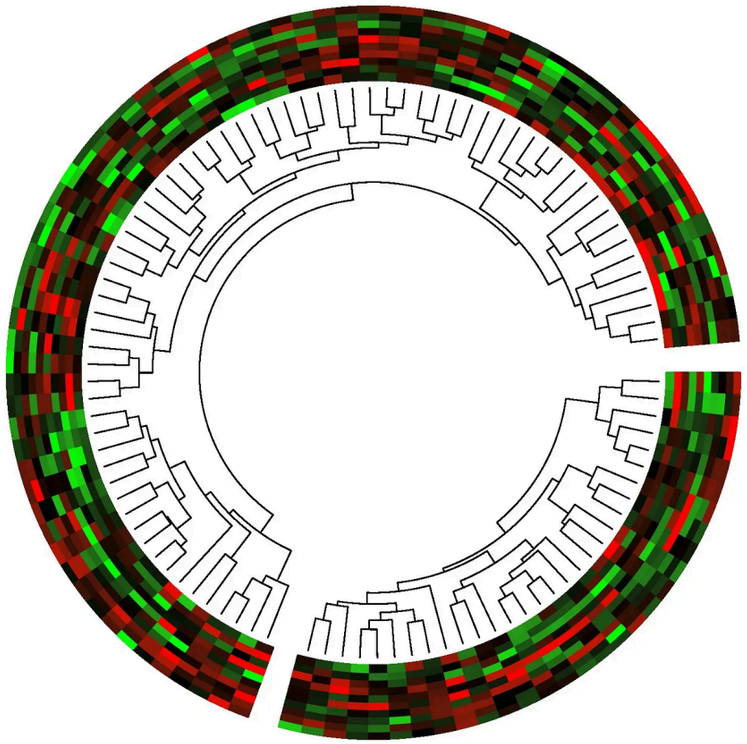
Full correlation matrix



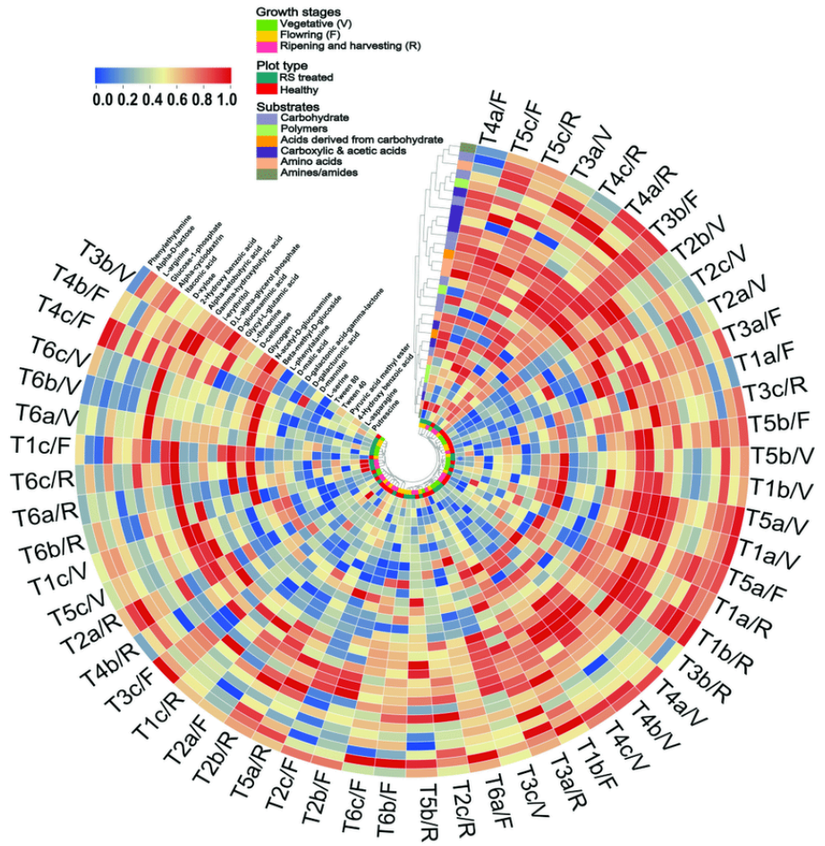
Triangle plus numeric values

Circular heatmaps

A circular layout can combine annotations and trees; compare its readability with a rectangular alternative.



Circular layout



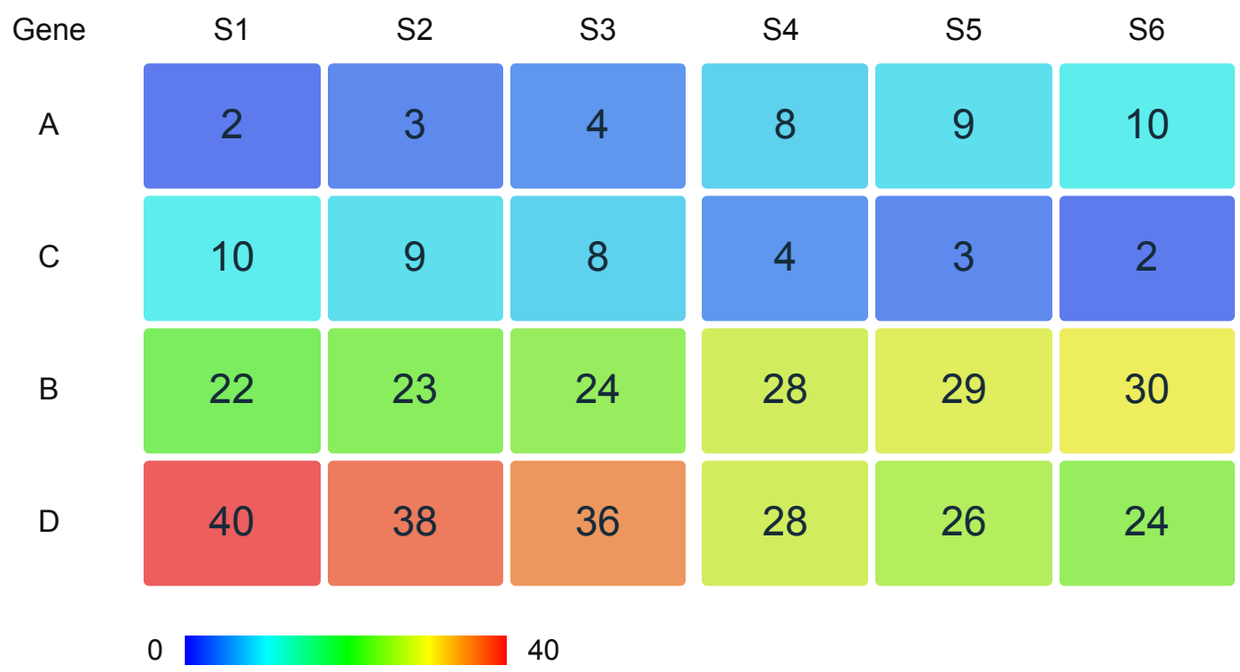
Circular annotations

Build a final heatmap

One dataset, one decision at a time.

1 / 7 · Starting figure

Same 4 genes × 6 samples · original values



Same observations throughout; each step changes one display decision.

Previous improvement

Next improvement

Show R / Python change

Original course example

Before you share: six heatmap checks

Can someone understand the figure without guessing?

▦

Values

What does one cell represent?

↦

Transformation

Raw, log or scaled, and why?

◐

Colour scale

Label units, limits and midpoint.

⑂

Ordering

Explain distance, linkage or design order.

▰

Annotations

Show conditions and missing values.

Aa

Readability

Check labels and colour key at final size.

› **Original course example**

Hands-on · build together, then explore in groups

Day 2 · Part 2 · Hands-on with ComplexHeatmap

First we build a simple heatmap together. Then you apply the same steps to a larger dataset.

1 · Together

Follow the live walkthrough

Dataset 1 · two bacteria
Akkermansia + Bacteroides

We fill the empty notebook, run each chunk and inspect the plot together.

`day2_walkthrough_empty.Rmd`
Reference: `day2_walkthrough.Rmd`

2 · Groups of 3–5

Try the exercise yourselves

Dataset 2 · adds Turicimonas
The original two bacteria + one more

Repeat the steps, inspect missing values and choose a clear design. AI assistance is welcome.

`day2_exercises.Rmd`
Start with Explore Data + Simple Heatmap.

3 · Share & review

One heatmap from each group

Show your plot and explain **one design choice** and **one challenge or surprise**.

We compare approaches, then walk through the relevant solution sections together.

`day2_exercises_solution.Rmd`

Keep the colour scale at 0–1 and show NA distinctly. Later, if time allows: add annotations and extend to dataset 3 with all six genomes.

> **Compare the supplied datasets**

Start the hands-on task → Open setup notebook ↗

Your task · build, compare, explain

Day 2 Part B · From the concepts to the R assignment

Next: set up RStudio together. The following slides walk through launching the server, connecting, finding the data, making an editable copy and opening the supplied files, with screenshots. Press **Next** to begin.

Dataset 1

24 SNPs × 64 samples
0 missing cells

Live demonstration · follow along.
I will showcase a heatmap with Akkermansia and Bacteroides.

Dataset 2

48 SNPs × 64 samples
186 missing cells

Your first assignment · work in groups.
Build a heatmap from dataset 2: it already includes Turicimonas alongside the original two bacteria.

Dataset 3

71 SNPs × 64 samples
443 missing cells

Optional extension.
Build a heatmap with all six supplied genomes.

Dataset 2 is ready to use: load its supplied CSV files to include Turicimonas in your heatmap. You do not need to add rows or combine files manually.

Keep allele-frequency colours fixed at **0–1**. Keep missing entries as **NA**, with a distinct colour. Explain a visible pattern without assuming a treatment effect.

> **Where is the code in the R Markdown notebooks?**

> **R · draw and export**

Practise the R functions → Guided notebook ↗ Your exercise ↗ Annotation walkthrough ↗

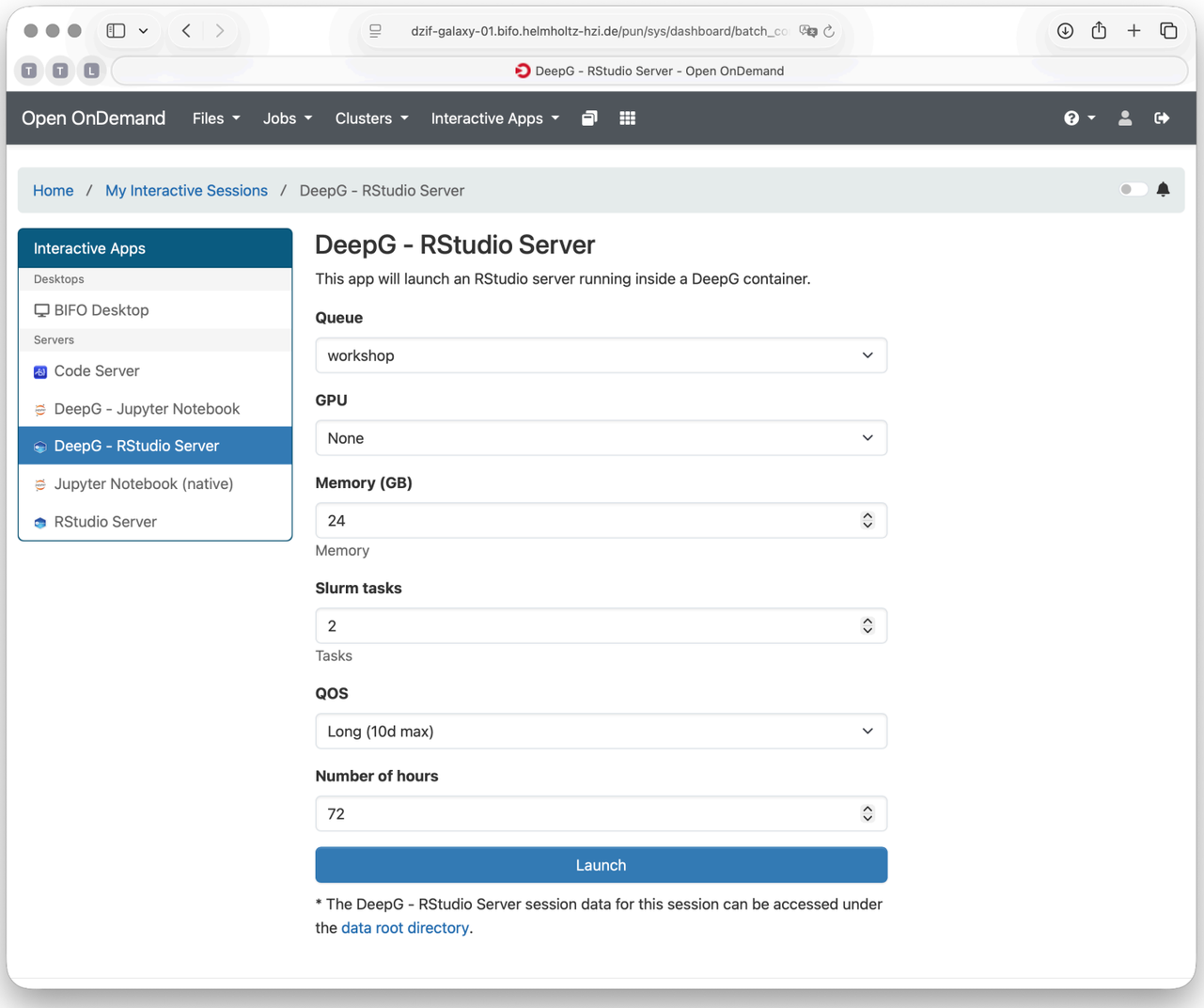
Open DeepG · RStudio Server

Day 2 · Part 2 · Hands-on with ComplexHeatmap

1. Go to dzif-galaxy-01.bifo.helmholtz-hzi.de ↗ and sign in.
2. Open **Interactive Apps → DeepG - RStudio Server**.
3. Leave the default settings unchanged, then click **Launch**.

Default settings · leave unchanged

Queue	**workshop**
GPU	**None**
Memory	**24 GB**
Slurm tasks	**2**
QOS	**Long (10d max)**
Number of hours	**72**



The launch page should look like this. Click to enlarge.

› FAQ · trouble opening RStudio?

Wait for your session · then connect

Day 2 · Part 2 · Hands-on with ComplexHeatmap

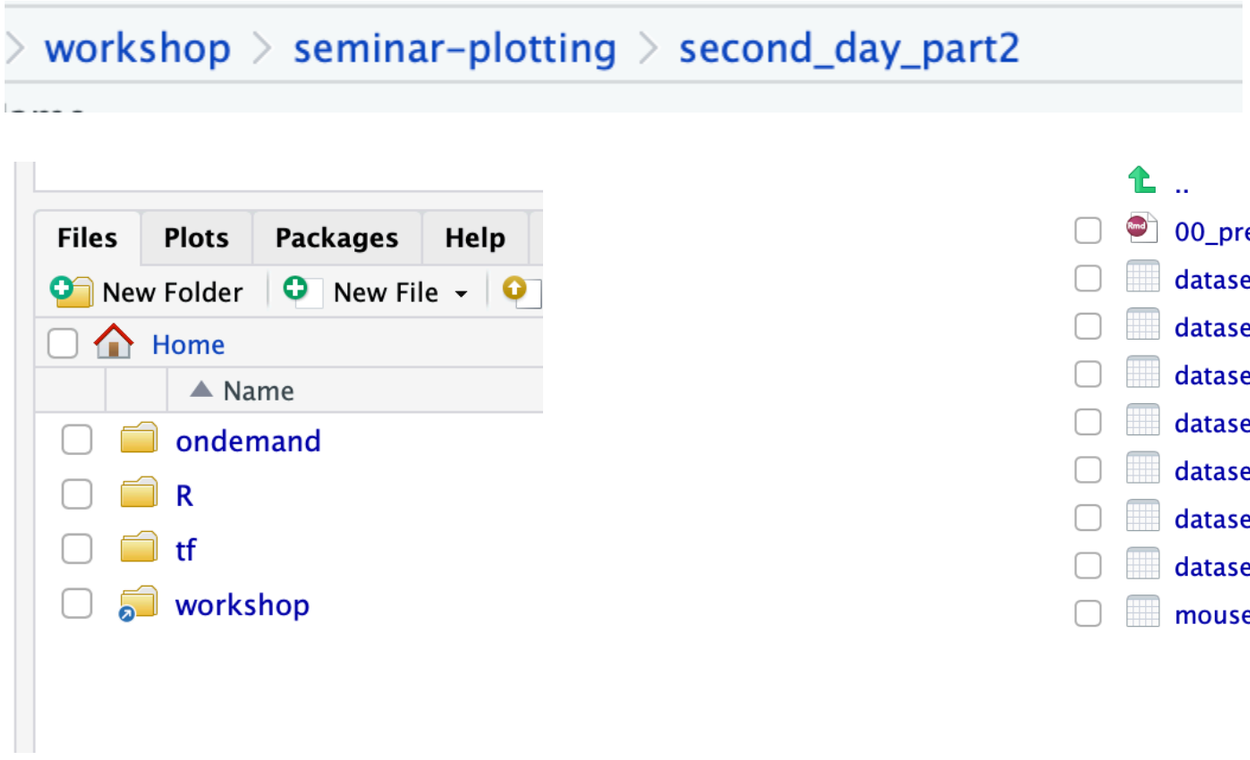
After clicking Launch: you will see “Your session is currently starting... Please be patient as this process can take a few minutes.”

Wait until the session is ready, then click **Connect to RStudio Server** to open RStudio.

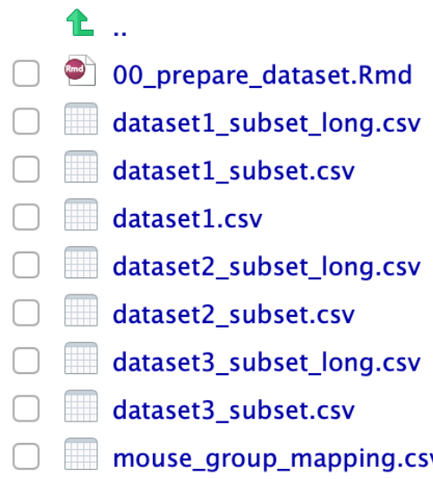
Find the workshop data in RStudio

Day 2 · Part 2 · Hands-on with ComplexHeatmap

In the **Files** pane, start at **Home** and open **workshop** → **seminar-plotting** → **second_day_part2** → **data**.



1 · Open the workshop folder, then follow the path above.



2 · Open data and inspect the supplied files.

Look for: `dataset1_subset.csv` (wide table), `dataset1_subset_long.csv` (long table), and the corresponding files for datasets 2 and 3. `mouse_group_mapping.csv` supplies mouse/group metadata.

Make your own editable copy

Day 2 · Part 2 · Hands-on with ComplexHeatmap

The shared workshop folder may be read-only. Before editing notebooks, copy **the whole second_day_part2 folder** into your own home folder. This keeps the notebook and data paths together.

1 · Run once in the R Console

```
dir.create("~/my-plotting", showWarnings = FALSE)

file.copy(
  from = "~/workshop/seminar-plotting/second_day_part2",
  to = "~/my-plotting",
  recursive = TRUE,
  copy.mode = FALSE,
  overwrite = FALSE
)
```

Wait for copying to finish. Check for warnings; on the first copy, the result should be `TRUE`. `overwrite = FALSE` protects files you have already copied and edited.

2 · Work in your copy

In the Files pane, open **Home** → **my-plotting** → **second_day_part2**.

- **scripts:** open `day2_walkthrough.Rmd` or your exercise notebook here and save your edits.
- **data:** inspect the CSVs here and use this folder for the next console examples.

`copy.mode = FALSE` avoids copying the original file permission bits.

Already made edits? Use **File** → **Save As** to save your open notebook into the copied `scripts` folder before closing it.

Continue saving PDF exports to `~/pdf`.

Install once · load in each R session

Day 2 · Part 2 · Hands-on with ComplexHeatmap

R packages add functions to R and can come from different repositories (collections of packages).

1 · Install

```
if (!requireNamespace("BiocManager", quietly = TRUE)) {
  install.packages("BiocManager")
}
BiocManager::install("ComplexHeatmap")
install.packages(c("circlize", "viridisLite"))
```

CRAN is R's main package repository. Use `install.packages()` for packages such as circlize and viridisLite.

Bioconductor provides packages for biological data analysis, including ComplexHeatmap. Install these with `BiocManager::install()`.

BiocManager is the installer, not the repository. It is itself installed from CRAN in the first block above. Run installation before class.

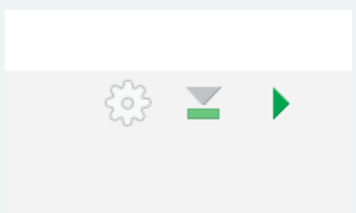
2 · Load

```
library(ComplexHeatmap)
library(circlize)
library(viridisLite)
```

Installing saves a package on your computer; `library()` loads it for use in the current R session, regardless of where it came from.

R is case-sensitive: `Heatmap()` has a capital H.

Official package and installation page ↗



Run a code chunk: in an open R Markdown notebook, click the green ▶ button at the top right of the chunk. It runs all the code in that chunk. Work from top to bottom so that objects created earlier are available to later chunks.

Knitting an HTML notebook also needs Pandoc; exporting a plot with pdf() does not.

Start with the supplied files

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Ready-to-use CSV files

`dataset1_subset.csv`
Wide: SNP rows, sample columns.

`dataset1_subset_long.csv`
Long: one SNP/sample pair per row, with metadata.

The datasets are already prepared. Rebuilding the publication table is optional.

Next: practise the R functions with dataset 2 →

Compare datasets 1, 2 and 3 →

Read and inspect

```
data_dir <- path.expand("~/my-plotting/second_day_part2/data")
wide_df <- read.csv(file.path(data_dir, "dataset1_subset.csv"),
                    check.names = FALSE)
long_df <- read.csv(file.path(data_dir, "dataset1_subset_long.csv"))
head(wide_df[, 1:6])
dim(wide_df)
str(wide_df)
```

These Console examples use your copied data folder explicitly; no working-directory change is needed. The notebooks can keep their original relative paths.

Problems importing data?

read.csv() · bring the data into R

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Start with dataset 2 so we can practise with real missing values.

Hover, focus or click a code fragment to explain it. Only the selected fragment is highlighted.

Try it in R

```
data_dir <- path.expand("~/my-plotting/second_day_part2/data")
long_df <- read.csv(
  file.path(data_dir, "dataset2_subset_long.csv"),
  check.names = FALSE
)
wide_df <- read.csv(
  file.path(data_dir, "dataset2_subset.csv"),
  check.names = FALSE
)
dim(long_df)
```

> **Reveal expected output**

> **Copy the complete R code**

> **Problems importing data?**

What does this code do?

The long table has 3,072 SNP/sample rows and seven columns. check.names = FALSE preserves sample names such as 1683-0 in the wide table.

The explicit data_dir points to your personal copy, so these Console examples do not require setwd(). Run these examples in order; they reuse long_df, wide_df and later mat.

Your turn

Which file would you use to inspect metadata, and which to build the heatmap matrix?

head() · look before you plot

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Inspect a few rows and check what one measurement represents.

Hover, focus or click a code fragment to explain it. Only the selected fragment is highlighted.

Try it in R

```
head(long_df[, c("snp_id", "mouse_id", "day", "value")], 3)
```

> **Reveal expected output**

> **Copy the complete R code**

What does this code do?

head(..., 3) shows the first three rows. c(...) selects four columns by name; the empty space before the comma keeps all rows before head() selects the first three.

Each row is one SNP measured in one mouse/day sample. A frequency of 0 is an observed value.

Your turn

Change 3 to 6. Can you identify the SNP, sample and frequency in one row?

str() · check the column types

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Numbers and labels have different jobs in a heatmap.

Hover, focus or click a code fragment to explain it. Only the selected fragment is highlighted.

Try it in R

```
str(long_df)
```

Shortened str() output · select a column type to explore it.

```
3072 observations; 7 variables
Genome          : character
snp_id          : character
Position        : integer
value           : numeric
mouse_id        : integer
day             : integer
treatment_group : character
```

> **Copy the complete R code**

What does this code do?

This is a shortened summary of str() output. value is numeric; genome and treatment are labels. A numeric mouse ID still identifies a mouse, it is not a measurement.

Only sample frequency columns belong in the heatmap matrix. Numeric metadata such as Position must also be excluded.

Your turn

Which column provides the cell colours? Which columns describe the rows and samples?

[rows, columns] · select the observations you need

Hover or click the code: two conditions, and the rows they share.

& AND · both | OR · either ! NOT · exclude Turicimonas

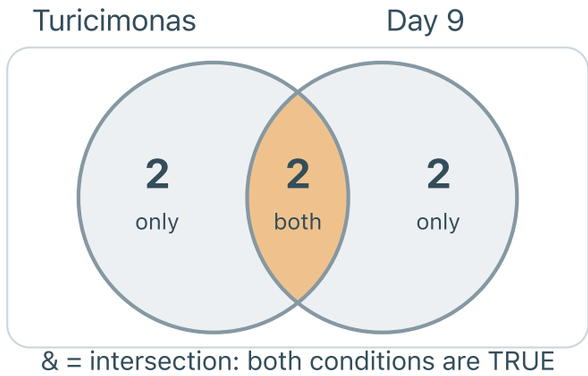
Keep rows from Turicimonas that were measured on day 9.

Read the code in pieces

```
selected <-  long_df[
  long_df$Genome == "Turicimonas_muris_YL45" &
    long_df$day == 9 ,
]
nrow(selected)
```

AND: intersection

Keep a row if both conditions are TRUE.



AND: just the overlap = 2 rows

Each number counts a separate region, not the whole circle. Six example rows in total.

Copy the complete R code

Which rows pass this test?

Illustrative excerpt · names shortened · each row is one SNP/sample measurement.

Genome	Day	Pass?
Akkermansia	9	FALSE
Turicimonas	0	FALSE
Turicimonas	9	TRUE · keep
Bacteroides	9	FALSE
Turicimonas	9	TRUE · keep
Turicimonas	30	FALSE

2 of these 6 example rows pass both conditions.

Reveal the full dataset result

Your turn: predict what changes if you use `day == 30`, then try it in R.

table() · count rows by group

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Check how many measurement rows belong to each genome.

Hover, focus or click a code fragment to explain it. Only the selected fragment is highlighted.

Try it in R

```
table(long_df$Genome)
```

> **Reveal expected output**

> **Copy the complete R code**

What does this code do?

Each SNP has 64 sample rows: 5 × 64, 19 × 64 and 24 × 64. table(Genome) counts the genome labels, including rows with a missing frequency.

These are measurement-row counts, not biological replicate counts. The dataset has four mice measured on 16 days.

Your turn

Try table(long_df$day). Why does each day have 192 rows?

is.na() · find missing measurements

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Missing is different from a measured zero.

Hover, focus or click a code fragment to explain it. Only the selected fragment is highlighted.

Try it in R

```
is.na(long_df$value)  # TRUE where a value is missing
sum(is.na(long_df$value))
sum(is.na(selected$value))
```

> **Reveal expected output**

> **Copy the complete R code**

What does this code do?

is.na() returns TRUE/FALSE for each entry; sum() counts TRUE values. The second count assumes selected still contains Turicimonas on day 9.

Use is.na(x), not x == NA. Preserve these NA values. Later, na_col = "grey80" gives missing cells a distinct colour; it does not fill them in.

Your turn

Compare sum(is.na(long_df$value)) with sum(long_df$value == 0, na.rm = TRUE). Why are these different questions?

as.matrix() · prepare numeric heatmap input

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Use the wide table, removing metadata before conversion.

Hover, focus or click a code fragment to explain it. Only the selected fragment is highlighted.

Try it in R

```
sample_cols <- setdiff(
  names(wide_df), c("Genome", "snp_id", "Position")
)
mat <- as.matrix(wide_df[, sample_cols])
rownames(mat) <- paste(wide_df$Genome, wide_df$snp_id, sep = " | ")
dim(mat)
is.numeric(mat)
```

> **Reveal expected output**

> **Copy the complete R code**

What does this code do?

setdiff() keeps column names that are not metadata. as.matrix() converts those frequency columns to a matrix; missing values remain NA.

Including text metadata can turn the entire matrix into text. Keep Genome and Position separately for annotations, and SNP identities in row names.

Your turn

Why does wide_df have 67 columns, while mat has only 64?

colSums() · count missing cells per sample

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Summarise data completeness before choosing a heatmap design.

Hover, focus or click a code fragment to explain it. Only the selected fragment is highlighted.

Try it in R

```
missing_per_sample <- colSums(is.na(mat))

head(missing_per_sample, 4)

sum(missing_per_sample)
```

> **Reveal expected output**

> **Copy the complete R code**

What does this code do?

is.na(mat) creates a TRUE/FALSE matrix. colSums() counts missing SNP frequencies in each sample column. The total matches the long-table count.

These counts can later become an annotation with anno_barplot(missing_per_sample). Keep their order aligned with the heatmap columns. A count of zero means no missing entries, not zero allele frequencies.

Your turn

Try rowSums(is.na(mat)). What does each number describe now?

Wide and long · the same data

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Same nine measurements, arranged two ways. Click any value to find it in the other table.

Genome: Akkermansia muciniphila YL44 · **Mouse:** 1683 · **Treatment:** Control

Wide · one row per SNP

Each sample gets its own column.

SNP	1683-0	1683-4	1683-9
239840-C-G	0	0	0
241793-A-G	0.049587	0.031414	0.076271
355328-A-T	0.138182	0.132275	0.138211

3 SNP rows × 3 sample columns = **9 values**

Select only the sample columns to make the numeric matrix for Heatmap(). Keep SNP identities as row names.

Long · one row per measurement

Sample identity and value get their own columns.

SNP	Mouse	Day	Value
239840-C-G	1683	0	0
239840-C-G	1683	4	0
239840-C-G	1683	9	0
241793-A-G	1683	0	0.049587
241793-A-G	1683	4	0.031414
241793-A-G	1683	9	0.076271
355328-A-T	1683	0	0.138182
355328-A-T	1683	4	0.132275
355328-A-T	1683	9	0.138211

9 rows = the same 9 values

Choose a value. Nothing is averaged or lost when reshaping.

> **Where is the rest of the dataset?**

From a table to a matrix

Day 2 Part B · From the concepts to the R assignment

Long table · inspect and filter

SNP	Mouse	Day	Value
239840-C-G	1683	0	0
241793-A-G	1683	0	0.049587
355328-A-T	1683	0	0.138182

Actual rows for Akkermansia from dataset 1. One row per SNP/sample pair.

Wide table → numeric matrix

Rows = genome + SNP

Columns = mouse/day sample IDs

Cells = allele frequencies

Keep Genome, SNP ID and Position as row metadata. Keep mouse, day and treatment as column metadata.

R · the matrix-building step

Your first Heatmap() call

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Create → draw

```
# 1. Create the heatmap object and store it in ht
ht <- Heatmap(mat, name = "Allele frequency")

# 2. Display the stored heatmap
draw(ht)
```

`<-` assigns the result to a variable. Here, `ht` holds the heatmap object: its data and plotting settings. You can choose another name, but use the same name in `draw()`.

Assignment alone does not display the plot. Run `draw(ht)` to show it in RStudio's **Plots** pane. Storing the object in `ht` does not save a PDF or image file.

Use `Heatmap()` with a capital **H** for ComplexHeatmap; lowercase `heatmap()` is a different R function.

`mat` is a numeric matrix. Row and column names become labels.

Defaults cluster rows and columns. They are a starting point; choose colours and ordering for your question.

What the matrix contains

Akkermansia · actual subset of dataset 1

SNP / mouse-day	1683-0	1683-4	1683-9	1683-14
239840-C-G	0.000	0.000	0.000	0.000
241793-A-G	0.050	0.031	0.076	0.048
355328-A-T	0.138	0.132	0.138	0.113
356291-C-A	0.000	0.000	0.000	0.000

Displayed values rounded to 3 decimals; colour uses the full value on a fixed 0–1 scale.

Preview of input values, not a rendering of the default Heatmap() output.

Inside Heatmap() · settings have different jobs

Day 2 · Part 2 · Hands-on with ComplexHeatmap

```
ht <- Heatmap(
  mat,
  name = "Allele frequency",
  col = col_fun, na_col = "grey80",
  cluster_rows = TRUE,
  cluster_columns = FALSE,
  show_row_names = FALSE,
  column_names_rot = 90,
  top_annotation = ha
)
draw(ht)
```

Data: the numeric matrix.

Colour: values, missing cells and legend title.

Order: cluster genes; preserve the prepared sample order.

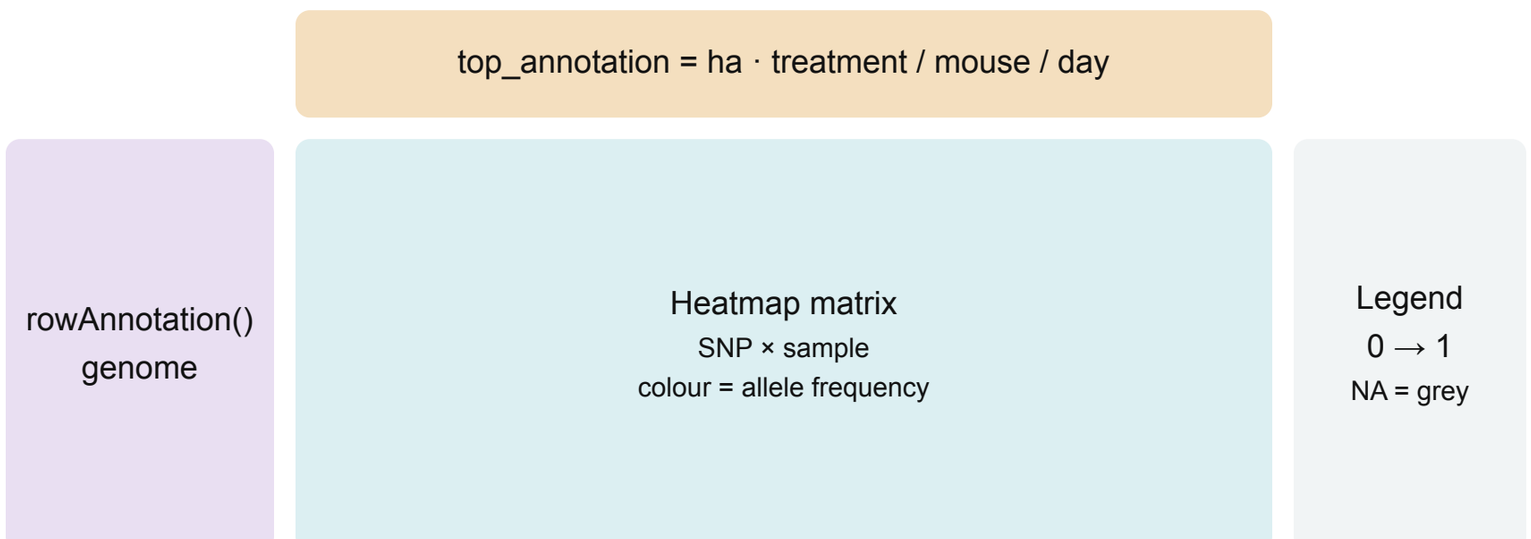
Labels: visibility and rotation.

Context: sample annotations above columns.

> **Define the colour function first**

Annotations describe the rows and columns

Day 2 · Part 2 · Hands-on with ComplexHeatmap



Annotation colours describe metadata; cell colours describe measurements. Keep their legends separate.

`column_split` creates sample blocks; `row_split` creates row blocks. An annotation strip alone does not split or reorder the matrix.

Keep sample annotations attached to the right columns

Day 2 Part B · From the concepts to the R assignment

Matrix column names → match sample IDs → reorder metadata → add annotations

Design order

Order by mouse, then day to follow trajectories.

Use treatment, mouse and day as annotation strips.

Clustering answers a similarity question; it will not necessarily preserve time order.

Check alignment before plotting

```
sample_info <- sample_meta[
  match(colnames(mat), sample_meta$sample_id), ]
stopifnot(identical(
  colnames(mat), sample_info$sample_id))

ha <- HeatmapAnnotation(
  treatment = sample_info$treatment_group,
  mouse = sample_info$mouse_id,
  day = sample_info$day)
```

Reordering a matrix without reordering its metadata can attach the wrong treatment label to a sample.

Use the illustrated reference when you need a setting

Day 2 · Part 2 · Hands-on with ComplexHeatmap

Single heatmap

Colours, labels, clustering and splits.

Open illustrated chapter ↗

Annotations

Sample strips, row metadata and legends.

Open annotation examples ↗

Help inside R

```
?ComplexHeatmap::Heatmap
browseVignettes("ComplexHeatmap")
```

Look up one argument, then change it and inspect the result.

Hands-on heatmaps

Continue with the existing Day 2 Part 2 notebooks.

Explore the data ↗

Build a simple heatmap ↗

Add annotations ↗

Build the complete heatmap ↗

Exercises ↗

From plot to publication

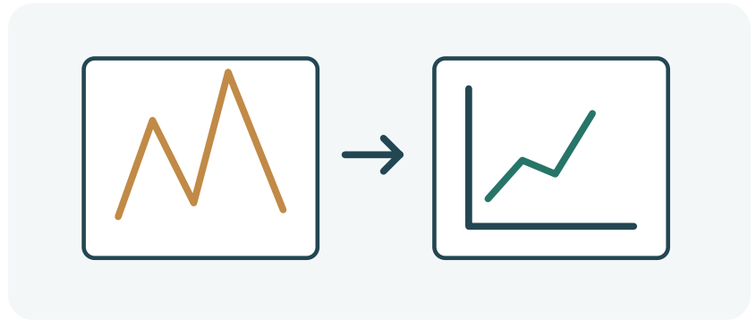
Choose a topic to explore, improve and assemble scientific figures.

1 What kinds of plots are there?

Explore plot families and the questions they answer.

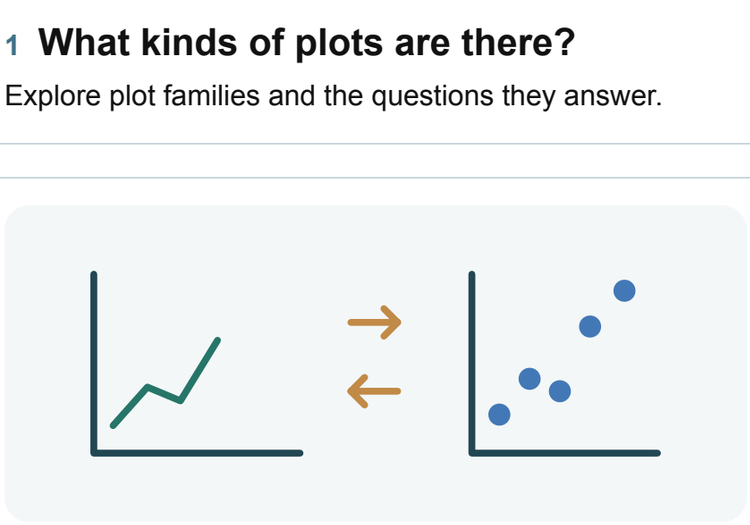
2 Examples for good figures

Inspect design choices in real biological research.



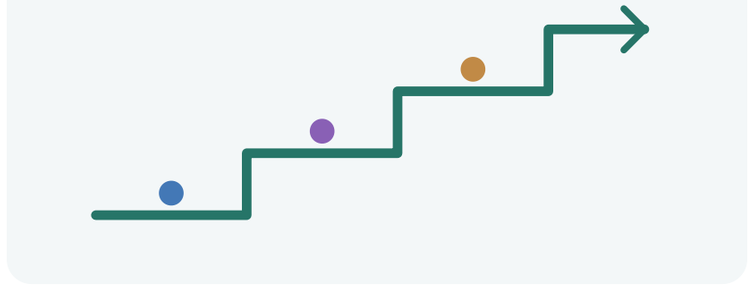
3 Good and bad plots

See the issue, then reveal an improved figure.



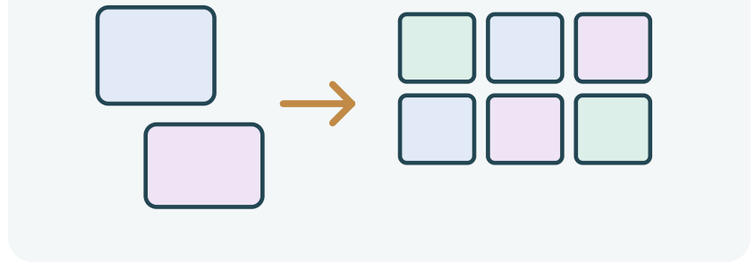
4 Same data, different plots

Compare two views and discuss which fits the question.



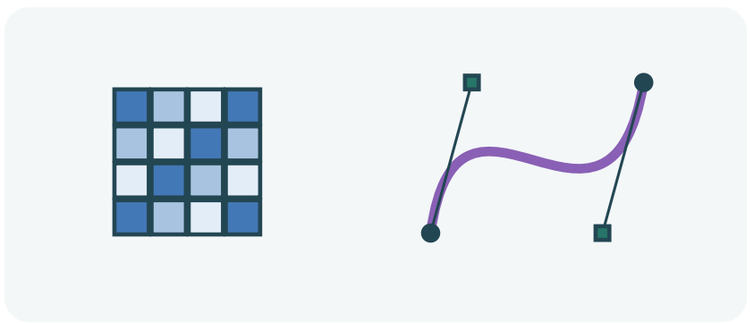
5 Improve a figure step by step

Build consistent panels and untangle crowded point plots.



6 Assemble a multi-panel figure

Combine plots with code or a vector editor.

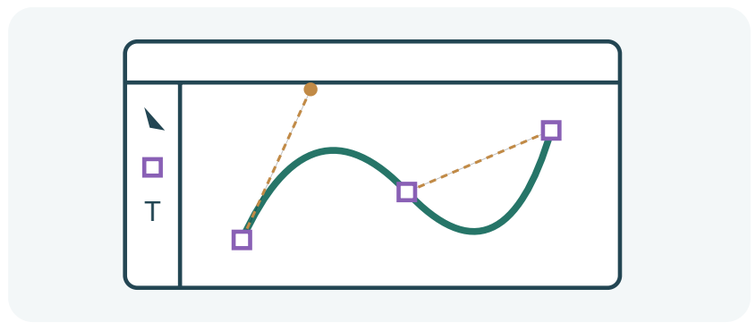


7 File formats and resolution

Raster vs vector, zooming, and DPI.

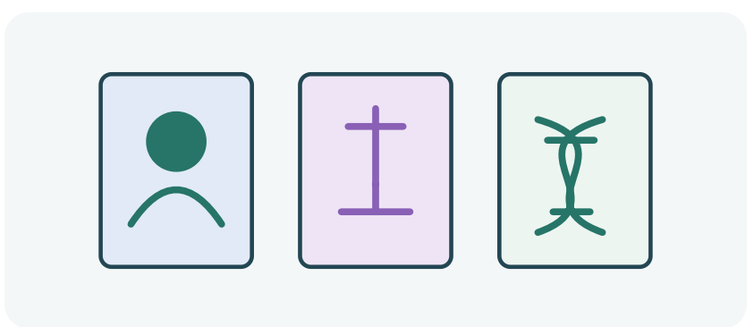
8 Post-processing tools

Edit layouts, handle fonts, and preserve the evidence.



9 Tools: Inkscape

Install the editor, arrange panels, add labels and export a figure.

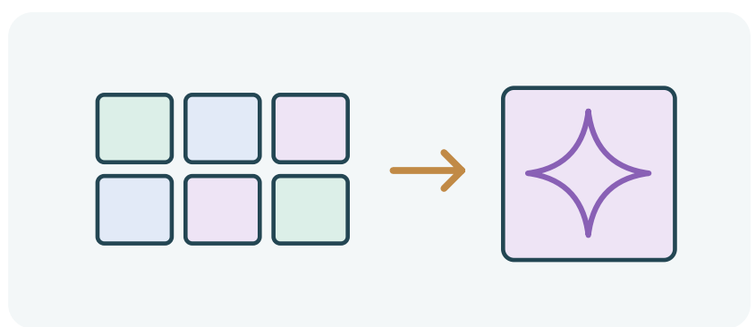


10 Include icons in the plot

Generate, compare and place clear scientific icons.

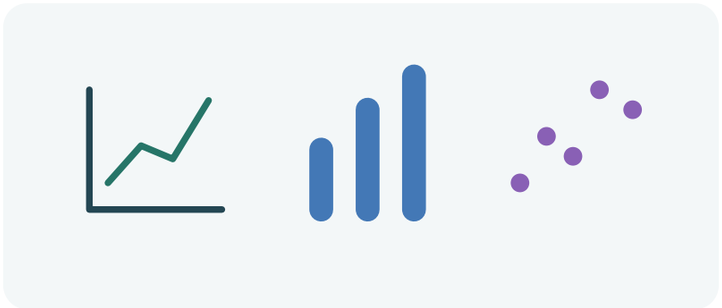
11 Submit the manuscript

Embed previews, upload figure files and prepare for revisions.



12 Use generative tools

Redesign six mismatched panels from the Day 1 and Day 2 data.



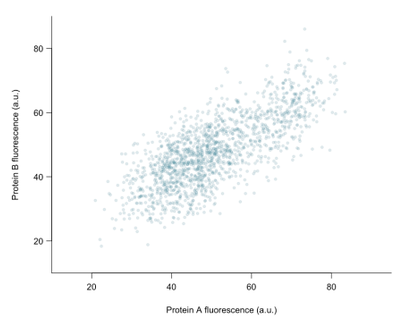
Day 3 · Chapter 1 of 12

What kinds of plots are there?

Explore plot families and the questions they answer.

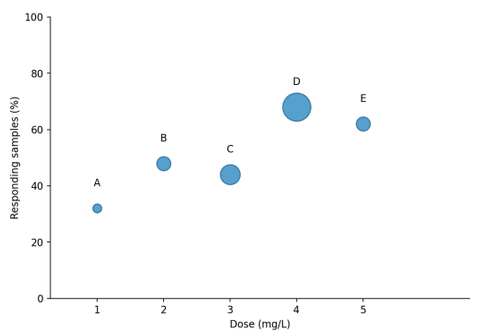
What kinds of plots are there?

Category: All



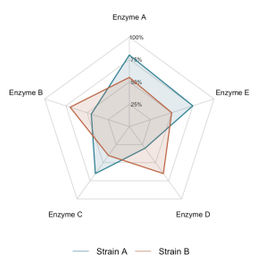
Scatter plot

Explore the relationship between two numeric measurements.



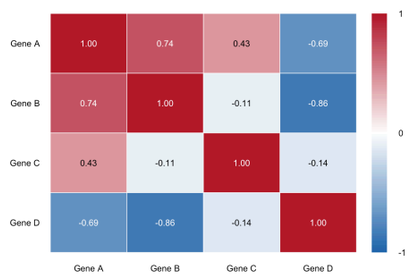
Bubble plot

Compare dose and response, with circle area showing how many samples were measured.



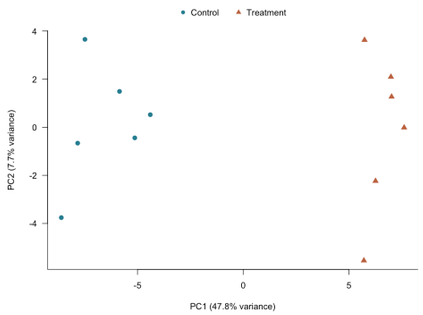
Radar / spider chart

Compare multivariable profiles on radial axes with a common, meaningful scale.



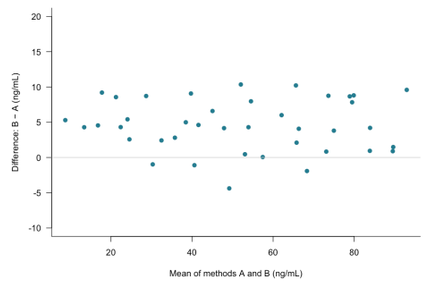
Correlation matrix

Summarise pairwise associations among several numeric measurements.



PCA / ordination plot

Summarise variation across many measured features in a few coordinates.

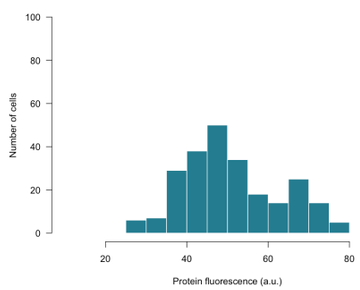


Bland–Altman plot

Inspect agreement between two methods measuring the same samples.

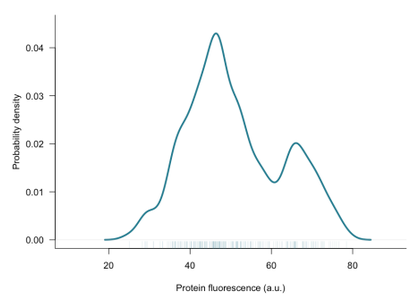
What kinds of plots are there?

Category: All



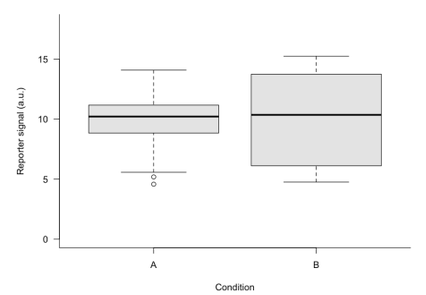
Histogram

See the shape, spread, and concentrations of one numeric variable.



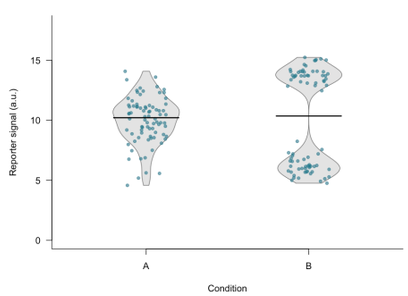
Density plot

Show a smoothed distribution of a numeric measurement.



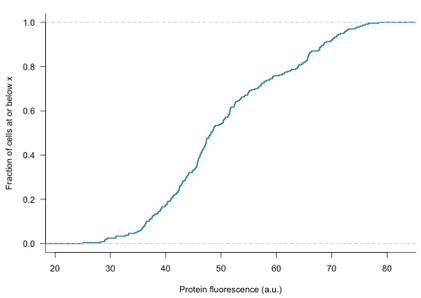
Box plot

Compare medians and spread across groups in a compact display.



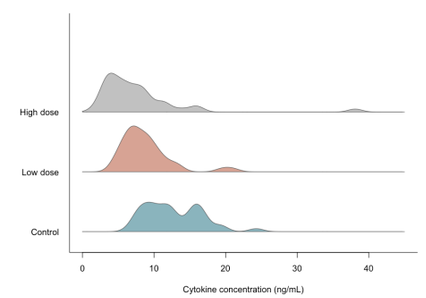
Violin plot

Compare distribution shapes when there are enough observations to estimate density.



Cumulative distribution (ECDF)

Read the fraction of observations at or below any threshold, without bins or smoothing.

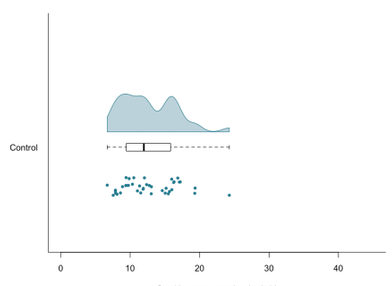


Ridgeline plot

Compare several distribution shapes on a shared horizontal scale.

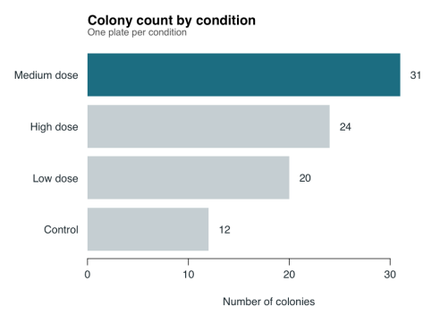
What kinds of plots are there?

Category: All



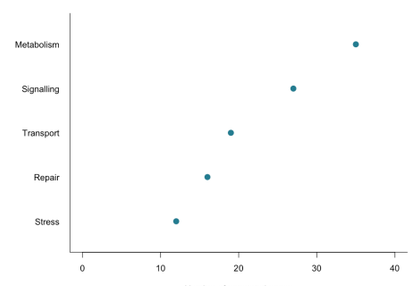
Raincloud plot

Combine a distribution estimate, compact summary, and individual observations.



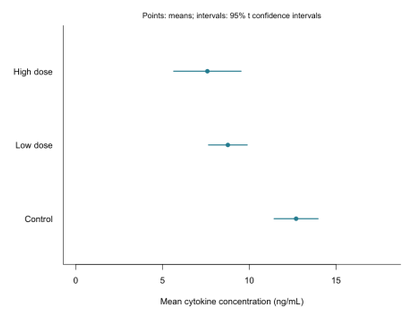
Bar plot

Compare counts, proportions, or other quantities across categories.



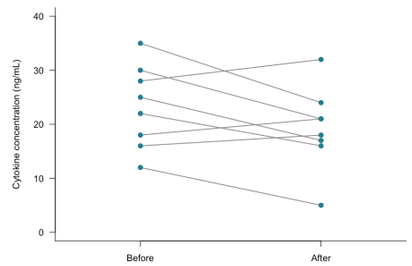
Dot / lollipop plot

Compare or rank category values using positions on a shared scale.



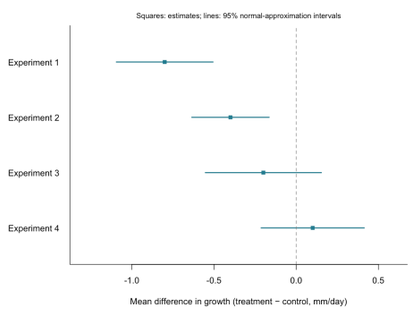
Point-and-interval plot

Show estimates together with explicitly defined uncertainty intervals.



Paired plot

Show repeated measurements from the same donors, samples, or experimental units.

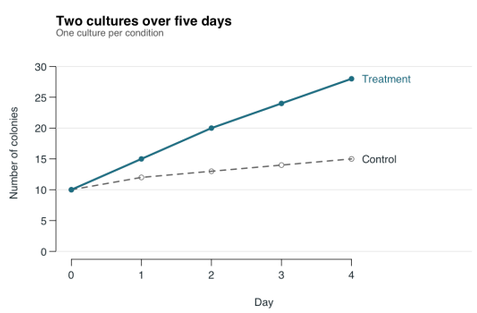


Forest plot

Compare effect estimates and uncertainty across studies, experiments, or subgroups.

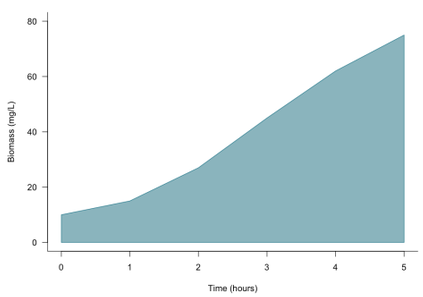
What kinds of plots are there?

Category: All



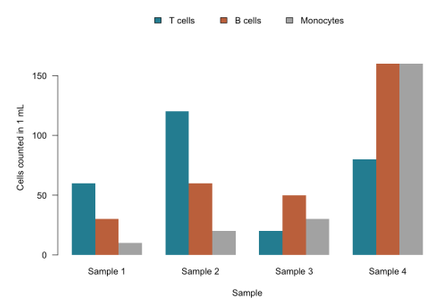
Line plot

Show change along an ordered variable, usually time.



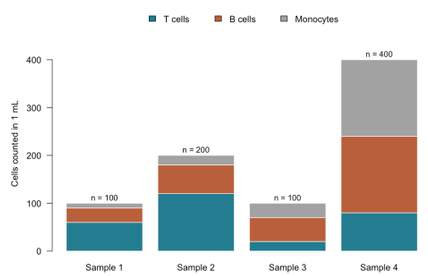
Area / stacked area plot

Show a total changing over time, optionally broken into additive components.



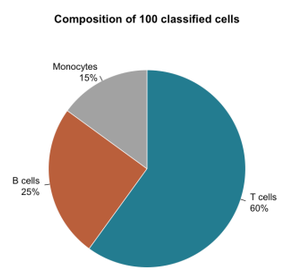
Grouped / side-by-side bar plot

Compare several categories within each sample, condition, or discrete time point.



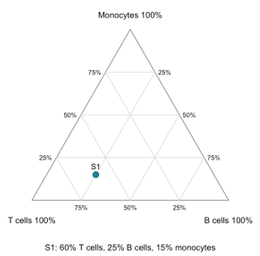
Stacked bar plot

Show how categories contribute to each sample’s total, as counts or as proportions.



Pie / donut chart

Show a few mutually exclusive parts of one whole.

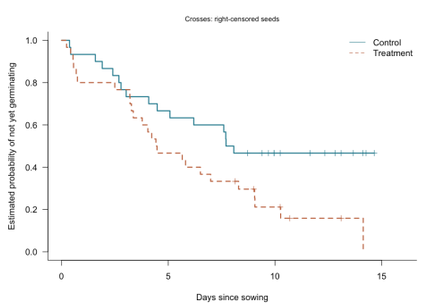


Ternary plot

Compare three components that sum to a fixed total, such as 100%.

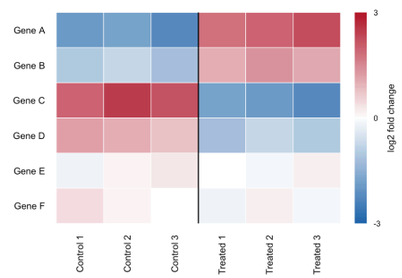
What kinds of plots are there?

Category: All



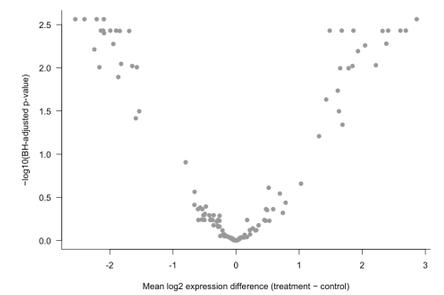
Survival / Kaplan–Meier curve

Show time until an event while retaining right-censored observations.



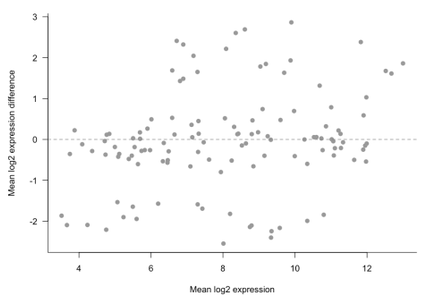
Heatmap

Find patterns in a matrix, such as measurements across genes and samples.



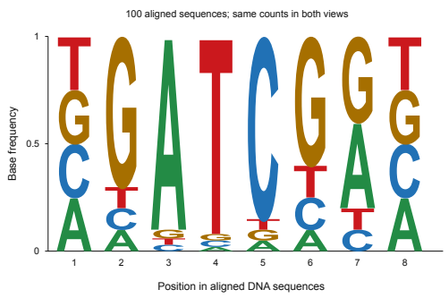
Volcano plot

Compare effect magnitude with statistical evidence across many features.



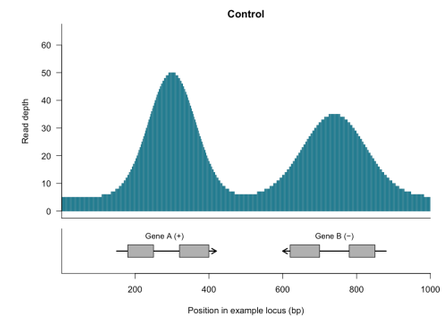
MA plot

Inspect expression differences against average expression level.



Sequence logo

Show base or amino-acid preferences across aligned sequence positions.

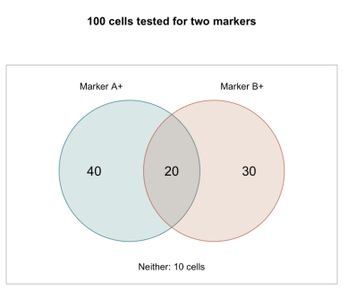


Genome / coverage tracks

Align measurements and annotations along a shared genomic coordinate.

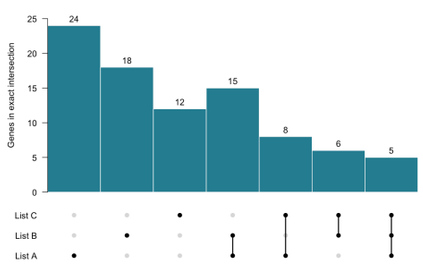
What kinds of plots are there?

Category: All



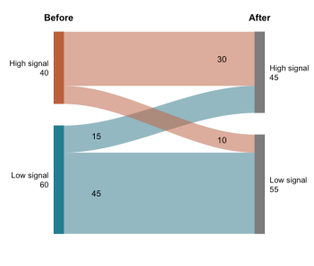
Venn diagram

Show membership and overlap between a small number of sets.



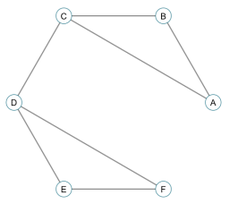
UpSet / intersection plot

Compare exact intersections among several sets with bars and a membership matrix.



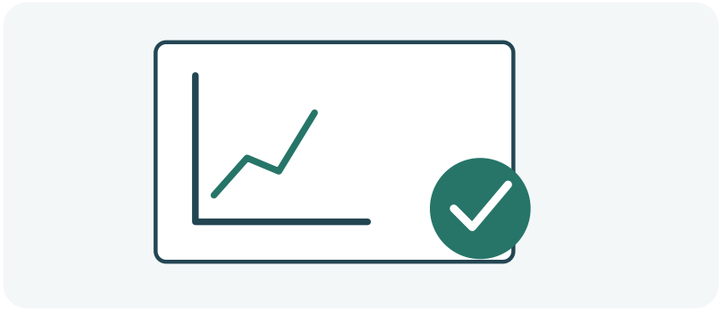
Sankey diagram

Show how quantities flow or split between categories; ribbon thickness represents amount.



Network diagram

Show entities and relationships, such as proteins connected by recorded associations.

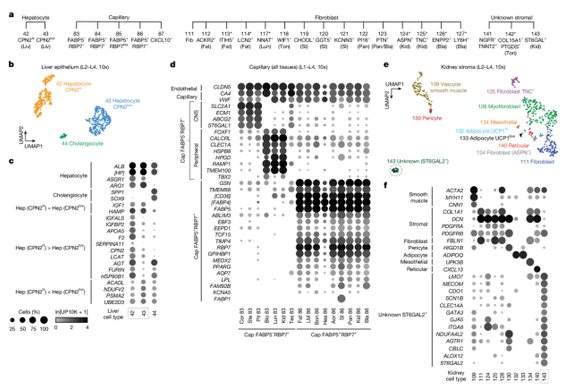


Examples for good figures

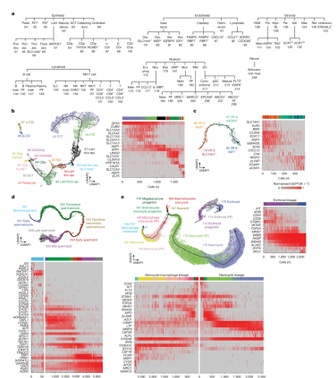
Inspect design choices in real biological research.

Examples for good figures

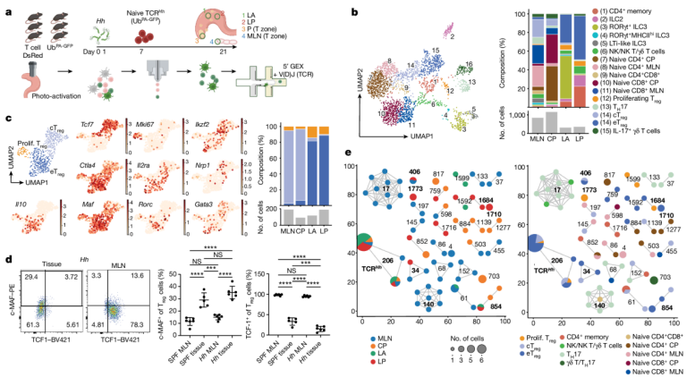
Select a figure to explore it.



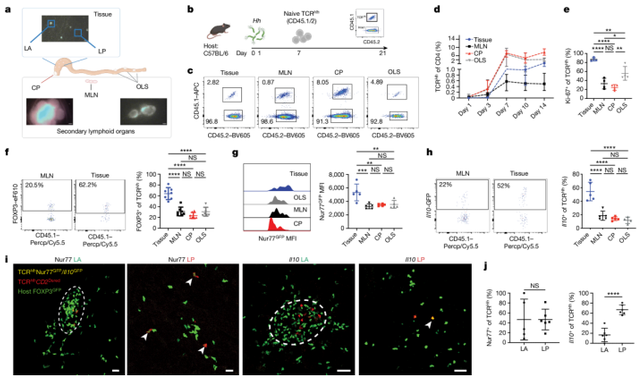
Size and shade answer different questions



Use colour to connect two views



Show proportions and totals together

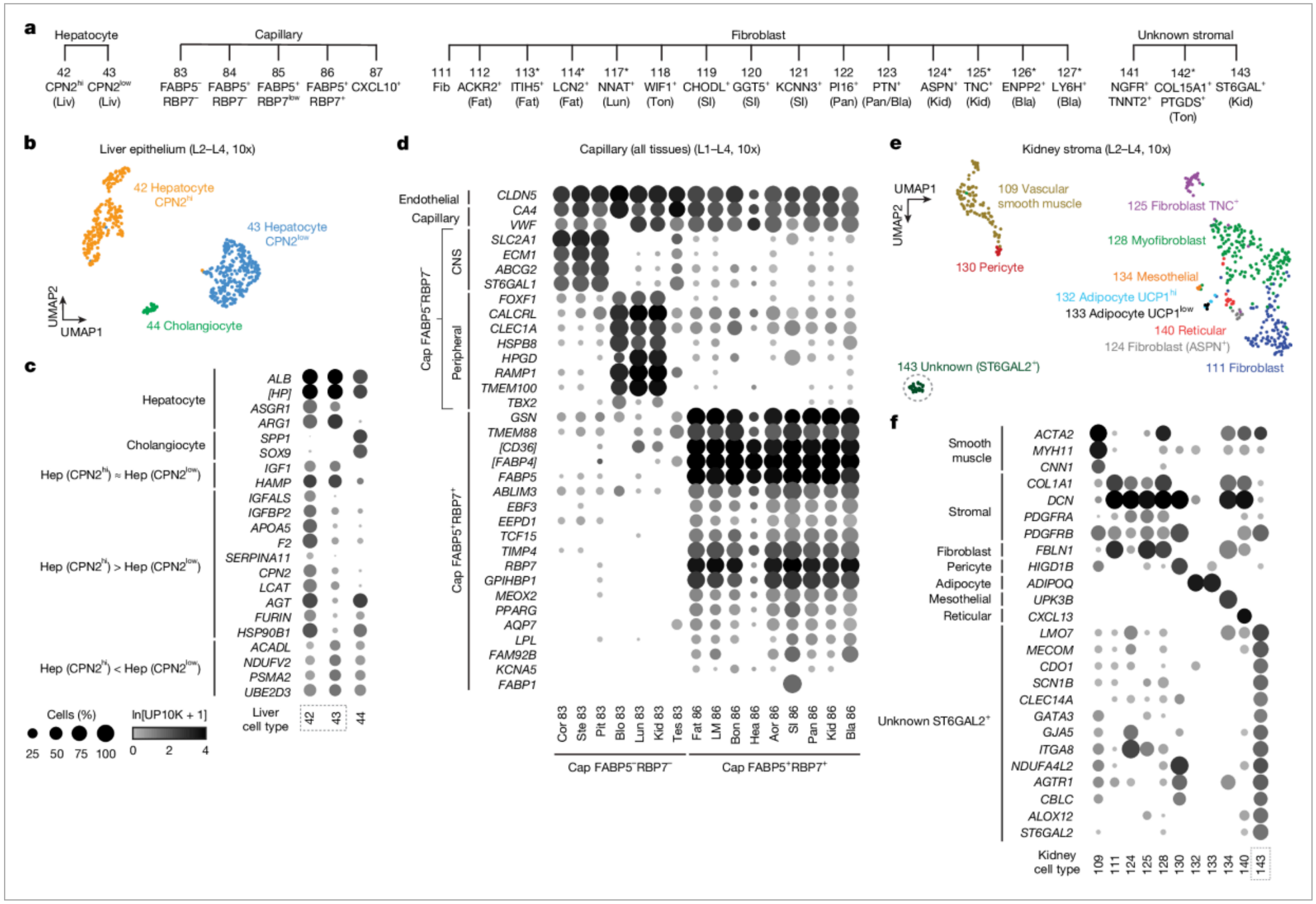


Pair an example with its summary

> **More examples**

Good plot

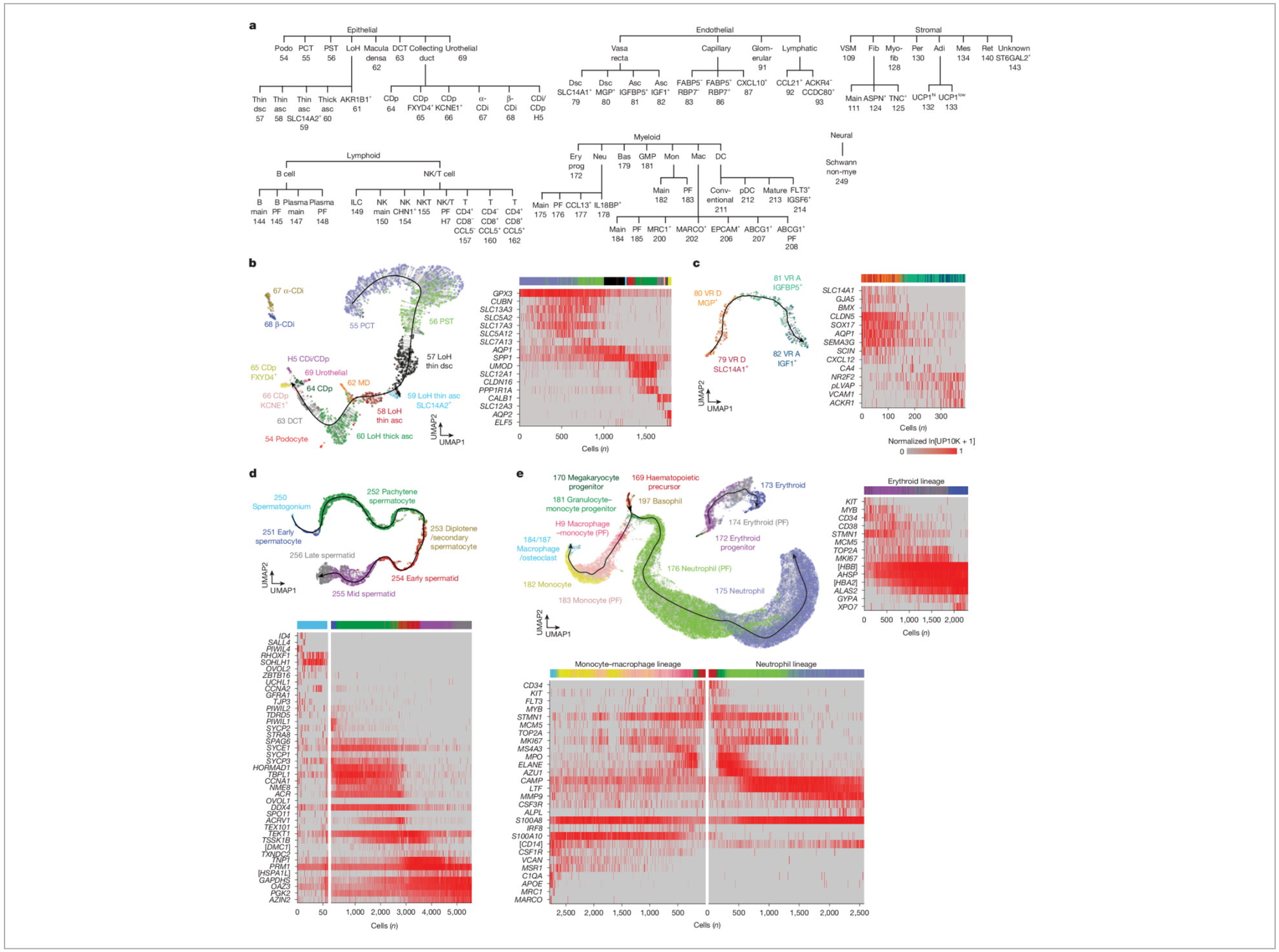
Panel c: dot size and shade distinguish how many cells express a gene from how strongly they express it.



Source & credits

Good plot

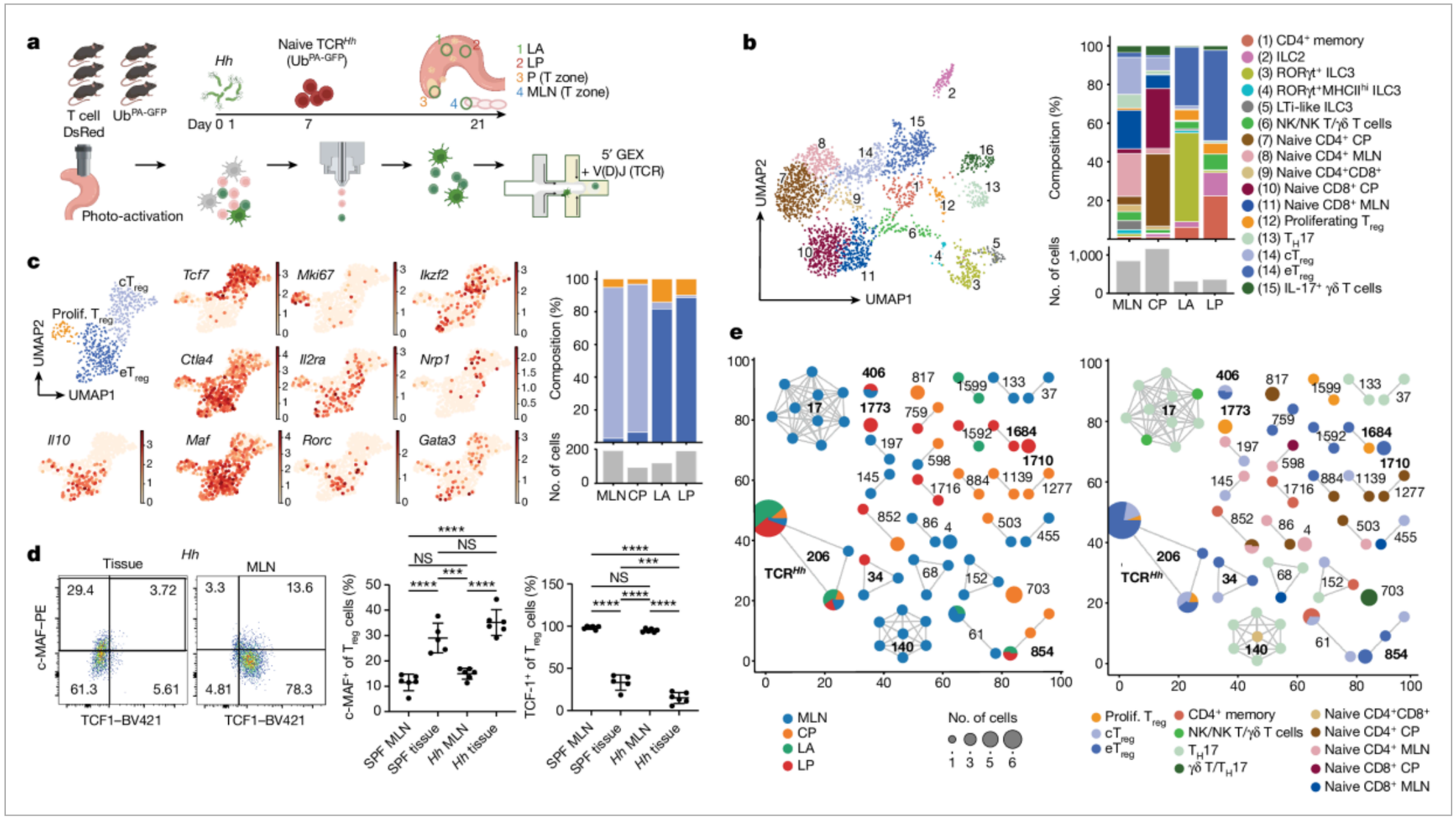
Panel b: matching cell-type colours connect the cell map to the heatmap.



Source & credits

Good plot

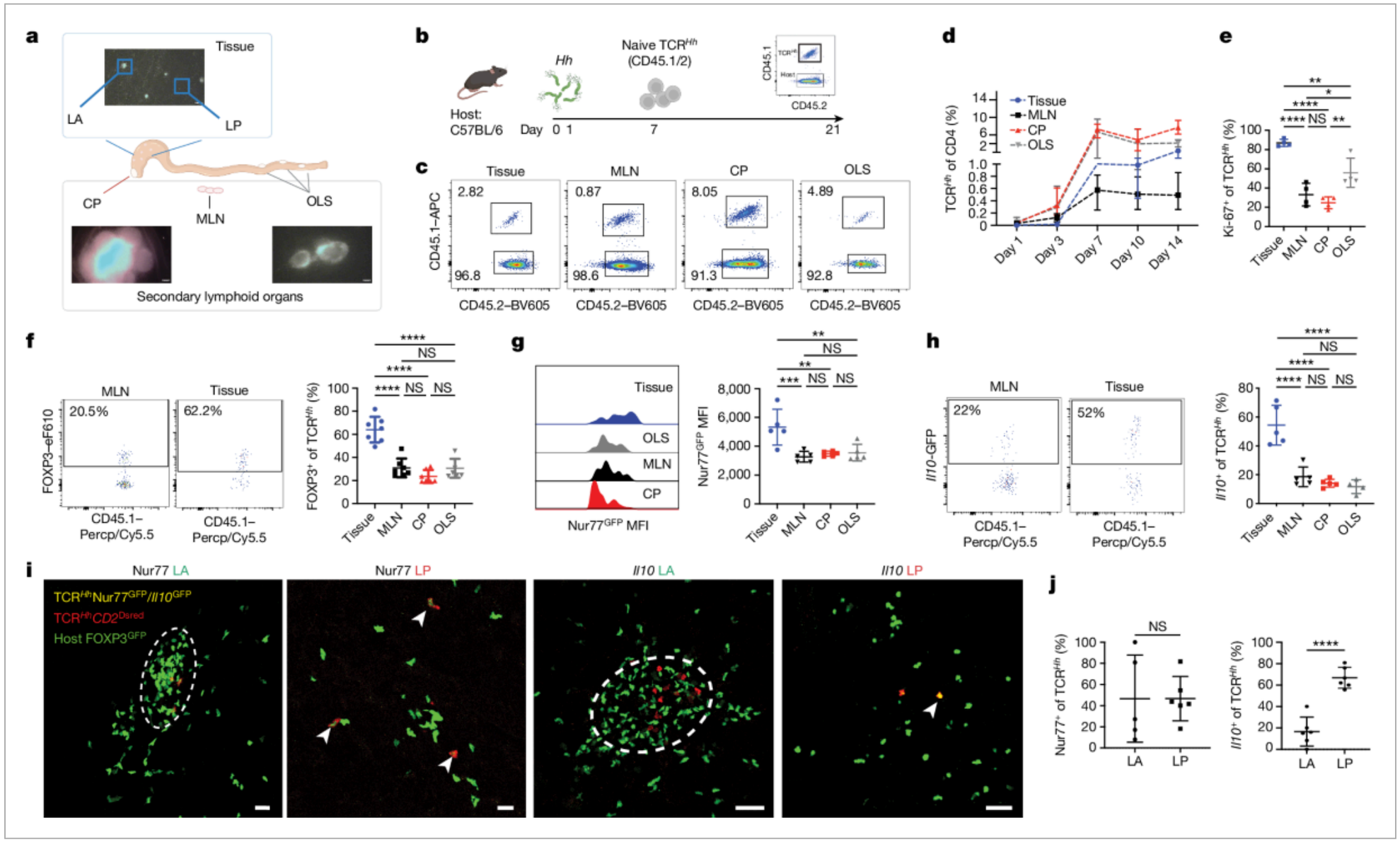
Panel b: aligned count bars show the totals hidden by percentages.



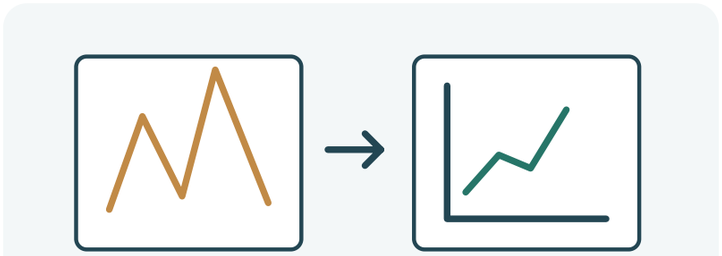
Source & credits

Good plot

Panels f–h: examples sit beside their summaries, with consistent colours linking tissue sites.



Source & credits

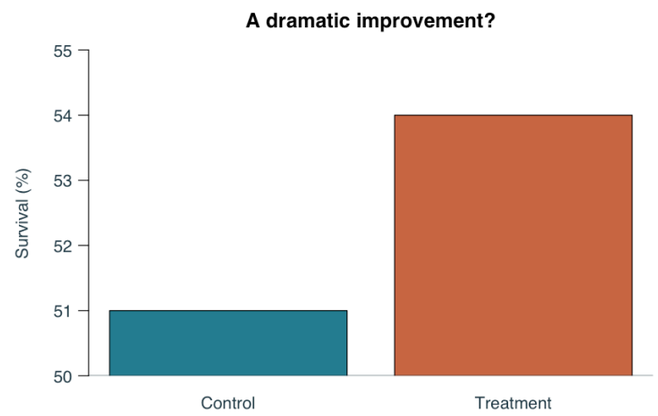


Good and bad plots

See the issue, then reveal an improved figure.

Good and bad plots

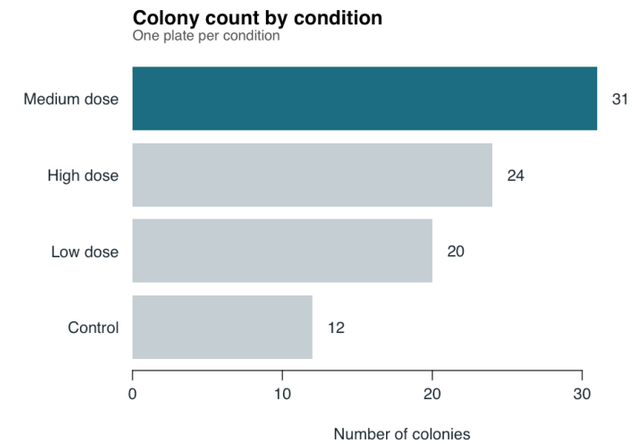
Explore what works, what gets in the way, and how a different design helps.



Spot the problem

Look at the starting plot. What would you change?

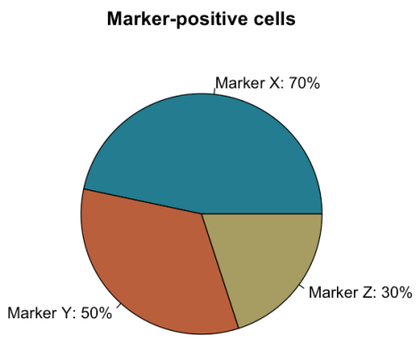
Explore plots →



Good plots: simple examples

See how clear labels and thoughtful design help.

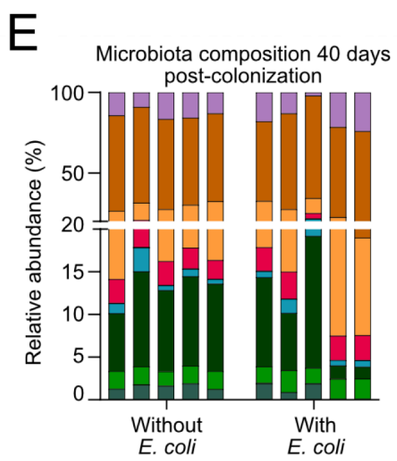
Explore plots →



Biology design comparisons

Compare designs and choose the view that answers the question.

Explore plots →



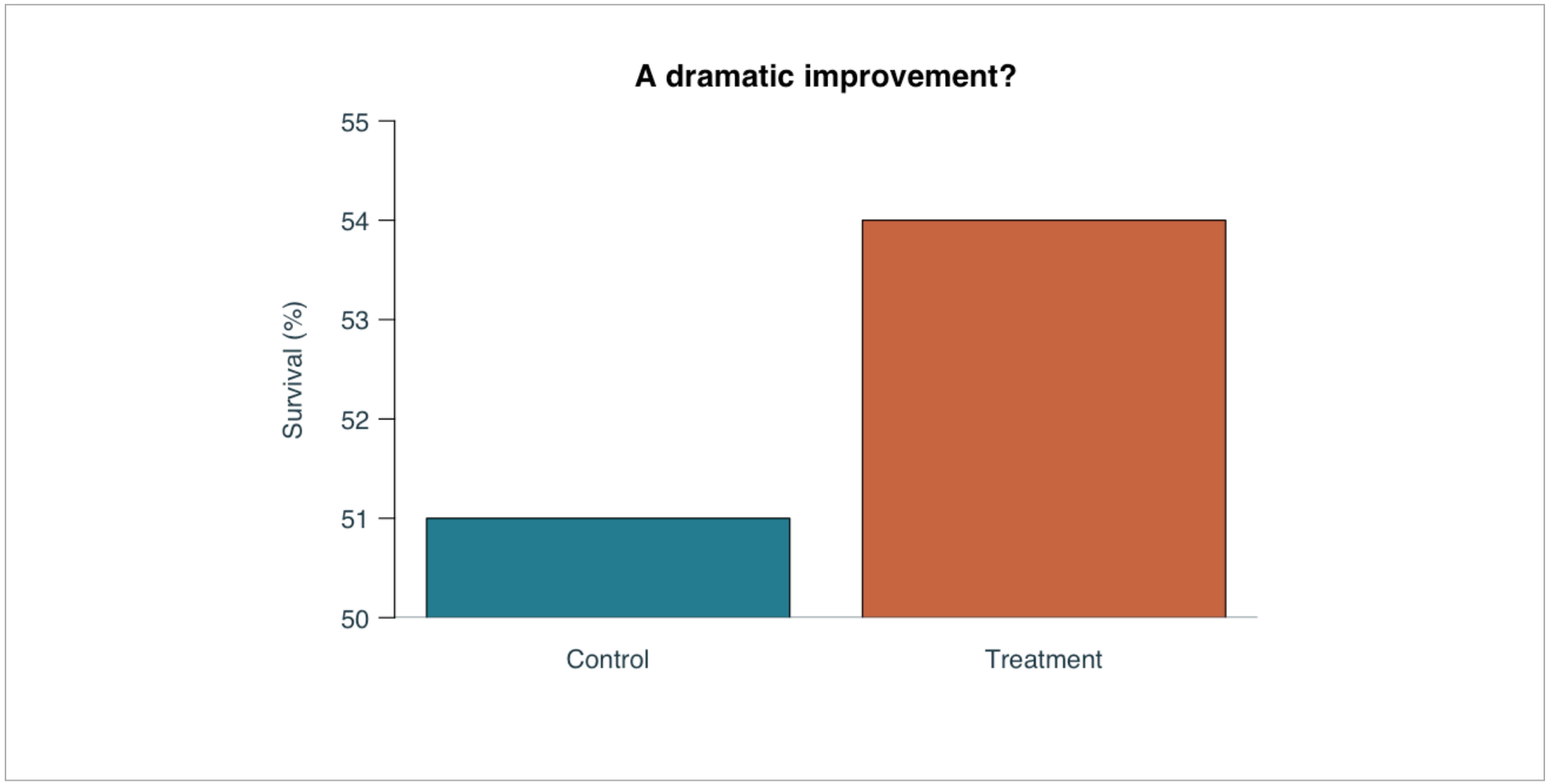
Microbiota composition

Explore the supplied stacked-bar example across samples.

Open example →

How big is the improvement?

The bars begin at 50%, exaggerating the difference.

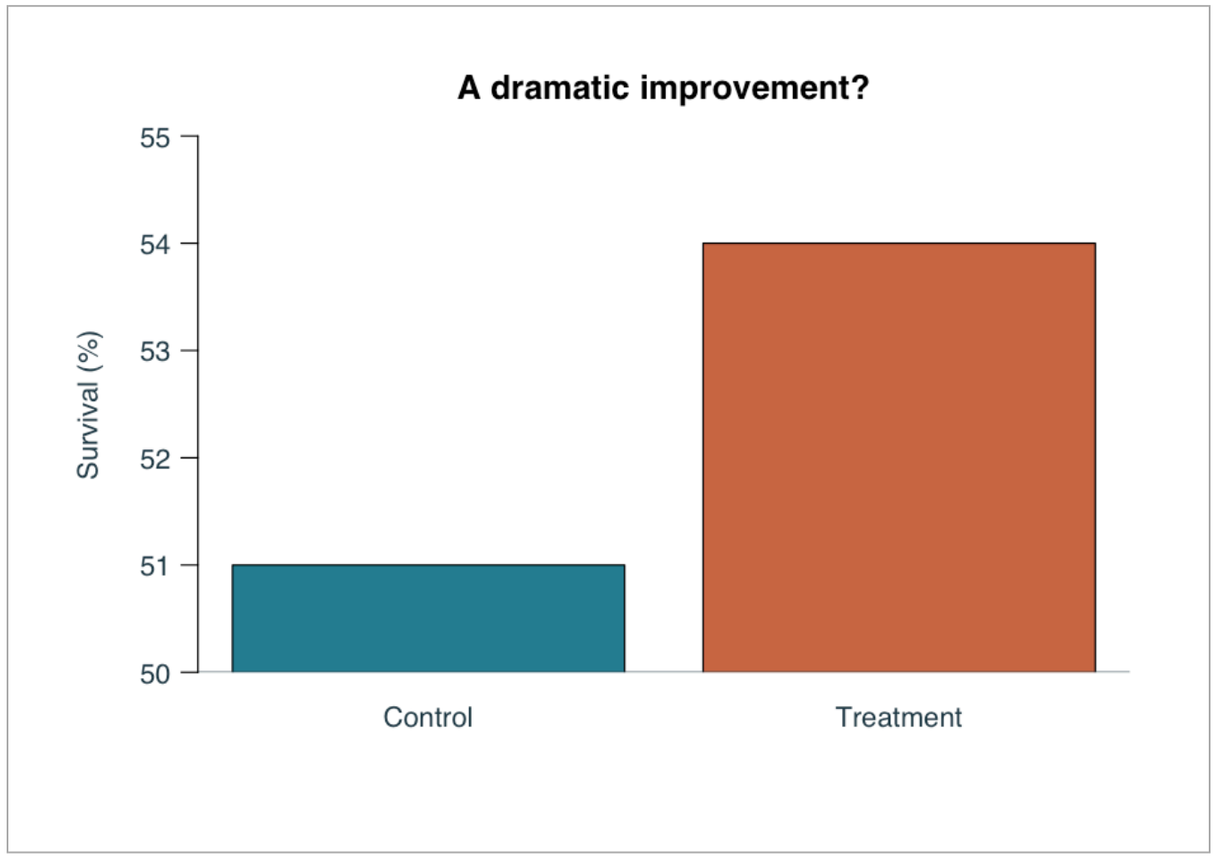


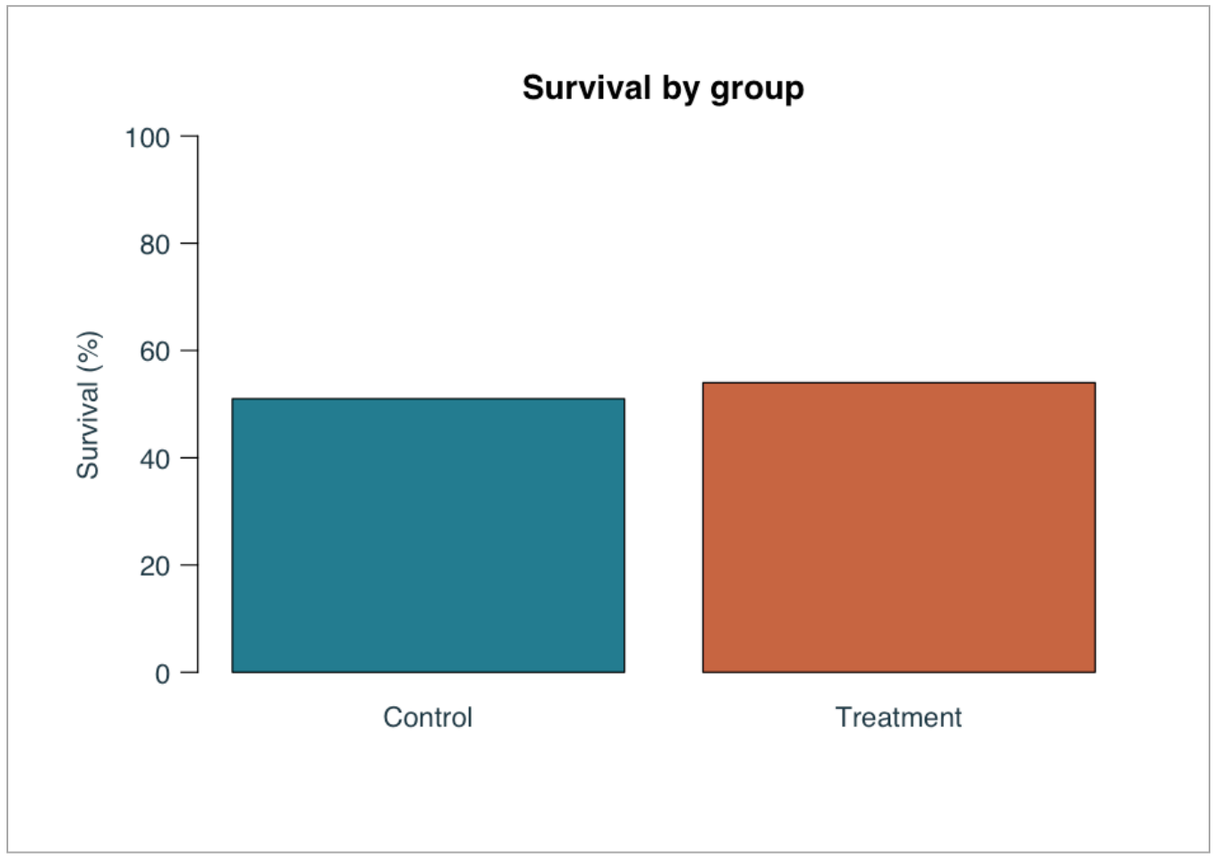
Reveal alternative figure

Context & source

How big is the improvement?

The bars begin at 50%, exaggerating the difference.



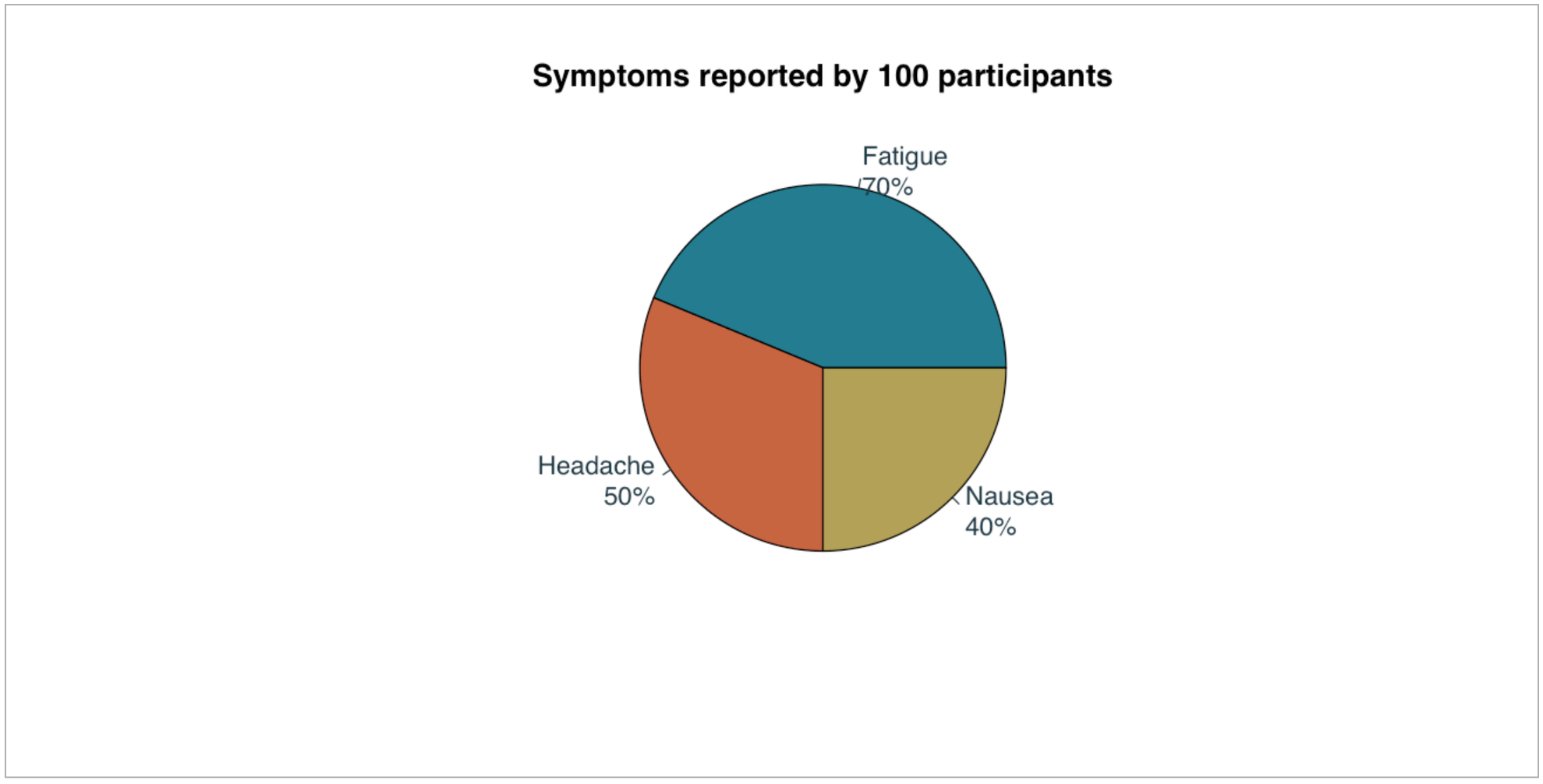


The visible heights are 1 and 4, although the values are 51 and 54. The difference is 3 percentage points; relative to 51%, it is about 5.9%.

Context & source

Can these slices form one whole?

The categories overlap; their percentages add to 160%.

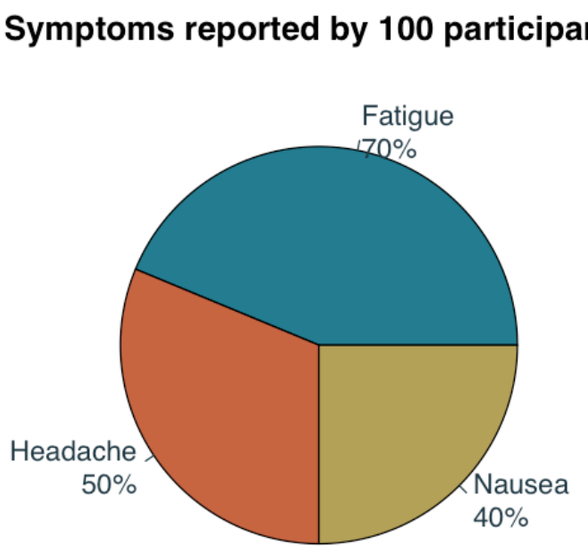


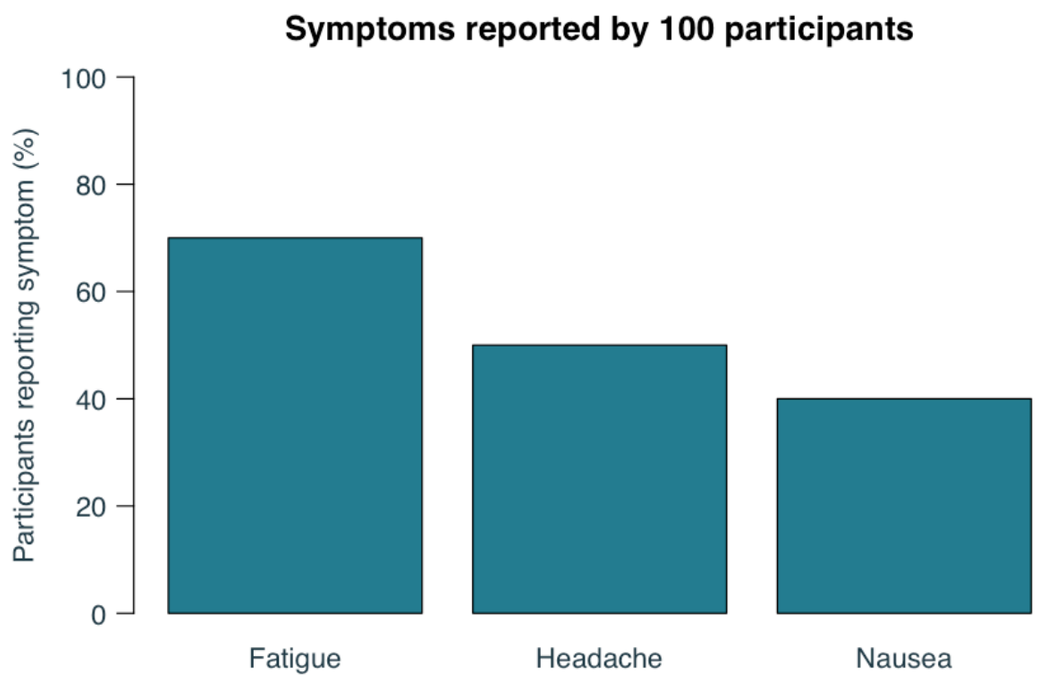
Reveal alternative figure

Context & source

Can these slices form one whole?

The categories overlap; their percentages add to 160%.



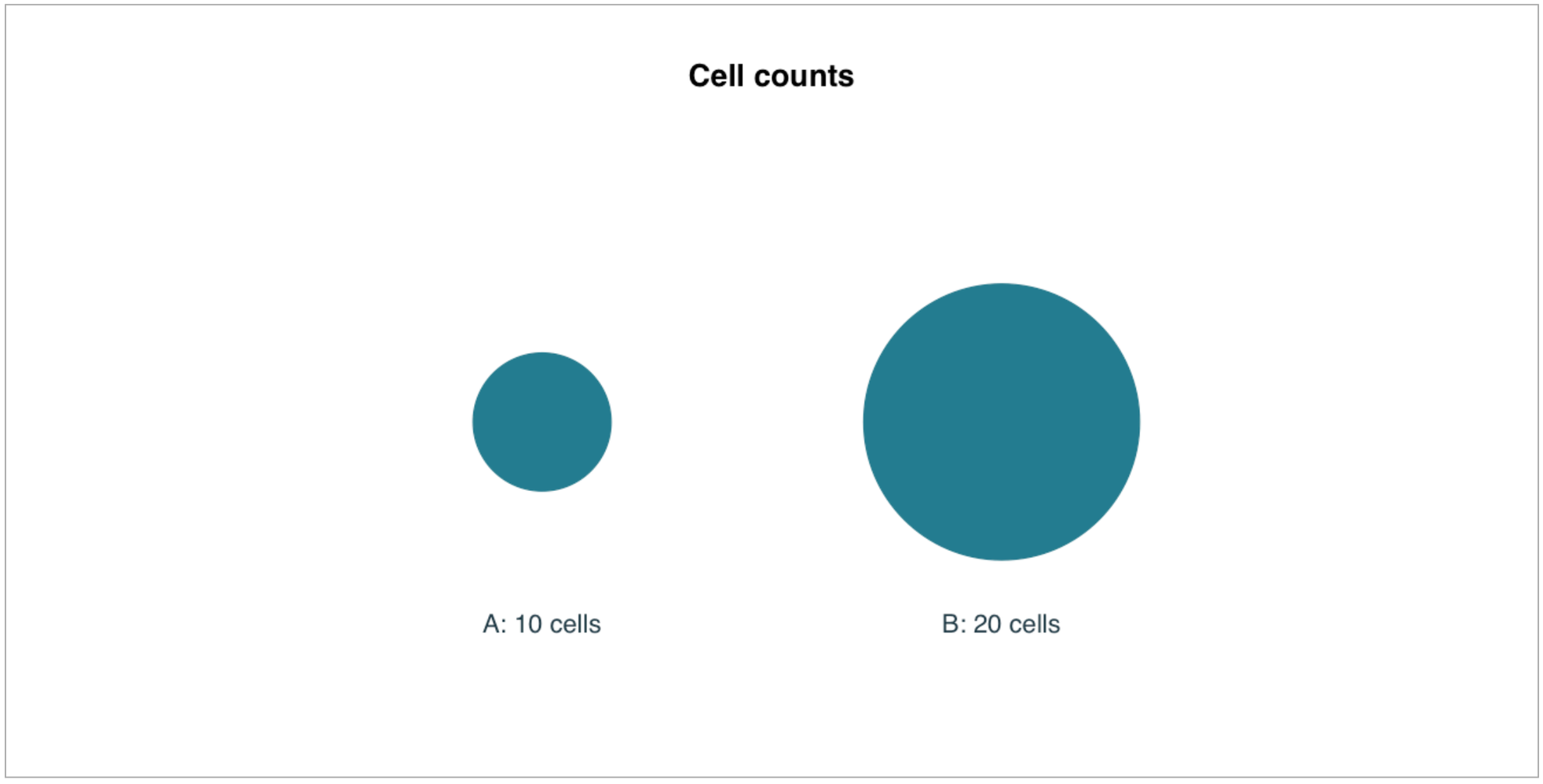


A pie implies one partition of a whole. These are overlapping responses from the same 100 people; separate bars would preserve the percentages.

Context & source

Is B twice as much?

The larger circle has four times the area, but the count only doubles.

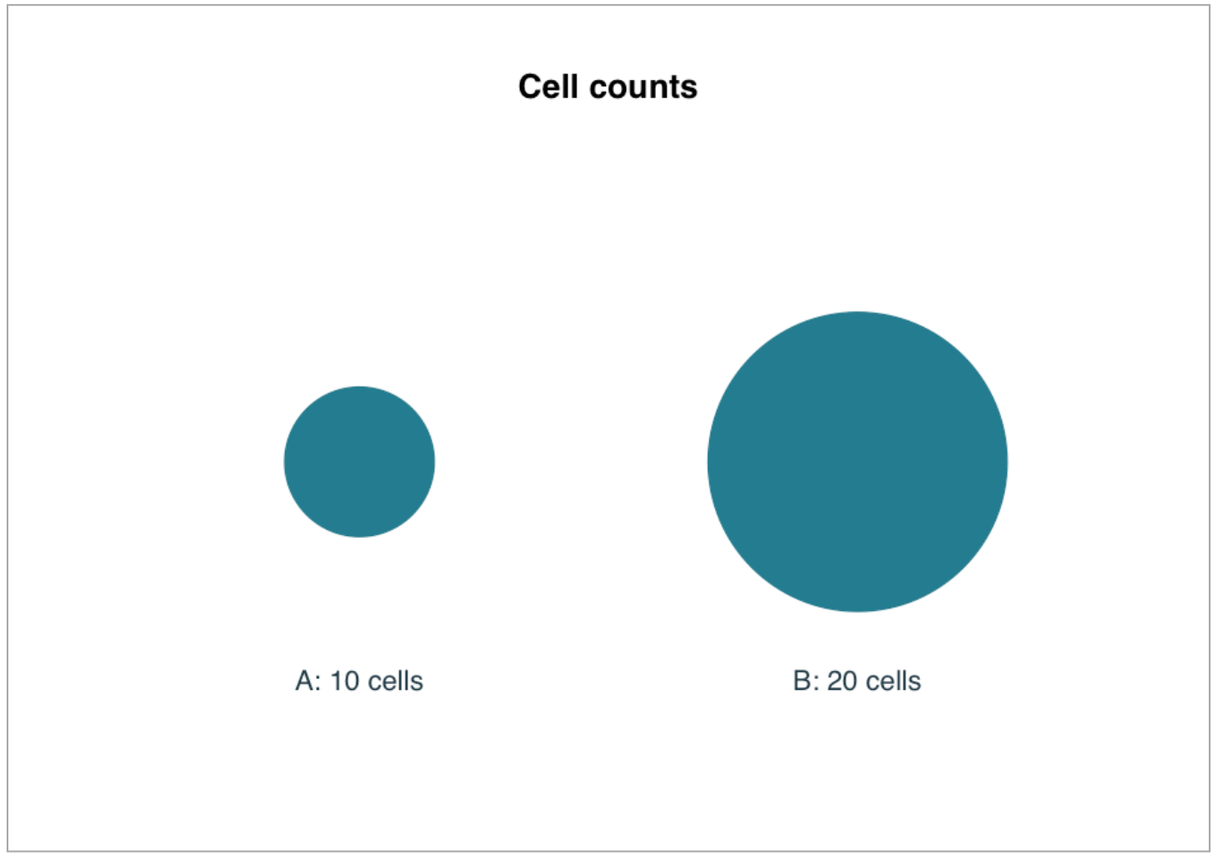


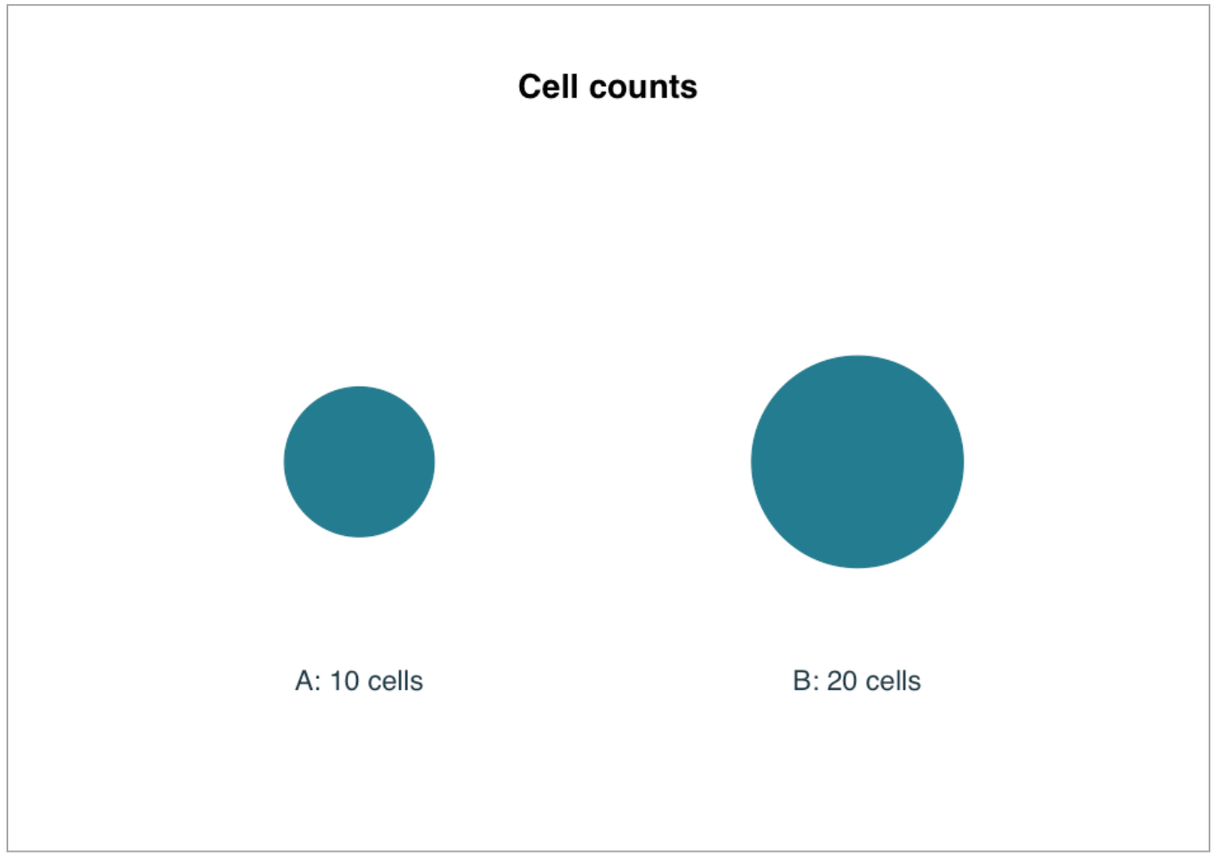
Reveal alternative figure

Context & source

Is B twice as much?

The larger circle has four times the area, but the count only doubles.



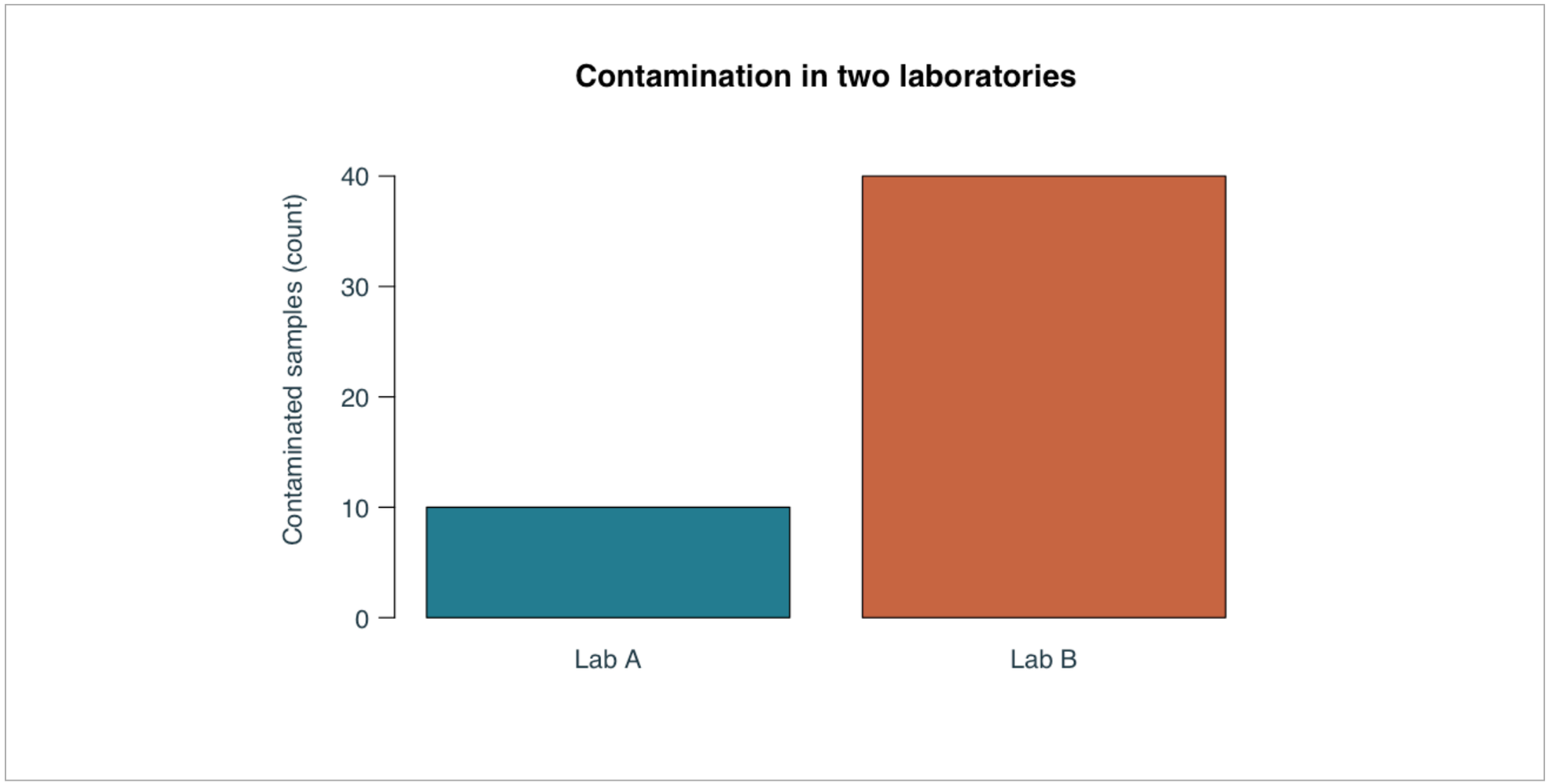


Doubling the radius quadruples the area. If area represents count, doubling the count should double the area.

Context & source

Which lab has the higher rate?

Contaminated samples as a percentage of all samples tested.

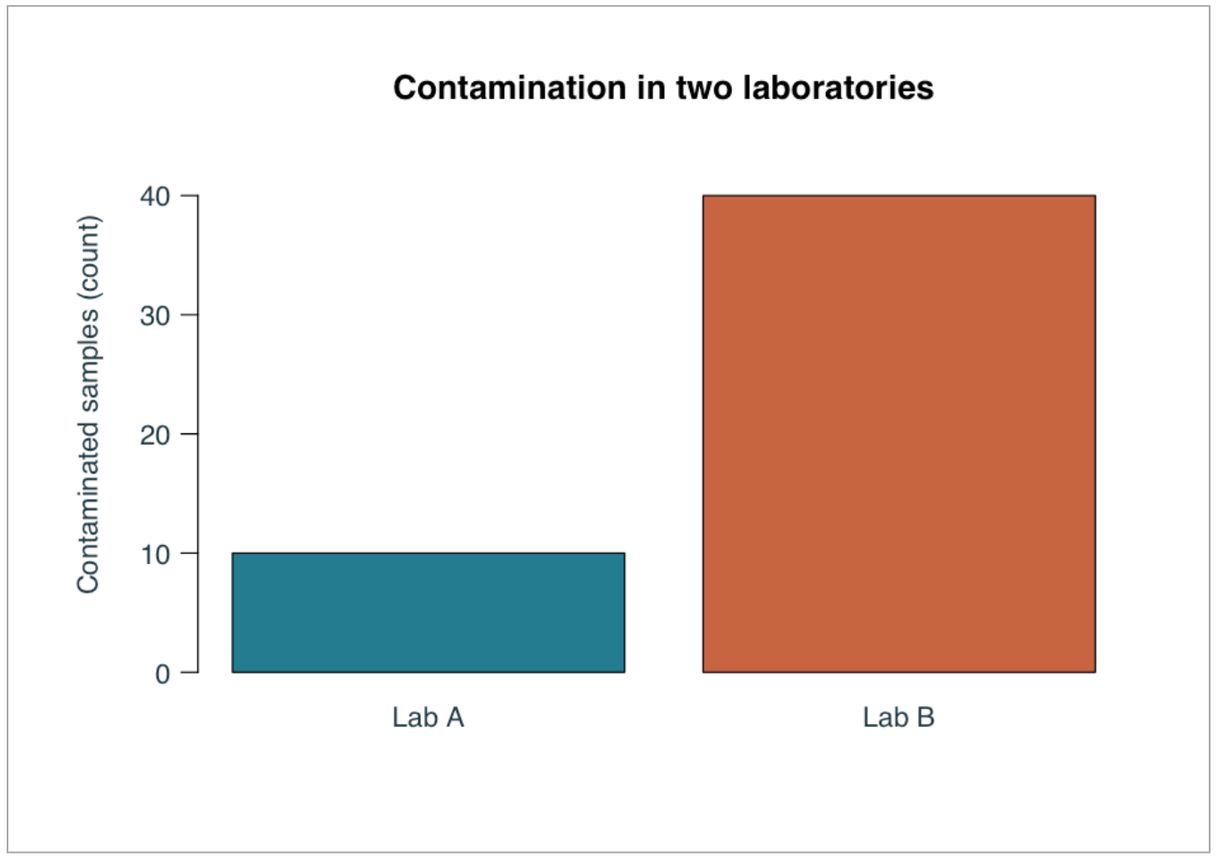


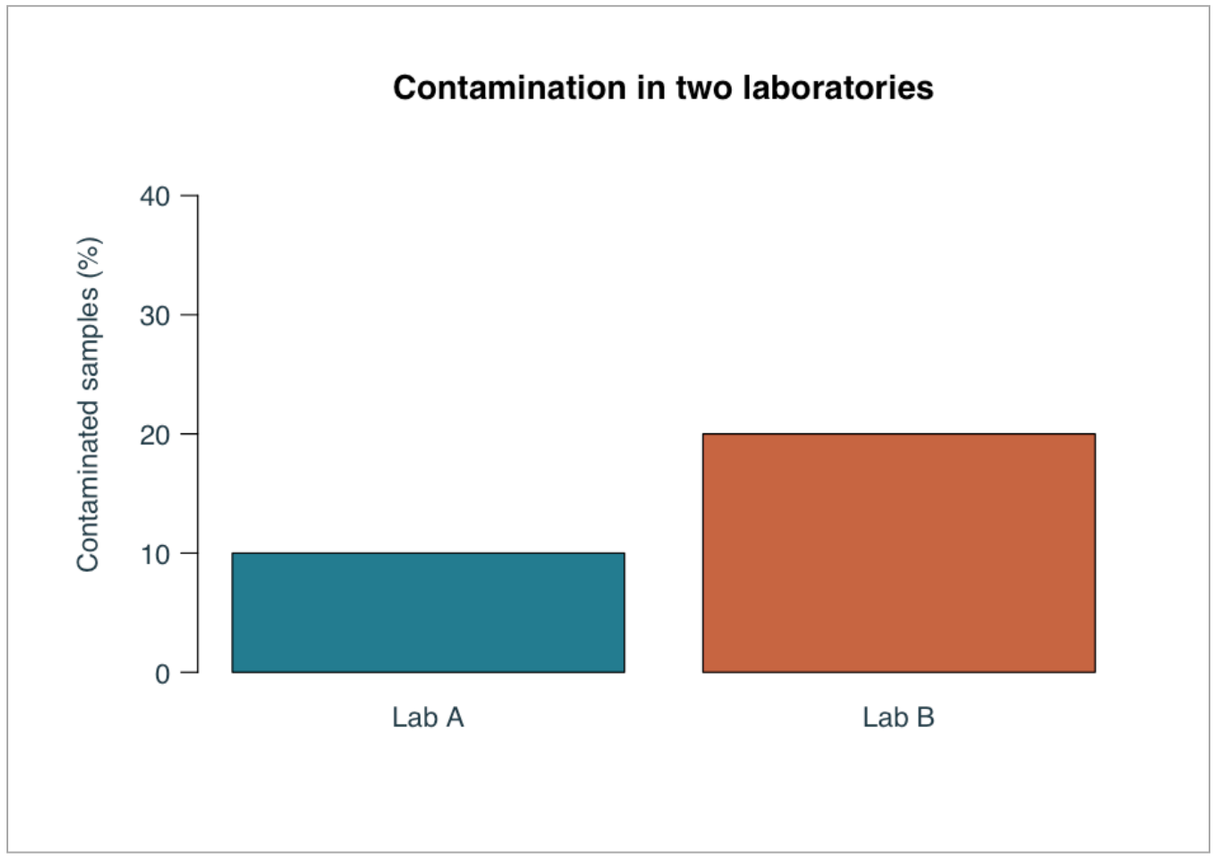
Reveal alternative figure

Context & source

Which lab has the higher rate?

Contaminated samples as a percentage of all samples tested.



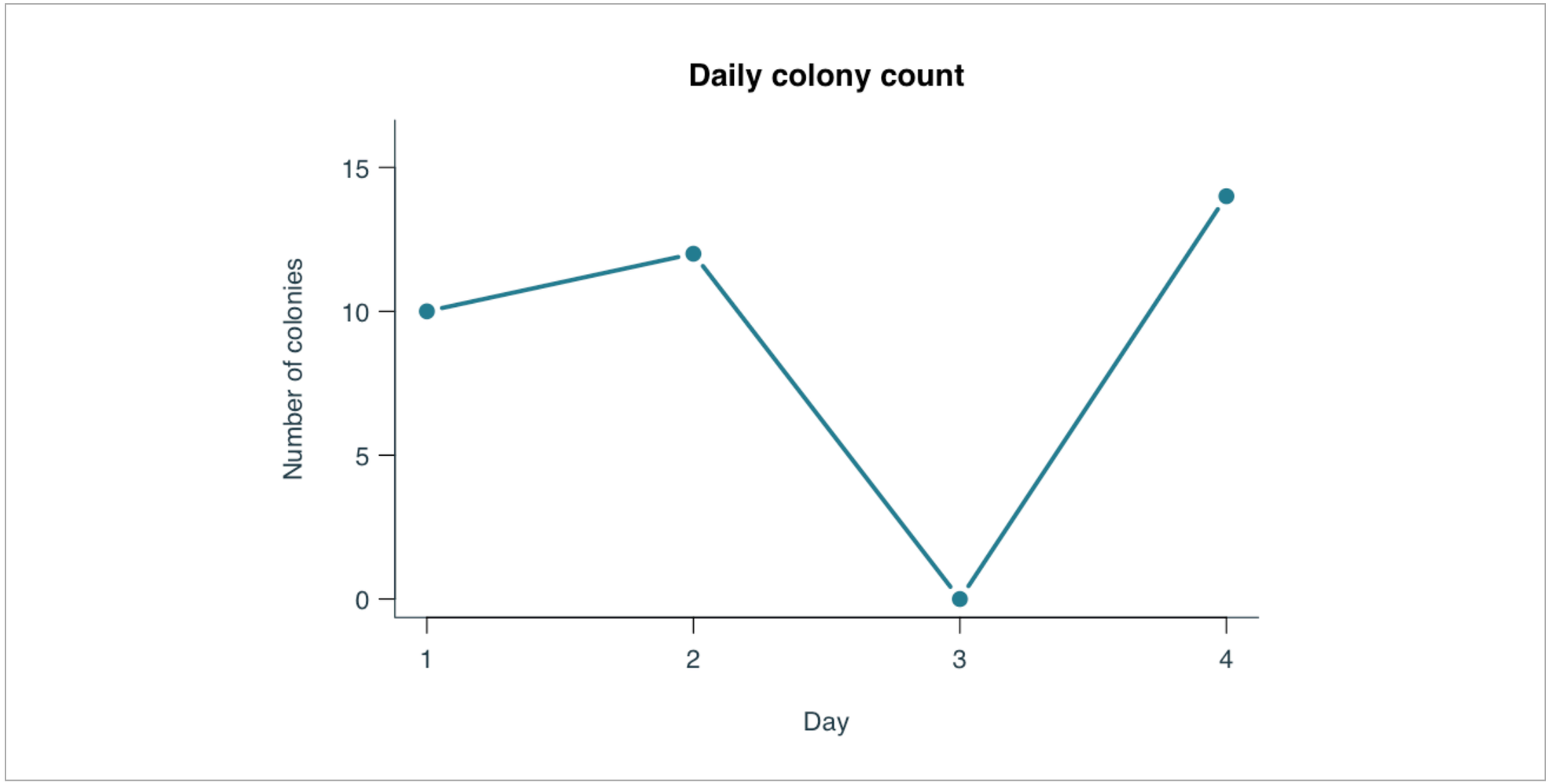


The rates are 10% and 20%. Lab B has twice the rate, although it has four times the number of contaminated samples.

Context & source

What happened on day 3?

The missing measurement is shown as zero.

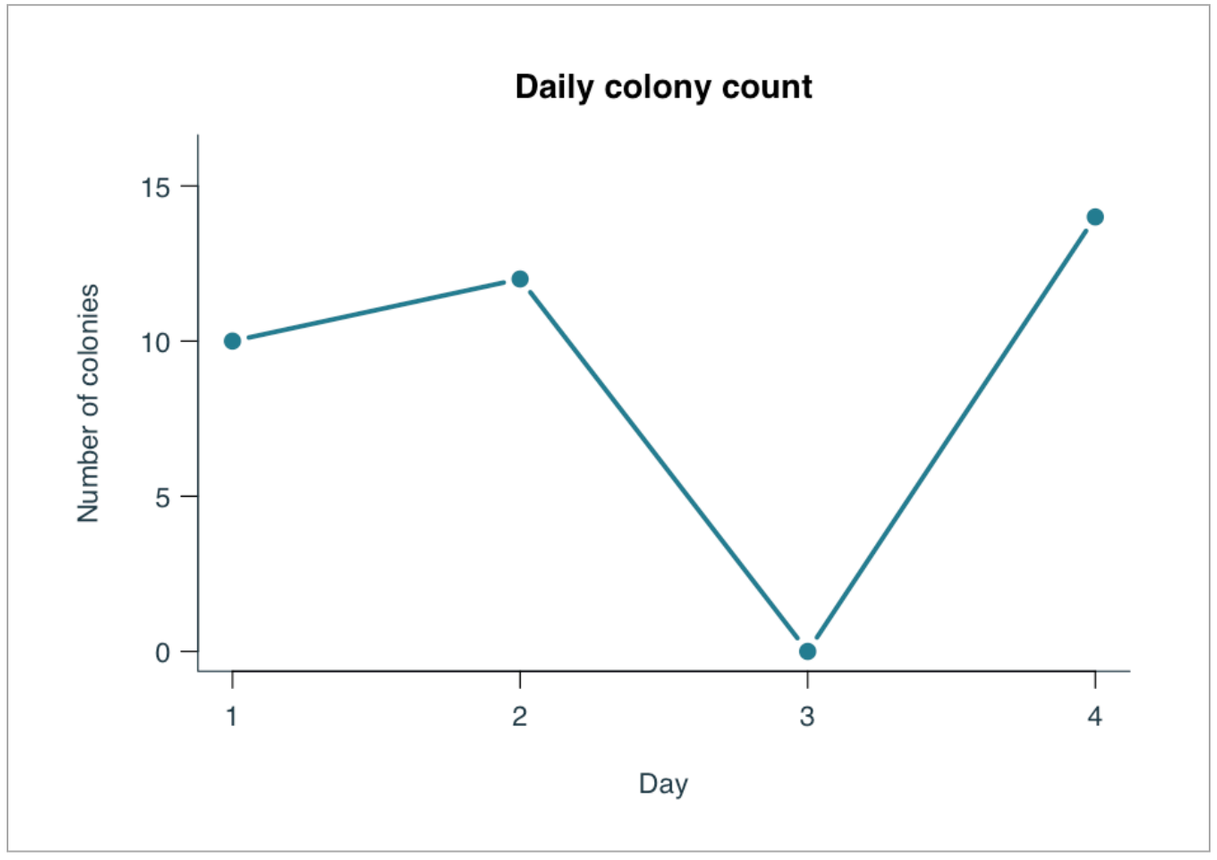


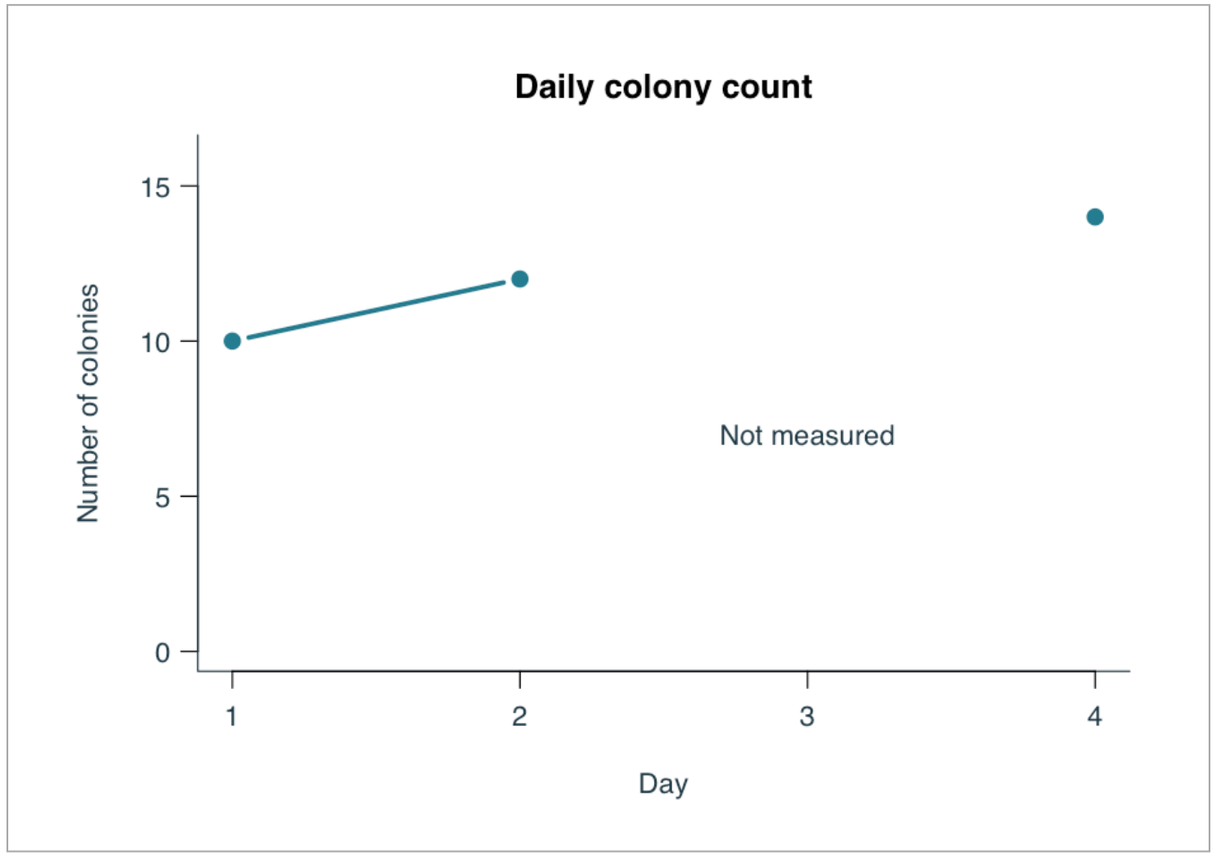
Reveal alternative figure

> **Context & source**

What happened on day 3?

The missing measurement is shown as zero.



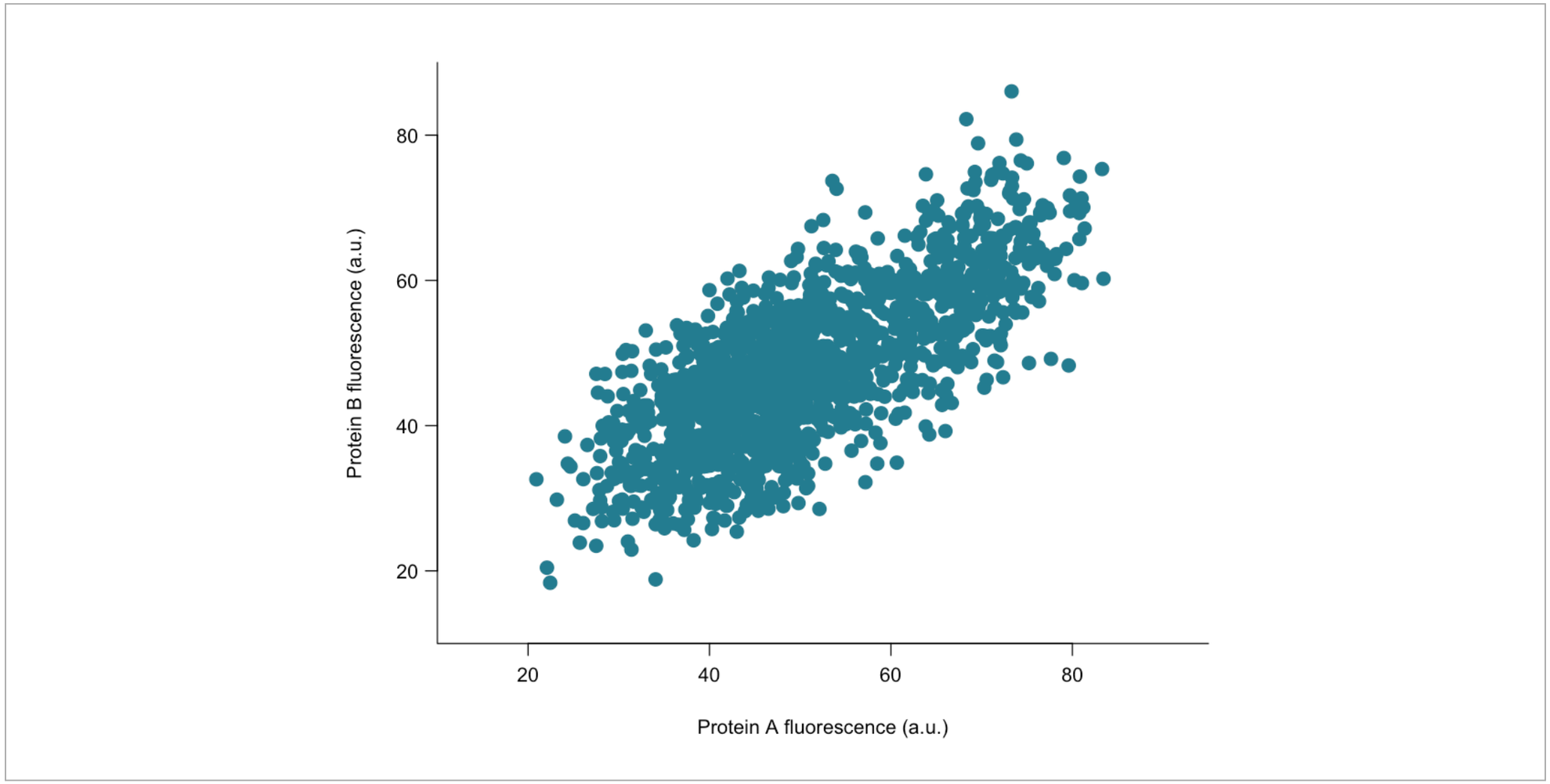


Not measured does not mean zero. Leave a gap and label the missing observation.

Context & source

What is hidden in this cloud of cells?

Not reliably: points cover other points.

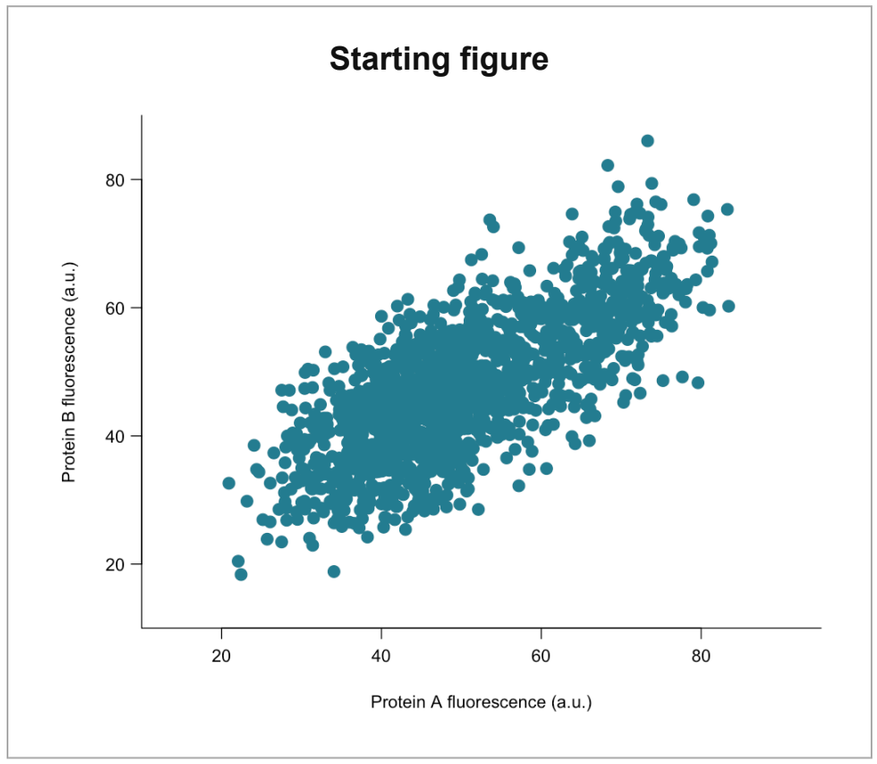


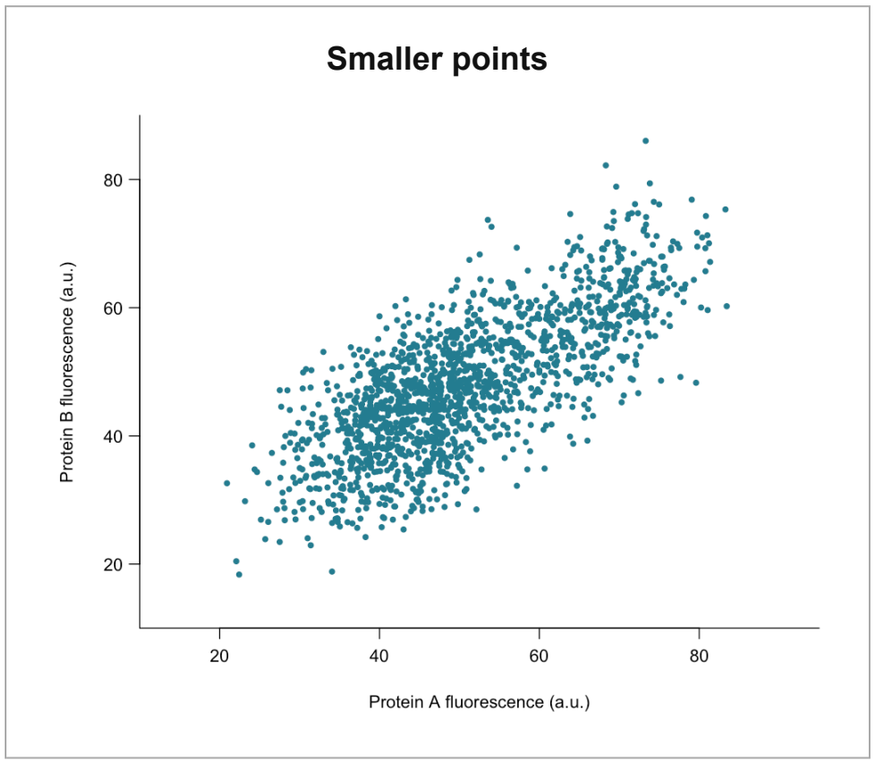
Reveal first improvement

> **Context & source**

What is hidden in this cloud of cells?

Not reliably: points cover other points.





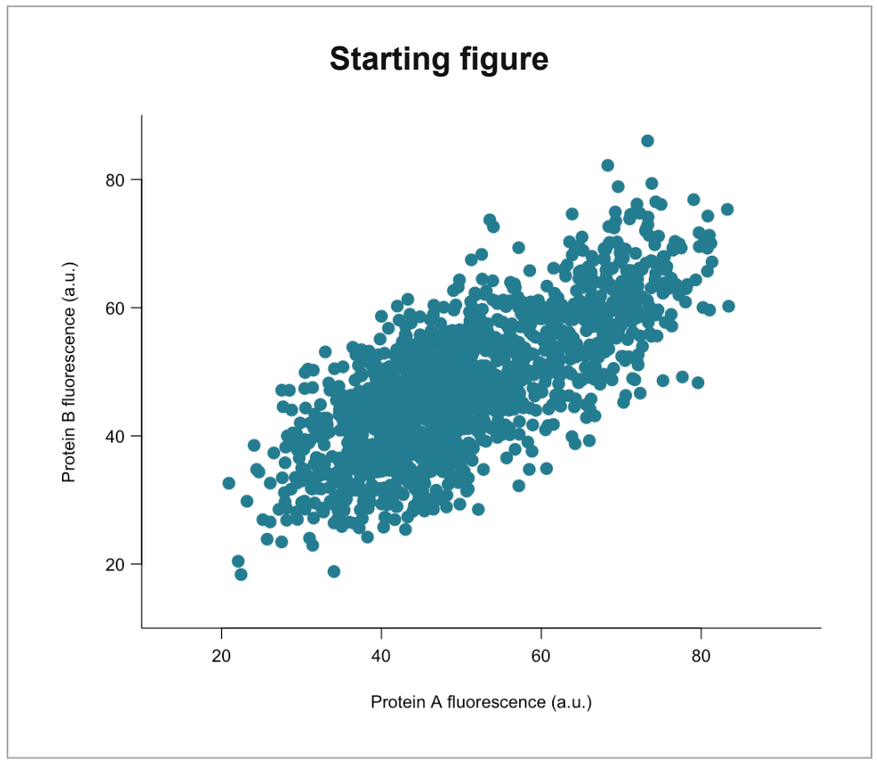
Less ink reveals gaps and individual cells near the edges. The crowded centre is still difficult to read.

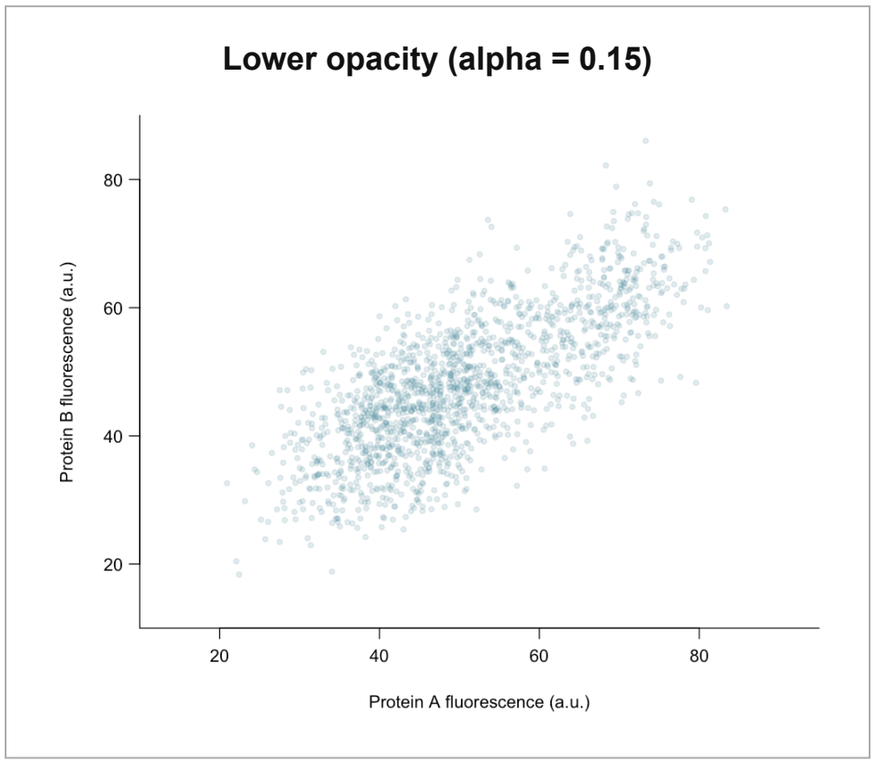
Previous step
Step 2 / 5
Next step

Context & source

What is hidden in this cloud of cells?

Not reliably: points cover other points.





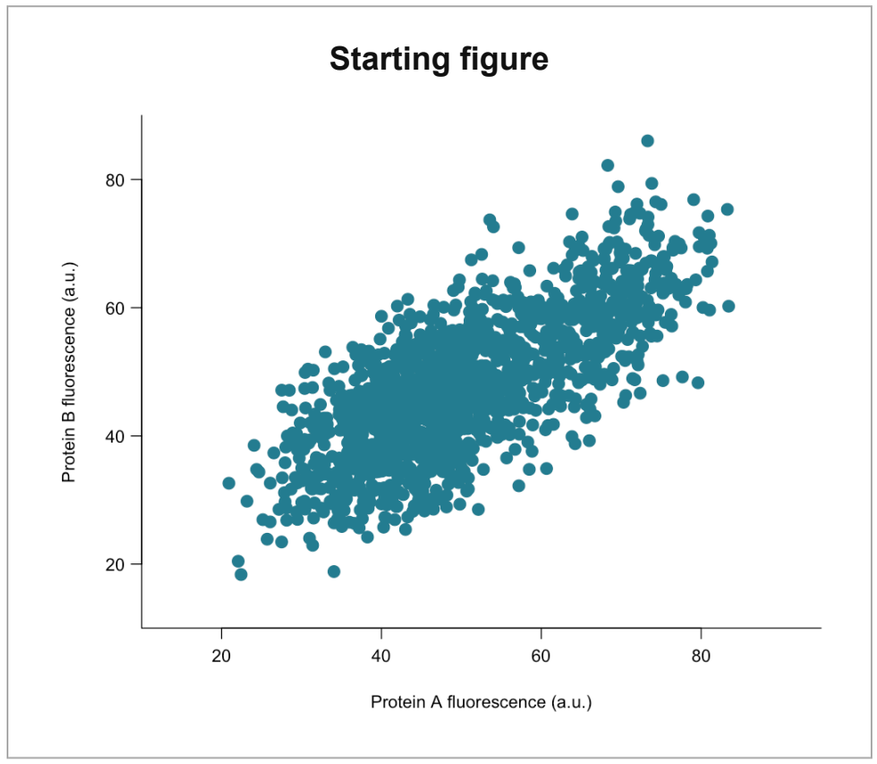
Keep the smaller points and make them transparent. Overlap now appears darker, revealing concentrations. Very dense regions can still saturate; darkness is not an exact cell count.

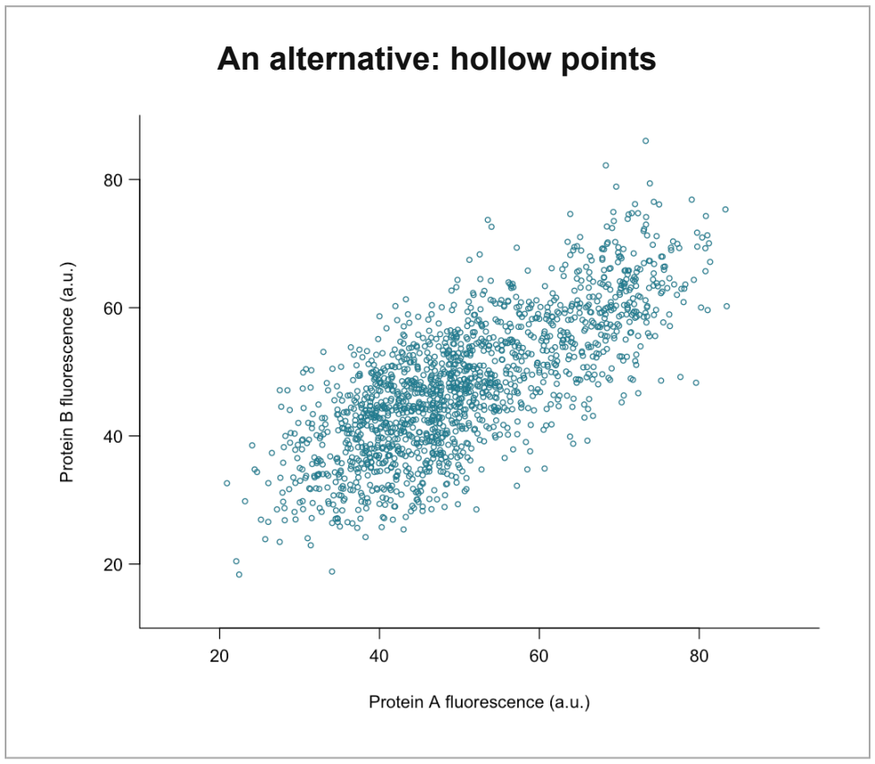
Previous step Step 3 / 5 Next step

Context & source

What is hidden in this cloud of cells?

Not reliably: points cover other points.





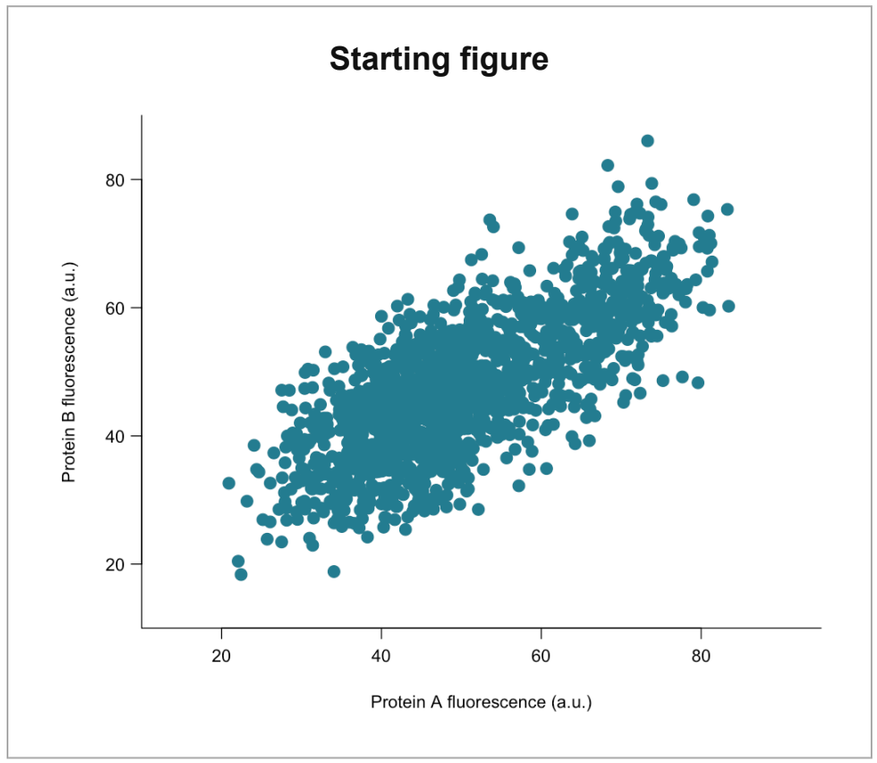
Draw only the borders, at full opacity. Empty centres make some overlaps easier to see, but many outlines can still become cluttered. Compare with transparency: hollow points are not automatically better.

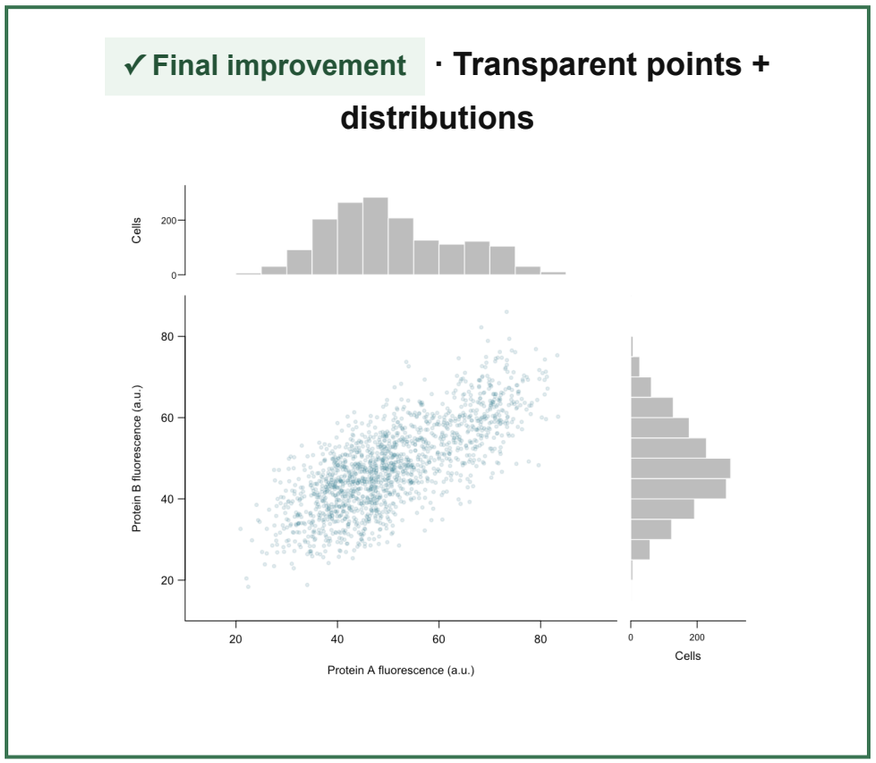
Previous step Step 4 / 5 Next step

Context & source

What is hidden in this cloud of cells?

Not reliably: points cover other points.





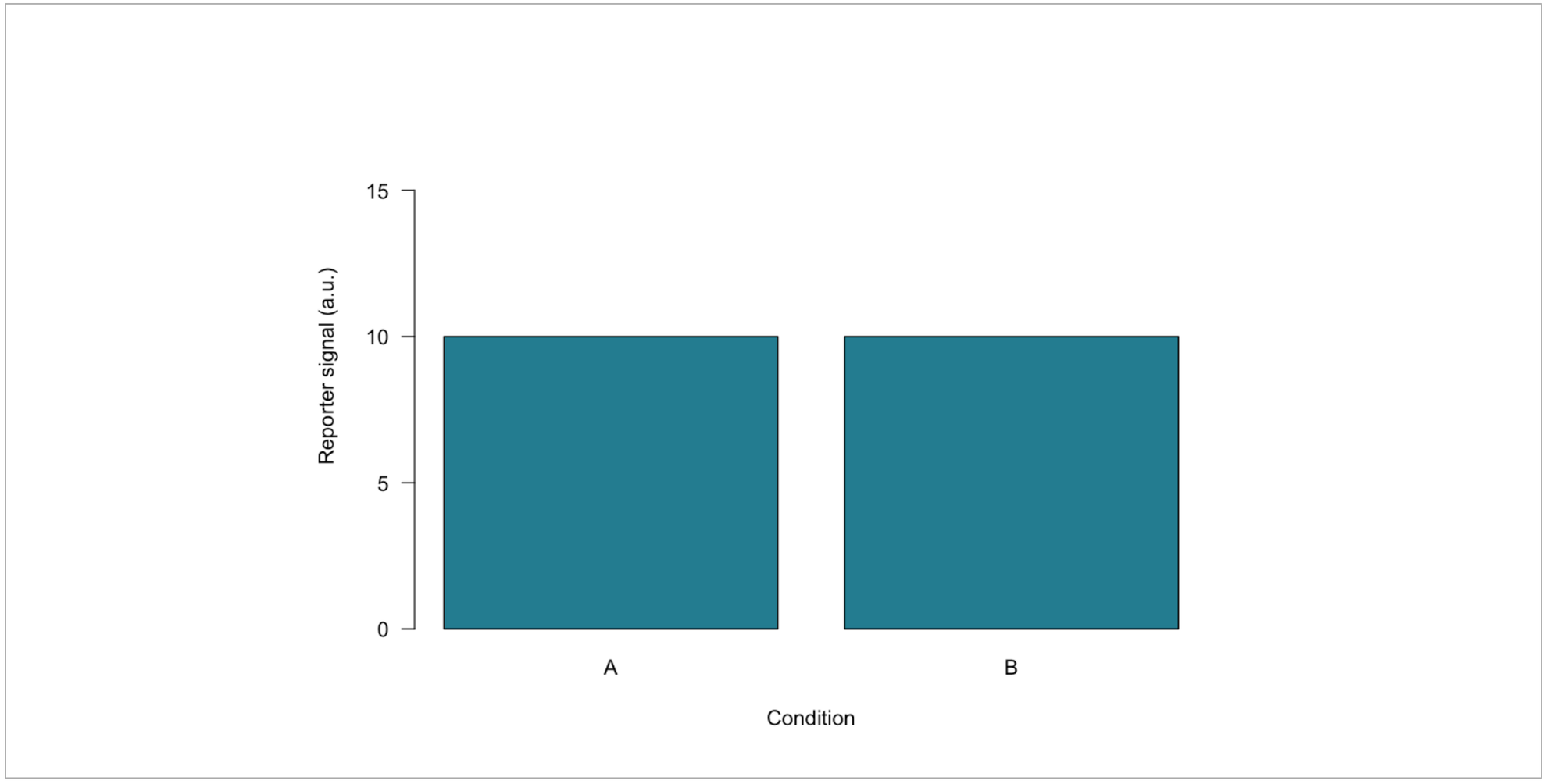
Return to transparent points and add histograms. The top shows the distribution of protein A; the right shows protein B. Bar length counts cells. These help read each protein separately; the scatterplot is still needed to see how the two relate.

Previous step | Step 5 / 5 | Next step

Context & source

Do similar averages mean similar cells?

The averages are similar, but the distributions may differ.

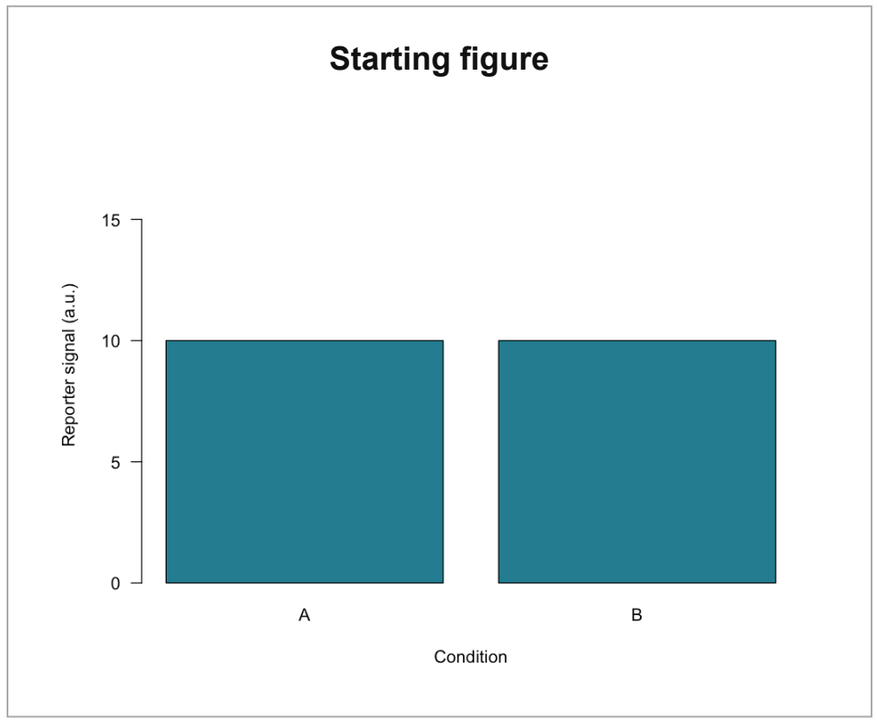


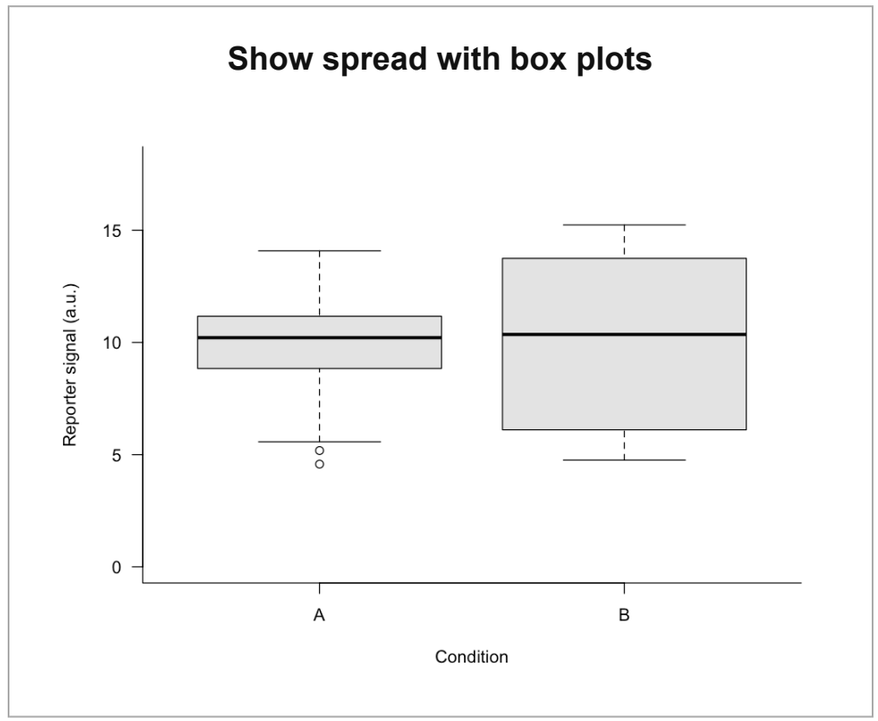
Reveal first improvement

Context & source

Do similar averages mean similar cells?

The averages are similar, but the distributions may differ.





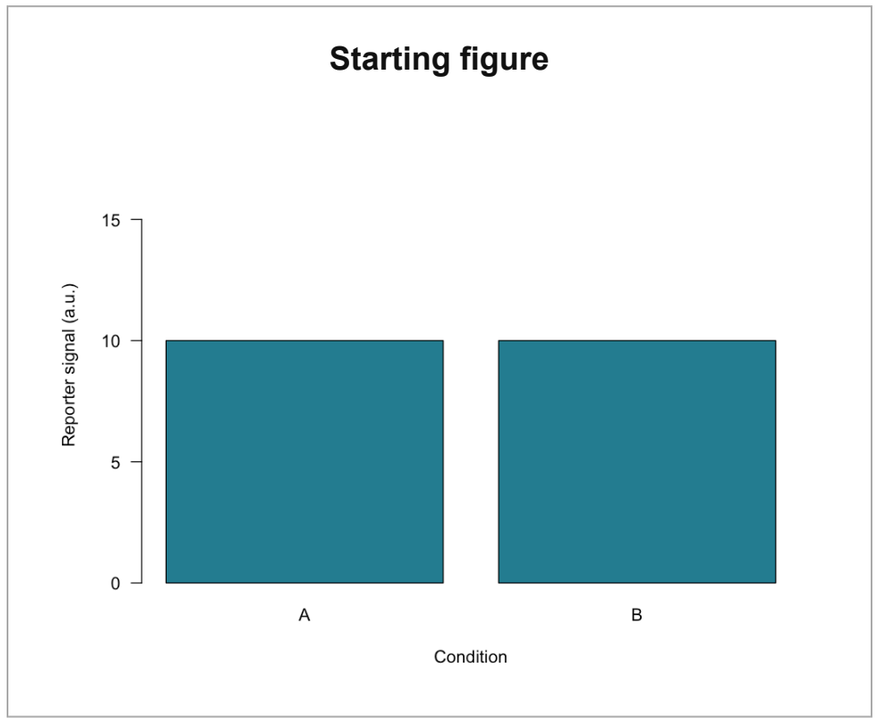
The box spans the middle 50% of values; the line is the median. Whiskers extend to observations within 1.5 interquartile ranges. B is more spread out, but its internal shape remains hidden.

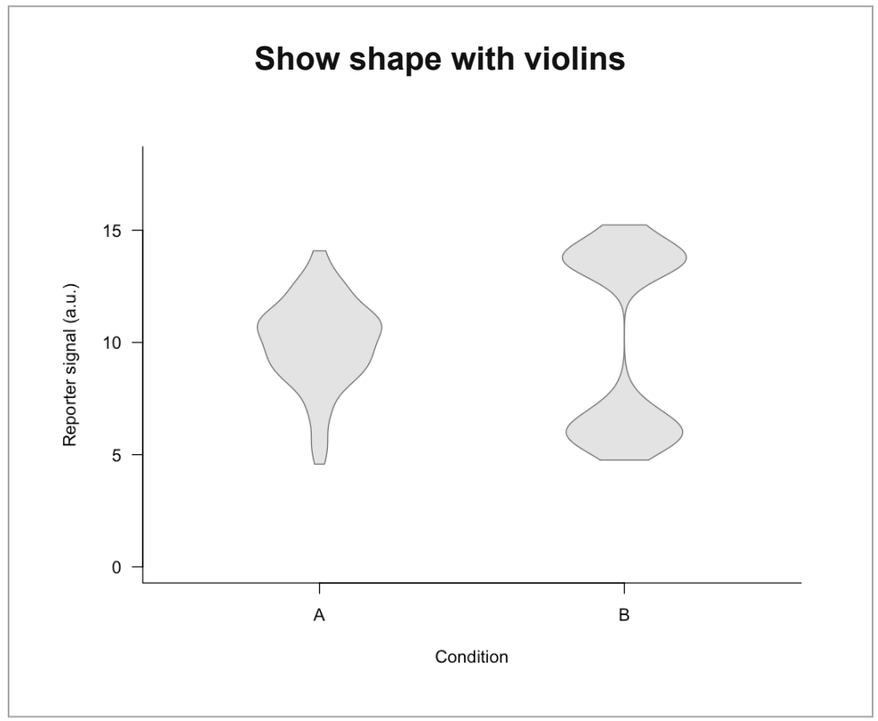
Previous step Step 2 / 4 Next step

Context & source

Do similar averages mean similar cells?

The averages are similar, but the distributions may differ.





Width shows estimated density at each signal level. B has two concentrations of values. Both violins use the same smoothing bandwidth and width rule; changing the smoothing can change the apparent shape.

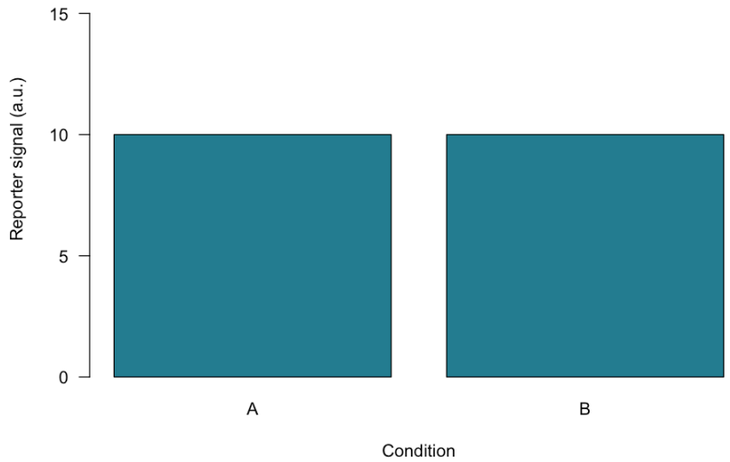
Previous step Step 3 / 4 Next step

Context & source

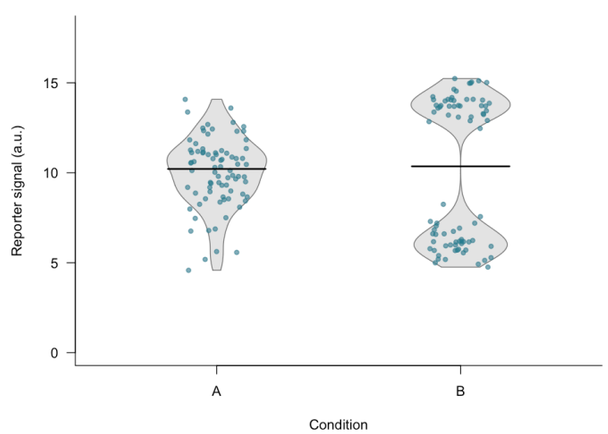
Do similar averages mean similar cells?

The averages are similar, but the distributions may differ.

Starting figure



✓ Final improvement · Keep the individual cells visible



Add dots and a median line. Horizontal jitter separates points without changing signal values. The dots let us check the smoothed outline. With only a few observations, dots alone would be preferable.

Previous step

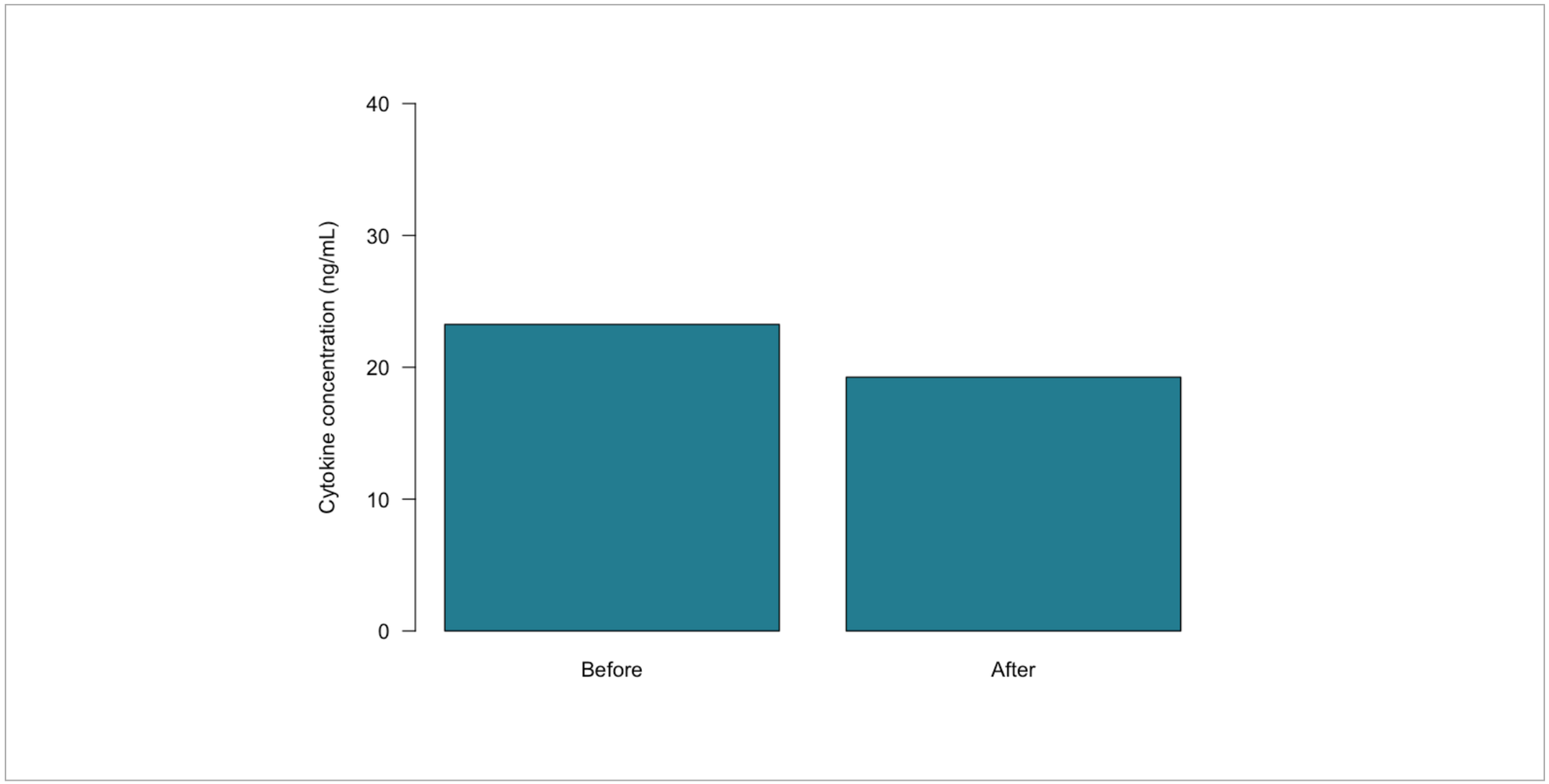
Step 4 / 4

Next step

Context & source

Did every donor improve?

No: individual changes may differ.

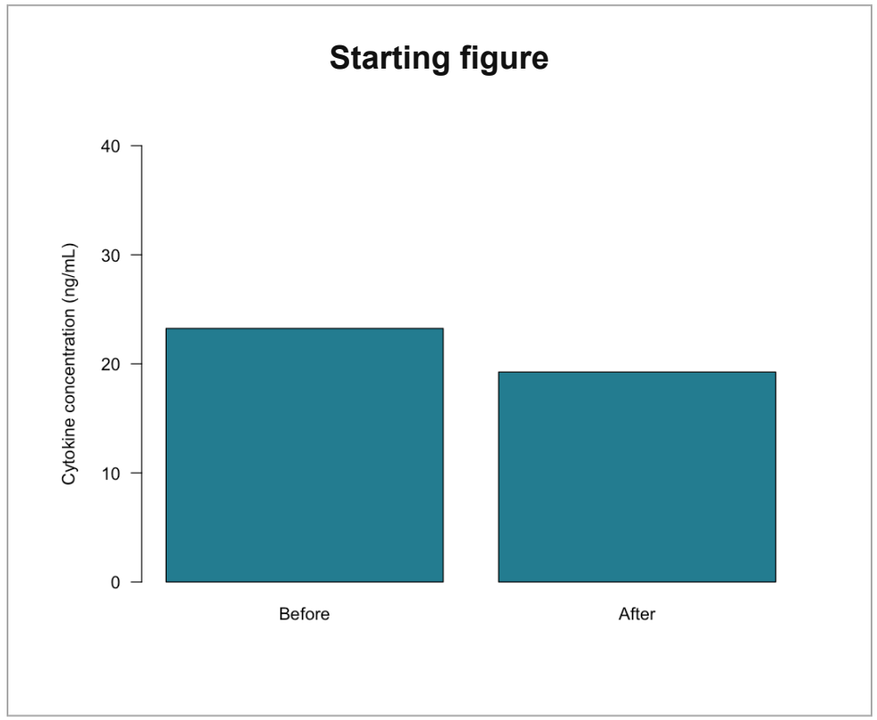


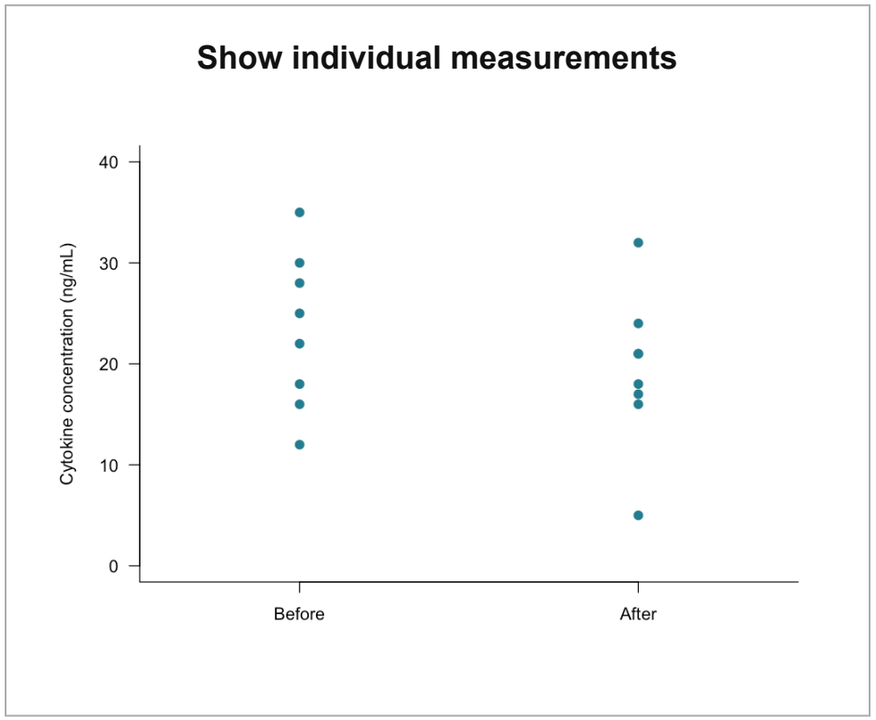
Reveal first improvement

Context & source

Did every donor improve?

No: individual changes may differ.





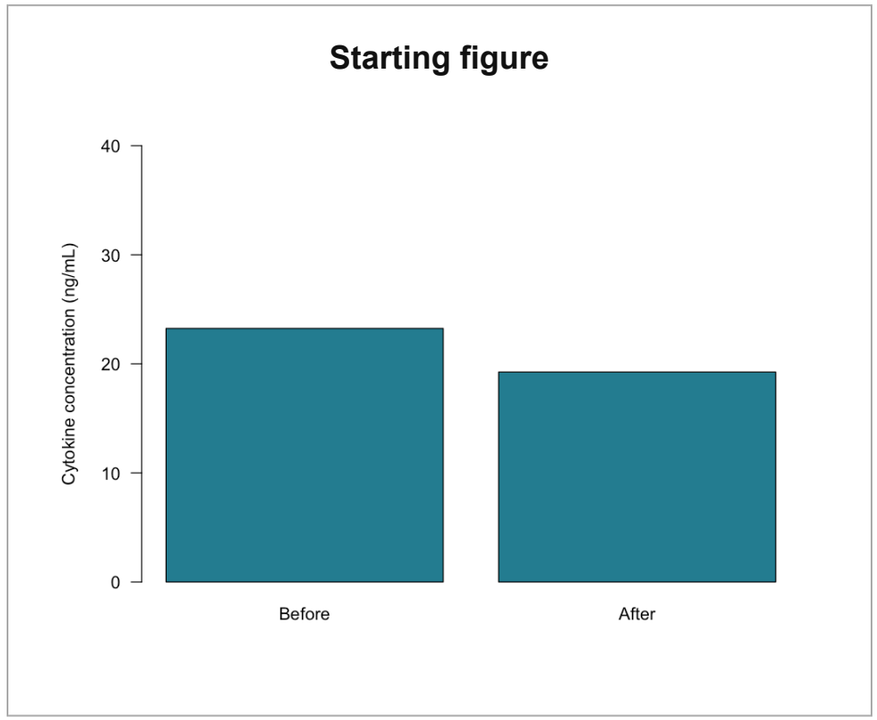
Dots reveal the range of concentrations. We still cannot identify which before and after measurements belong together.

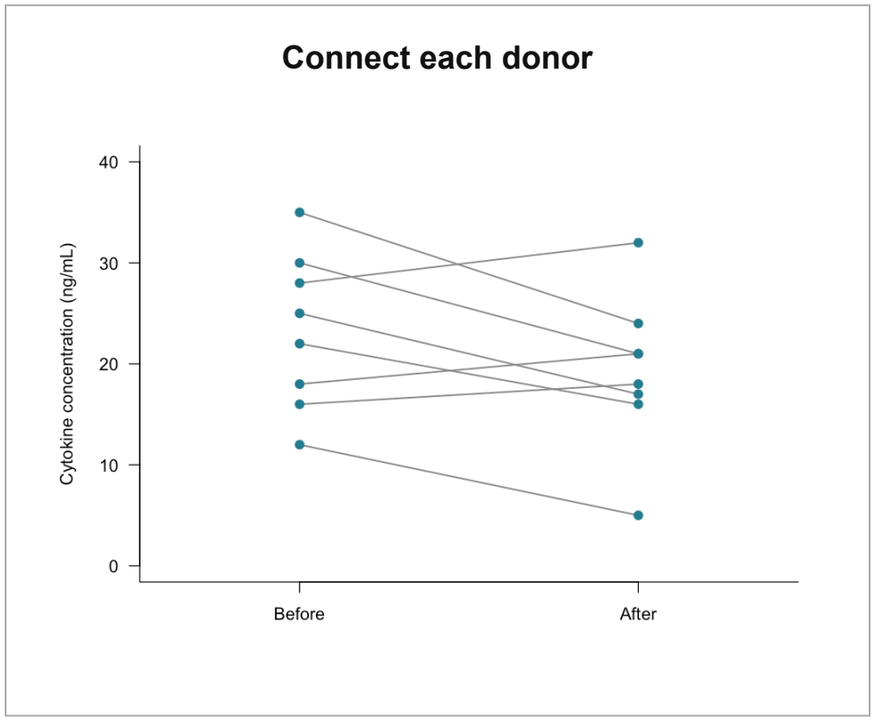
Previous step | Step 2 / 4 | Next step

Context & source

Did every donor improve?

No: individual changes may differ.





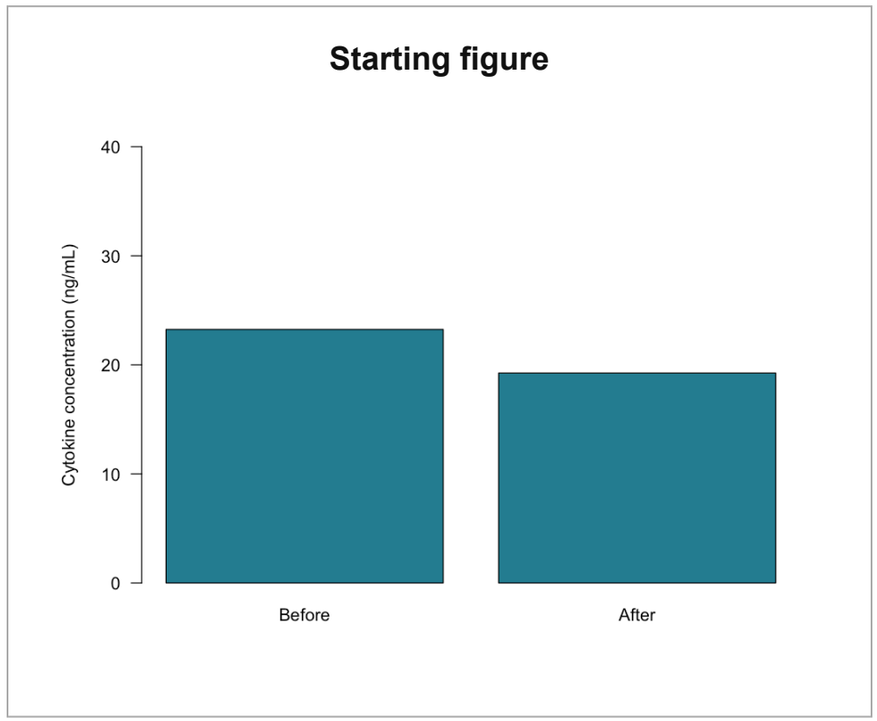
Each line joins measurements from the same donor. Five donors decrease and three increase. The pairing reveals what the separate groups of dots could not.

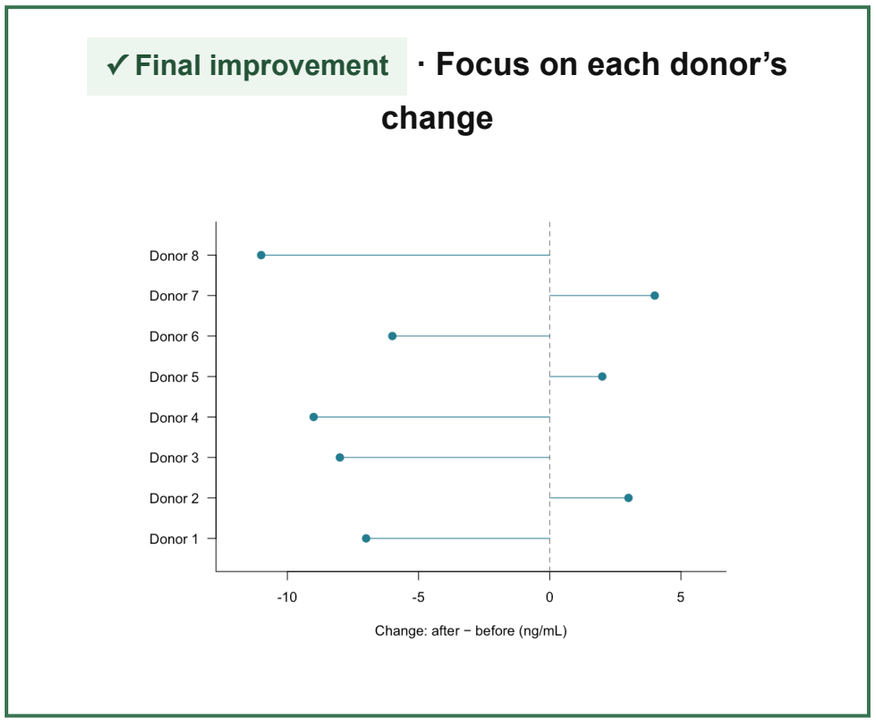
Previous step
Step 3 / 4
Next step

> **Context & source**

Did every donor improve?

No: individual changes may differ.





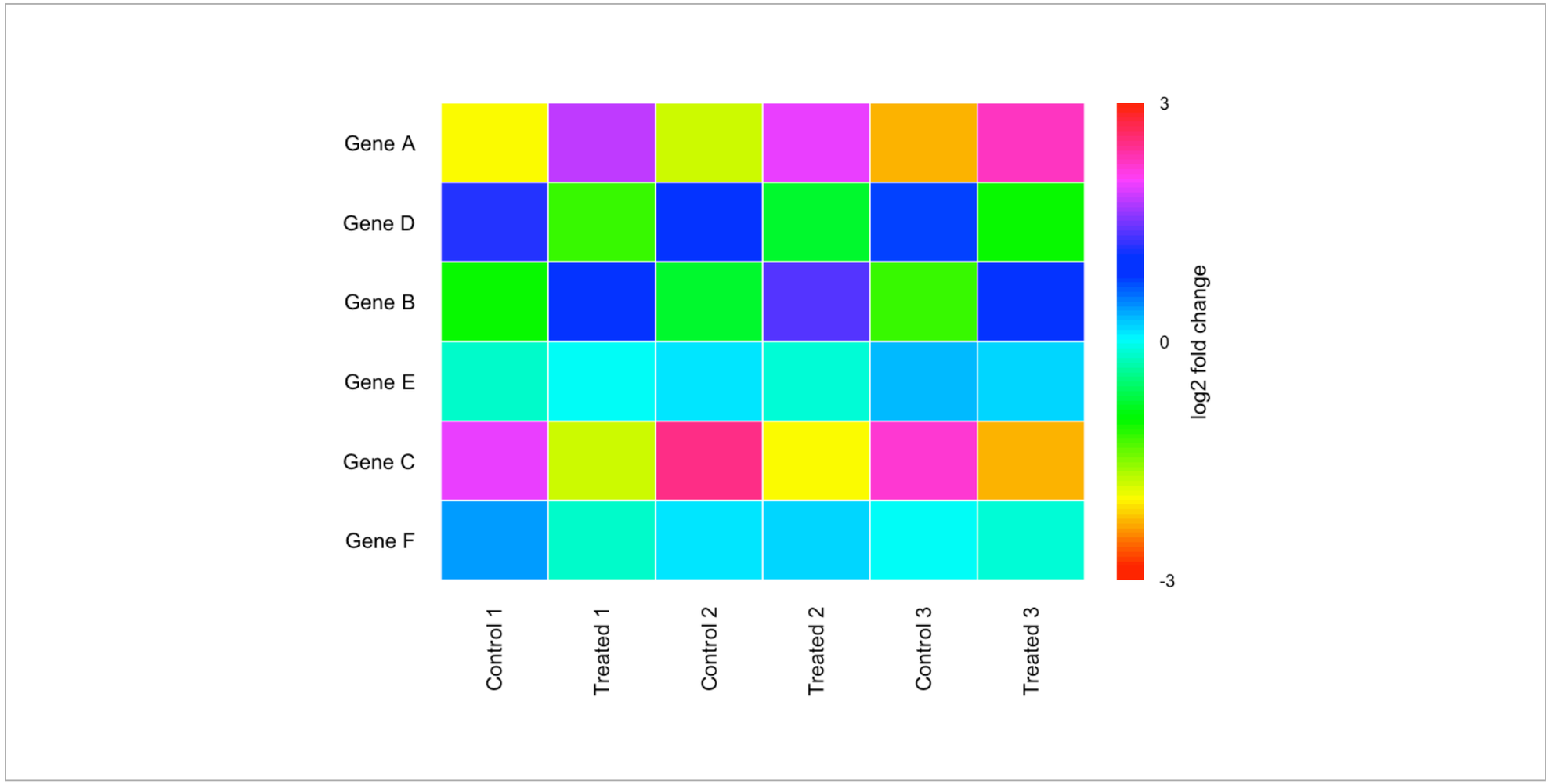
Now the horizontal axis is after minus before; rows identify donors. Negative values indicate decreases. This makes the direction and size of each change easy to compare, while the previous view preserves starting levels. Neither view alone establishes a treatment effect without an appropriate study design.

Previous step Step 4 / 4 Next step

Context & source

Which genes respond together?

Use a meaningful colour scale and organise the samples.

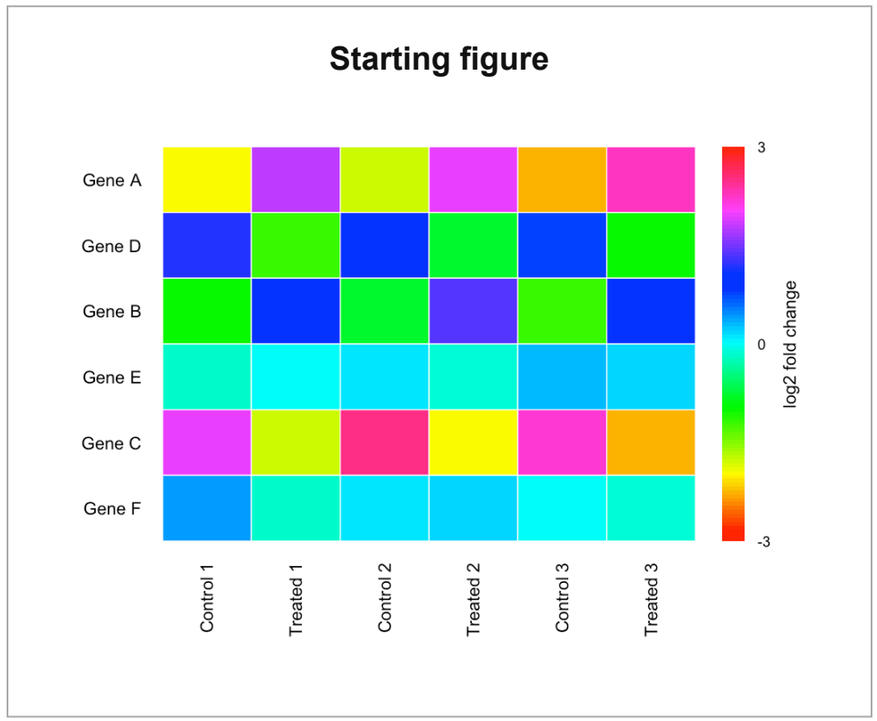


Reveal first improvement

Context & source

Which genes respond together?

Use a meaningful colour scale and organise the samples.



Use white at zero, blue for negative values, and red for positive values. The symmetric limits are unchanged. A diverging scale suits this meaningful reference point; it is not the default for every heatmap.

Previous step

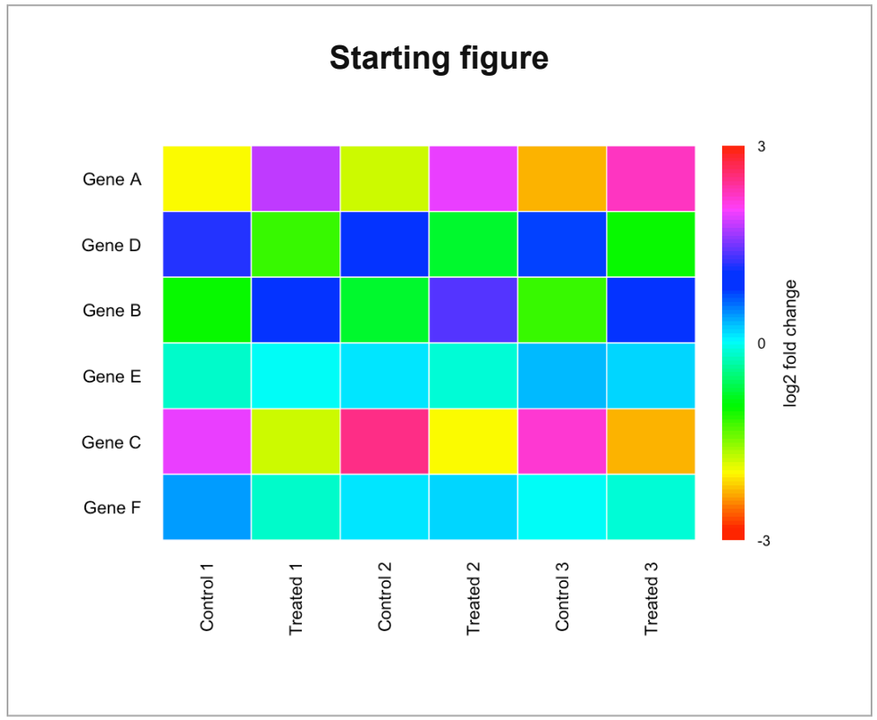
Step 2 / 5

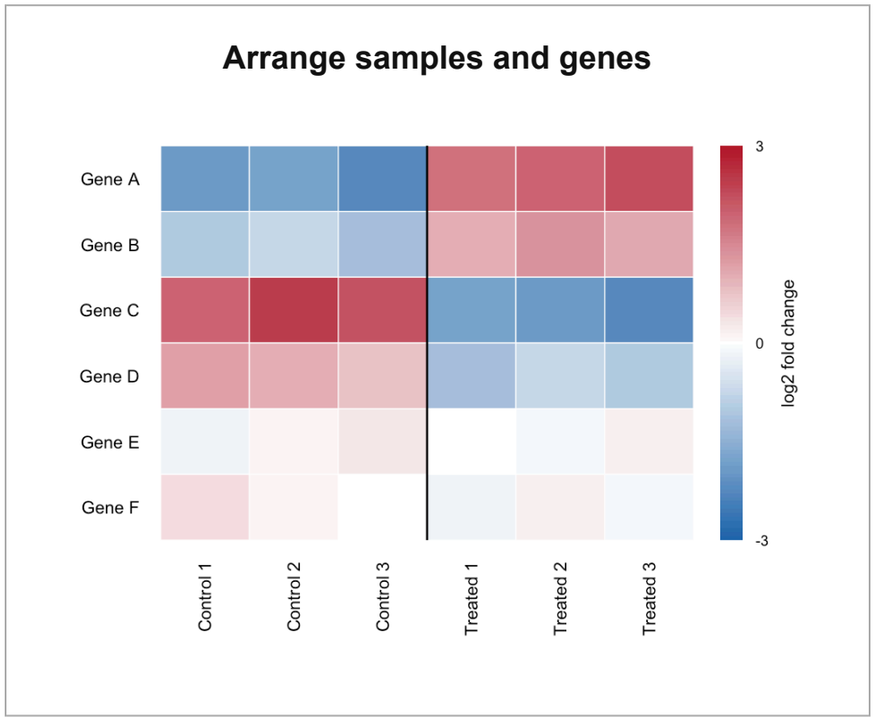
Next step

Context & source

Which genes respond together?

Use a meaningful colour scale and organise the samples.





Group control and treated samples, then place genes with similar responses together. Only the order changes: no values are rescaled. The divider marks the two sample groups. This is a teaching order, not a statistical clustering result.

Previous step

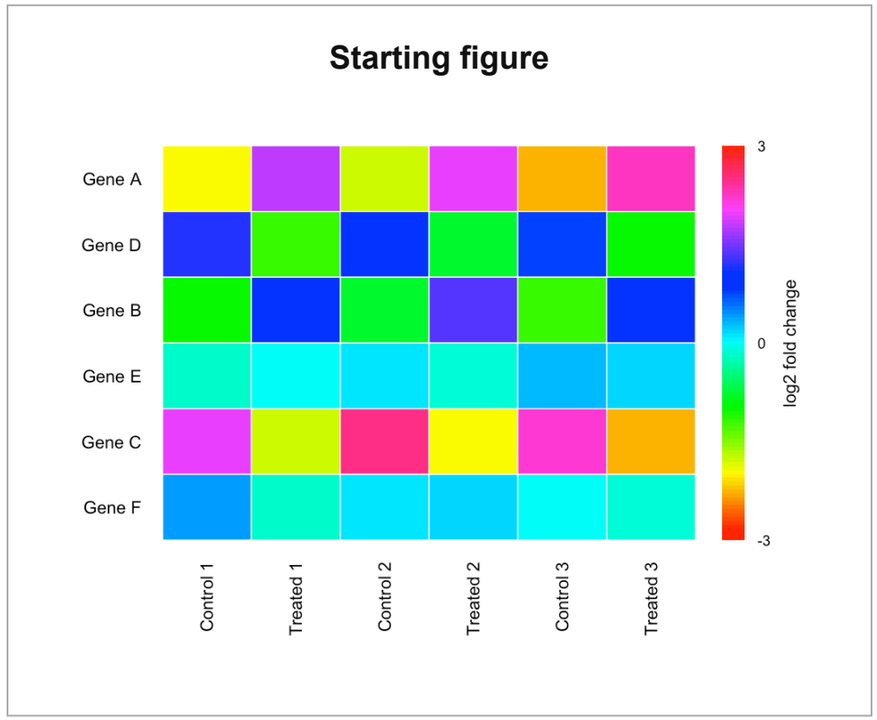
Step 3 / 5

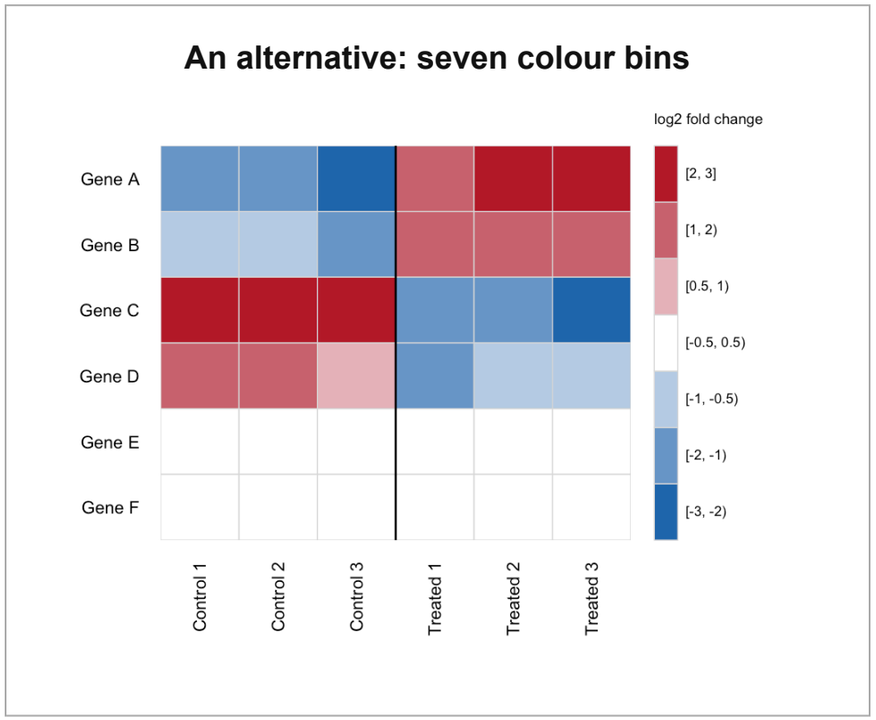
Next step

Context & source

Which genes respond together?

Use a meaningful colour scale and organise the samples.





Group values into seven labelled colour ranges. This makes broad categories easier to distinguish, but hides differences within each bin and creates sharp visual boundaries. White now means −0.5 to below +0.5, not exactly zero. In the legend, [includes a boundary and) excludes it. These cutoffs are teaching choices, not biological thresholds; use bins when ranges matter more than fine differences.

Previous step

Step 4 / 5

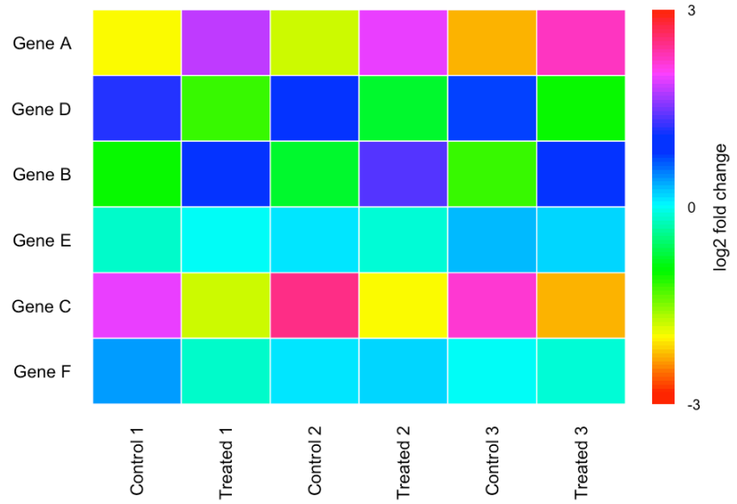
Next step

Context & source

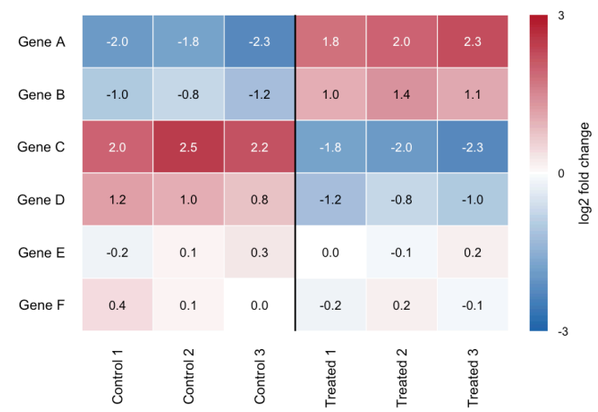
Which genes respond together?

Use a meaningful colour scale and organise the samples.

Starting figure



✓ Final improvement · Add readable values when useful



Return to the continuous colour scale and add exact values. This small matrix has room for numbers. White text on dark cells and black text on light cells keeps them readable. On a much larger heatmap, omit the numbers and let colour show the overall pattern.

Previous step

Step 5 / 5

Next step

Context & source

What made the figure clearer?

Match the display to the question. Check scales and visual encodings. Keep useful detail, and remove distractions.

Choose one figure from your work. What would you change, and why?

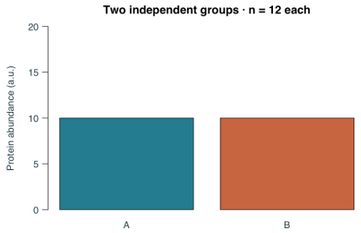
Continue to the next section →

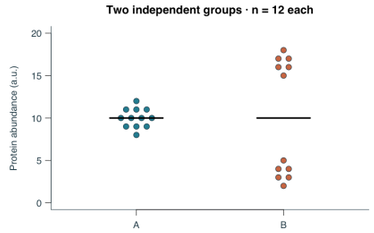
Same data, different plots

Compare two views and discuss which fits the question.

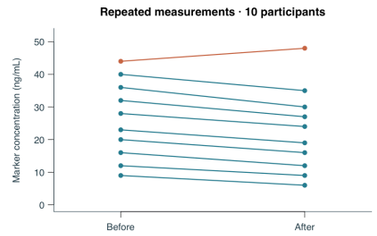
Same data, different plots

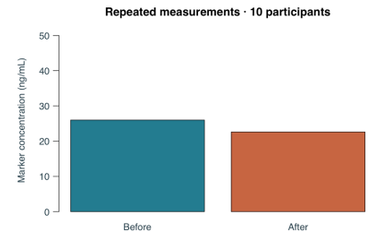
Which view fits the question, and what does each view hide?



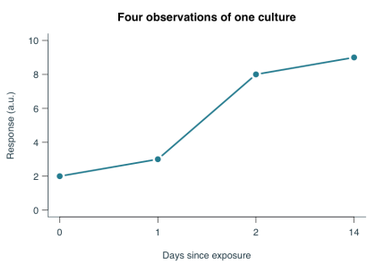


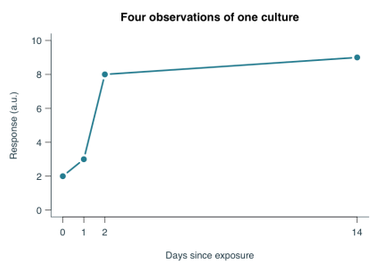
Does the mean describe the group?



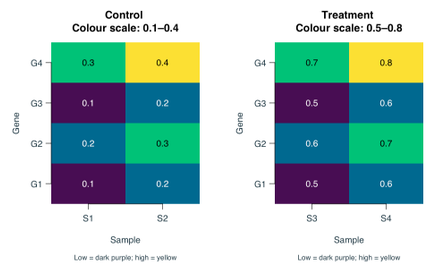


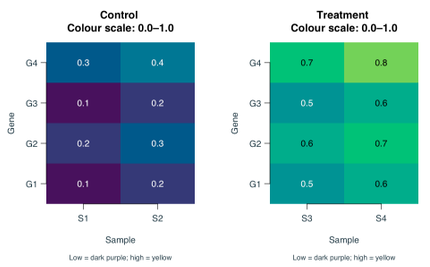
Did everyone improve?



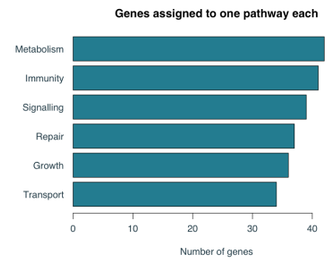


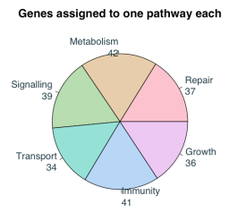
When did the response rise fastest?



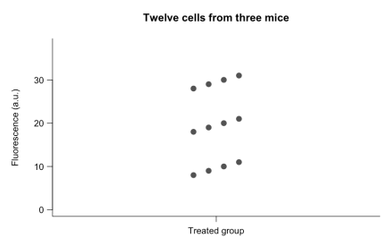


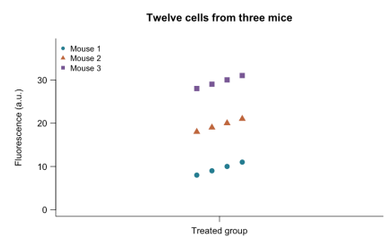
Can colours be compared across panels?



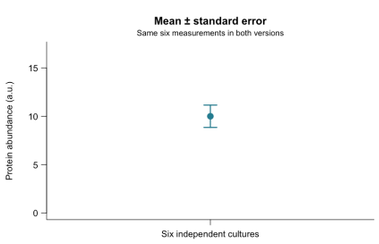


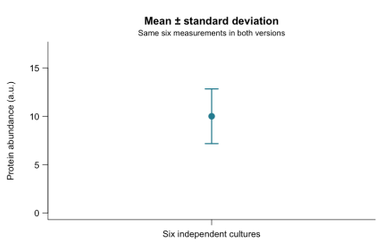
Which pathway ranks third?





Twelve cells or three mice?



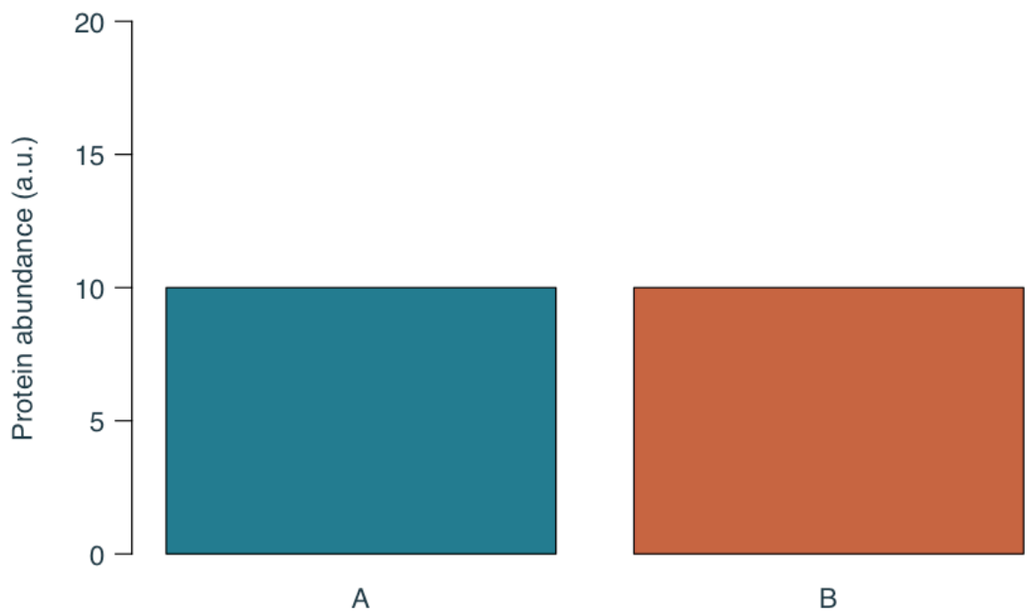


What do the error bars describe?

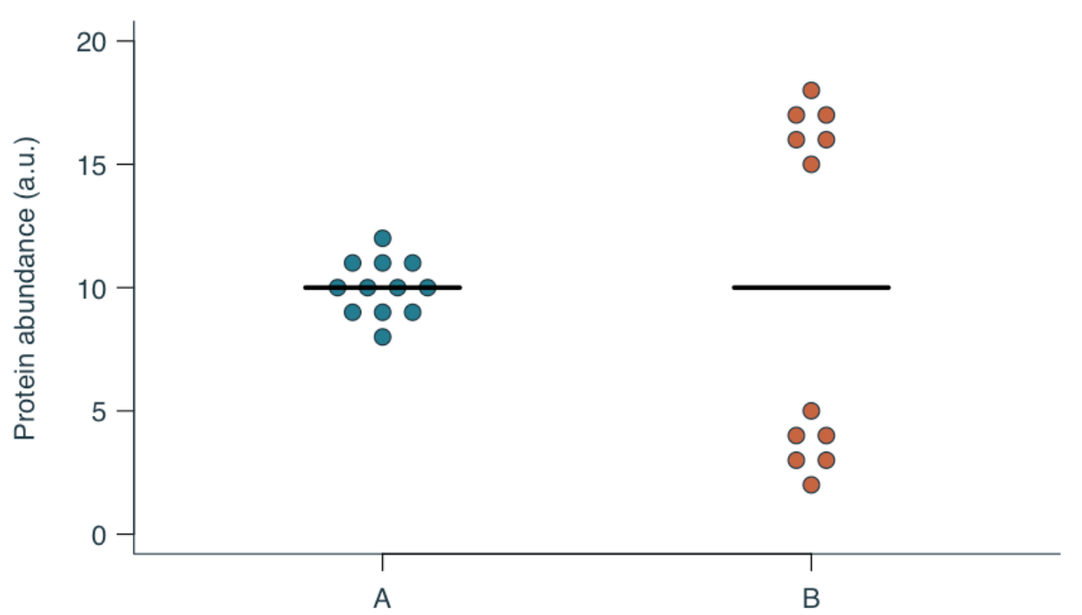
Does the mean describe the group?

Which view better answers: “Do the groups have similar distributions?”

View 1



View 2



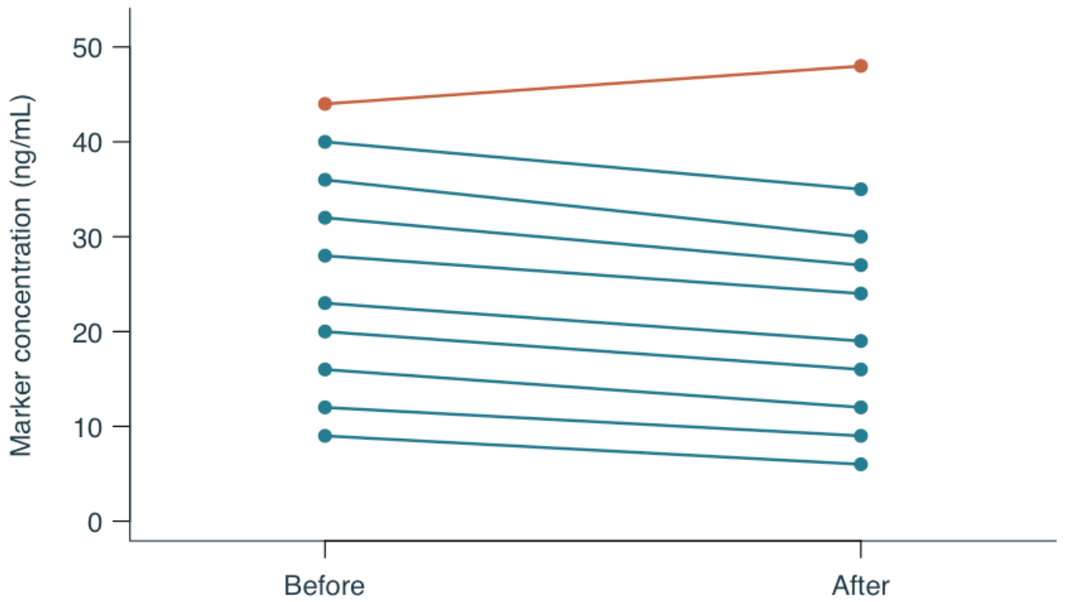
Discuss: which would you use, and why?

> **Data context**

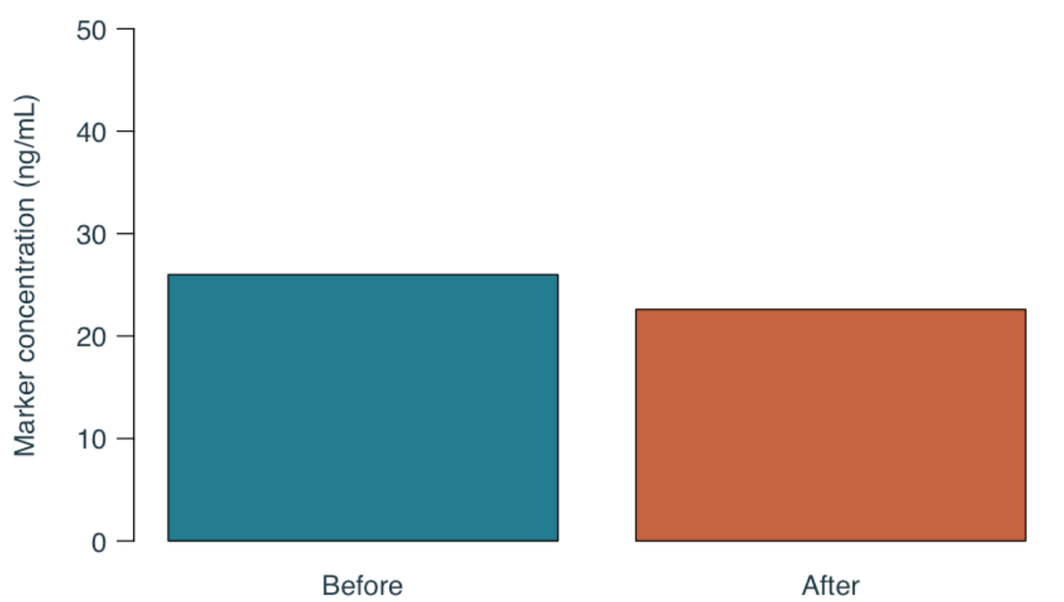
Did everyone improve?

Lower concentration is the desired direction. Which view reveals whether each participant decreased?

View 1



View 2



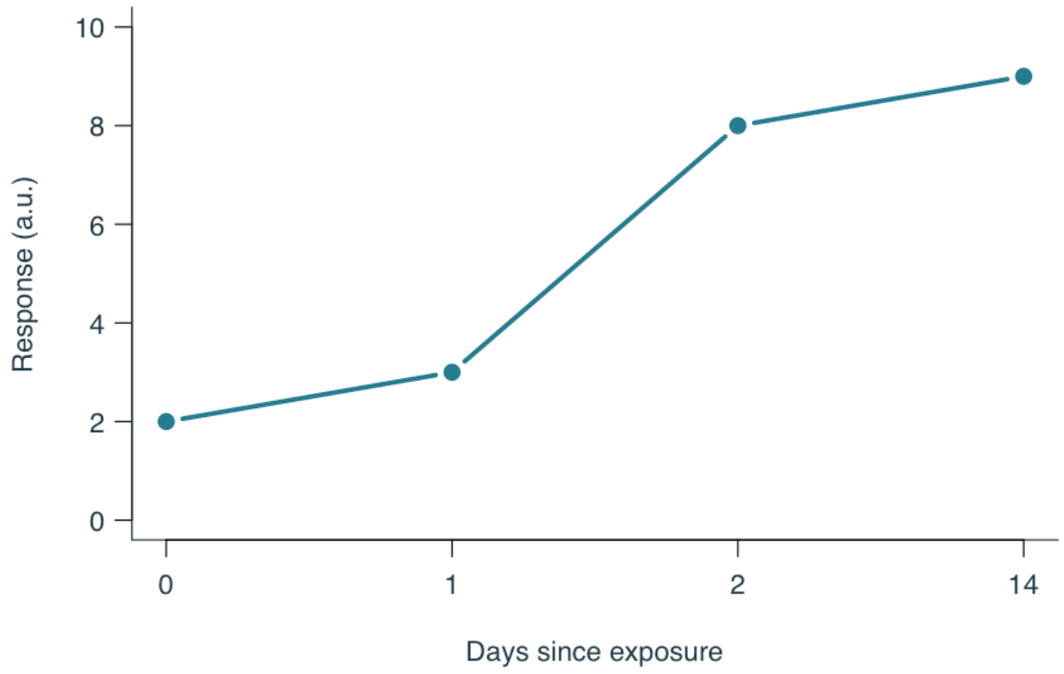
Discuss: which would you use, and why?

> **Data context**

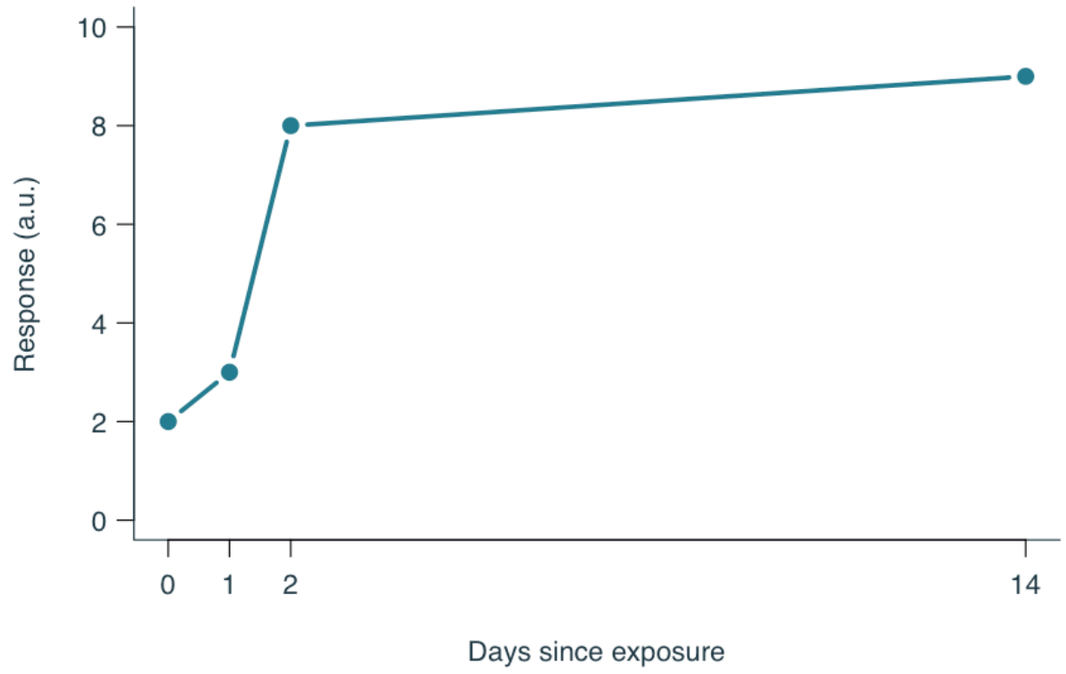
When did the response rise fastest?

Which view represents change per day faithfully?

View 1



View 2

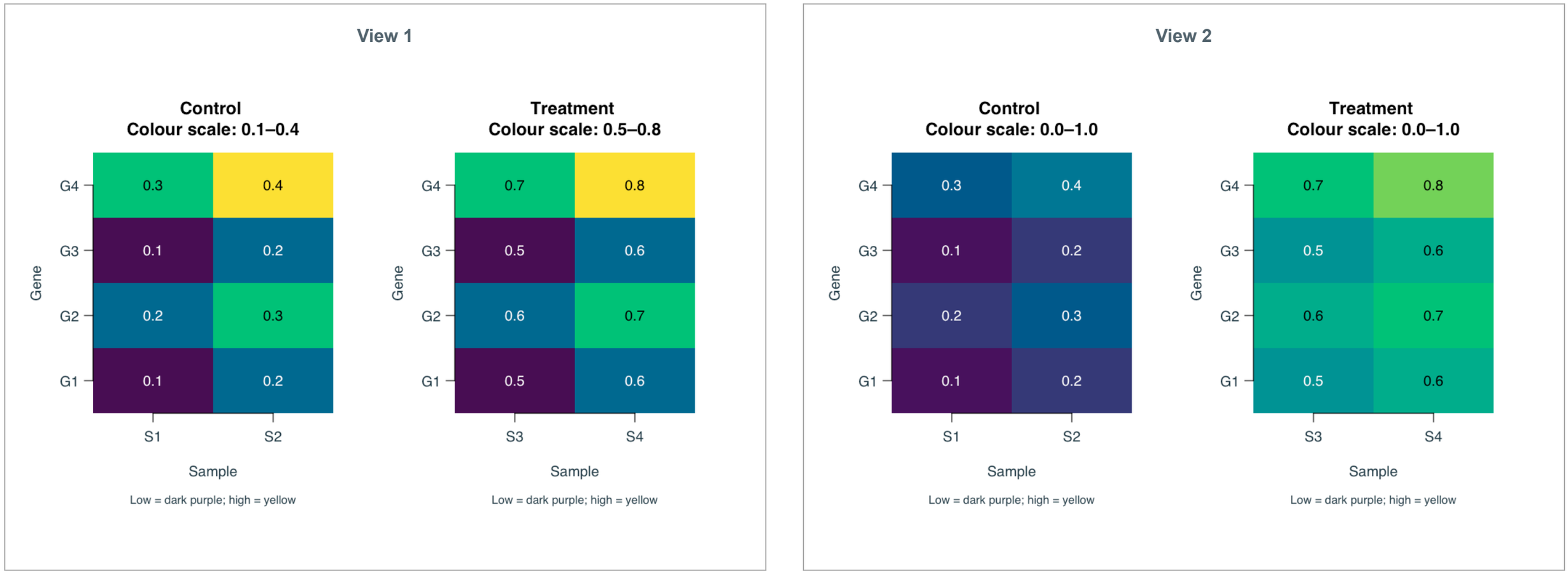


Discuss: which would you use, and why?

Data context

Can colours be compared across panels?

Which view better supports comparing absolute allele frequencies between conditions?



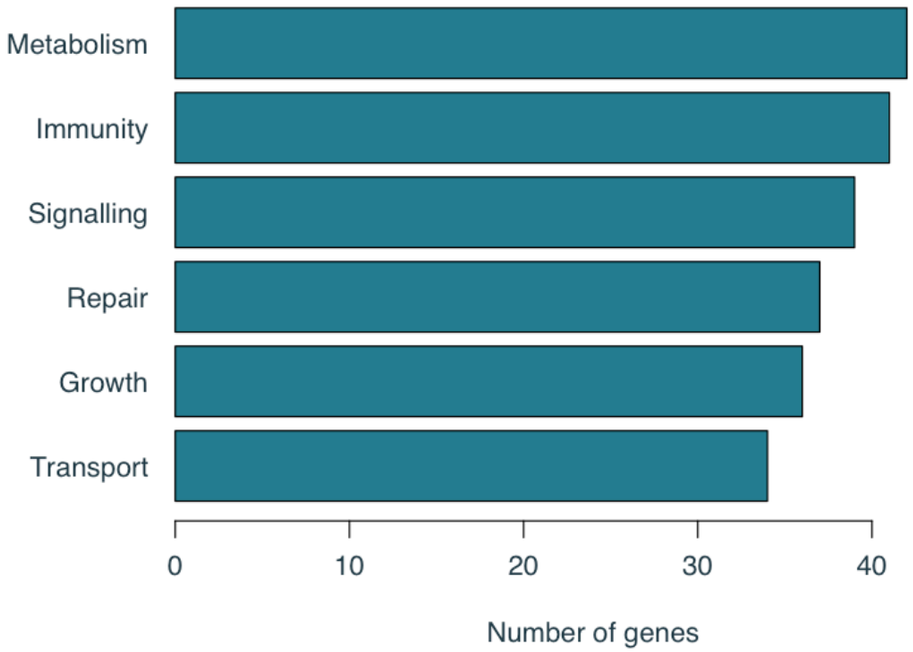
Discuss: which would you use, and why?

> **Data context**

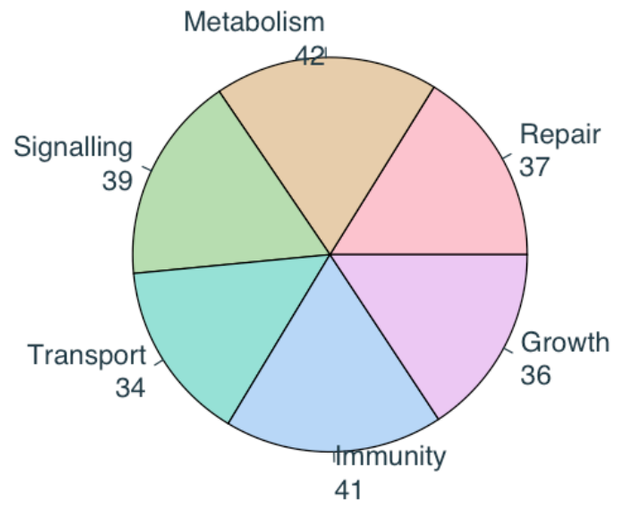
Which pathway ranks third?

Which display makes it easiest to rank these six similar counts?

View 1



View 2



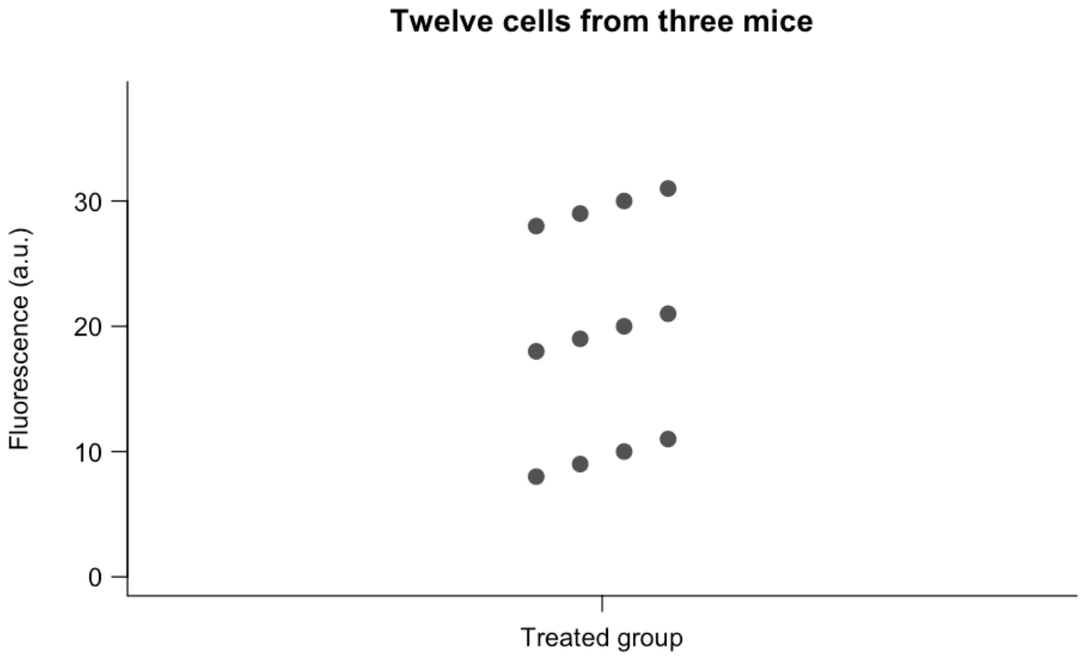
Discuss: which would you use, and why?

> **Data context**

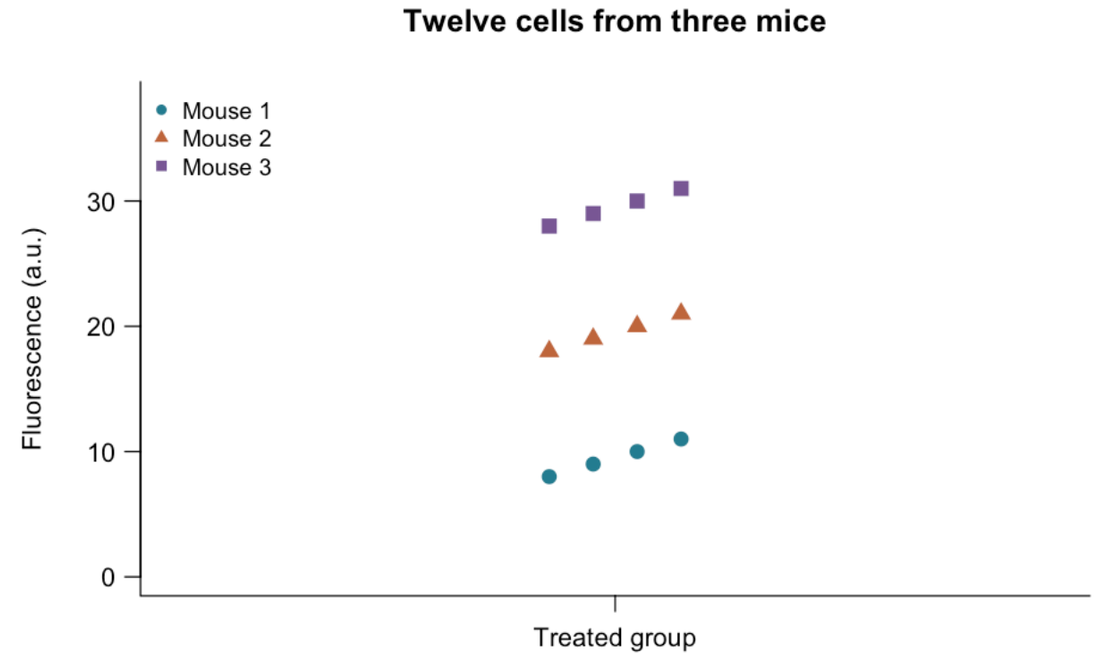
Twelve cells or three mice?

Which version lets you distinguish variation within a mouse from differences between mice?

View 1



View 2



Discuss: which would you use, and why?

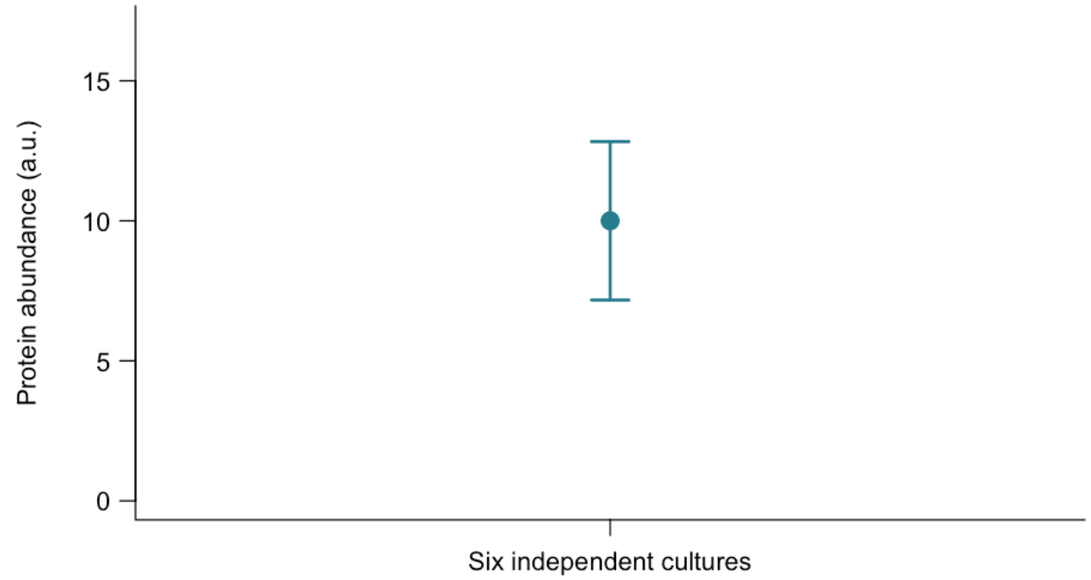
> **Data context**

What do the error bars describe?

Which version describes the spread of measurements among the cultures?

View 1

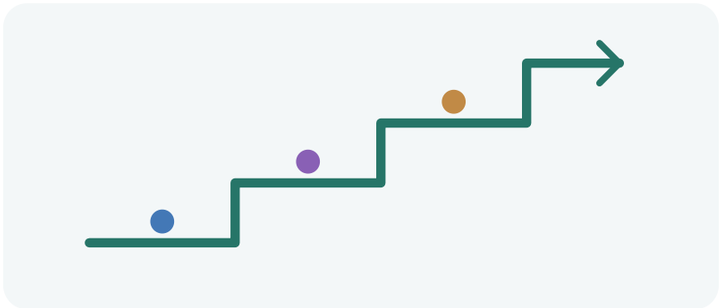
View 2



Discuss: which would you use, and why?

> **Reveal solution**

> **Data context**

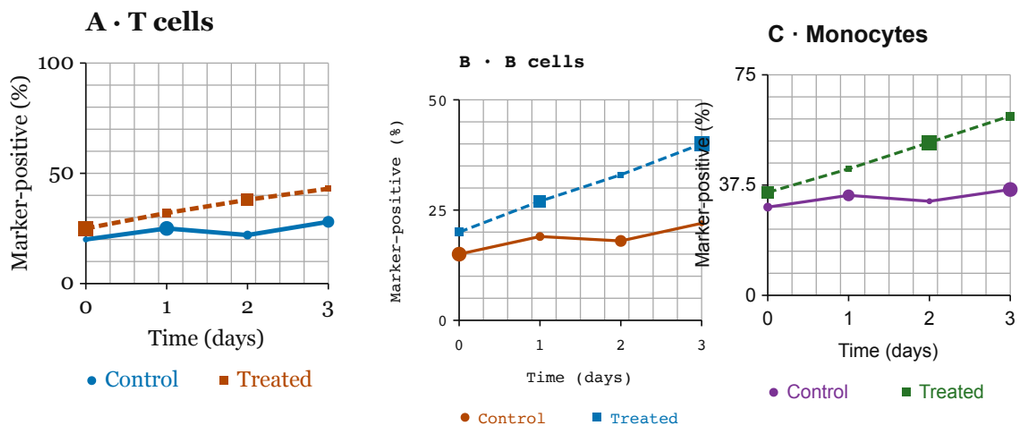


Improve a figure step by step

Build consistent panels and untangle crowded point plots.

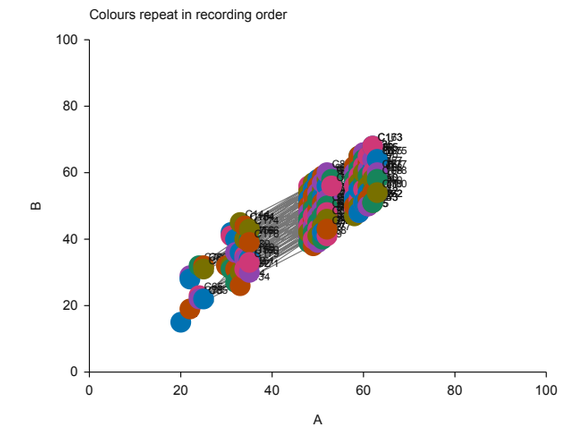
Improve a figure step by step

Keep the data fixed. Make one change, compare the result, and inspect the code.



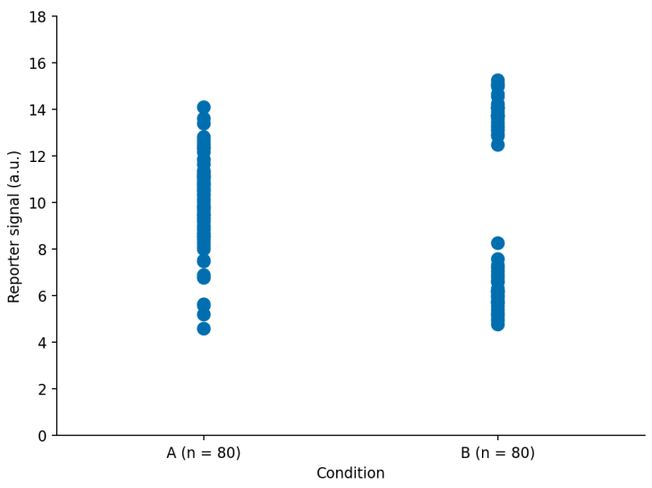
Consistent multi-panel figures

Fonts, spacing, colours, axes, legends and grid lines.



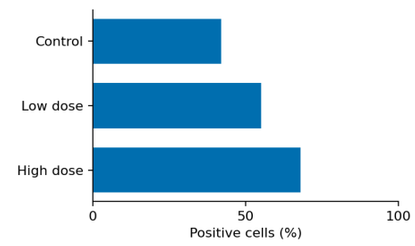
Crowded point plots

Overplotting, colour, labels and carefully placed annotations.



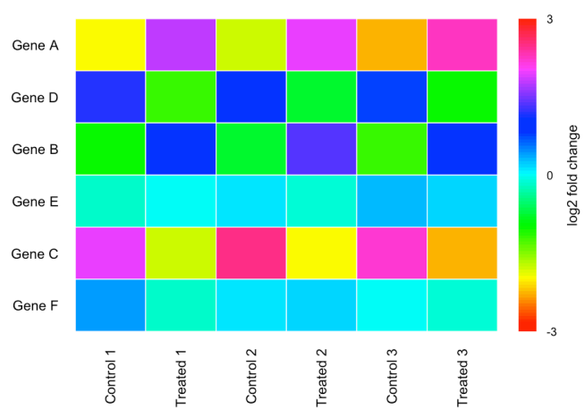
Points, boxes or violins?

Compare what each view reveals, and what it hides.



How small can a figure be?

Trim whitespace, test smaller sizes, and find the readability limit.



Clearer heatmaps

Meaningful colour, sample ordering, colour bins and readable values.

> **More step-by-step lessons**

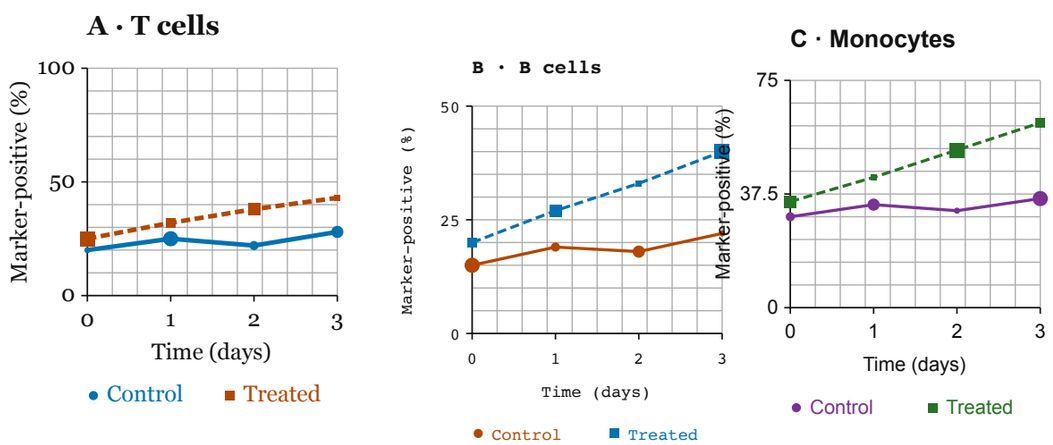
Do and don’t: consistent figures

Spot the problems

Start with the whole figure

Three cell populations measured over four days. What makes these panels difficult to compare?

Starting figure



Improved version

Click “Fix font family” to reveal the first improvement.

> **Show code changes**

> **Show full Python code for these plots**

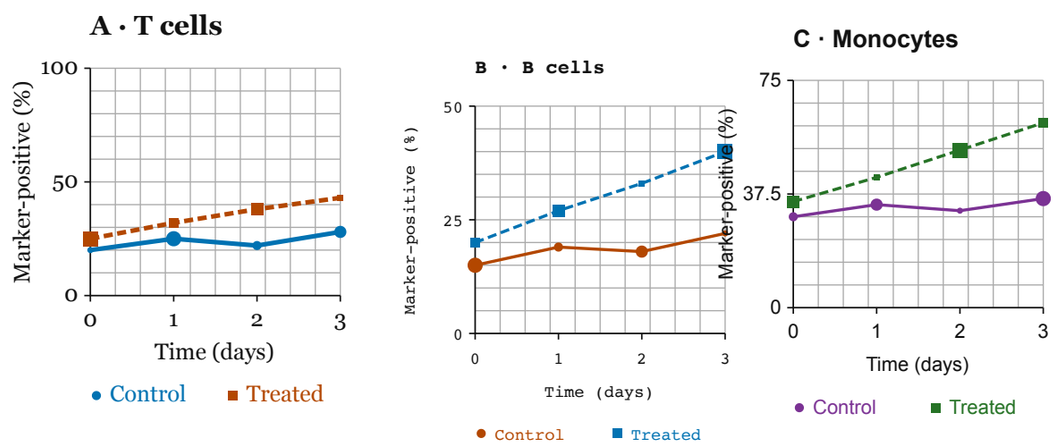
Do and don’t: consistent figures

Improvement 1 of 10

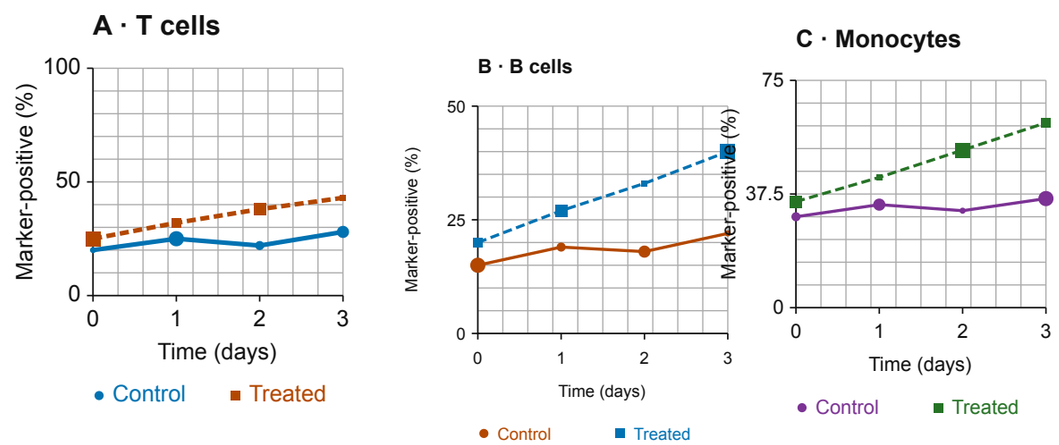
Use one font family

Use Arial throughout. Mixed typefaces no longer suggest different kinds of information.

Starting figure



Improved version



The original stays on the left; the right includes all improvements so far.

Show code changes

Show full Python code for these plots

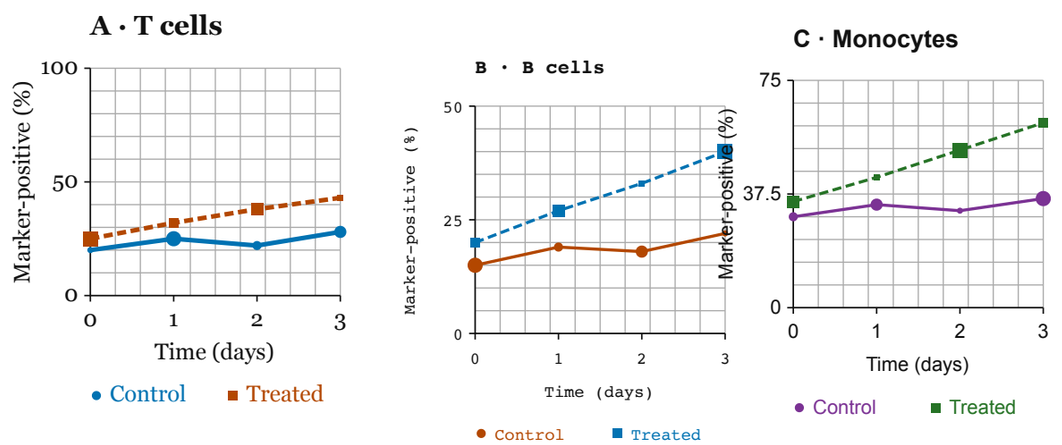
Do and don’t: consistent figures

Improvement 2 of 10

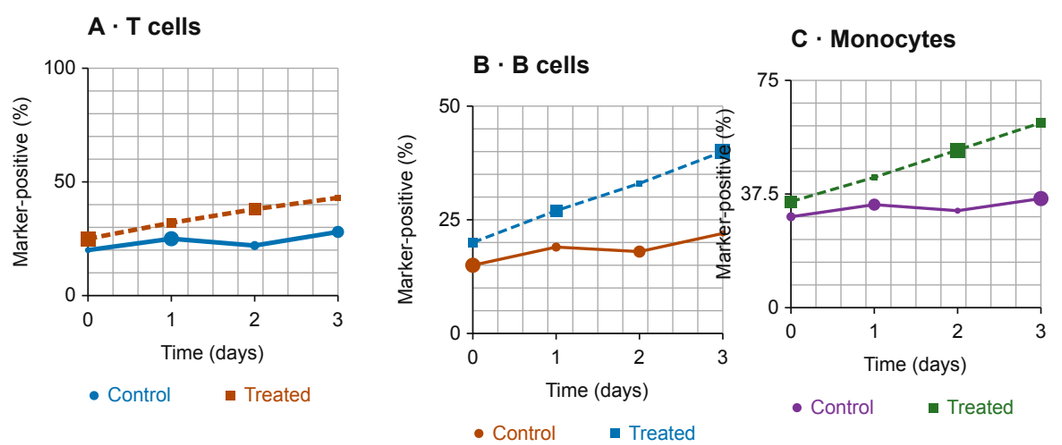
Use consistent font sizes

Give equivalent labels the same size across panels, with a consistent larger size for panel headings.

Starting figure



Improved version



The original stays on the left; the right includes all improvements so far.

Show code changes

Show full Python code for these plots

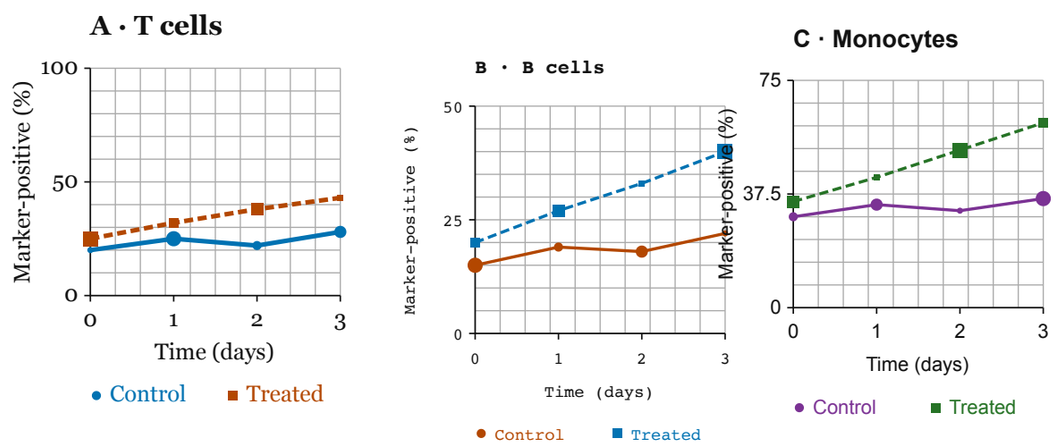
Do and don’t: consistent figures

Improvement 3 of 10

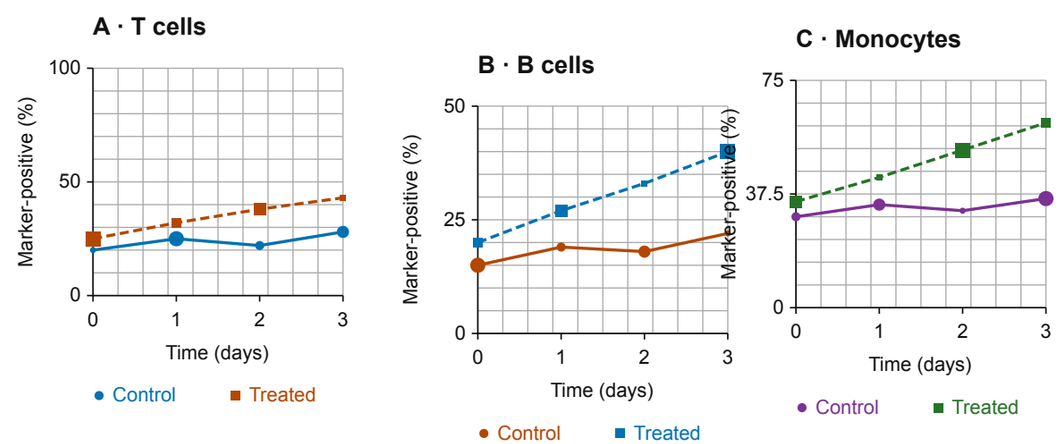
Match equivalent line widths

Panel A had slightly thicker traces than B and C. All condition traces now have the same width.

Starting figure



Improved version



The original stays on the left; the right includes all improvements so far.

Show code changes

Show full Python code for these plots

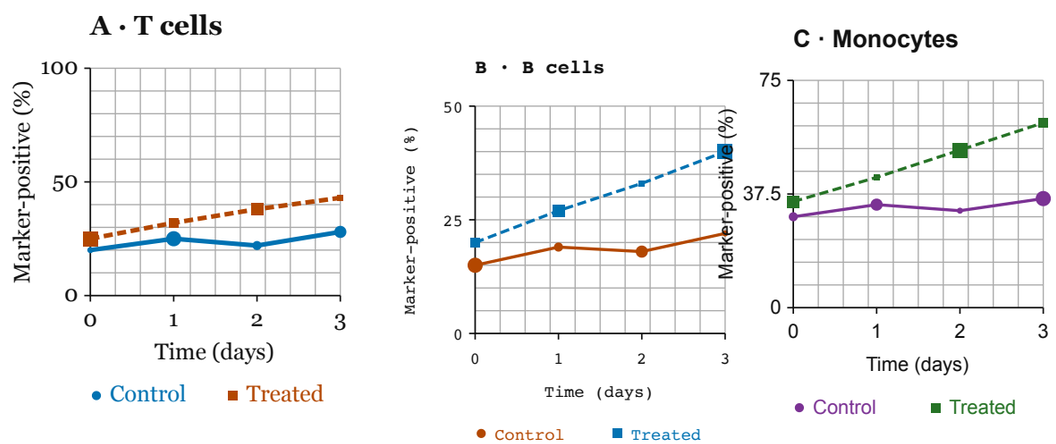
Do and don’t: consistent figures

Improvement 4 of 10

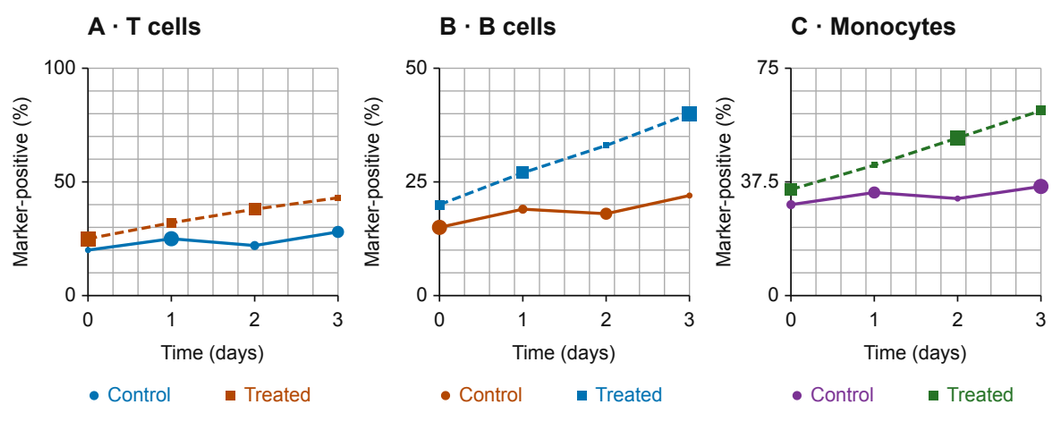
Align panels and repeat spacing

Align the plotting areas, repeat the gaps, and keep titles, axis labels and legends at consistent offsets.

Starting figure



Improved version



The original stays on the left; the right includes all improvements so far.

Show code changes

Show full Python code for these plots

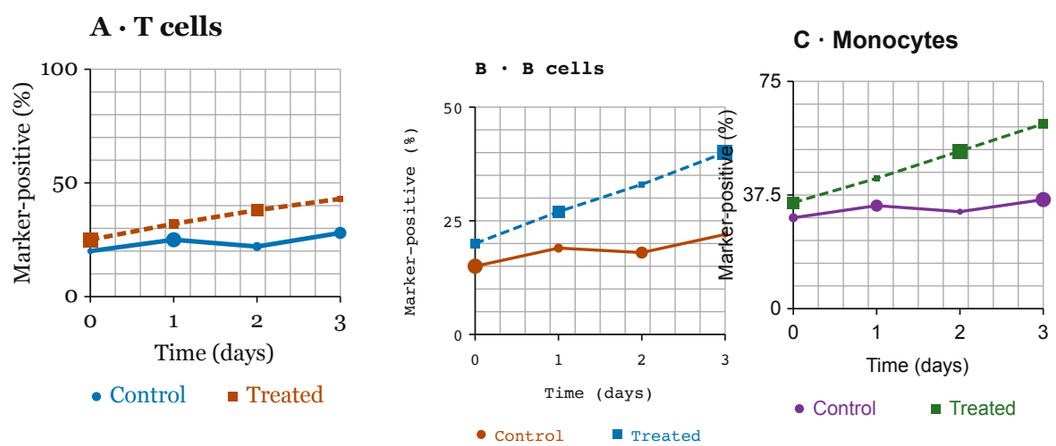
Do and don’t: consistent figures

Improvement 5 of 10

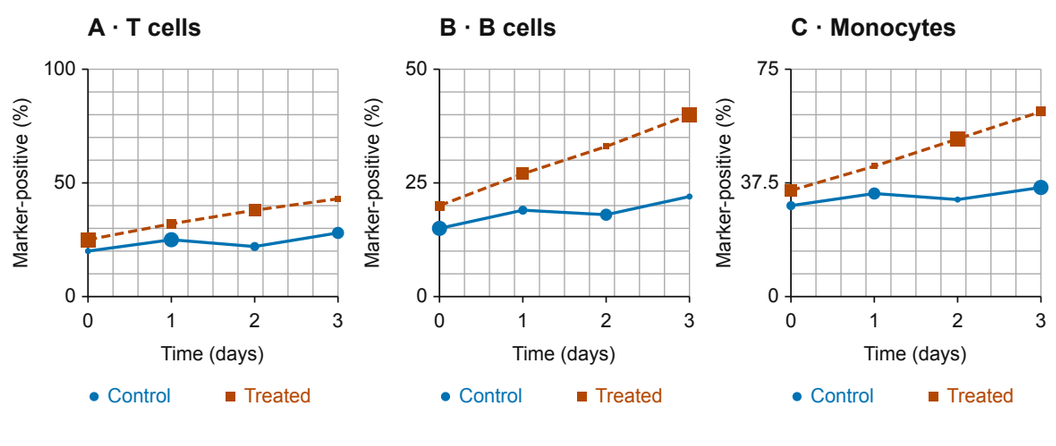
Keep colour meanings consistent

Control is blue and treated is orange in every panel. Circle and square markers also distinguish the conditions.

Starting figure



Improved version



The original stays on the left; the right includes all improvements so far.

Show code changes

Show full Python code for these plots

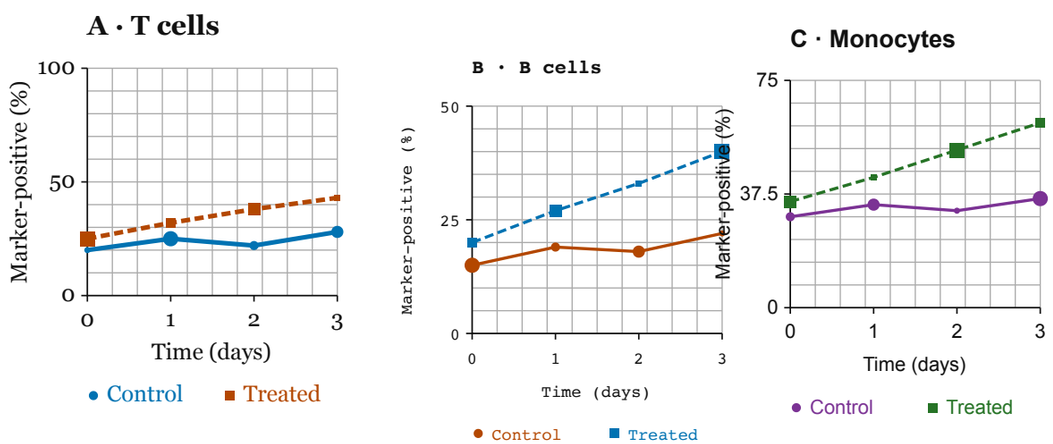
Do and don’t: consistent figures

Improvement 6 of 10

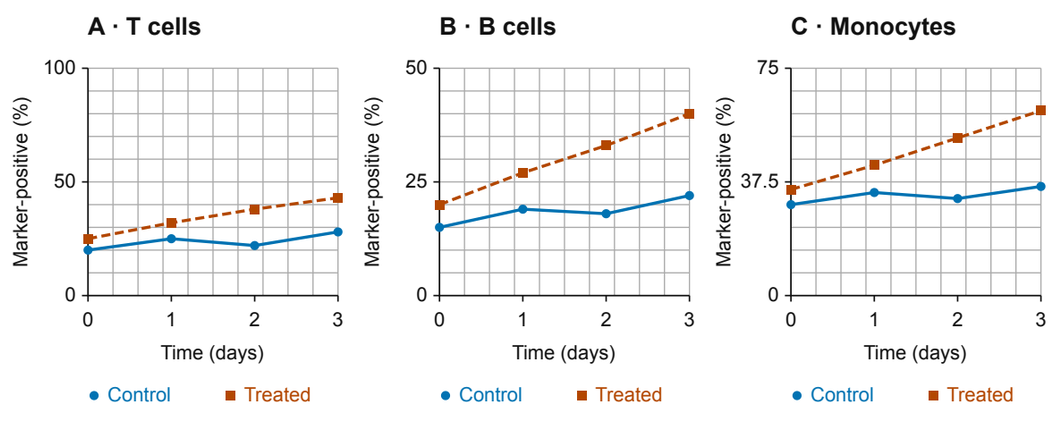
Match equivalent marker sizes

Use a consistent marker size for each condition across panels and days. Marker size does not encode a measurement here.

Starting figure



Improved version



The original stays on the left; the right includes all improvements so far.

Show code changes

Show full Python code for these plots

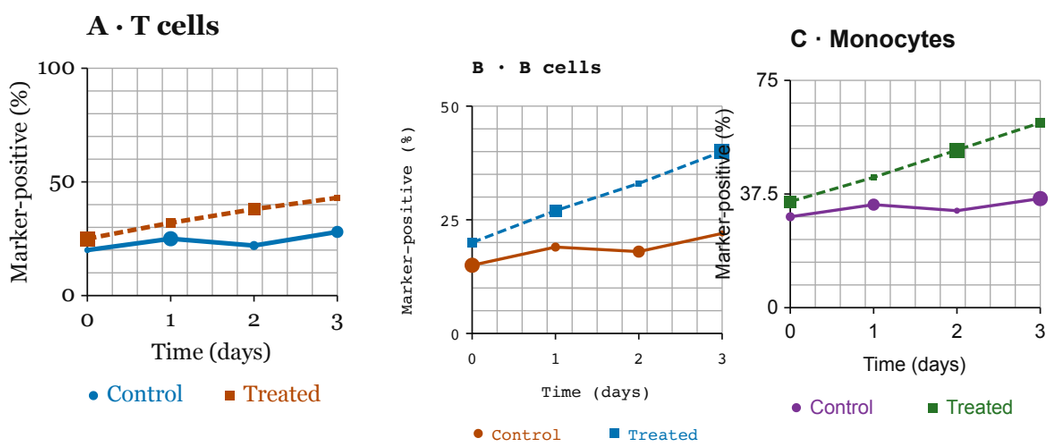
Do and don’t: consistent figures

Improvement 7 of 10

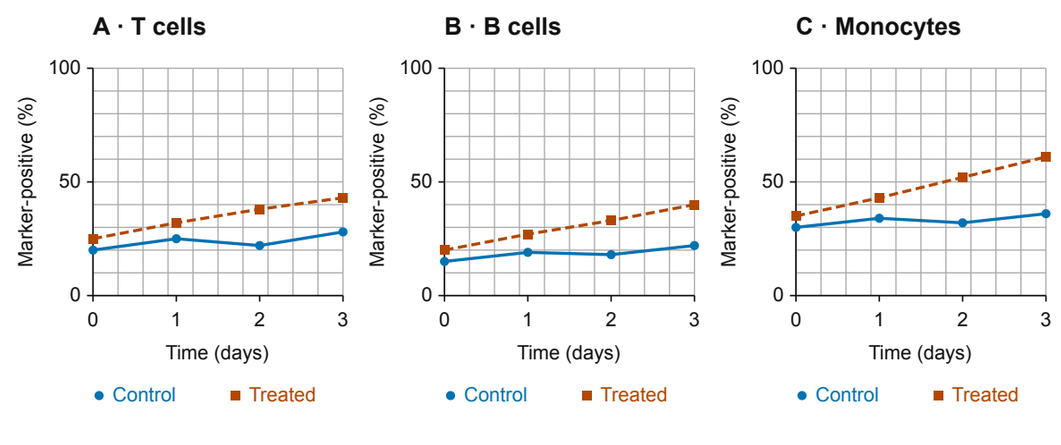
Use comparable axes

Use the same 0–100% range and ticks for all panels. Values are unchanged; heights can now be compared directly.

Starting figure



Improved version



The original stays on the left; the right includes all improvements so far.

Show code changes

Show full Python code for these plots

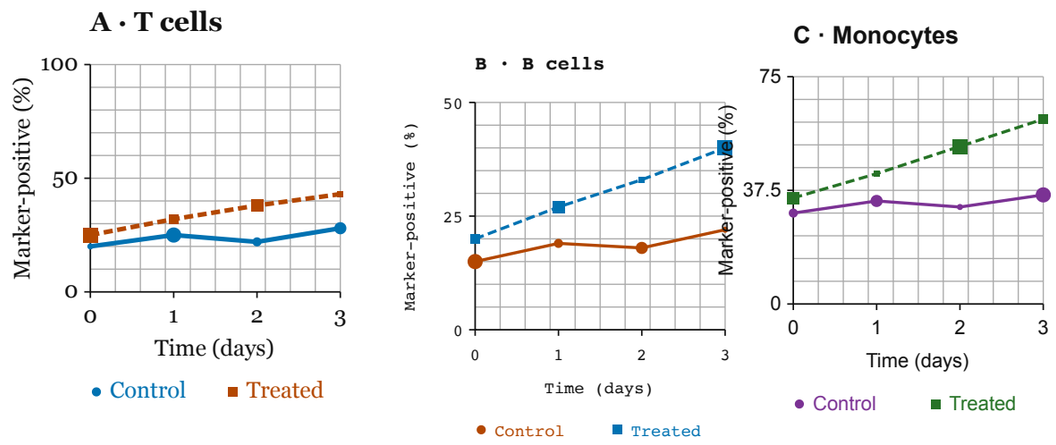
Do and don’t: consistent figures

Improvement 8 of 10

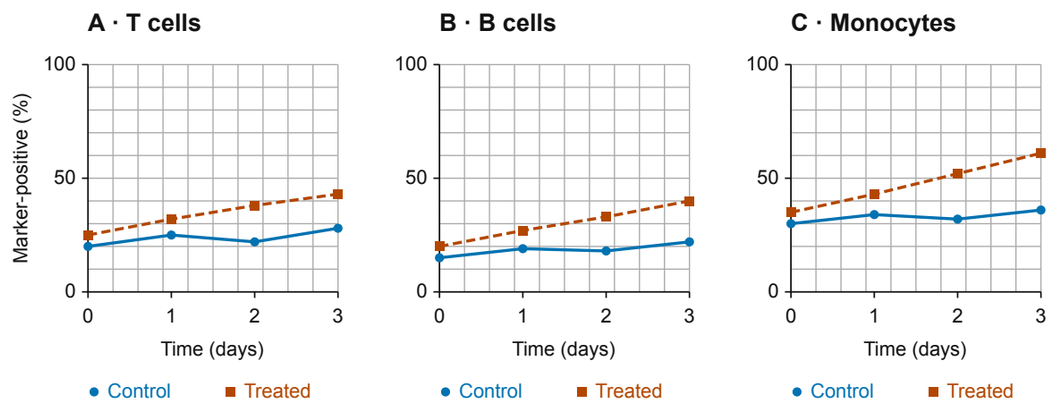
Share the y-axis title

For this aligned row with identical units and scales, keep Marker-positive (%) on the left panel only. Retain tick numbers on every panel. Repeated titles can be preferable when panels are separated or must stand alone.

Starting figure



Improved version



The original stays on the left; the right includes all improvements so far.

Show code changes

Show full Python code for these plots

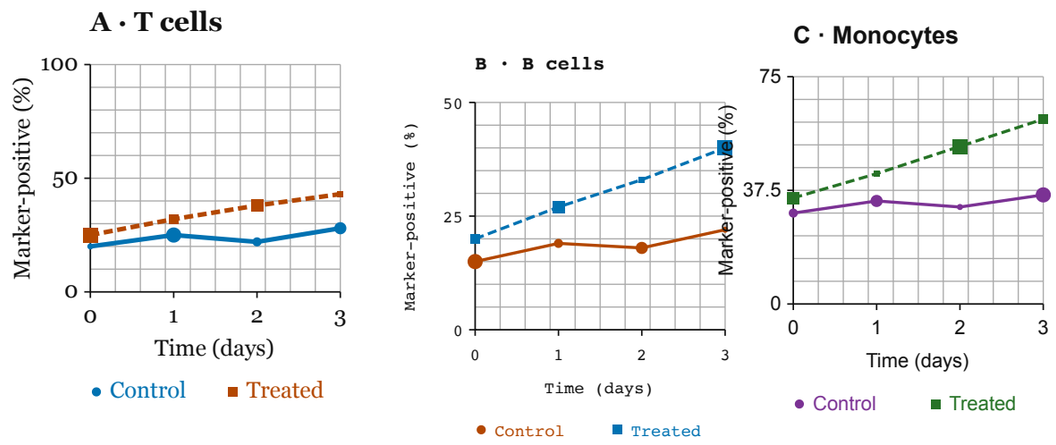
Do and don’t: consistent figures

Improvement 9 of 10

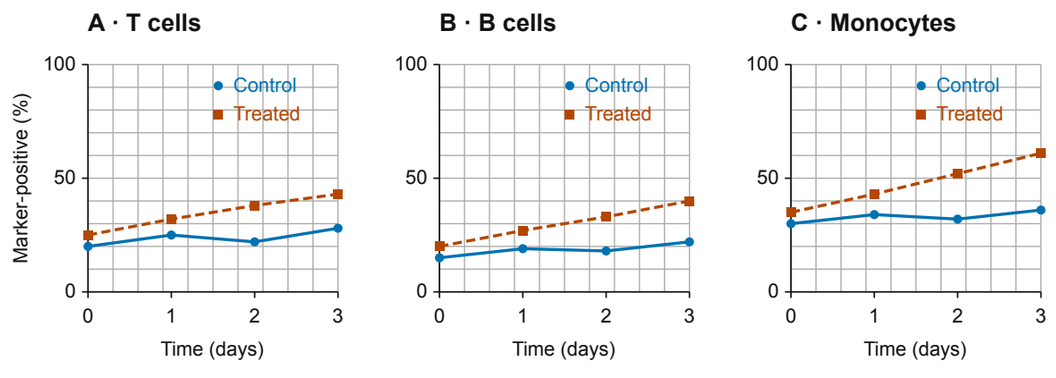
Move legends into unused space

Place Control and Treated inside the upper-right corner of each panel. Here that space is clear of the measurements; an inside legend is useful only when it does not obscure data.

Starting figure



Improved version



The original stays on the left; the right includes all improvements so far.

Show code changes

Show full Python code for these plots

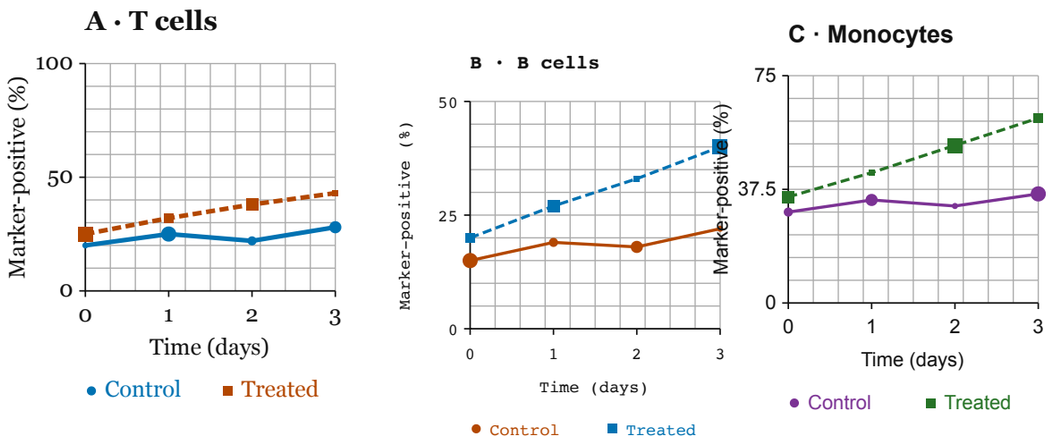
Do and don’t: consistent figures

Final comparison

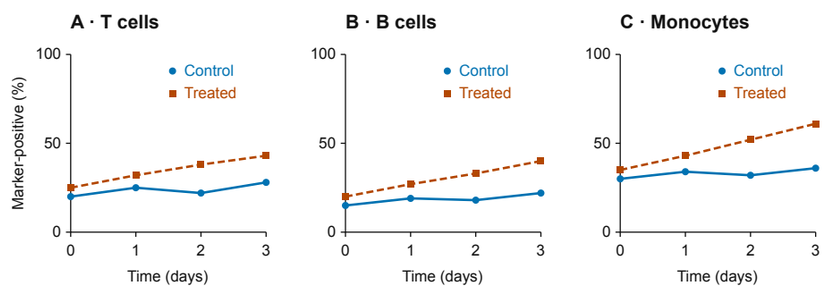
Remove the distracting grid

Remove the dense background grid so the measurements stand out. Keep the axes and ticks for orientation. Light major grid lines can still help when readers need to compare exact values.

Starting figure



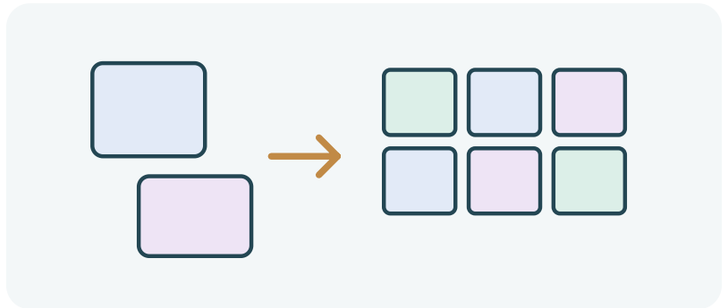
✓ Final improvement



Different styles can be useful when they communicate a deliberate meaning: a bold summary, a size-encoded quantity or a clearly explained alternative scale.

Show code changes

Show full Python code for these plots



Assemble a multi-panel figure

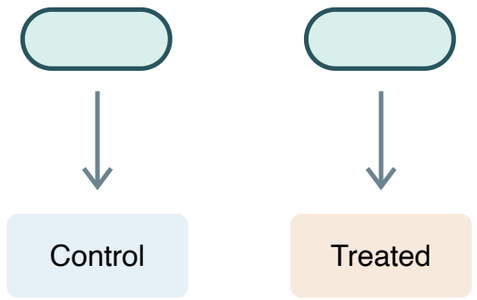
Combine plots with code or a vector editor.

Figures that communicate · Static edition · 28 October 2026 · 150

One manuscript figure, several panels

A figure in a manuscript often combines several panels. Together, they answer one scientific question.

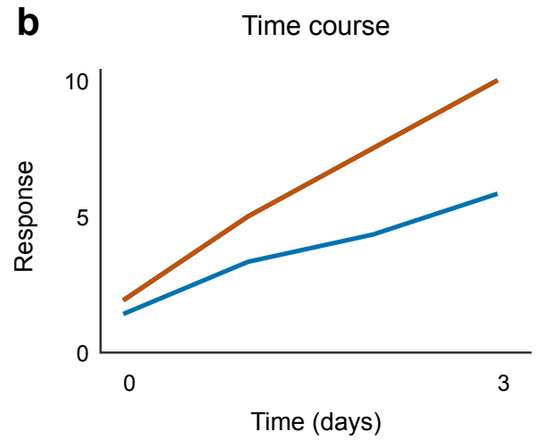
a · Experimental context



What was compared?

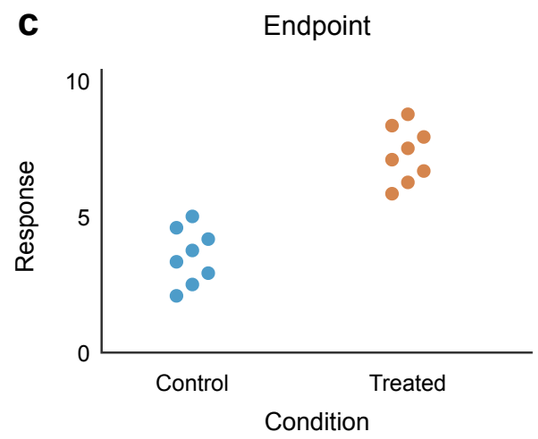
Introduce the experiment.

b · Measurements



Show the response over time.

c · Focused comparison



Compare the endpoint observations.

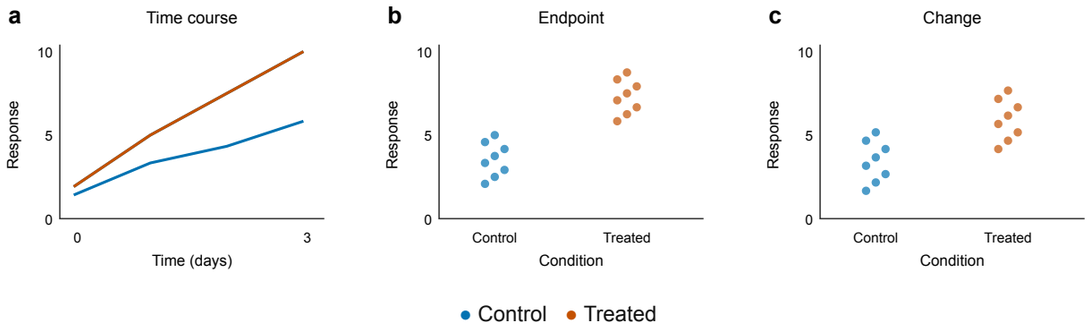
Context → evidence → comparison. A simple overview can prepare the reader for more detailed panels.

An illustrative reading order, not a fixed rule: organise panels around the scientific question, not increasing complexity alone.

How do we build this figure?

Two routes to the same multi-panel figure:

1 · Generate the whole figure in code

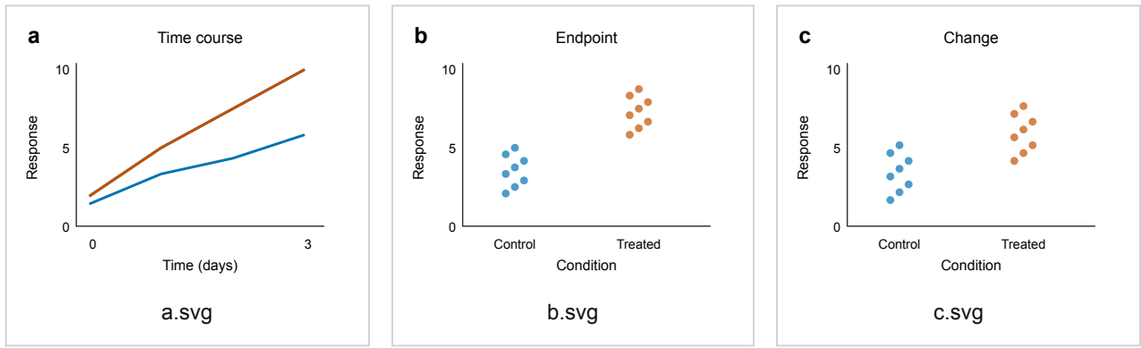


Draw panels → arrange with a library → export one figure

R: ggplot2 + patchwork
Python: Matplotlib

Rebuild the whole figure when your data change.

2 · Export panels, then assemble manually



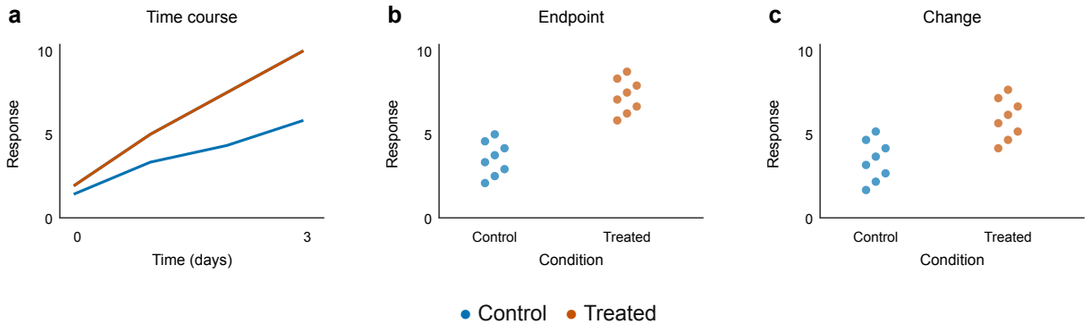
Save a.svg, b.svg, c.svg → place in an editor → export one figure

Inkscape or Adobe Illustrator

Match fonts, line widths and colours before exporting; use the editor for placement.

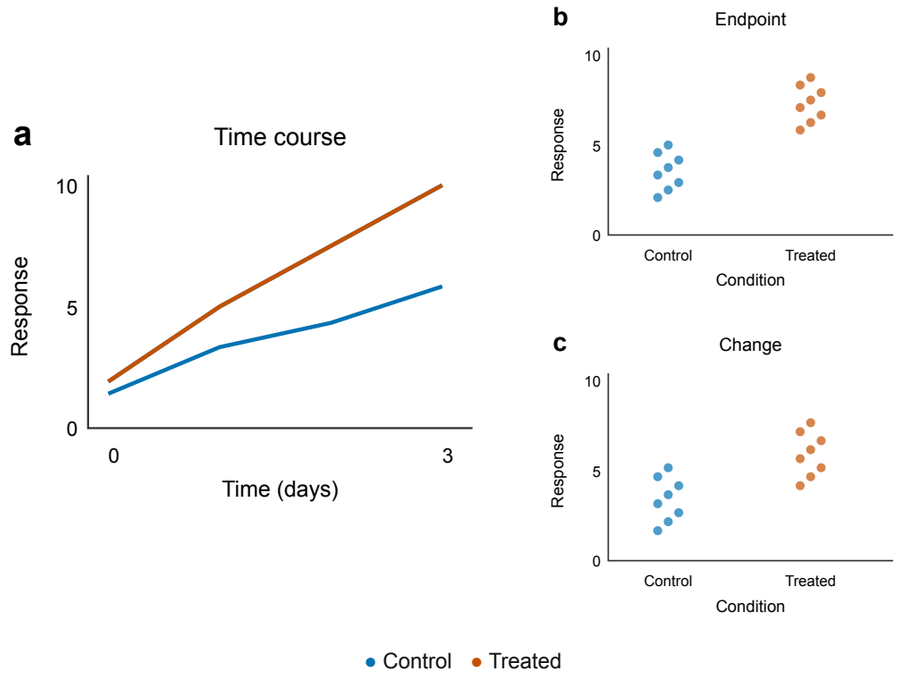
Choose a layout for the story

Equal importance



Three panels in one row.

Give the main result more room



A larger panel a; supporting panels b and c.

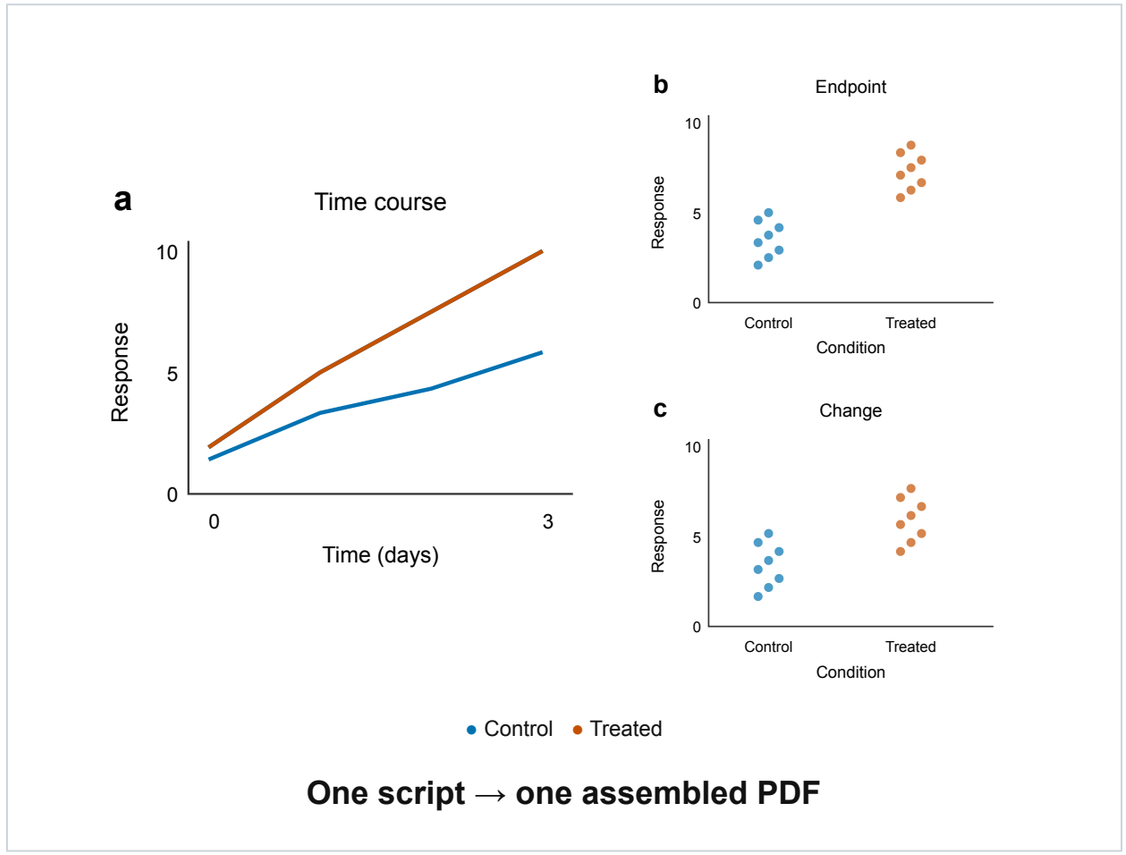
Choose the reading order before adjusting individual details.

Route 1 · Assemble with code in R

ggplot2 draws the panels; patchwork arranges them. Export one complete figure.

R · ggplot2 + patchwork

```
library(ggplot2)
library(patchwork)
# p_a, p_b, p_c: your existing ggplots
figure <- (p_a + p_b + p_c) +
  plot_layout(design = "AB\nAC",
              guides = "collect") +
  plot_annotation(tag_levels = "a")
figure <- figure & theme(
  text = element_text(family = "sans", size = 10),
  legend.position = "bottom")
ggsave("figure.pdf", figure,
       width = 7, height = 4, units = "in")
```



One script → one assembled PDF

Use identical colour scales for a shared legend. Other R layout tools include cowplot and ggpubr.

patchwork documentation

R · Other libraries for combining plots

Patchwork is one option. These examples all arrange three existing plots in a row.

patchwork · compose layouts

```
library(patchwork)
p_a + p_b + p_c + plot_layout(ncol = 3)
```

Combine ggplots; control layouts and collect guides.

cowplot · align and label

```
cowplot::plot_grid(
  p_a, p_b, p_c, nrow = 1,
  labels = "auto", align = "h")
```

Align panels and add panel letters.

Documentation

ggpubr · shared legends

```
ggpubr::ggarrange(
  p_a, p_b, p_c, ncol = 3,
  common.legend = TRUE, legend = "bottom")
```

Convenient layouts and a common legend when scales match.

Documentation

gridExtra · plots and tables

```
gridExtra::grid.arrange(
  p_a, p_b, p_c, ncol = 3)
```

Arrange plots, tables and other grid graphics.

Documentation

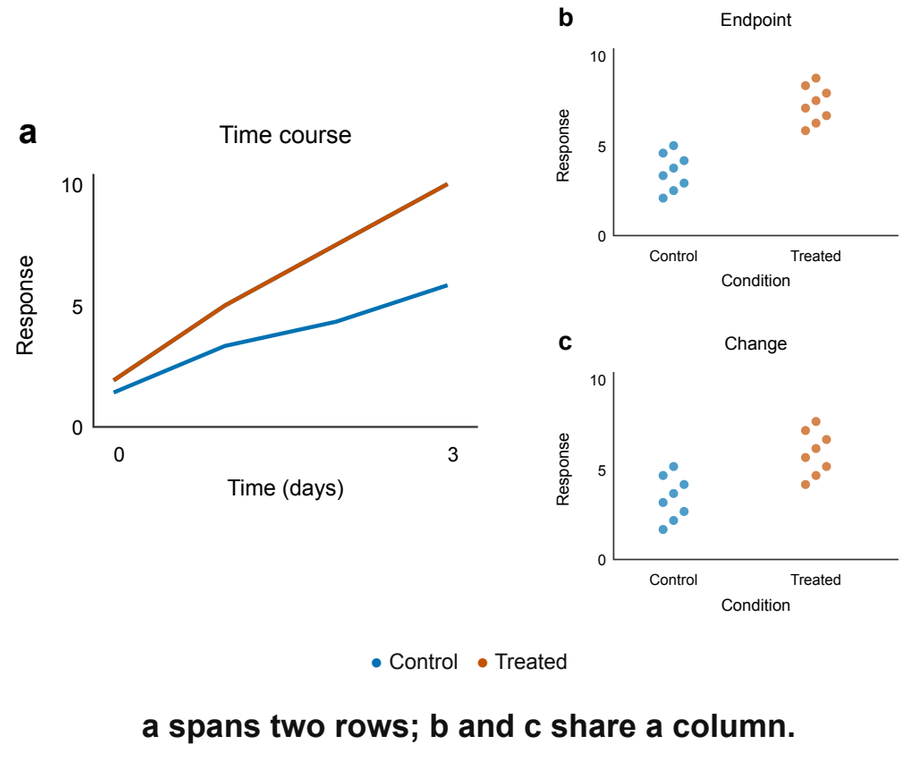
Choose one workflow; you do not need all four packages.

Route 1 · Assemble with code in Python

Matplotlib draws into a shared layout. Use subplots for a regular grid or subplot_mosaic for panels of different sizes.

Python · Matplotlib

```
import matplotlib.pyplot as plt
plt.rcParams.update({"font.family": "sans-serif",
                     "font.size": 10, "lines.linewidth": 1.2})
fig, axes = plt.subplot_mosaic(
    [["a", "b"], ["a", "c"]],
    figsize=(7, 4), layout="constrained")
# Replace these variables with your data:
axes["a"].plot(time, response)
axes["b"].scatter(group, endpoint, s=16)
axes["c"].scatter(group, change, s=16)
for label, ax in axes.items():
    ax.set_title(label, loc="left", fontweight="bold")
fig.savefig("figure.pdf")
```



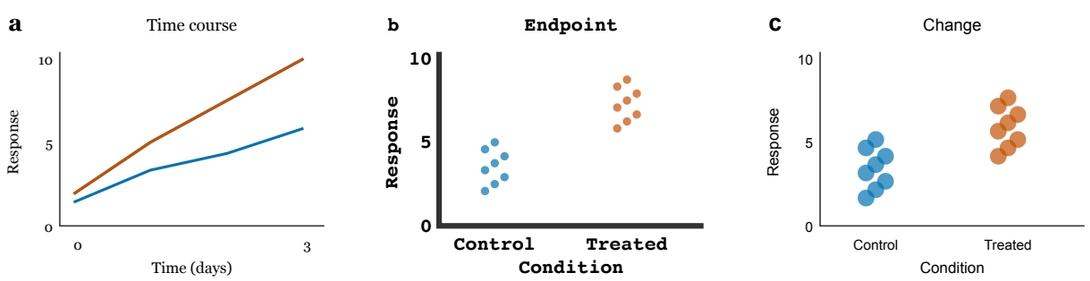
a spans two rows; b and c share a column.

Recipe for the layout; add your axis labels, colour mapping and legend in the script. Seaborn can also draw into these axes with ax=axes["b"].

Matplotlib documentation

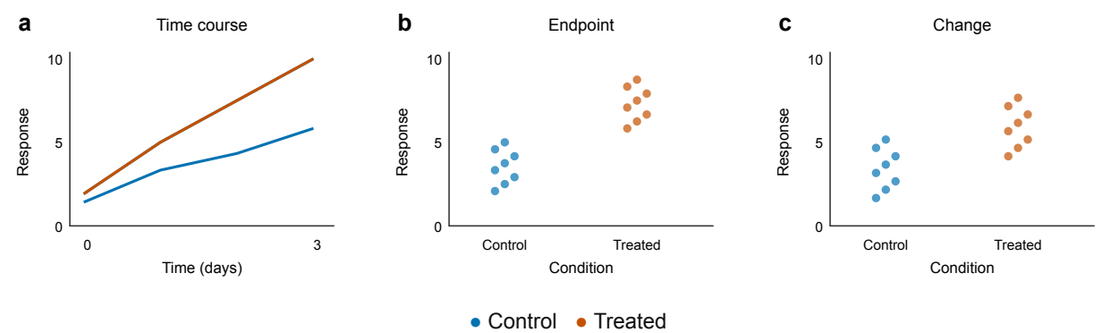
Route 2 · Prepare matching panels

Different settings → extra editing



Mixed fonts, heavy axes and oversized markers.

Shared settings → ready to assemble



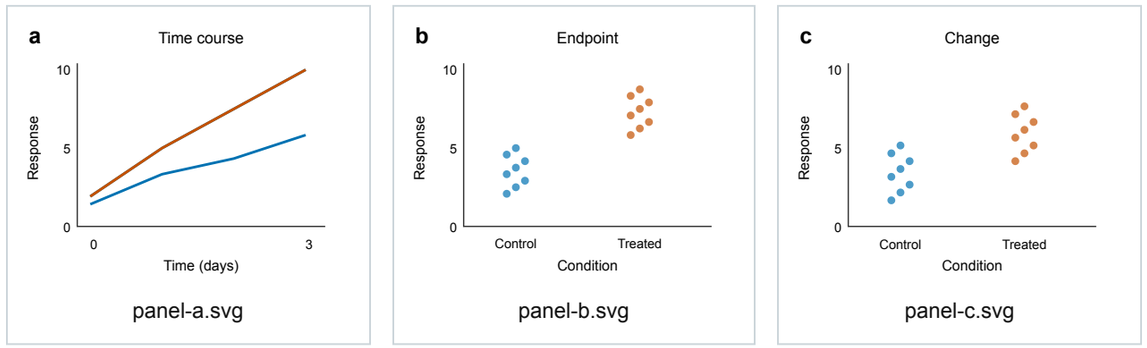
Match equivalent elements before exporting.

One font family Same text sizes by role Same line widths by role Same colour meanings Same marker sizes by role

For example: labels 10 pt, ticks 8 pt, data lines 1.2 pt, axes 0.6 pt. These are starting values, check readability at the final size.

Route 2 · Save three separate files

Three separate exports



Save each plot as its own file.

R · three ggplot objects

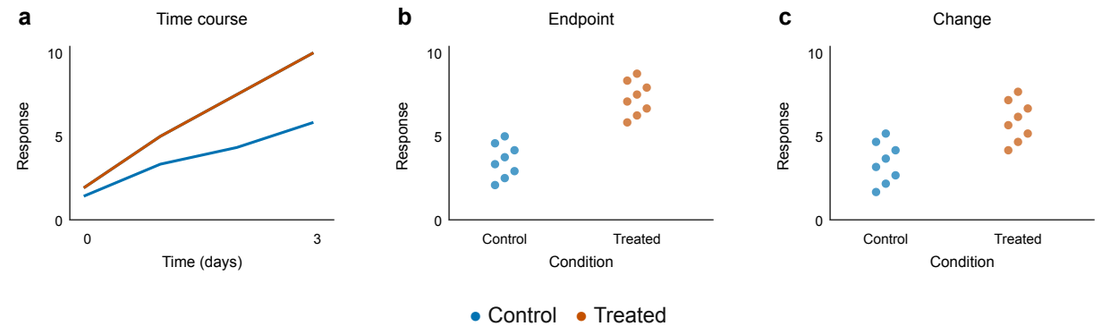
```
ggsave("panel-a.svg", p_a, width = 3, height = 2.5)
ggsave("panel-b.svg", p_b, width = 3, height = 2.5)
ggsave("panel-c.svg", p_c, width = 3, height = 2.5)
```

Python · three Matplotlib figures

```
fig_a.savefig("panel-a.svg")
fig_b.savefig("panel-b.svg")
fig_c.savefig("panel-c.svg")
```

Use the same style and intended panel size when creating the plots. R dimensions above are in inches.

One reading order: a → b → c



One layout, consistent labels and a shared colour key.

Illustrative panels throughout this lesson: the layout changes, the plotted marks stay fixed.

Route 2 · Set the style and export size

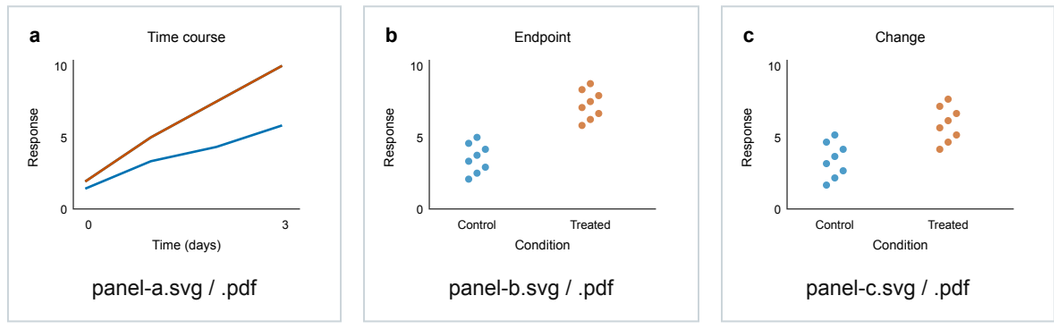
Style in R/Python → export separate SVG/PDF files → assemble in Inkscape or Illustrator.

R · reuse one theme

```
shared <- theme_classic(base_size = 10,
                        base_family = "sans")
# Apply the same colours and geom widths too.
ggsave("panel-a.pdf", p_a + shared,
       width = 3, height = 2.5, units = "in")
# Repeat for p_b and p_c at their target sizes.
```

Python · set style before drawing

```
plt.rcParams.update({"font.size": 10,
                     "font.family": "sans-serif",
                     "lines.linewidth": 1.2,
                     "axes.linewidth": 0.6})
fig, ax = plt.subplots(figsize=(3, 2.5),
                       layout="constrained")
# Draw panel a; repeat for b and c.
fig.savefig("panel-a.svg")
```

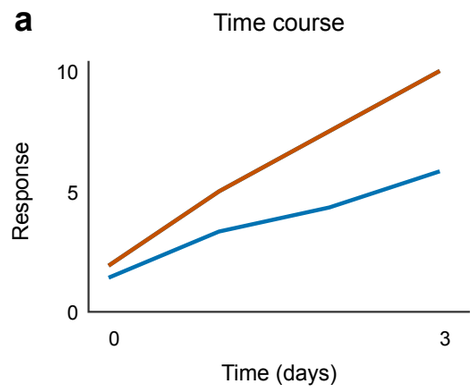


Export at the intended placement size.

Shrinking a panel later also shrinks its labels, markers and lines. Equal export text sizes only stay equal if the scaling stays equal.

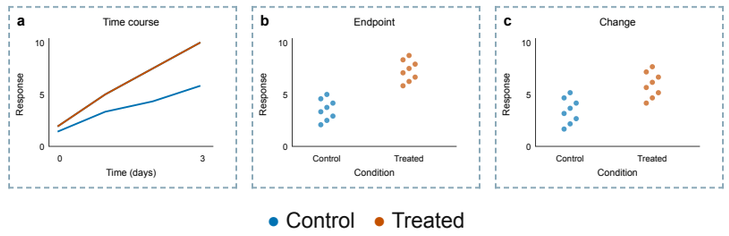
Route 2 · Place, align and save

1 · Export



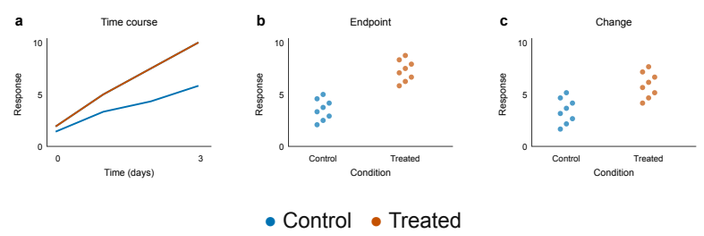
SVG / PDF panels
Same font and target size

2 · Place



One page or artboard
Align and distribute

3 · Save

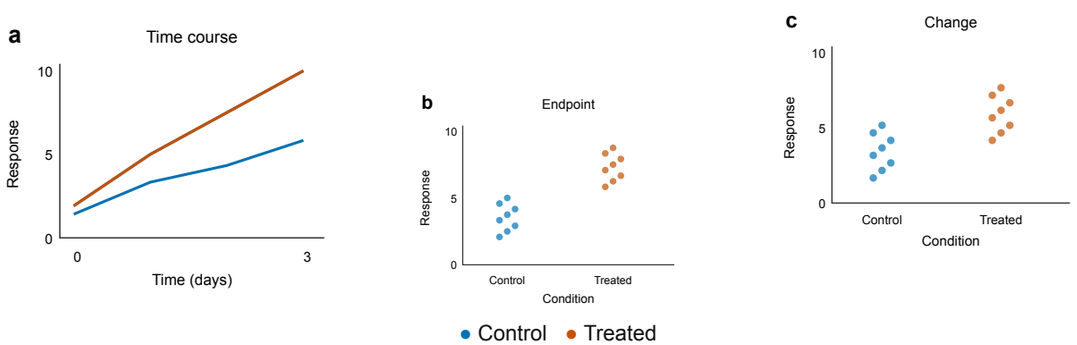


Editable master + final PDF
Check the exported result

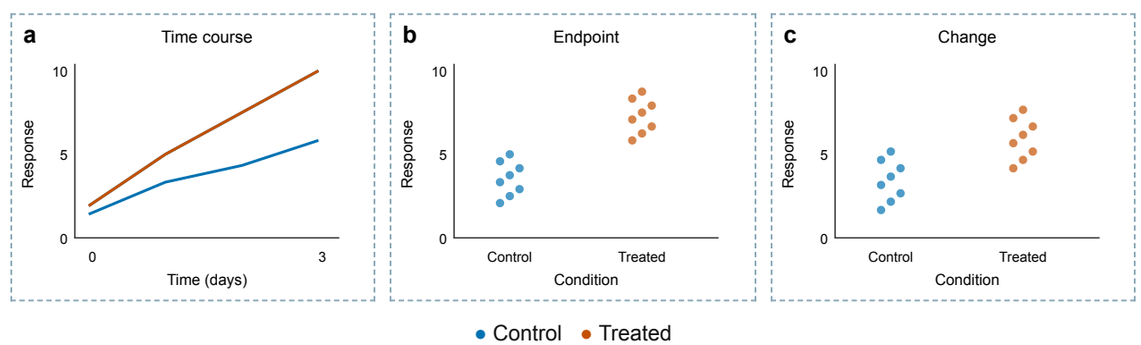
Steps in Inkscape and Illustrator

Use guides, not guesswork

Uneven placement



Aligned panels

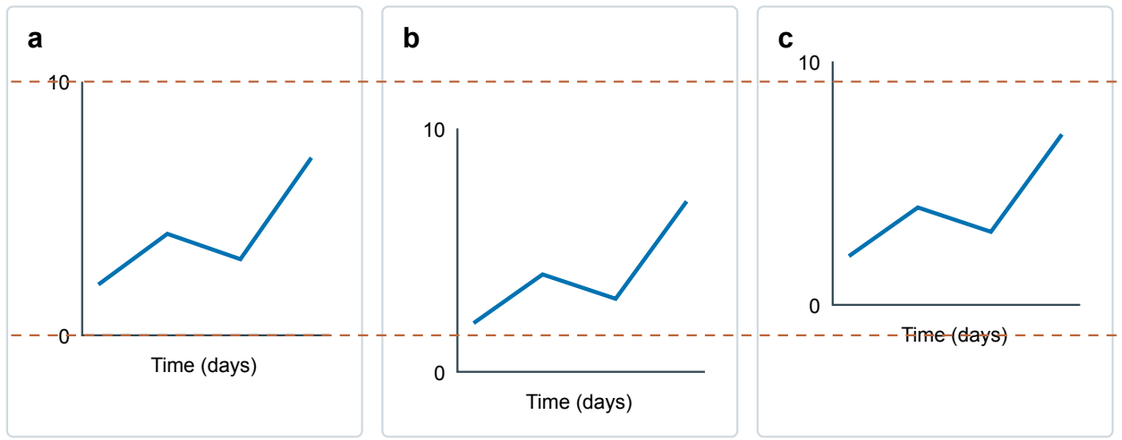


Align plotting areas, keep equal gaps, and place a, b, c at the same offset.

Keep colour meanings and text sizes consistent. Share a legend only when its meaning is identical.

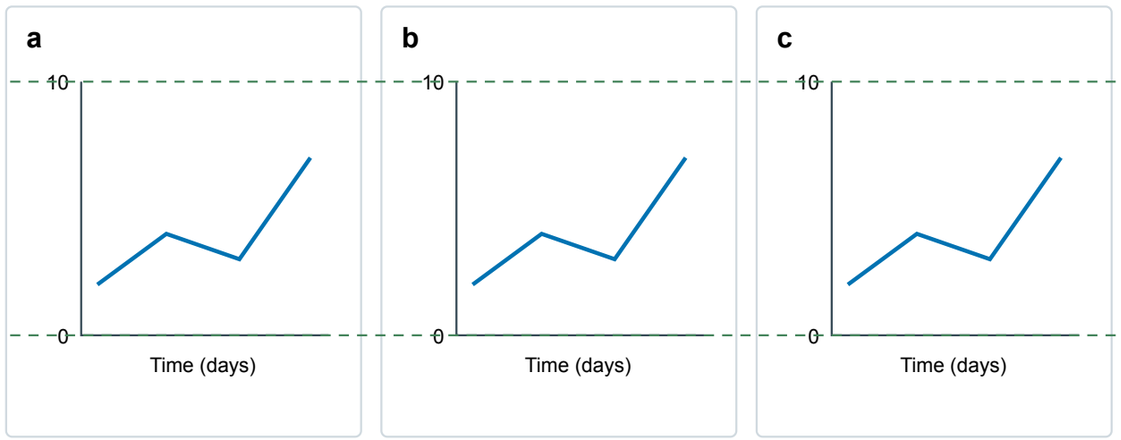
Align the axes, not just the image boxes

Boxes aligned, plots misaligned



Different margins put the axes at different heights.

Plotting areas aligned



Top and bottom edges line up; the gaps are equal.

The dashed guides mark the plotting-area edges. For stacked panels, align their left and right edges too.

With code

Use a shared grid. In cowplot, align horizontal panels explicitly:

```
cowplot::plot_grid(p_a, p_b, p_c,
  nrow = 1, align = "h", axis = "tb")
```

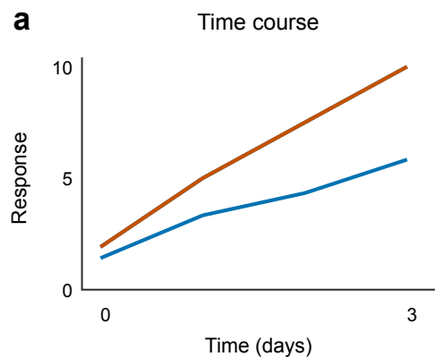
In Inkscape / Illustrator

Place guides on the axes, then move each grouped panel to meet them. Distribute gaps between plotting areas.

Keep complete panels grouped. If plot heights differ, regenerate them with matching plotting areas; stretching the exports distorts the figure. Share axis limits only when the quantities are comparable.

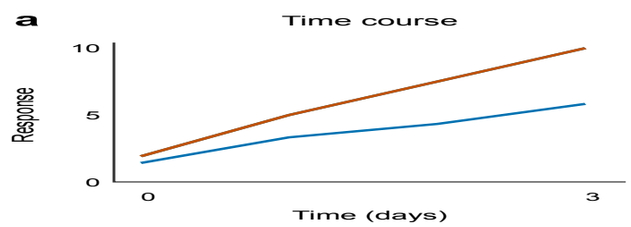
Resize the panel, not the data

Do · preserve proportions



Group the complete panel and lock its aspect ratio.

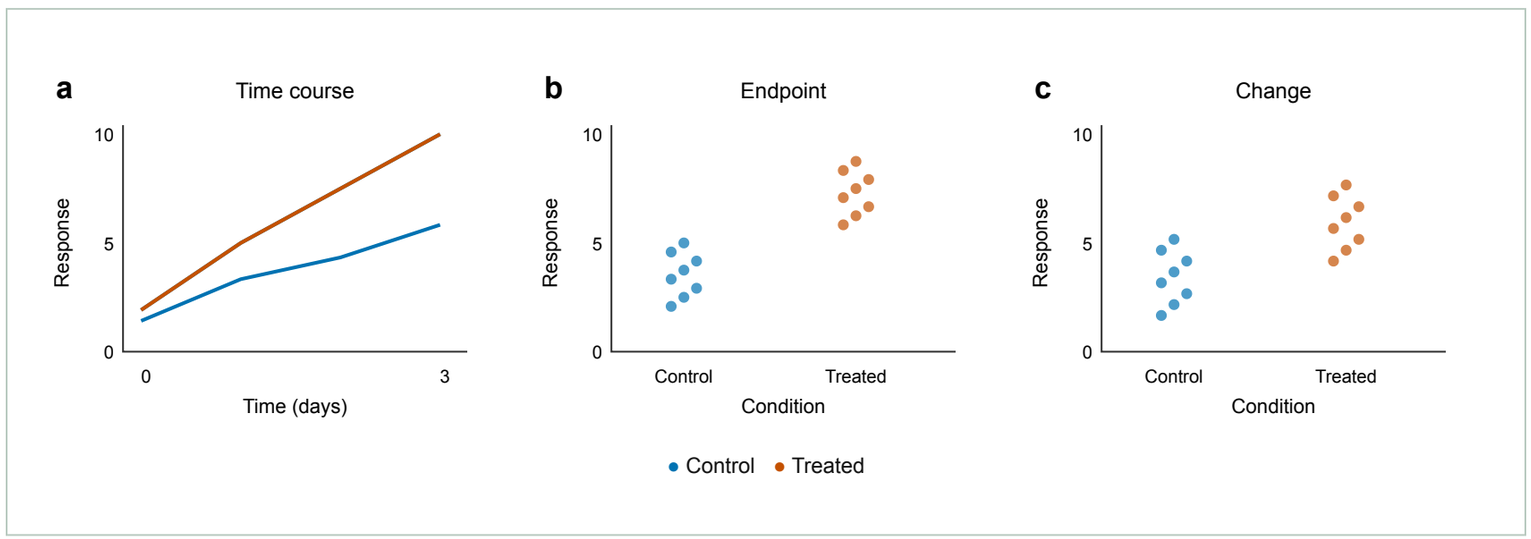
Don’t · stretch to fill a gap



Distorted axes change the visual impression.

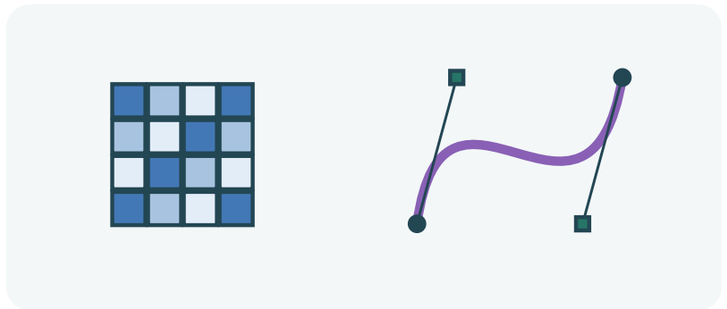
For a different shape, regenerate the plot at the required dimensions. Never move points or change bar lengths in the editor.

Check the whole figure at final size



✓ Reading order ✓ Aligned panels ✓ Consistent text ✓ Clear legends ✓ No clipping

Inspect the exported file at its intended size. Keep the plotting scripts, original exports and editable master.

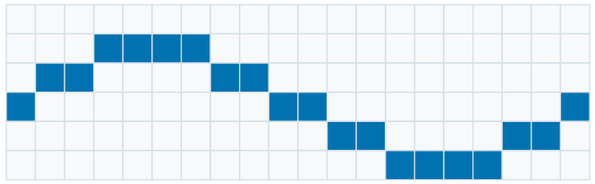


File formats and resolution

Raster vs vector, zooming, and DPI.

Two ways to store an image

Raster · a grid of pixels



Fixed grid · each square stores a colour

Extension	Full name
`.png`	Portable Network Graphics
`.jpg / .jpeg`	Joint Photographic Experts Group
`.tif / .tiff`	Tagged Image File Format
`.bmp`	Bitmap
`.gif`	Graphics Interchange Format
`.webp`	WebP (format name, not an acronym)

Photographs, microscopy and pixel-based plot exports. Enlarging the image eventually reveals its pixels.

Vector · shapes and paths

Coordinates + paths · redrawn at each size

Extension	Full name
`.svg`	Scalable Vector Graphics · Inkscape's native format
`.ai`	Adobe Illustrator Artwork · Illustrator's native format
`.eps`	Encapsulated PostScript
`.pdf`	Portable Document Format

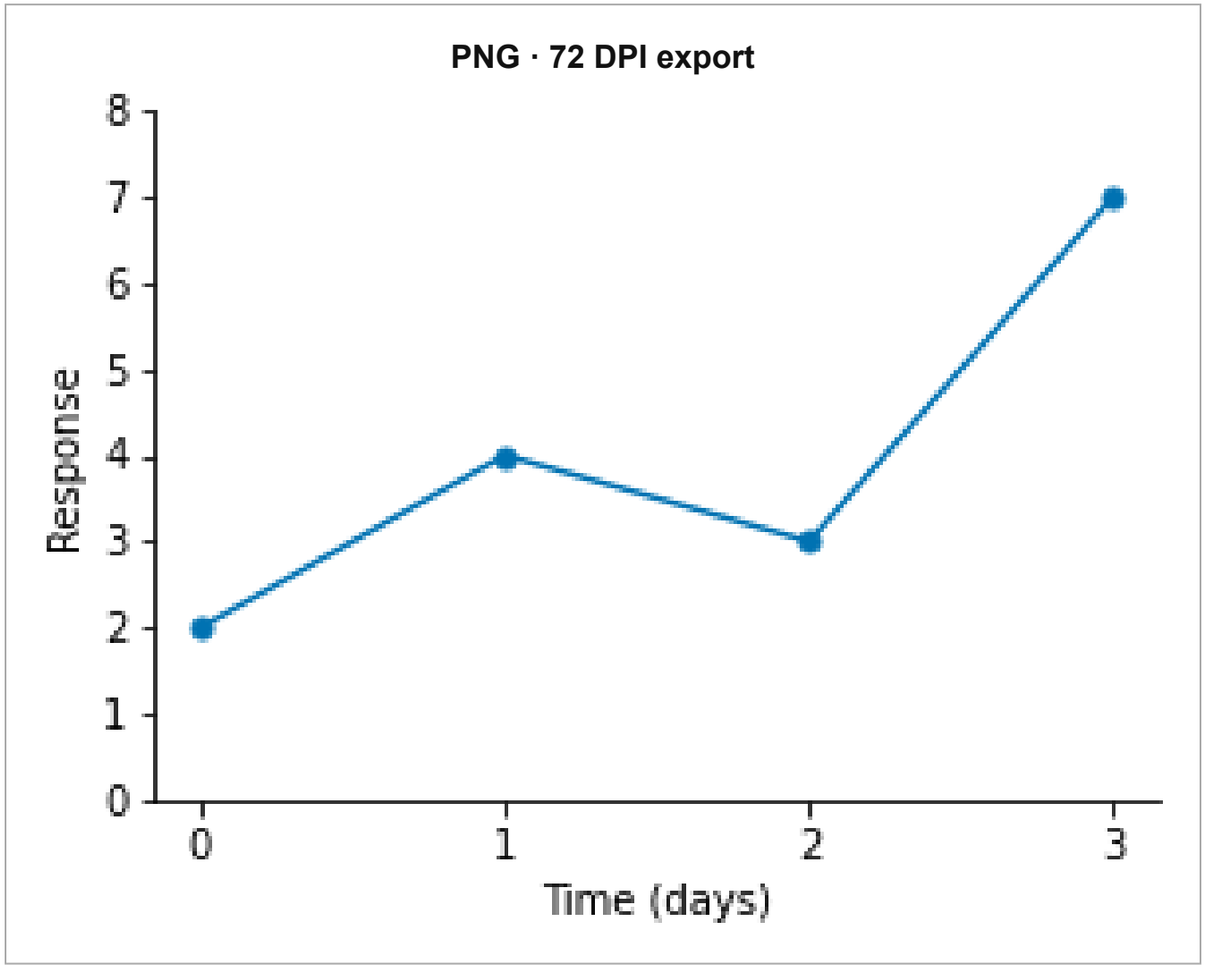
Lines, text and shapes can be drawn at different sizes without a fixed pixel grid.

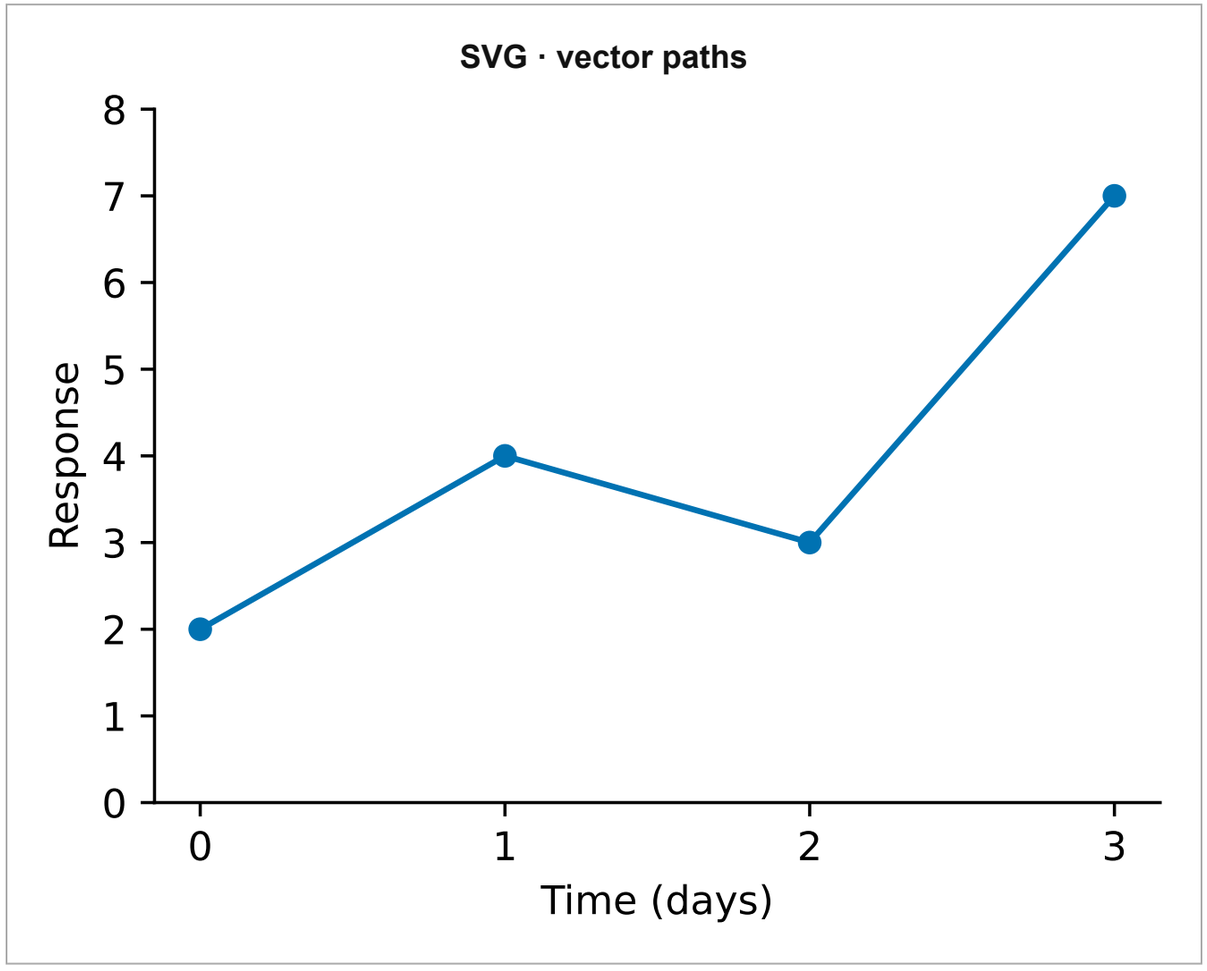
Inkscape SVG keeps extra editing information; Plain SVG is useful for exchange. Both use `.svg`. Illustrator uses `.ai` for its editable source files.

A PDF can contain vectors, raster images, or both. Saving a PNG inside a PDF does not turn it into vector artwork.

> **Python export settings**

Same plot, different formats

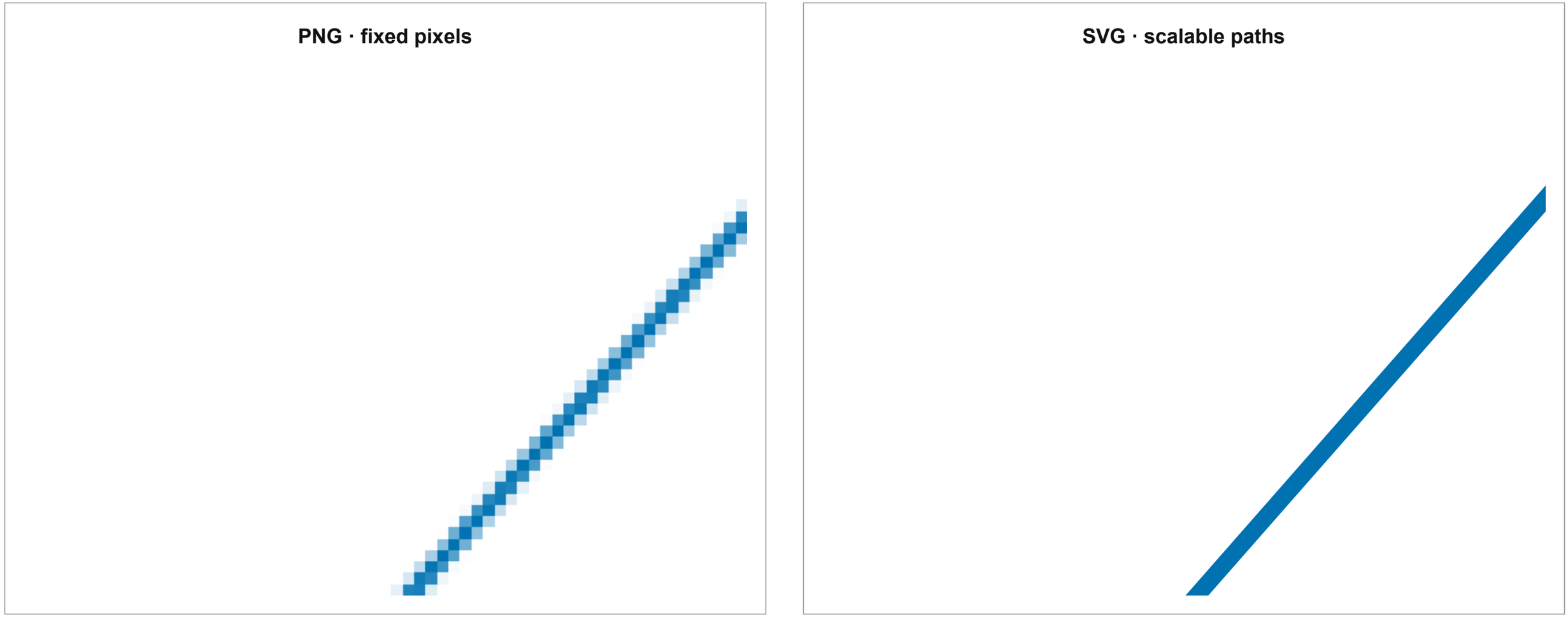




At a modest display size, both can look acceptable. Inspect a small detail to see the difference.

Python export settings

Zoom into the same detail



PNG stores a fixed grid of pixels: zooming enlarges that grid. SVG stores drawing instructions, for example, a line’s start and end coordinates, stroke width, or a curve’s control points. At each zoom level, the viewer renders those shapes again onto the screen’s pixels. That is why vector edges stay sharp.

Python export settings

What does DPI mean?

Pixels ÷ inches = PPI

A 1,200-pixel-wide image printed 4 inches wide has **300 pixels per inch**.

Printed 8 inches wide, those same pixels give **150 PPI**.

Export DPI controls raster resolution

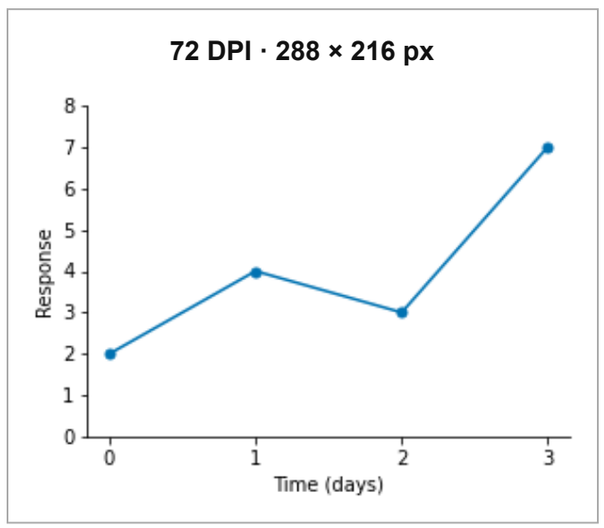
A 4 × 3 inch plot exported at 300 DPI produces **1,200 × 900 pixels**.

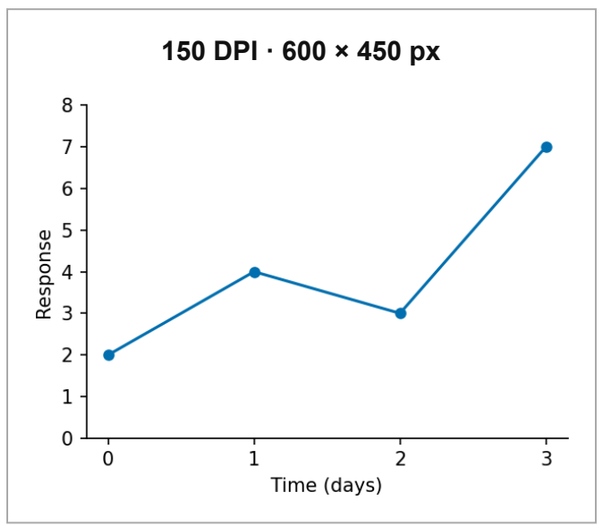
Strictly, PPI describes image pixels and DPI describes printer dots. Plotting software commonly calls its raster export setting “DPI”.

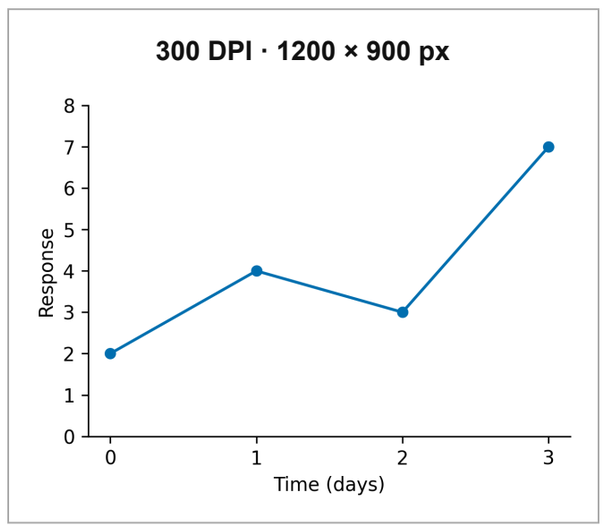
Changing resolution metadata alone adds no detail. Screen appearance depends on pixel dimensions and display scaling; vector paths do not have a fixed DPI.

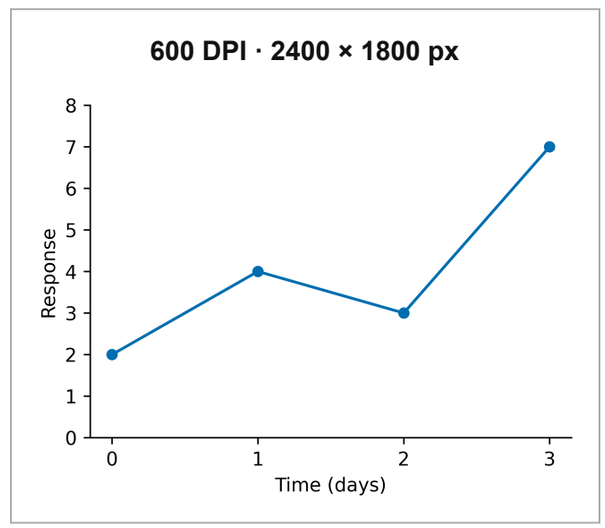
> **Python export settings**

Compare four DPI exports









Zoom into the same detail

All exports use a 4 × 3 inch canvas and identical font sizes. Comparing the same detail reveals differences that may disappear when full images are scaled down on screen.

Python export settings

Is there a “Retina DPI” threshold?

For a **600-pixel-wide display area on a 2× screen**, aim for a **1,200-pixel-wide image**.

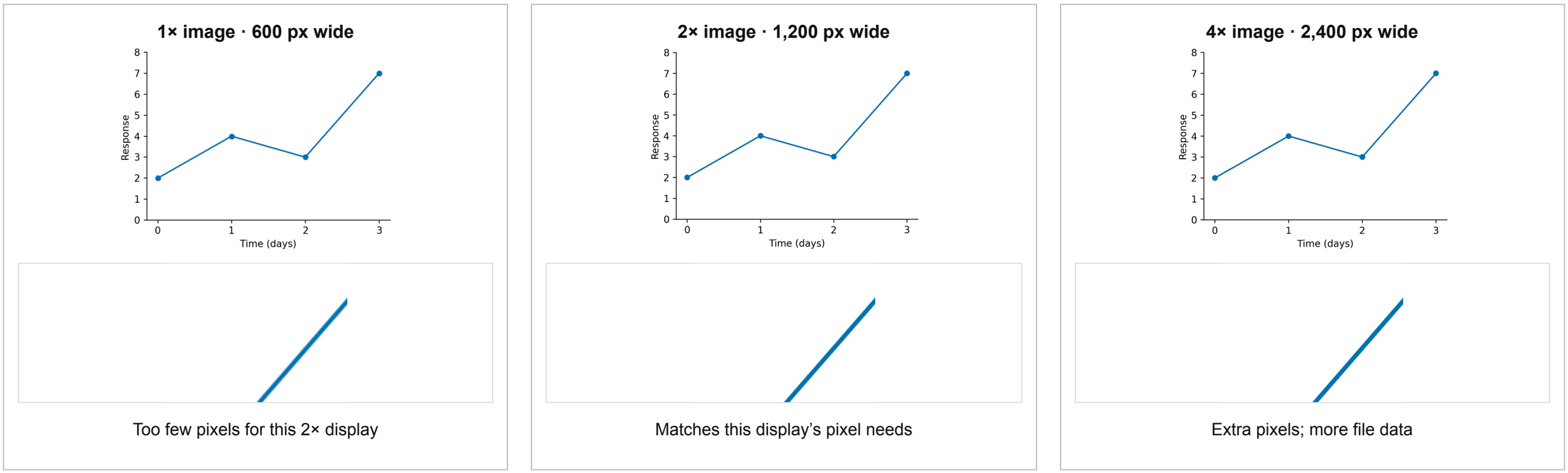
600 px
display width

×

2×
screen density

=

1,200 px
image width



Too few pixels for this 2× display

Matches this display’s pixel needs

Extra pixels; more file data

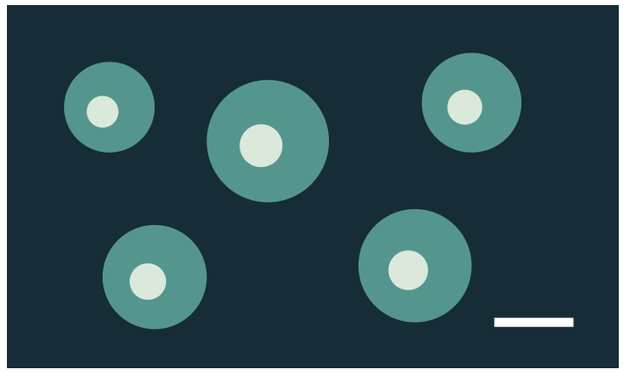
Top: same plot at the same display size. Bottom: enlarged pixel detail. Actual sharpness depends on your screen and browser zoom. “Retina” is not a universal DPI setting. About screen density.

Python export settings

What should go to a journal?

Match the export to the figure, and check the journal’s requirements at final size.

Image / microscopy

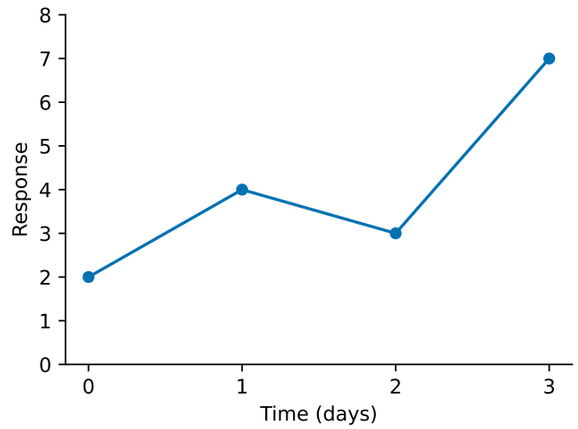


Raster · 300–600 DPI

Nature’s photographic-image guidance.

Illustration only · keep labels and scale bars editable.

Plot / line drawing

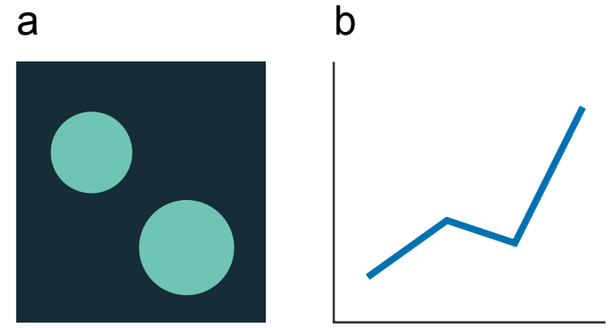


Vector PDF / EPS

Keep lines and text editable.

Nature prefers vector artwork for graphs.

Image + plot + labels



Mixed vector + raster

Keep labels sharp; embed sufficient image pixels.

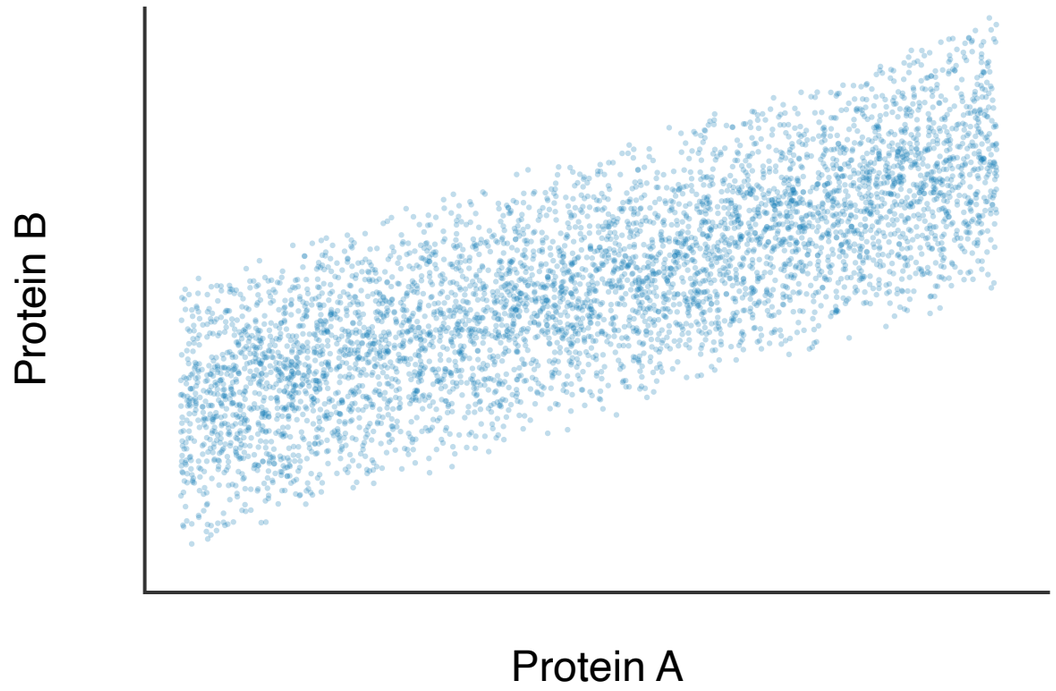
Cambridge: at least 600 DPI for raster combination artwork.

> **Journal-specific requirements & sources**

> **Python export settings**

What is inside a vector file?

5,000 points



One SVG circle per observation.

5,005 lines · 358 KiB of uncompressed SVG source.

Download the full SVG

The code that draws it

Full source below, scroll to see all 5,000 circles.

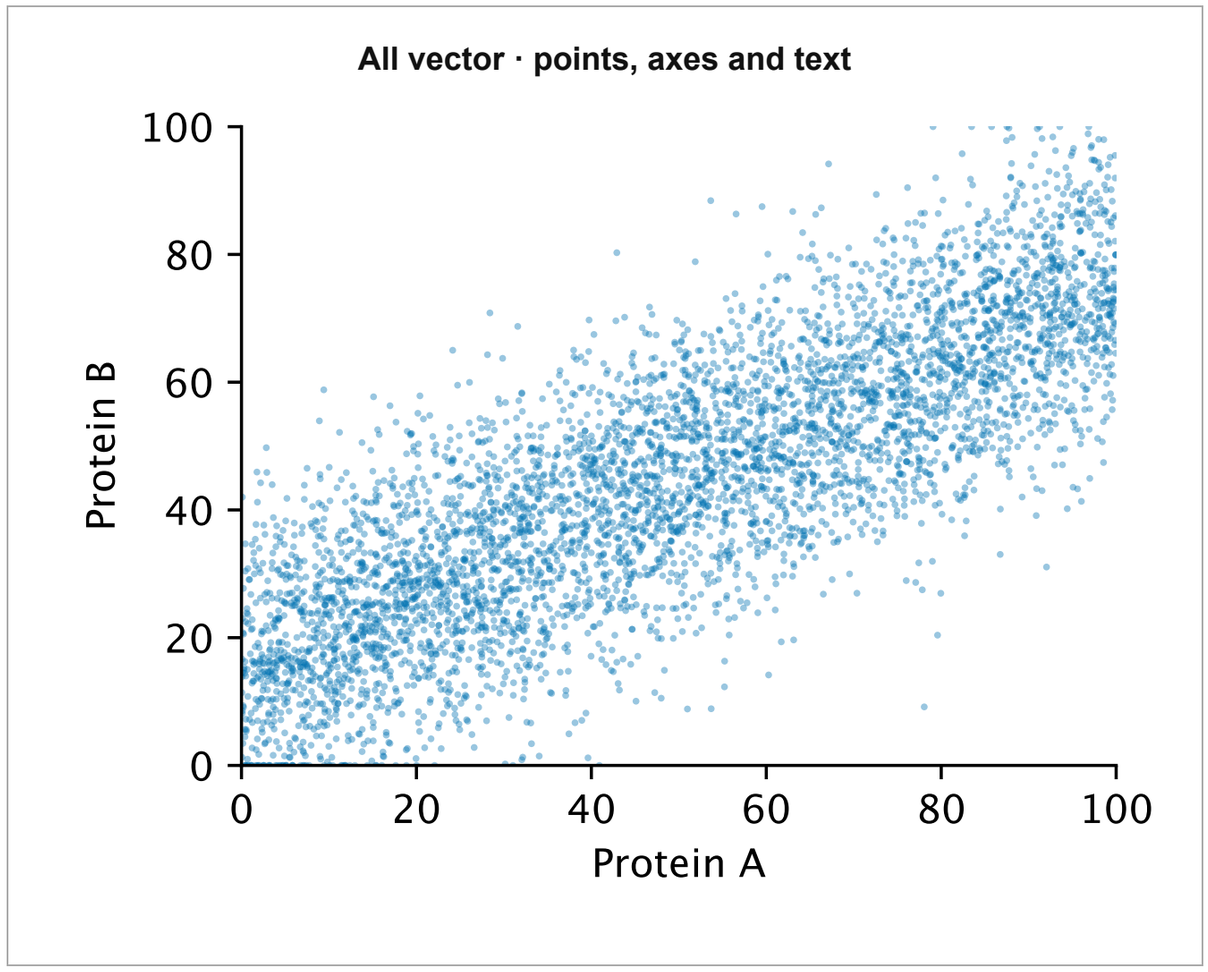
```
<svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 320 210" role="img" aria-
label="Illustrative scatter plot with 5000 observations">
  <path d="M45 10 V175 H300" stroke="#333" fill="none"/>
  <circle cx="113.04" cy="93.04" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="106.19" cy="84.67" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="157.87" cy="91.77" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="283.77" cy="65.91" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="202.66" cy="96.07" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="75.87" cy="121.93" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="267.65" cy="74.55" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="96.52" cy="104.08" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="271.92" cy="54.66" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="245.76" cy="45.06" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="60.16" cy="138.04" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="249.43" cy="65.25" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="189.70" cy="98.21" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="220.28" cy="89.71" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="77.52" cy="103.35" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="84.21" cy="130.15" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="240.07" cy="49.99" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="117.57" cy="76.86" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="131.81" cy="133.97" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="99.32" cy="128.96" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="132.44" cy="77.97" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="280.39" cy="53.85" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="240.32" cy="71.42" r="0.8" fill="#0072B2" opacity="0.25"/>
  <circle cx="125.53" cy="87.54" r="0.8" fill="#0072B2" opacity="0.25"/>
```

This illustrative export stores every circle separately. Other exporters may reuse marker definitions; compression and rasterized layers can change the file size substantially.

Python export settings

Rasterize only the dense layer

Rasterize means converting shapes into a fixed grid of pixels. Here, only the 5,000 points become an embedded image; the axes and text stay vector.



Zoom into points and axis

The two versions use identical measurements and styling. Zoom in to see which layer has a fixed resolution.

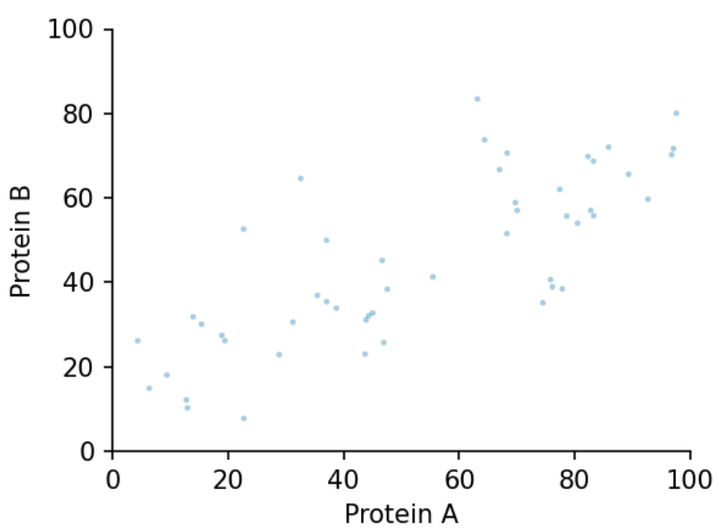
The point layer is intentionally rasterized at 100 DPI to make the effect visible. Choose the resolution for the intended output size; this is not a submission recommendation.

Show the one-line change

Python export settings

More points: SVG versus PNG file size

Same canvas, axes, marker size and transparency. Every PNG is **600 × 450 pixels** (4 × 3 inches at 150 DPI).



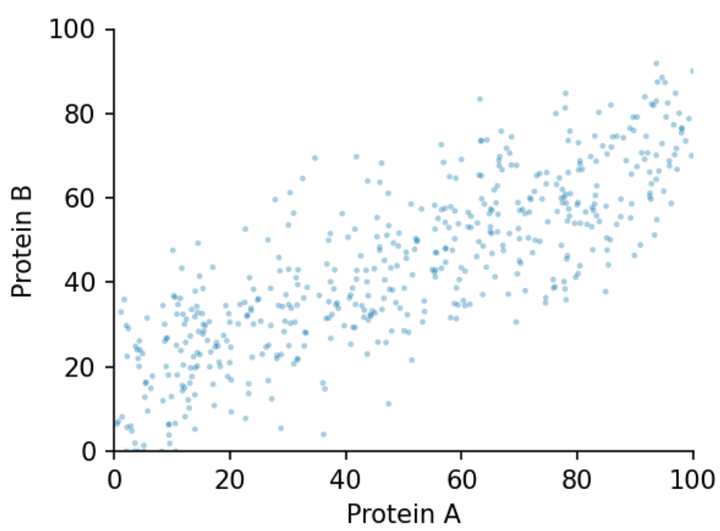
PNG 18.3 KB

SVG 23.9 KB

0 5 MB 10 MB

SVG is **1.3× larger**

PNG · SVG



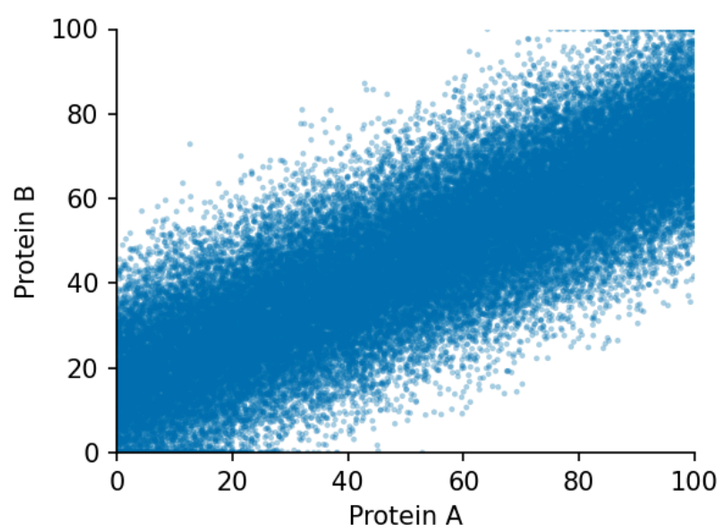
PNG 48.0 KB

SVG 95.9 KB

0 5 MB 10 MB

SVG is **2.0× larger**

PNG · SVG



PNG 168.4 KB

SVG 8.02 MB

0 5 MB 10 MB

SVG is **47.6× larger**

PNG · SVG

All bars use the same 0–10 MB scale. **Fixed pixels ≠ fixed file size.** PNG compression depends on the rendered image. SVG stores geometry for the points, so dense plots can grow much more quickly.

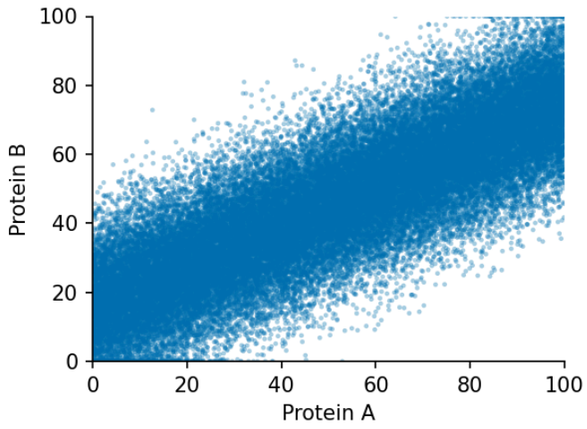
> **Why can a dense PDF take time to appear?**

> **How this comparison was measured**

> **Python export settings**

Many points can mean a heavy vector file

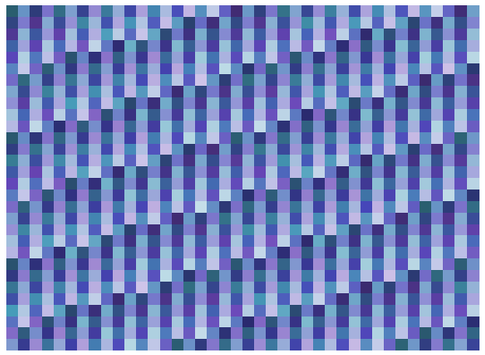
Thousands of markers



Each marker needs geometry or a reference.

Try rasterizing the point layer.

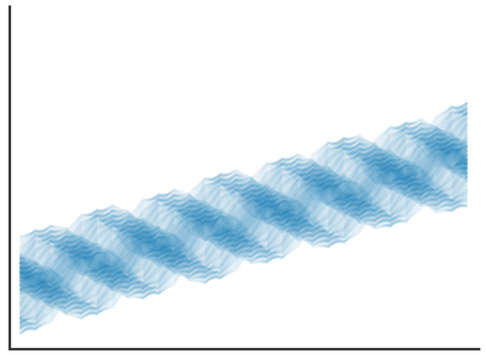
Thousands of heatmap cells



A mesh export may store a shape for every cell.

An image-based heatmap can be more compact; keep labels vector.

Many transparent traces



Long paths, transparency and clipping add rendering work.

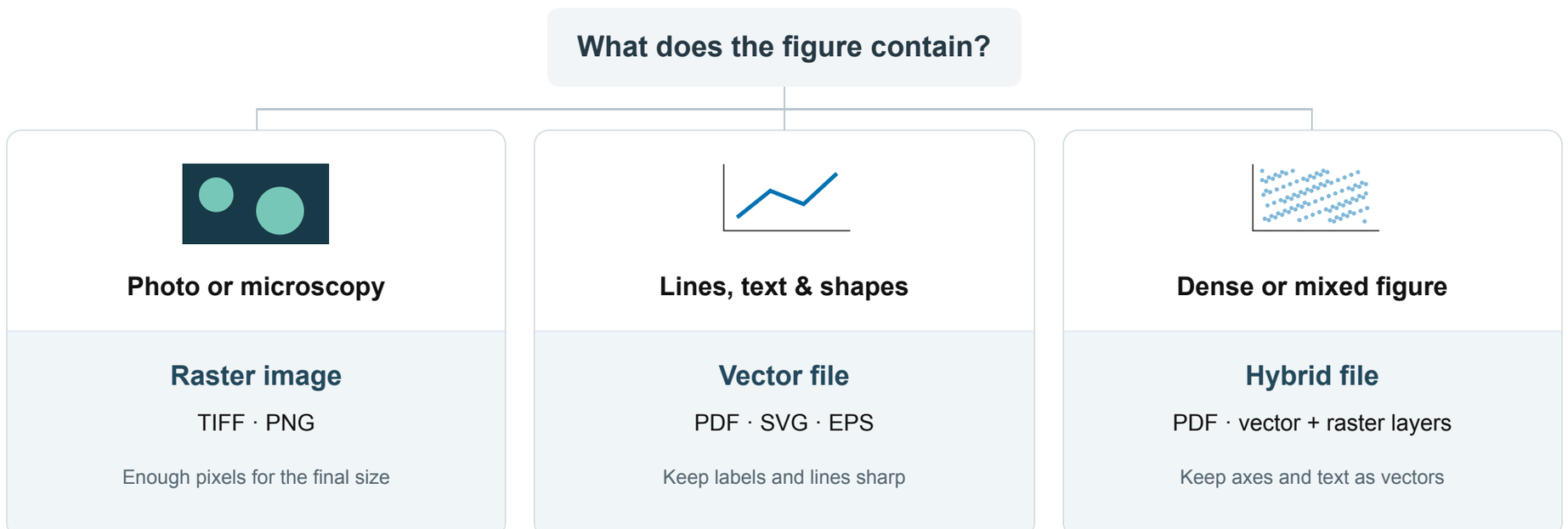
Rasterize background traces if needed; retain summary lines as vectors.

Keep text and axes as vectors

Heatmap and trace panels are schematic illustrations. Vector export does not necessarily preserve every original application layer.

Python export settings

Which format should I choose?

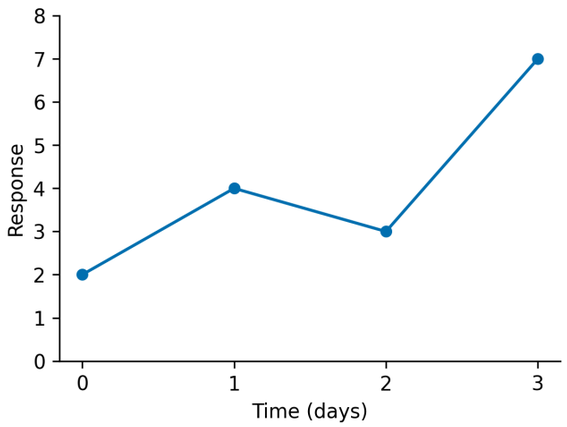


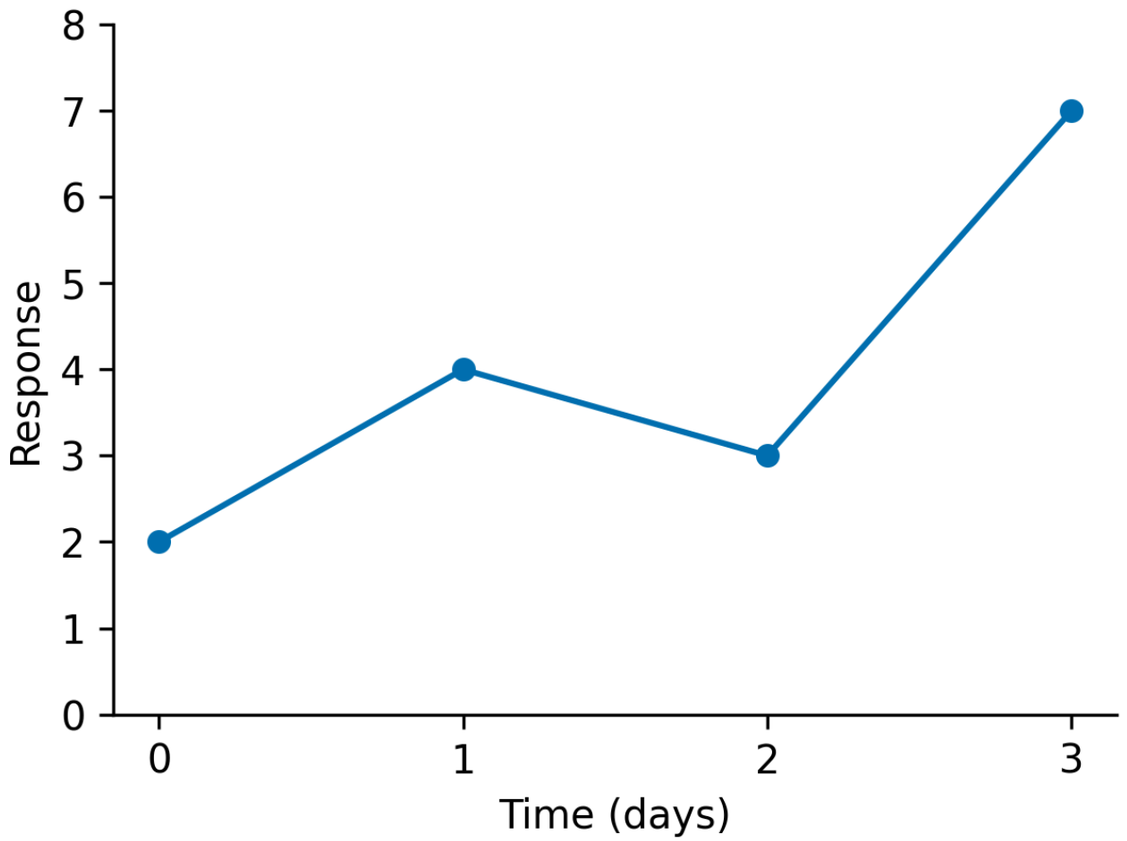
For dense plots, start with vector. If file size or rendering becomes a problem, rasterize only the dense layer. Always check the destination’s accepted formats.

> **Python export settings**

Resizing changes effective resolution

The same **1,200-pixel-wide PNG**, placed at two different widths. No pixels have been added.



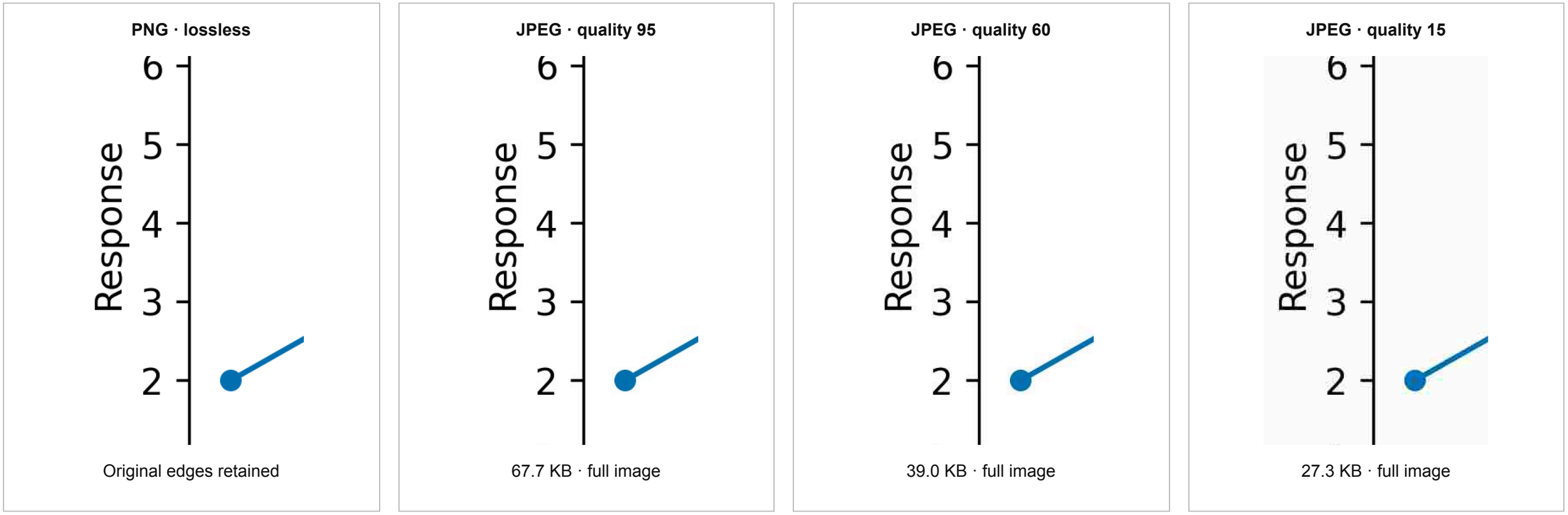


Doubling placement width halves effective PPI. These previews show the size ratio, not calibrated physical inches on your screen.

Python export settings

JPEG versus PNG: what does compression change?

PNG compresses losslessly. Standard JPEG uses lossy compression: smaller files can lose fine detail. These images have identical pixel dimensions.



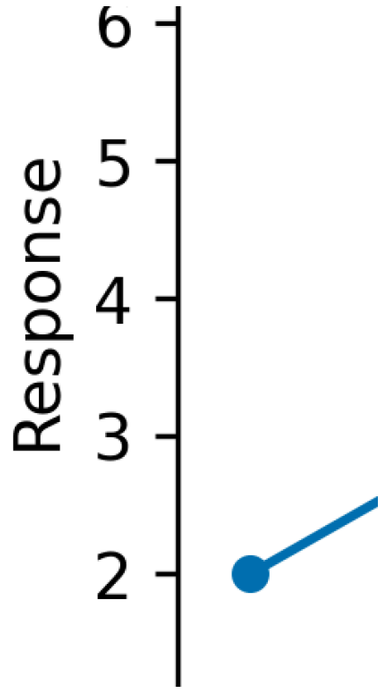
For plots, prefer vector or PNG when possible. JPEG can create halos, block patterns and blurred text. It is often suitable for photographs.

Same enlarged crop in each panel. Quality settings are encoder-specific, not percentages of retained information. File sizes refer to the complete JPEG, not these enlarged PNG previews. Follow the journal’s required formats.

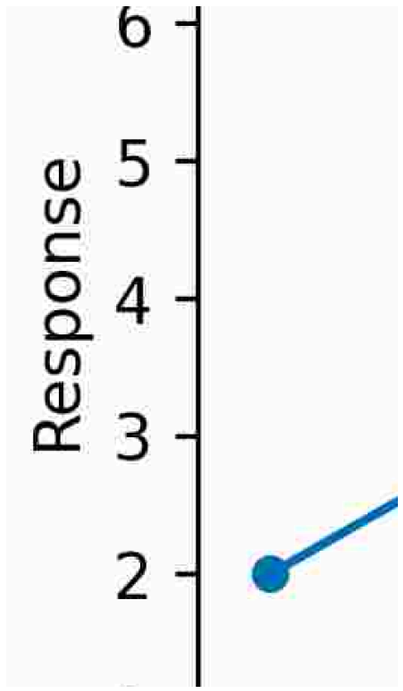
▸ **Python export settings**

JPEG can damage text and fine lines

PNG · lossless



JPEG · deliberately strong compression



Look for halos and speckles around the letters and axis. Both exports have identical pixel dimensions; DPI alone cannot prevent compression damage.

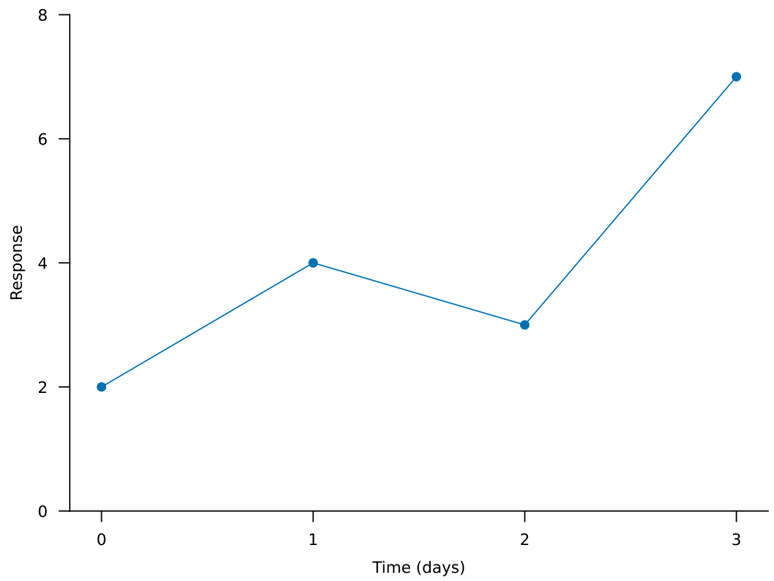
JPEG quality 12/100 is deliberately low to expose the effect. JPEG can suit photographs; use lossless raster or vector output for plot text and line art when supported.

Python export settings

Sharp is not the same as readable

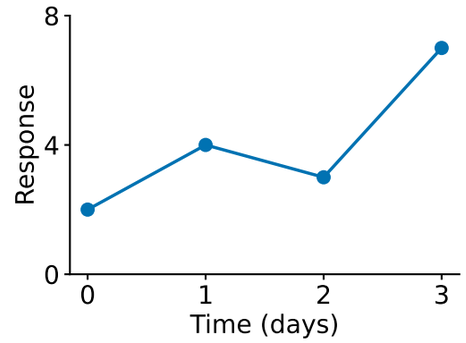
Same data and 4 × 3 inch source canvas. Stronger text, markers and lines let the right-hand figure remain readable at a smaller display size.

Vector, but elements are too small



Text: **5 pt** · markers: **2 pt** · lines: **0.4 pt**

Larger elements, easier to read



Source settings · text: **16 pt** · markers: **8 pt** · lines: **2 pt**

Right figure width: 100% | 75% | 60% | 50% | **40%** | 30%

At 40% and 30%, the plot is re-exported with larger text and markers, adjusted margins and fewer y-axis tick labels (0, 4, 8). Data and axis limits are unchanged. Percentages refer to display width; listed point sizes are source settings. Vector keeps edges sharp, it does not make tiny labels readable. Adjust font size, marker size and line width for the final display or publication size.

Illustrative settings, not universal minimum sizes. Check the intended viewing distance and publication requirements.

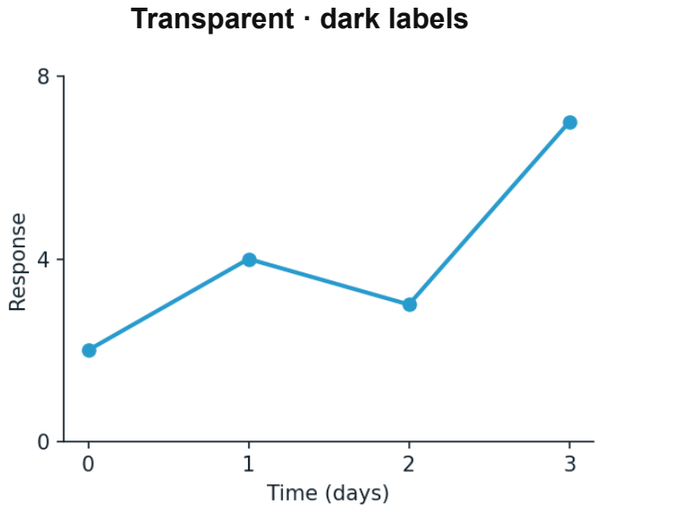
> **Export a smaller figure · R & Python**

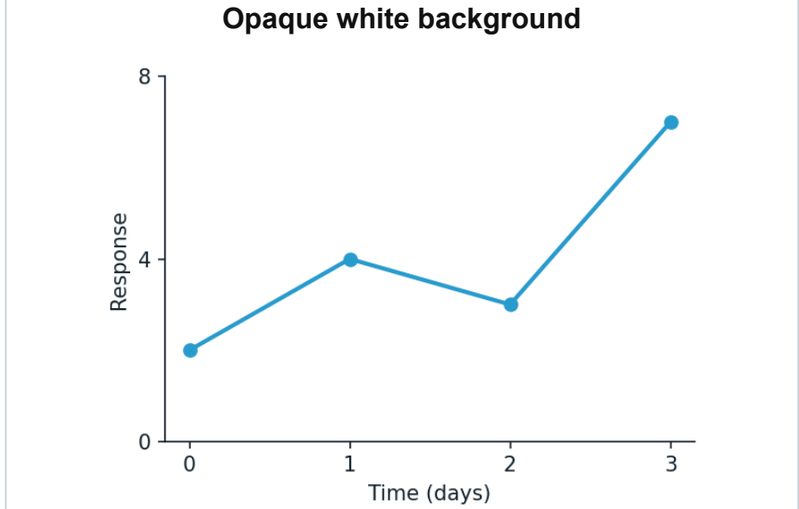
> **Python export settings**

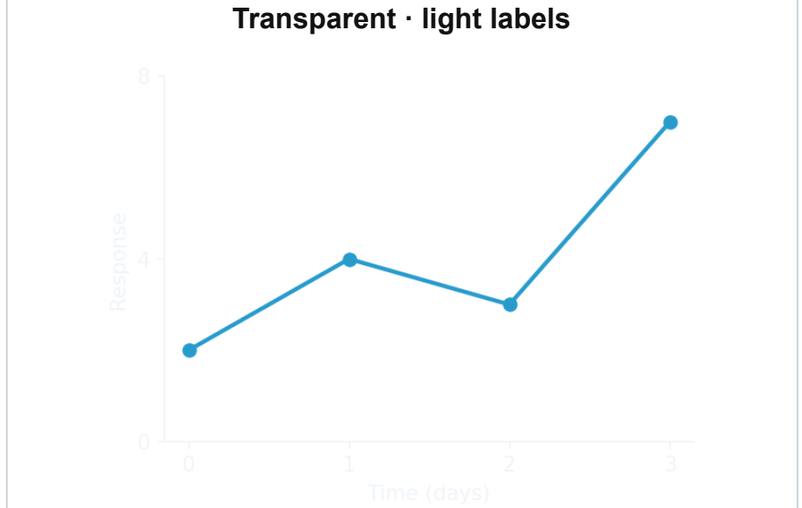
Transparent background: try it on your slide

A transparent background lets the slide show through. It does **not** automatically adapt the colour of labels or lines.

White | Dark | Grey | Warm | Blue | Animate backgrounds — White







Dark labels vanish on dark slides; light labels vanish on white slides. A white export stays readable here, but brings a visible white rectangle onto a coloured slide.

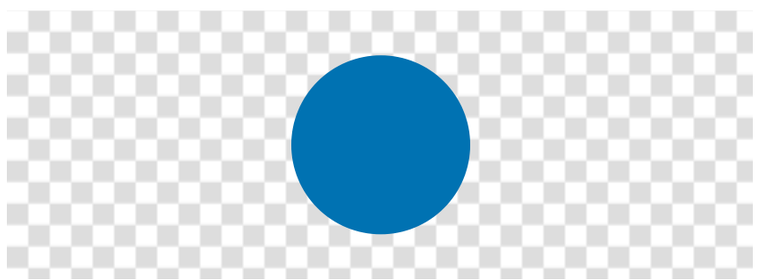
Export settings

Python export settings

What is an alpha channel?

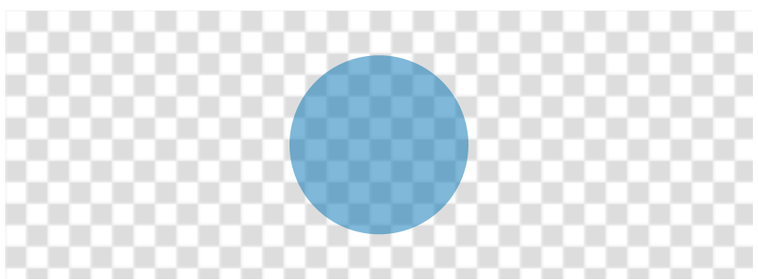
RGB stores red, green and blue. **Alpha** stores opacity, how much of the background shows through. Together: **RGBA**.

α = 1 · Opaque



Foreground covers the background

α = 0.5 · Half transparent



Foreground and background blend

α = 0 · Invisible

Only the background is visible

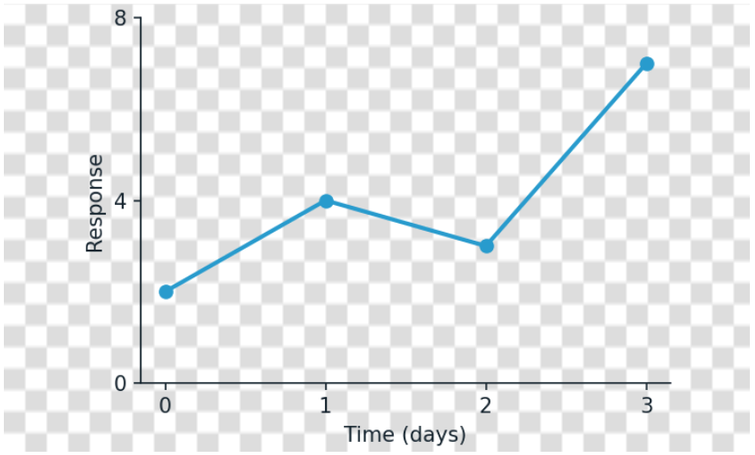
Common conventions: alpha 0–1, or 0–255 in an 8-bit channel. In both, zero means fully transparent. The checkerboard is an editor’s transparency indicator, not part of the image.

PNG supports alpha transparency. Ordinary JPEG does not; transparent regions must be flattened against a background. An opaque PNG may have no alpha channel, or alpha set fully opaque.

- **Which file formats support transparency?**
- **Python export settings**

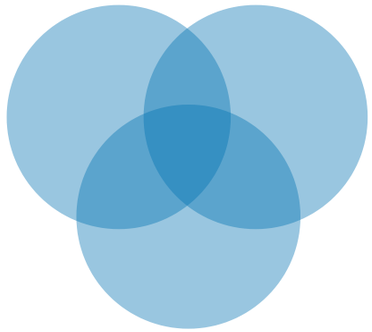
Background transparency is not point transparency

Transparent canvas



Controls what appears behind the plot.

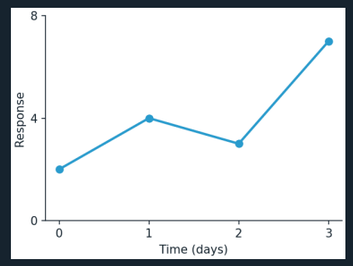
Transparent marks



Controls overlap within the data layer.

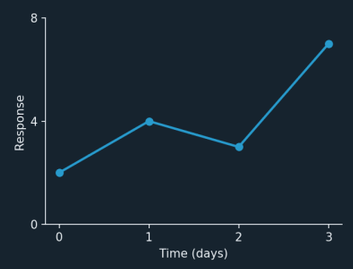
Check both in context

Results · response over time



Awkward: white background pasted onto a dark slide

Results · response over time



Integrated: transparent background + light labels

A white panel can be intentional, but check the choice against the slide design. Transparency alone will not fix dark labels.

Transparency changes perceived colours. Preserve contrast, group distinctions and the meaning of the data when changing backgrounds.

Python export settings

Choose an export for its purpose

Plots with text and lines

Prefer SVG or vector PDF when supported. Keep a vector master for editing and resizing.

Vector containers may still include raster layers, such as a heatmap or photograph.

Images and required raster exports

PNG is lossless and suits plots. JPEG is usually better suited to photographs; lossy compression can leave artefacts around text and lines.

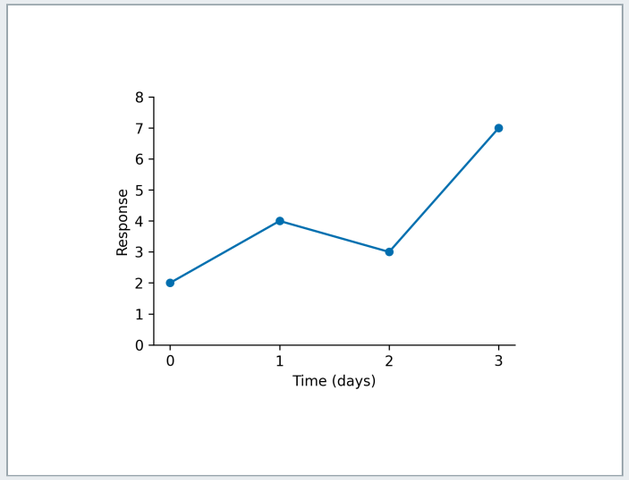
Export at the intended size and required resolution. Higher DPI increases pixel count and file size; it does not add information to the measurements.

Check the destination’s requirements and inspect the export at its intended size. There is no single DPI setting for every journal, printer or screen.

Python export settings

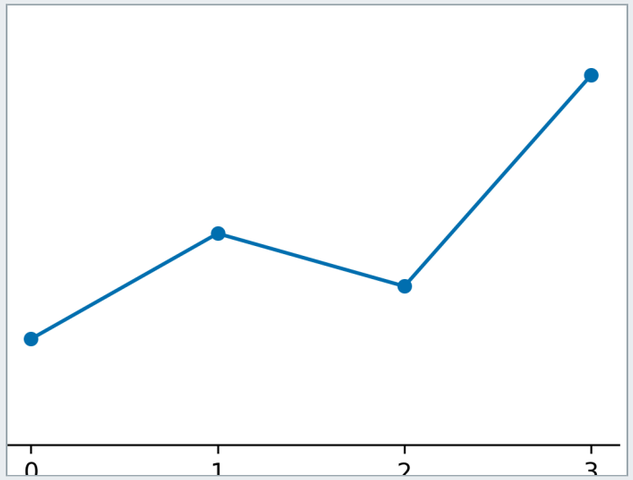
Extra whitespace or clipped content?

Excess margins



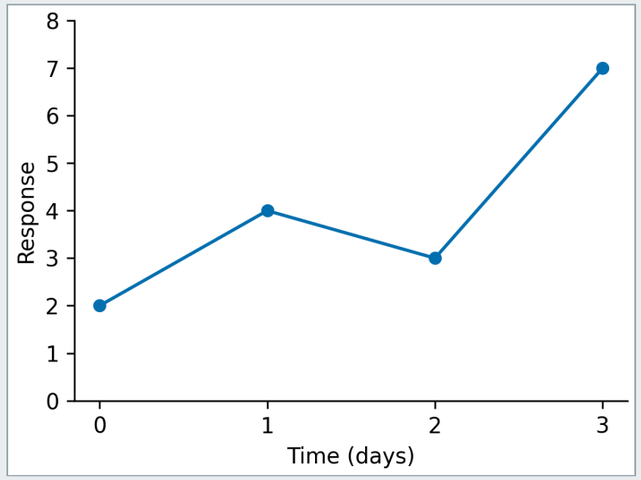
The data are intact, but the page is larger than the figure.

Clipping



The boundary cuts through labels. Make the page or margins larger.

Enough room, little waste



Fit the page around the whole figure, leaving a small margin.

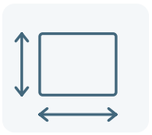
Crop empty space, not evidence. Check all labels, legends, points and annotations before tightening the boundary.

- **Fix the page boundary**
- **Python export settings**

Before you export: six checks

Format

Vector for lines and text; raster for images. Use mixed layers when helpful. Check the destination’s requirements.



Dimensions

Set the intended width and height. Inspect at the size readers will see.

Readable labels

Check fonts, markers and line widths. Keep units, legends and scale bars clear.

Resolution

For raster layers, match pixel dimensions to placement size and required PPI. Increasing metadata alone adds no detail.

Background

Check white or transparent export on the actual slide or page background. Keep enough contrast.

Margins & clipping

Trim excess whitespace without cutting off labels or data. Open the export and check for distortion or changed fonts.

Export → open → inspect → adjust. Keep the source code and an editable master.

› **Python export settings**

Post-processing tools

Edit layouts, handle fonts, and preserve the evidence.

Tools for the final polish

Inkscape

Free and open source.

Vector editing and panel assembly.

inkscape.org

Editable master: SVG

Useful for labels, alignment and annotations.

Adobe Illustrator

Paid subscription · can be costly.

Check whether your group or institution provides a licence.

adobe.com/illustrator

Editable master: AI

Useful for multi-panel artwork and typography.

Other options

Affinity Designer for vector layouts; PowerPoint or Keynote for simple assembly.

Check the exported file: appearance, fonts and editability can change.

Both offer similar core tools for figure editing: paths, text, alignment and layers. Illustrator may feel more polished to some users; usability depends on experience and workflow.

Keep the plotting script and an untouched export. Save the edited layout as a separate master.

Why can a label arrive as separate characters?

One text object

```
<text x="30" y="50">Control</text>
```

Edit the whole word together.

Individually positioned characters

```
<text x="30" y="50">C</text>
<text x="64" y="50">o</text>
<!-- …one object per character -->
```

Editing can leave uneven spacing or move letters separately.

PDF/SVG export or import can fragment labels or position glyphs individually. It depends on the exporter, font and editor. Inspect the imported text before changing it.

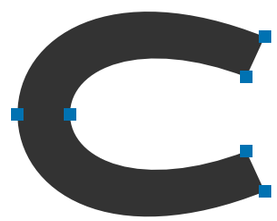
Editable text or outlines?

Text objects

Easy to edit, search and copy. The correct font must be available or embedded as supported.

```
<text font-family="Arial">Control</text>
```

Letters converted to paths



Blue squares: anchor nodes on the outline

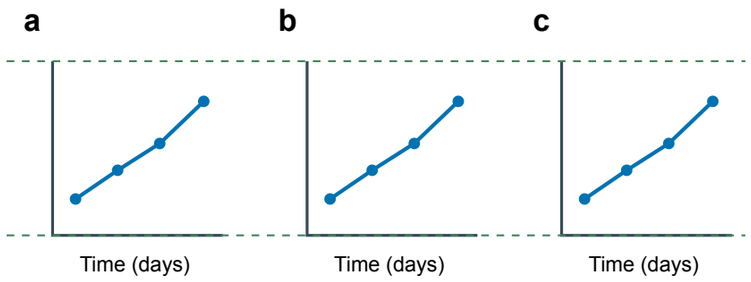
Appearance is preserved without the font, but normal text editing and search are lost. A journal may require editable text.

```
<path d="M90 23 C35 2 … Z" />
```

If a label imports badly, prefer re-exporting with suitable font settings. If you replace it manually, preserve its exact wording, units and scientific meaning.

Polish the layout, preserve the evidence

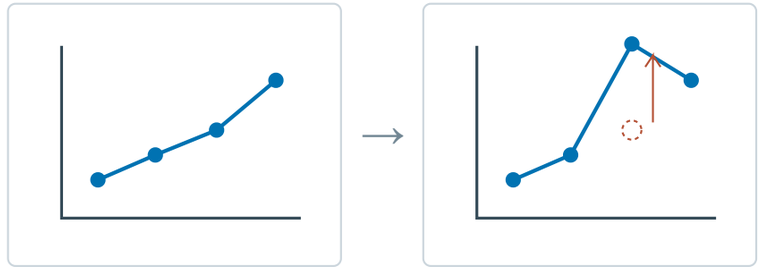
Do



Align axes and labels; keep every observation.

- Align panels and standardise fonts.
- Adjust spacing and add panel letters.
- Add clear annotations without covering observations.
- Resize complete panels proportionally.
- Keep scale bars, units and colour meanings correct.

Don’t



Moving one point changes the evidence.

- Move points, resize individual bars or delete inconvenient observations.
- Stretch a panel in only one direction.
- Relabel ticks or recolour values so the encoding changes.
- Crop away results or alter images to strengthen a finding.

If an edit changes data, scales or analysis, make it in the plotting script and regenerate the figure.

Before sending the edited figure

Compare with the original

Same observations, axes, units, legends and scale bars?

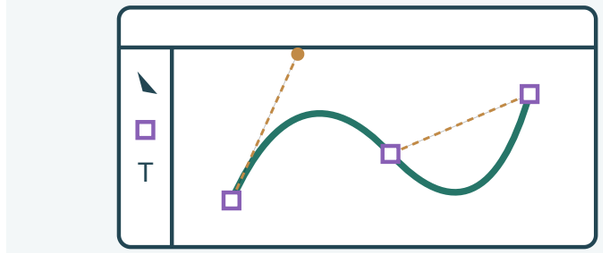
Nothing missing, distorted, hidden or clipped?

Inspect the final export

Fonts correct? Labels readable at final size?

Raster layers sharp enough? File format accepted?

Save the script, source data, untouched export and editable layout. Keep a short record of manual changes.



Tools: Inkscape

Install the editor, arrange panels, add labels and export a figure.

Install Inkscape

A free desktop vector editor for arranging panels, adding labels and preparing figure layouts.

Windows

1. Open the official download page and choose Windows.
2. Download the installer that matches your system.
3. Run it, keep the default options and launch Inkscape.

macOS

1. Choose macOS, then Apple silicon (arm64) or Intel.
2. Open the downloaded DMG and drag Inkscape into Applications.
3. Launch it from Applications.

Check your processor in Apple menu → About This Mac.

Linux

1. Search for Inkscape in your distribution’s software manager.
2. Install and launch it.

The official download page lists other installation options.

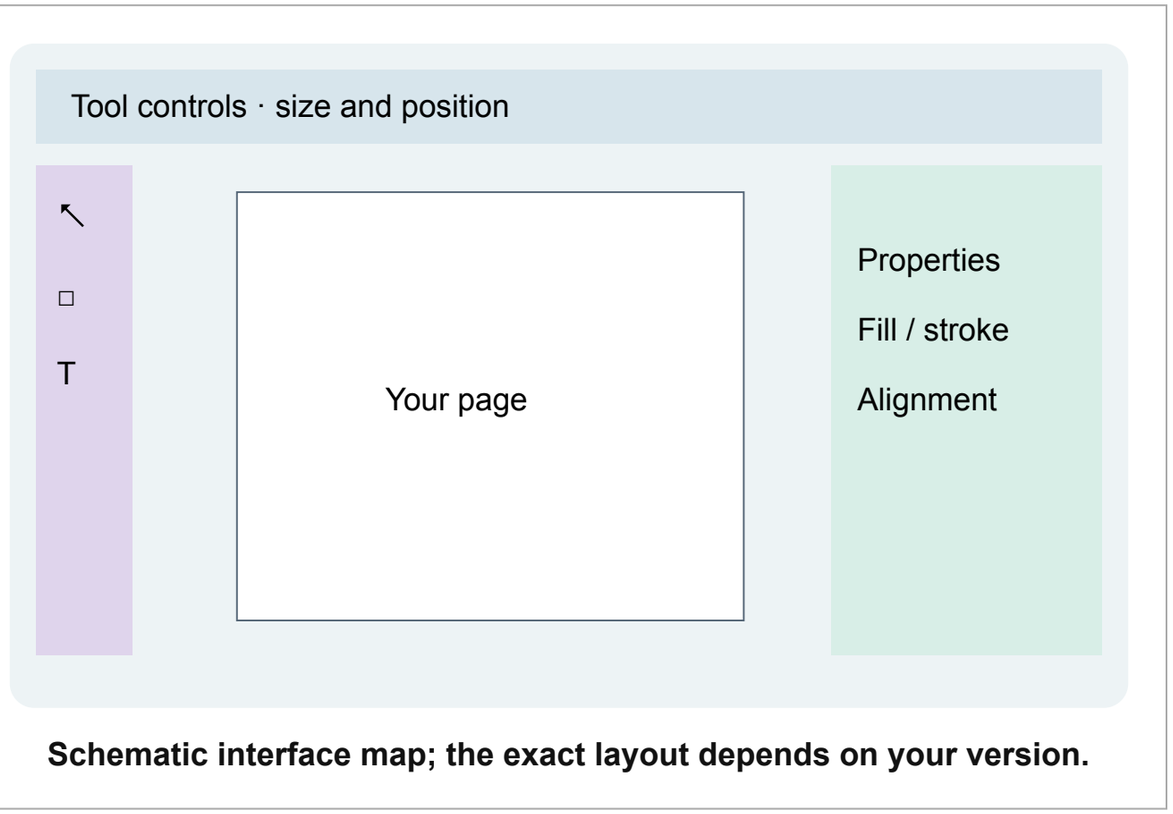
https://inkscape.org/release/ ↗

Already installed? Open it and create a blank document. On a managed computer, ask IT if installation is restricted.

Official installation instructions ↗

Find your way around

Think of the page as your final figure, and the surrounding canvas as your workspace.



Schematic interface map; the exact layout depends on your version.

Start with three tools

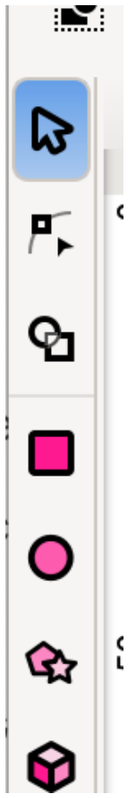
Selector: select, move and resize objects.

Text: add panel letters and labels.

Zoom: inspect details without changing the figure's size.

Use File → Document Properties to set the page size and units.

The two arrows · objects versus nodes



Start with the top arrow.

Top arrow: Selector

Your main tool for figure assembly. Select, move, resize and rotate an entire object or panel group. Its internal shape stays together.

Below it: Node tool

Edit the nodes and curve handles of a path. Moving a node changes the shape itself.

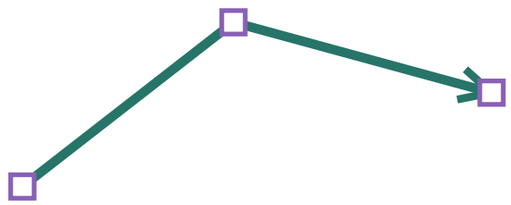
A **path** is a sequence of points (nodes) joined by straight or curved segments. A drawn arrow, an outline or an exported plot line can be a path.

> Use the Selector to arrange plot panels. Use the Node tool to adjust a schematic shape, not to move measured data points.

What is a path? · points joined by segments

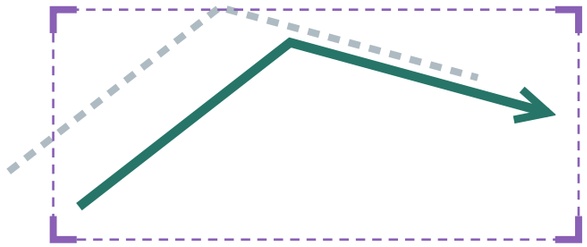
Imagine a simple bent arrow in a workflow diagram. Its line is a path with three nodes.

A path



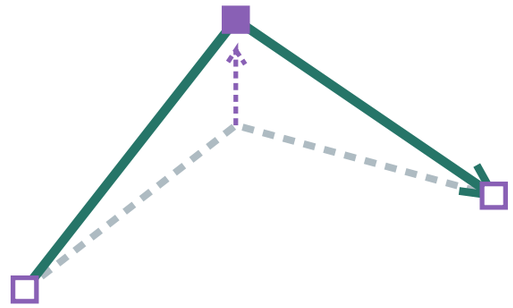
Squares mark the nodes; the lines between them are segments.

Selector: move the object



The bounding box moves with the whole object. The shape is unchanged.

Node tool: edit one node



Only the middle node moves. The bend changes.

Simplified teaching diagrams: grey dashed lines show the original position. The squares in the first example identify nodes; they are not always visible in Inkscape.

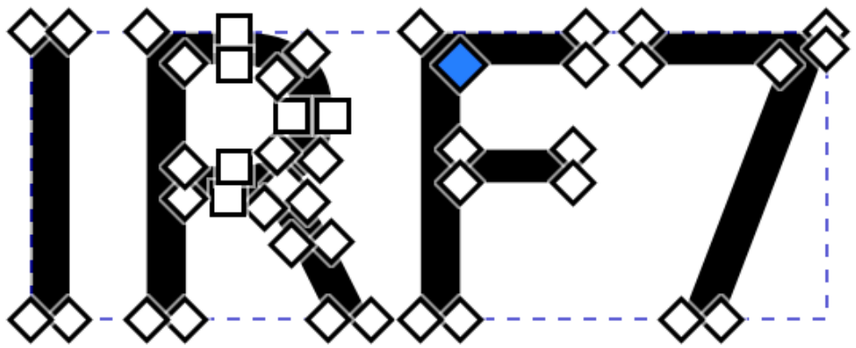
Curved segments also have handles that control their direction and curvature.

Try it: draw a simple line with the Bézier tool, then switch to the Node tool and move one node. Undo, switch to the Selector, and move the whole line.

A PNG is made of pixels, not editable path nodes. Shapes such as rectangles have their own editing handles until converted to paths.

When letters are paths · text you cannot retype

It looks like text, but each letter is now a vector outline.



Your example: the Node tool reveals points around each letter.

Editable text

The Text tool lets you change the wording, font and size.

Letters converted to paths

The Selector can move or resize them; the Node tool changes their outlines. The Text tool cannot retype the label or change its font.

This can happen when exporting or importing a figure. Outlines preserve the letter shapes without needing the original font.

Need different wording? Re-export with editable text, or replace the outlined label with a new text object. Keep its scientific meaning and placement correct.

> **Can I turn it back into text?**

Inkscape guide: converting text to paths ↗

Open a plot, then import another

Open = a document

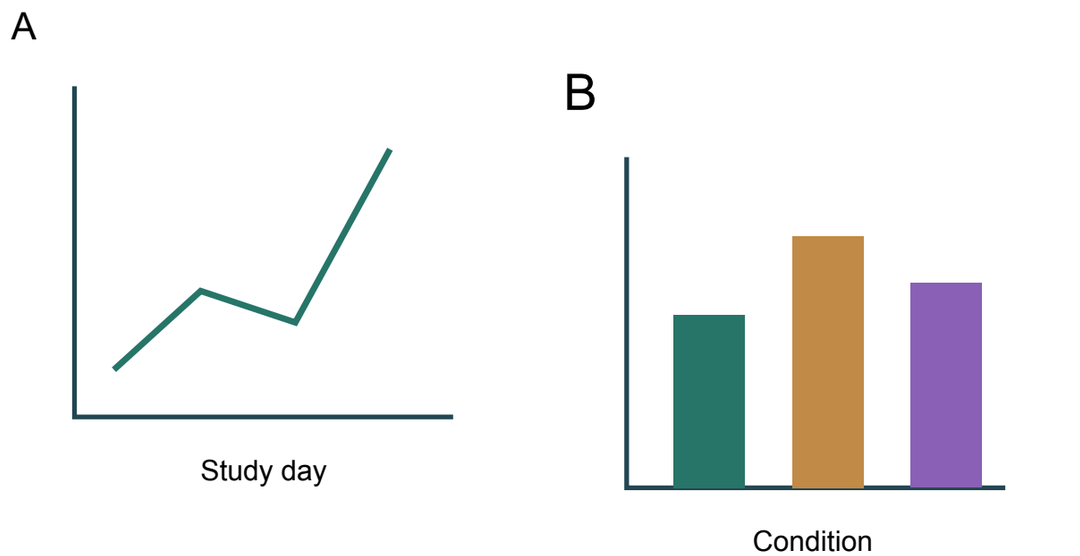
Use **File → Open** for an SVG you want to edit.

Import = add to this document

Use **File → Import** to place another panel in your layout.

SVG usually preserves vector objects. PDF import depends on the export and import options; inspect fonts, symbols and clipping.

A PNG remains one raster image. Its plotted points are not individually editable.



Practice figure: simple shapes, labels and two panels.

Download the practice SVG

Try it: open the practice SVG and save a working copy as `figure-layout.svg`.

Select, group and resize

1 · Select

Click an object with the Selector. Shift-click to add another object to the selection.

Click empty canvas to deselect.

2 · Group

Use **Object → Group** to keep a panel together.

Use **Object → Ungroup** only when you need its individual objects. Imported figures can contain nested groups.

3 · Resize

Lock the width-to-height ratio in the Selector toolbar before changing the size.

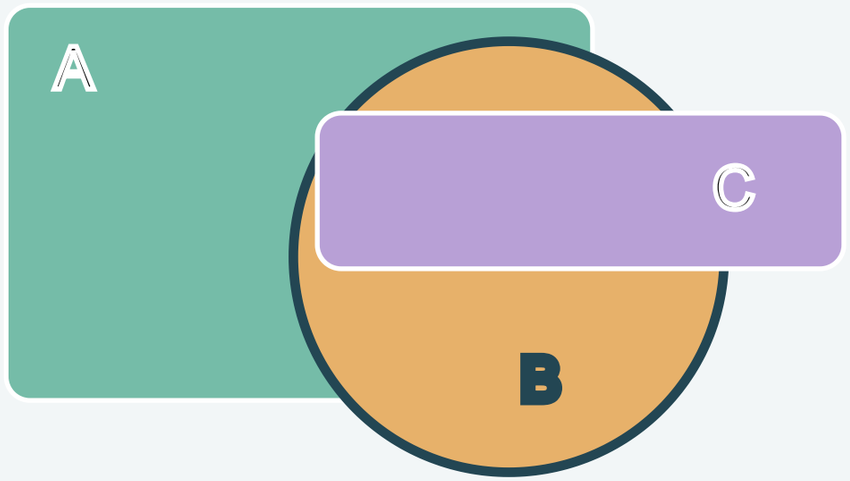
Move and resize whole panels together. Check whether stroke widths and text have scaled as intended.

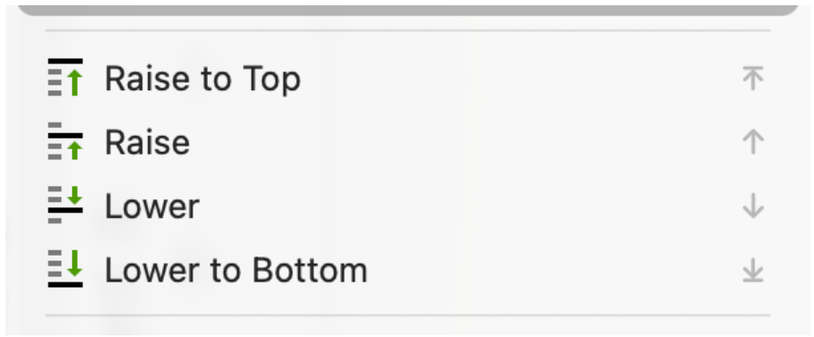
Try it: select one practice panel, resize it proportionally, then undo. Keep the data marks in their original relationship to the axes.

Objects overlap · change what is in front

Stacking order decides which object covers another. Raising or lowering changes the order, not the object’s position or size.

Selected object: B · orange circle





Raise to Top | Raise | Lower | Lower to Bottom | Reset

B starts between A and C. Try Raise to Top, then Lower one step.

Raise / Lower: one step in this overlapping stack.

Top / Bottom: go to the front or back of the current stack.

Back → A → B → C → Front

Figure example: send a background rectangle behind the plots, or bring an annotation label in front of a schematic.

This example uses one layer. Objects inside groups or layers have their own stacking context; change the group or layer order when needed.

Keep backgrounds safe · hide and lock

Use **Layer → Layers and Objects** to find, name and organise the objects in your figure.

Arrange the stack

Labels and annotations
above
Plot panels
above
Background

Select a covered object in the panel when you cannot reach it on the canvas.

Hide · eye control

Temporarily hide an object or layer to inspect what is underneath.

Show it again before export if it belongs in the figure. Hidden content is not included in the normal exported view.

Lock · padlock control

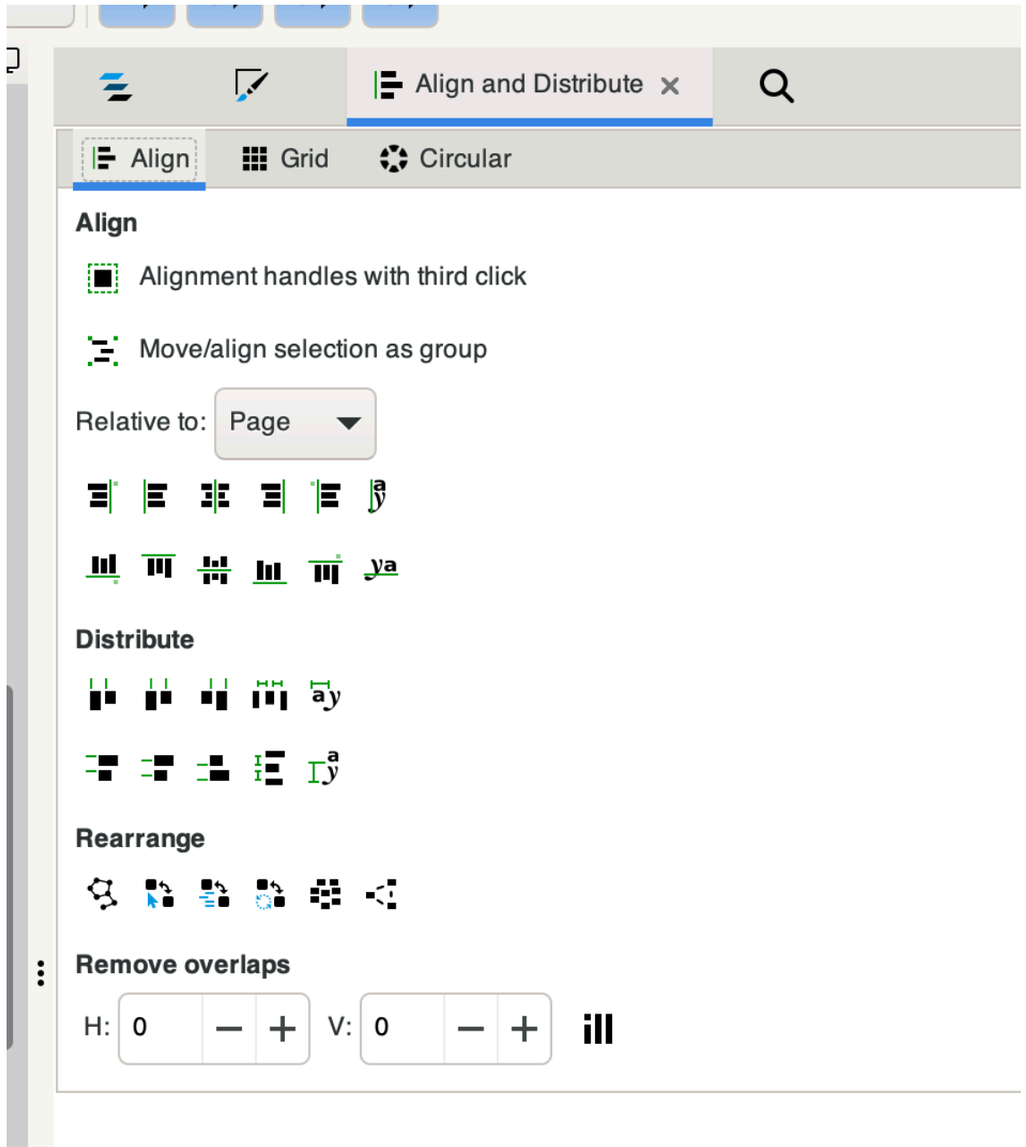
Lock a background after placing it. It stays visible, but you cannot accidentally select and drag it on the canvas.

Unlock it in the panel when you need to edit it.

Try it: draw a pale rectangle, lower it to the bottom, then lock it in Layers and Objects. Select and move a panel above it.

Do not cover unwanted data with white shapes. Correct the plot in R or Python instead.

Align panels · choose the reference first



Your Align and Distribute panel. Hover over a button in Inkscape to see its exact action.

Open **Object → Align and Distribute**, then select your panel groups with the Selector.

What does “Relative to” mean?

Page: align against the page. This is what your screenshot currently shows.

Selection Area: use the bounds of the selected objects to line them up with each other.

Last selected: keep one panel as the reference. Click the moving panel first, then Shift-click the reference panel last.

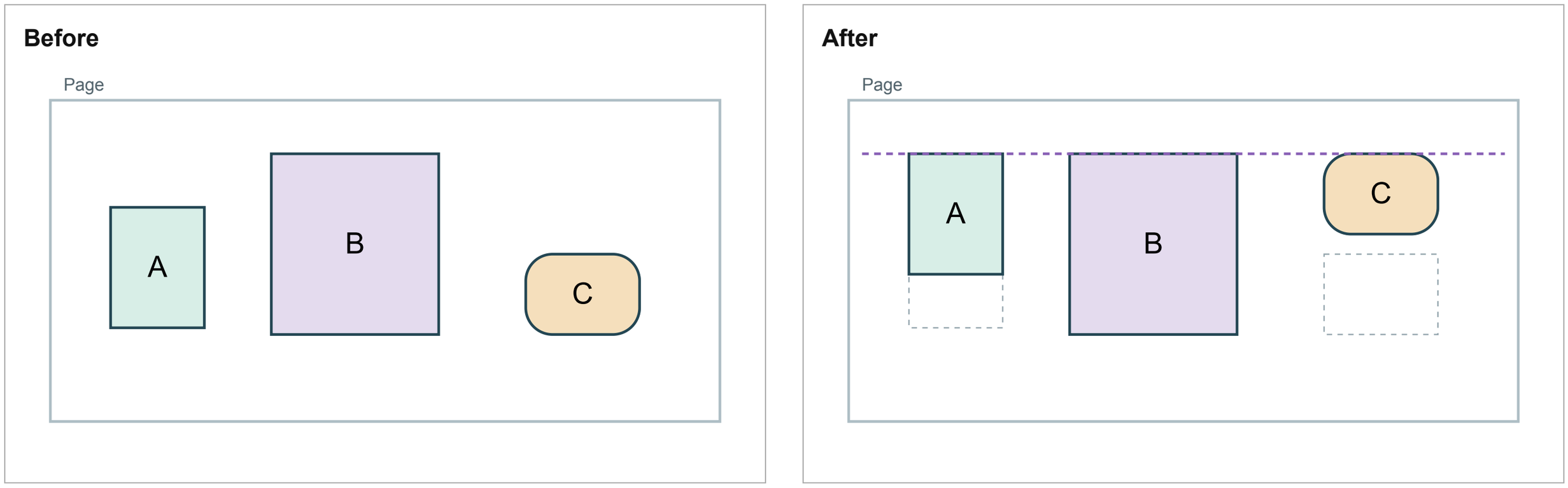
> **Try it:** select the two practice panels, choose Selection Area, then hover to find **Align top edges** and click it.

Leave **Move/align selection as group** off when aligning the panels with each other. Turn it on to move the whole arrangement together, for example to centre it on the page.

Align versus distribute · try different references

Same three objects, different instructions. Select A first, then B, then C last.

Top → Selection Area | Top → Page | Top → First selected (A) | Top → Last selected (C) | Centre group on page | Equal horizontal gaps | Equal horizontal centres

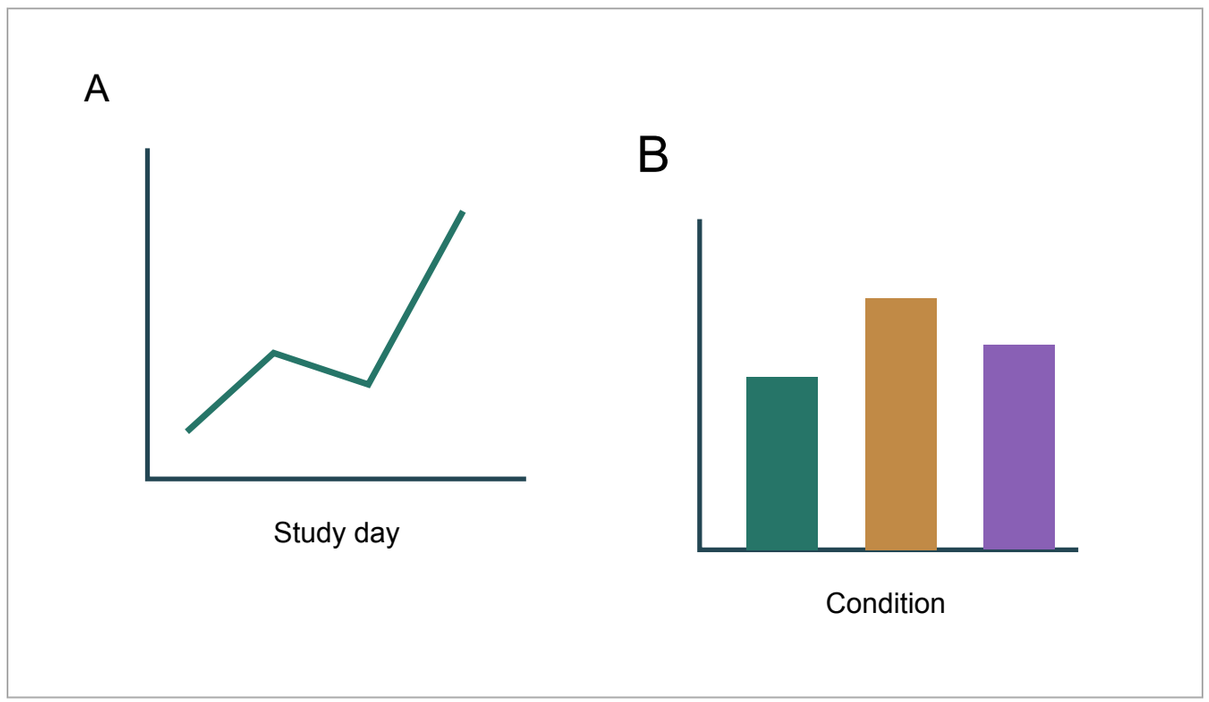


Align top edges · Selection Area

The top of the original selection is B’s top edge. B stays still; A and C move up. Move/align selection as group is off.

Illustrative object bounds. Dashed outlines show the original positions; purple guides show the alignment target or spacing measurement.

Add panel labels · check the visible plot areas



Finish the layout

1. Make the panel letters the same font, size and weight.
2. Place them consistently at the top left of each panel.
3. Check the axes and visible plot areas, not only the outer object boxes.
4. Inspect the figure at its intended publication size.

Long axis labels and legends change the bounds of a panel. Aligned boxes can still leave the plotting areas uneven.

Use guides for visual alignment. Keep each panel’s data marks and axes together.

Try it: align the practice panels, choose a gap, and make A and B consistent. Save a new SVG copy.

Hex colours · six digits, plus optional alpha

A hex code is a compact way to write a colour. Each pair gives the amount of red, green or blue, from `00` to `ff`.

00	9e	73	ff
Red	Green	Blue	Alpha

In Inkscape's **RGBA** field, `009e73ff` means green `#009e73` at full opacity. The last two digits are alpha.

009e73ff

Fully opaque

009e7380

About 50% opaque

009e7300

Fully transparent

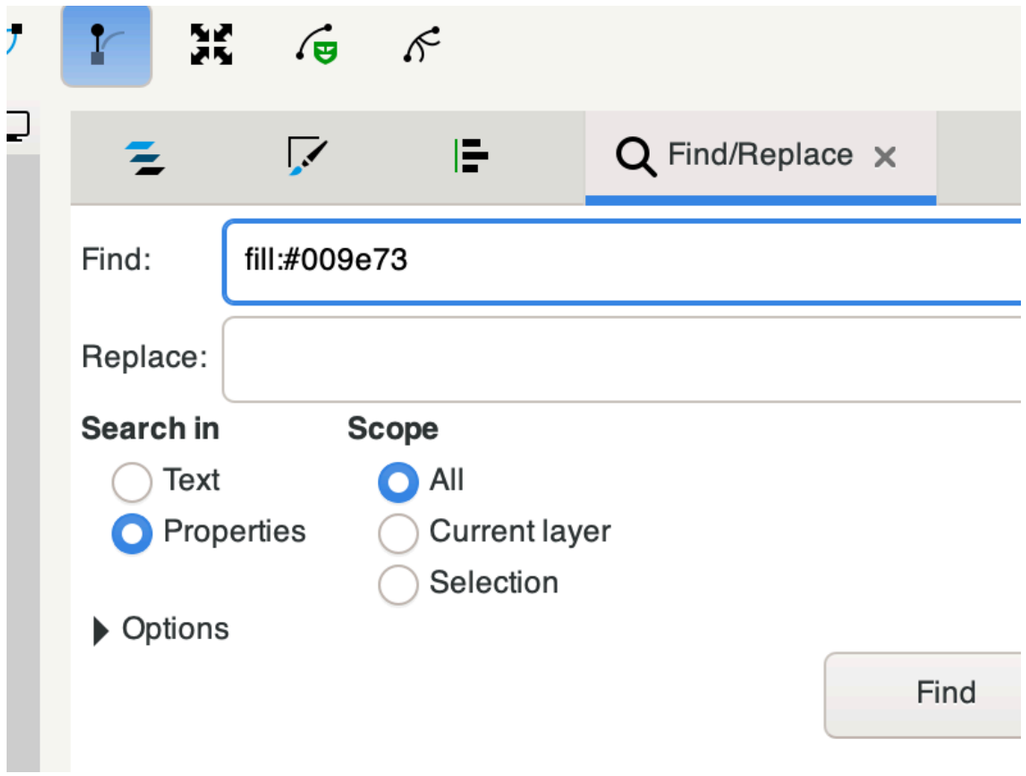
Black
#000000
RGBA: `000000ff`

White
#ffffff
RGBA: `ffffffff`

Blue
#0000ff
RGBA: `0000ffff`

The checkerboard shows transparency. Object or group opacity can further reduce how opaque the colour looks.

Find by colour · search object properties



Search for a fill colour in object properties.

Select matching objects

1. Open **Edit** → **Find/Replace**.
2. Enter `fill:#009e73` in Find.
3. Choose **Properties**, then the scope: All, Current layer or Selection.
4. Under Options, enable **Style** and leave Exact match off.
5. Click **Find**, check the selected objects, then change their colour in Fill and Stroke.

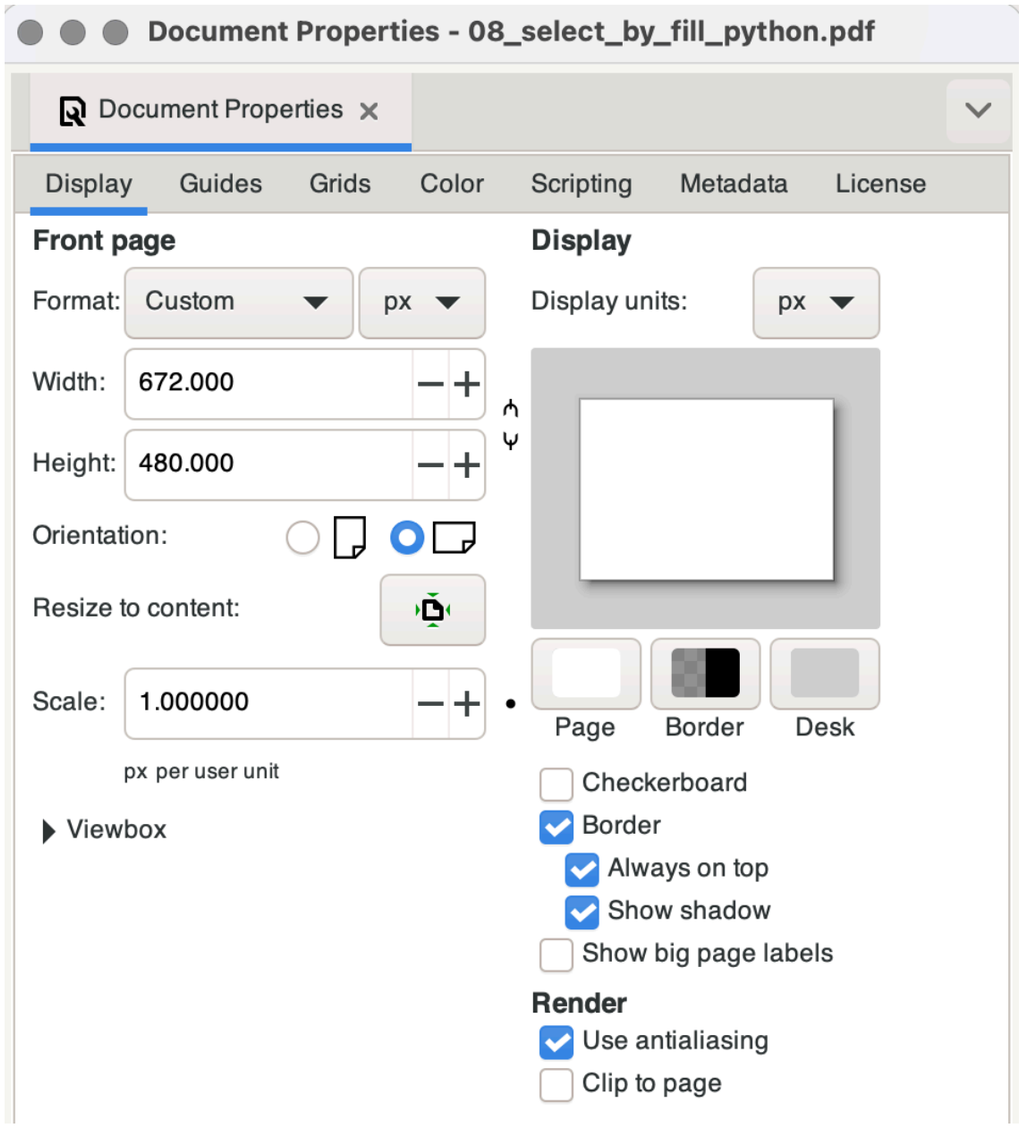
For line colours: search `stroke:#009e73`.

Why no ff? Inkscape commonly stores the six-digit colour in the SVG style and opacity separately, for example `fill:#009e73;fill-opacity:1`. The picker's RGBA text is not the literal style string.

For white background rectangles, try `fill:#ffffff`. Other white objects can match too; the page background is not an object.

No results?

Fit the page to the figure · Resize to content



The Resize to content button is below Orientation.

File → Document Properties

1. Open the **Display** tab.
2. Find **Resize to content**, below Orientation.
3. Click the button with the page and inward-pointing arrows.
4. Check that the fitted page includes all labels, legends and strokes.

This fits the page around the artwork. It does not resize the plot itself. Leave **Scale** unchanged.

For the whole figure, deselect first. To fit selected objects instead, select the complete figure and use **Resize Page to Selection** (default shortcut Ctrl + Shift + R).

Export area: use **Page** to export the fitted page. Drawing and Selection instead use the bounds of the drawing or selected objects.

> **Still too much whitespace?**

Save the editable figure and export a copy

SVG · keep editing

Save your working document as Inkscape SVG. Keep original plot exports alongside it.

PDF · share the figure

Save a copy as PDF. Inspect the exported PDF for changed fonts, missing symbols and clipped labels.

PNG · raster preview

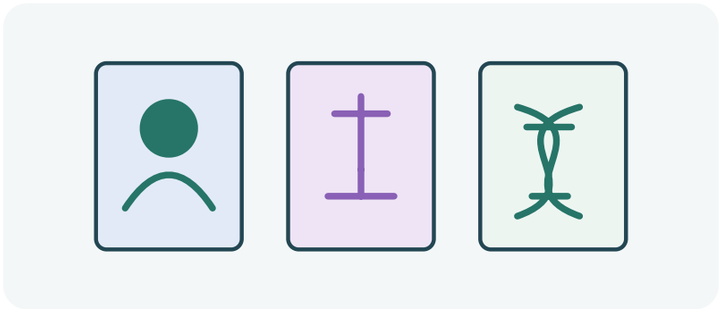
Use File → Export. Choose the page or selection, then the required pixel dimensions or resolution.

Check transparency and the intended background.

Mini task: export your aligned two-panel figure, reopen it and check readability at its final size.

Return to R or Python to change data, statistical summaries or axis scales; use Inkscape for the layout.

Inkscape beginners' guide ↗

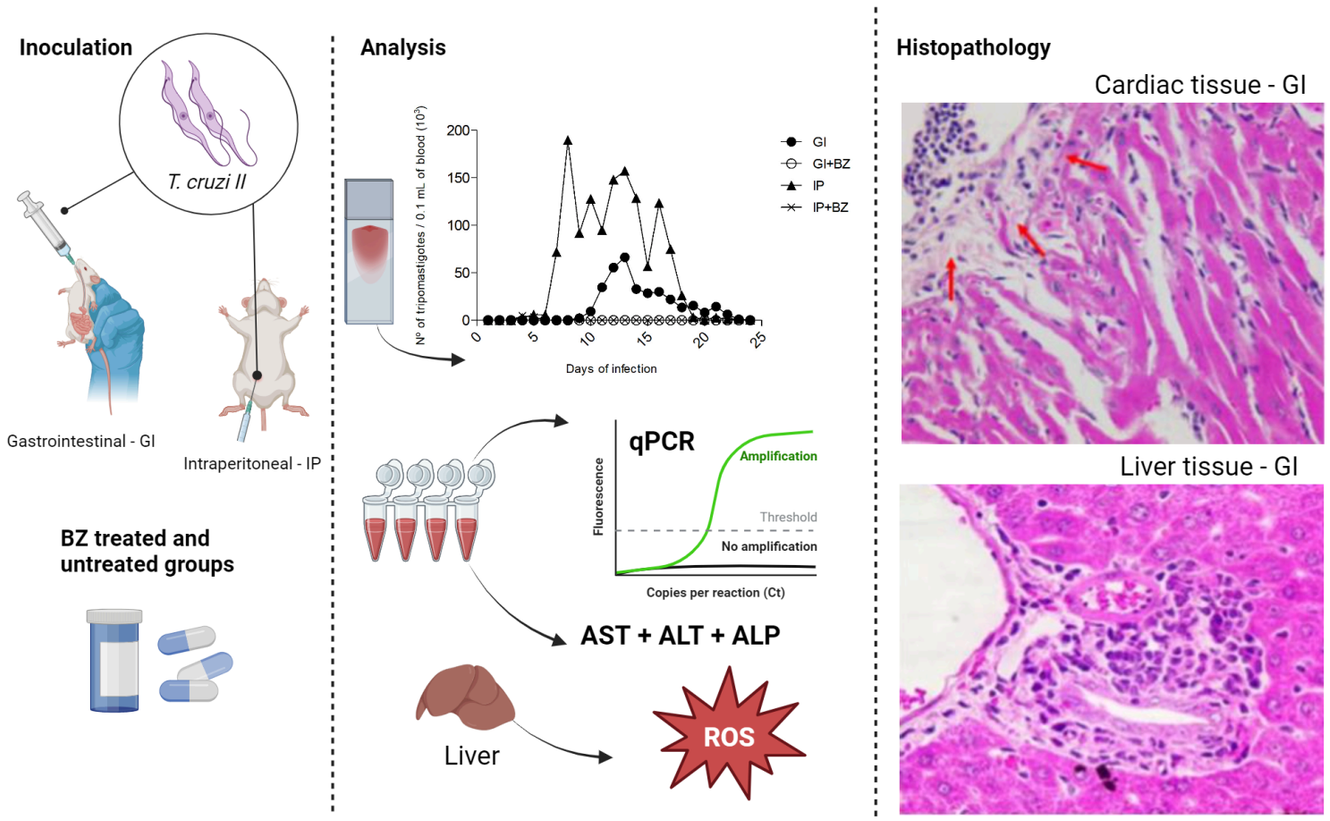


Include icons in the plot

Generate, compare and place clear scientific icons.

Graphics can explain the study at a glance

A graphical abstract or overview figure can connect the experimental setup, measurements and results.



Example from the original course deck · BioRender template cited in that slide

Graphics have a job

Setup: show organisms, treatments and sampling.

Workflow: connect steps in the experiment.

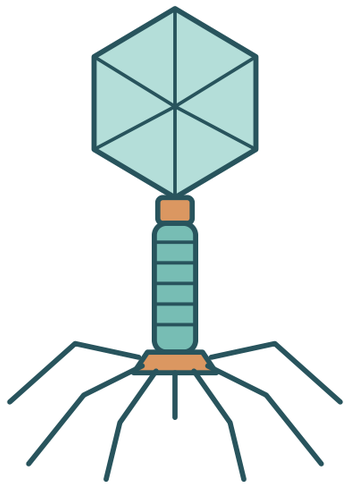
Overview: help readers locate the measurements and results.

Use graphics where they explain something. Keep measured evidence distinct from schematic illustrations.

Source & context

The same export rules apply to graphics

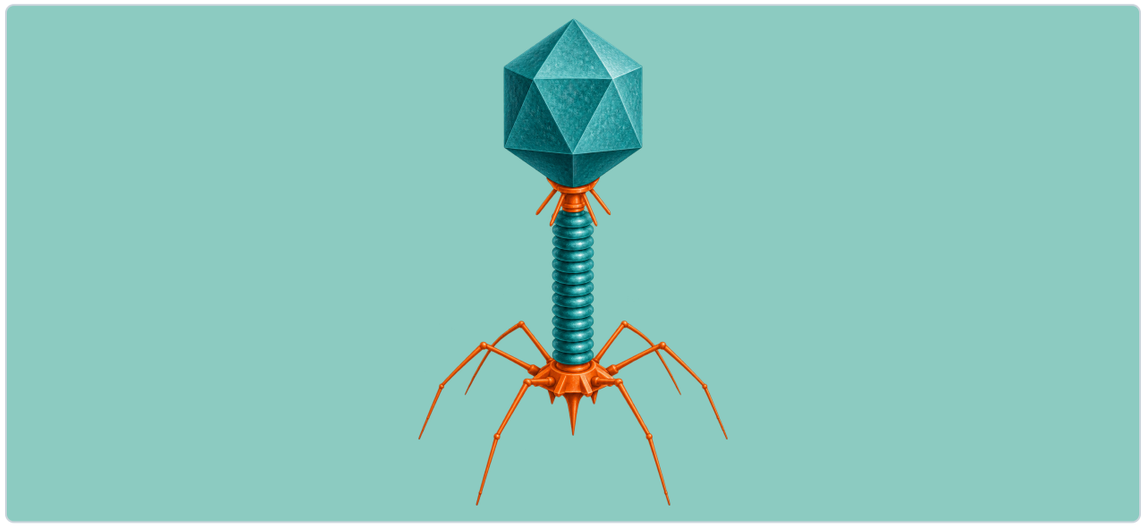
Editable shapes and text



SVG or vector PDF keeps lines sharp and labels editable where supported.

Keep font sizes, line widths and colours consistent with the plots.

Raster illustrations and microscopy



Use enough pixels at the final size. Check transparency and background contrast.

A PDF can contain raster images. Placing a PNG inside it does not turn the PNG into vector paths.

Combine graphics and plots, then inspect the complete figure at its intended size.

Start with a scientific icon library

You do not have to generate every icon. Start with a source that identifies the creator and reuse terms.

BioRender

Scientific icons, templates and a figure editor: cells, anatomy, microbes and laboratory equipment.

Publication rights depend on your licence.

Browse the BioRender library ↗

Bioicons

Scientific vector illustrations from multiple contributors.

Check each icon’s licence and author. The collection includes CC0, CC BY-SA and MIT assets.

Browse Bioicons ↗

Ongoing Science

Science drawings covering organisms, cells, microbes, molecules and instruments.

AI-generated SVG schematics. Check the graphics terms and credit requirements.

Open the graphics page ↗

NIH BioArt Source

Scientific and medical visuals: viruses, bacteria, equipment and more.

Read the asset’s usage and credit terms.

Browse BioArt Source ↗

An image-search result is a discovery tool, not proof of permission. Follow it to the original source.

Examples from Ongoing Science

Browse all 3,054 drawings from the local library. Click an icon to open its SVG.

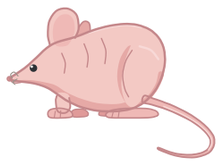
Search Phage, bacteria, micr Category All categories 3,054 / 3,054 drawings

Graphics provided by Ongoing Science. Local snapshot · 26 October 2026. AI-generated schematics; check scientific details. Saved graphics terms · CC BY 4.0 where applicable.

Organisms 726

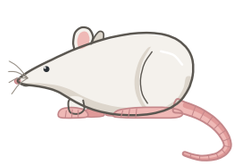
Mouse
Mammals

Mouse (C57BL/6, black)
Mammals



Mouse (athymic nude)
Mammals

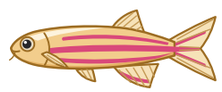
Mouse (top view)
Mammals



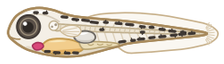
Rat
Mammals

Rat (hooded, Long-Evans)
Mammals

Rat (top view)
Mammals



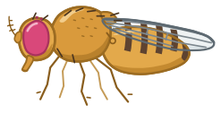
Zebrafish
Fish



Zebrafish larva (5 dpf)
Fish

Zebrafish (casper, transparent)
Fish

Fruit fly
Insects and arthropods



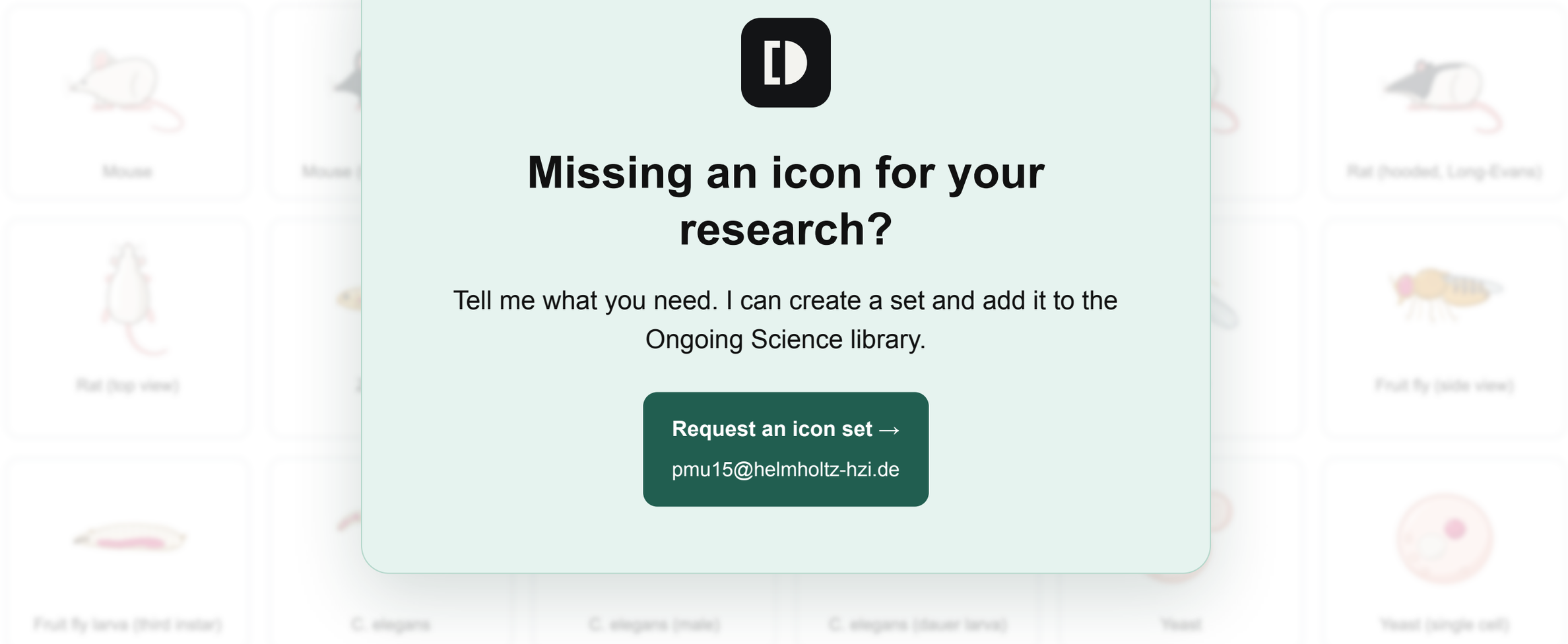
Fruit fly (side view)
Insects and arthropods

PDF preview: 12 drawings. Browse the complete library in the online slides.

Help grow the Ongoing Science library

Examples from Ongoing Science

Browse the drawings from the local lib



Missing an icon for your research?

Tell me what you need. I can create a set and add it to the Ongoing Science library.

Request an icon set →

pmu15@helmholtz-hzi.de

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Publication and export instructions ↗

Credit it in the figure legend

> **Figure 2. Experimental workflow.**
>
> [Scientific description of the panels.]
>
> [Paste the citation and unique figure URL generated by BioRender here.]

Follow the exact citation supplied with the publication licence and your journal's instructions. Keep the licence with your figure files.

BioRender citation guidance ↗

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Does it allow publication, modification and your intended distribution? Check any share-alike, non-commercial or other conditions.

3 · Give the required credit

Record creator, source link, licence and any modifications. Follow the asset’s exact requirements.

4 · Keep a record

Save the licence or permission with your editable figure. If the terms are unclear, choose a clearly licensed alternative or ask the owner.

Caption template · adapt to the licence

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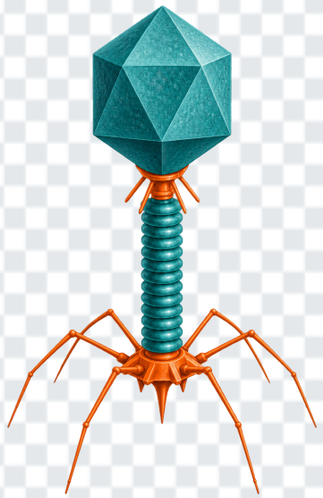
“Found online” is not a source credit. Free download does not automatically mean unrestricted reuse.

Two ways to generate an icon

Try a background: Transparency grid | White | Dark | Teal

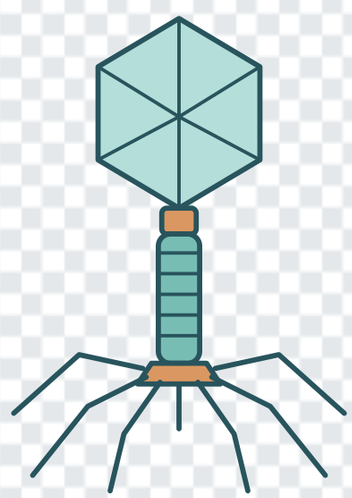
Raster · image generation



PNG with an alpha channel. Transparent pixels let the slide background show through.

Support varies by generation tool. In ChatGPT, explicitly ask: “Generate a PNG with a transparent background (alpha channel).” Check the saved file.

Vector · SVG code



SVG with no background shape. Unpainted areas stay transparent on any background.

SVG supports opacity without a raster pixel alpha channel. A white background rectangle would cover the slide; dark outlines can still lose contrast.

The grid is a preview aid, not part of either file. A generated checkerboard picture is not real transparency. Both examples here have transparent backgrounds.

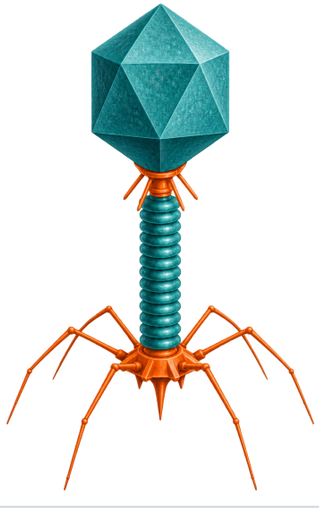
> **Reveal both prompts**

> **Tool capabilities and sources**

Start with a simple request: generate a phage

```
Generate an image of a tailed bacteriophage for a scientific teaching
slide. Show a recognisable faceted capsid, tail sheath, baseplate and
tail fibres. Clean detailed scientific illustration, restrained teal
and muted orange colours, centred with the whole phage visible.
Transparent background. No text, labels or decorative background. A
schematic illustration, not microscopy or a species-specific structural
model.
```

Copy prompt



Phage illustration generated with this prompt.

For an editable vector version, ask for SVG code instead of a raster image.

Build an overview figure · choose your tool

Arrange icons, arrows, labels and plots into a clear workflow or graphical abstract.

Presentation software

PowerPoint · Keynote

Works well for arranging graphics, connecting steps and aligning labels.

BioRender

Web-based · paid publication licence

Scientific icons, templates and an editor. Check your group’s access and publication rights.

biorender.com ↗

Ongoing Science

Free web figure editor

Combine scientific graphics, arrows and labels in your browser.

ongoingscience.com/os/editor ↗

Live demo · Ongoing Science

Let’s build a small workflow: add graphics → connect steps → label → export.

Open the figure editor ↗

Submit the manuscript

Embed previews, upload figure files and prepare for revisions.

Submit the manuscript · keep the editable figures

A practical workflow I use:

1 · Manuscript

PNG **Readable figure previews**

Embed raster versions in Word or Google Docs; check the exported manuscript PDF.

2 · Submission system

PDF / AI **Separate figure files**

Upload the full-quality figures in the manuscript tracking system (MTS), in formats the journal accepts.

3 · Revision archive

SOURCE **Originals + editable versions**

Keep raw images, plotting code and adjusted layouts ready for changes.

This is a working practice, not a universal submission rule. Initial, revised and final submissions can have different requirements.

In the manuscript · make review easy

Readable preview beside its caption

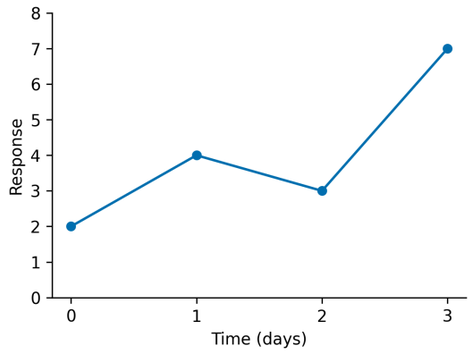


Figure 1. Response over time.

Embed a raster preview

Use a clear PNG at the intended placement size.

Keep figure numbers, captions and in-text references consistent.

Open the resulting Word/PDF export: labels must remain readable.

The preview is a reading copy. Keep the full-quality source separately; do not reconstruct it from a screenshot.

In the MTS · upload the figure files

One file per complete figure

`a + b + c` **Fig1.pdf**

`a + b` **Fig2.pdf**

Use PDF or AI when accepted, retaining editable vector content where appropriate.

Check the journal and submission stage

Some journals accept PDF/AI; others require TIFF, EPS or other formats.

Match filenames to the manuscript and select the correct upload category.

Inspect the submission system’s compiled proof before completing submission.

Examples of journal requirements

Keep everything needed for the next revision

A figure folder you can return to

```
figure-01/
  raw/            untouched images and data
  code/           plotting scripts
  original/       unedited plot exports
  editable/       adjusted AI or SVG master
  submitted-v1/   exact files sent to journal
  revision-v2/    revised exports
  changes.txt     what changed and why
```

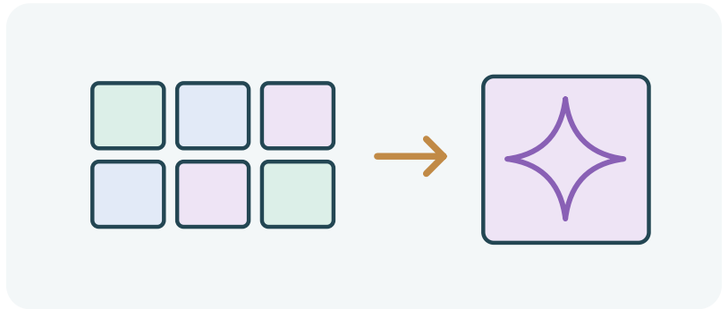
“Please increase the labels…”

Reviewers or editors may request small changes to labels, colours or panel order.

Reopen the script or editable master, make the change, and regenerate the preview and submission file.

Keep raw and adjusted versions. Save revisions separately and back up the folder, including linked assets.

Update both the manuscript preview and the separately uploaded figure so they show the same version.

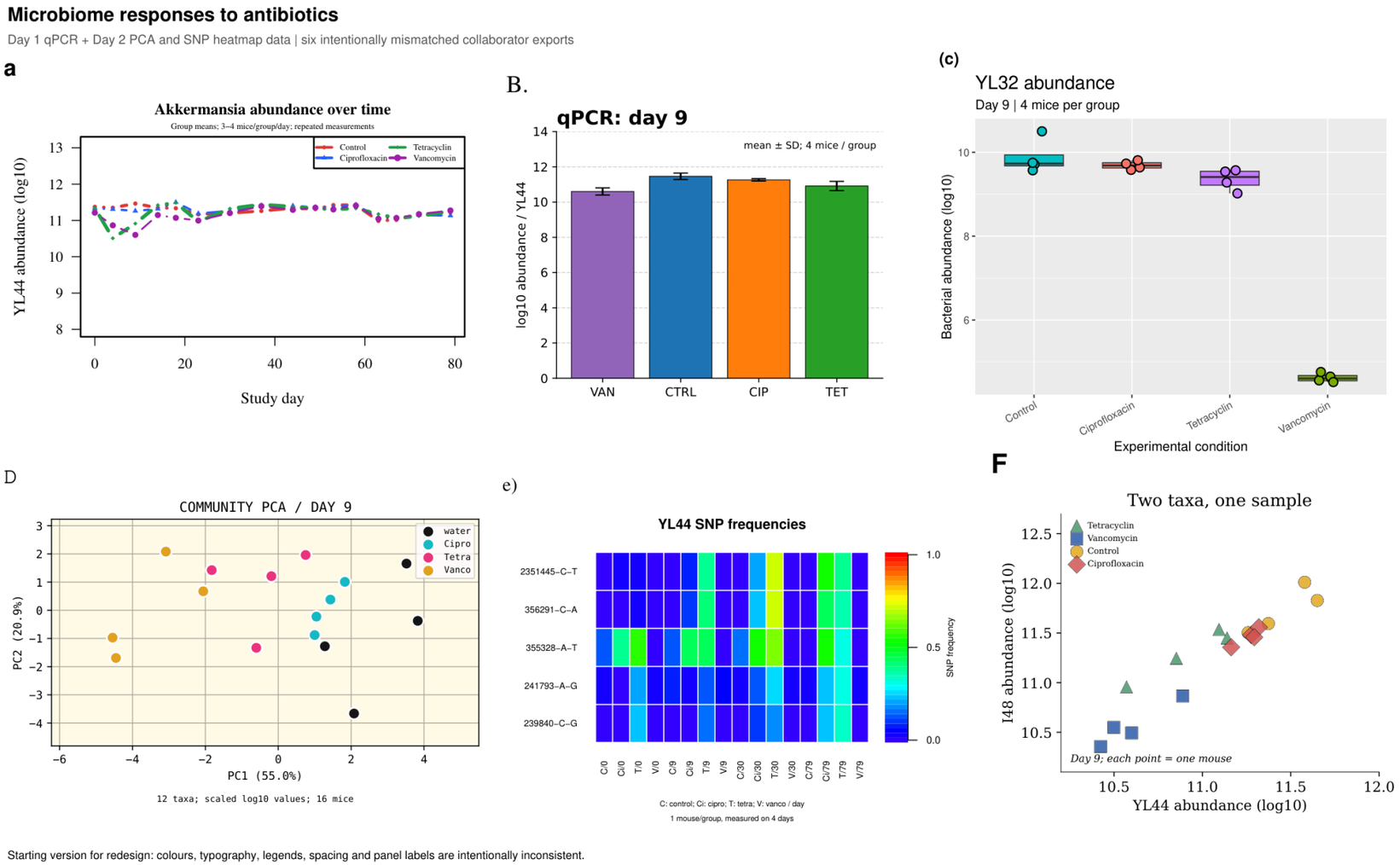


Use generative tools

Redesign six mismatched panels from the Day 1 and Day 2 data.

Use generative tools

One project, six collaborator exports. Can you make them look like one figure?



Starting version for redesign: colours, typography, legends, spacing and panel labels are intentionally inconsistent.

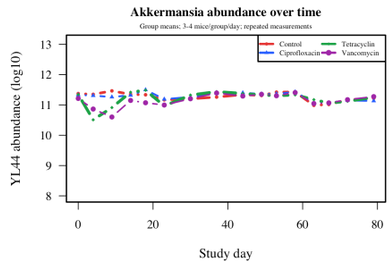
Starting version · existing Day 1 & Day 2 data · intentionally inconsistent design

Open the generated design guide in Claude ↗ Download the full exercise (ZIP) Open figure PDF ↗ Read the instructions ↗

What should we improve?

Every panel comes with its data and code

Open a panel and inspect its data and code. Use these files to inform the design guide before changing the plots.

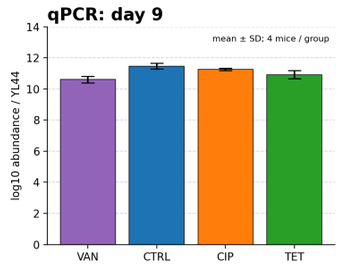


a · qPCR over time

R · base graphics

Day 1 · YL44 group means across study days.

Data CSV Code PDF ↗ README ↗

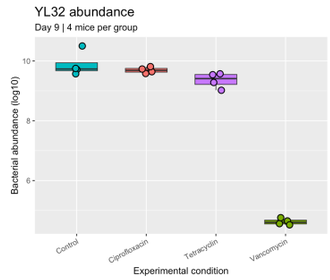


b · qPCR at day 9

Python · Matplotlib

Day 1 · YL44 mean ± SD, four mice per group.

Data CSV Code PDF ↗ README ↗

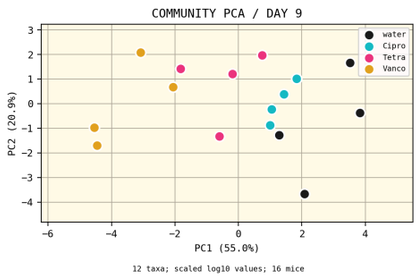


c · qPCR distributions

R · ggplot2

Day 1 · YL32 boxes and individual mice.

Data CSV Code PDF ↗ README ↗

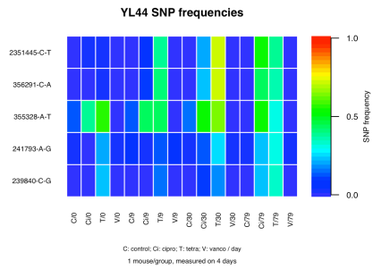


d · Community PCA

Python · NumPy

Day 2 · 12 taxa, scaled log10 abundances.

Data CSV Code PDF ↗ README ↗

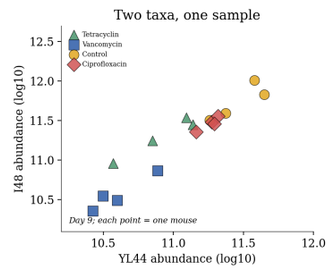


e · SNP-frequency heatmap

R · base graphics

Day 2 · Five SNPs, four mice, four time points.

Data CSV Code PDF ↗ README ↗



f · Two taxa, one sample

Python · Matplotlib

Day 1 · Matched YL44 and I48 abundances.

Data CSV Code PDF ↗ README ↗

› **Recreate the whole figure**

Create a design-guide artifact first

Give the tool the complete folder. Ask for a visual guide to the intended style, using schematics only. Review it before asking the tool to replot the data.

A starting prompt

Help plan the redesign of the six-panel figure in example_plot for a scientific manuscript.

First read the README, datasets and plotting scripts to understand the panels. Your first deliverable is a separate visual design-guide artifact showing how the finished figure should look.

Include treatment names and colours (with hex codes), marker shapes, line styles and widths, fonts and text sizes, axes and grid styling, uncertainty styling, a heatmap colour scale, legends, panel letters and spacing. Plan two columns and three rows: a/b at the top, c/d in the middle, e/f at the bottom. Design for a final width of 180 mm and specify sizes at that final scale.

Use schematics only: colour swatches, marker and line samples, example labels, a schematic legend and empty panel frames. Label these as design examples. Do not replot the datasets, recreate the real panels, invent data patterns or edit the plotting scripts at this stage.

Explain the choices and which axes could share limits versus which measure different quantities. Preserve the meaning of groups, shapes, transformations and uncertainty.

Provide the guide as a downloadable, standalone design_guide.html with inline CSS and SVG schematics, no external assets or dependencies. It must open offline and contain the written design specifications as well as the visual examples. Deliver this artifact first and stop for review. Apply the agreed design to the actual plots only in a later step.

Copy prompt

Review and download the guide

1. **Look for a visual artifact.** Colour swatches, marker shapes, line samples, typography, a schematic legend and empty panel frames.
2. **Agree on the intended style.** Can you follow each treatment through the proposed colours and shapes? Are axes, text and spacing consistent?
3. **Keep this step schematic.** No reconstructed data plots or invented results. Discuss the guide before changing code.
4. **Save the handoff.** Download the reviewed artifact as `design_guide.html`. Open it offline to check that its schematics and written specifications are included.

Next, inspect the artifact, comment on it and request improvements. Our example adds editable SVG schematics before the guide goes to another agent.

Download data, code, figures and READMEs

Let’s look at the generated design guide

Our generated artifact · Claude

Open the design guide ↗

Open it in a new tab and inspect the proposed colours, shapes, typography and panel layout.

Discuss: What would you keep? What would you change?

Next: turn your observations into comments and ask the agent to improve the artifact.

Query 2 · Inspect, comment and improve

Inspect the first artifact and give concrete feedback. Our example improvement: add useful, editable SVG schematics to the design guide.

Open the design guide to inspect and comment ↗

Prompt to revise the artifact

I have inspected the first design-guide artifact. Revise the existing design_guide.html using my comments before we generate any plots.

My review comments: [replace with your specific comments on colours, shapes, label sizes, spacing or layout, or write "none"]. Apply these comments and briefly explain what changed in response to each one. Preserve design choices I have not asked to change.

Our example improvement is to add useful, editable SVG schematics to the guide. Read the panel READMEs, data descriptions and plotting scripts to understand the biological context before proposing graphics.

Decide which panels, or which shared part of the figure, would benefit from a small contextual graphic. You choose the subjects, number and placement; do not add an icon to every panel by default. Prefer graphics that explain a biological entity, sampling scheme, measurement or connection between panels. Explain briefly what each proposed graphic helps the reader understand.

Create original, editable SVG assets that fit the guide's palette, stroke widths and visual style. Use transparent backgrounds, simple vector shapes and paths, and no external dependencies or embedded raster images. Keep labels as editable text where practical. Check that the schematics are biologically plausible and do not imply an experimental step, causal relationship or result unsupported by the supplied materials. Label them as schematics.

Show two layout options in the HTML guide using schematic panel frames and the proposed SVG graphics. Compare them at the intended 180 mm figure width, choose the clearer option and refine it. Check legibility, consistency, contrast, whitespace and whether the graphics compete with data or legends. Remove anything that adds clutter rather than context.

Deliver a revised standalone design_guide.html with the SVG previews embedded, plus separate named files in graphics/. Include the chosen placement, dimensions in mm, relevant panel or shared location, and short integration notes for the plotting or assembly agent. Keep an untouched copy of the previous guide and summarise the changes. Do not modify datasets, analyses or plotting scripts at this stage. The deliverable is the refined guide and reusable SVG assets for review, ready to add to the real figure later.

Copy prompt

Review before moving on

1. **Inspect the artifact.** Look at colours, marker shapes, typography, line widths and the proposed panel arrangement.
2. **Make specific comments.** For example: “These two colours are hard to distinguish” or “Make the legend labels larger.” Add your feedback to the prompt.
3. **Request an improvement.** In our example, ask for SVG schematics that add biological context. Let the agent suggest where they help and show placement options.
4. **Inspect the revision.** Check the updated guide and editable SVG assets. Keep useful changes, comment again where needed, and agree on the design before plotting.

Download the revised `design_guide.html` and the separate `graphics/` SVG files. Hand these to the next agent with the exercise folder.

The artifact remains schematic at this stage; no data panels are replotted yet.

Query 3 · Give another agent the guide

Attach the reviewed `design_guide.html`, the `graphics/` SVG files and the complete `example_plot` folder to a new agent. Ask it to produce the six panels and their code.

Prompt for the plotting agent

```
You are implementing an agreed figure design. I have attached design_guide.html and the example_plot folder with all six datasets, original plotting scripts and READMEs.

If the reviewed guide includes contextual graphics, also use the supplied graphics/ SVG assets and placement notes. Add them only at the agreed locations, preserve their proportions, and keep them clear of data marks and labels.

Read the standalone HTML guide, including its written specifications and schematic examples. Treat it as the design specification, not as measured data. Inspect the CSVs and original scripts before making changes.

Create a new set of reproducible plotting scripts for panels a-f that follows the guide: treatment colours and names, marker shapes, line styles, fonts, final-size text, axes, grids, legends, heatmap scale and margins. Retain each panel's original R or Python language. Define reusable style settings in each language so that equivalent settings stay consistent across all panels.

Read the original CSVs unchanged. Preserve every mouse and time point, repeated-measure identities, supplied log10 values, mean ± sample SD, PCA centering and scaling, and the heatmap's original 0-1 frequencies. Do not add statistical tests, observations or significance claims. Do not force the same numerical axes on different quantities. Flag any conflict between the guide and these requirements instead of silently changing the analysis.

Save the new code and exports under optimized/panel_a through optimized/panel_f, leaving the originals intact. Run the scripts and export each panel as a vector PDF, SVG and PNG preview. Size panels for the guide's two-column layout at 180 mm total width; include space for labels and legends. Keep global panel letters for the assembly step.

Provide optimized/render_panels.py (or an equivalent single command), dependency instructions, a short README and a record of checks showing that the plotted values and summaries still match the originals. Inspect every export for clipping and legibility. Deliver the six separate panels and their code for review; do not assemble the final figure yet.
```

Copy prompt

From guide to reproducible panels

1. **Provide both inputs.** The guide specifies the appearance; the CSVs and original scripts specify the measurements and calculations.
2. **Generate new code.** Keep the starting files and save the revised R/Python scripts in `optimized/`.
3. **Review six separate outputs.** Check each PDF/SVG against the guide and compare values with the original panel.
4. **Keep the rebuild command.** The next agent should be able to regenerate a panel if the final layout needs different margins.

The deliverable is six consistent panels with editable code. Assembly is the next query.

Query 4 · Assemble the final figure

Provide the reviewed panels, their code and the same design guide. Give the agent an explicit reading order and layout.

Prompt for figure assembly

Assemble the six reviewed panels into one manuscript figure. I have supplied design_guide.html and optimized/ containing panels a-f, their source code and vector exports.

If the reviewed guide includes graphics/ SVG assets, include them at the agreed locations and sizes. Keep them as vectors, separate from measured evidence and clear of labels and legends.

Use the guide's colours, typography, spacing and final-size specifications. Arrange two columns and three rows, read from left to right:
Top row: a - YL44 qPCR time course; b - day 9 YL44 mean ± SD.
Middle row: c - day 9 YL32 distribution; d - community PCA.
Bottom row: e - YL44 SNP-frequency heatmap; f - YL44 versus I48 scatterplot.

Use 180 mm total width, equal column widths, consistent outer margins and approximately 4 mm gutters. Start around 180 mm high and increase the height if labels need room. Align plotting areas within rows and columns, not just file boundaries. Place bold lowercase panel letters a-f consistently at the upper left of each panel. Keep labels at the guide's final printed size; do not shrink text simply to force the layout to fit.

Use a shared treatment legend below the panel grid only where colours and shapes mean the same thing; retain any panel-specific keys and the heatmap's own frequency colourbar. Keep all axis labels, tick labels and units visible. Preserve aspect ratios and all data and analysis choices. If panel margins or label sizes need adjustment, update the plotting code and regenerate the affected panel instead of stretching or painting over it.

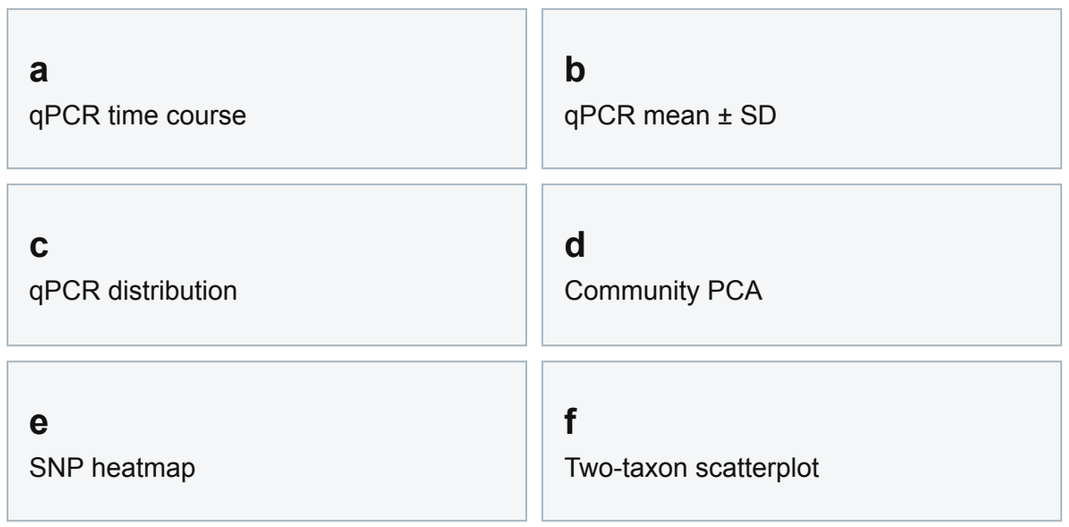
Assemble from the vector PDF or SVG exports so text and lines remain vector content. Save reproducible assembly code, final/figure_combined.pdf and a 300 dpi final/figure_combined.png. Include a separate caption explaining each panel, group sizes, repeated measurements, mean ± SD, PCA preprocessing and the heatmap subset. State that the heatmap contains one mouse per group measured on four days; do not imply that it has the same replication as the day 9 abundance panels.

Inspect the complete figure at its intended printed size. Check panel order, alignment, clipping, legend consistency, readability and that every panel matches its reviewed source. Provide the finished figure, caption and one command to rebuild it.

Copy prompt

Two columns · three rows

Schematic layout · 180 mm final width



Keep equal column widths, consistent gutters and aligned plotting areas. Allow enough height for readable labels.

Use a shared treatment legend where the encodings match; keep the heatmap colourbar with panel e.

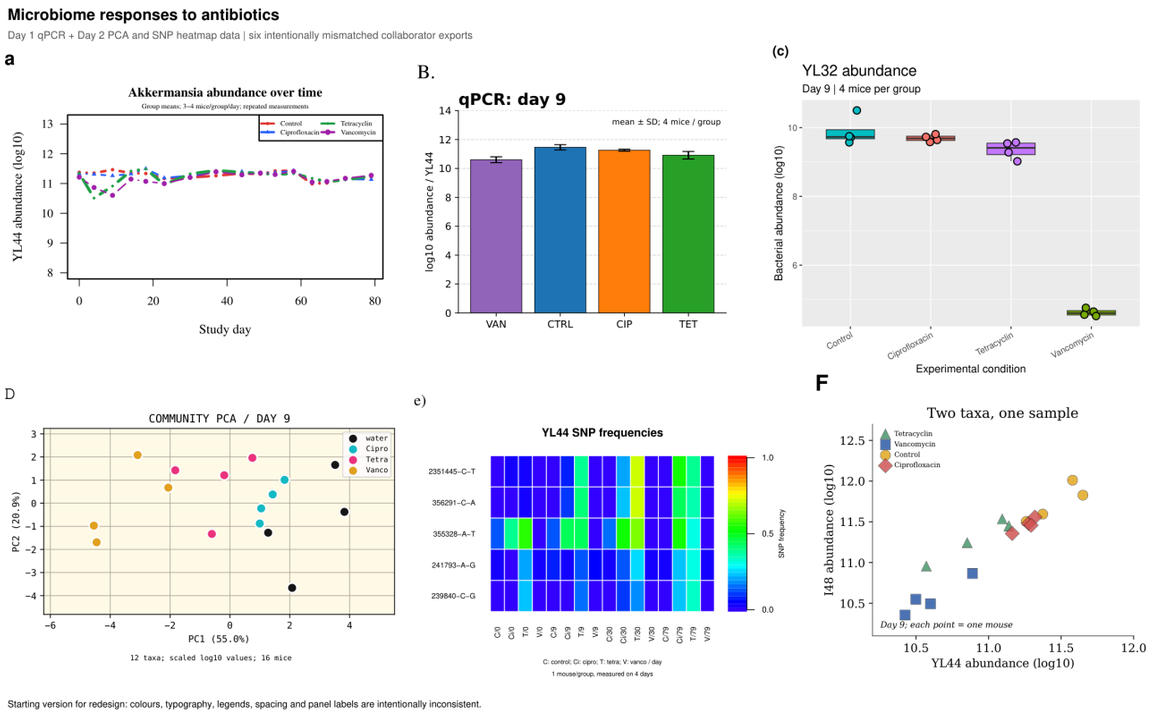
Final deliverables: vector PDF, 300 dpi PNG, separate caption and reproducible assembly code.

Before & after · From draft to consistent figure

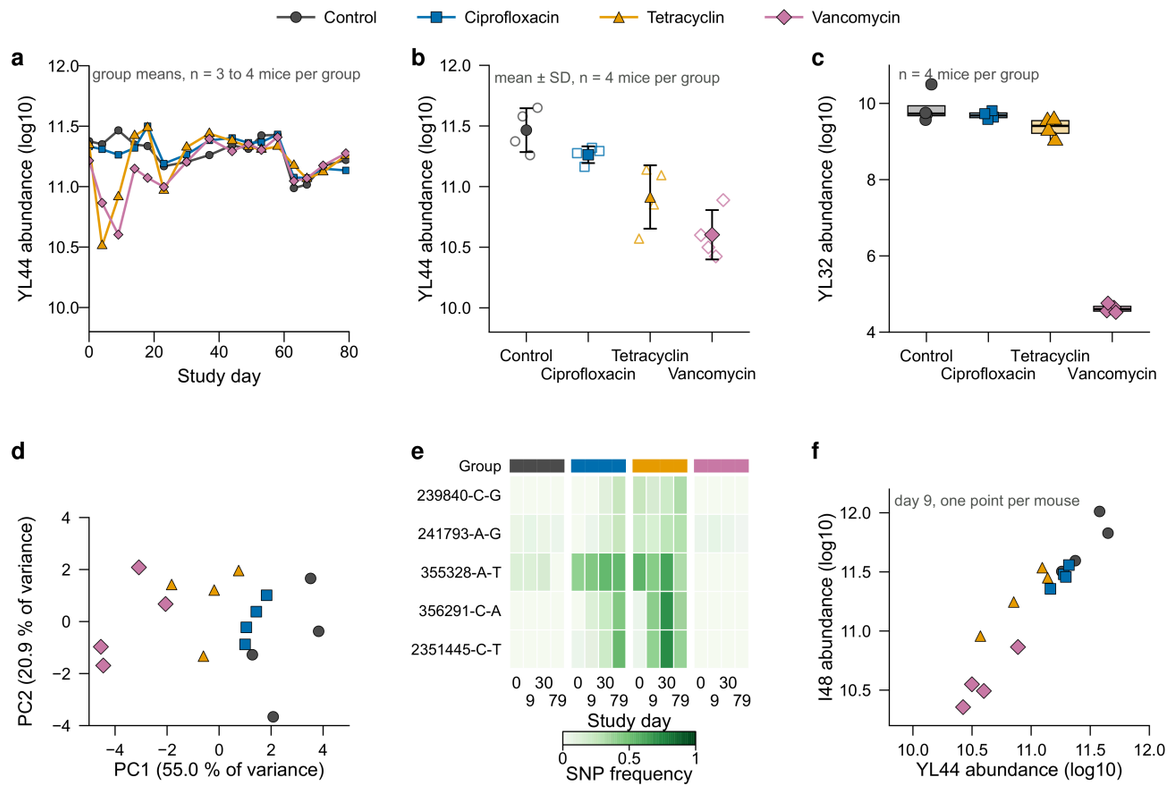
Compare the starting figure with our optimized version. Follow one treatment across all six panels.

Side by side | Before | After

Before · mixed styles



After · optimized version



Discuss: What changed in the colours, shapes, legends, axis ranges and heatmap ordering? What would you still improve?

Open before PDF ↗ Open after image ↗ Download optimized PNG

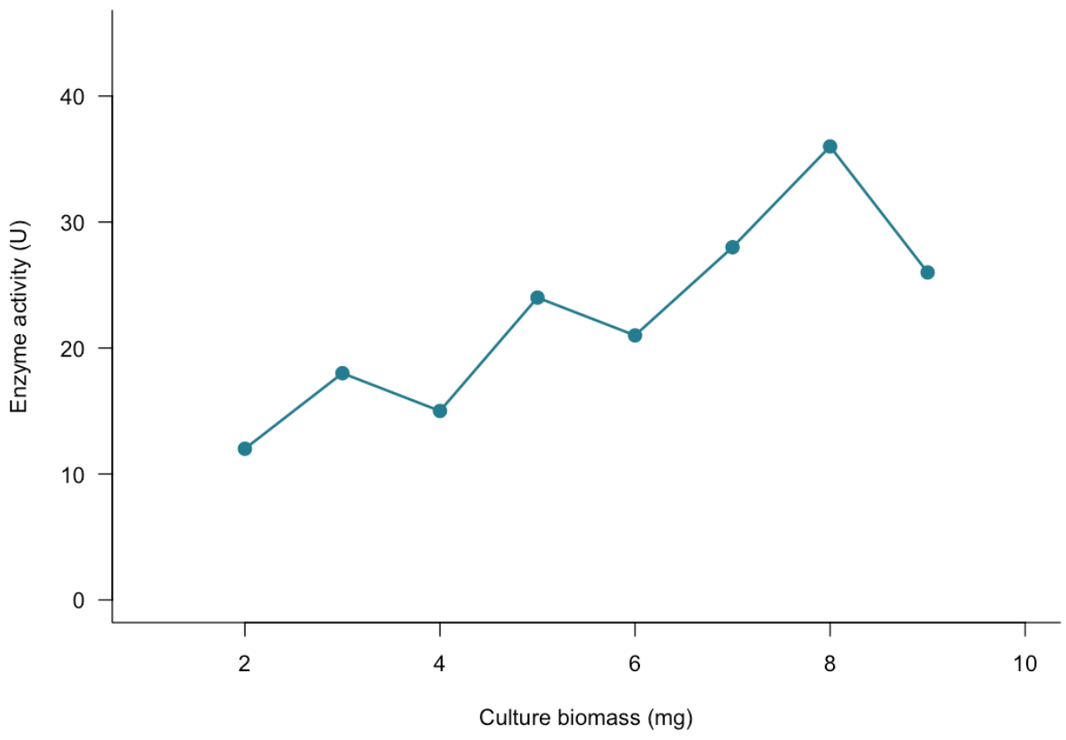
Scatter plot

Explore the relationship between two numeric measurements.

Few points: should we connect them?

Eight independent cultures: biomass versus enzyme activity.

Starting view



Alternative design

Discuss the first figure, then reveal the alternative.

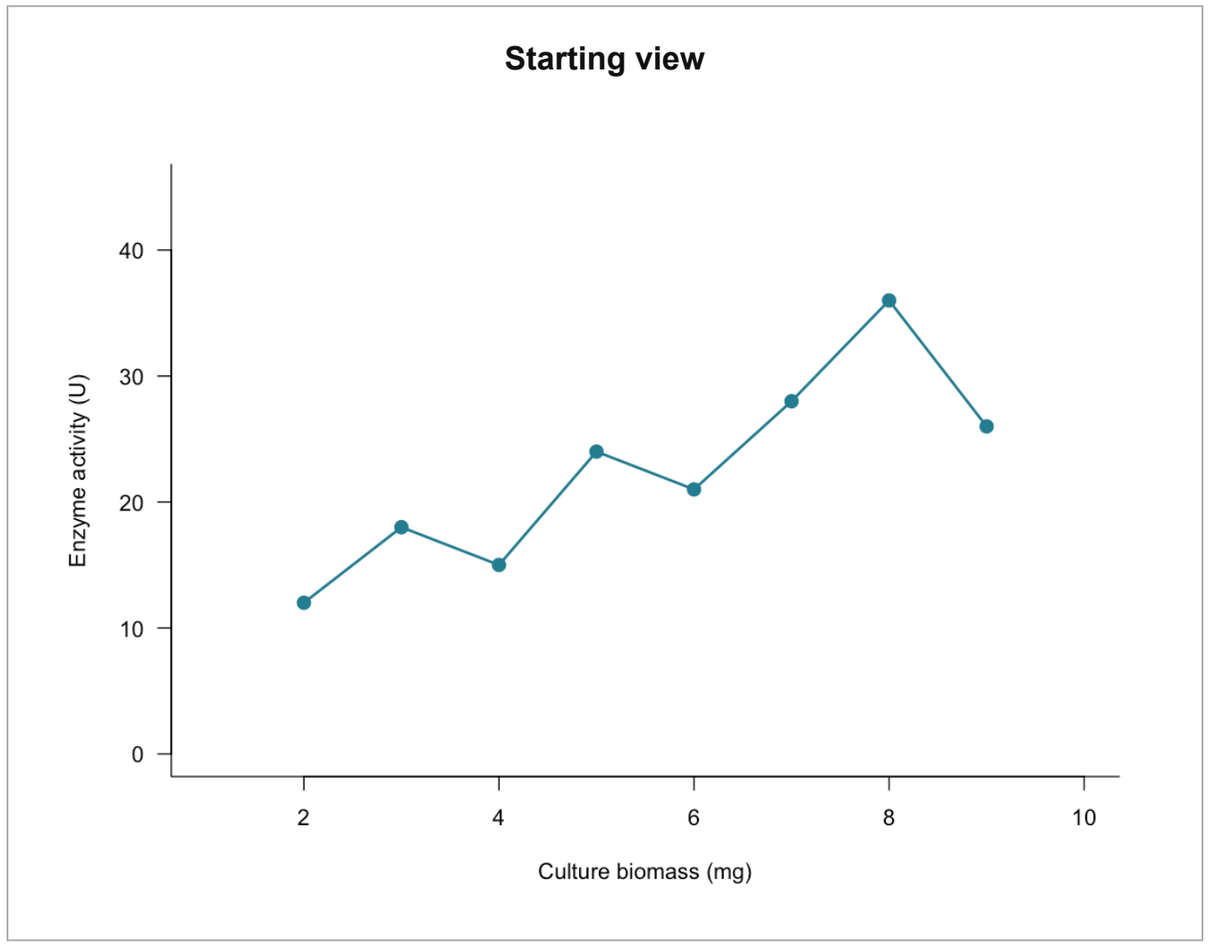
Advice, examples & further exploration

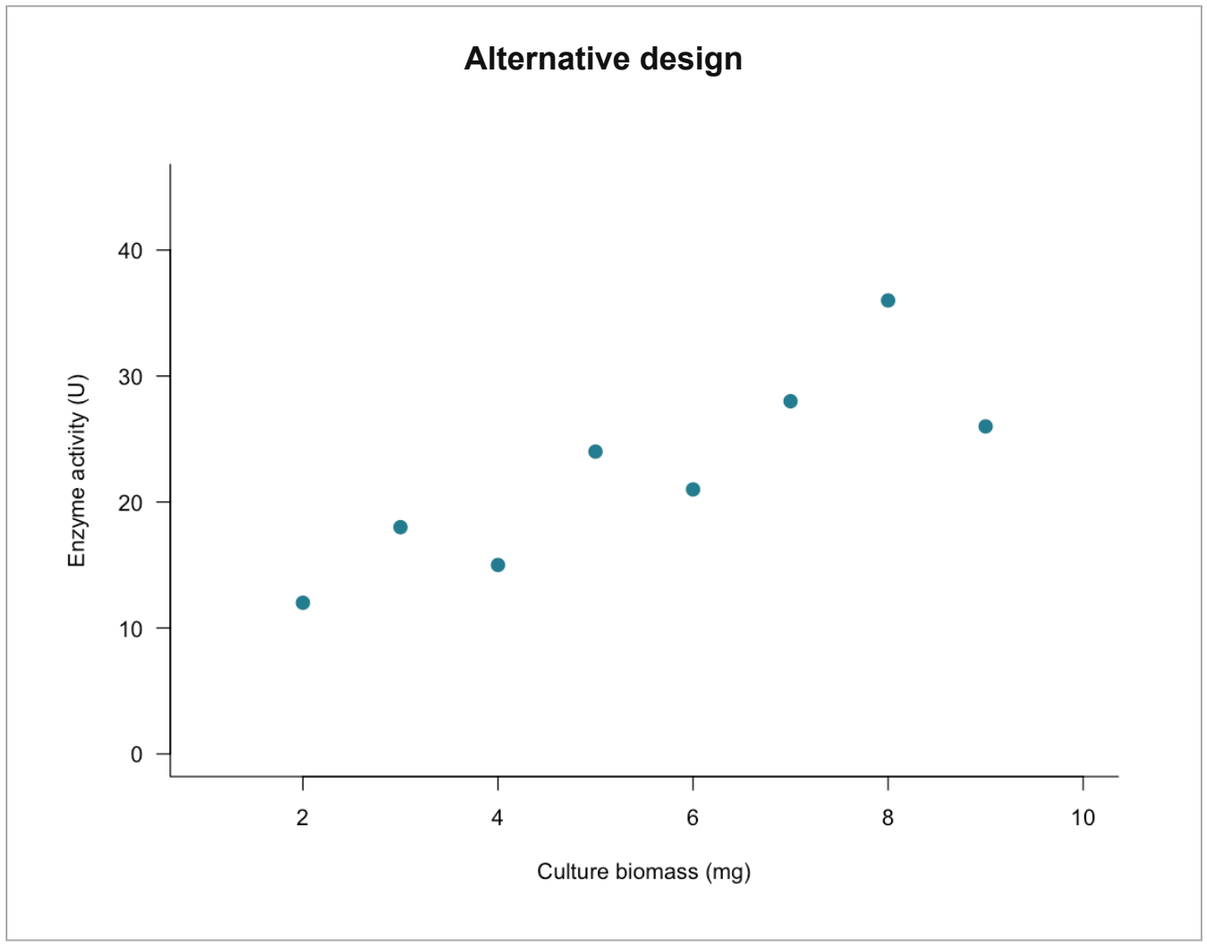
Scatter plot

Explore the relationship between two numeric measurements.

Few points: should we connect them?

Eight independent cultures: biomass versus enzyme activity.





The connecting line suggests a sequence or trajectory between independent cultures. Separate points show the association without implying that one culture turns into the next.

> **Advice, examples & further exploration**

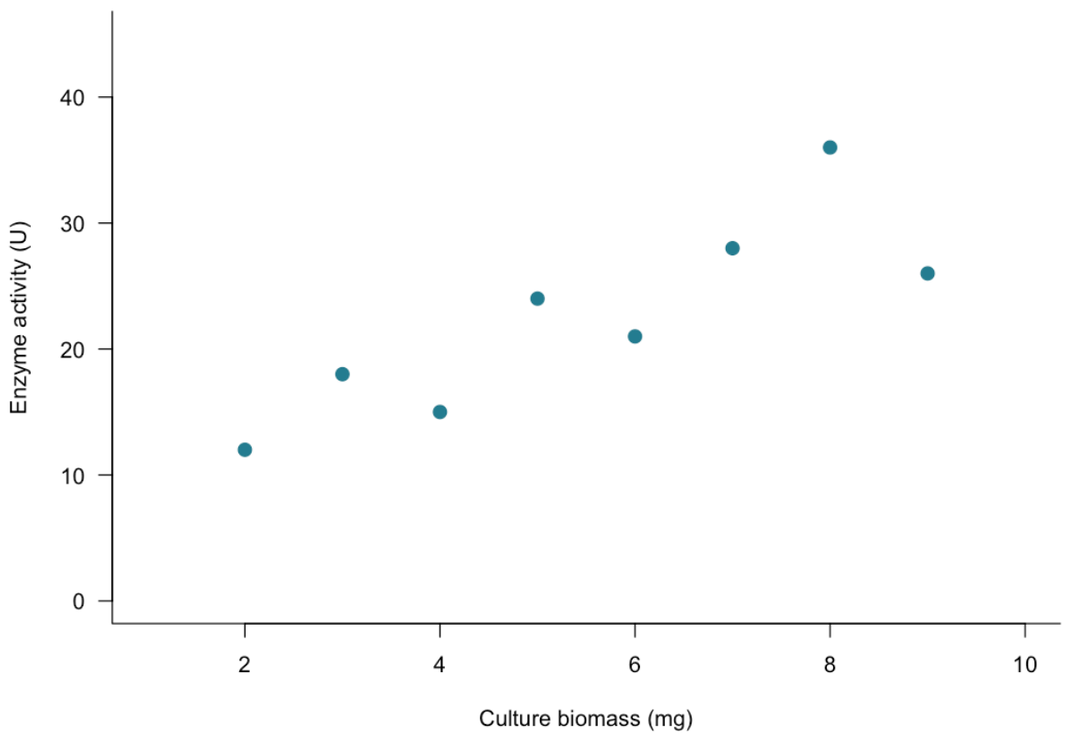
Scatter plot

Explore the relationship between two numeric measurements.

Add a useful callout

The same eight cultures.

Starting view



Alternative design

Discuss the first figure, then reveal the alternative.

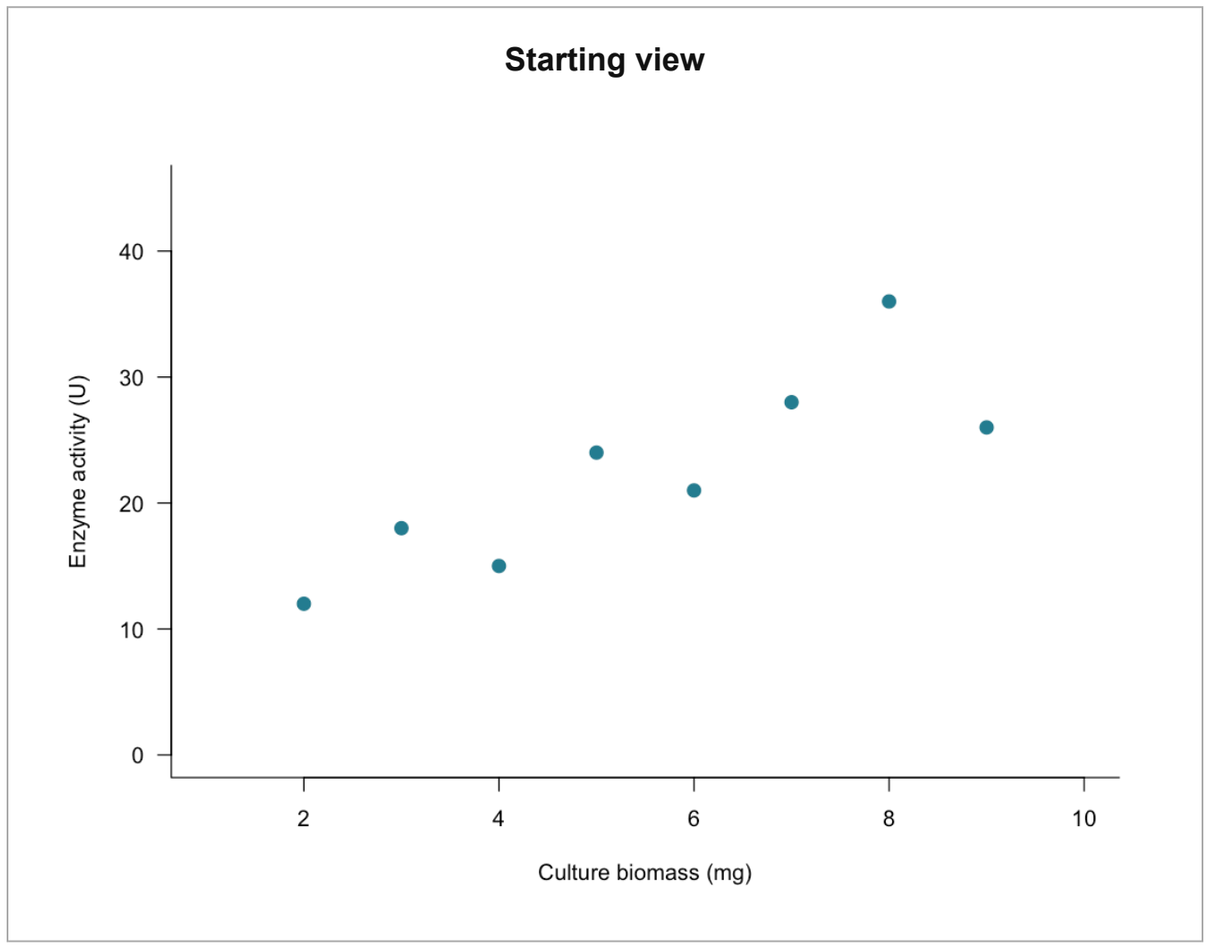
Advice, examples & further exploration

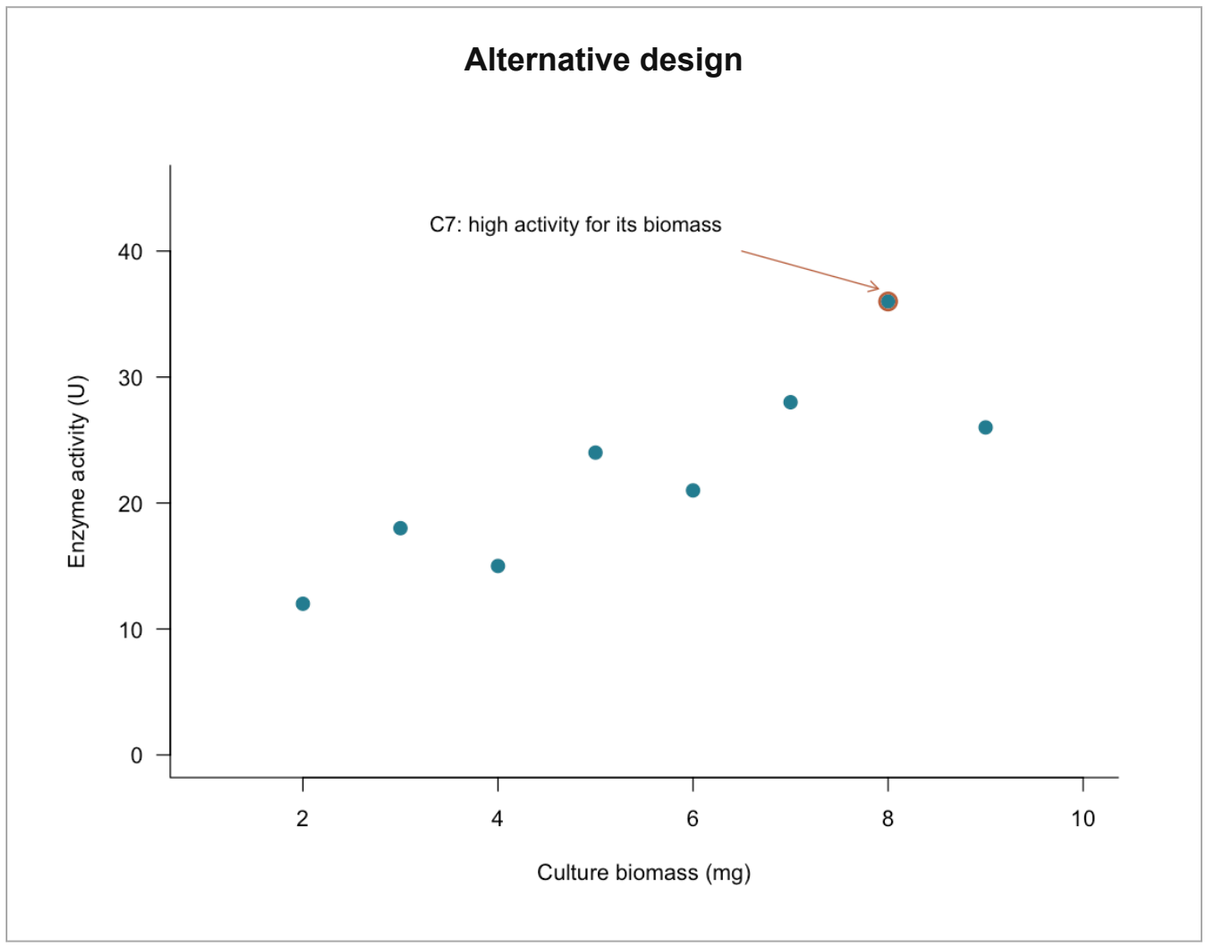
Scatter plot

Explore the relationship between two numeric measurements.

Add a useful callout

The same eight cultures.





A callout directs attention to C7 without changing or removing the other measurements. It is a descriptive observation, not a statistical outlier test.

> **Advice, examples & further exploration**

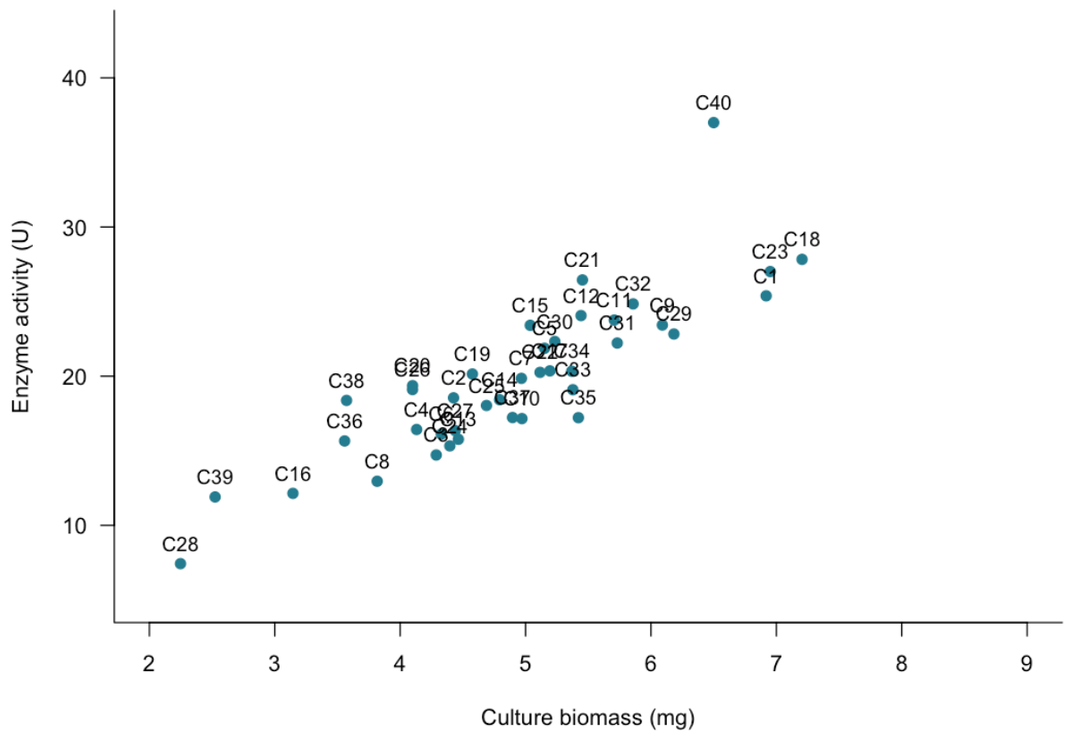
Scatter plot

Explore the relationship between two numeric measurements.

Which labels help?

Forty independent cultures: biomass versus enzyme activity.

Starting view



Alternative design

Discuss the first figure, then reveal the alternative.

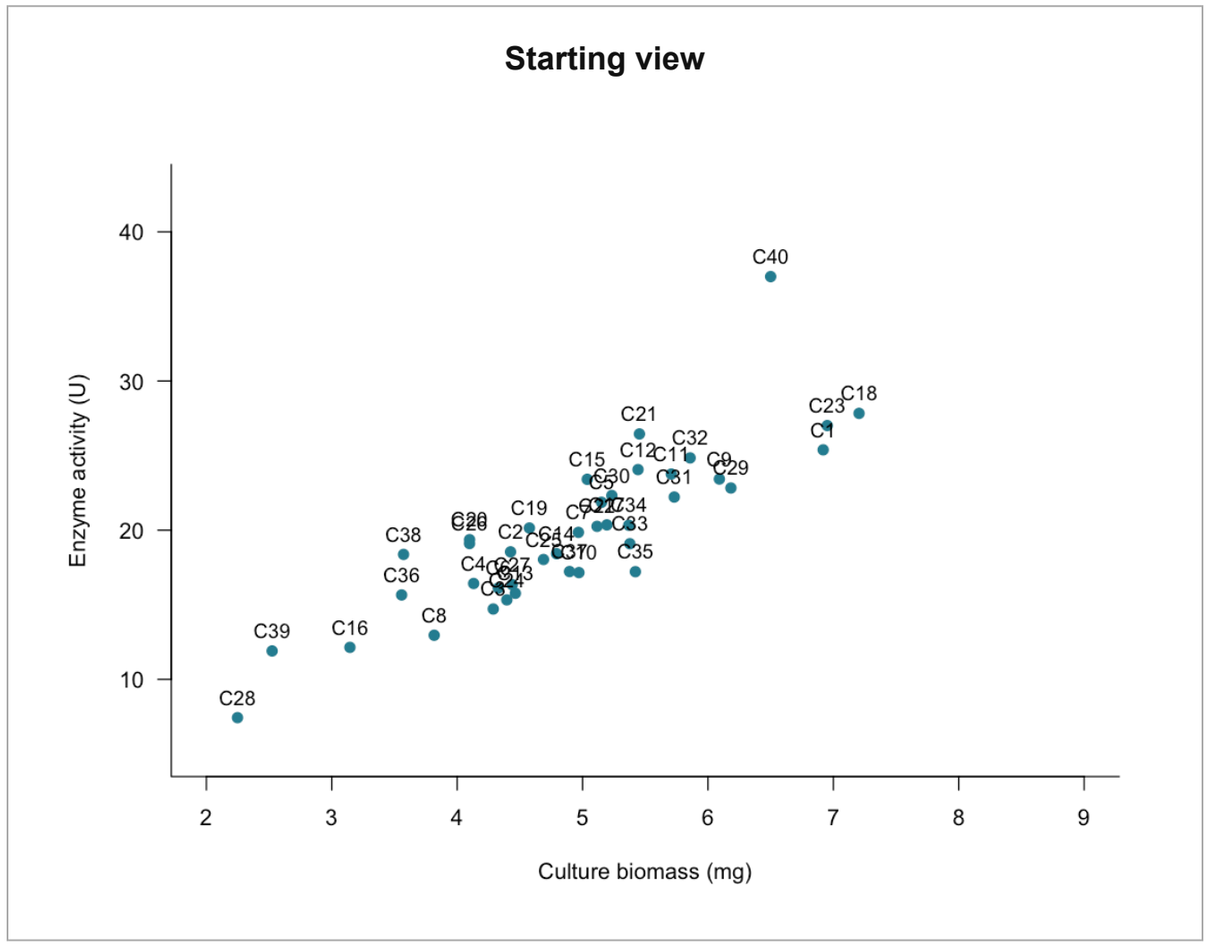
Advice, examples & further exploration

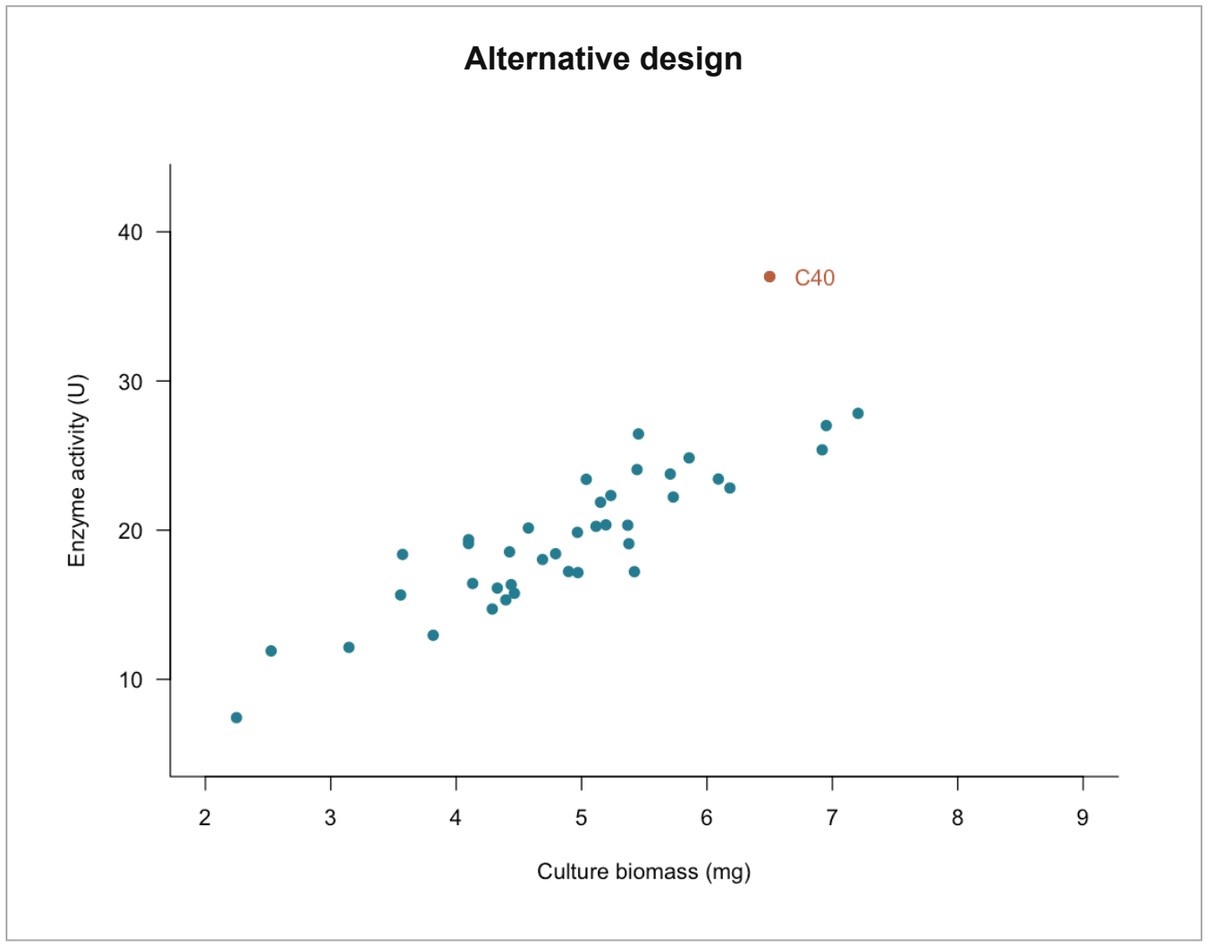
Scatter plot

Explore the relationship between two numeric measurements.

Which labels help?

Forty independent cultures: biomass versus enzyme activity.





Every label competes with the points. Naming just C40 makes the observation under discussion easy to find. Keep all identifiers in the data table; select labels for an explicit reason.

Advice, examples & further exploration

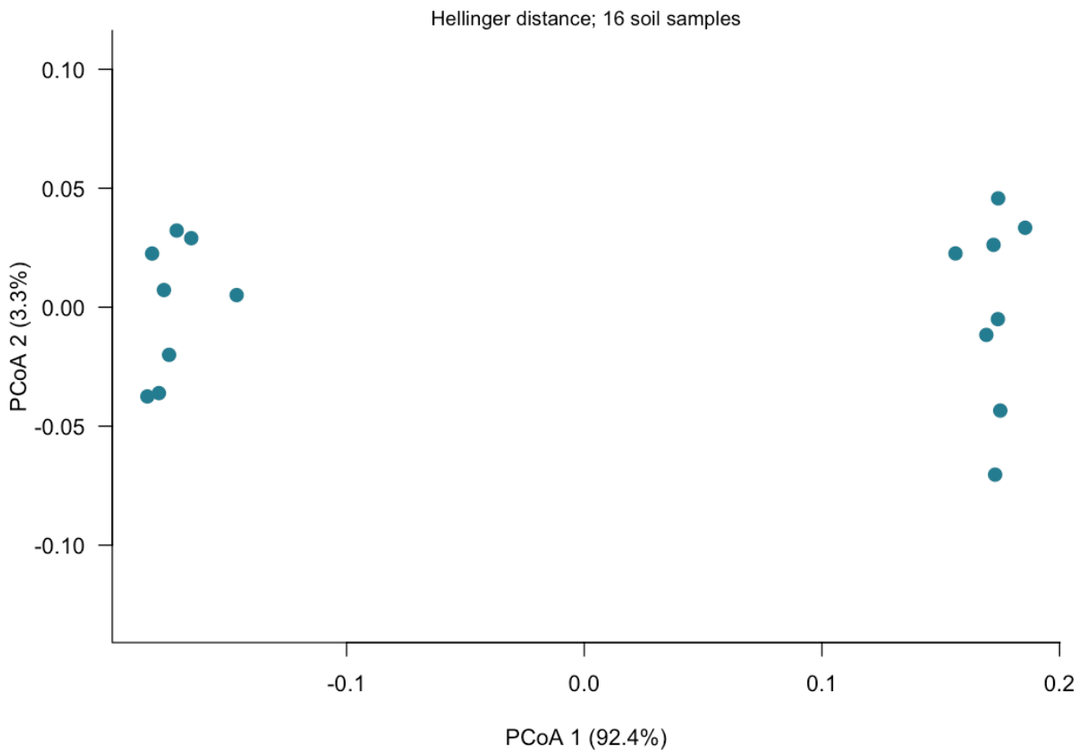
Scatter plot

Explore the relationship between two numeric measurements.

PCoA: add relevant sample information

Six taxon counts in each of 16 soil samples, converted to proportions. PCoA uses Hellinger distances between community profiles.

Starting view



Alternative design

Discuss the first figure, then reveal the alternative.

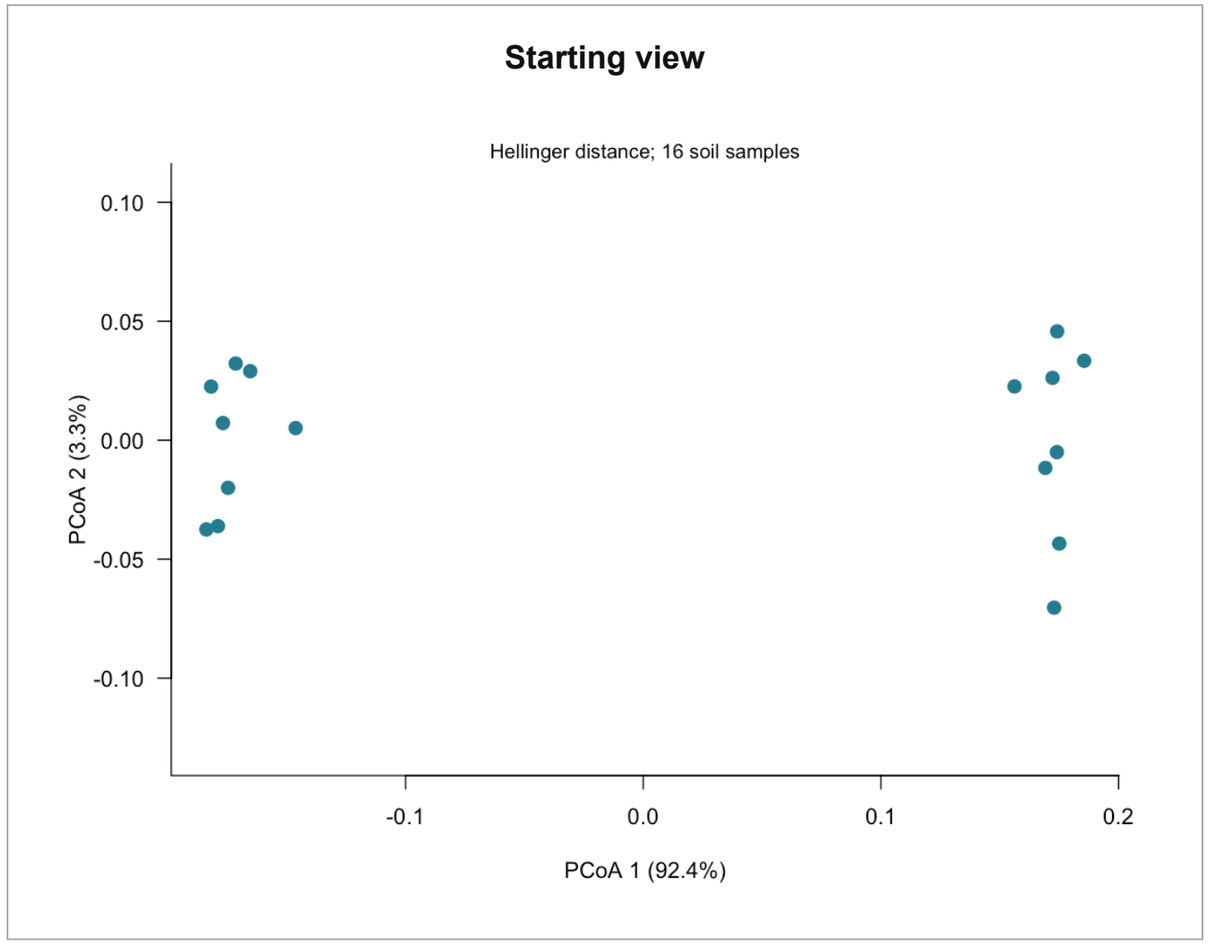
Advice, examples & further exploration

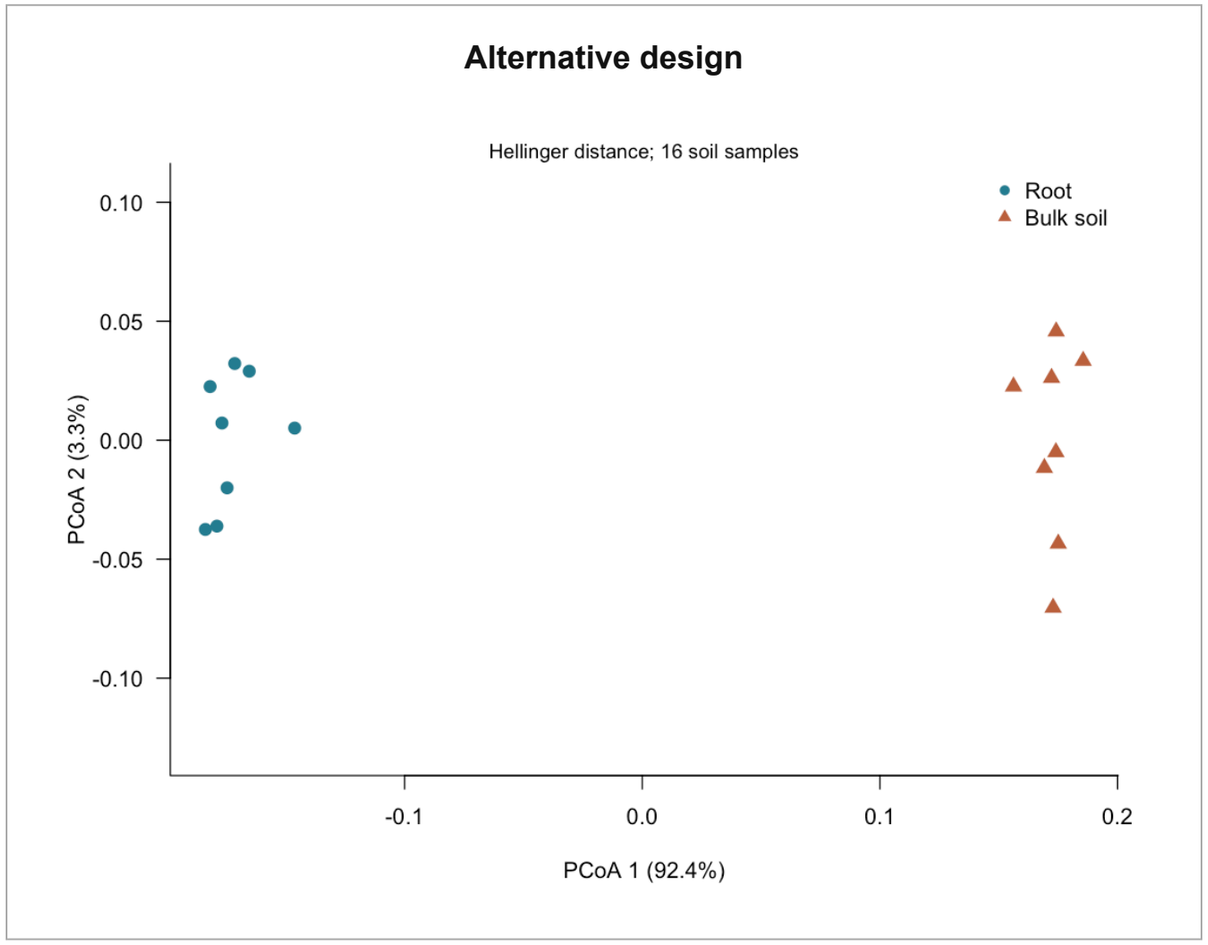
Scatter plot

Explore the relationship between two numeric measurements.

PCoA: add relevant sample information

Six taxon counts in each of 16 soil samples, converted to proportions. PCoA uses Hellinger distances between community profiles.





Colour and shape identify root and bulk-soil samples. Nearby points have more similar profiles in this two-axis approximation; these axes are not individual taxa. Percentages show distance-space variation represented. Separation alone does not establish significance or a causal habitat effect.

> **Advice, examples & further exploration**

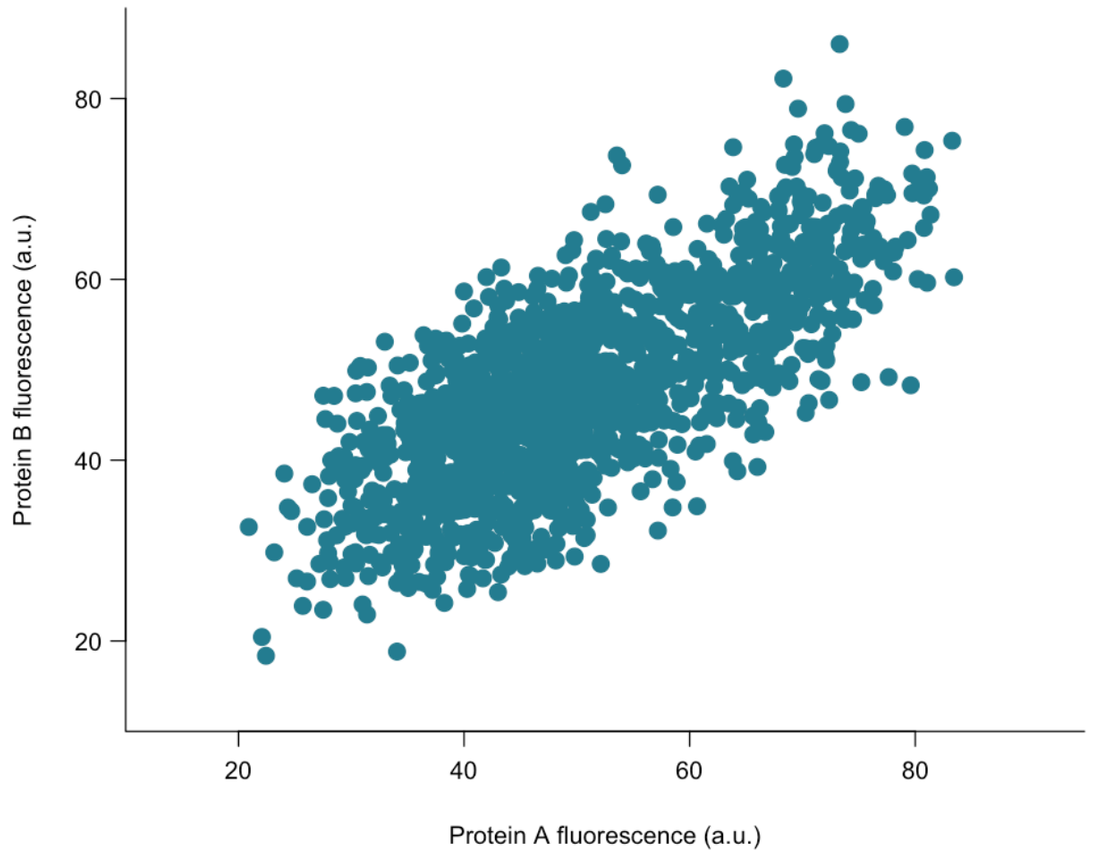
Scatter plot

Explore the relationship between two numeric measurements.

When points overlap

Protein A versus protein B in the same 1,600 cells.

Starting view



Alternative design

Discuss the first figure, then reveal the alternative.

Advice, examples & further exploration

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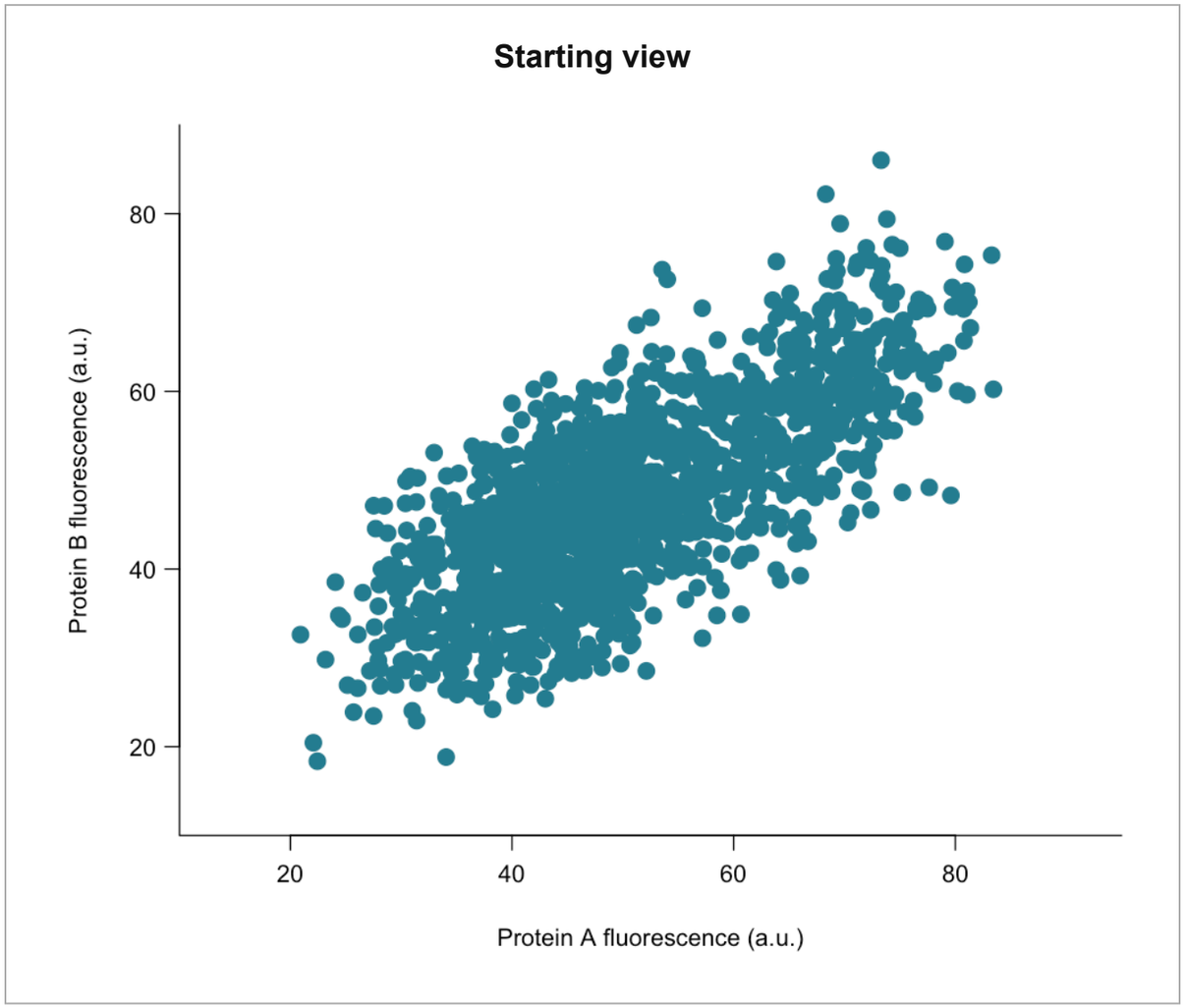
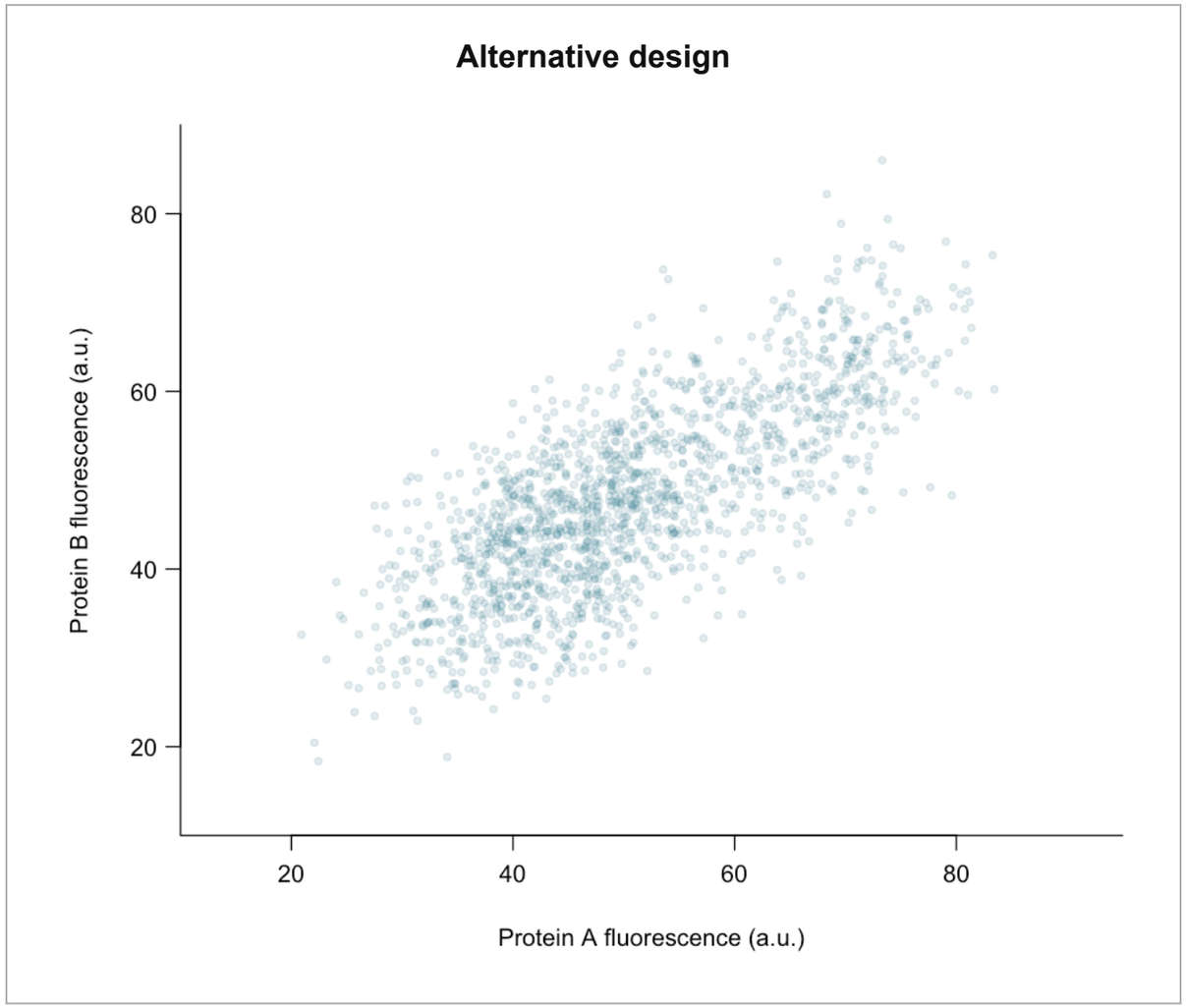




Large opaque points hide density. Smaller transparent points reveal crowded regions; opacity still saturates in very dense areas.

Advice, examples & further exploration

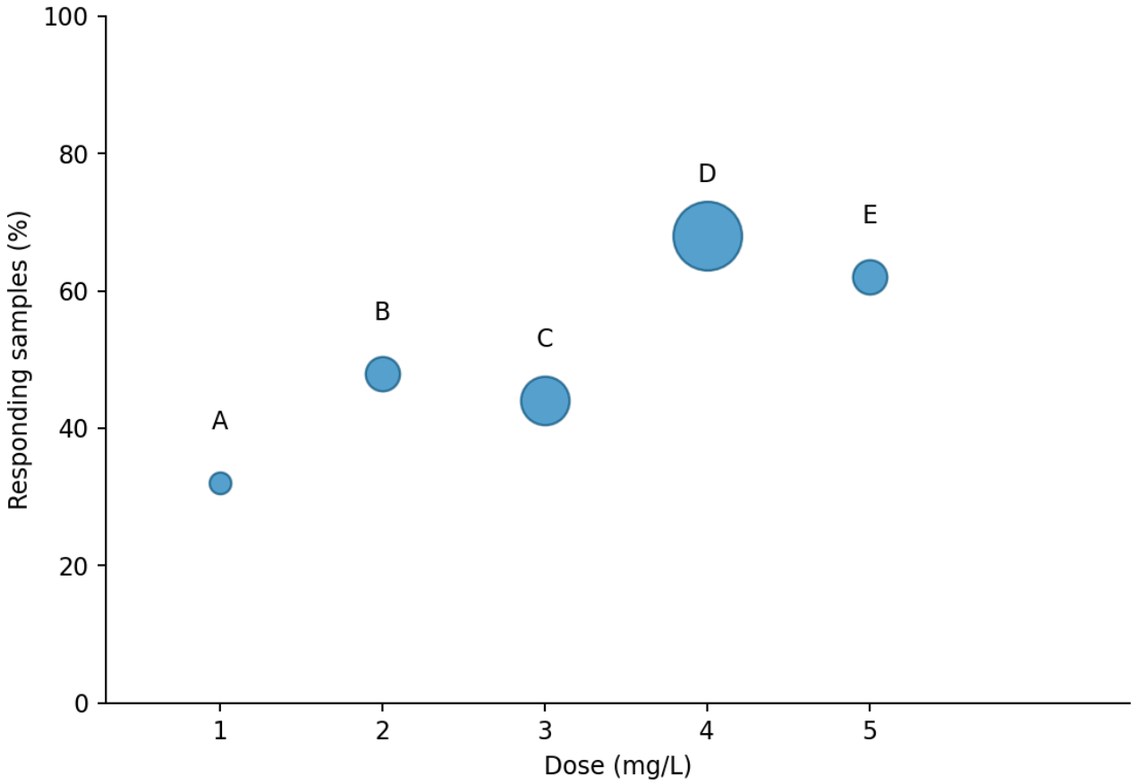
Bubble plot

Compare dose and response, with circle area showing how many samples were measured.

Explain sample size with a size key

Each circle is one simulated study. Position shows dose and response; circle area represents the number of independent samples.

Starting view



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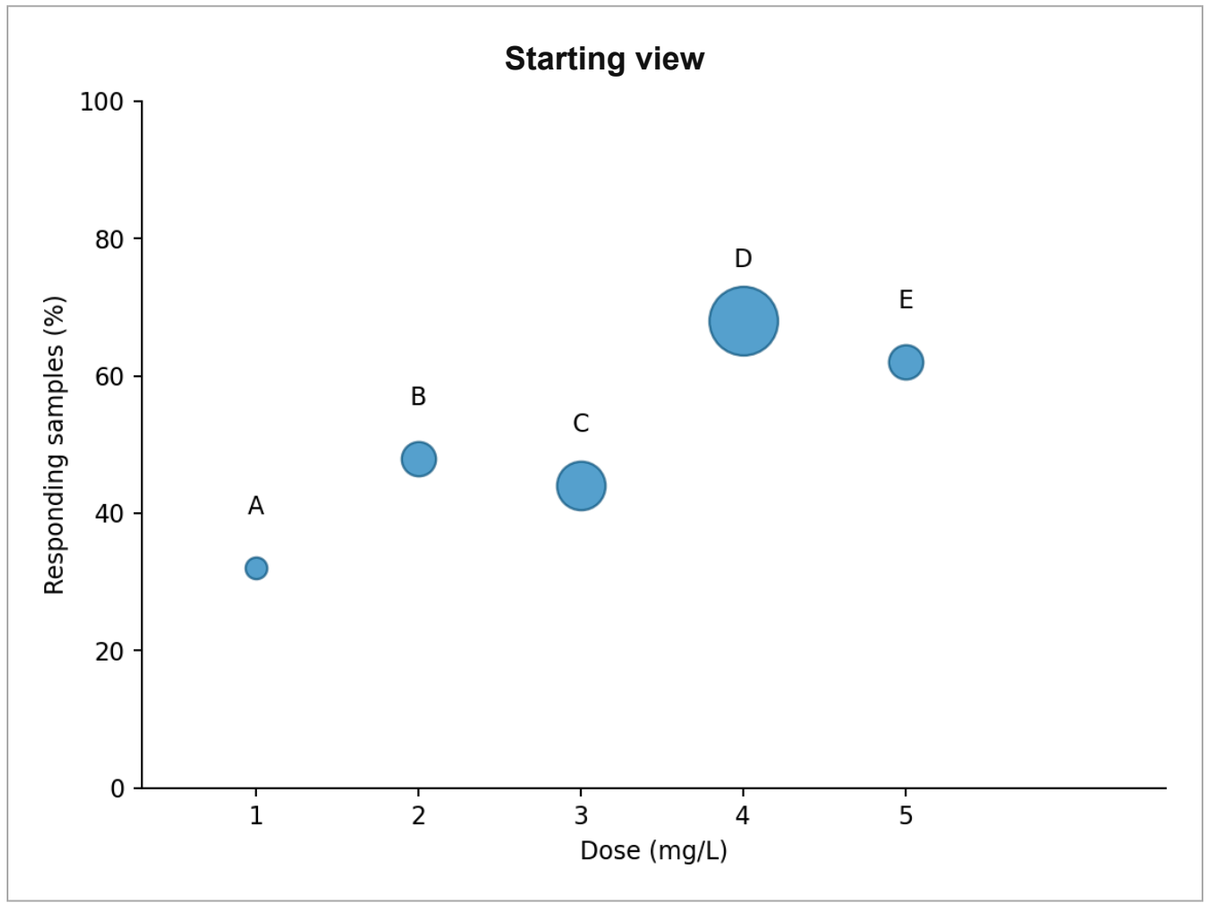
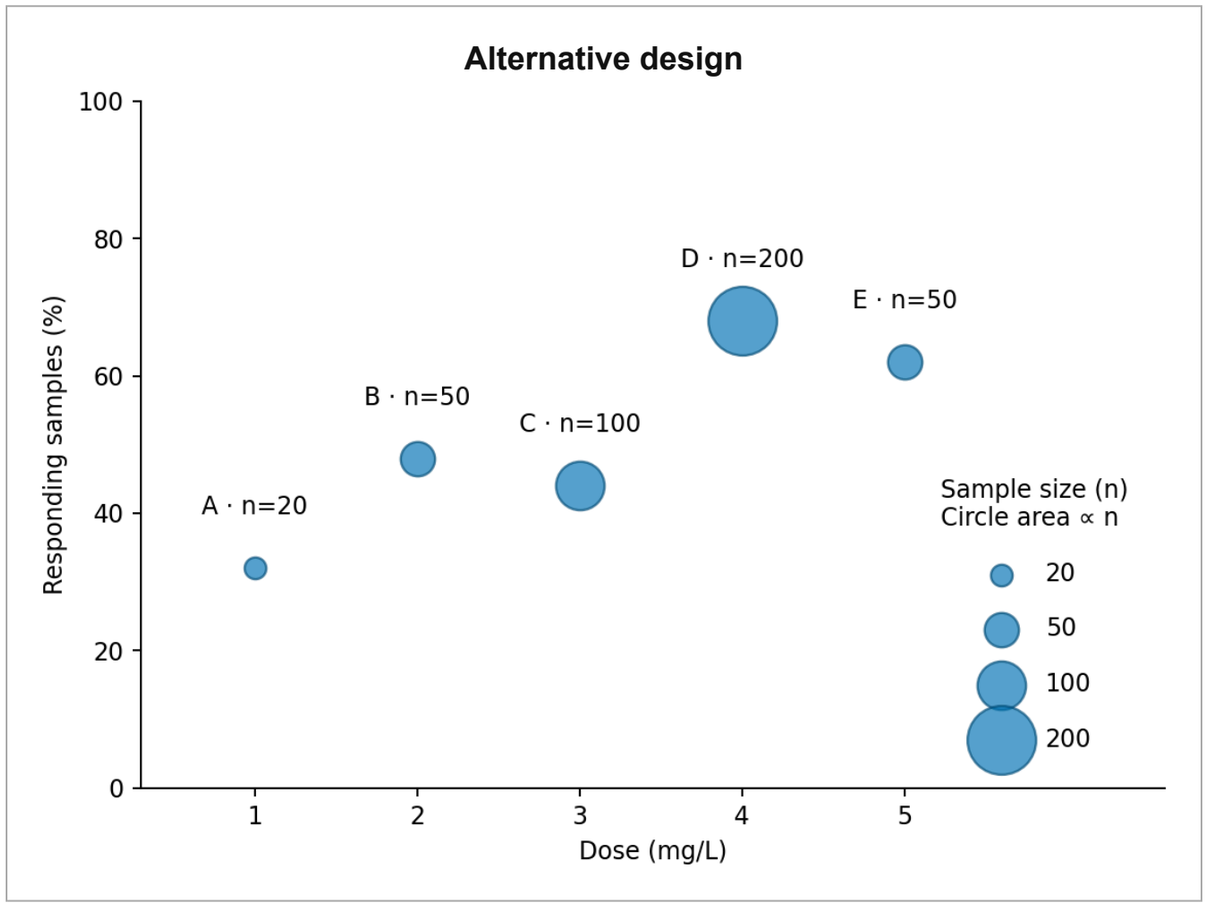




The size key and n labels make the third quantity explicit. Study D has n=200, so its circle has ten times the area of study A (n=20). More samples do not automatically mean a larger effect, better study quality or freedom from bias.

Advice, examples & further exploration

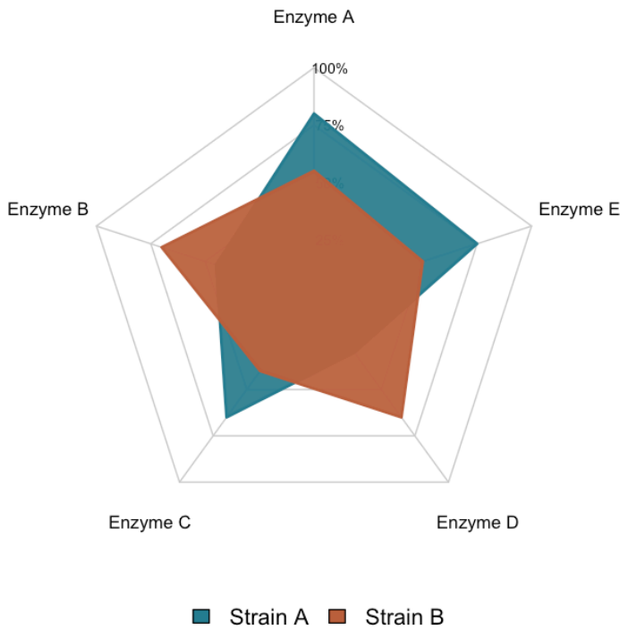
Radar / spider chart

Compare multivariable profiles on radial axes with a common, meaningful scale.

Do not hide one profile behind another

The same enzyme activities for two strains, on a common 0–100% scale.

Starting view



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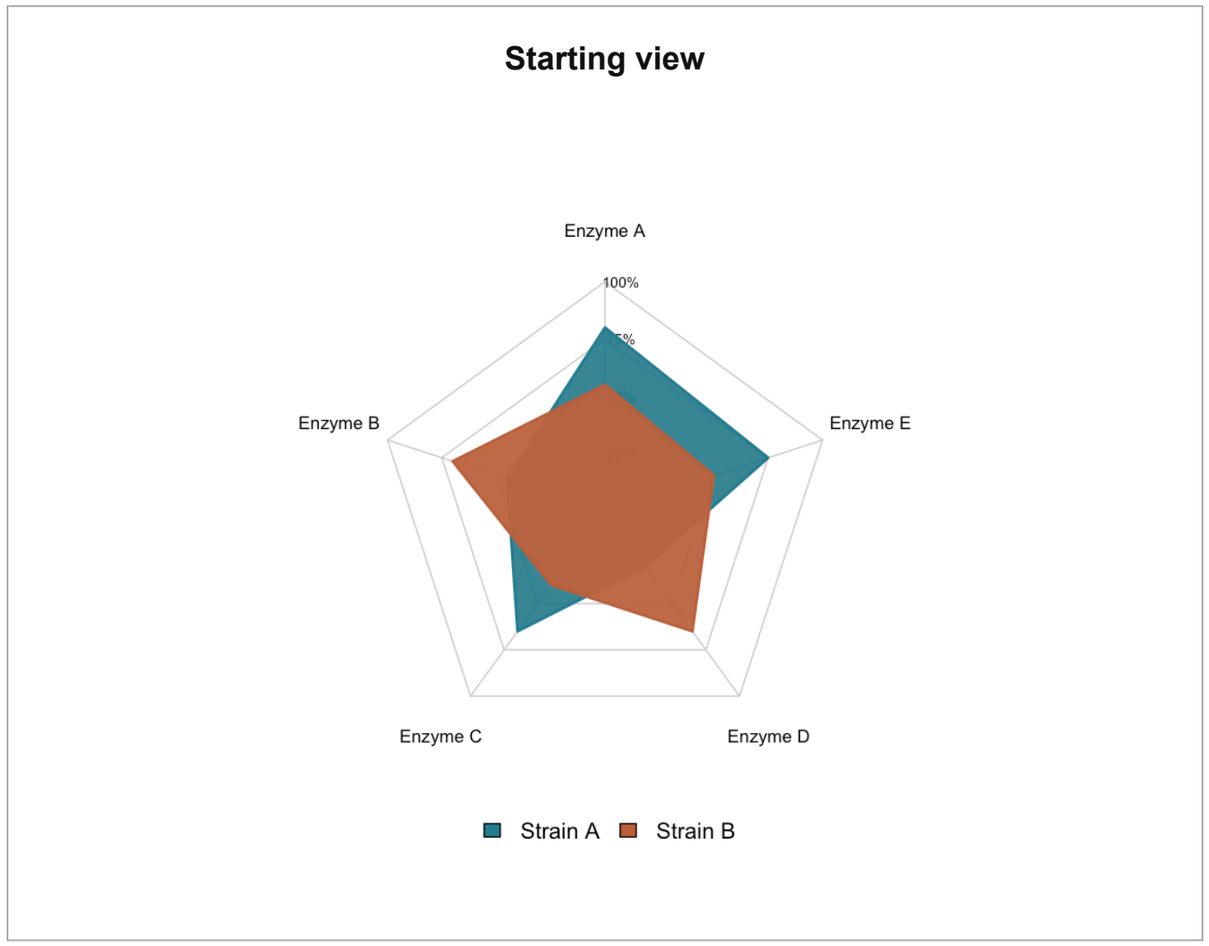
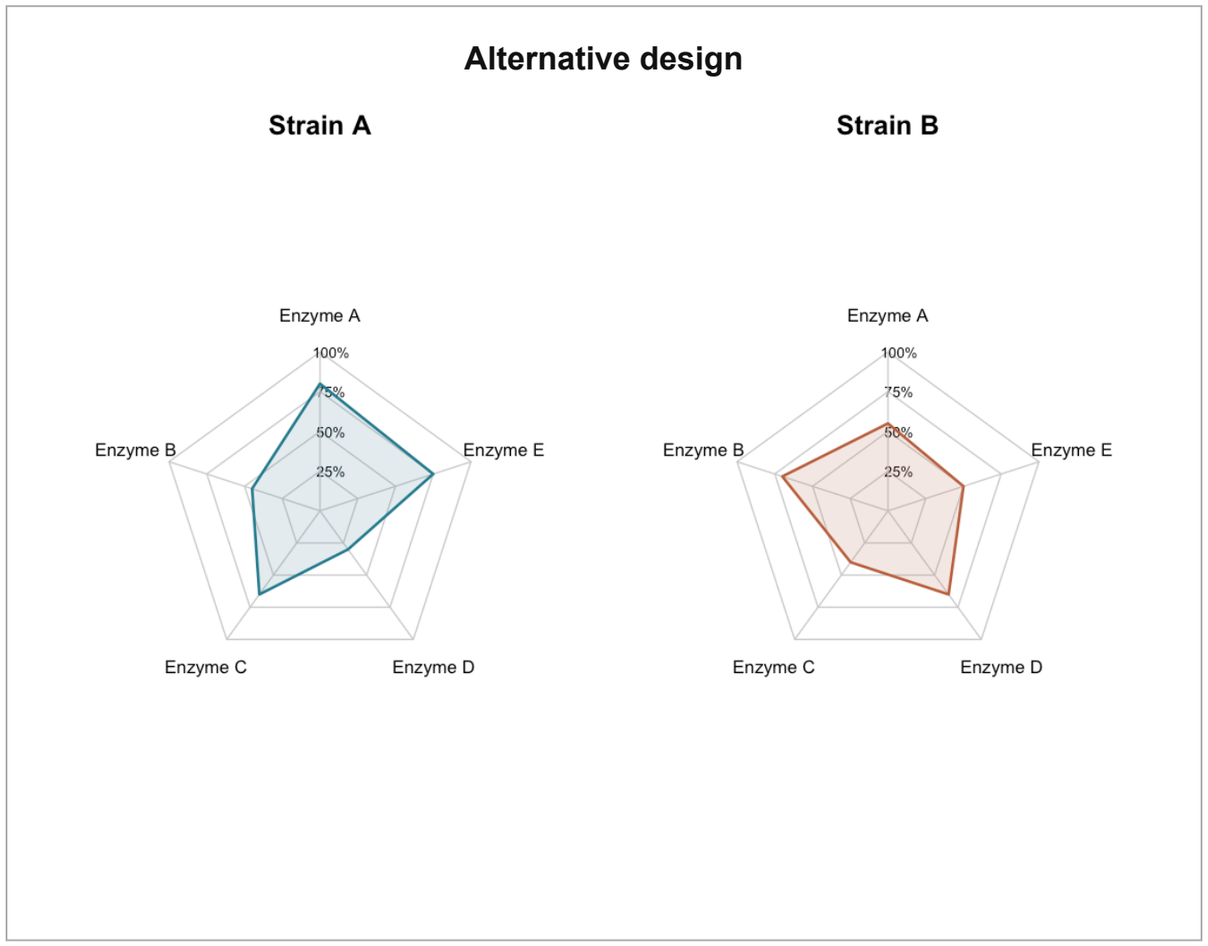




Opaque overlapping polygons obscure part of the first strain. Separate panels preserve both profiles with identical spoke order and scales. For precise comparisons of each enzyme, a dot plot may be simpler.

Advice, examples & further exploration

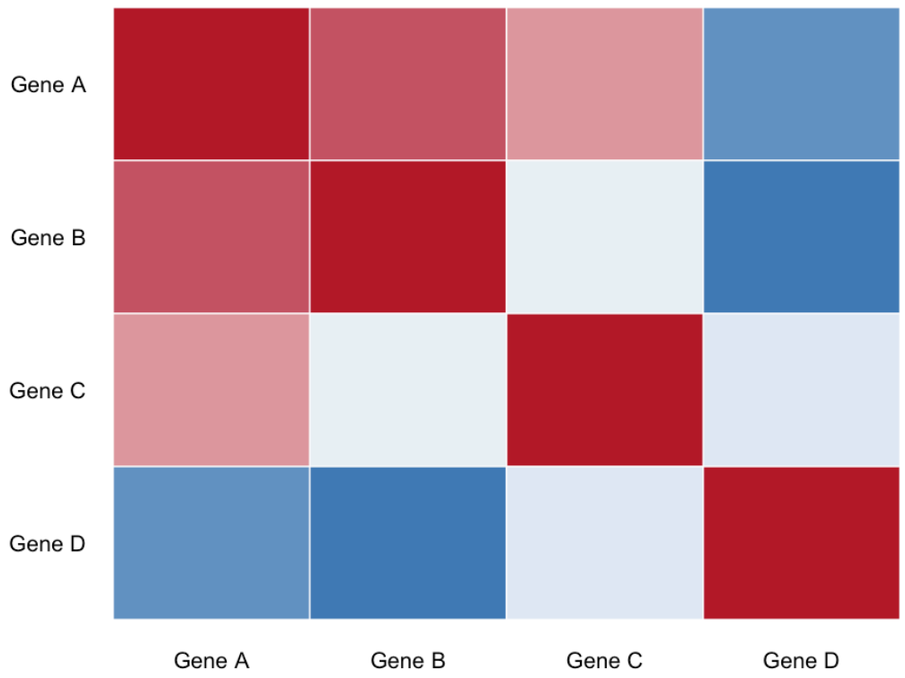
Correlation matrix

Summarise pairwise associations among several numeric measurements.

Show what the colours mean

The same Pearson correlations between four genes, measured in six samples.

Starting view



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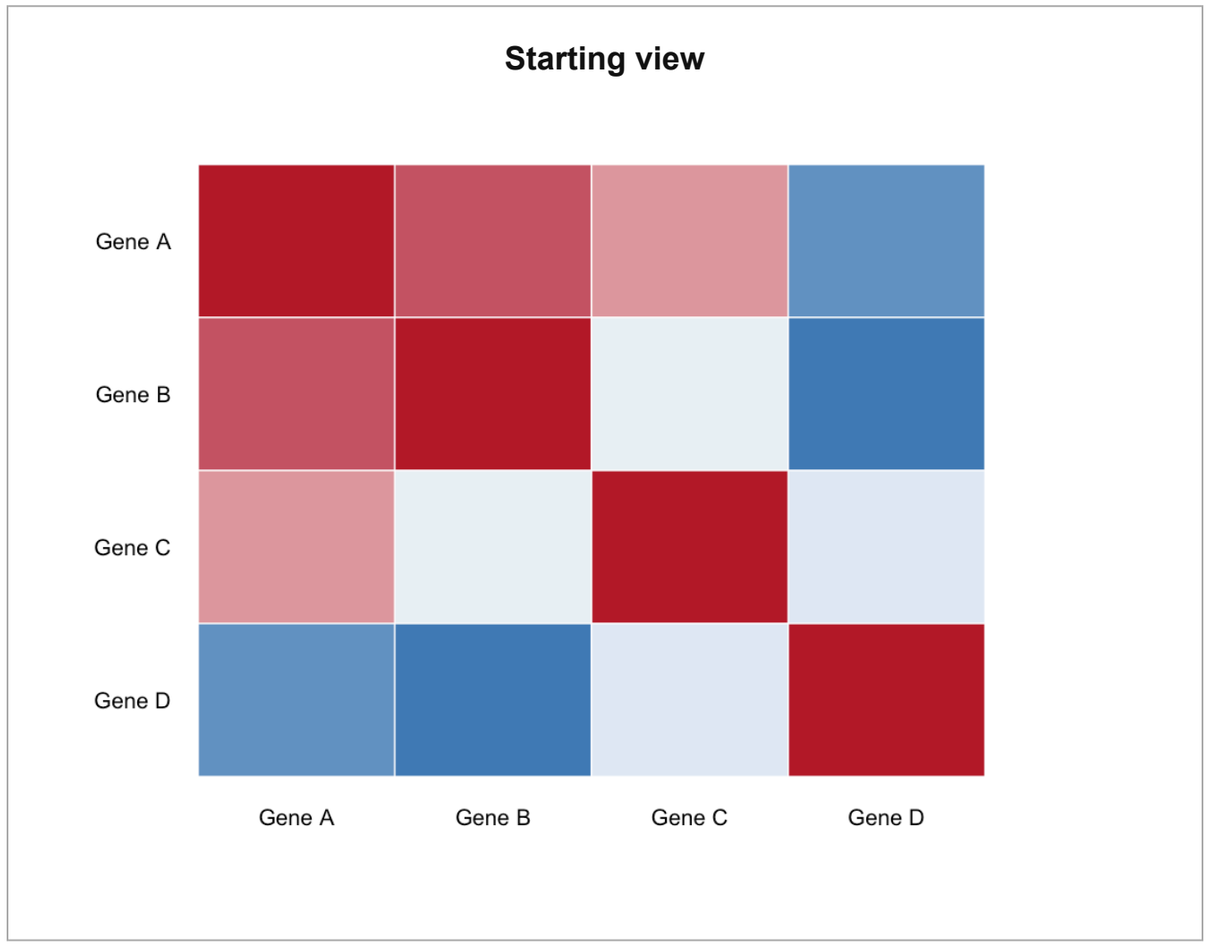
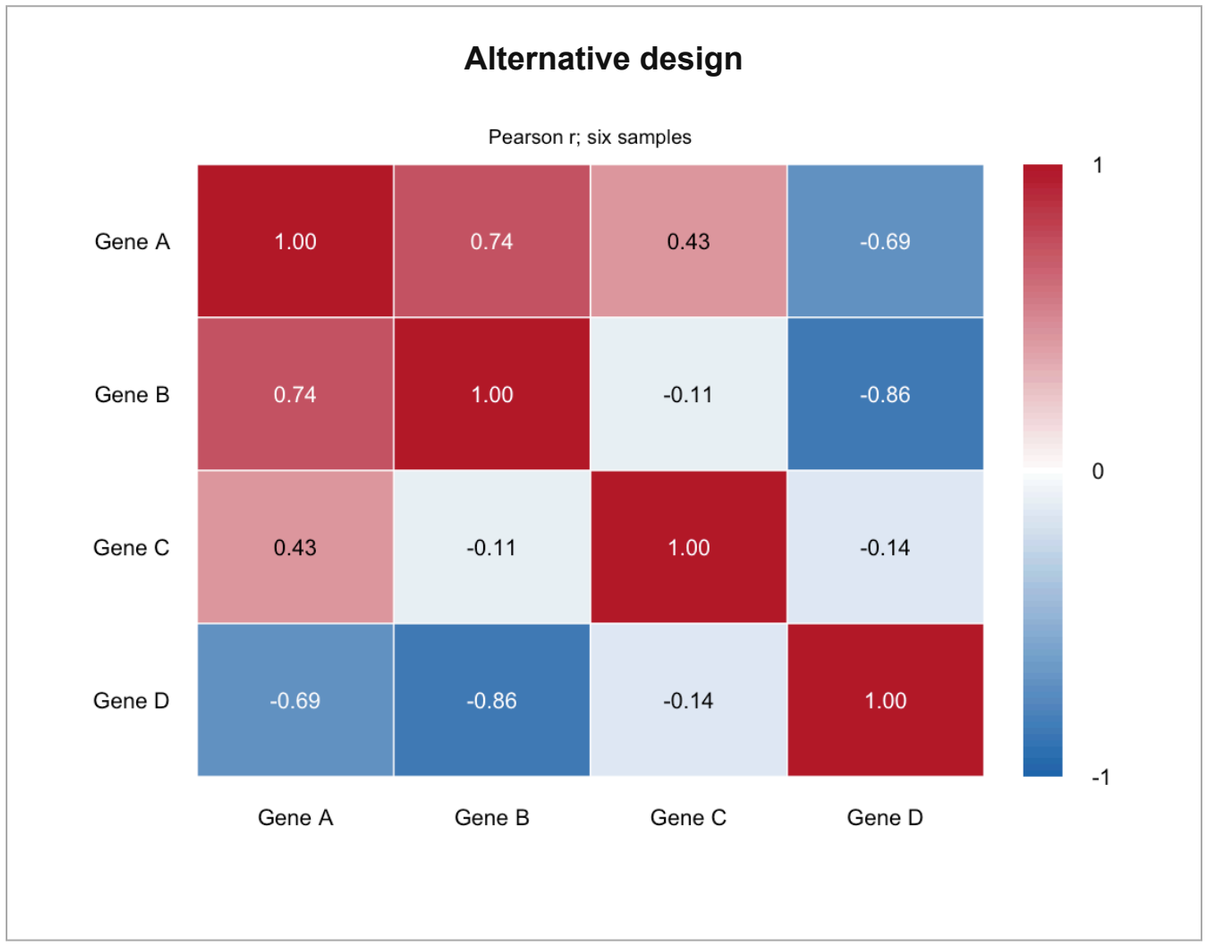




A matrix without a colour key leaves magnitude and sign uncertain. The −1 to +1 key, coefficients and sample size make the display readable while reminding students that these estimates have little supporting data.

Advice, examples & further exploration

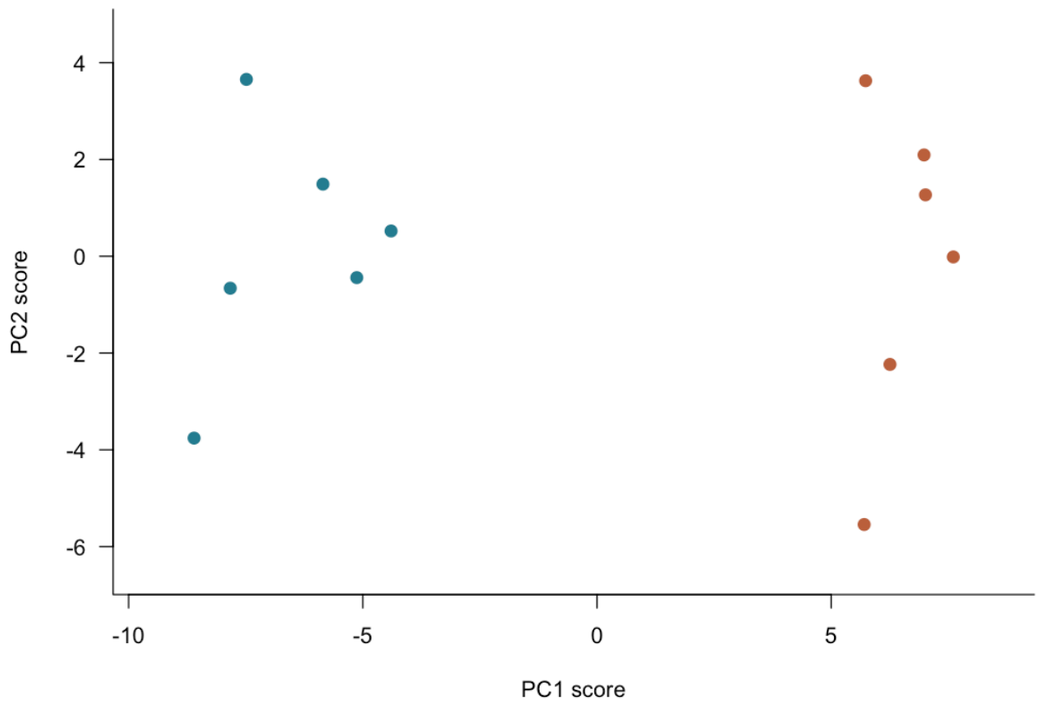
PCA / ordination plot

Summarise variation across many measured features in a few coordinates.

Identify groups and samples

The same 12 expression profiles projected onto their first two principal components.

Starting view



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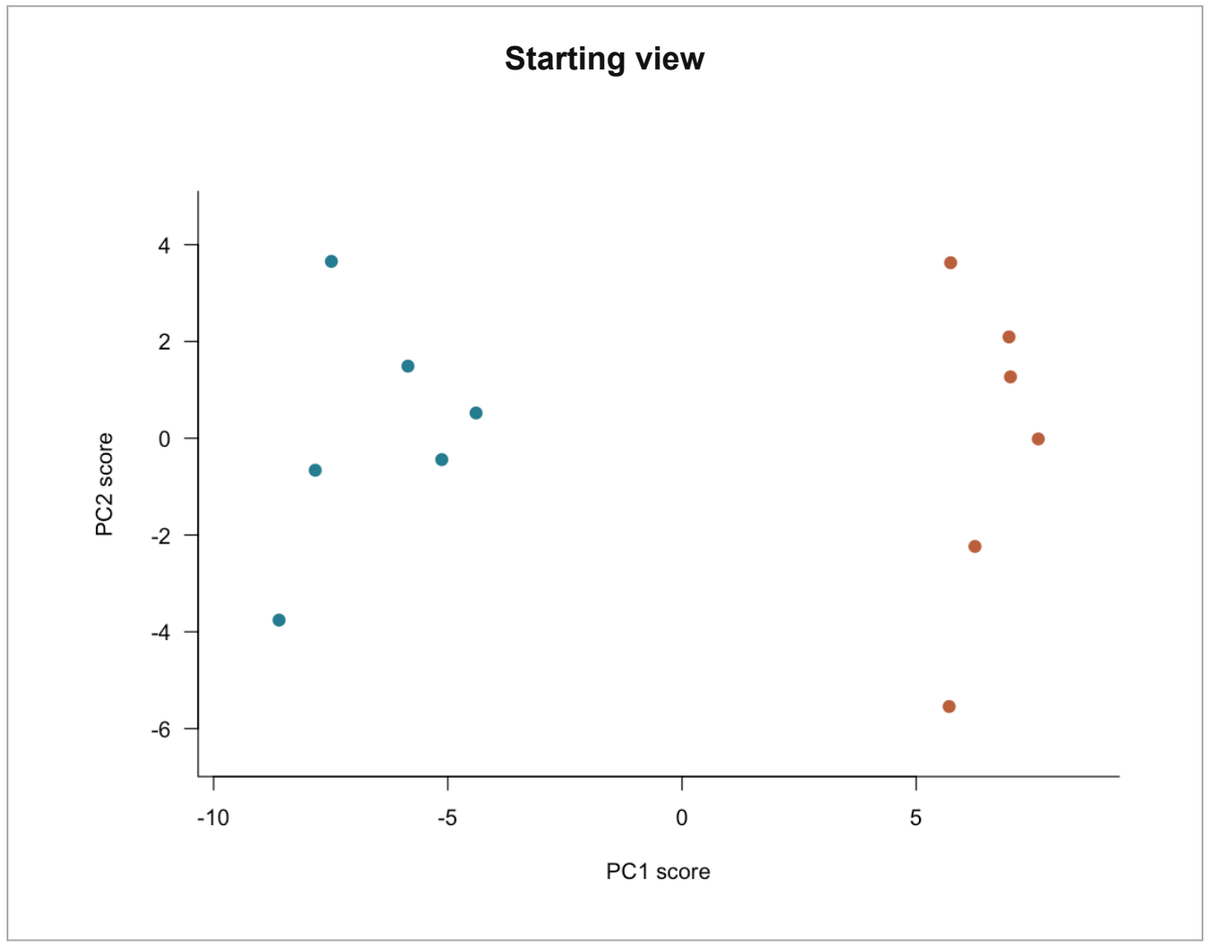
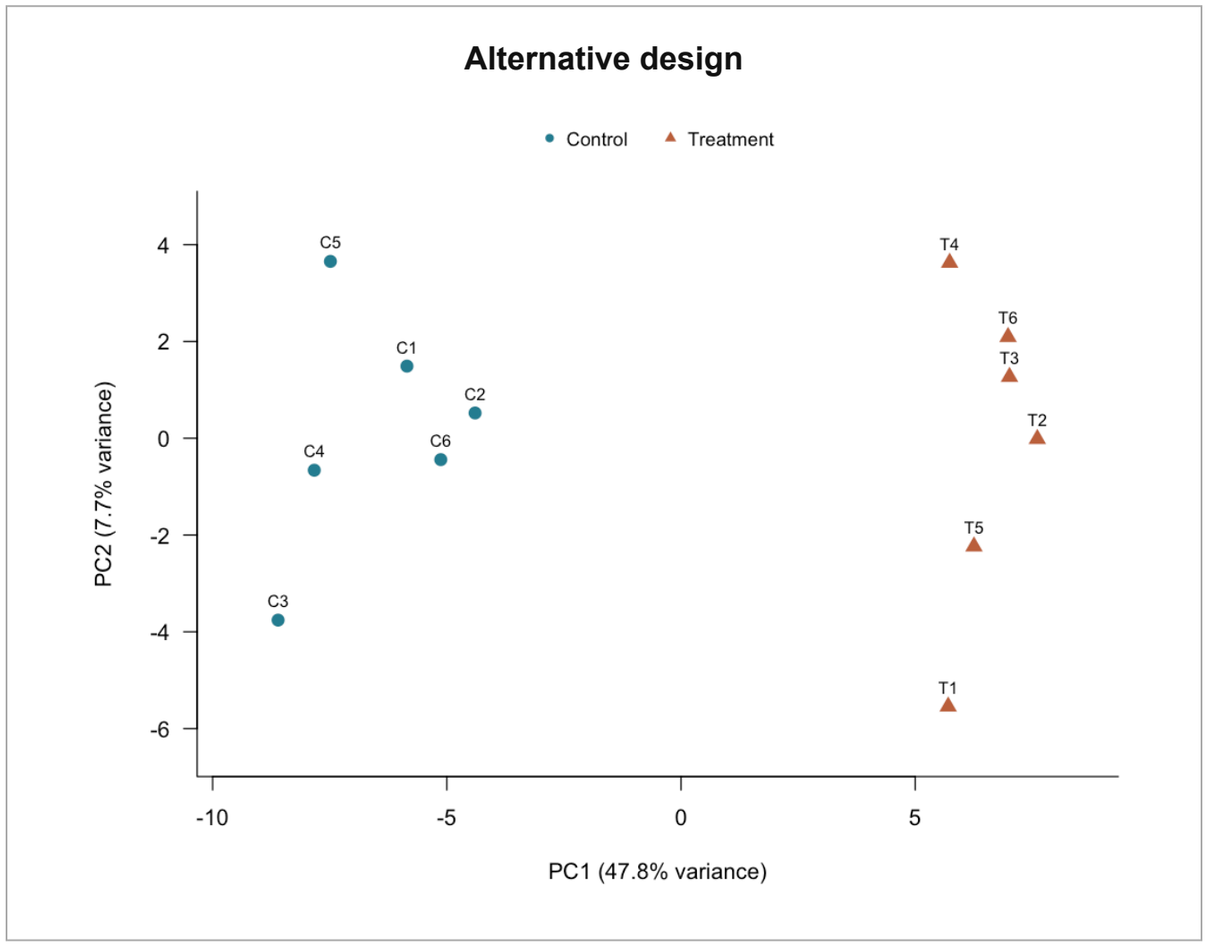




A colour split without a key is difficult to interpret. Group labels, redundant shapes, sample IDs and explained-variance labels make the coordinates interpretable. Inspect preprocessing and batch metadata before assigning a biological explanation.

> **Advice, examples & further exploration**

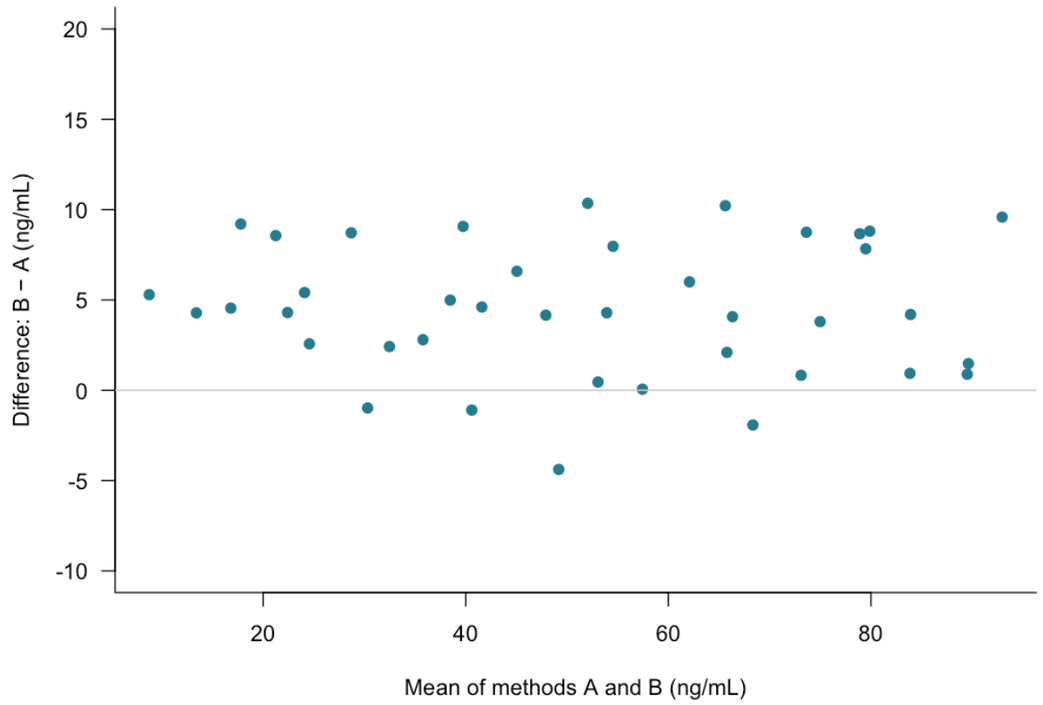
Bland–Altman plot

Inspect agreement between two methods measuring the same samples.

Give disagreement a reference

The same 40 samples measured by assays A and B; x is their mean and y is B minus A.

Starting view



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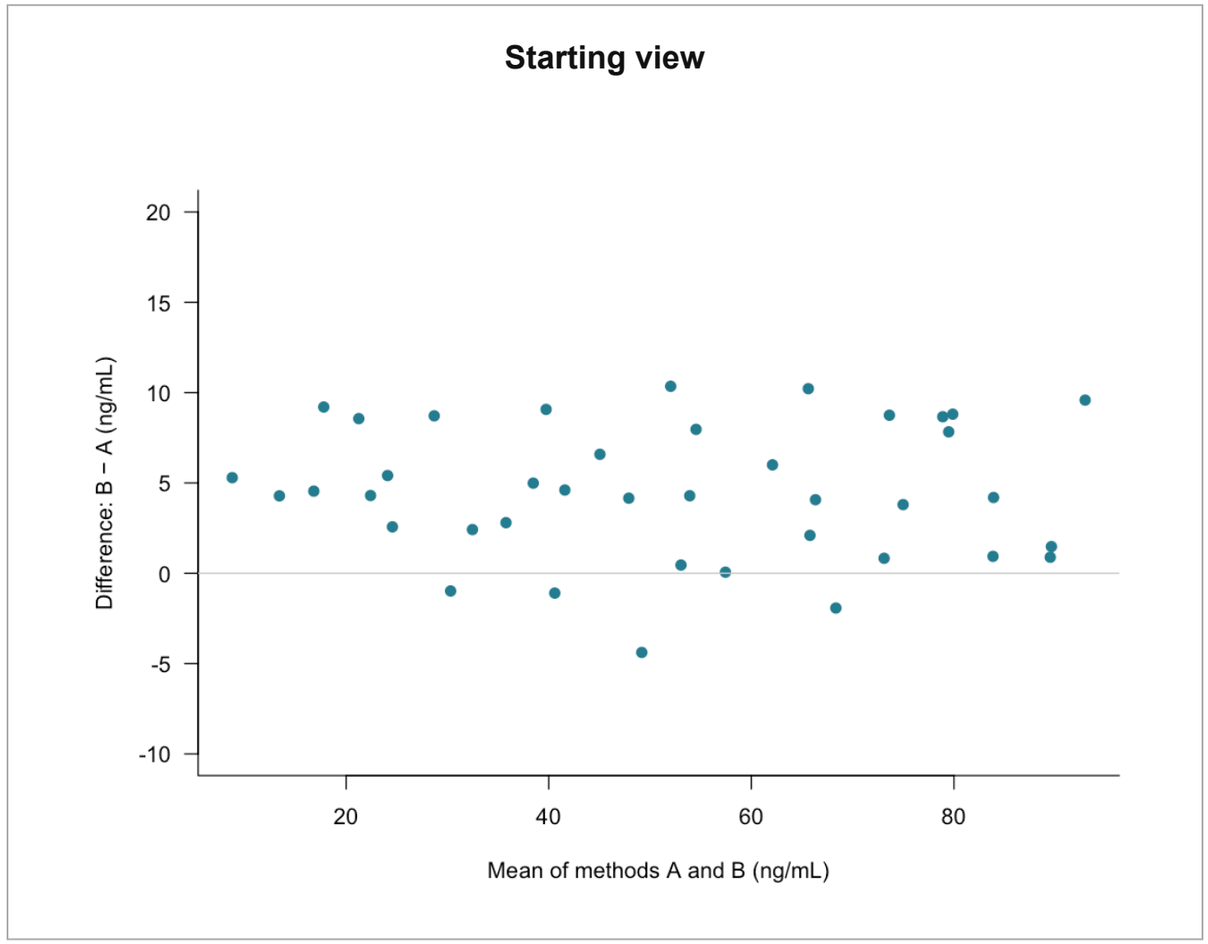
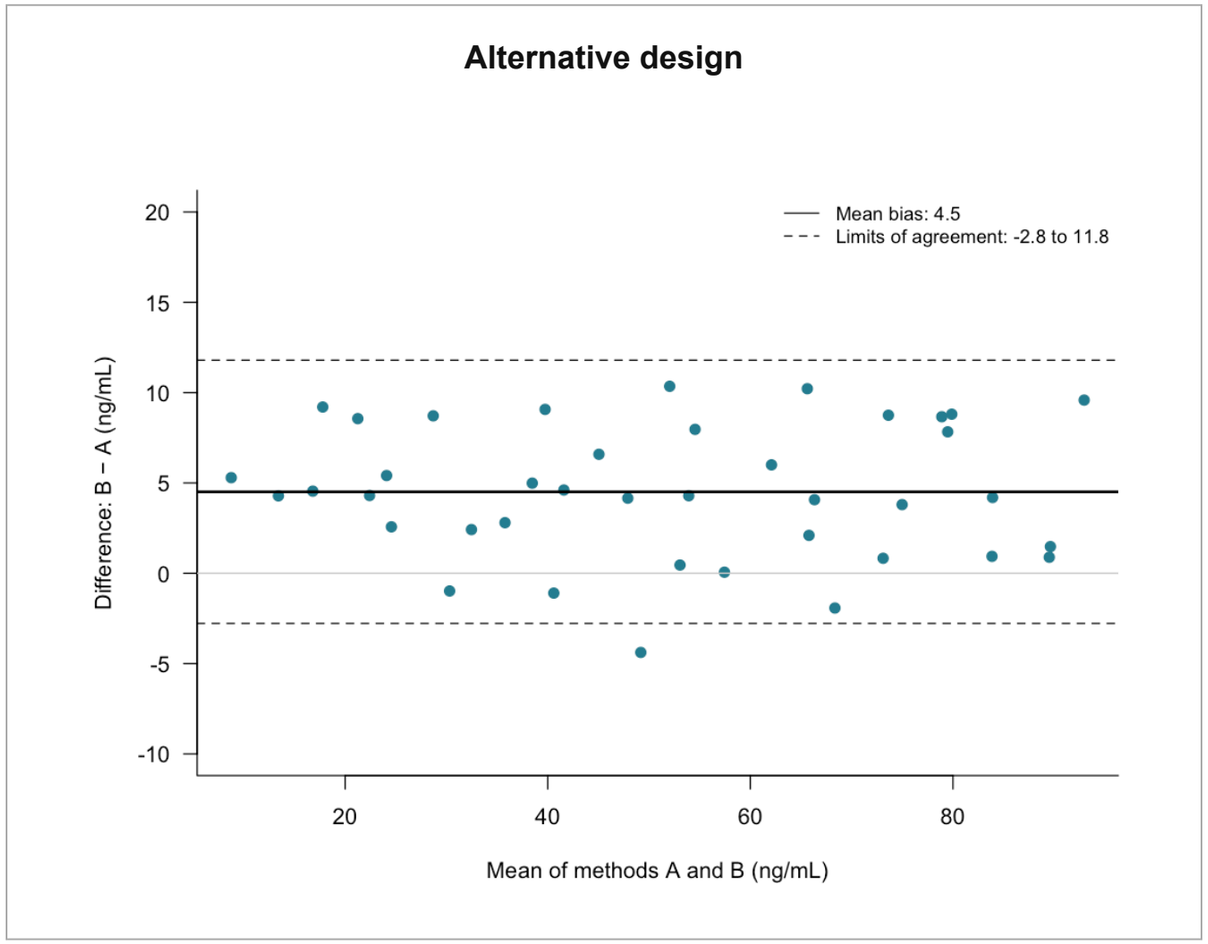




The cloud alone makes systematic bias and spread hard to judge. Mean bias and limits of agreement (bias ± 1.96 SD of the differences) summarise them. Decide acceptable assay disagreement separately; these lines are not acceptance limits or confidence intervals.

Advice, examples & further exploration

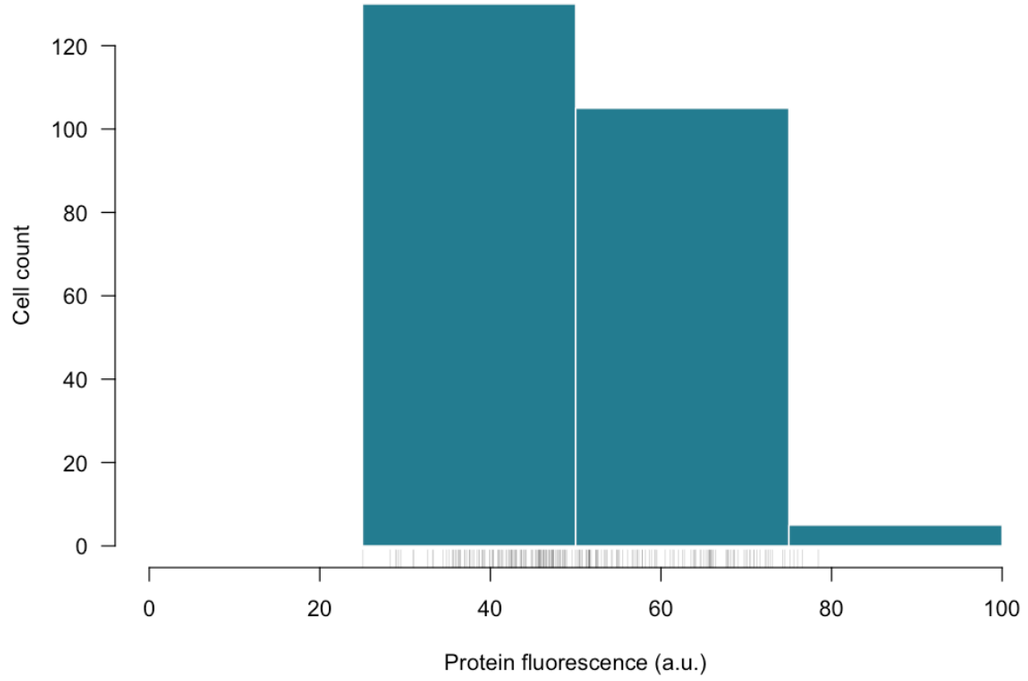
Histogram

See the shape, spread, and concentrations of one numeric variable.

Avoid bins that hide the structure

The same 240 cells, with fluorescence on the horizontal axis and cell counts on the vertical axis.

Starting view



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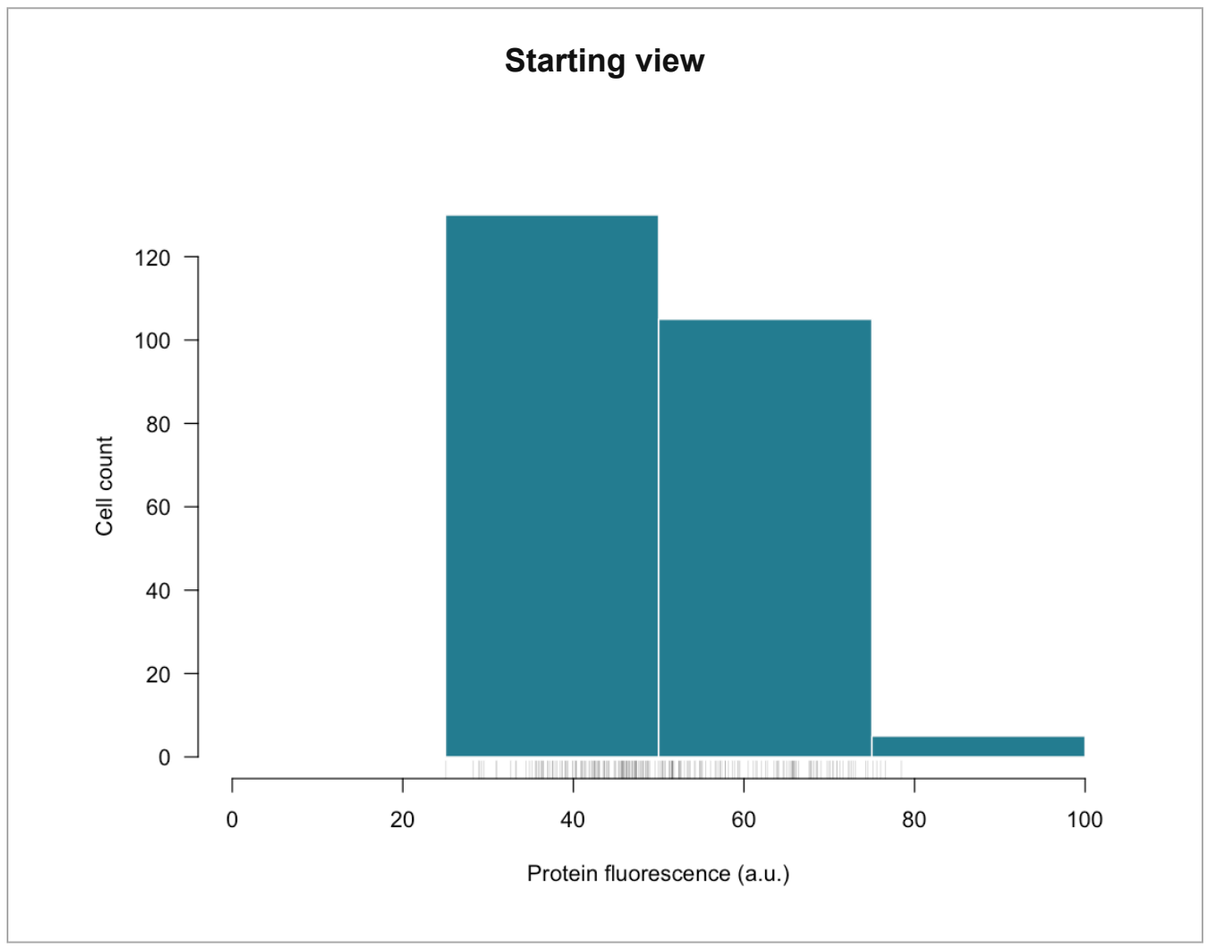
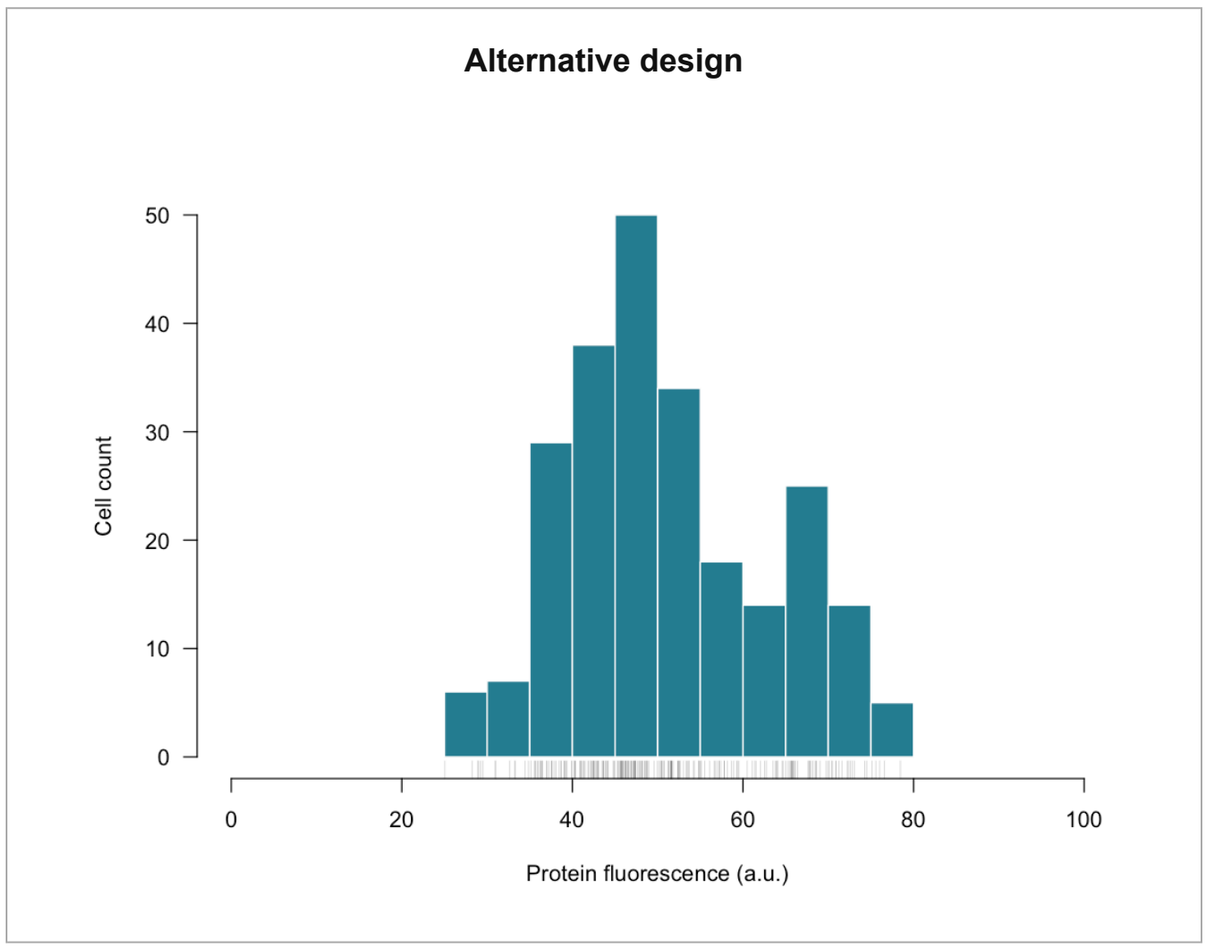




Very wide bins conceal much of the distribution. Five-unit bins reveal more detail here. This is not a universal ideal width: compare several choices and check the observations before claiming distinct populations.

> **Advice, examples & further exploration**

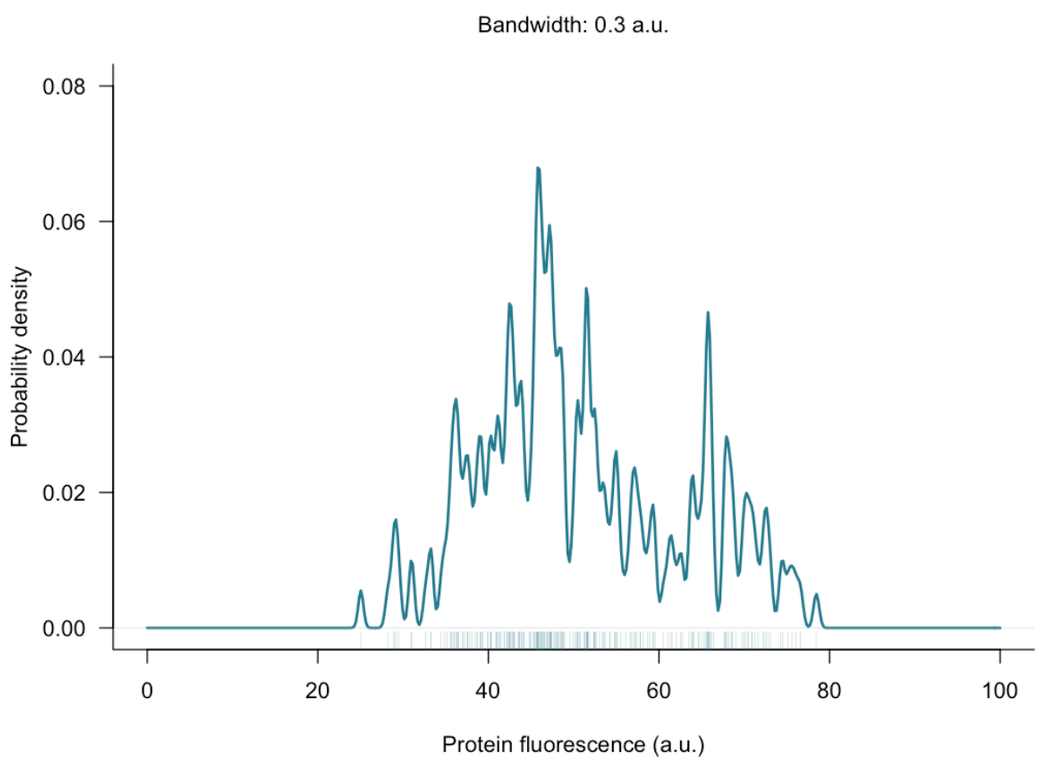
Density plot

Show a smoothed distribution of a numeric measurement.

Do not turn noise into peaks

The same 240 fluorescence measurements; the rug shows individual cells.

Starting view



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Advice, examples & further exploration

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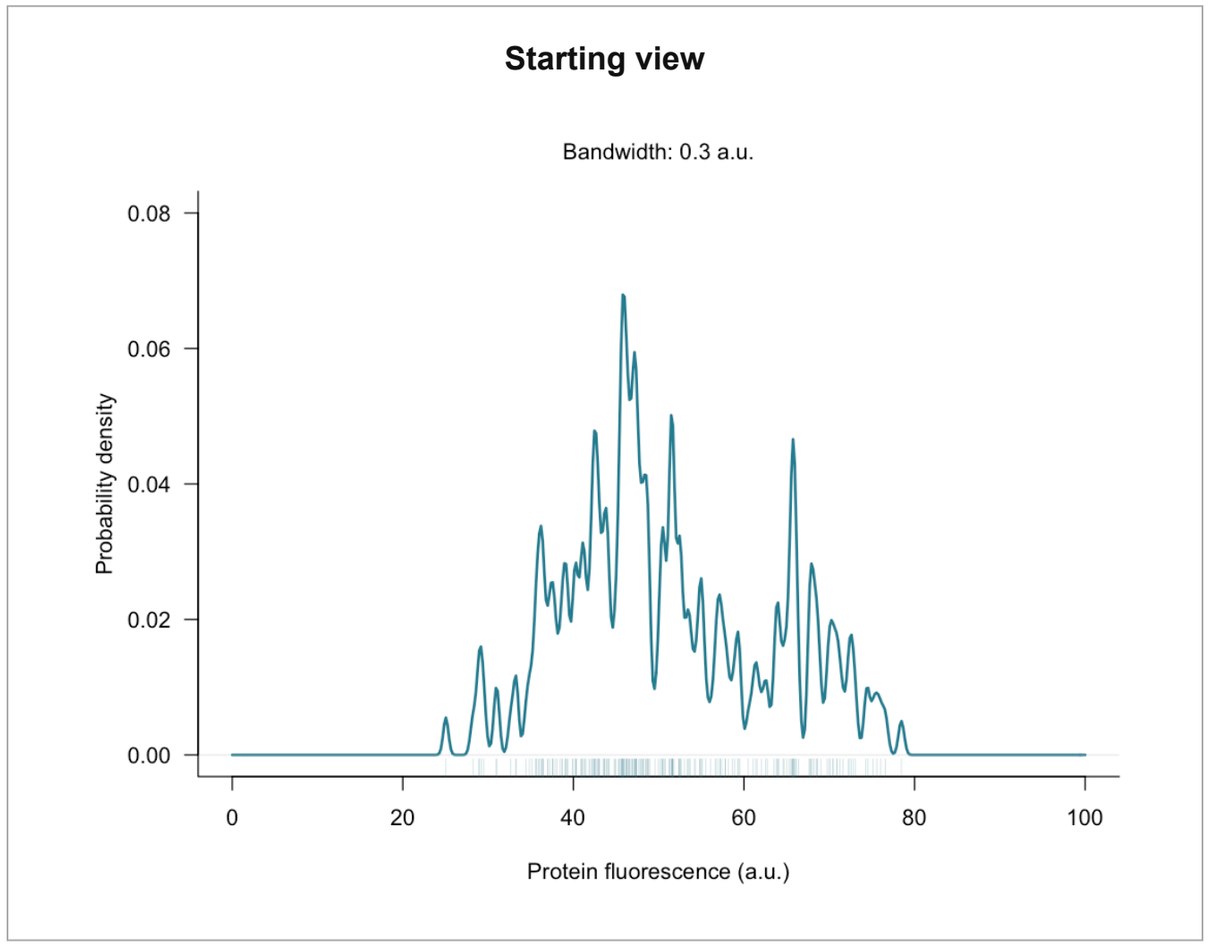
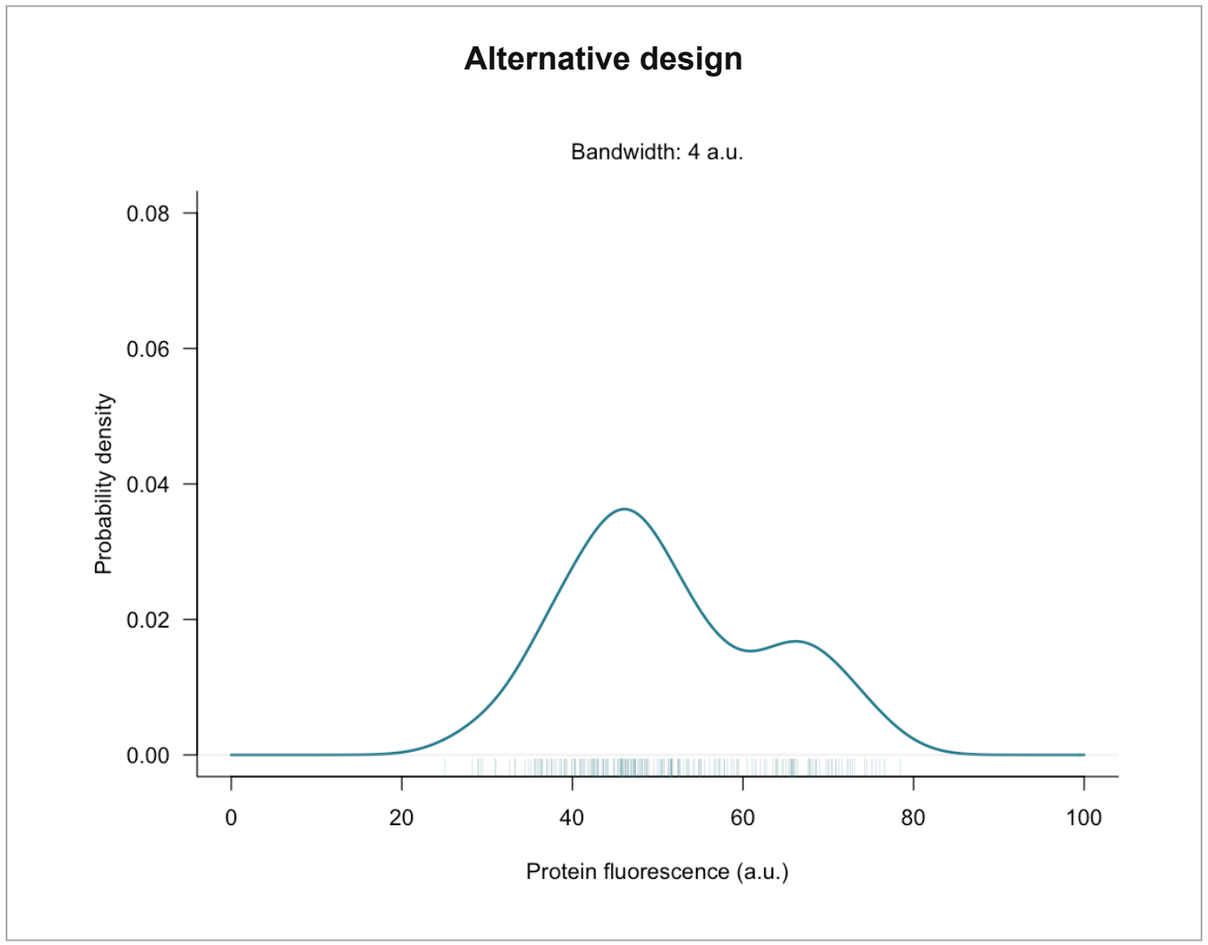




A bandwidth of 0.3 makes many small fluctuations look like peaks. A bandwidth of 4 gives a more readable summary. Smoothing is a choice, not proof of how many biological populations exist.

> **Advice, examples & further exploration**

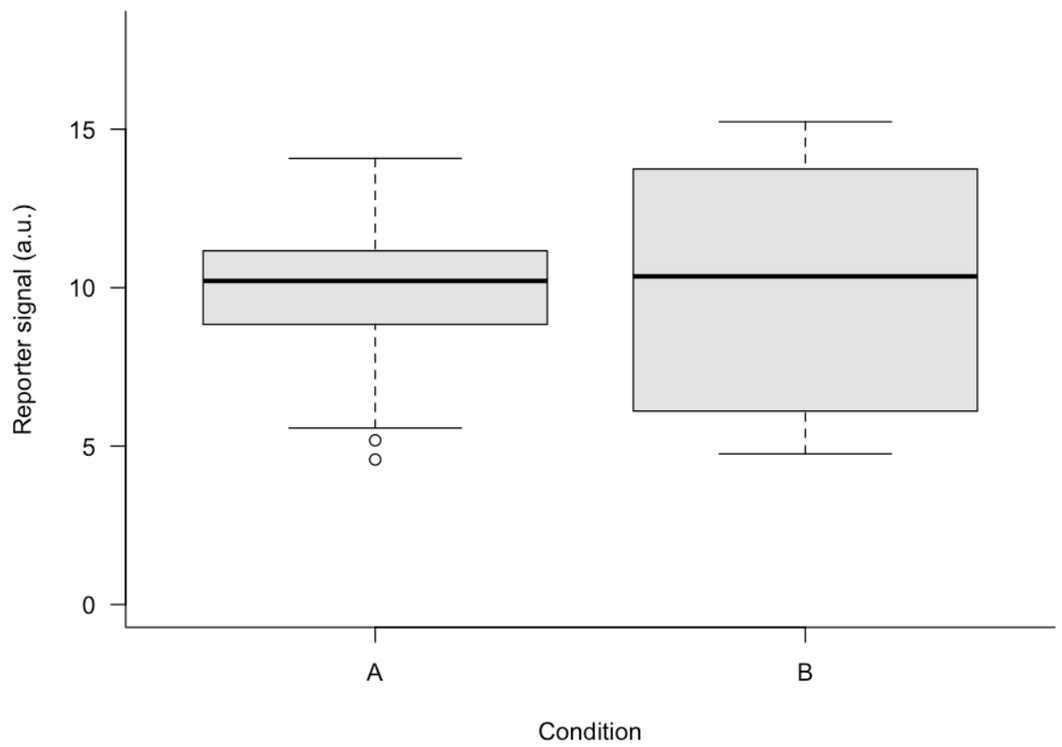
Box plot

Compare medians and spread across groups in a compact display.

Look beyond the box

The same 80 cells per condition.

Starting view



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Advice, examples & further exploration

Box plot

Compare medians and spread across groups in a compact display.

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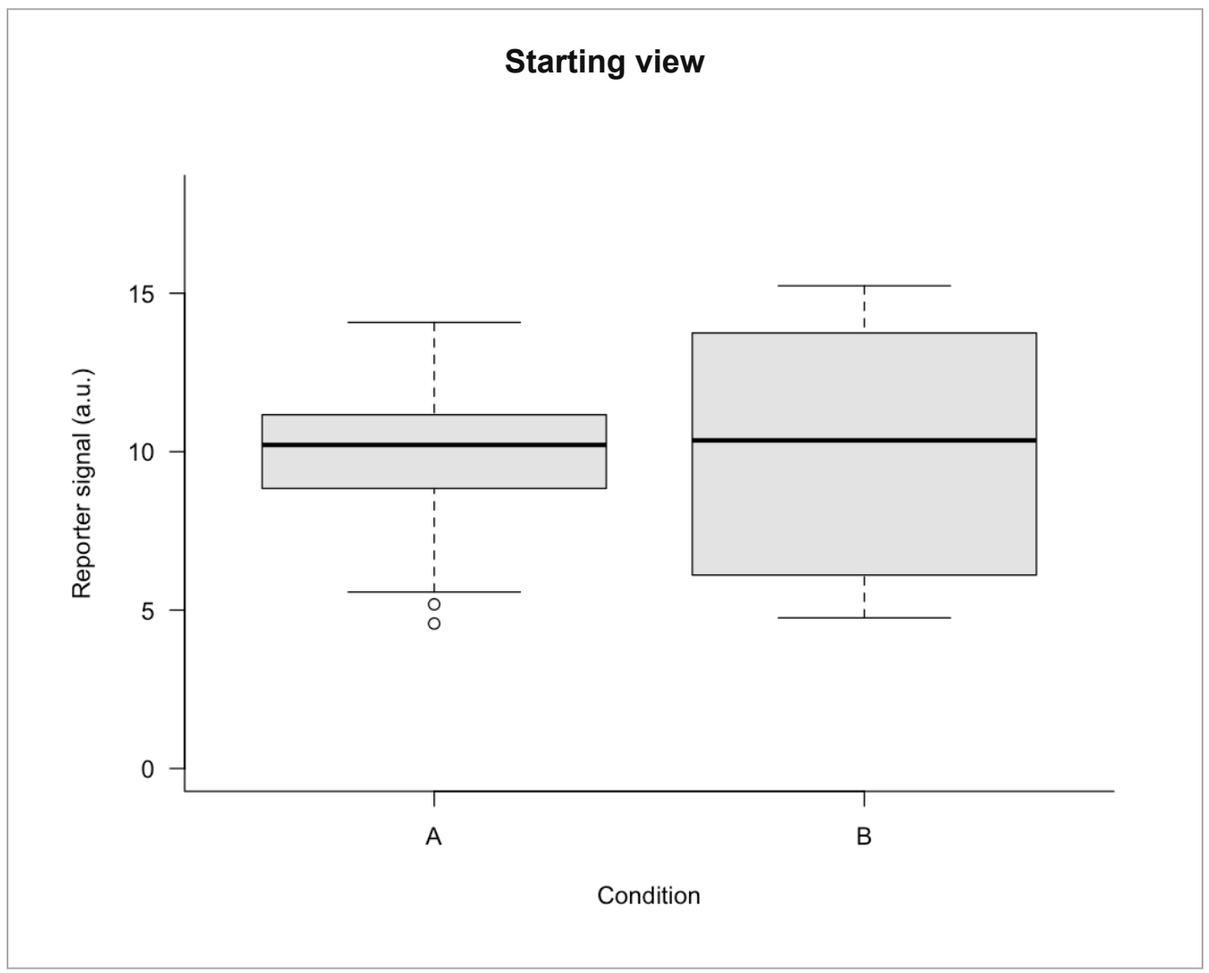
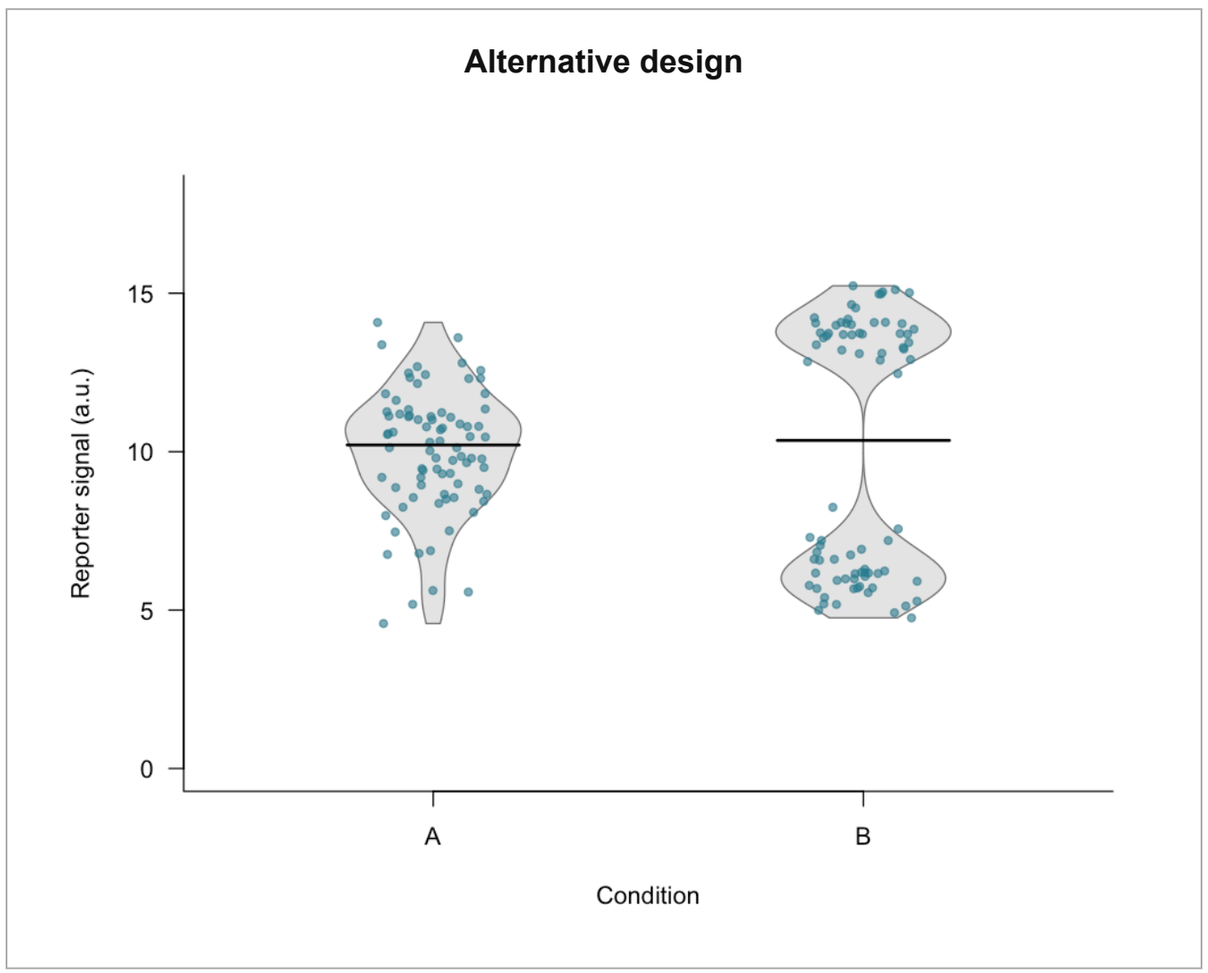




Boxes compactly summarise quartiles. The violin and dots expose distribution shape that quartiles can hide; the extra detail is useful when heterogeneity is the question.

> **Advice, examples & further exploration**

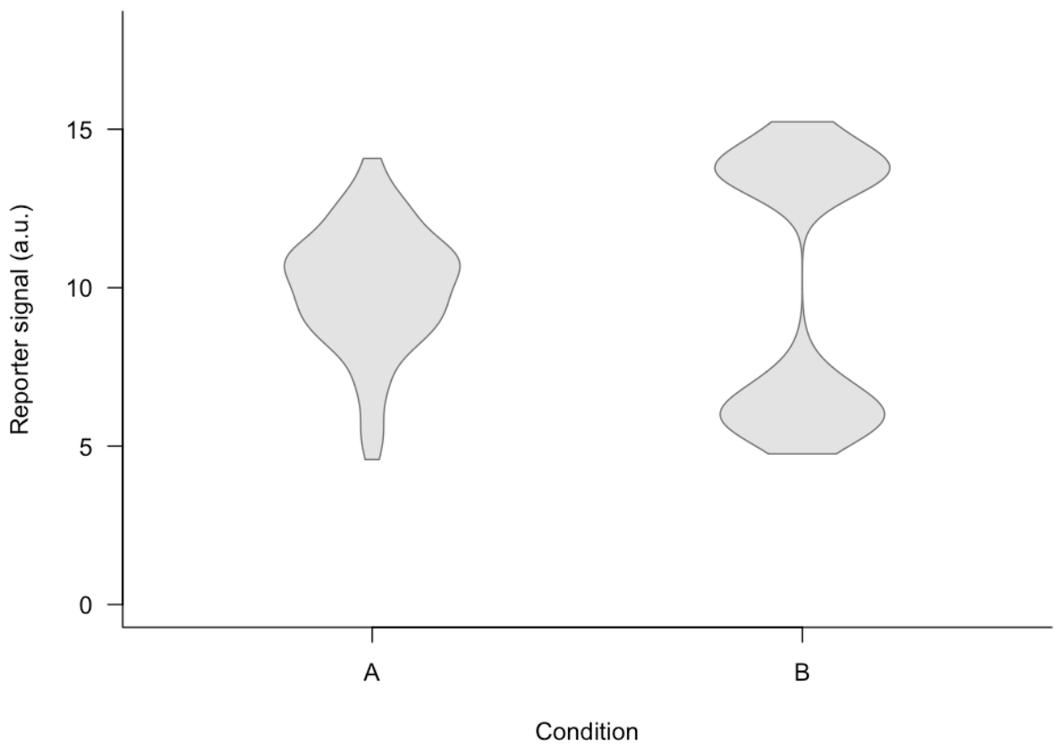
Violin plot

Compare distribution shapes when there are enough observations to estimate density.

Show the evidence behind the shape

The same 80 cells per condition.

Starting view



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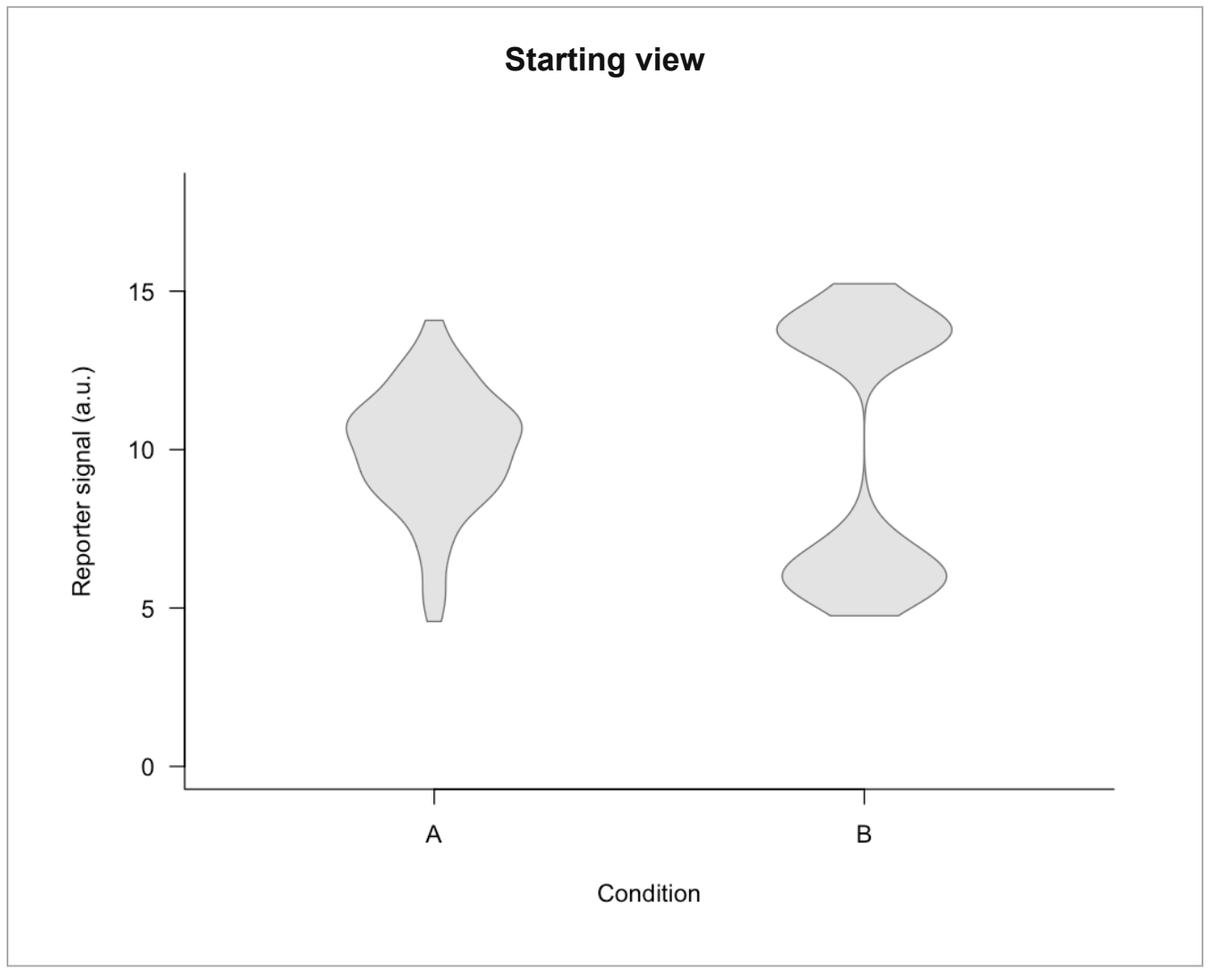
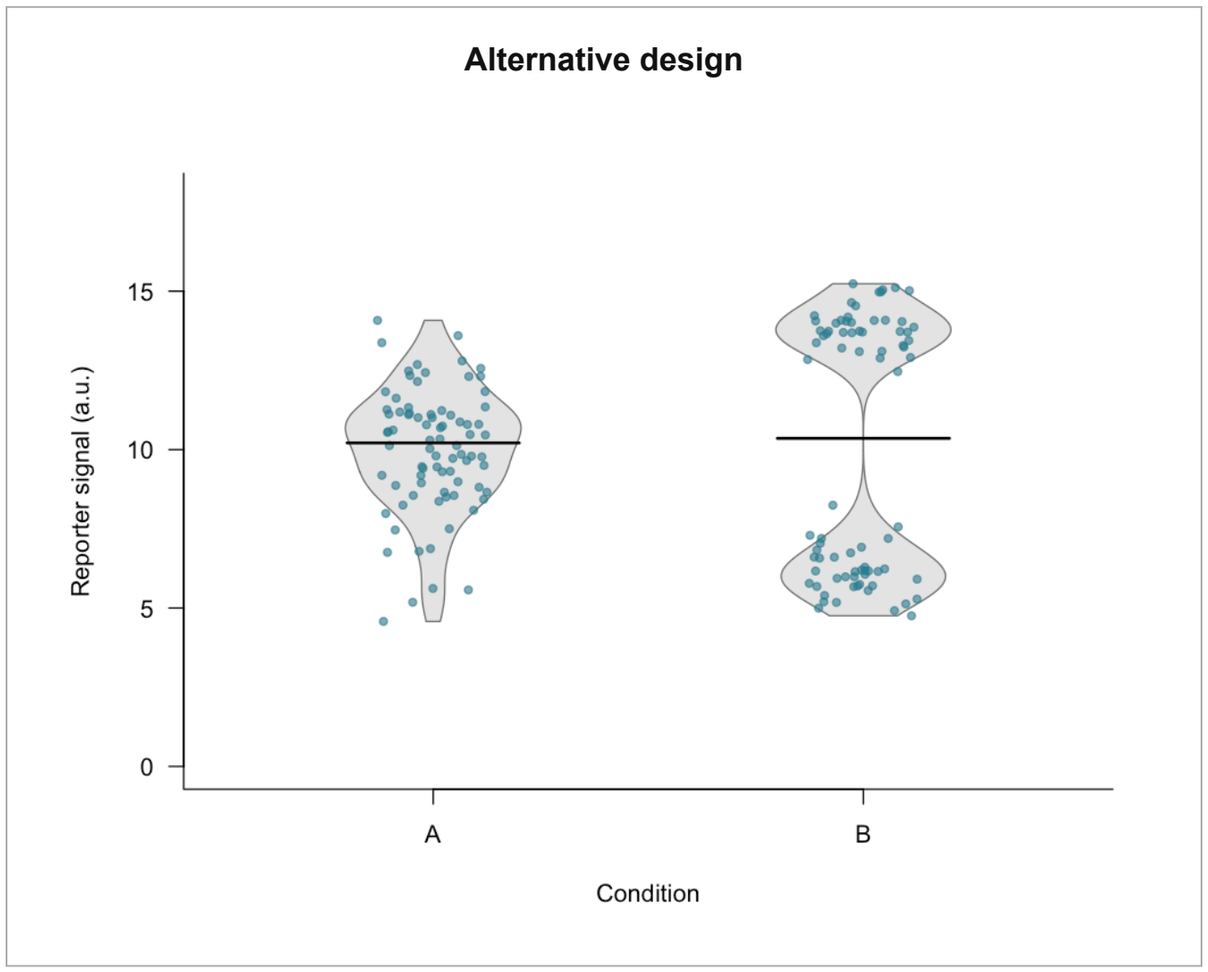




Adding observations lets readers check how much data supports the smoothed outline. With very few measurements, use the dots alone instead of implying a reliable density.

> **Advice, examples & further exploration**

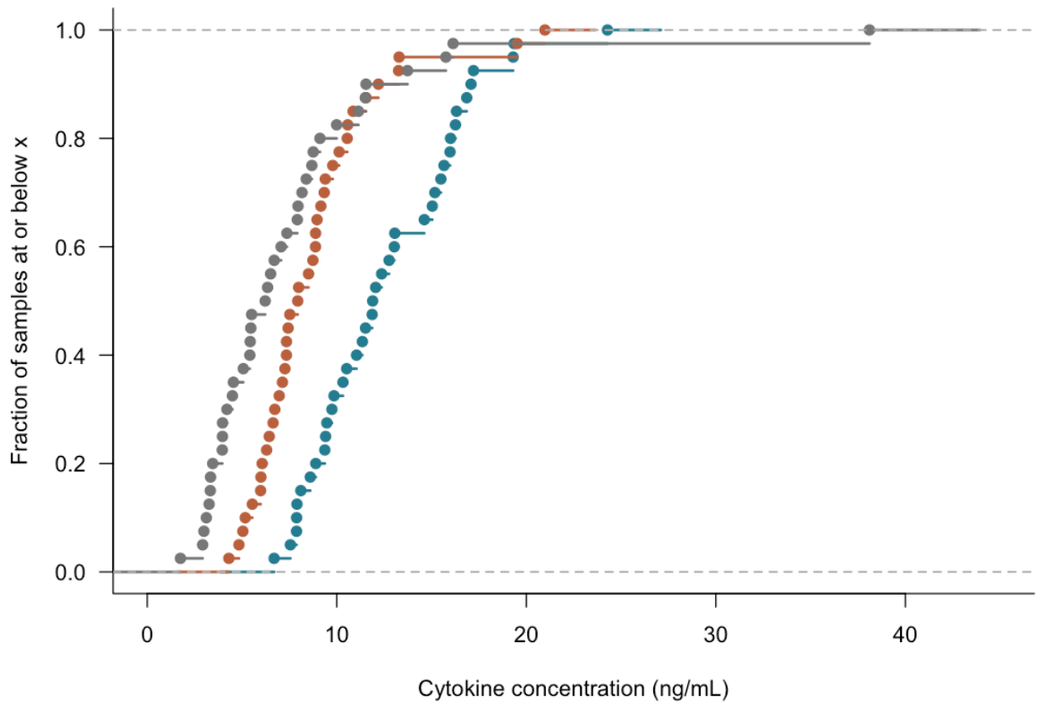
Cumulative distribution (ECDF)

Read the fraction of observations at or below any threshold, without bins or smoothing.

Identify the curves

The same cytokine measurements from 40 independent samples per dose group; height is the fraction at or below a concentration.

Starting view



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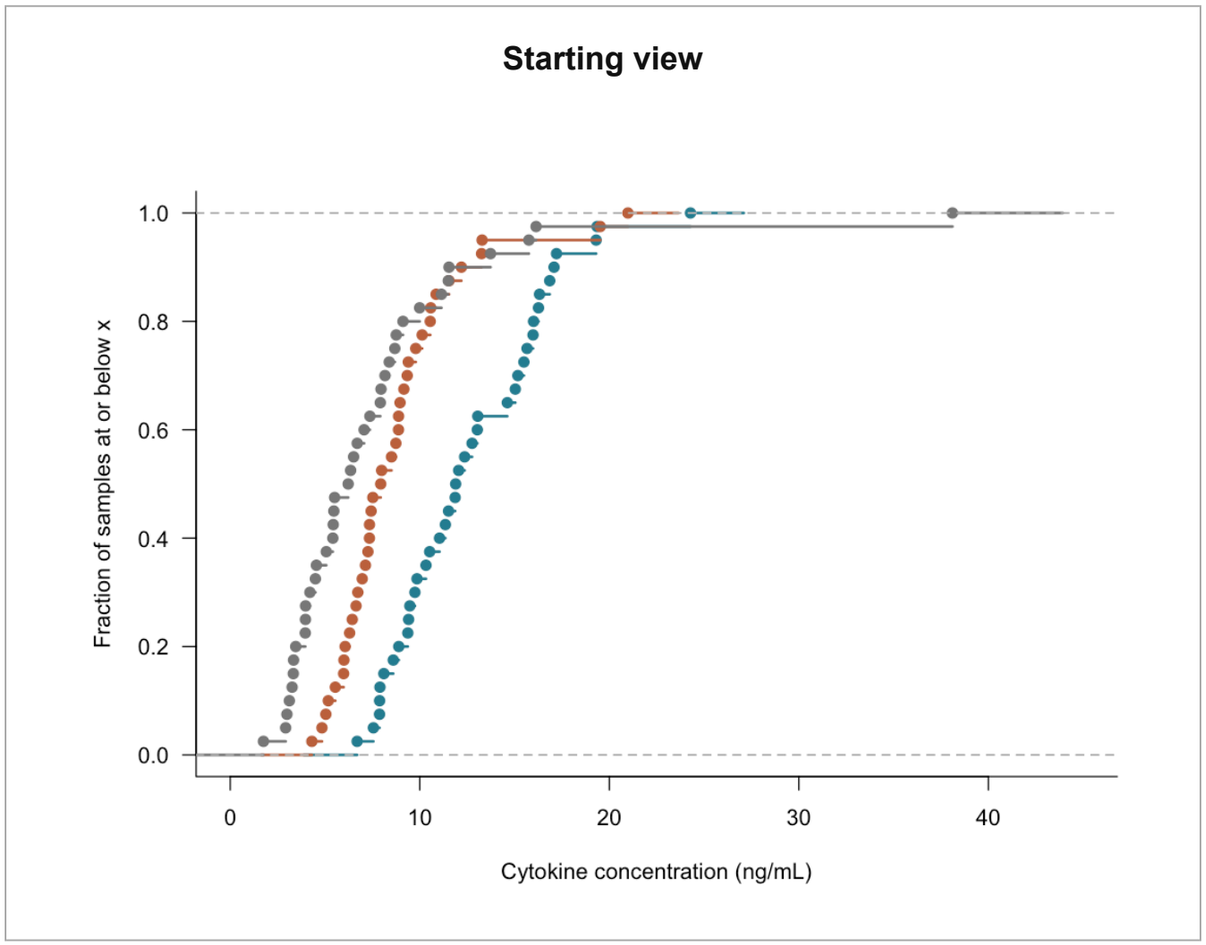
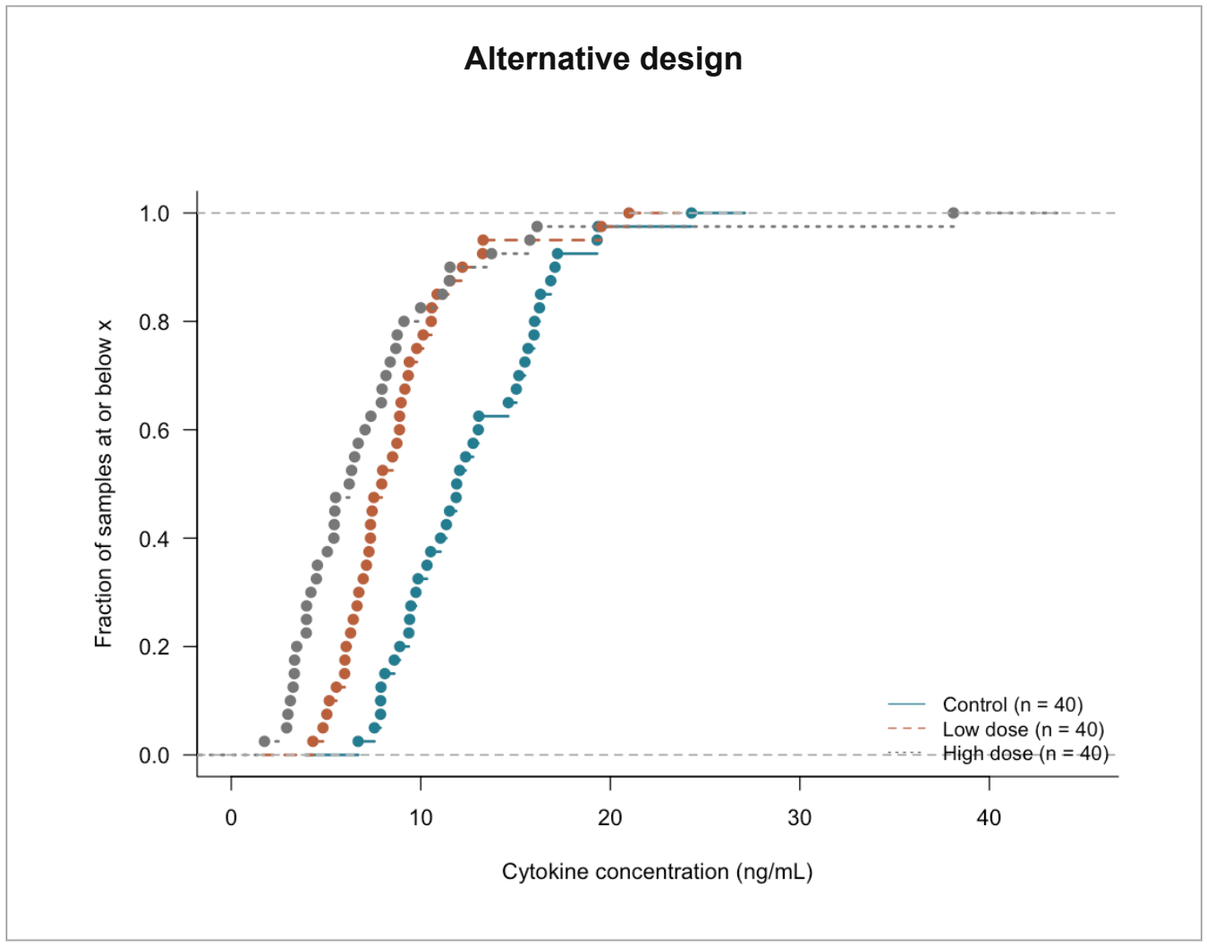




Unidentified colours cannot tell readers which group crosses a threshold first. A legend, group sample sizes and distinct line styles make the comparison interpretable, including without colour.

Advice, examples & further exploration

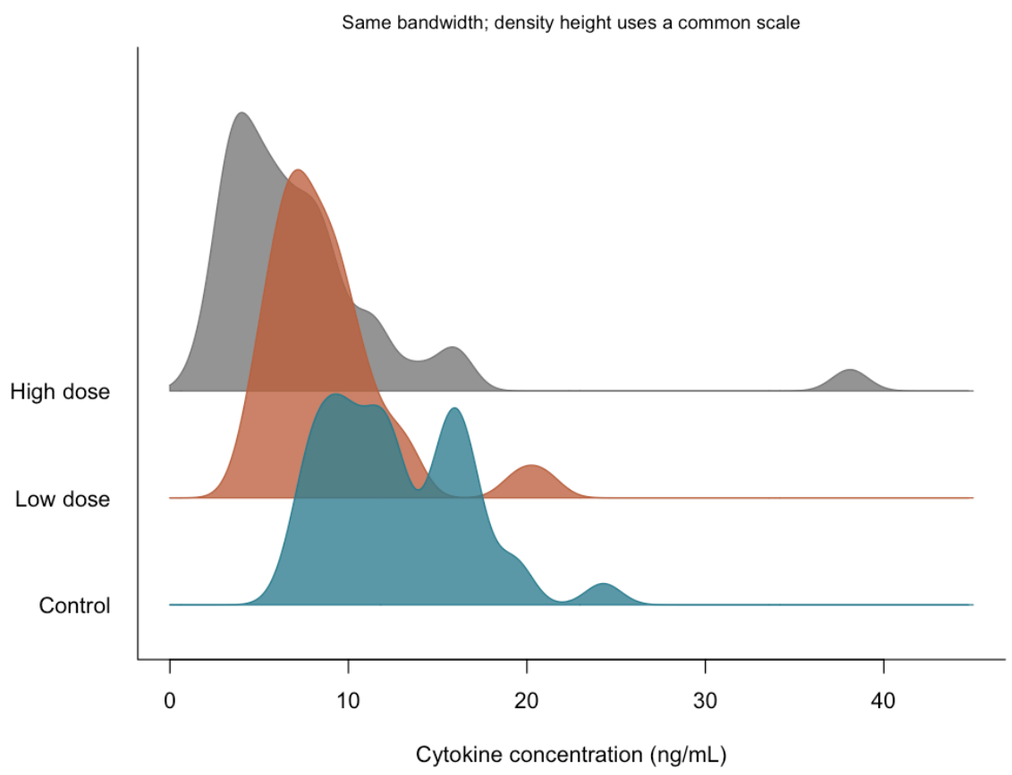
Ridgeline plot

Compare several distribution shapes on a shared horizontal scale.

Give distributions enough room

The same cytokine measurements in three dose groups; identical bandwidth and horizontal scale in both versions.

Starting view



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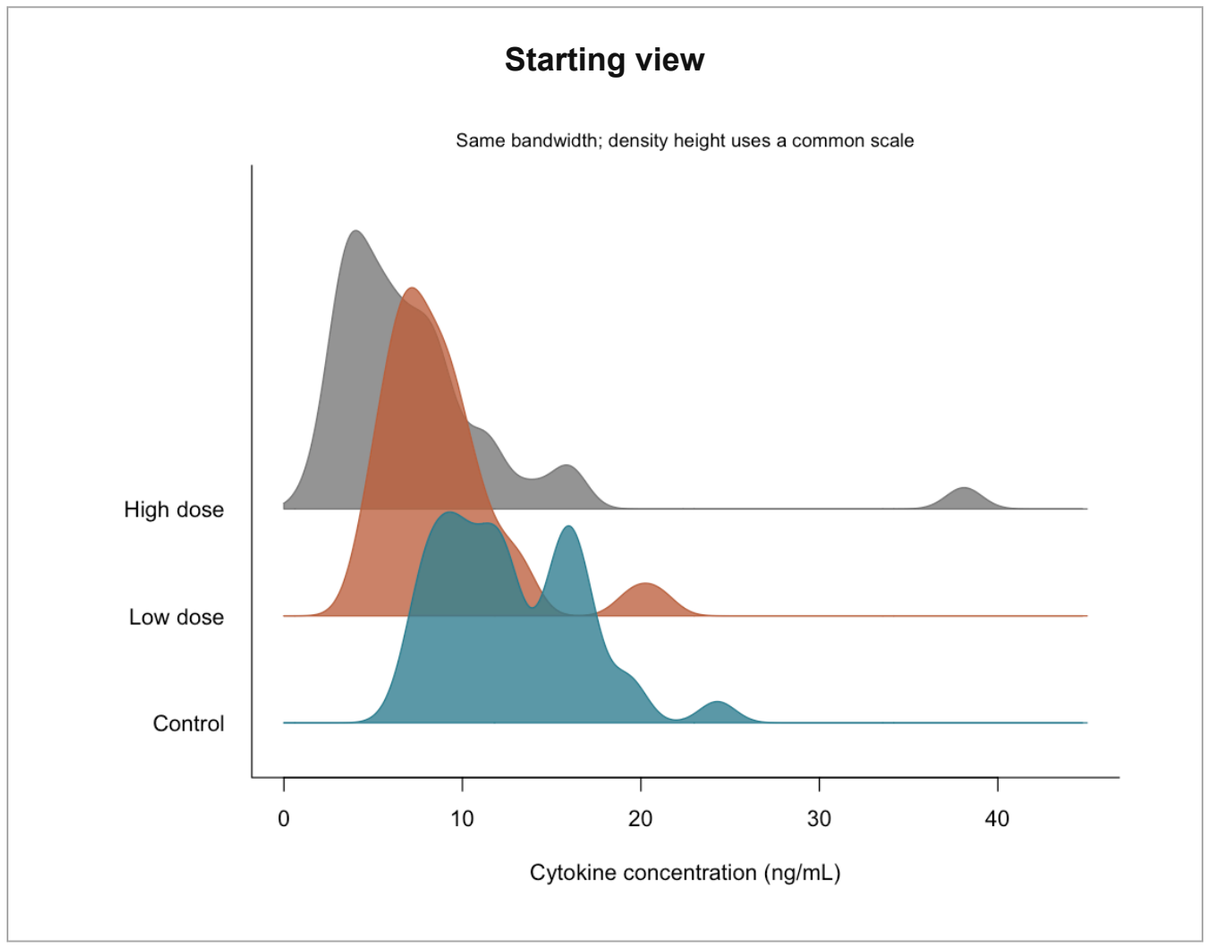
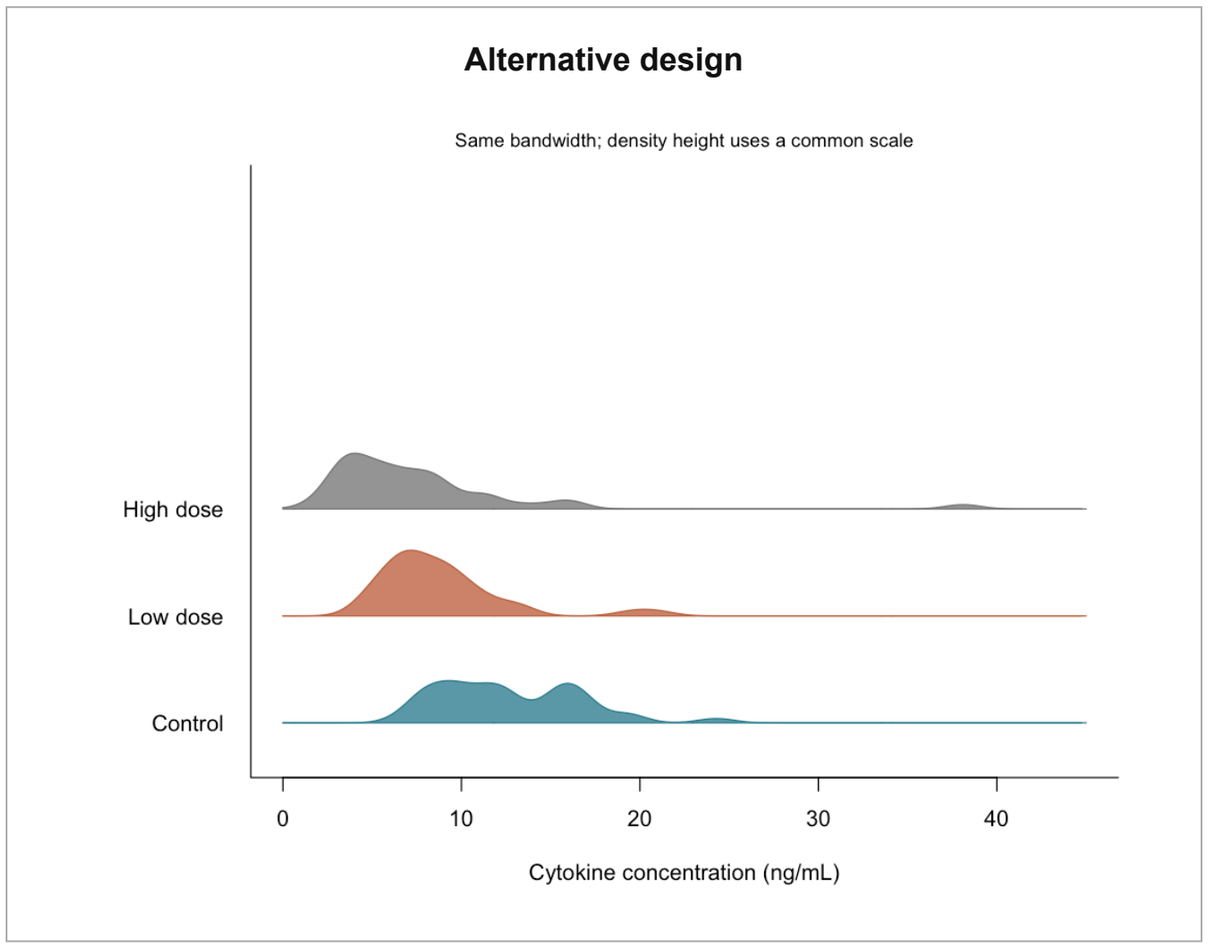




Tall overlapping ridges hide neighbouring distributions. Reducing the common density-height multiplier exposes all three. This changes vertical display scaling, not bandwidth or the concentrations.

> **Advice, examples & further exploration**

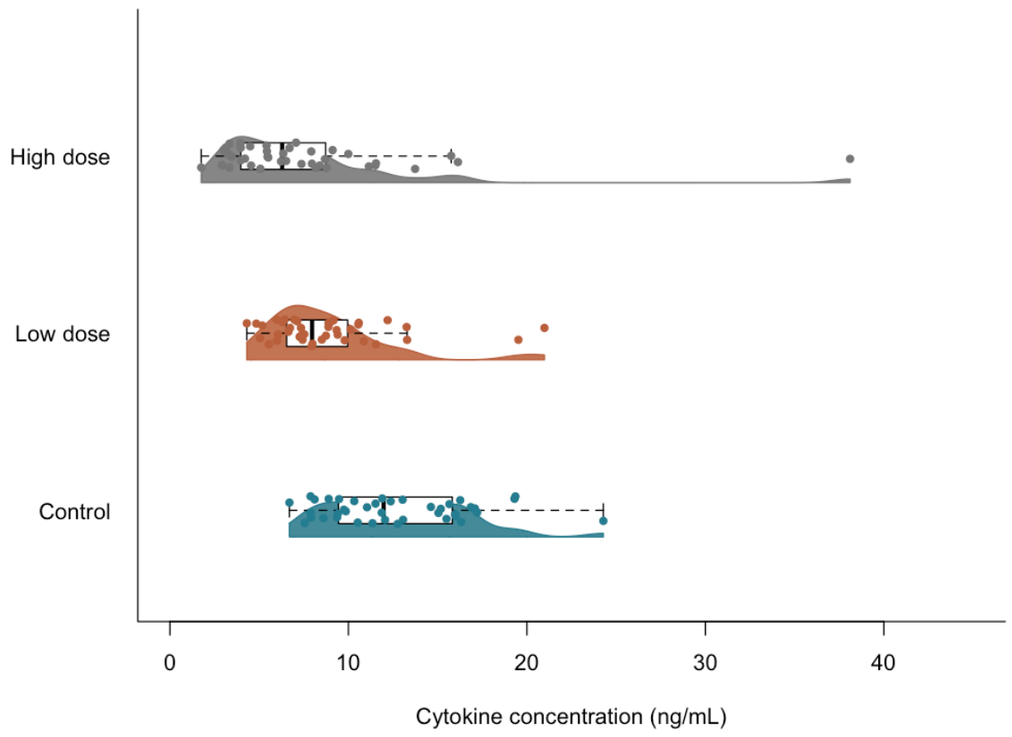
Raincloud plot

Combine a distribution estimate, compact summary, and individual observations.

Separate the layers

The same 40 cytokine measurements per dose group, using identical bandwidth, values and jitter in both versions.

Starting view



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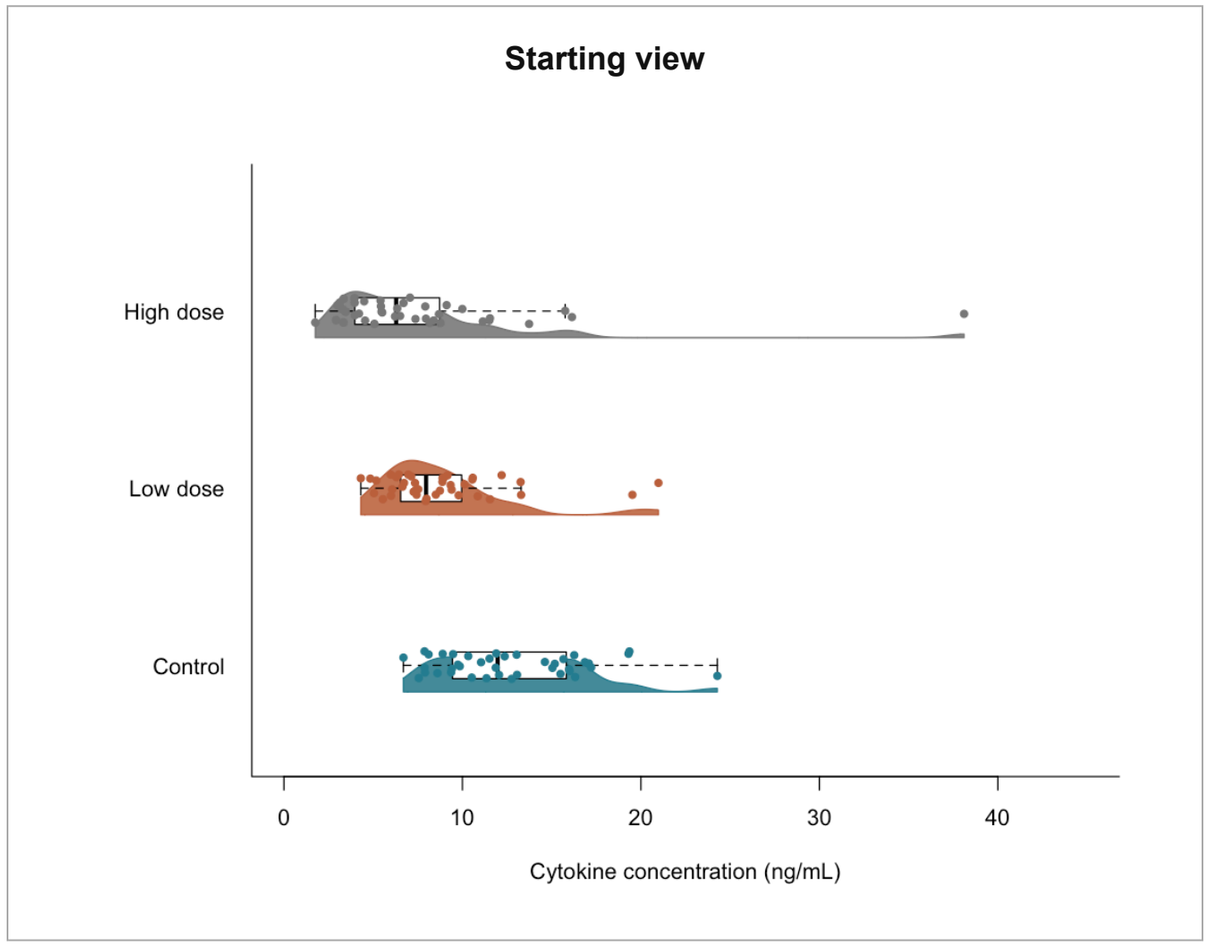
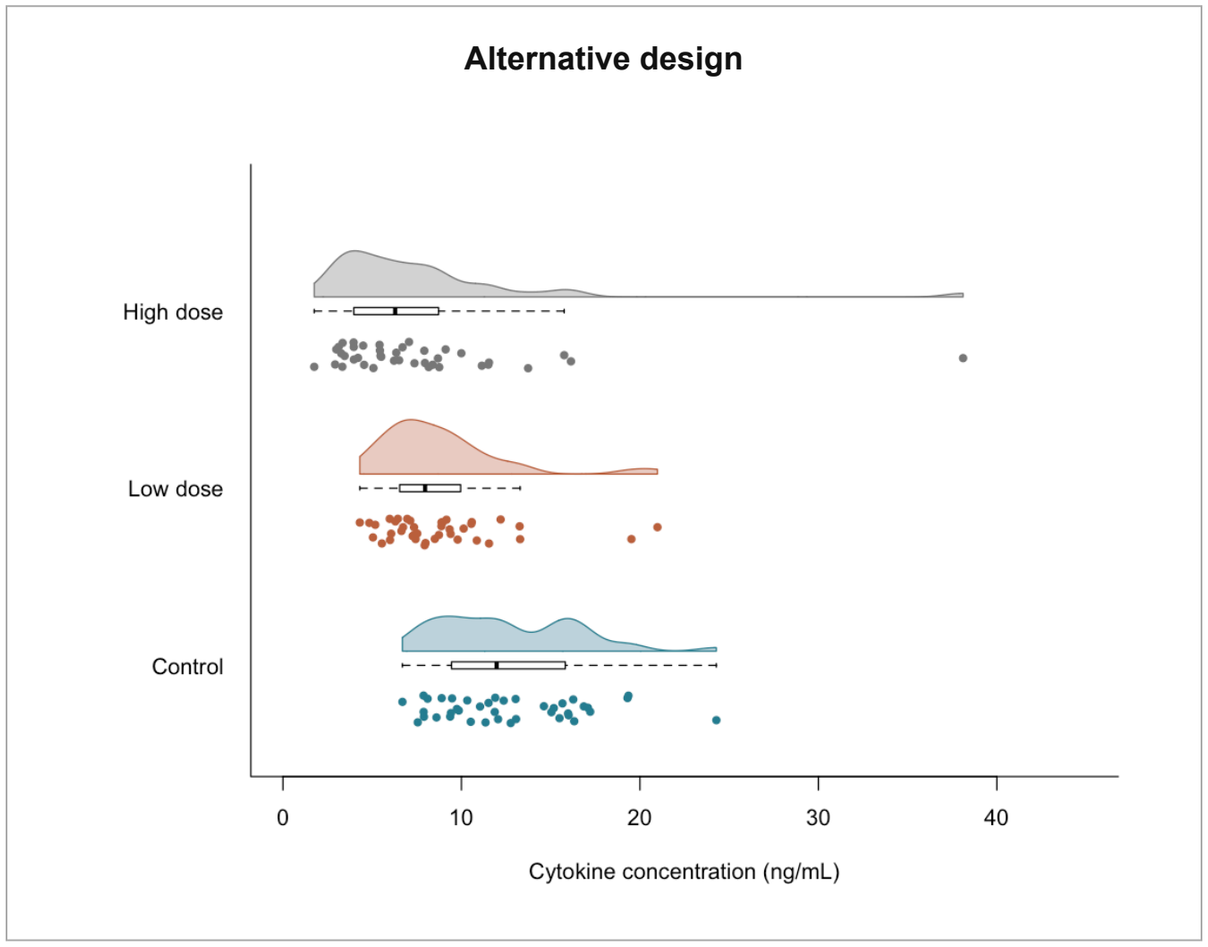




Putting dots, boxes and opaque densities on top of each other hides observations. Offsetting a light half-density and the dots lets each layer do a distinct job. Drop a layer if it does not help the question.

> **Advice, examples & further exploration**

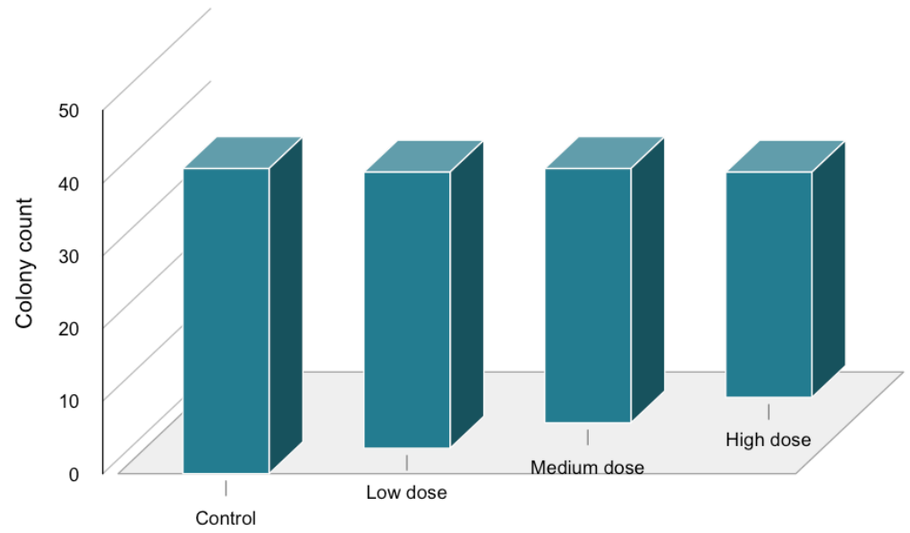
Bar plot

Compare counts, proportions, or other quantities across categories.

Remove decorative depth

The same four colony counts.

Starting view



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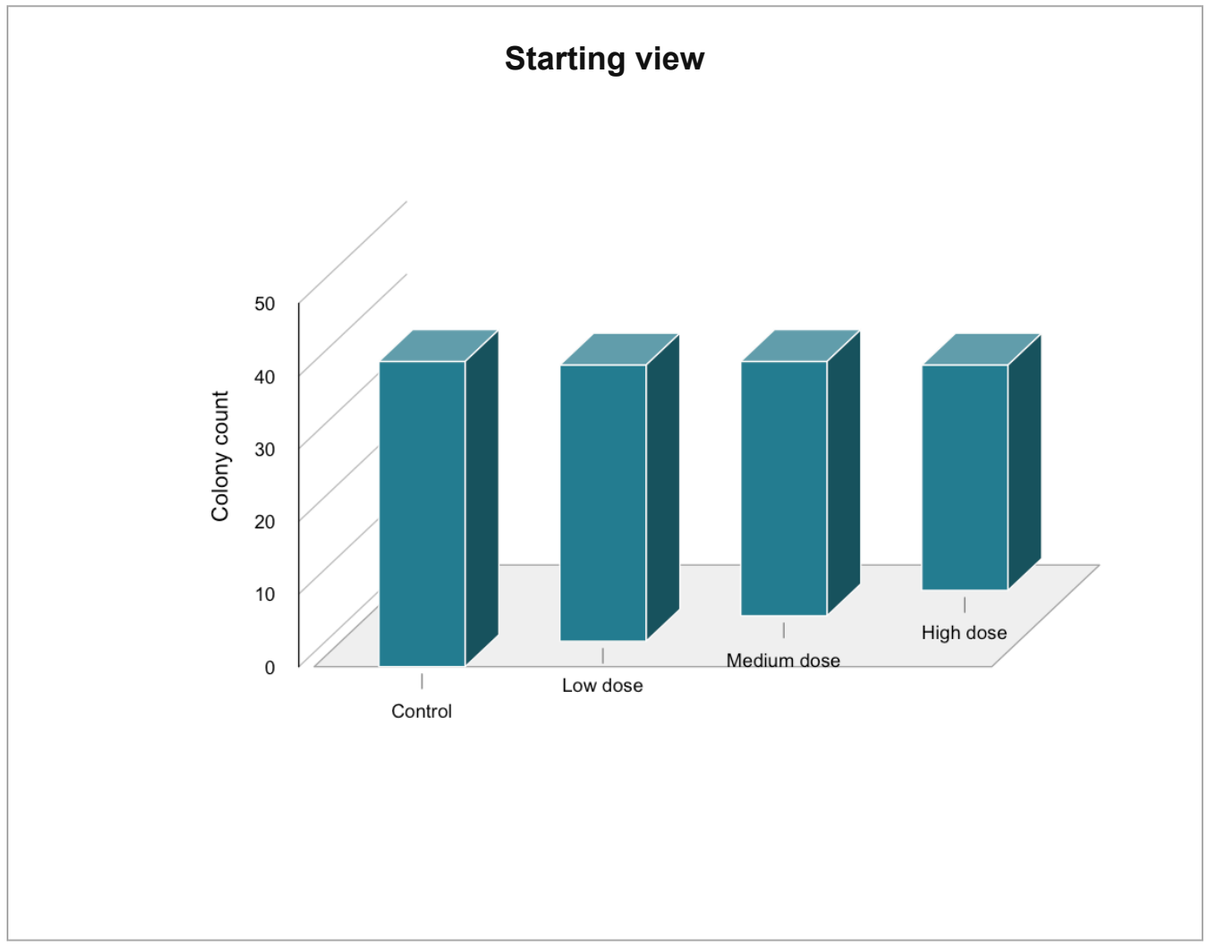
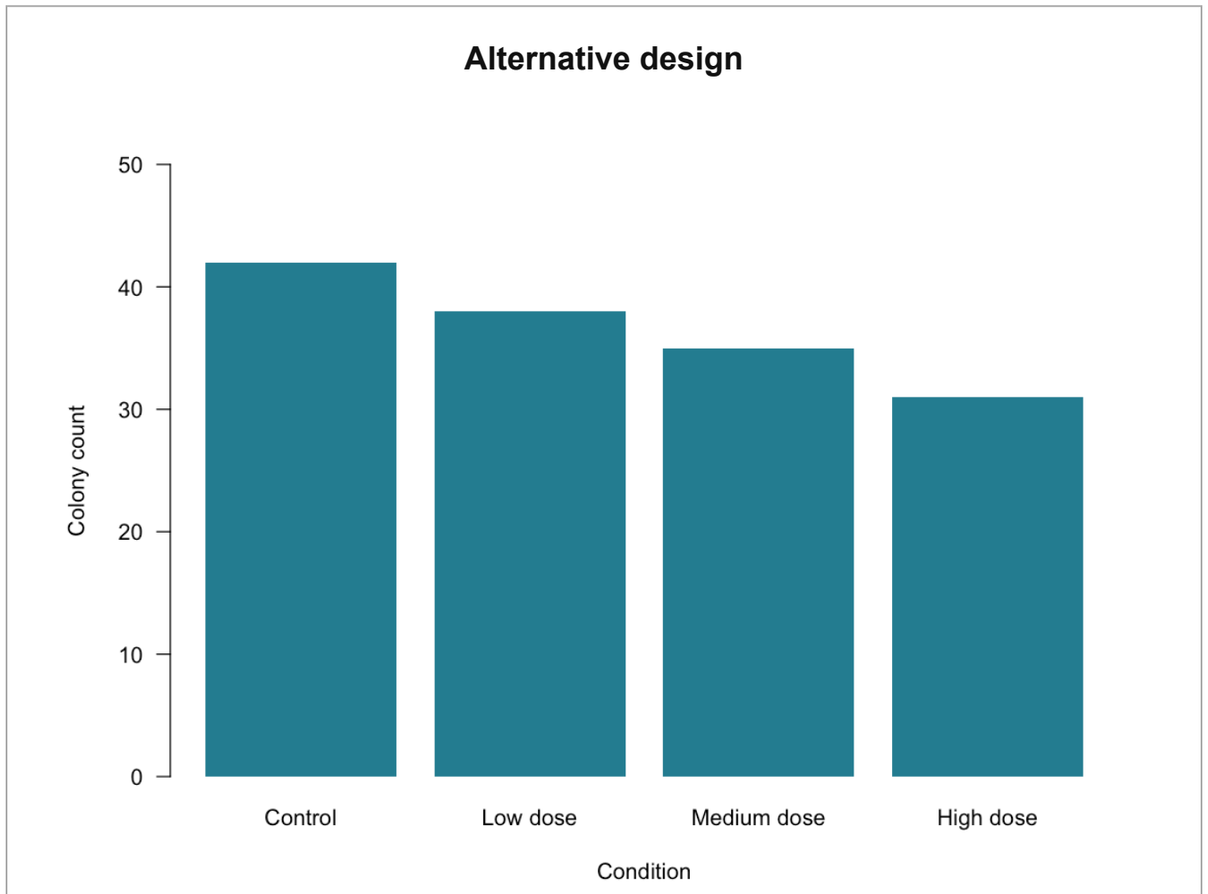
Remove decorative depth

The same four colony counts.





A shared zero baseline makes heights comparable. Perspective adds visual displacement without adding data.

> **Advice, examples & further exploration**

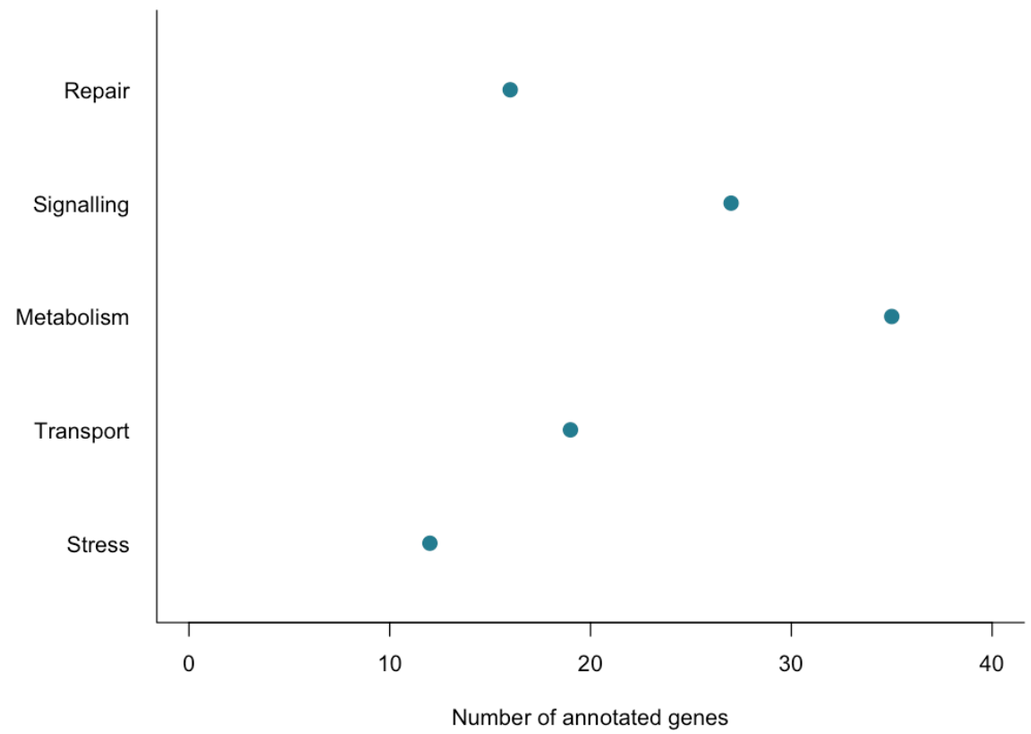
Dot / lollipop plot

Compare or rank category values using positions on a shared scale.

Order categories to help comparison

The same gene counts for five annotated pathways.

Starting view



Alternative design

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Advice, examples & further exploration

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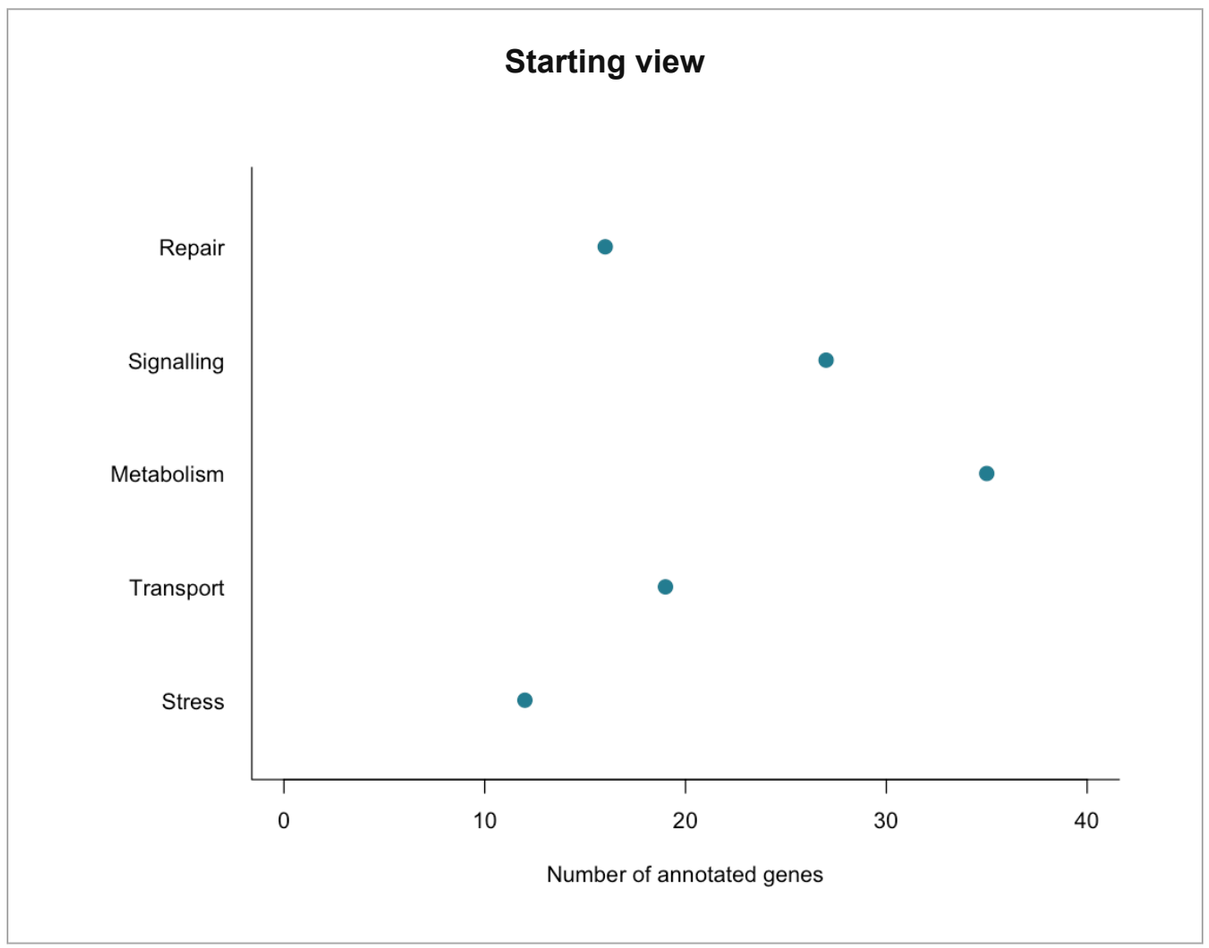
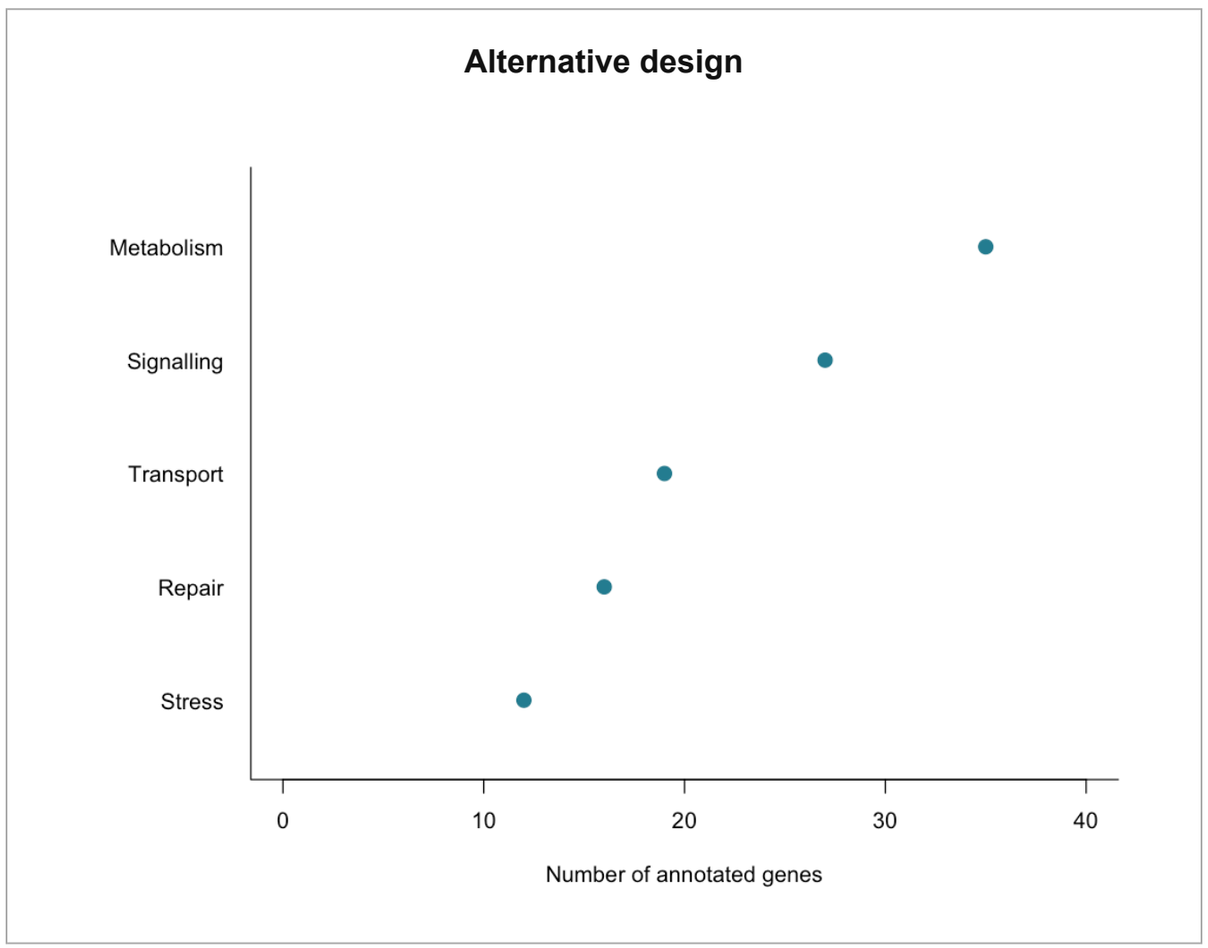




An arbitrary order makes the largest and smallest values harder to locate. Sorting supports ranking. Retain a fixed biological order instead when that order is meaningful across figures.

Advice, examples & further exploration

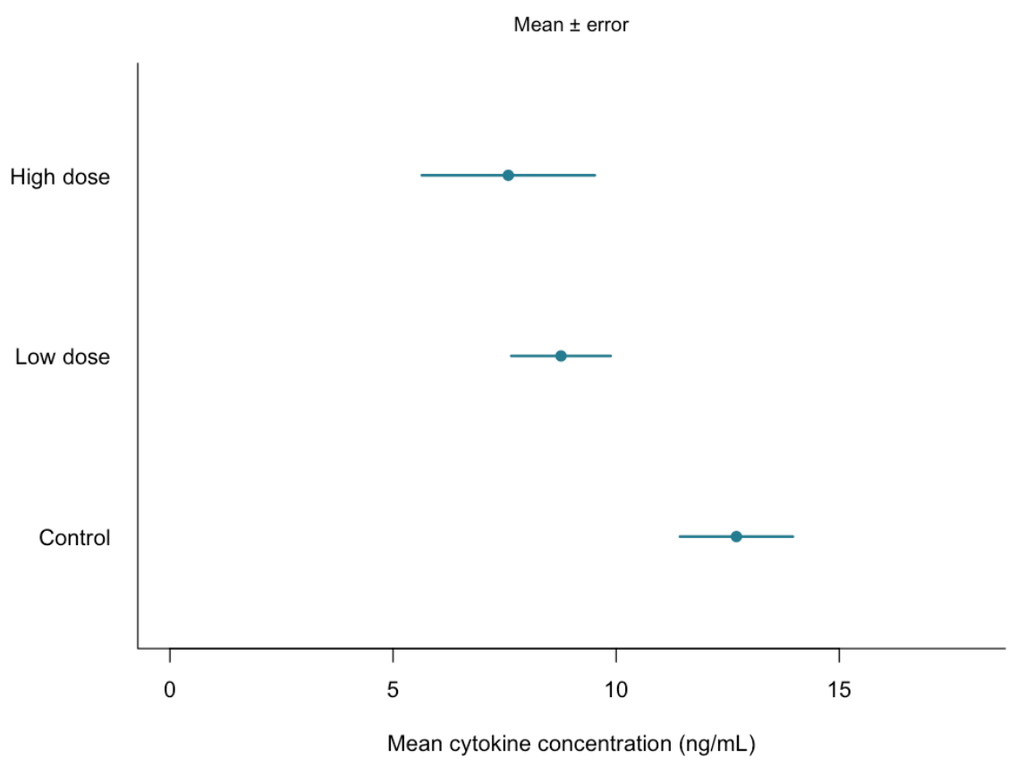
Point-and-interval plot

Show estimates together with explicitly defined uncertainty intervals.

Define the error bars

The same mean cytokine concentrations and intervals for 40 independent samples per dose group.

Starting view



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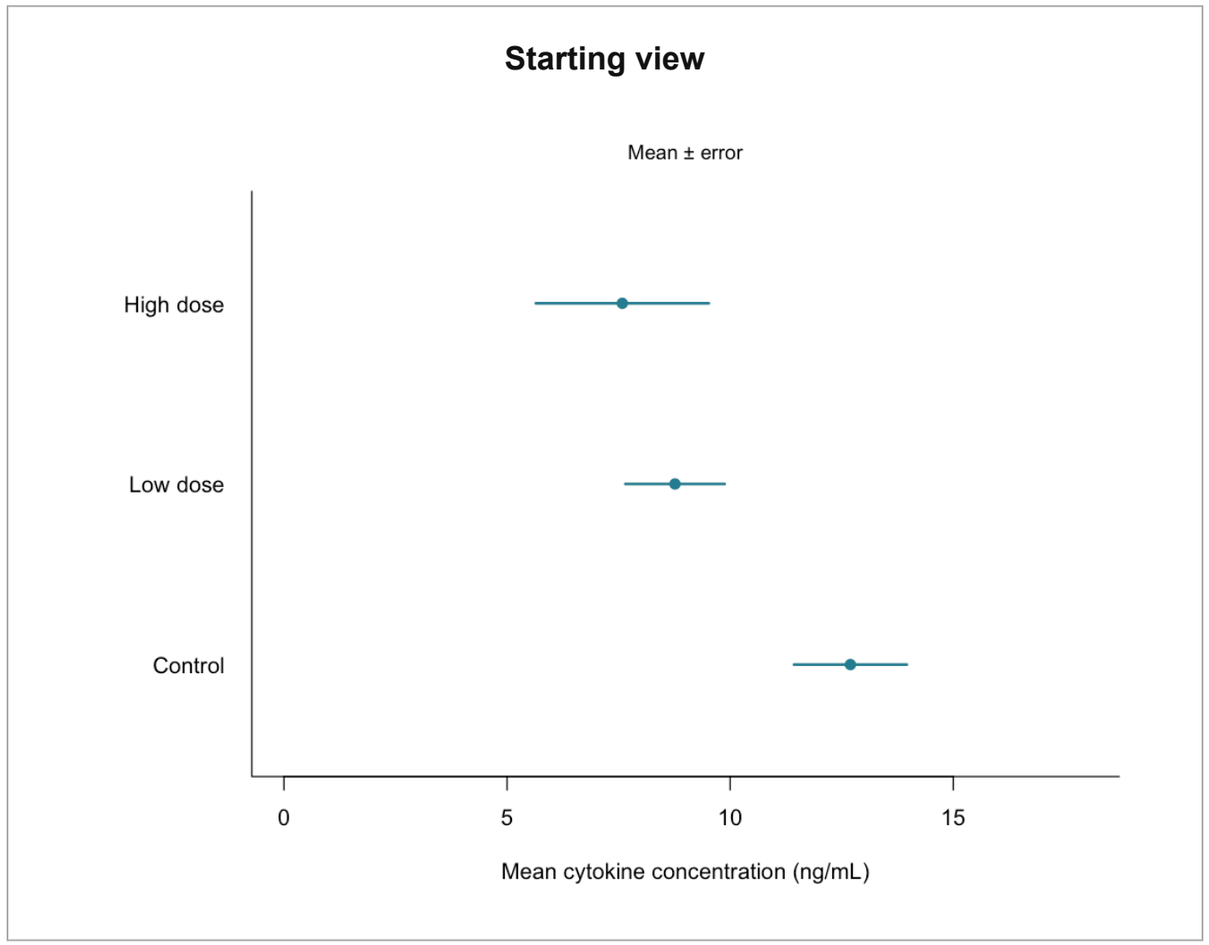
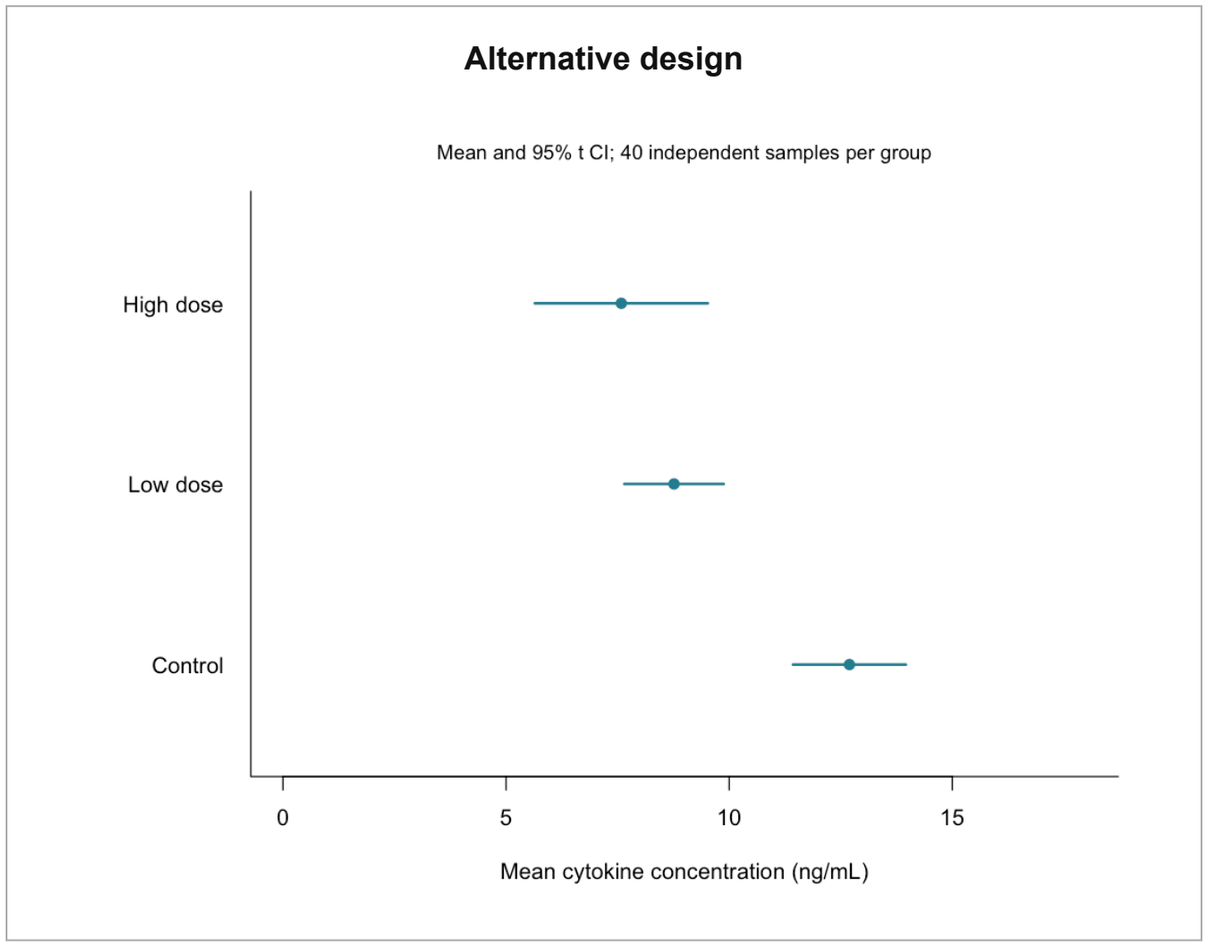




“Error” is ambiguous. Naming the 95% t confidence intervals and sample size tells readers that these describe uncertainty in group means, rather than the spread of individual measurements.

> **Advice, examples & further exploration**

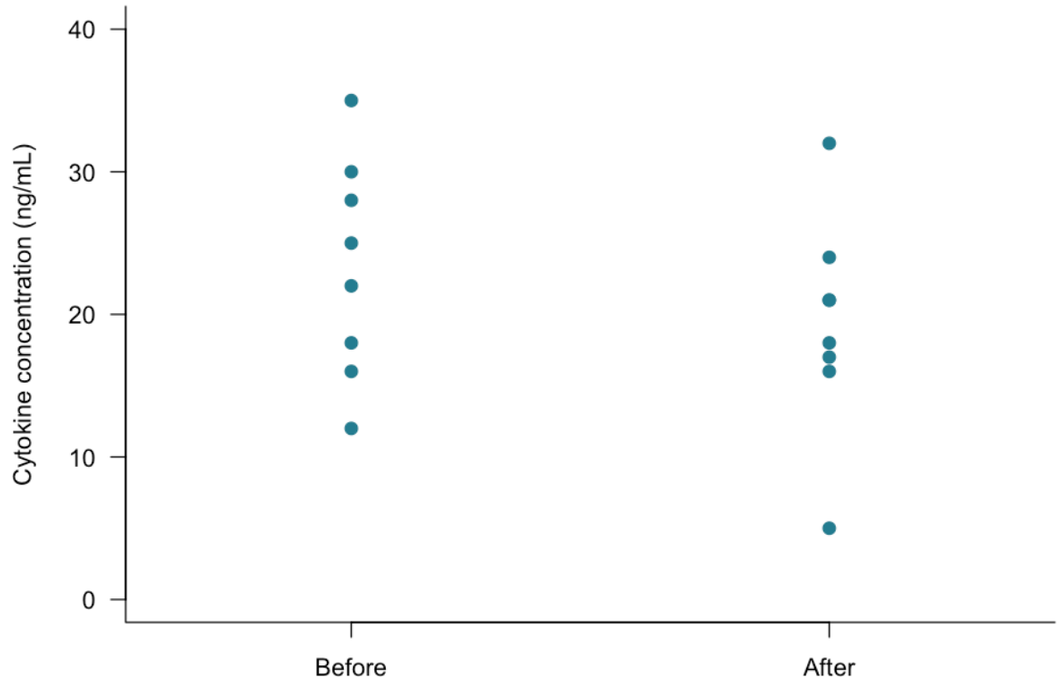
Paired plot

Show repeated measurements from the same donors, samples, or experimental units.

Keep the matching visible

The same before-and-after measurements in eight donors.

Starting view



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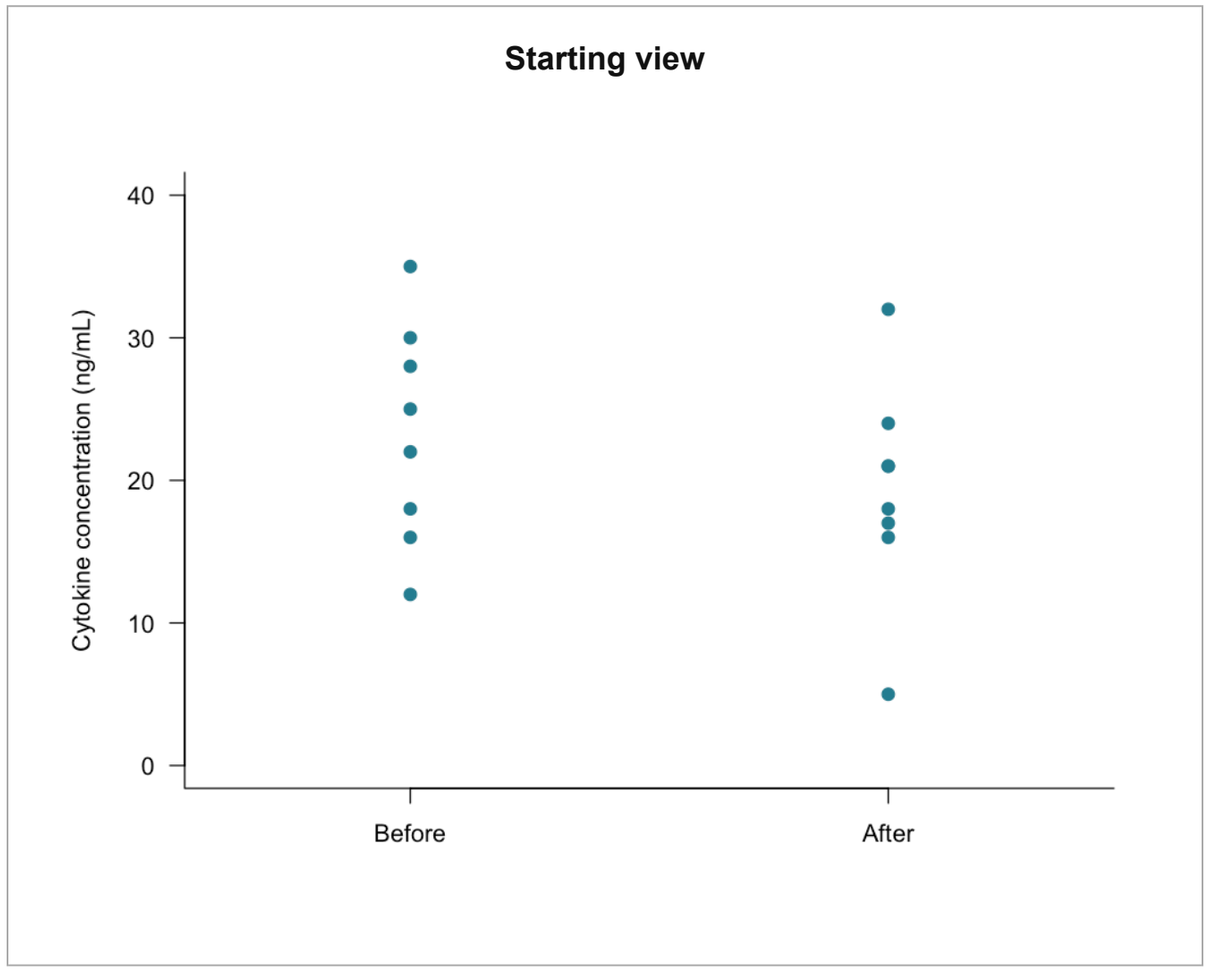
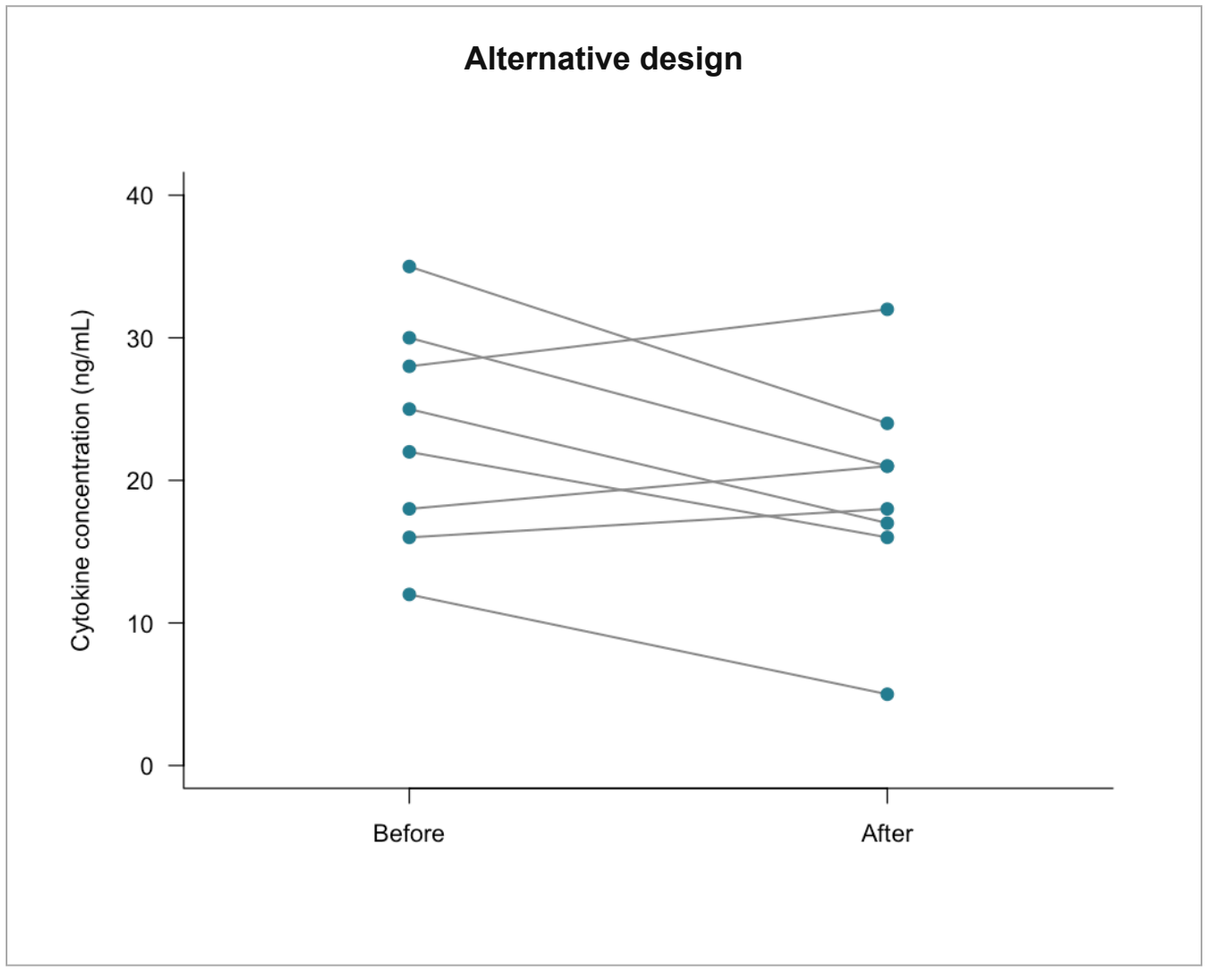




Connecting each donor reveals individual direction and magnitude. Separate groups conceal which observations belong together.

Advice, examples & further exploration

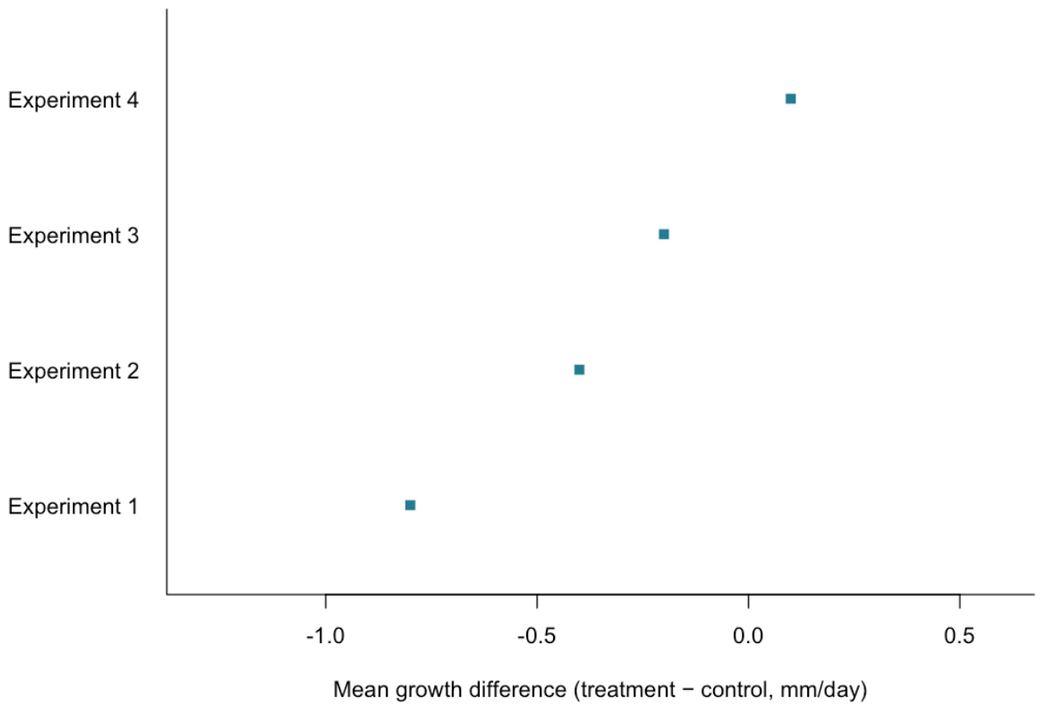
Forest plot

Compare effect estimates and uncertainty across studies, experiments, or subgroups.

Show how precise the estimates are

The same treatment–control growth differences from four experiments, in mm/day.

Starting view



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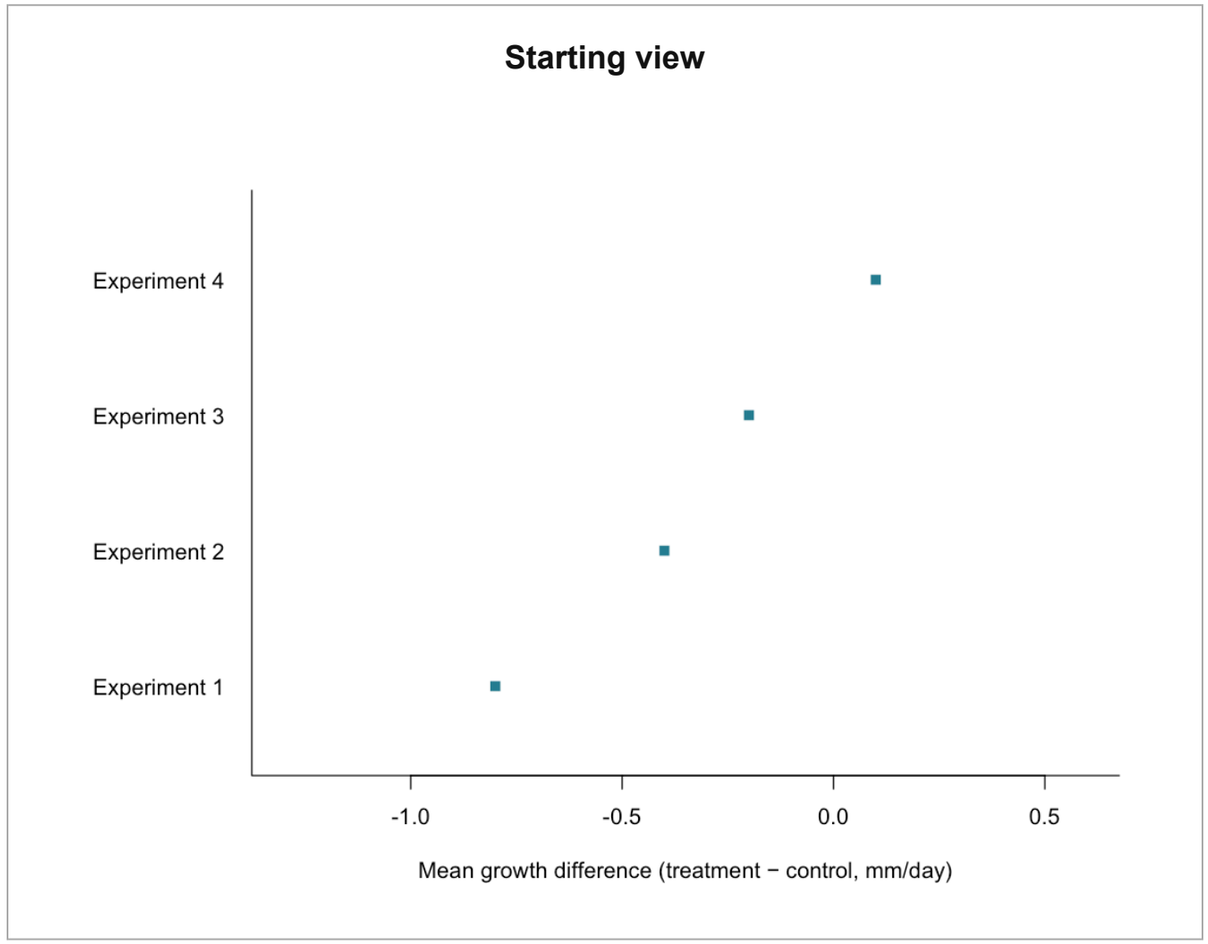
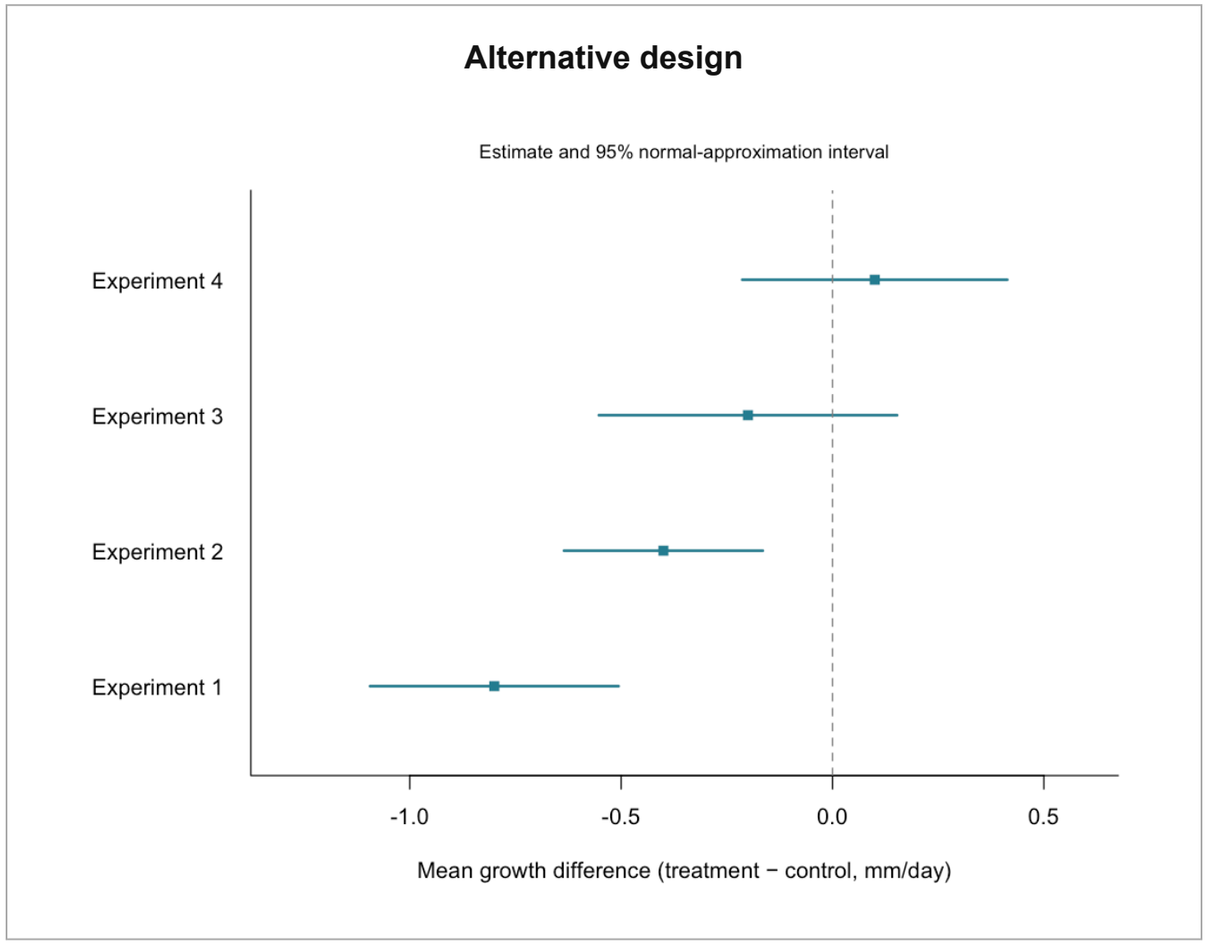




Estimates alone hide precision and their relationship to no effect. Adding 95% normal-approximation intervals and a zero reference makes both visible. An interval crossing zero does not establish equivalence.

> **Advice, examples & further exploration**

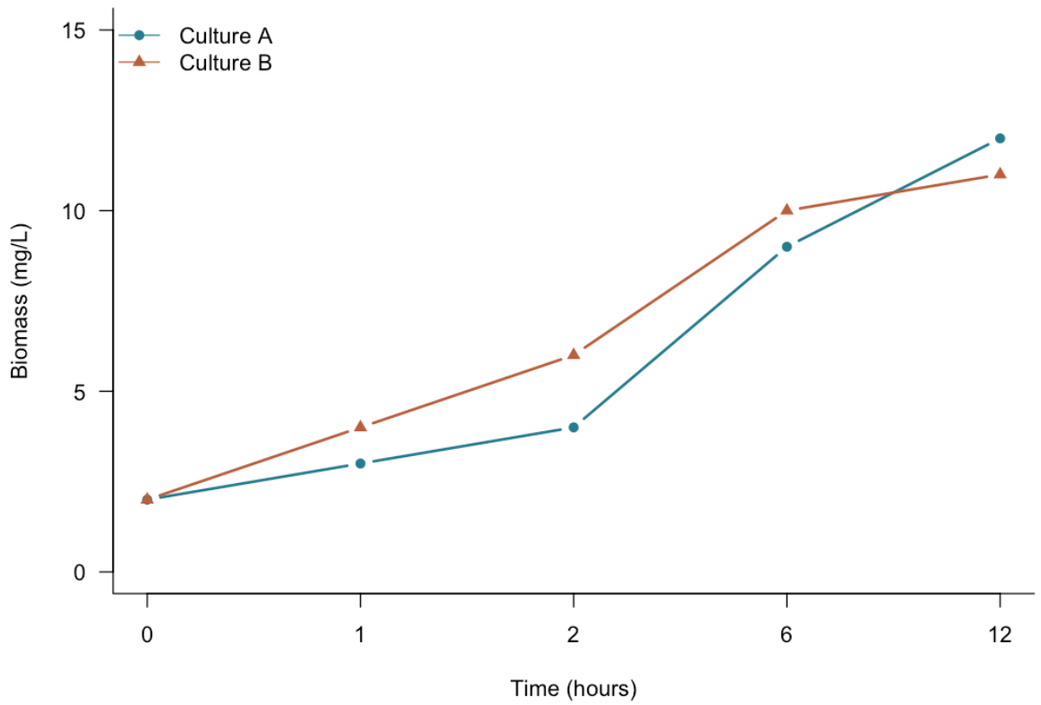
Line plot

Show change along an ordered variable, usually time.

Keep time intervals proportional

Two cultures measured at 0, 1, 2, 6 and 12 hours; the vertical axis shows biomass in mg/L.

Starting view



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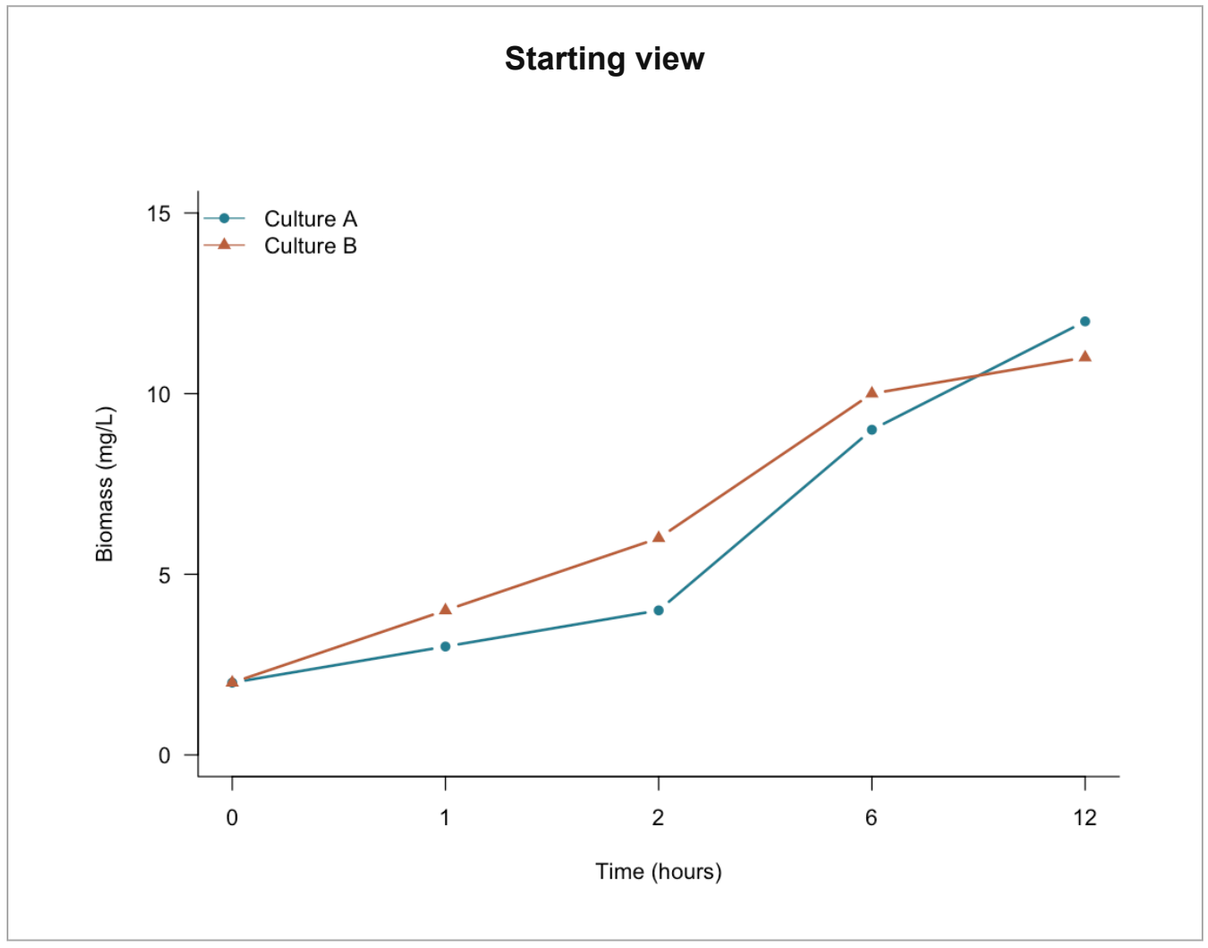
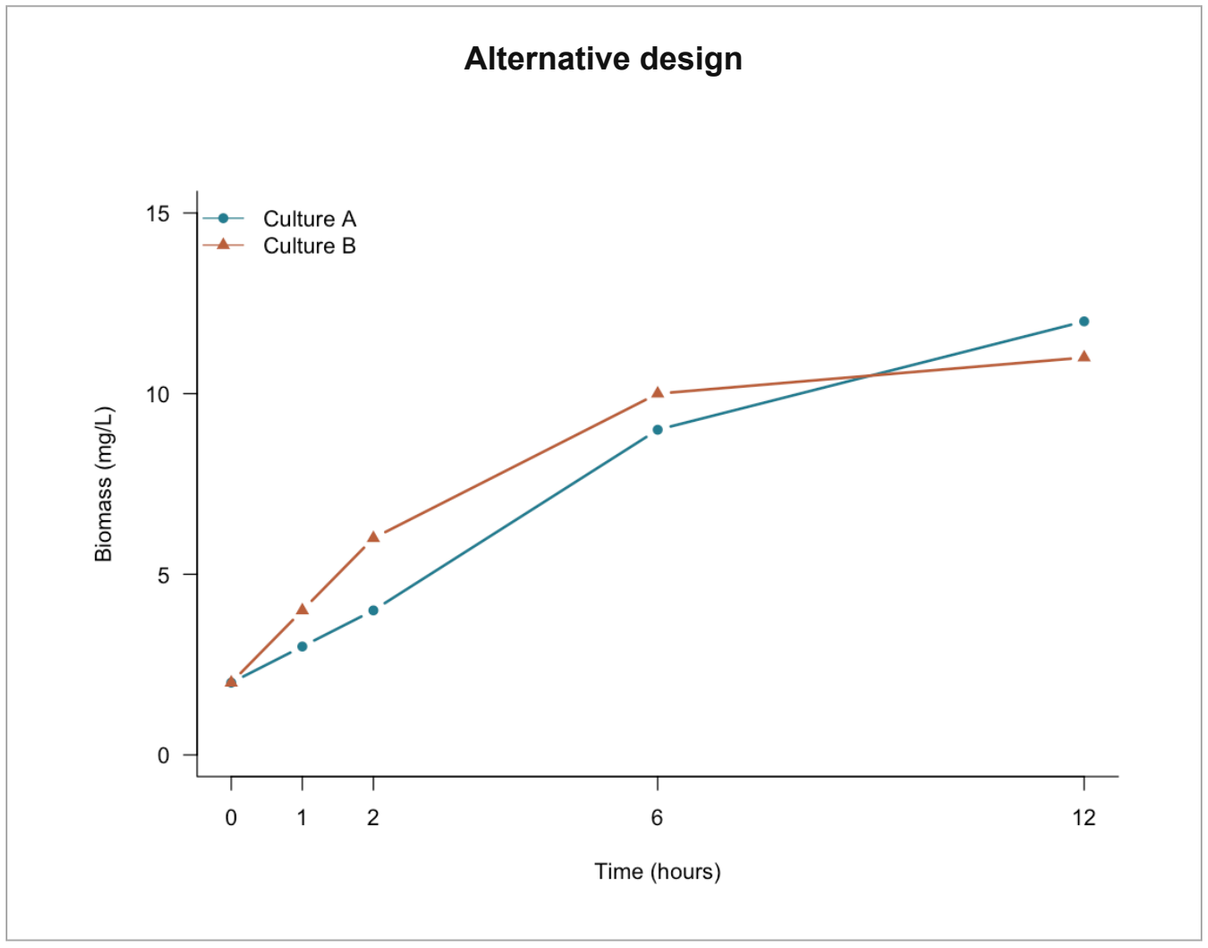




Equally spacing these time labels compresses the late intervals and distorts growth rates. Using elapsed hours as numeric positions preserves the slopes. Connect points only within the same culture.

Advice, examples & further exploration

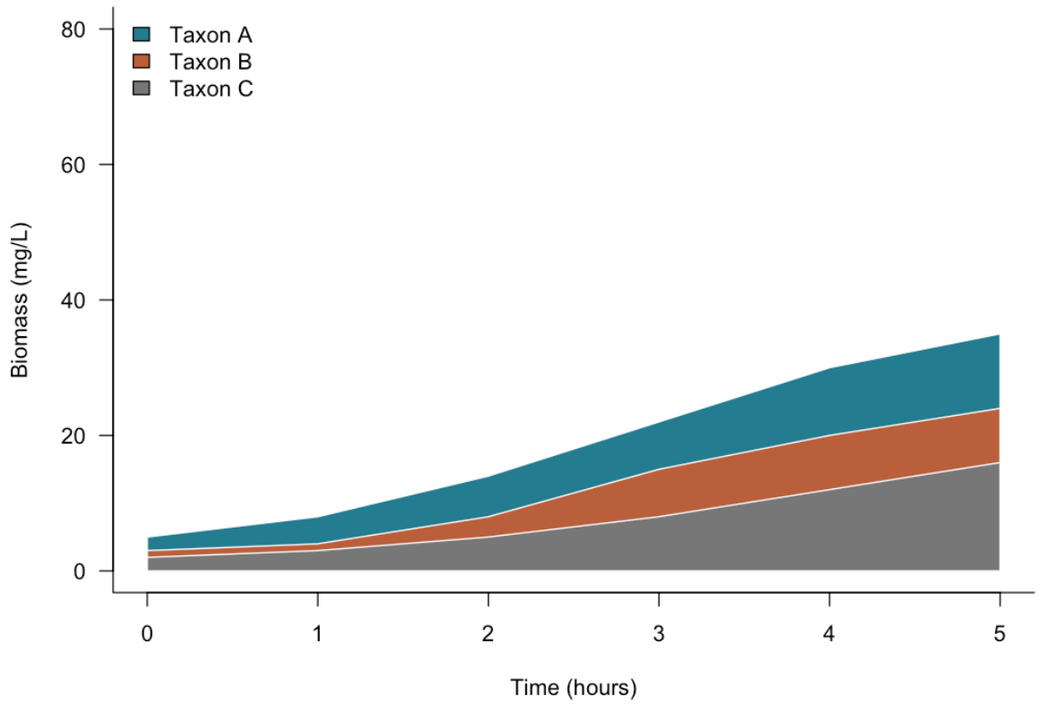
Area / stacked area plot

Show a total changing over time, optionally broken into additive components.

Do not cover one area with another

The same biomass measurements for three taxa over six time points.

Starting view



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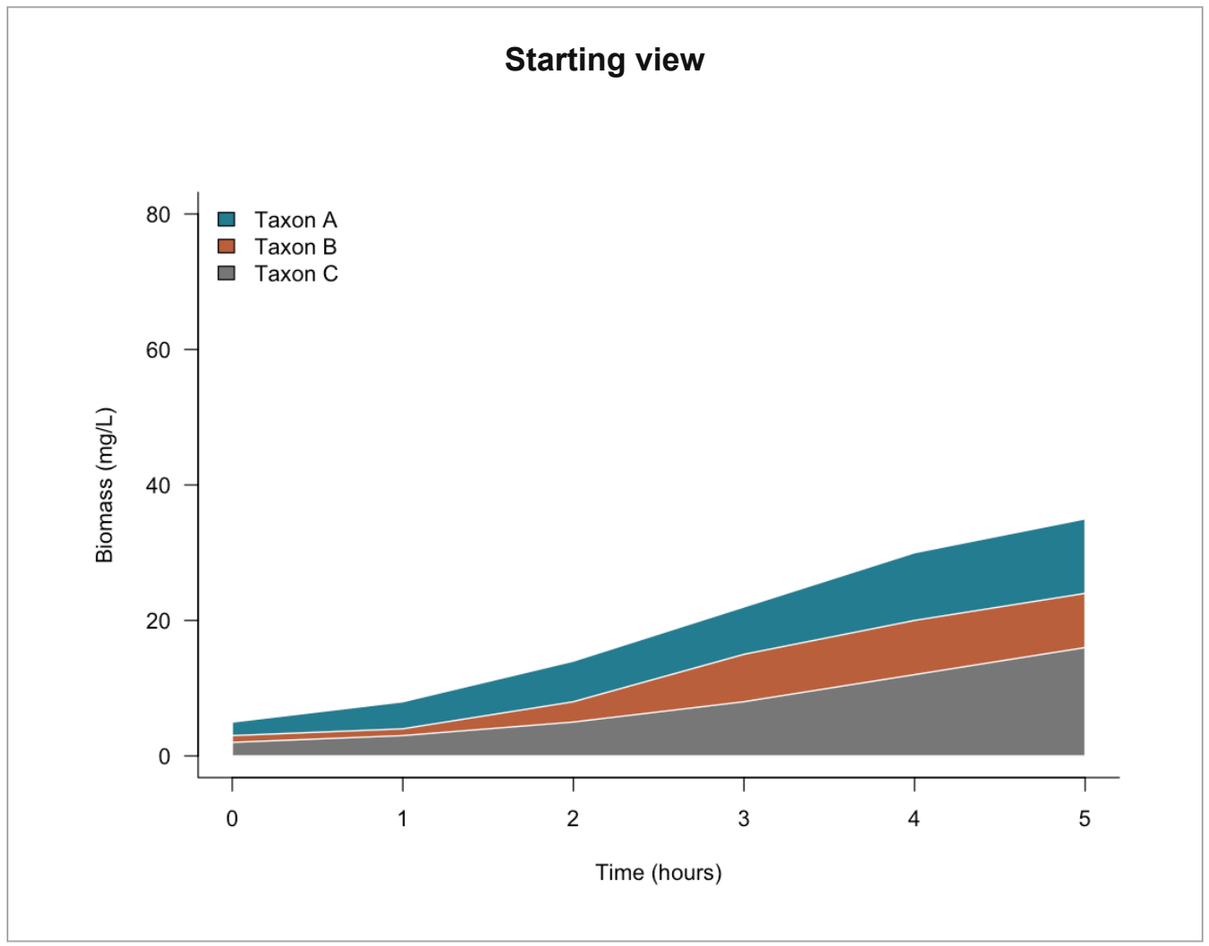
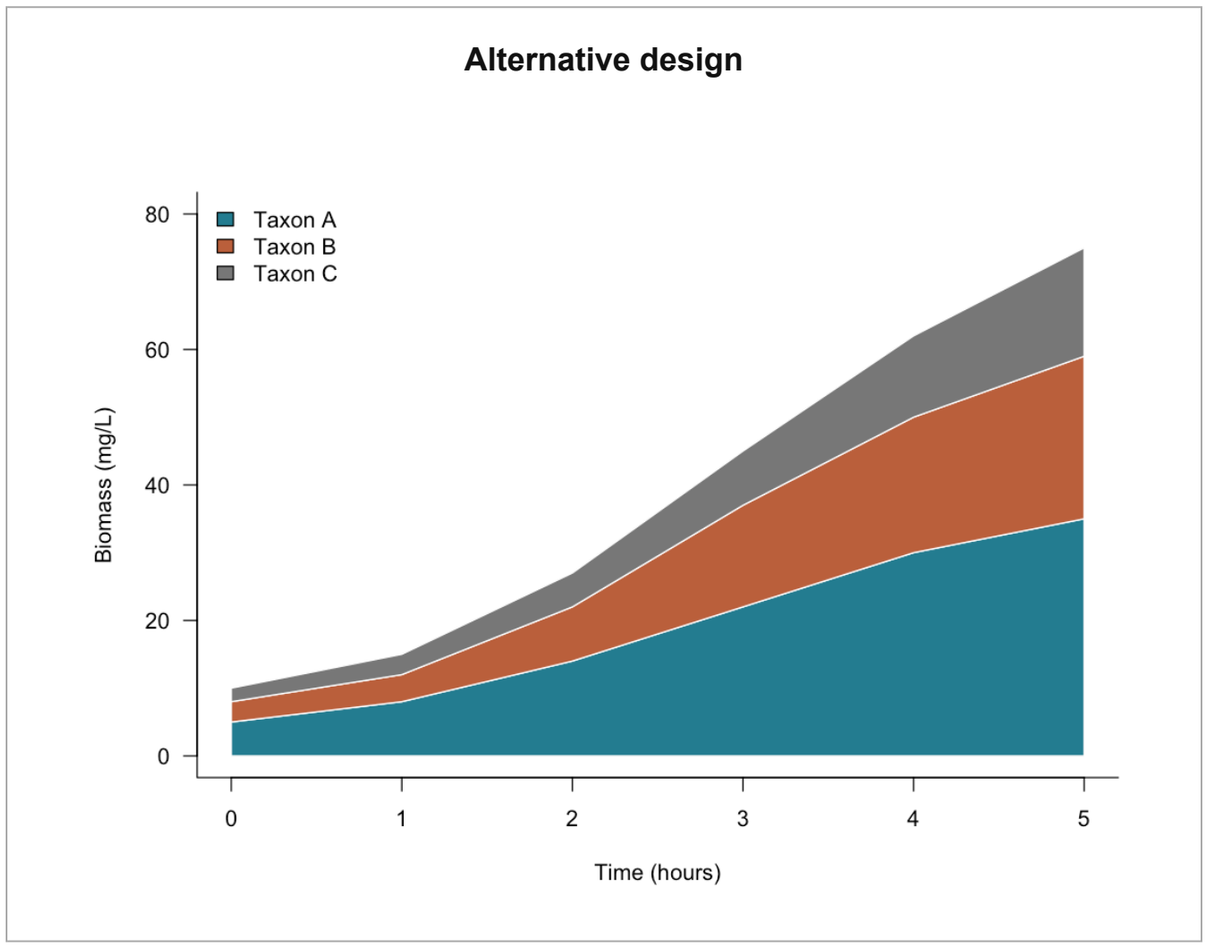




Opaque areas drawn from the same zero baseline obscure lower series. Stacking preserves their additive total and exposes each component. Middle components still lack a common baseline; use lines if their individual trajectories are the priority.

Advice, examples & further exploration

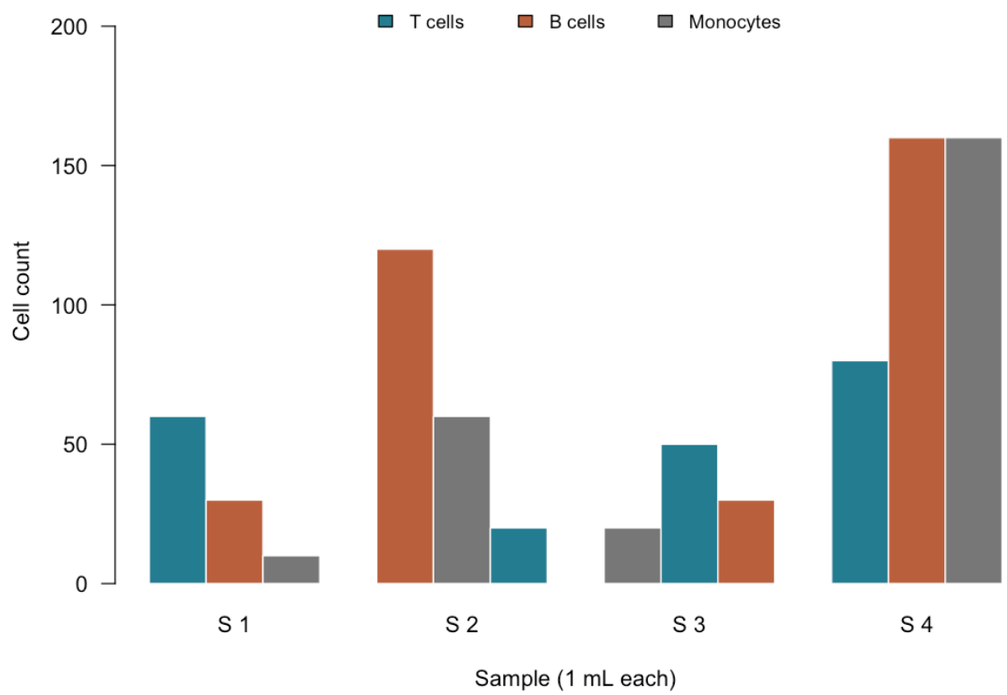
Grouped / side-by-side bar plot

Compare several categories within each sample, condition, or discrete time point.

Keep category colours consistent

The same T-cell, B-cell and monocyte counts in four 1 mL samples. Categories stay in the same bar order.

Starting view



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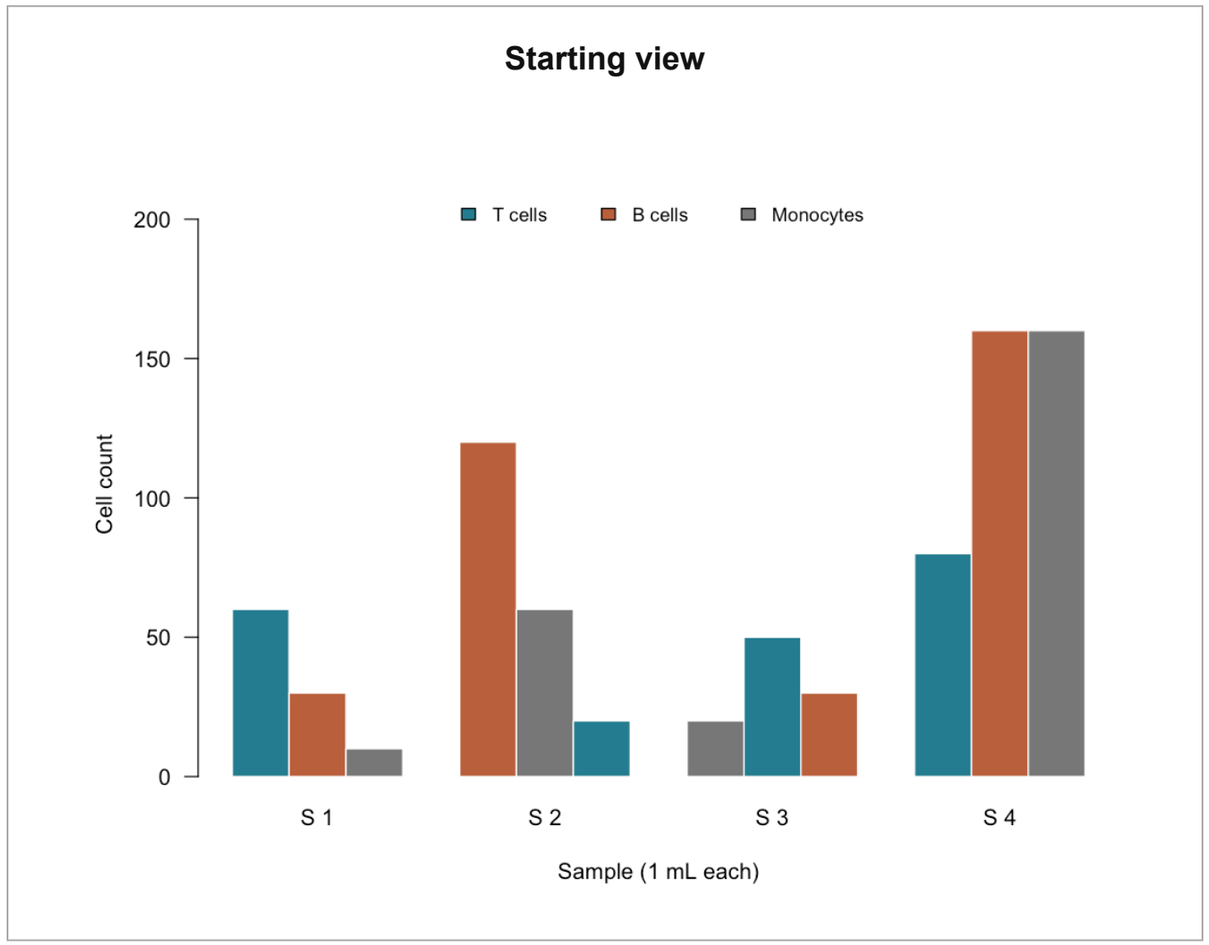
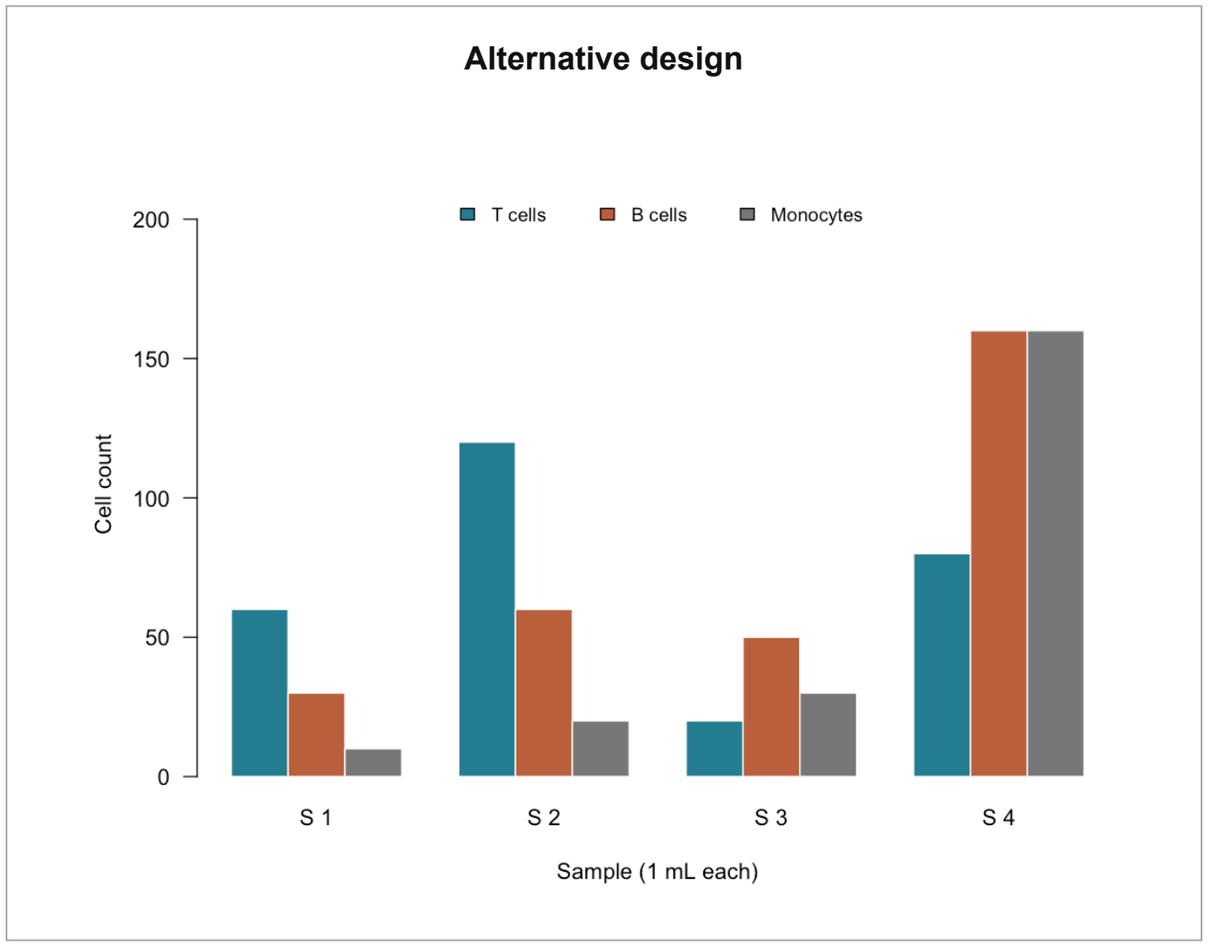




The first chart changes category colours between samples, conflicting with its legend. Fixed colours let readers trace each cell type across groups without repeatedly decoding positions.

> **Advice, examples & further exploration**

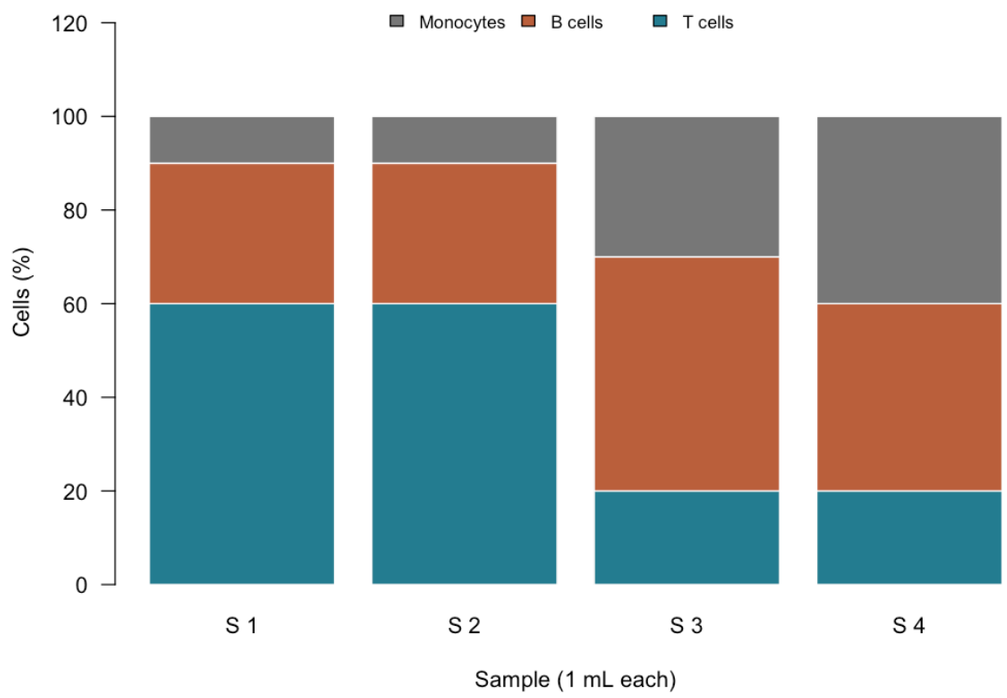
Stacked bar plot

Show how categories contribute to each sample’s total, as counts or as proportions.

Keep sample totals visible

The same cell counts from four 1 mL samples, displayed as percentages of T cells, B cells and monocytes.

Starting view



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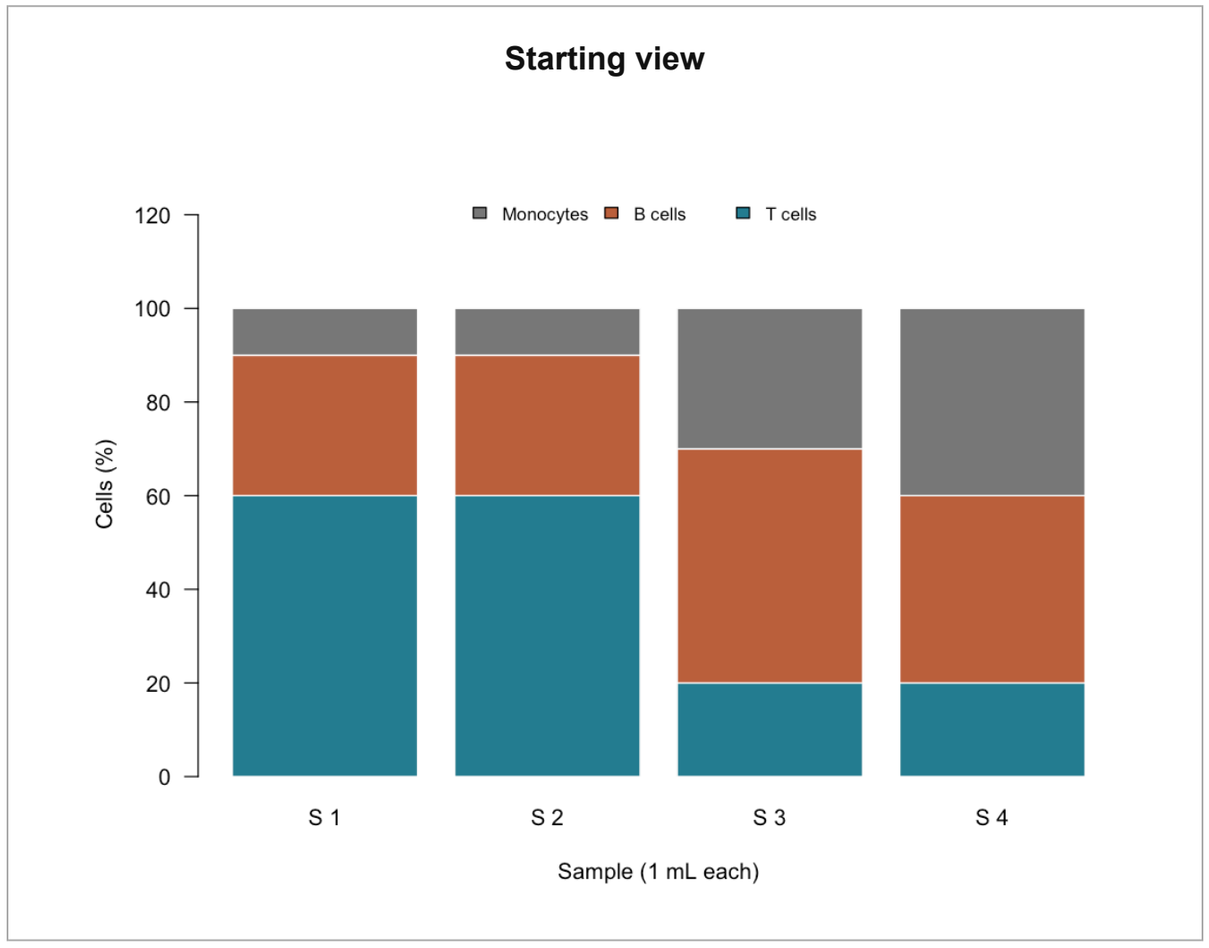
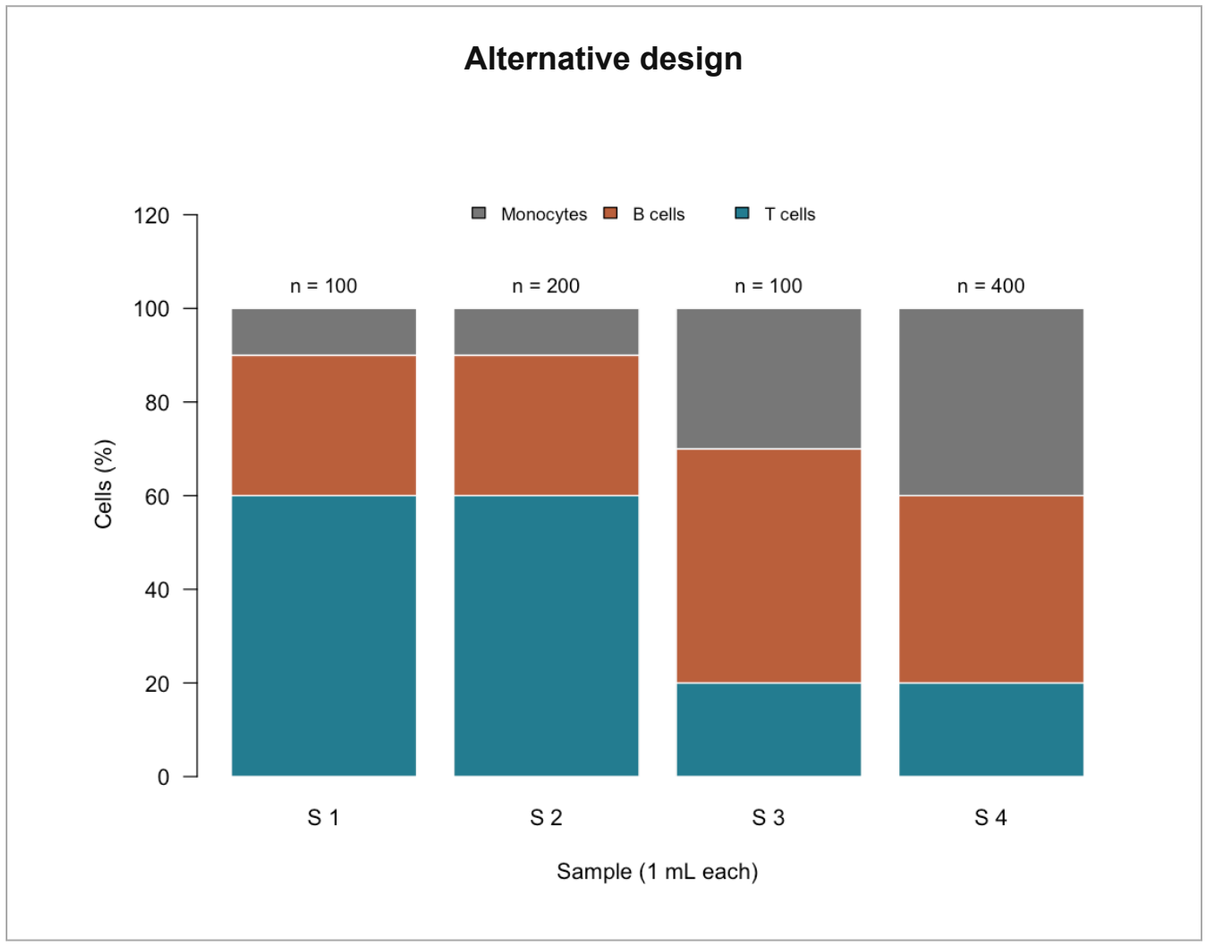




Both charts show the same proportions. Adding n exposes the different cell totals: S1 and S2 have identical composition but 100 versus 200 cells. Use absolute bars when the totals themselves are the main question.

Advice, examples & further exploration

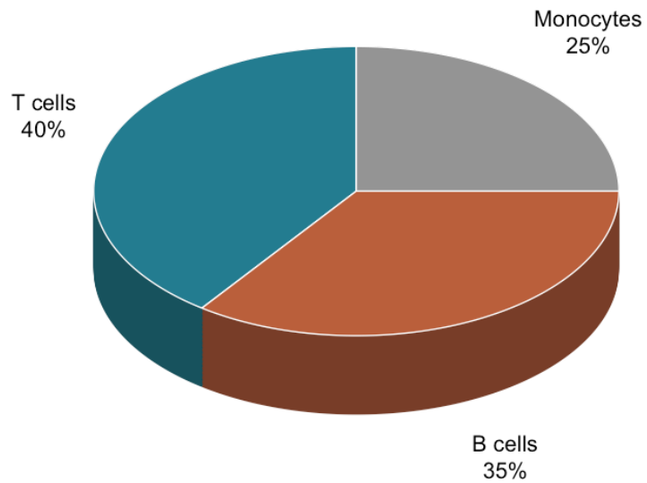
Pie / donut chart

Show a few mutually exclusive parts of one whole.

Remove 3D perspective

The same 40%, 35%, and 25% cell composition.

Starting view



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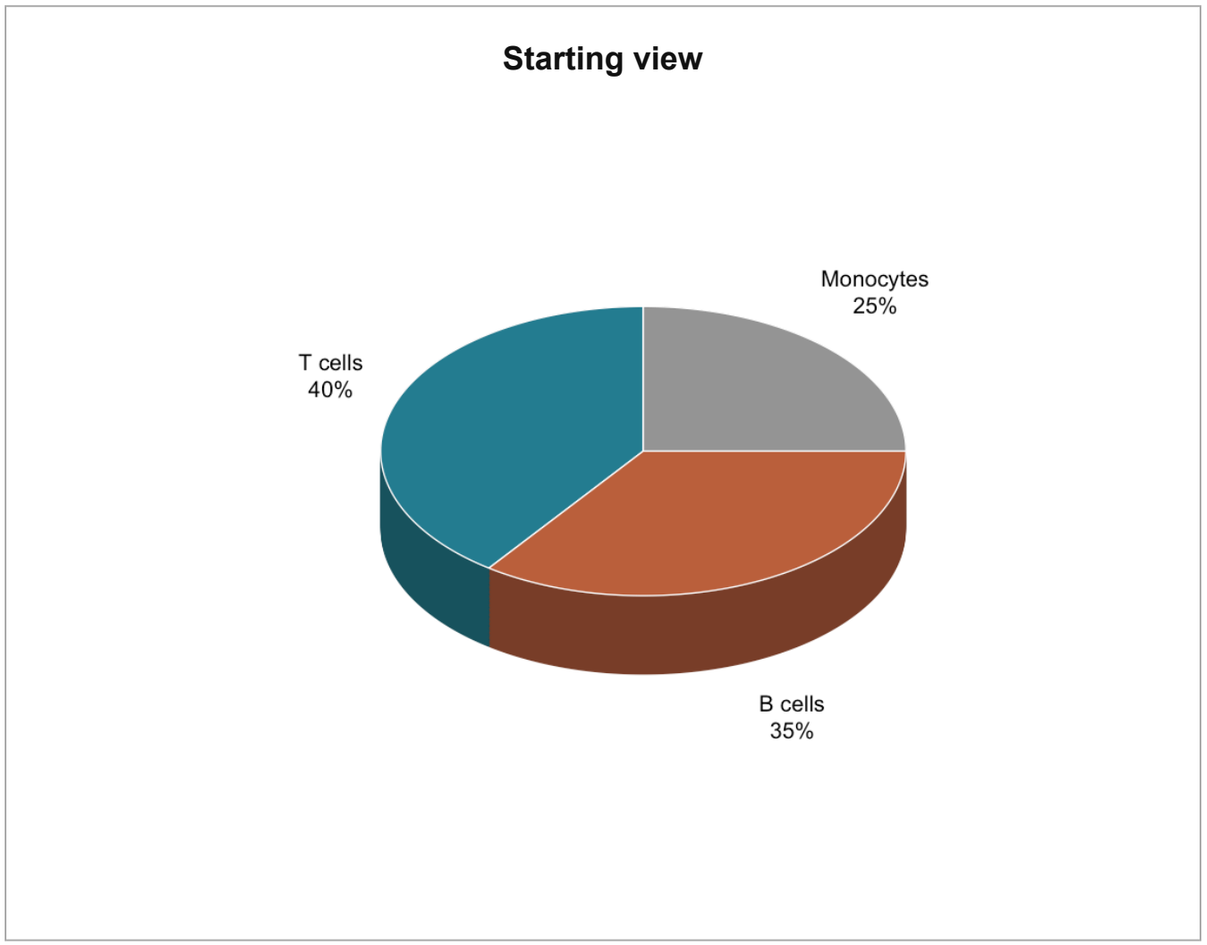
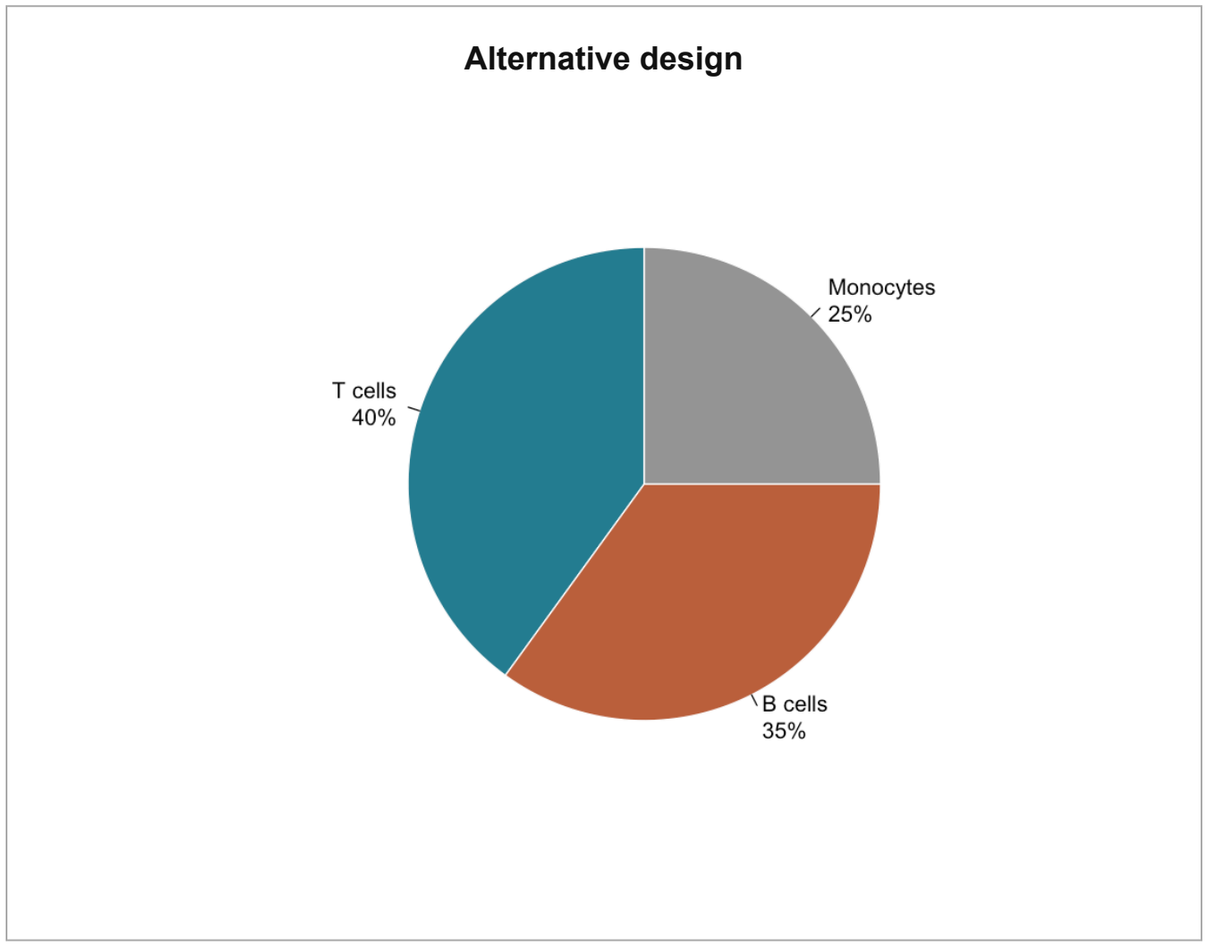




The foreground sidewall makes some slices look larger. Flat slices remove this distortion; aligned bars make close values still easier to compare.

> **Advice, examples & further exploration**

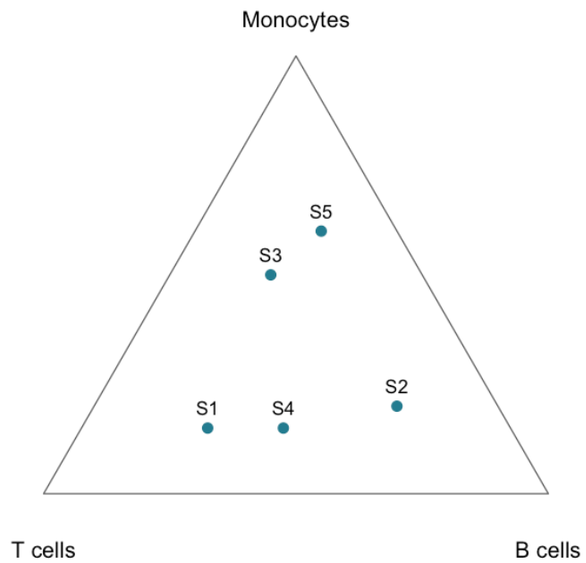
Ternary plot

Compare three components that sum to a fixed total, such as 100%.

Give readers a way to read proportions

The same five samples; T cells, B cells and monocytes sum to 100% within each sample.

Starting view



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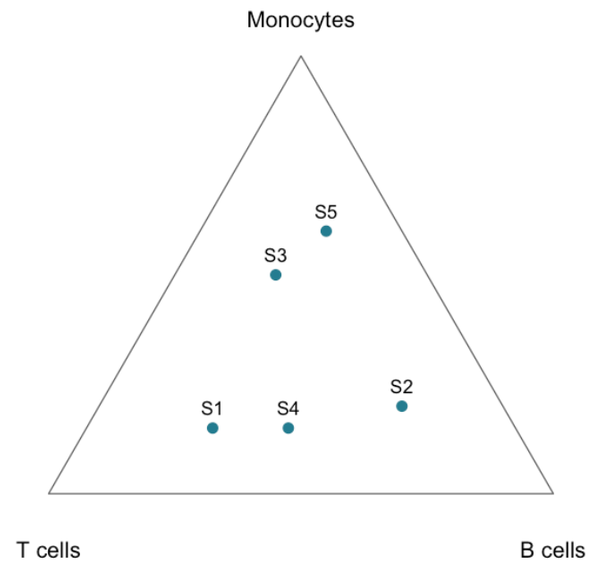

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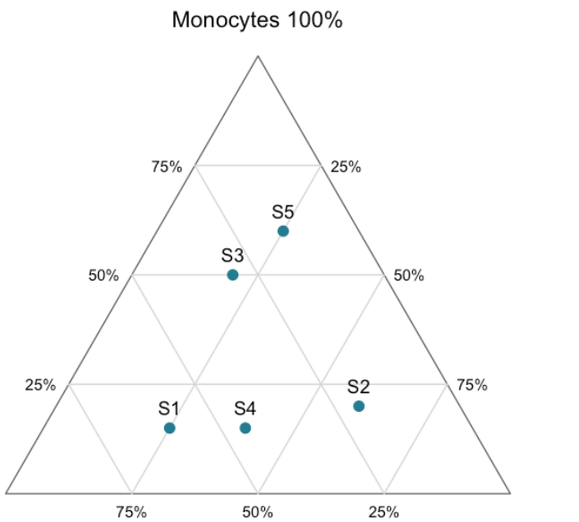


Vertices alone make intermediate proportions difficult to estimate. Grid lines, percentages and explicit 100% endpoints supply a reading guide without changing point locations.

Advice, examples & further exploration

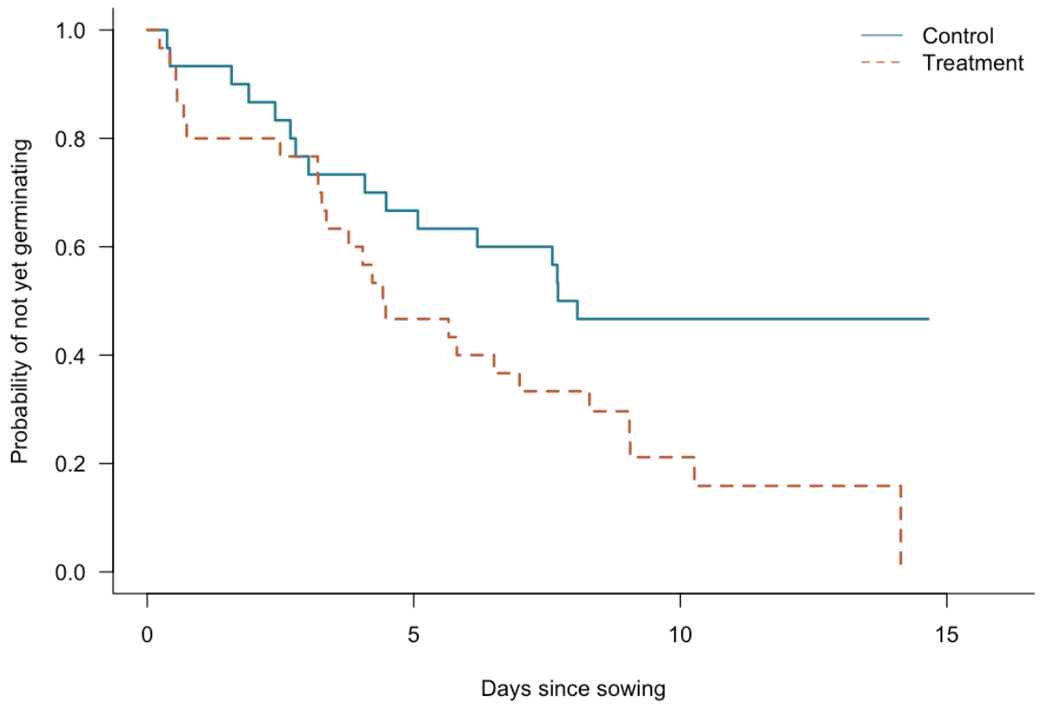
Survival / Kaplan–Meier curve

Show time until an event while retaining right-censored observations.

Make censoring and uncertainty visible

The same seed germination records from 30 seeds per treatment; height is the probability of not yet germinating.

Starting view



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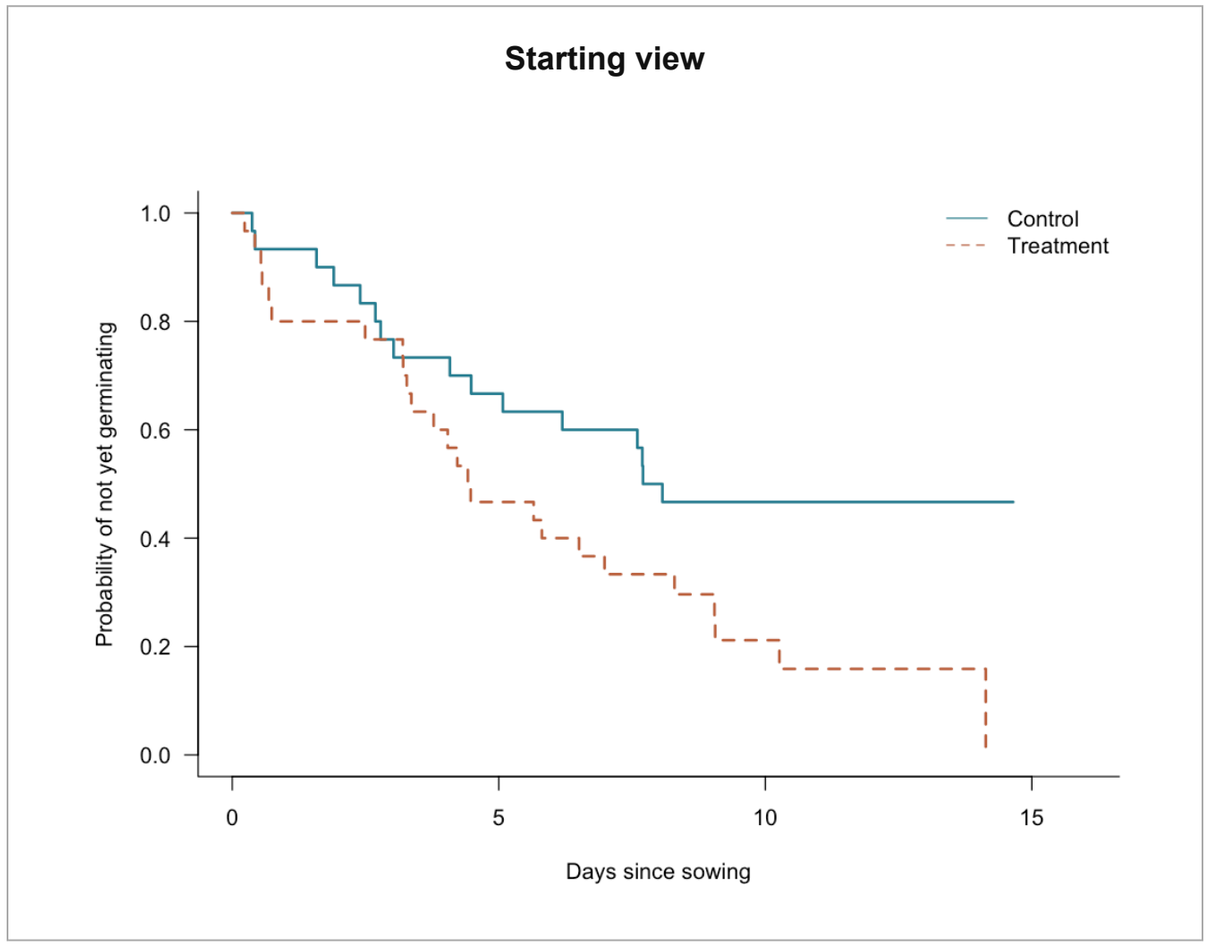
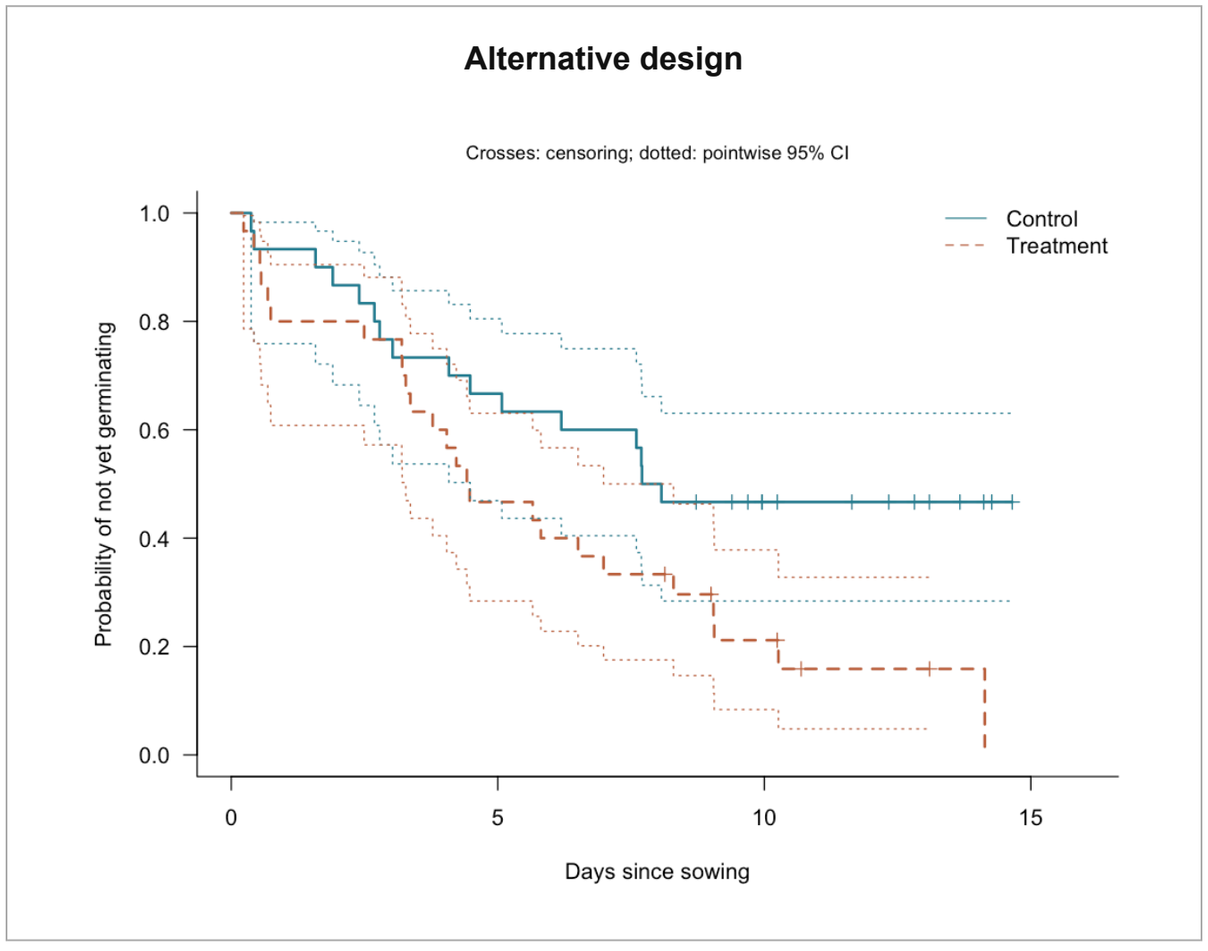




Bare curves look more certain than they are. Censoring marks and pointwise 95% confidence limits expose incomplete follow-up and estimation uncertainty. The sparse tails still need caution; at-risk counts would add useful detail.

> **Advice, examples & further exploration**

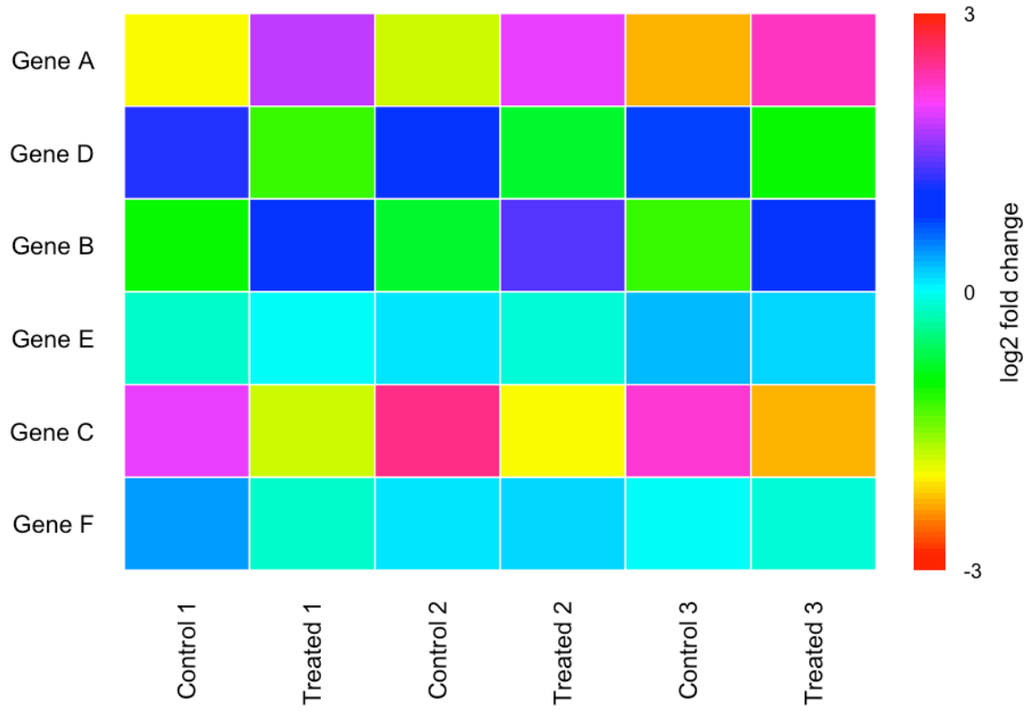
Heatmap

Find patterns in a matrix, such as measurements across genes and samples.

Make zero meaningful

The same gene-by-sample log2 fold changes.

Starting view



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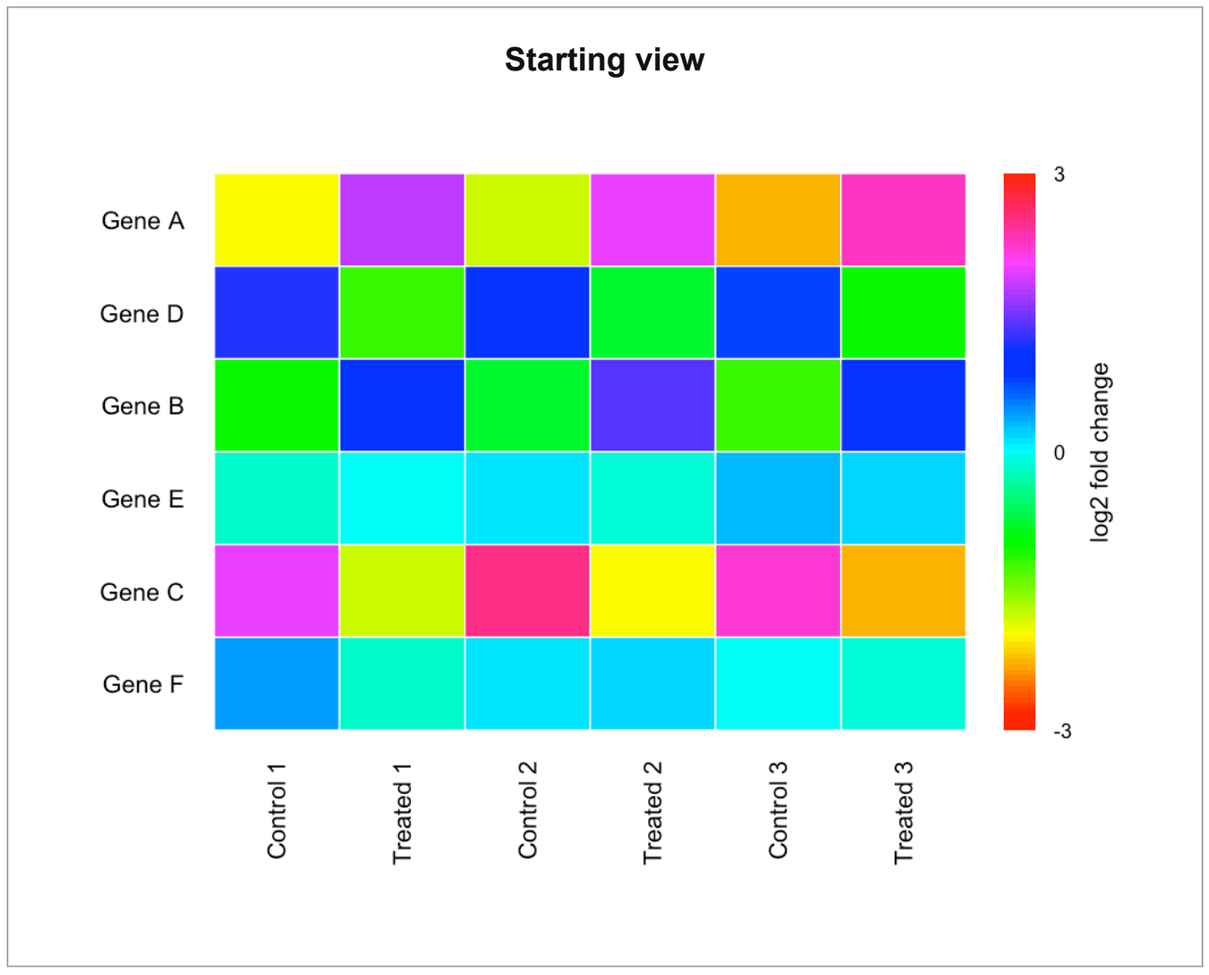
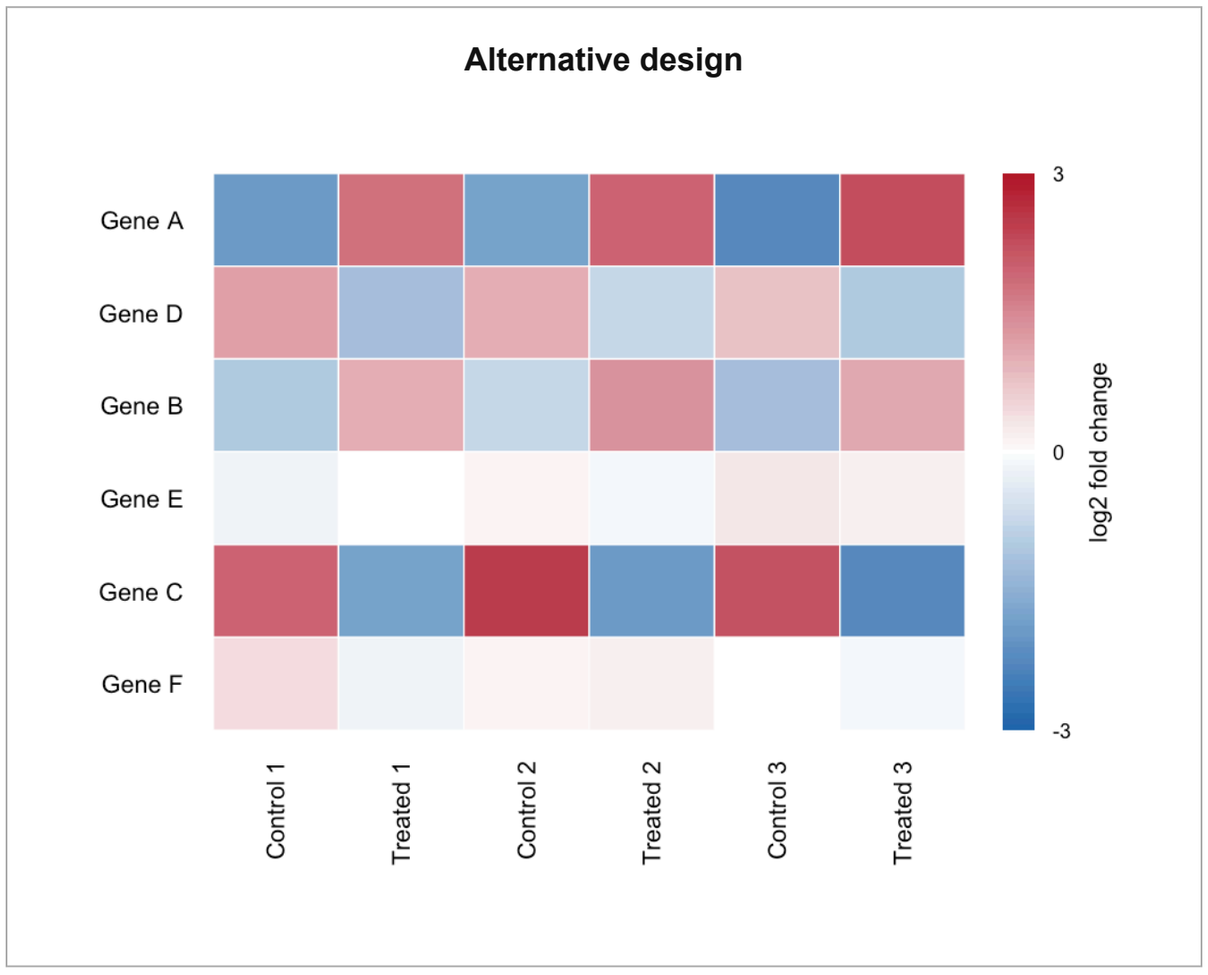




A diverging scale centred at zero separates decreases from increases. A rainbow introduces visual boundaries that do not correspond to a biological threshold.

Advice, examples & further exploration

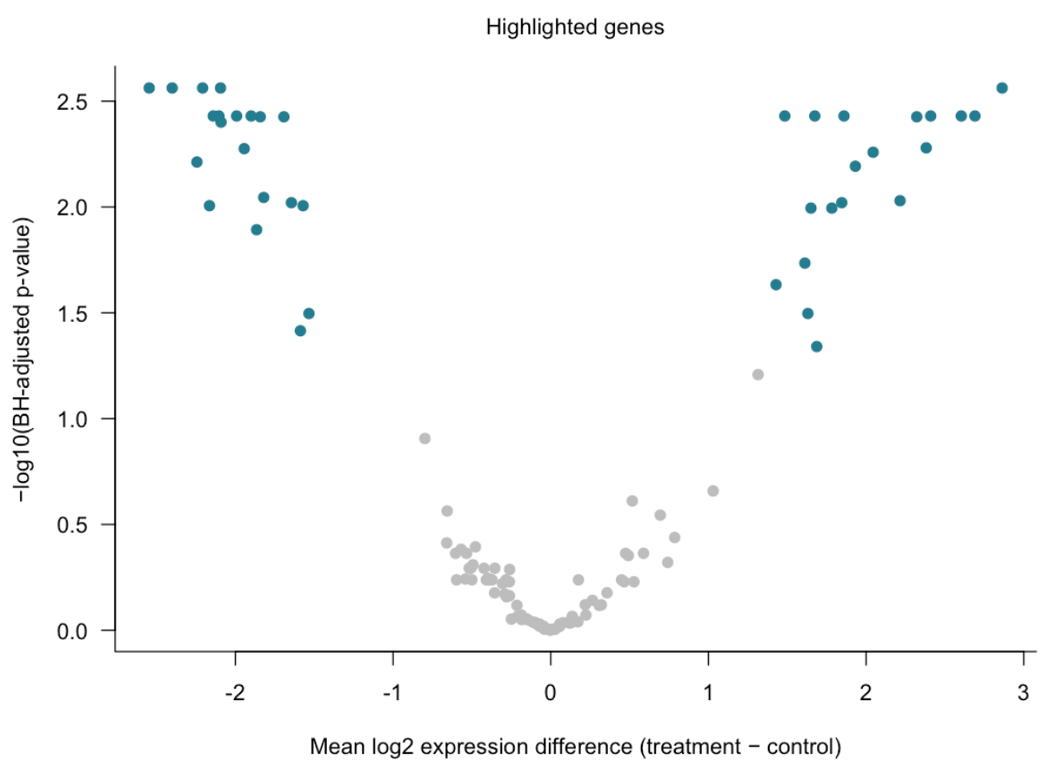
Volcano plot

Compare effect magnitude with statistical evidence across many features.

Explain why genes are highlighted

The same 120 genes, with expression difference versus −log10 BH-adjusted p-value.

Starting view



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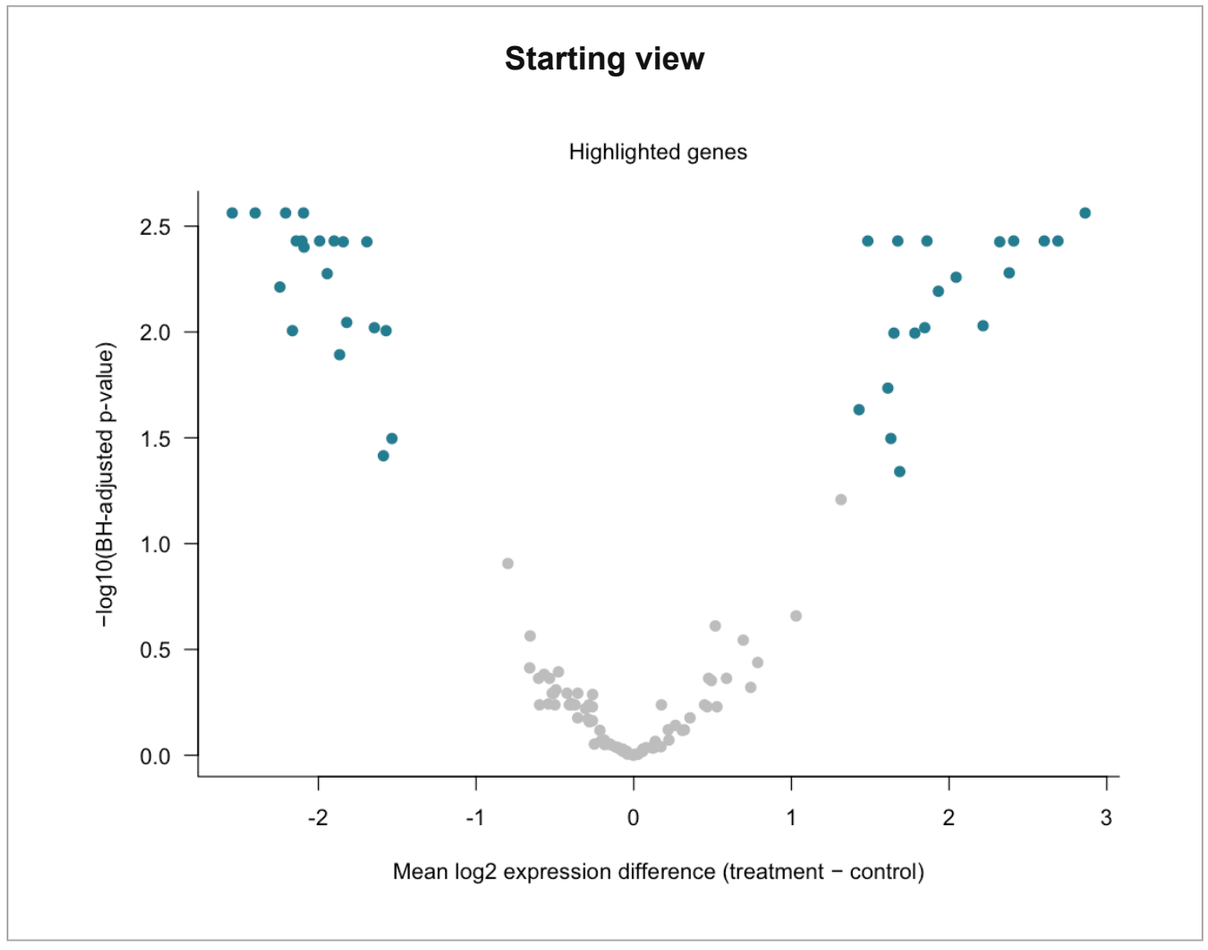
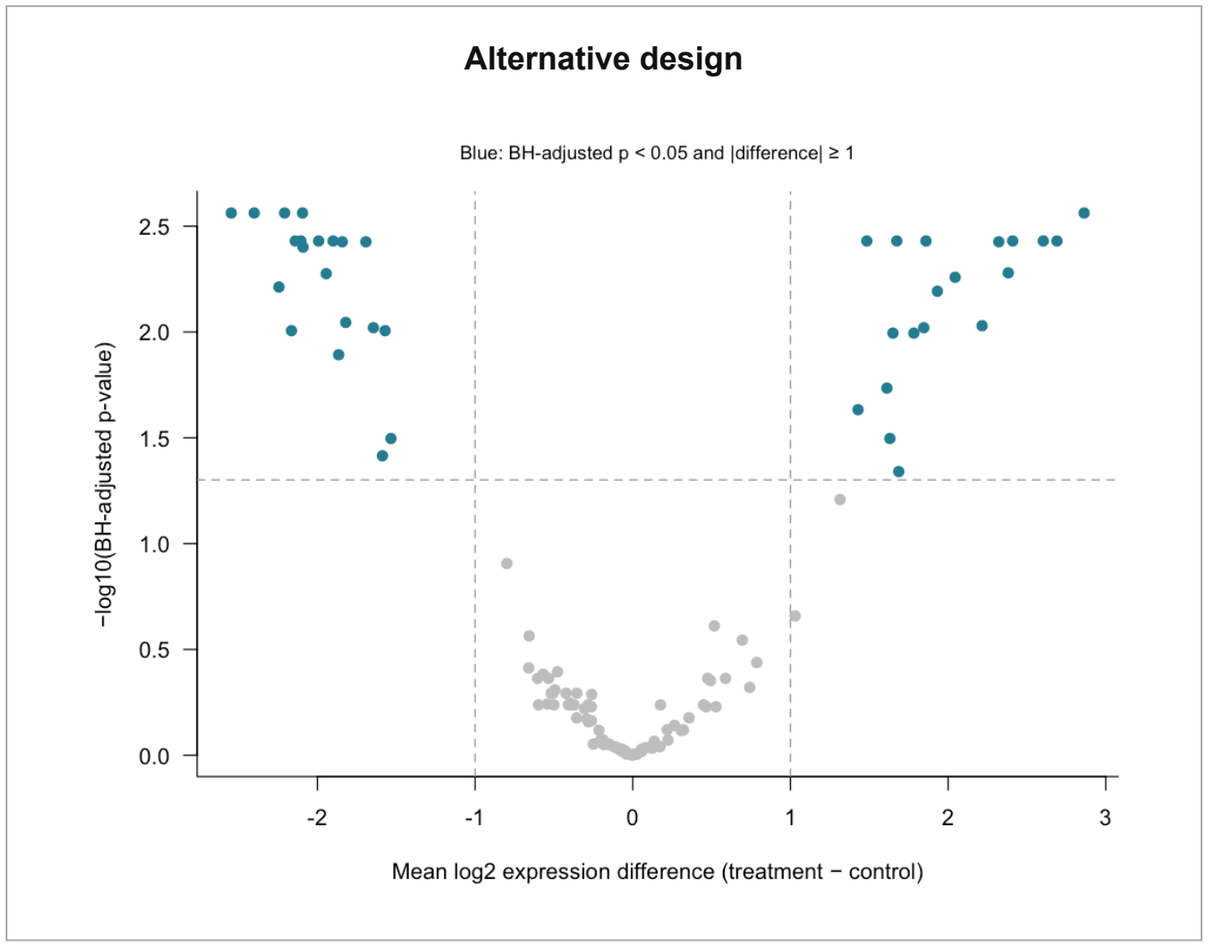




Unexplained coloured genes invite guesses about the selection rule. Threshold lines and an explicit rule make it reproducible. Crossing the thresholds does not establish biological importance.

Advice, examples & further exploration

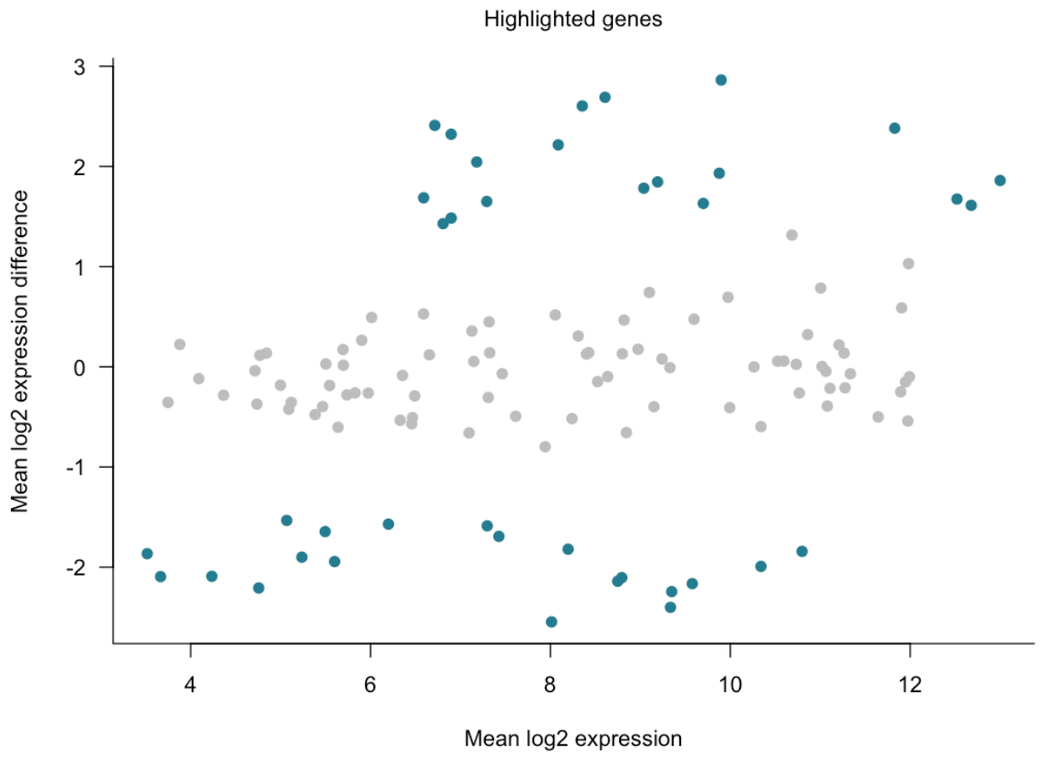
MA plot

Inspect expression differences against average expression level.

Explain the highlighted subset

The same 120 genes: mean log2 expression versus treatment–control log2 expression difference.

Starting view



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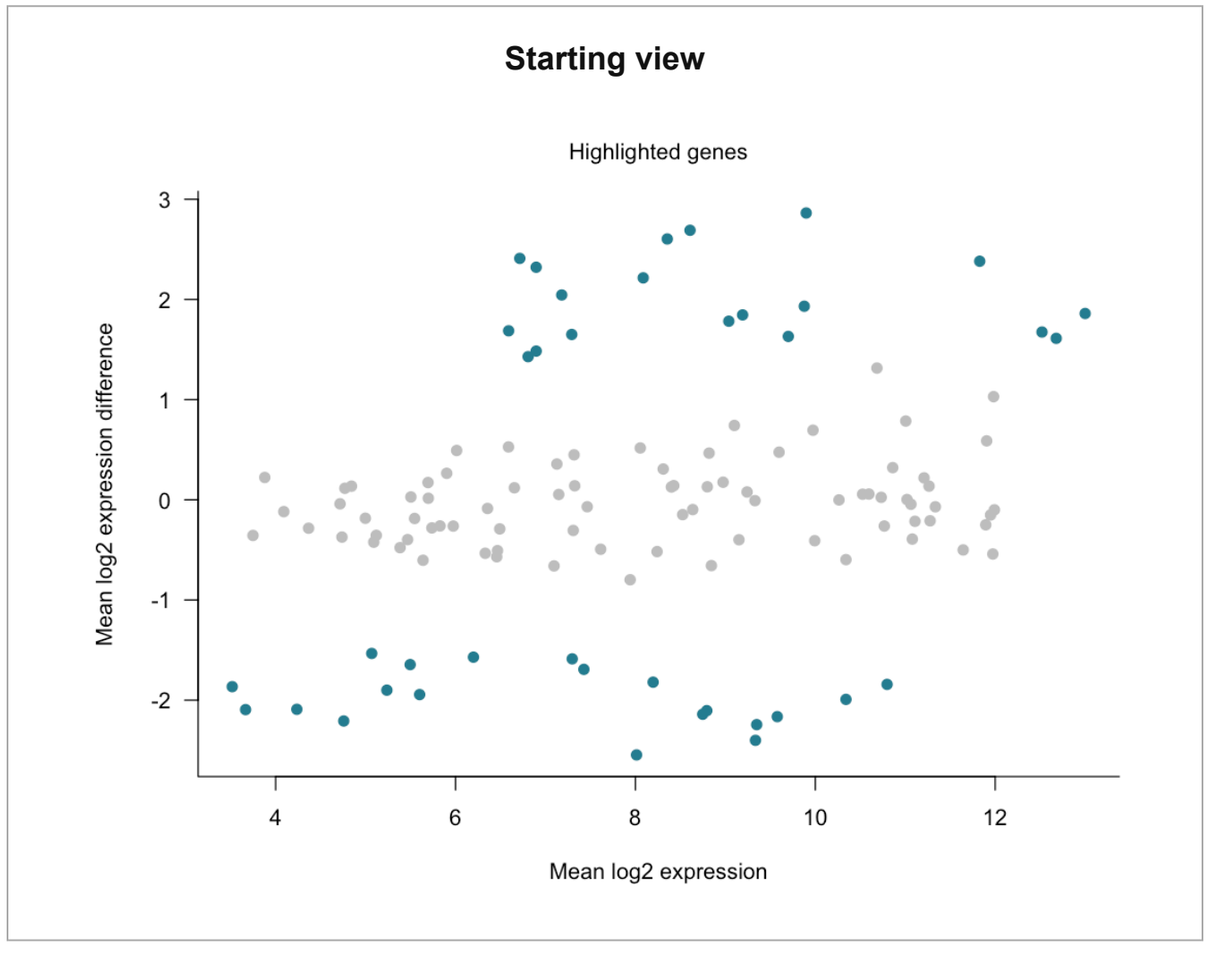
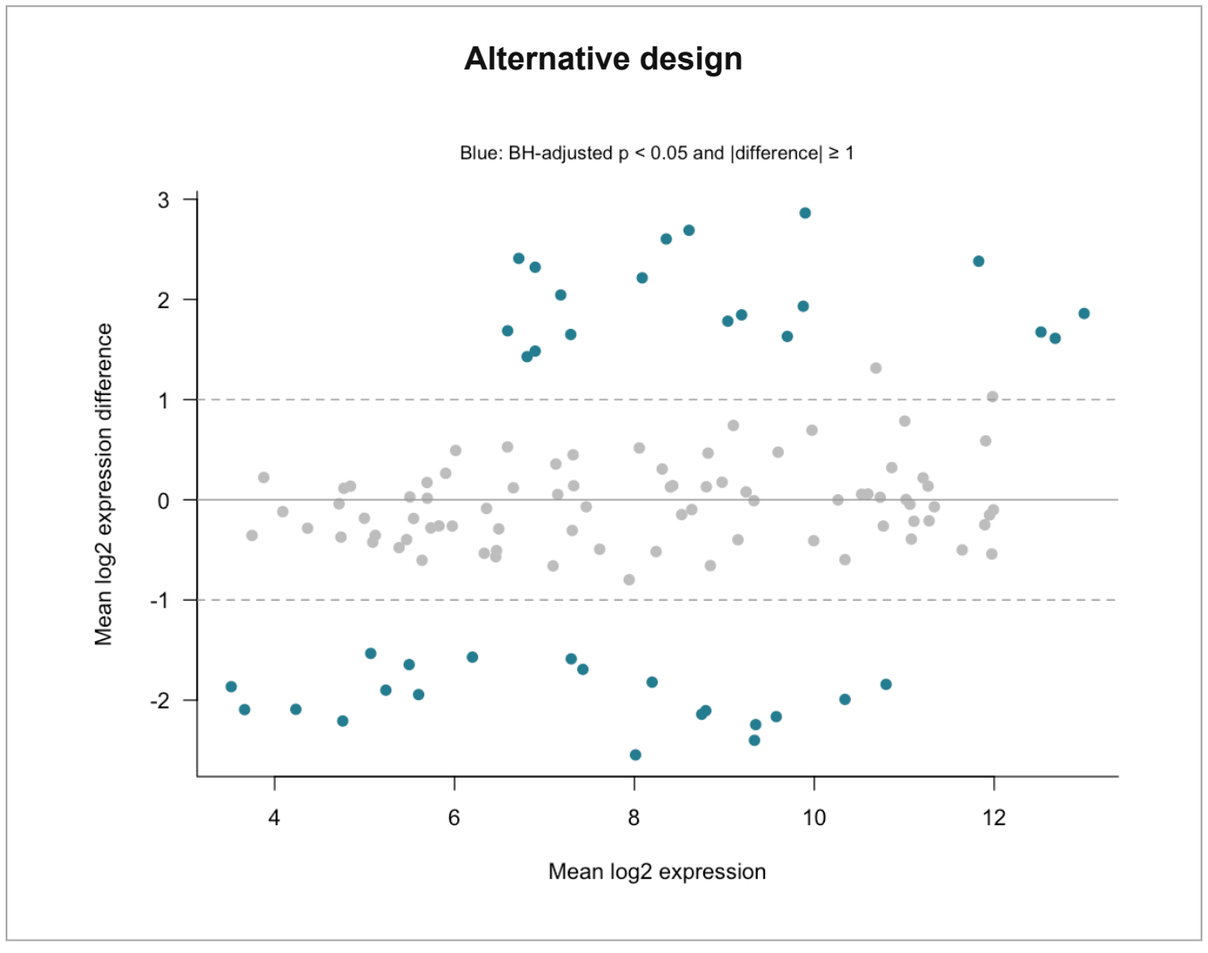
The same 120 genes: mean log2 expression versus treatment–control log2 expression difference.





The zero line anchors direction; effect thresholds and the stated adjusted-p rule explain the colours. The plot helps inspect abundance-dependent patterns but does not itself replace the expression-analysis model.

> **Advice, examples & further exploration**

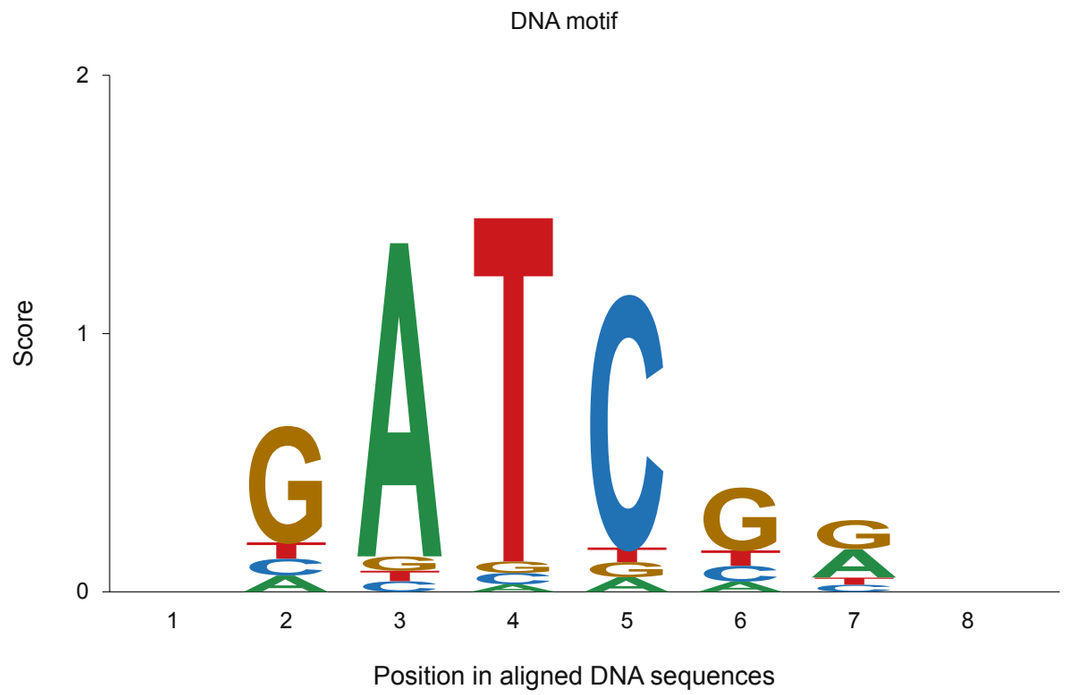
Sequence logo

Show base or amino-acid preferences across aligned sequence positions.

State what letter height means

The same base counts from 100 aligned DNA sequences; letter positions and heights are unchanged.

Starting view



Alternative design

Discuss the first figure, then reveal the alternative.

Advice, examples & further exploration

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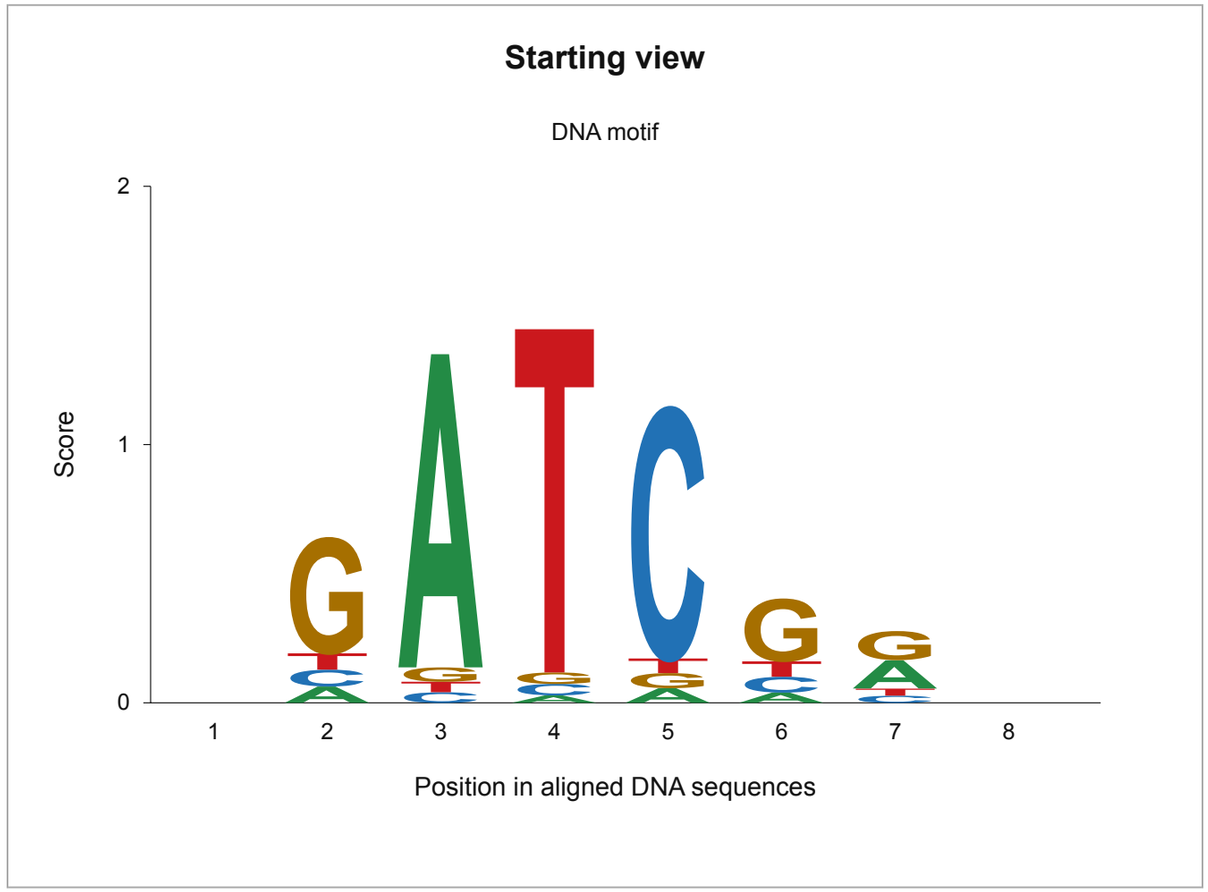
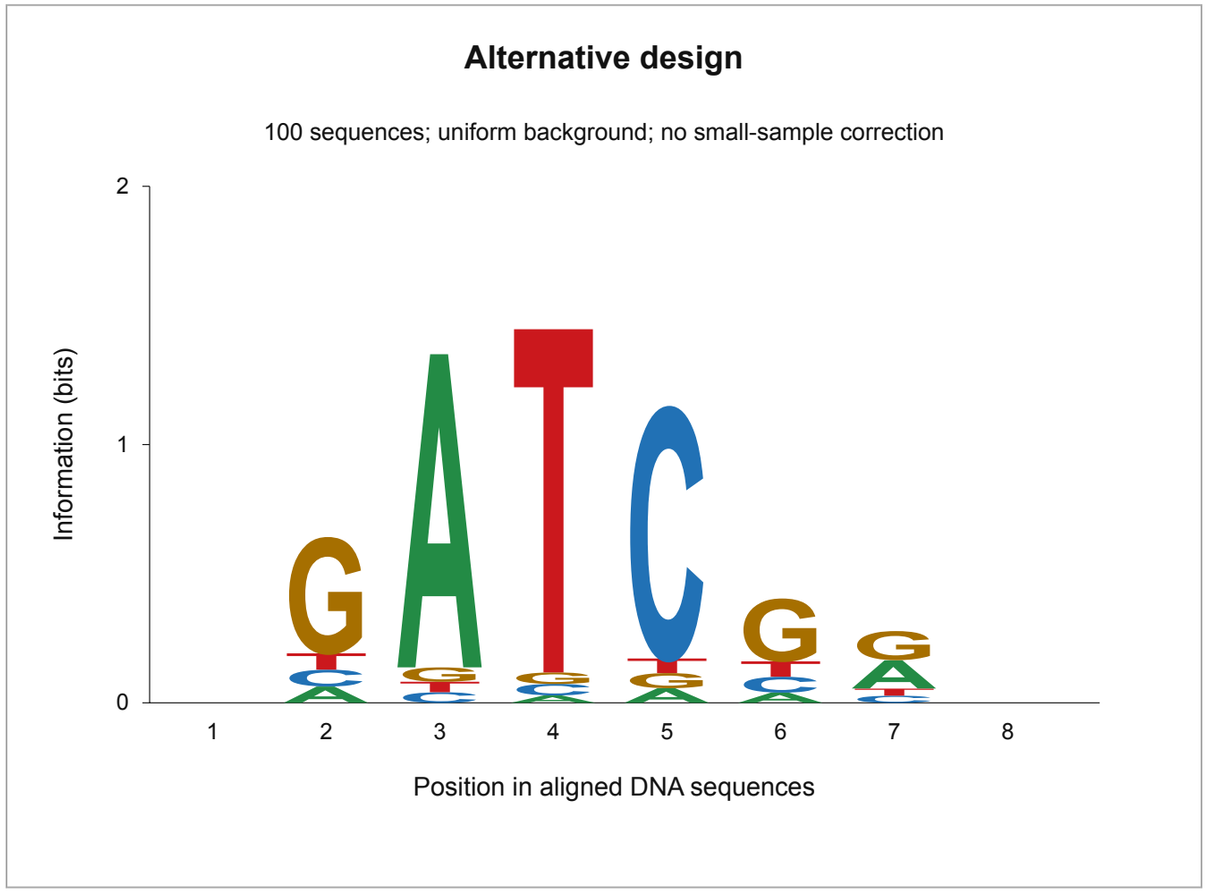




A generic “Score” label cannot distinguish frequency from information. Bits, background and correction details define the height convention. Uniform positions have zero information here; that does not mean no sequences were observed.

> **Advice, examples & further exploration**

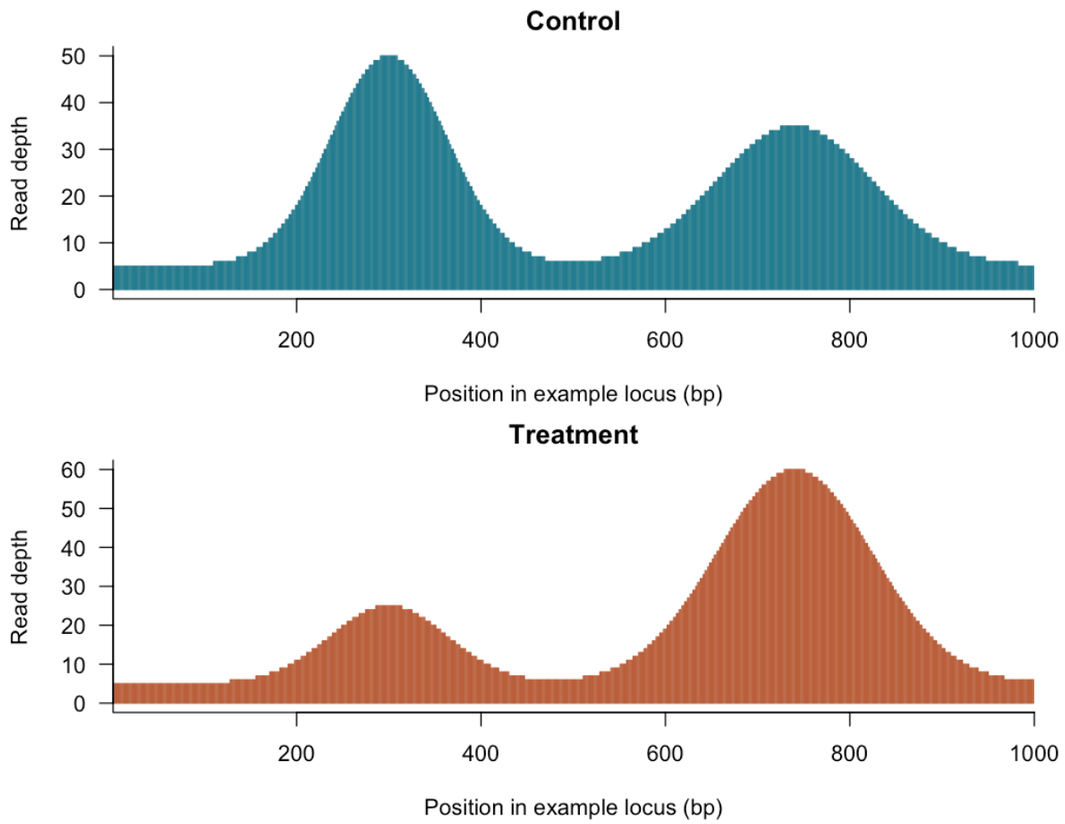
Genome / coverage tracks

Align measurements and annotations along a shared genomic coordinate.

Keep comparable tracks on one scale

The same control and treatment read depths across a 1,000 bp example locus.

Starting view



Alternative design

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Advice, examples & further exploration

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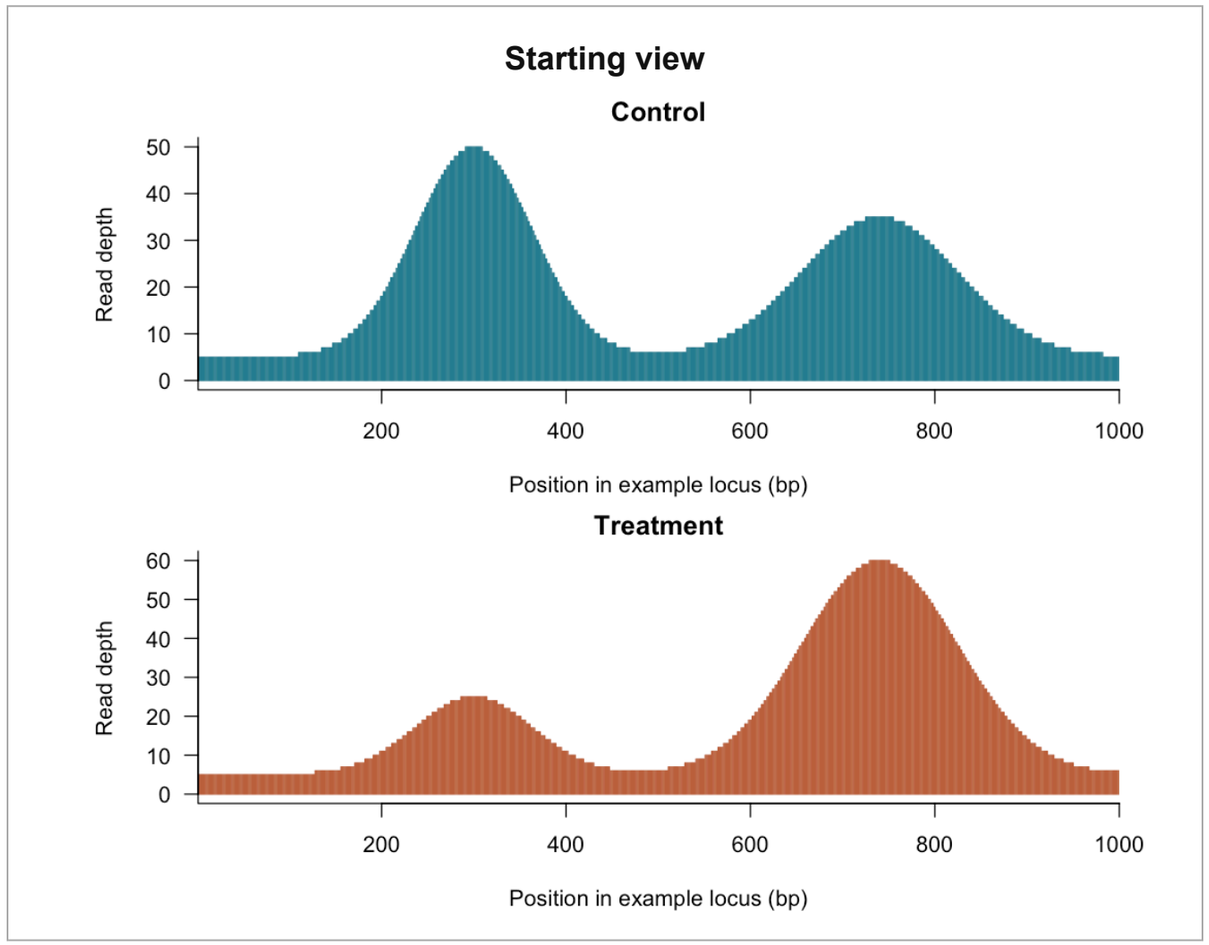
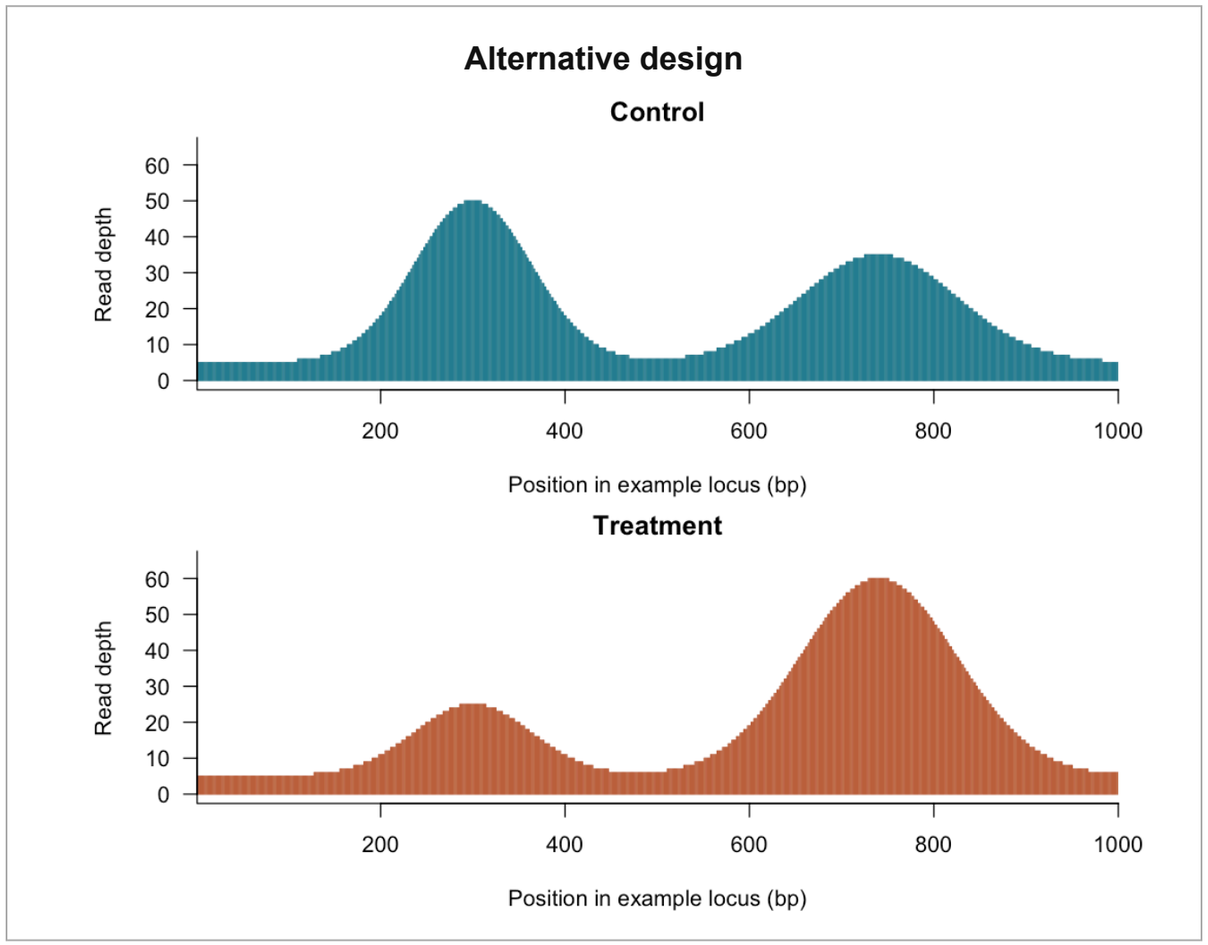




Independent vertical limits make peaks look more alike than their heights are. A shared 0–65 scale makes raw depths comparable. This does not correct sequencing-depth differences or turn coverage into a tested expression effect.

> **Advice, examples & further exploration**

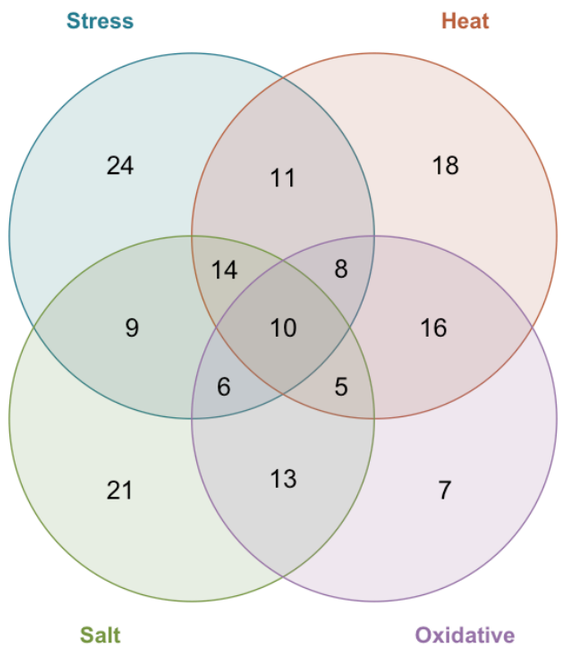
Venn diagram

Show membership and overlap between a small number of sets.

When circles become hard to read

The same exact gene memberships in four lists.

Starting view



Numbers count genes in each exact region; areas are not proportional

Alternative design

Discuss the first figure, then reveal the alternative.

Advice, examples & further exploration

Venn diagram

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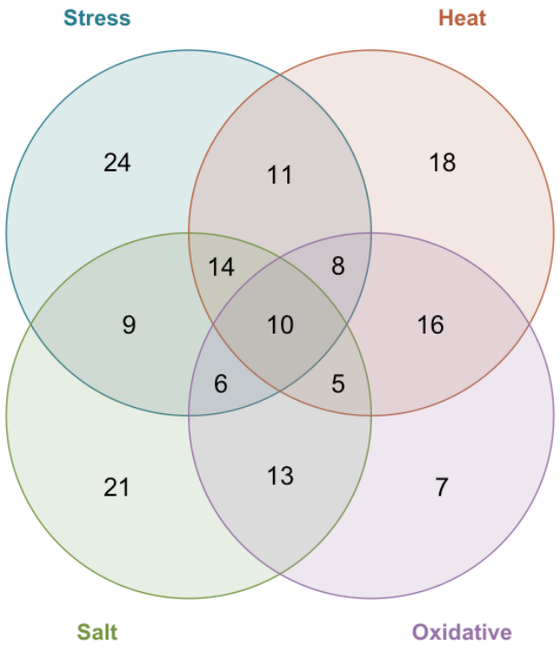
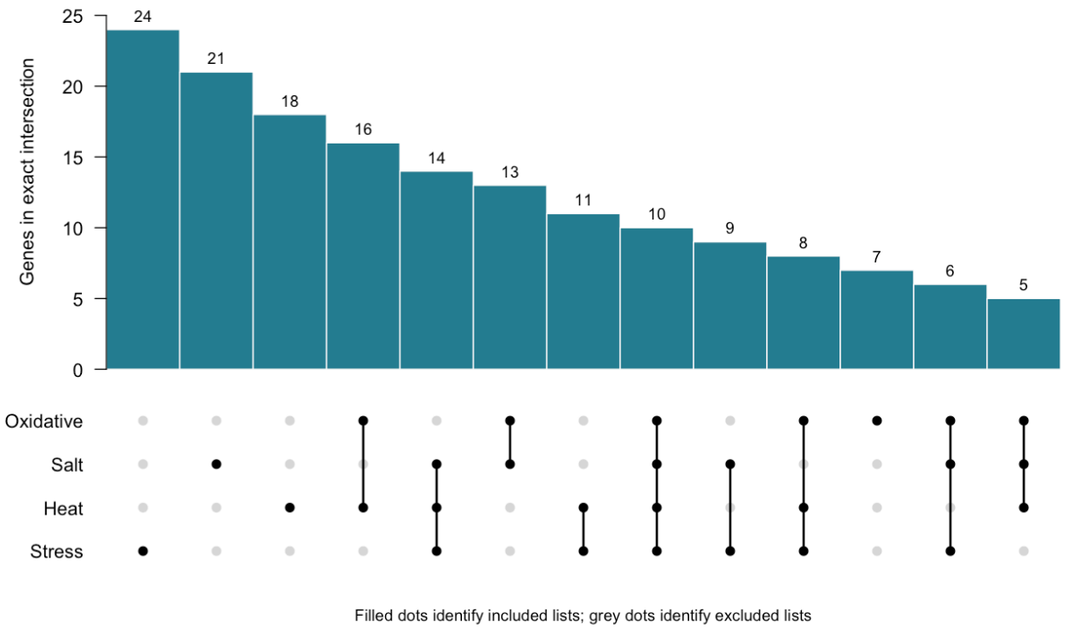


Numbers count genes in each exact region; areas are not proportional

Alternative design



Filled dots identify included lists; grey dots identify excluded lists

UpSet makes each included and excluded list explicit and aligns intersection counts. A small two-list Venn remains a useful simple alternative.

> **Advice, examples & further exploration**

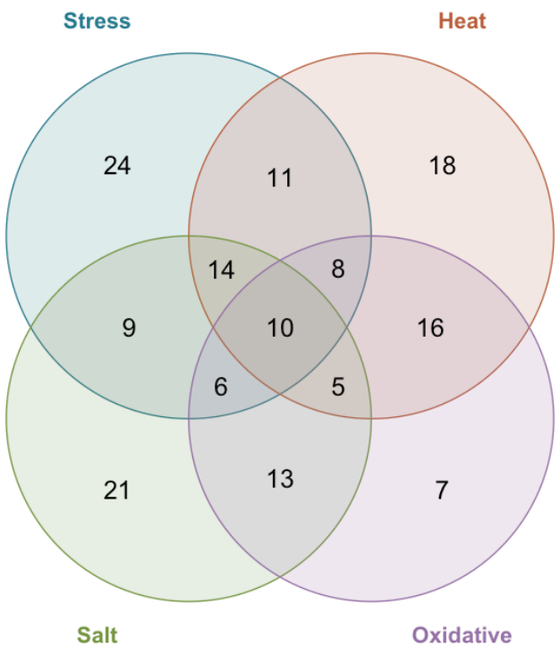
UpSet / intersection plot

Compare exact intersections among several sets with bars and a membership matrix.

Choose a readable intersection display

The same exact gene memberships in four lists.

Starting view



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Advice, examples & further exploration

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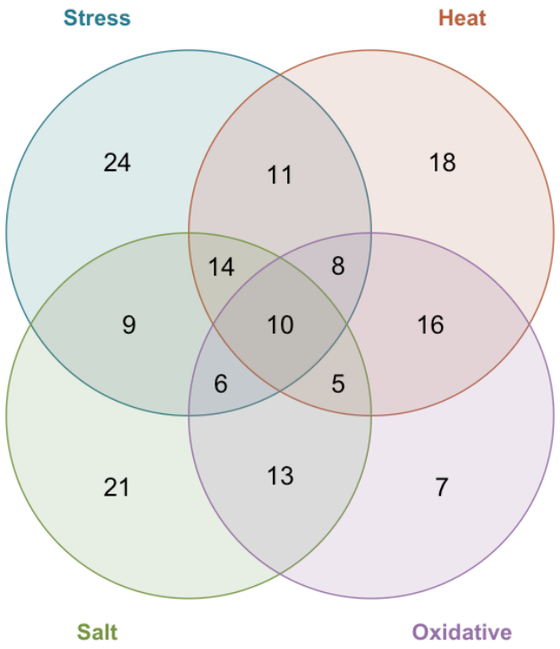

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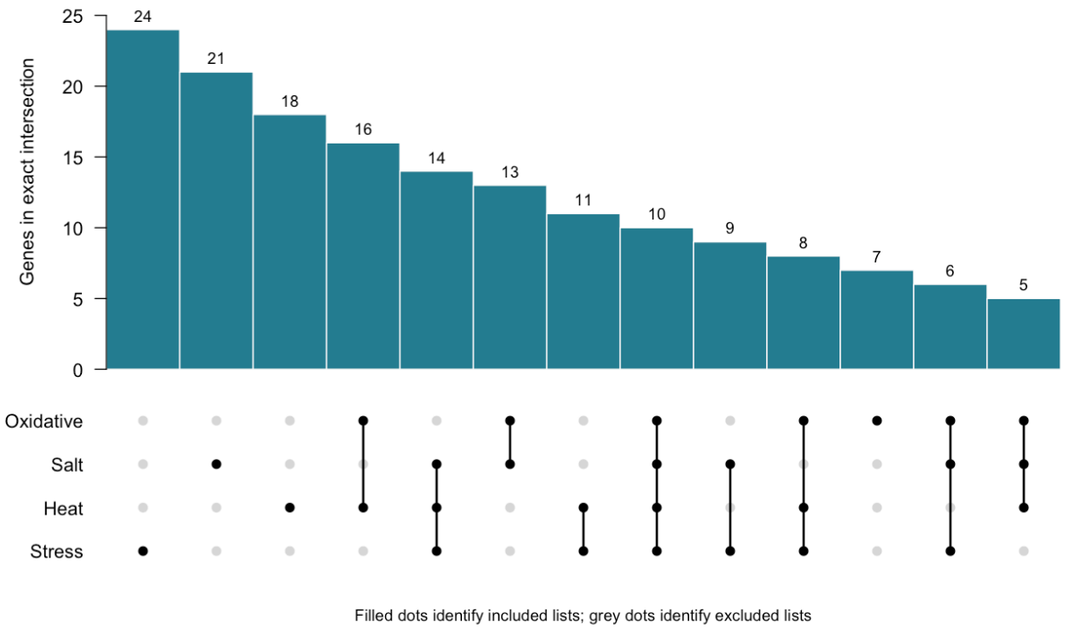


Filled dots identify included lists; grey dots identify excluded lists

The membership matrix removes the need to trace crowded region boundaries. Its bars show exact intersections, not inclusive overlaps.

> **Advice, examples & further exploration**

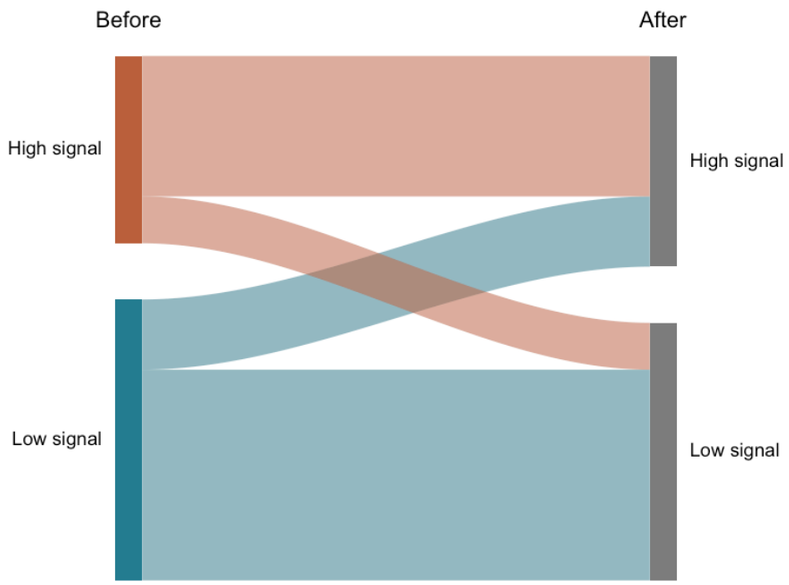
Sankey diagram

Show how quantities flow or split between categories; ribbon thickness represents amount.

Explain what the ribbons encode

The same 100 cells tracked from low/high signal before to low/high signal after treatment.

Starting view



Alternative design

Discuss the first figure, then reveal the alternative.

Advice, examples & further exploration

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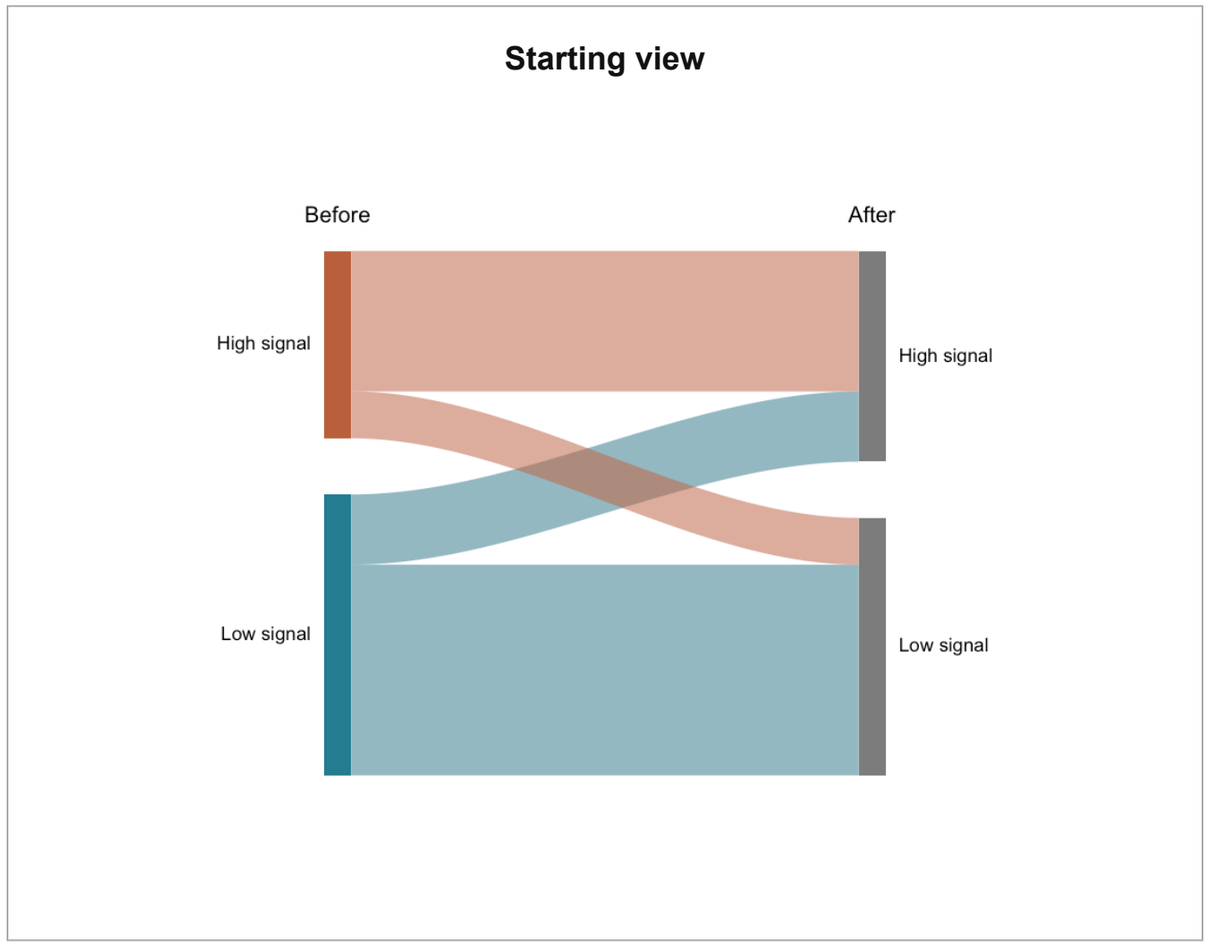
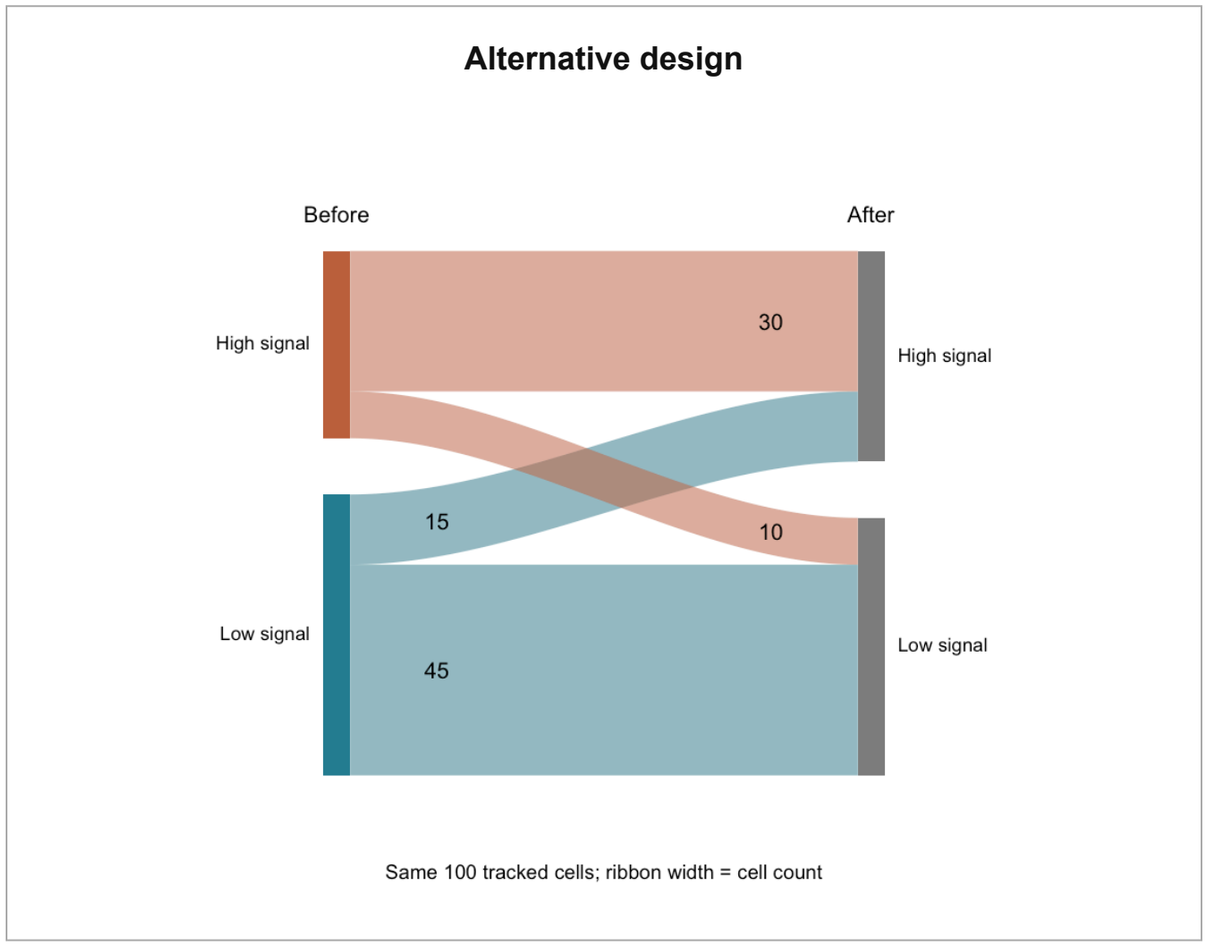




The first version leaves the denominator and width meaning implicit. Counts and a tracking description make the flows interpretable. Ribbon width counts cells; it does not show the size of a signal change.

Advice, examples & further exploration

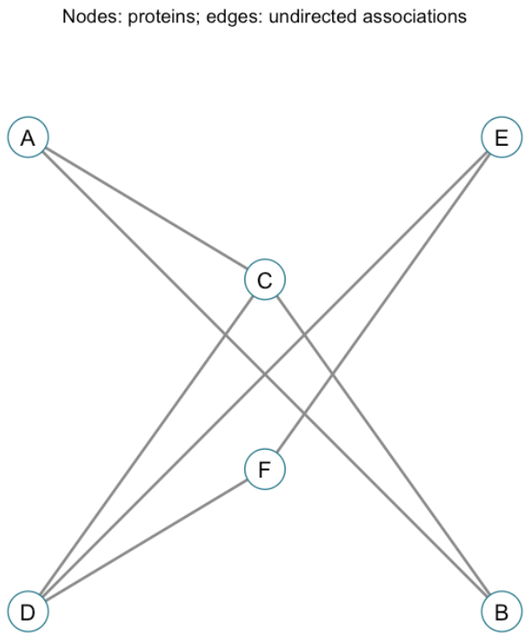
Network diagram

Show entities and relationships, such as proteins connected by recorded associations.

Reduce crossings, preserve connections

The same six proteins and seven undirected association edges.

Starting view



Alternative design

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Advice, examples & further exploration

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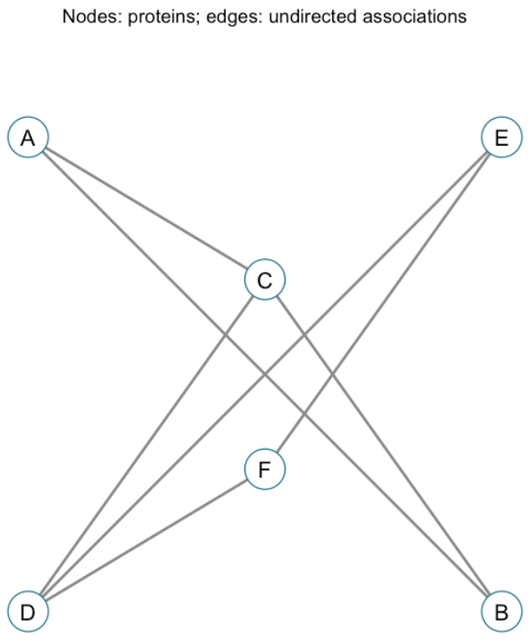
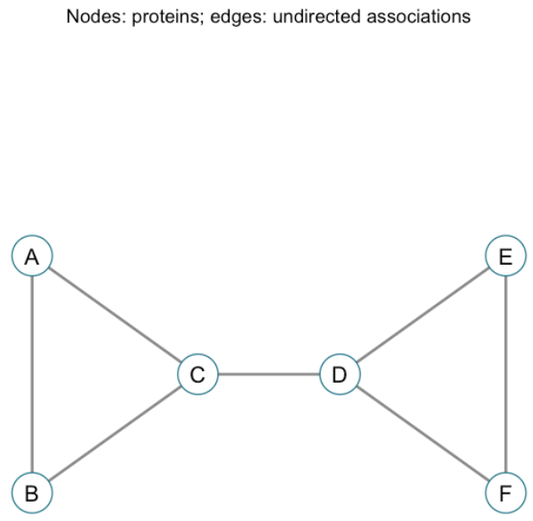


Alternative design



A tangled layout makes individual connections difficult to follow. Separating the connected groups reduces crossings without deleting edges. Node proximity is a layout choice, not another measured association.

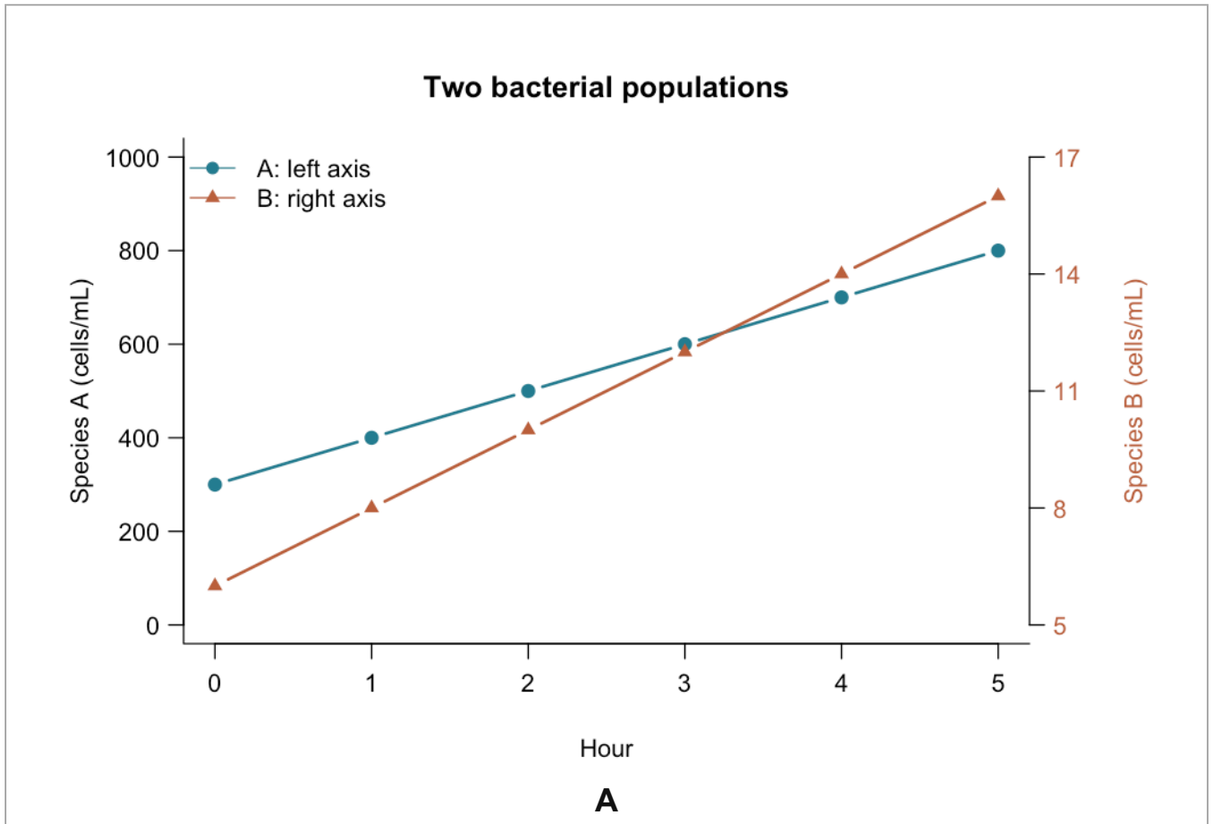
Advice, examples & further exploration

Did one species overtake the other?

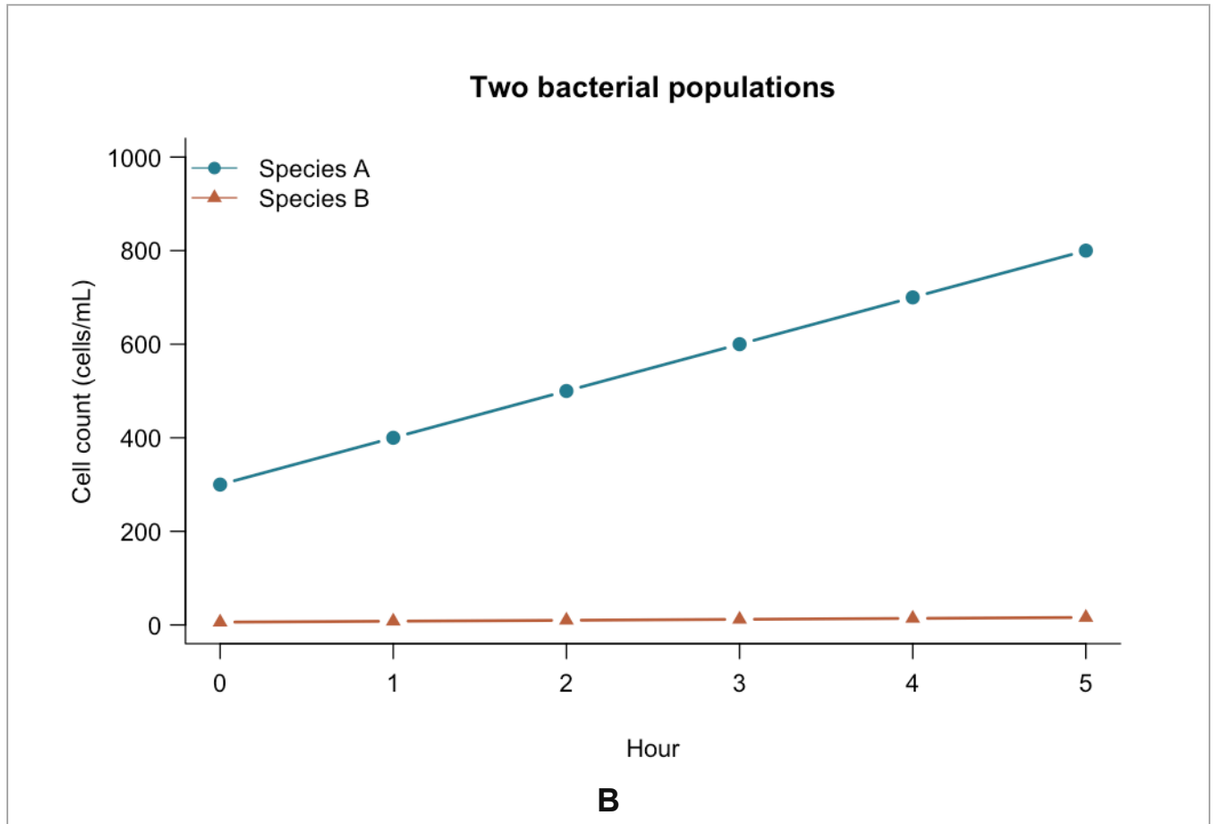
Two bacterial populations were counted hourly. The horizontal axes show hours; both populations are measured in cells per millilitre. A uses separate left and right scales; B uses one shared scale.

Source: Course example.

Which view supports comparing the absolute cell counts?



A



B

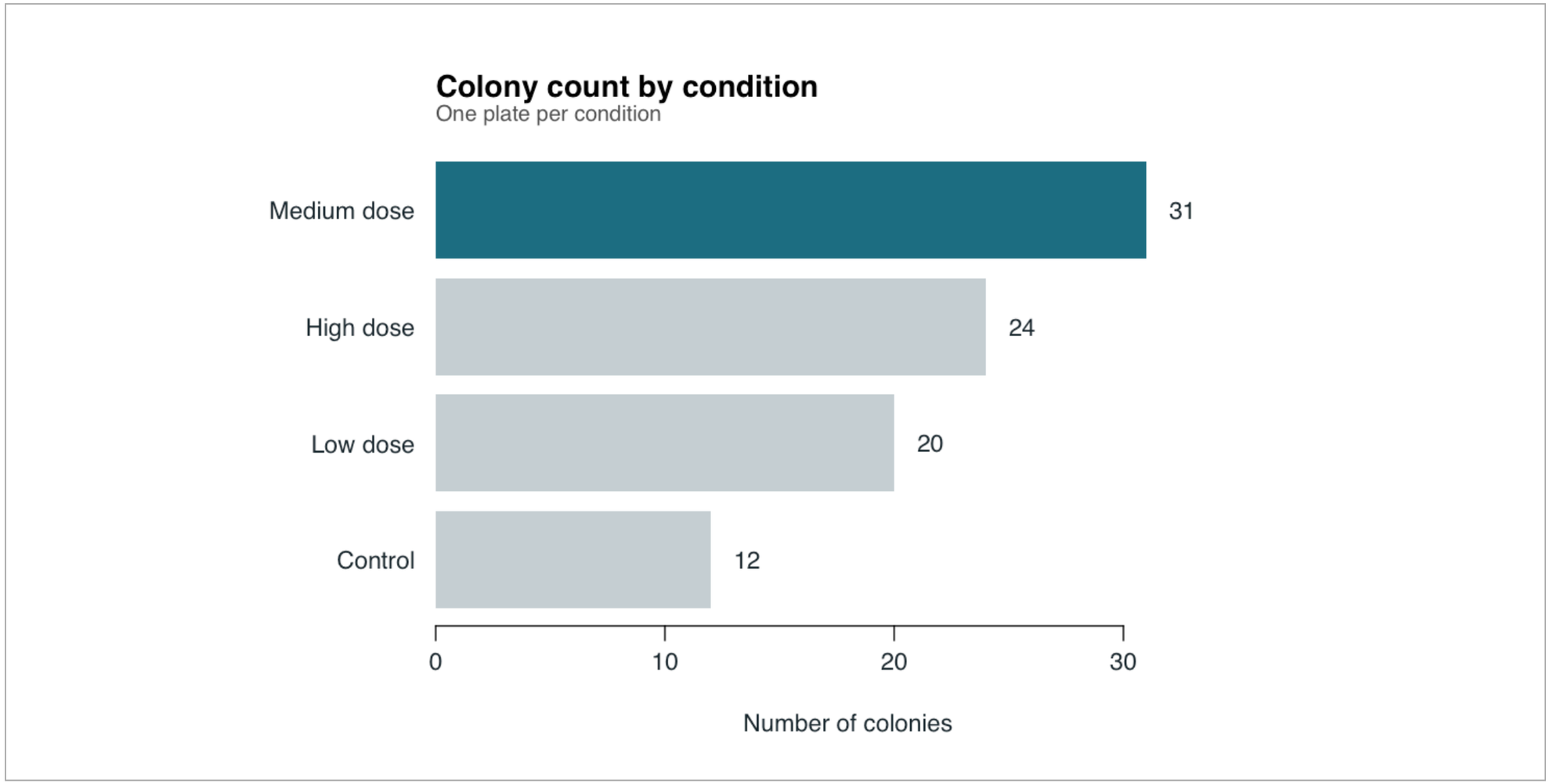
Choose A

Choose B

Further reading & data

A ranking you can read immediately

For a quick ranking, which choice helps most?

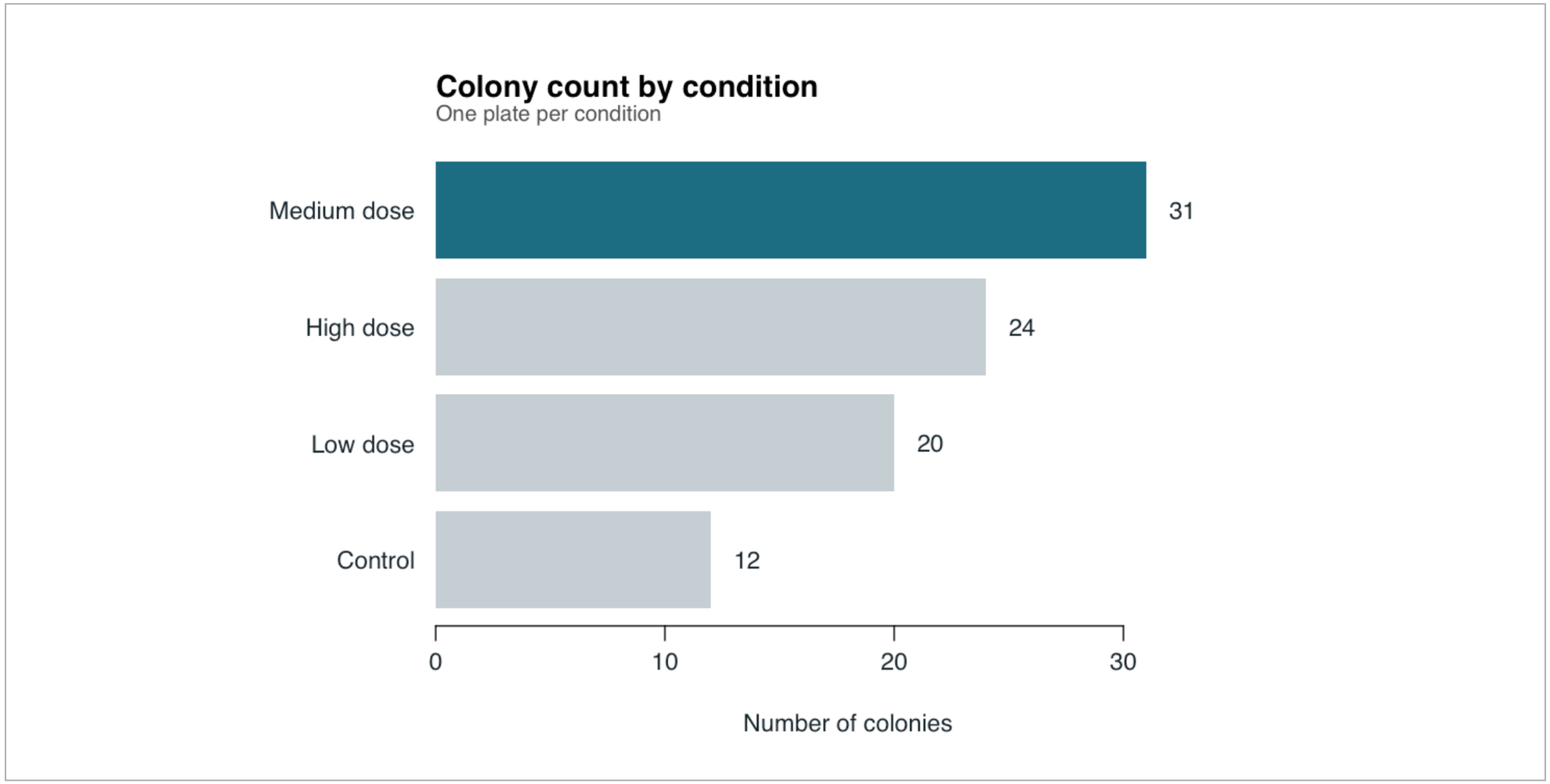


Reveal what works

Context & source

A ranking you can read immediately

For a quick ranking, which choice helps most?

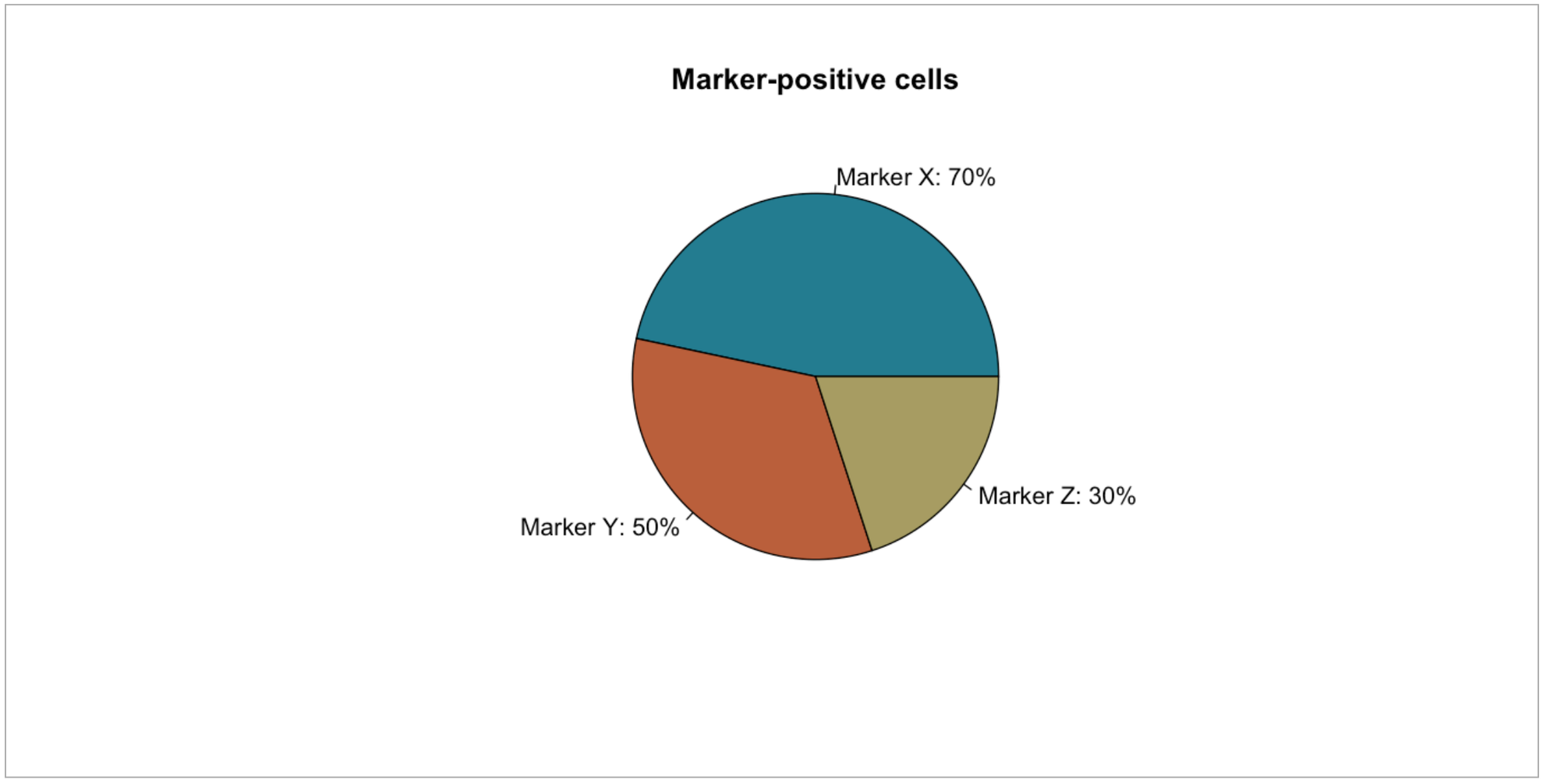


Ordering by count makes the rank visible immediately. Dose order would be useful for a different question: how counts change with dose.

Context & source

How many cells express each marker?

Which view represents the marker frequencies appropriately?

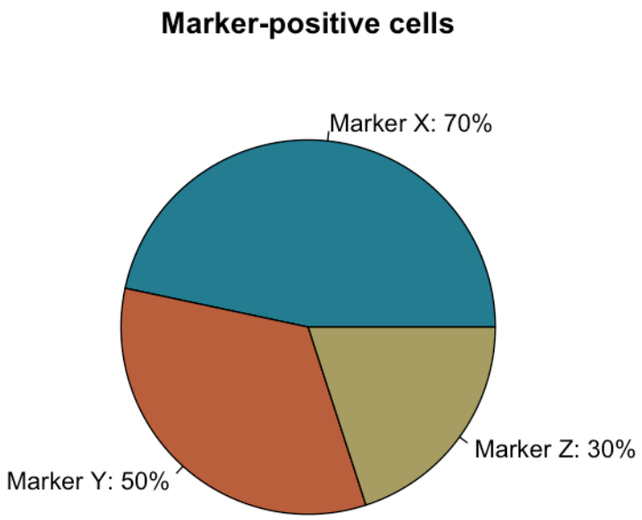


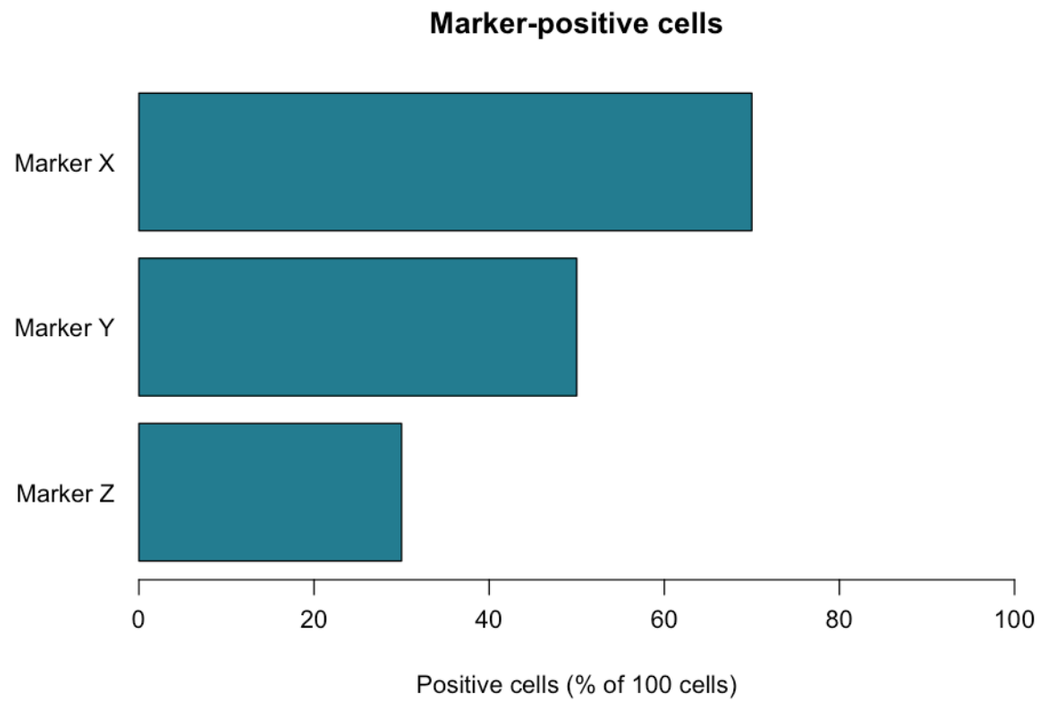
Reveal alternative figure

Context & source

How many cells express each marker?

Which view represents the marker frequencies appropriately?

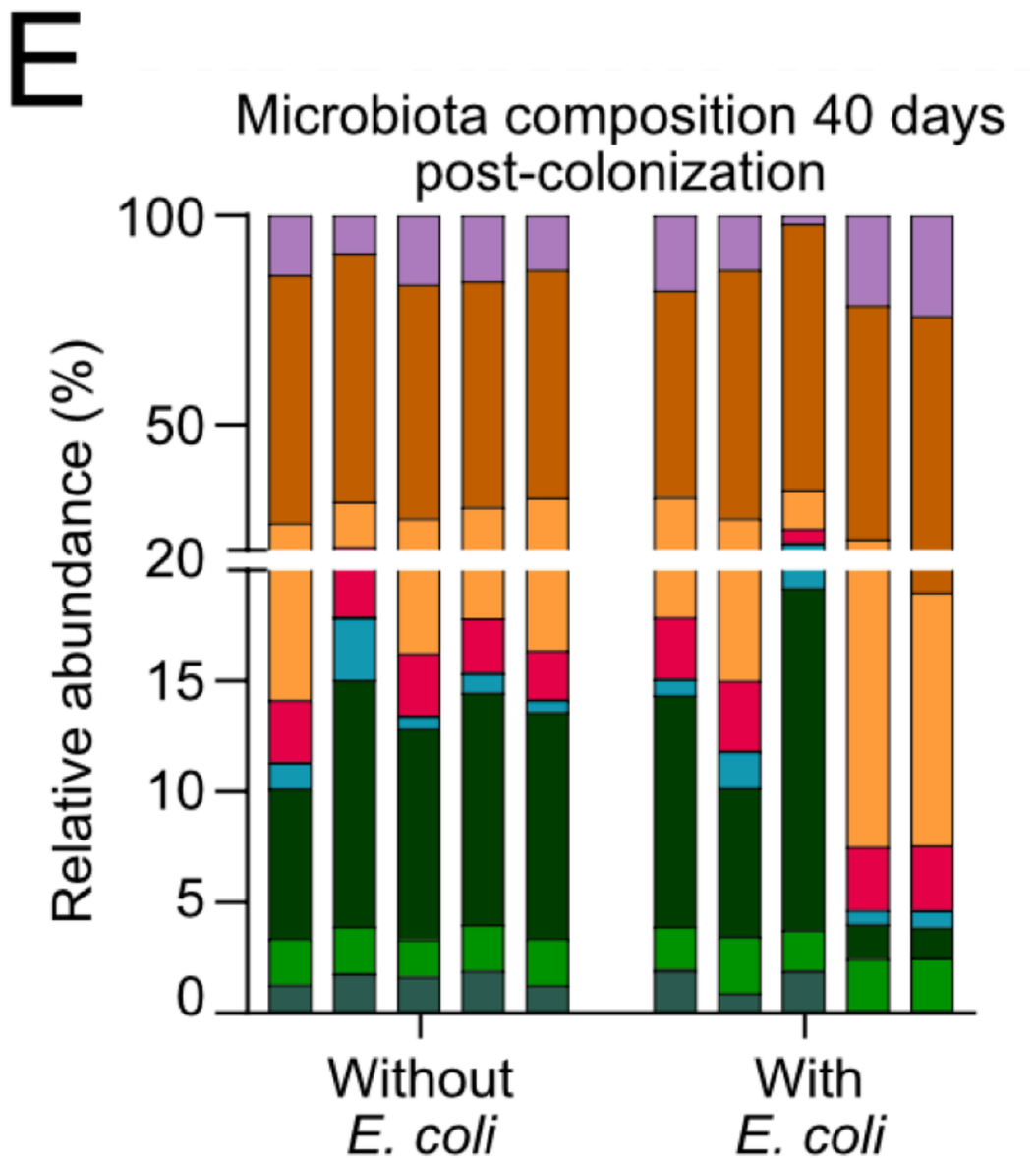




Separate bars preserve each marker’s percentage of the same 100 cells. Their percentages need not sum to 100%.

Context & source

Microbiota composition across samples



Supplied figure excerpt; a different dataset from taxa A–J above.

Read the composition

Each bar shows a sample. Coloured segments show its relative composition, adding up to 100%.

The groups compare samples **without** and **with *E. coli***, 40 days after colonization.

Watch the scale: the lower part expands 0–20%. The upper part uses a different scale, so segment heights across the split cannot be compared directly.

A full colour legend is needed to identify the taxa; it is not visible in this excerpt.

Discuss: what changes between samples are easy to see? Which small segments are harder to compare?

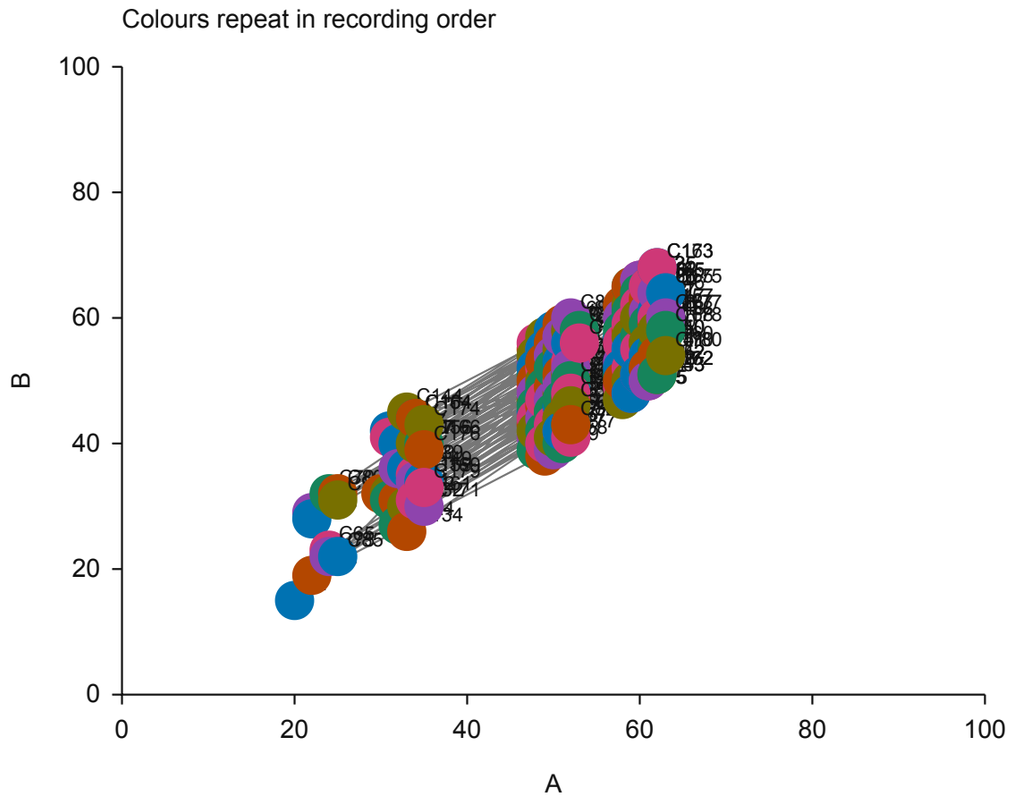
Crowded point plots

Spot the problems

A crowded point plot

Each point is one independent culture. Compare protein A and protein B fluorescence in 90 control and 90 treated cultures. What makes the relationship hard to read?

Starting figure



Improved version

Click “Remove connections” to reveal the first improvement.

Show code changes

Show full Python code for these plots

About the data

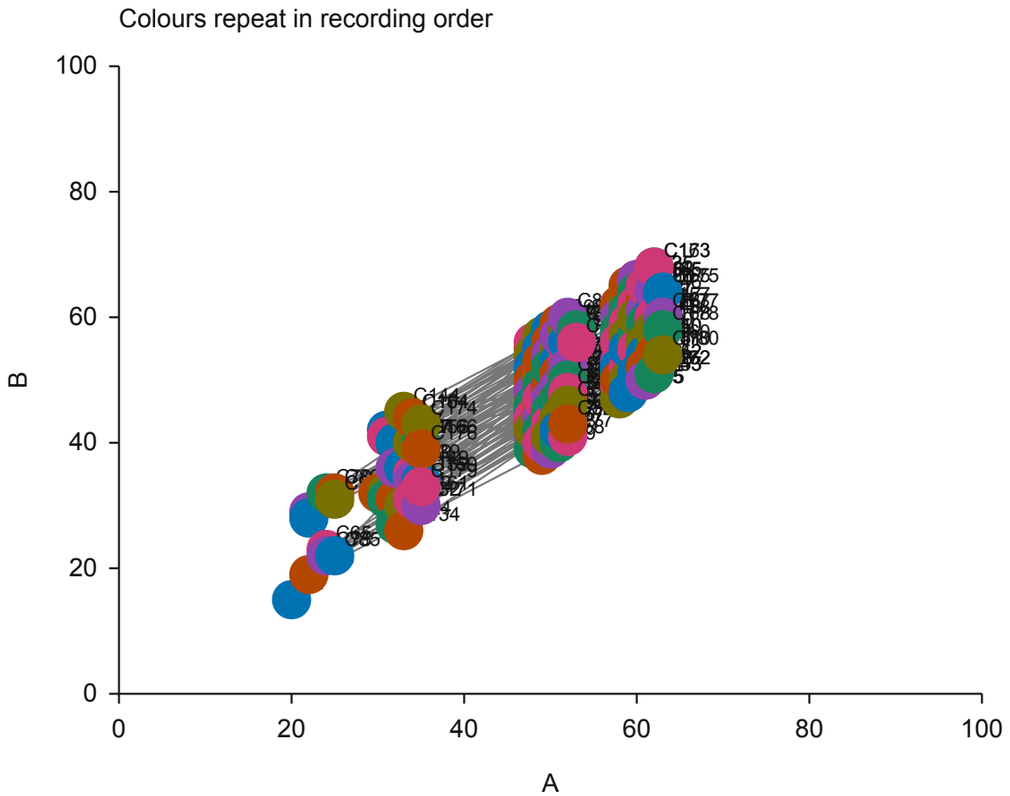
Crowded point plots

Improvement 1 of 9

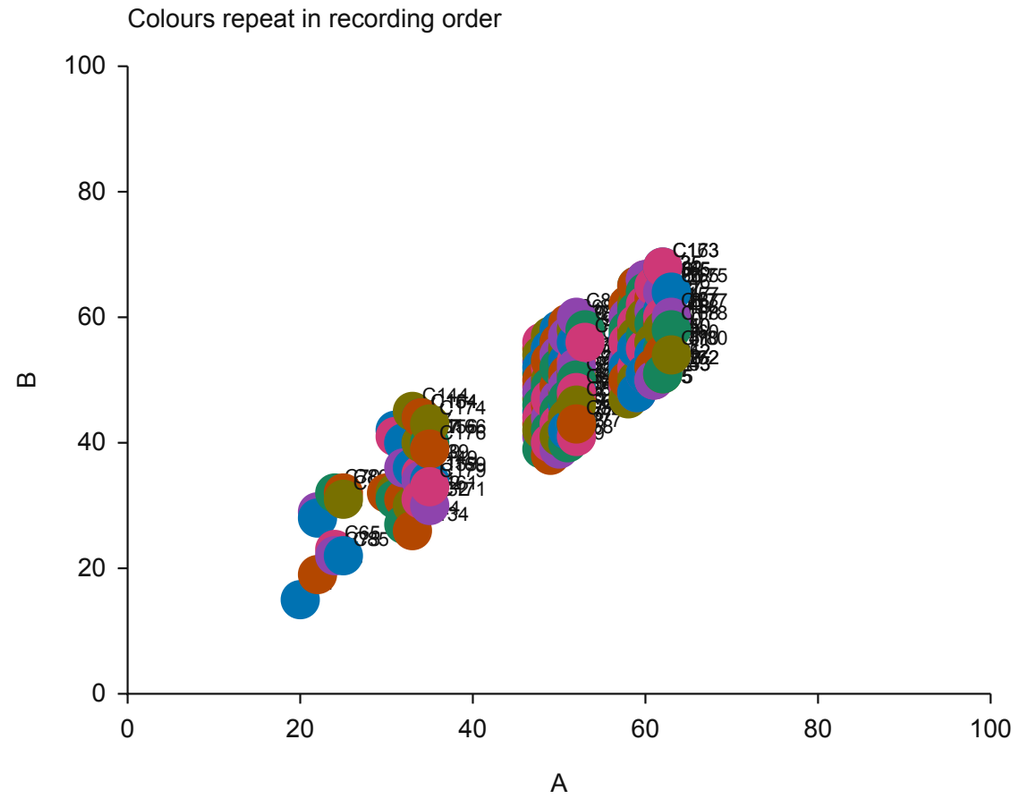
Remove meaningless connections

The cultures are independent. Joining them in recording order suggests a sequence that the experiment does not have.

Starting figure



Improved version



Show code changes

Show full Python code for these plots

About the data

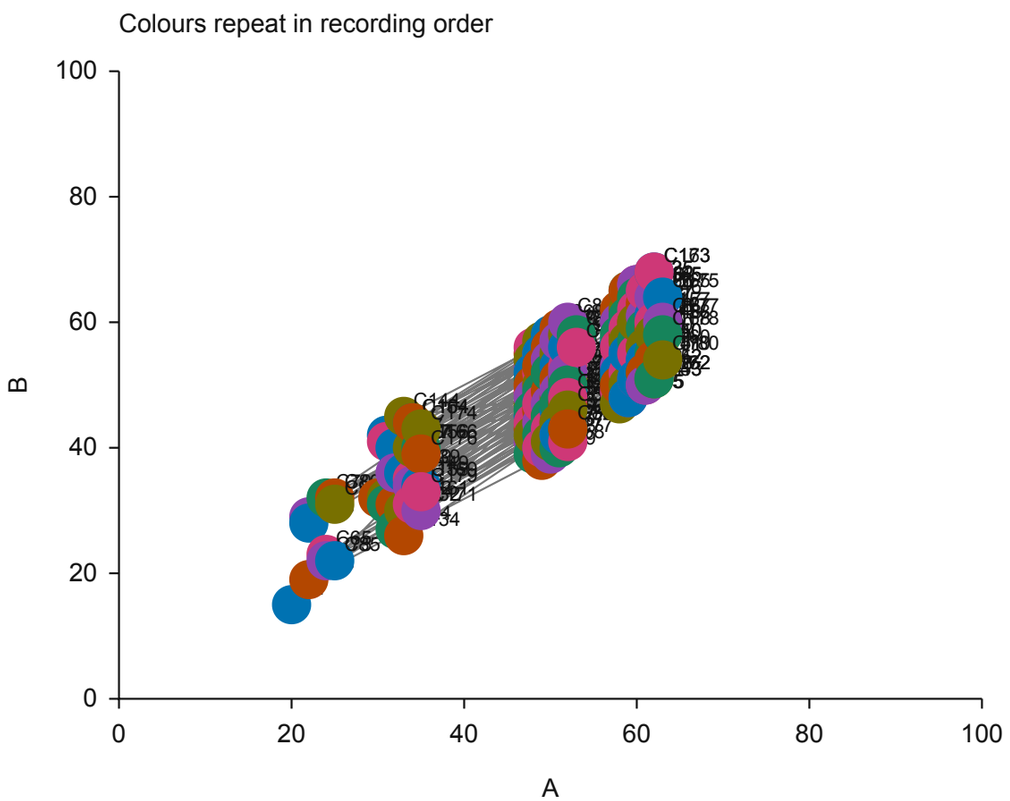
Crowded point plots

Improvement 2 of 9

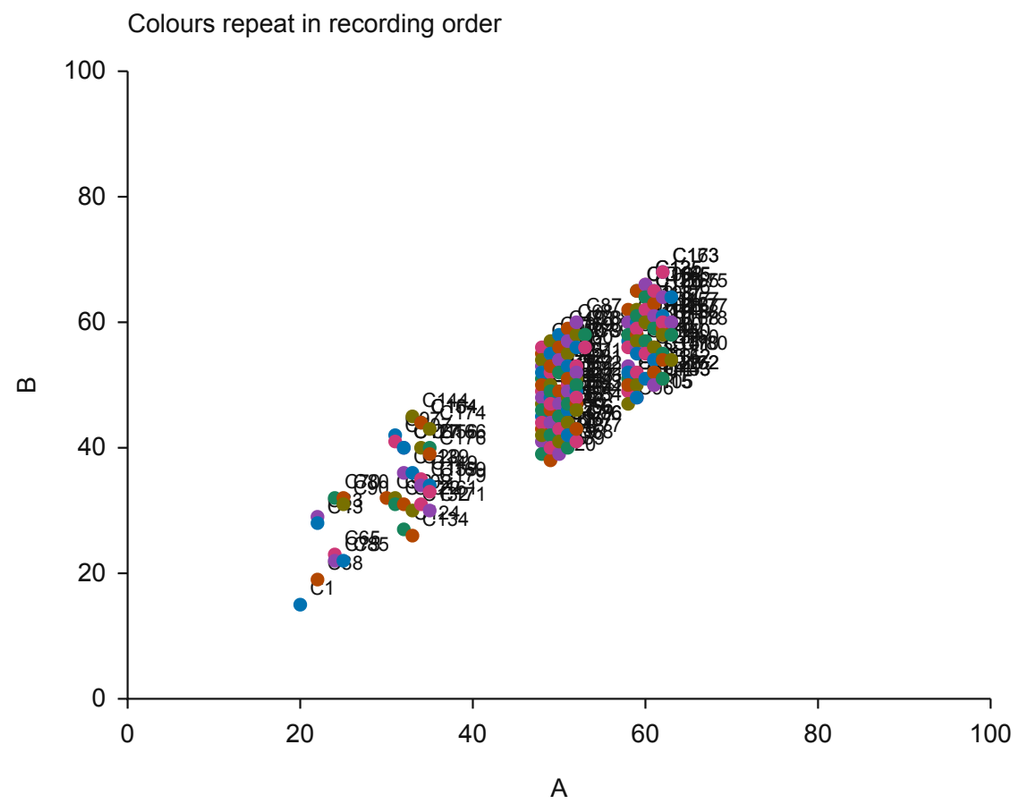
Reduce marker size

Smaller markers reveal more of the crowded regions. All 180 observations remain.

Starting figure



Improved version



Show code changes

Show full Python code for these plots

About the data

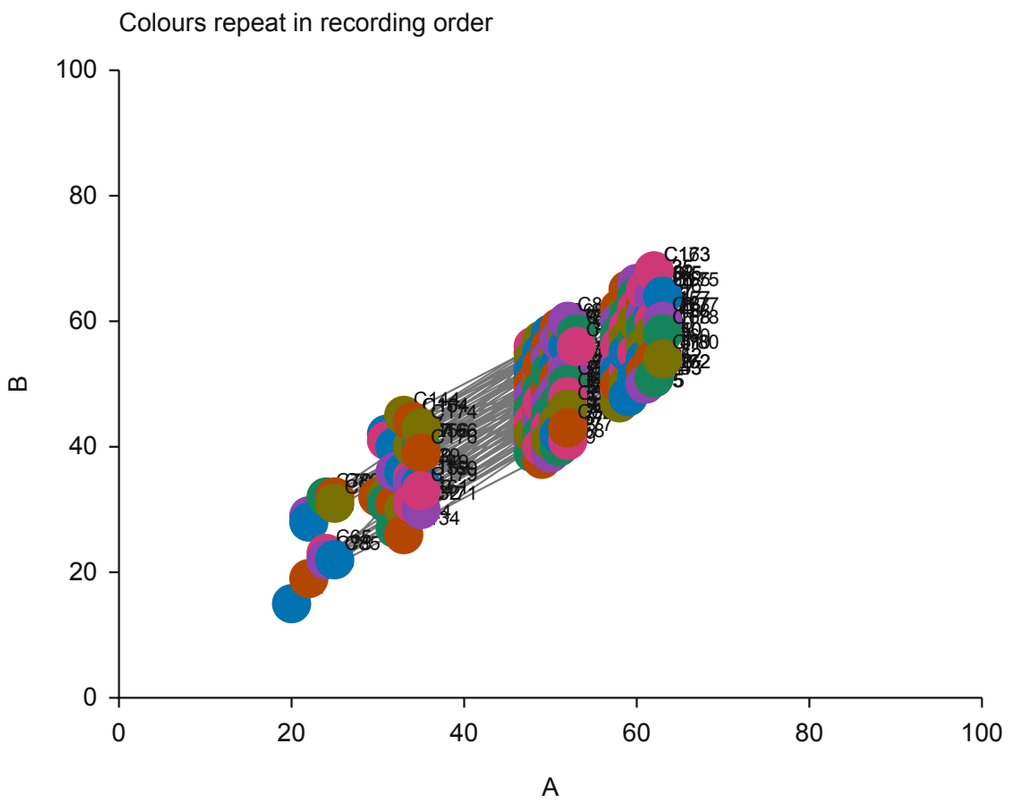
Crowded point plots

Improvement 3 of 9

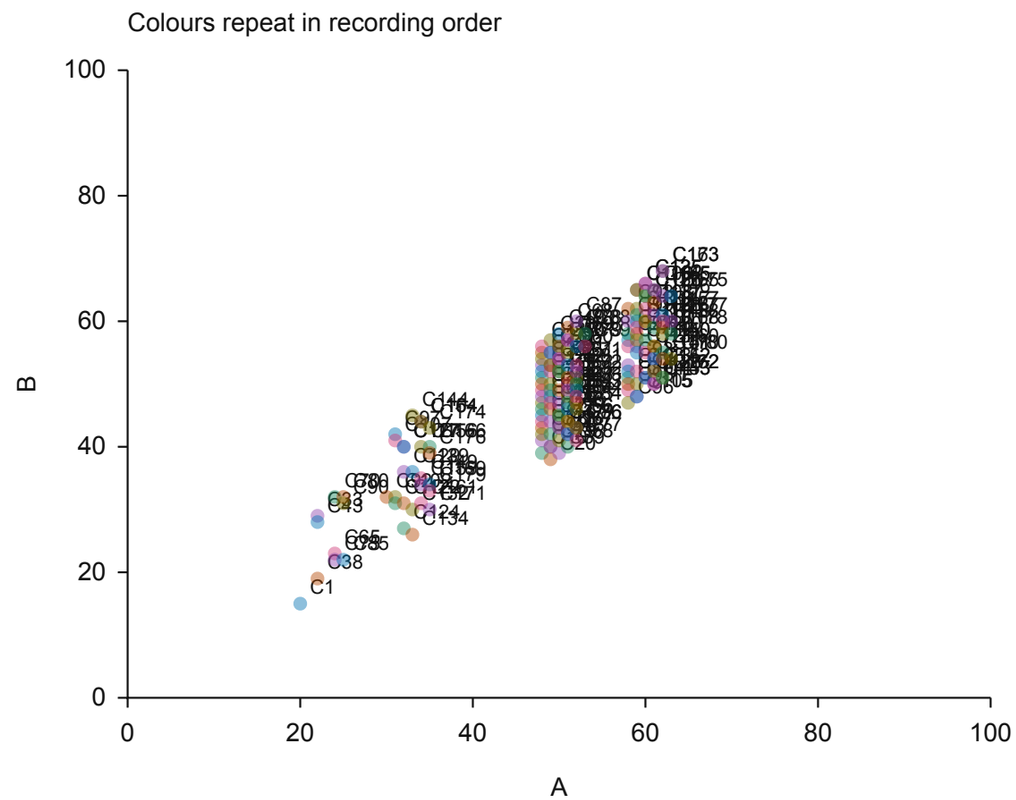
Use transparency to reveal overlap

Partly transparent markers make overlapping observations darker. Darkness suggests crowding; it is not an exact count.

Starting figure



Improved version



Show code changes

Show full Python code for these plots

About the data

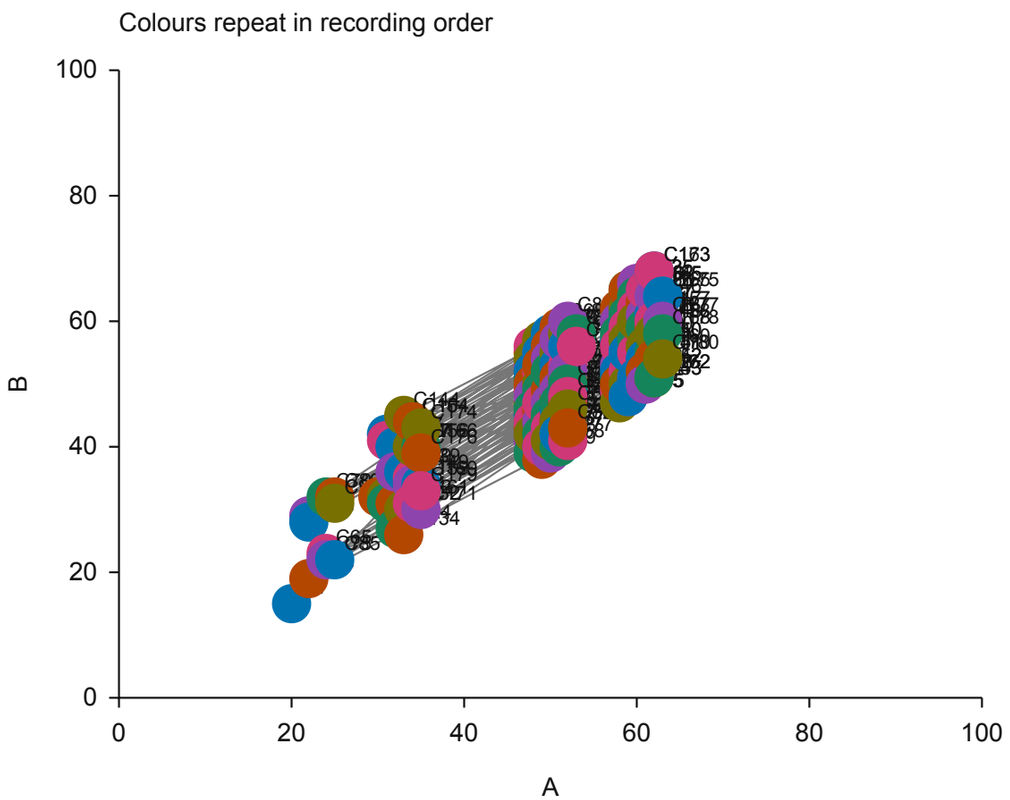
Crowded point plots

Improvement 4 of 9

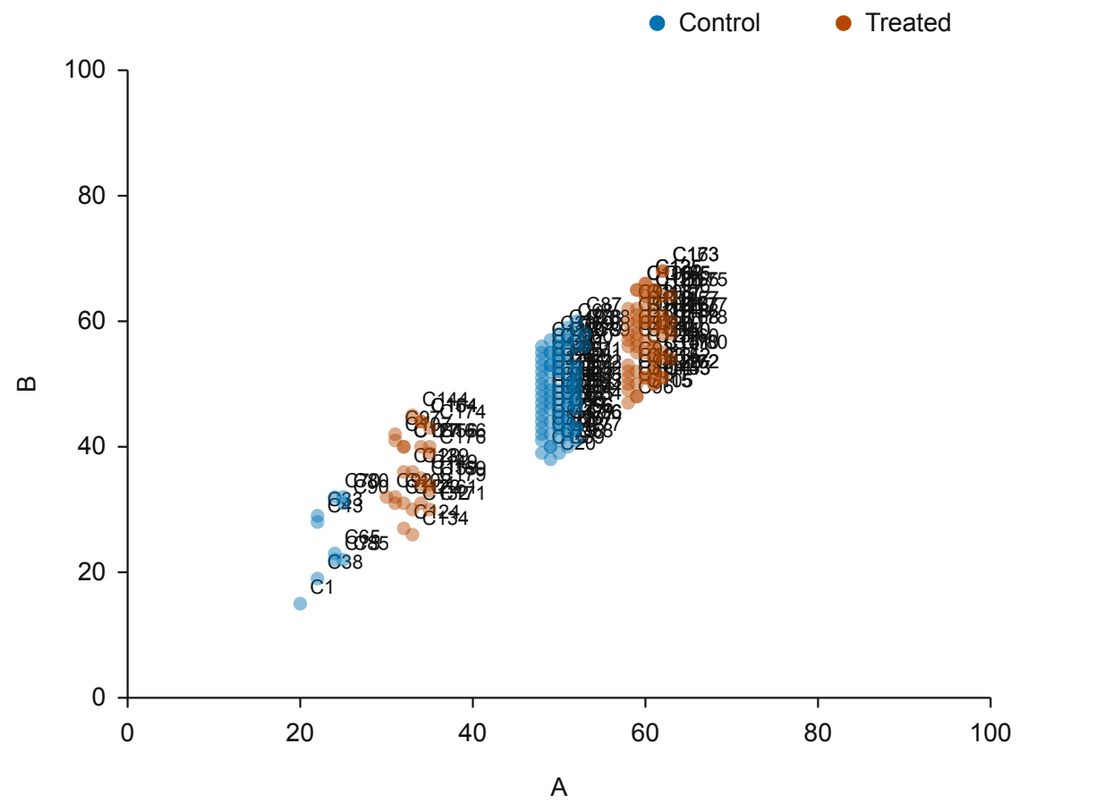
Give colour a useful meaning

Replace the repeating colours assigned by recording order with blue for control and orange for treated. The legend now explains the colour mapping.

Starting figure



Improved version



> **Show code changes**

> **Show full Python code for these plots**

> **About the data**

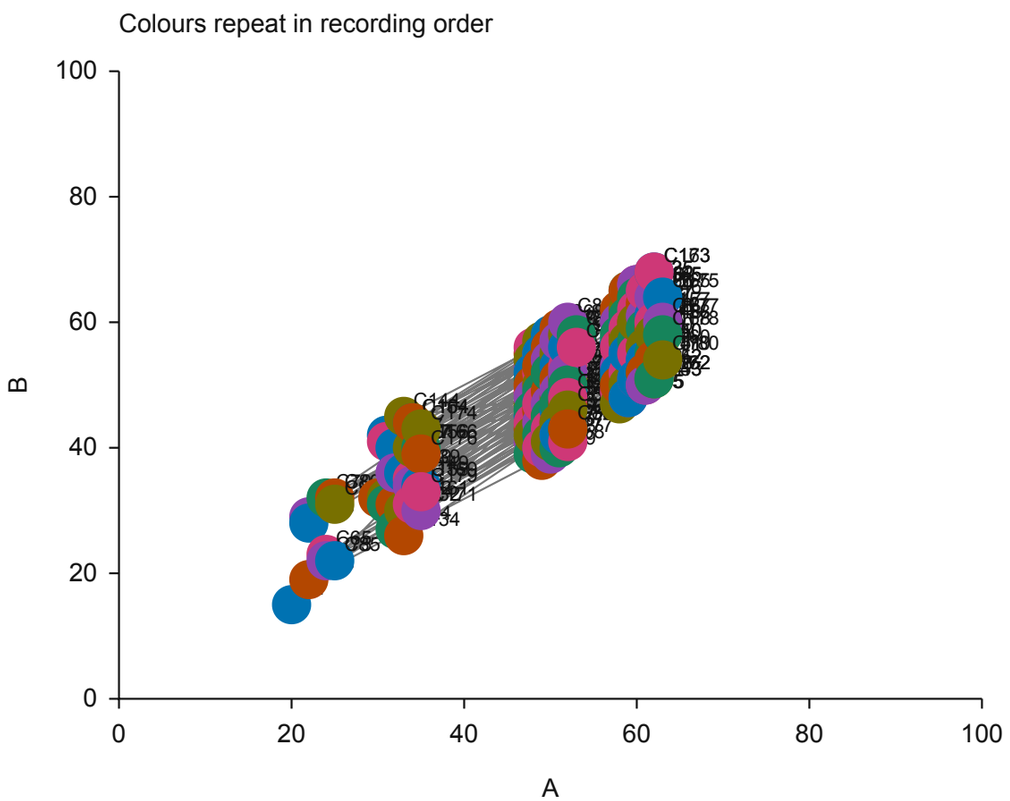
Crowded point plots

Improvement 5 of 9

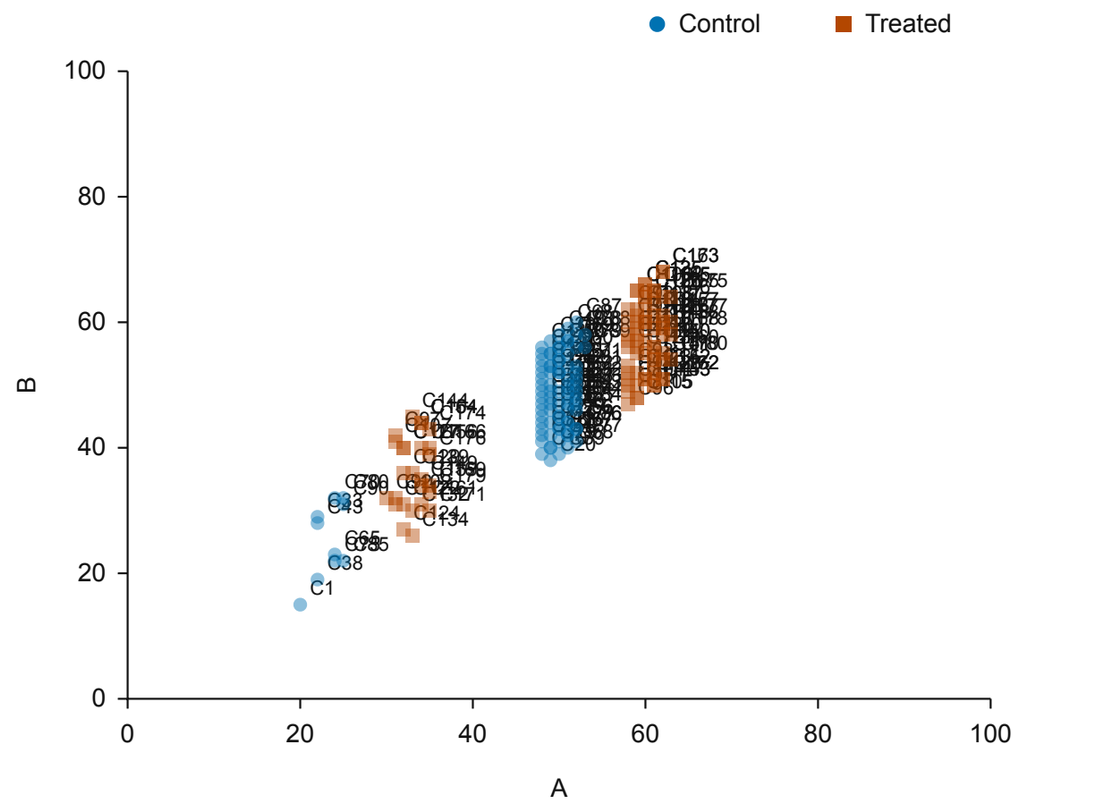
Add a second way to distinguish groups

Use circles for control and squares for treated, as well as colour. This helps readers distinguish groups when colours are hard to tell apart.

Starting figure



Improved version



> **Show code changes**

> **Show full Python code for these plots**

> **About the data**

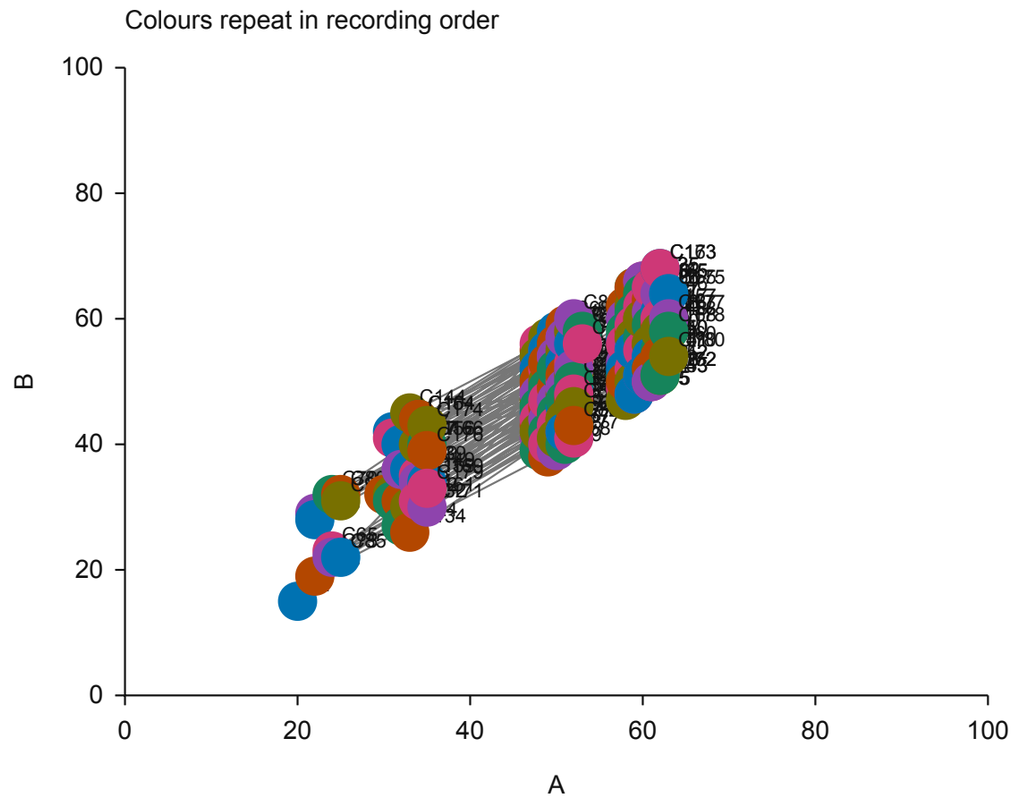
Crowded point plots

Improvement 6 of 9

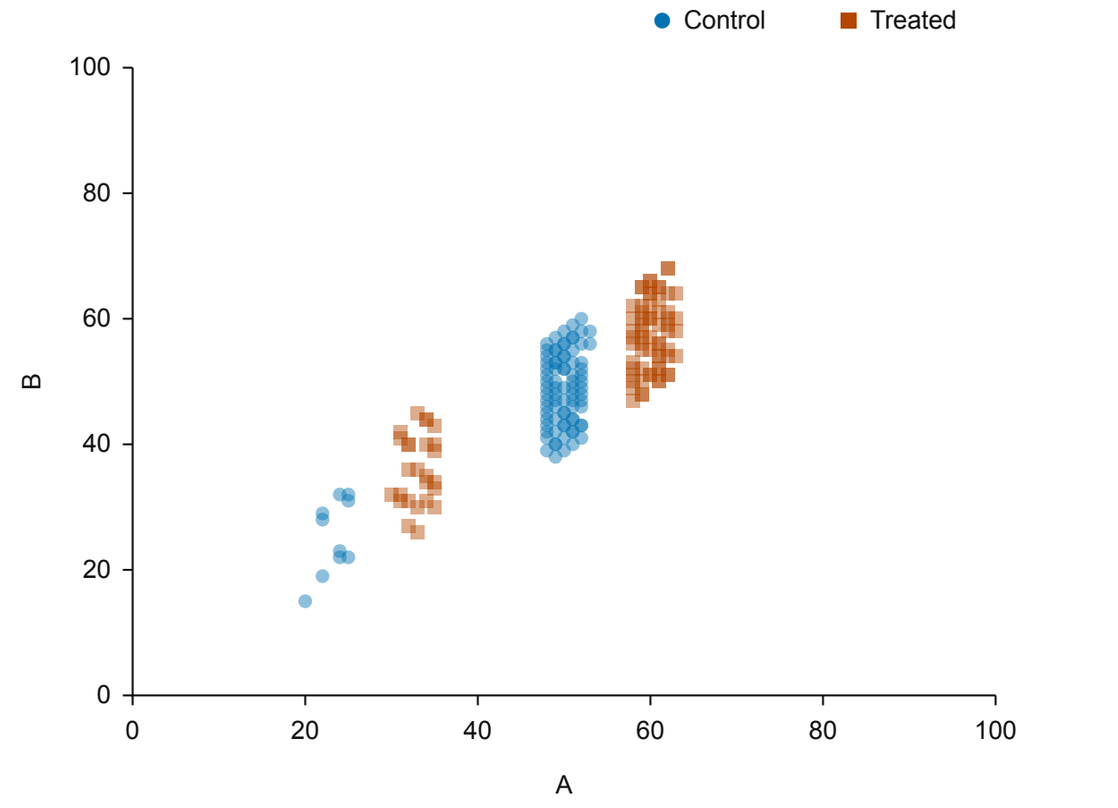
Remove unnecessary point labels

Hide the 180 culture IDs to expose the measurements. IDs remain in the downloadable data. Label individual observations only when they support a specific question.

Starting figure



Improved version



Show code changes

Show full Python code for these plots

About the data

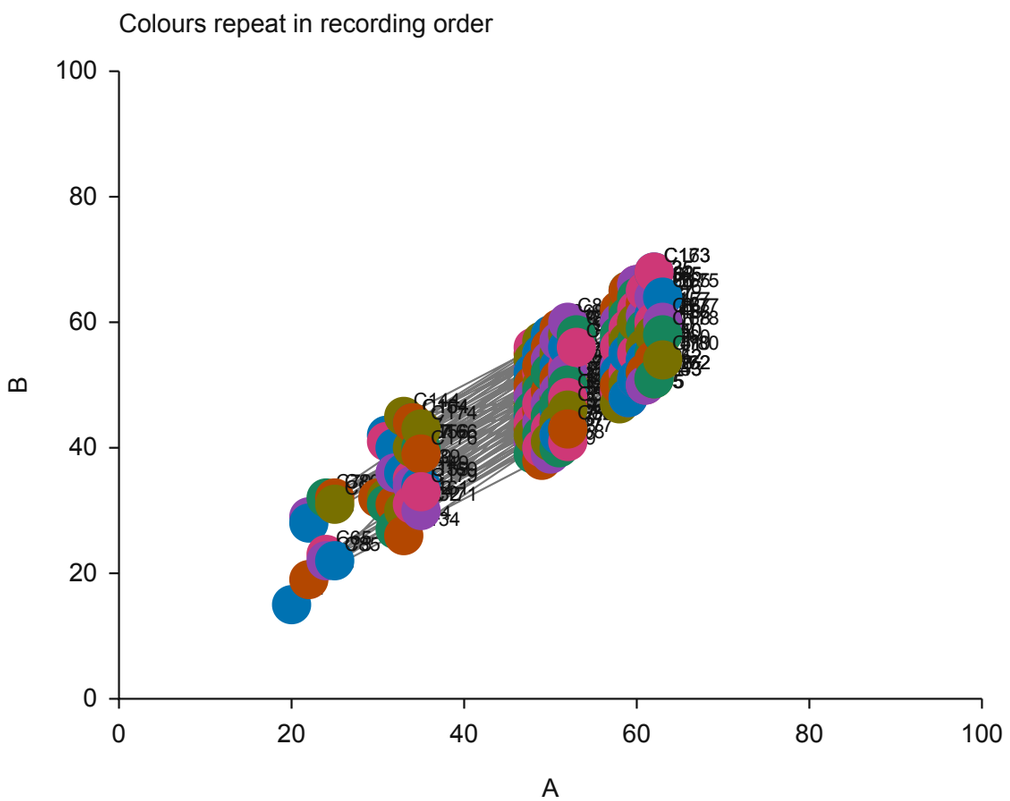
Crowded point plots

Improvement 7 of 9

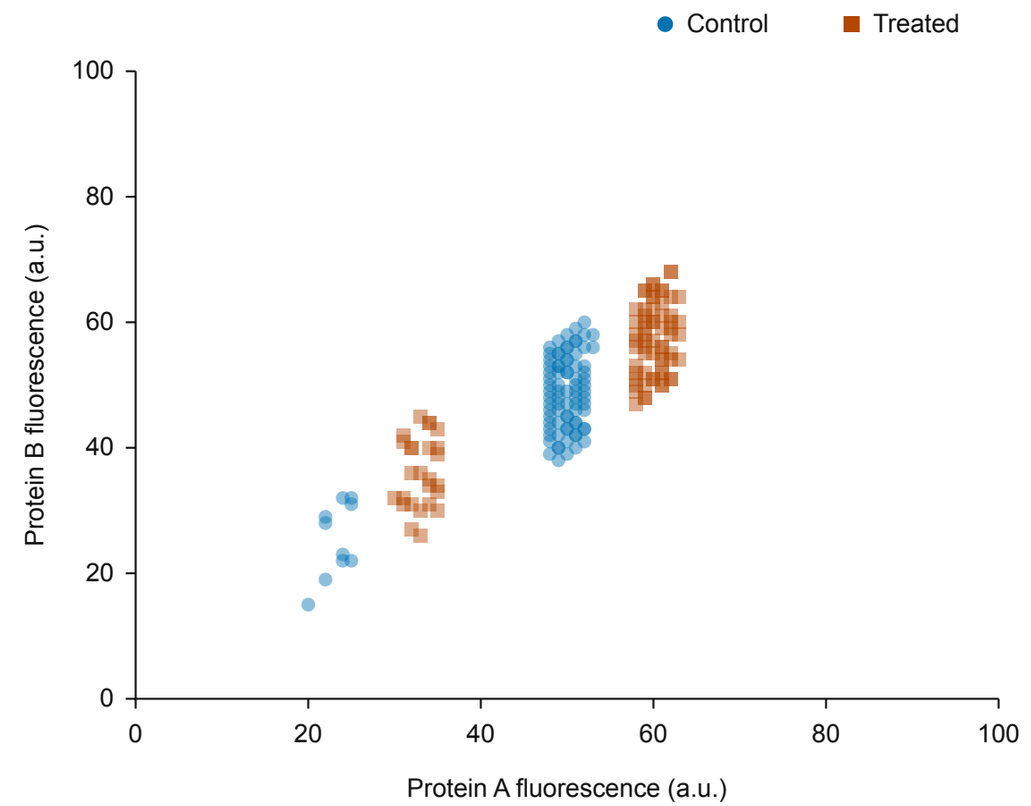
Explain both measurements

Replace cryptic variable names with protein names and units. The measurements are fluorescence intensities, not counts or concentrations.

Starting figure



Improved version



> **Show code changes**

> **Show full Python code for these plots**

> **About the data**

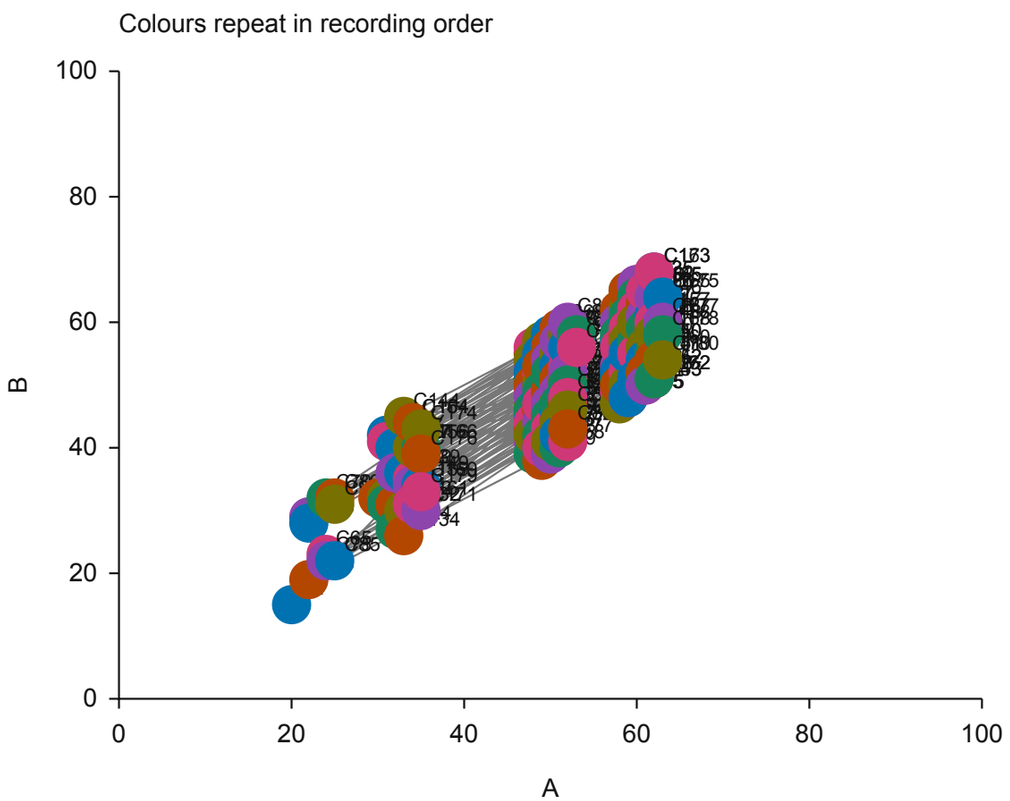
Crowded point plots

Improvement 8 of 9

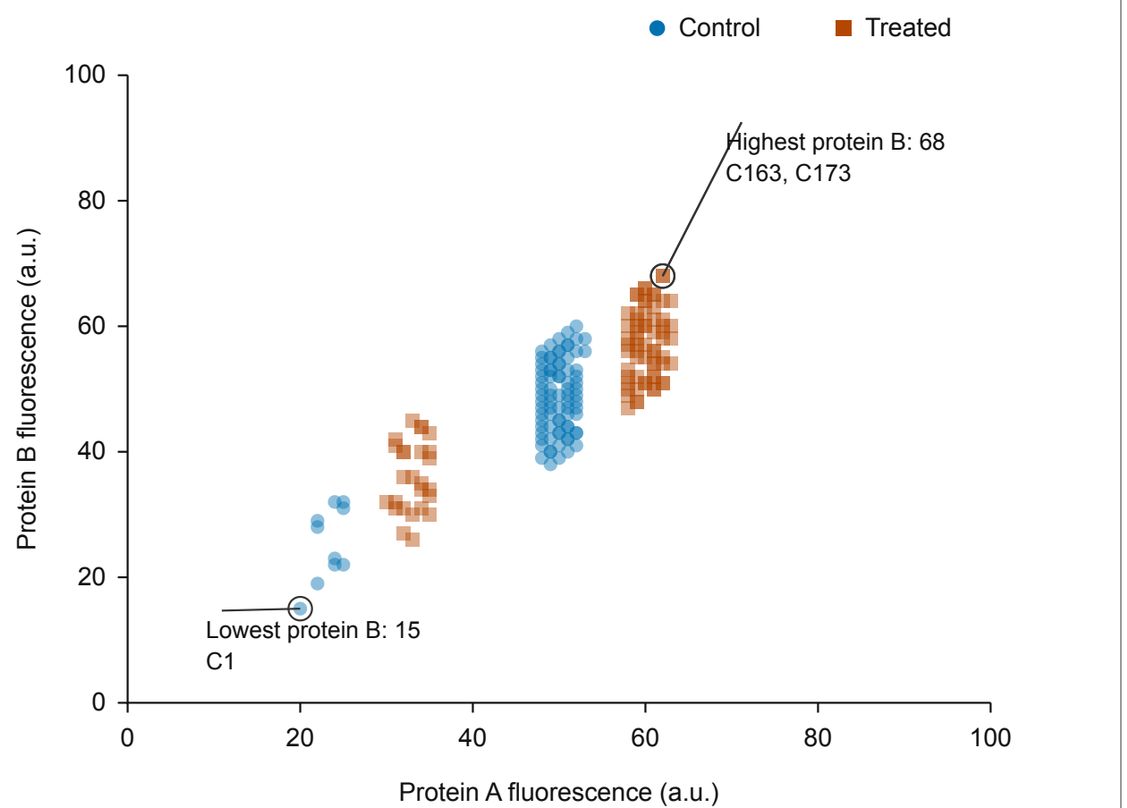
Annotate selected extremes

Label the lowest and highest protein B measurements with culture IDs and leader lines. These are observed extremes, not a statistical outlier classification.

Starting figure



Improved version



> **Show code changes**

> **Show full Python code for these plots**

> **About the data**

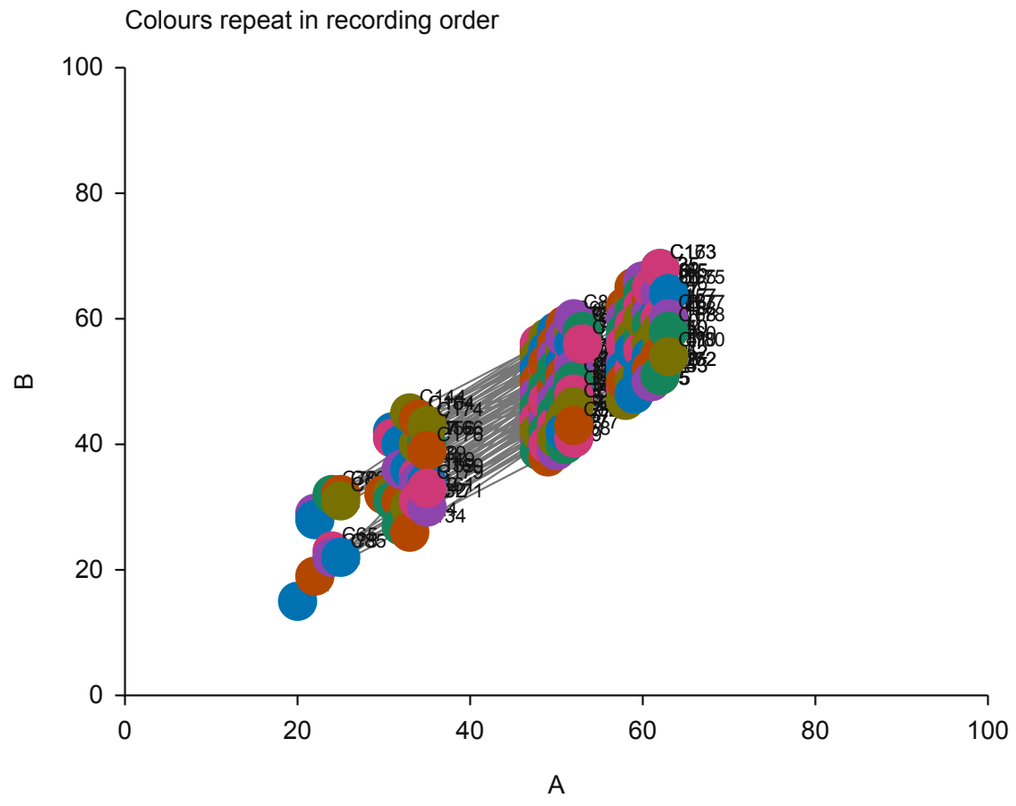
Crowded point plots

Final comparison

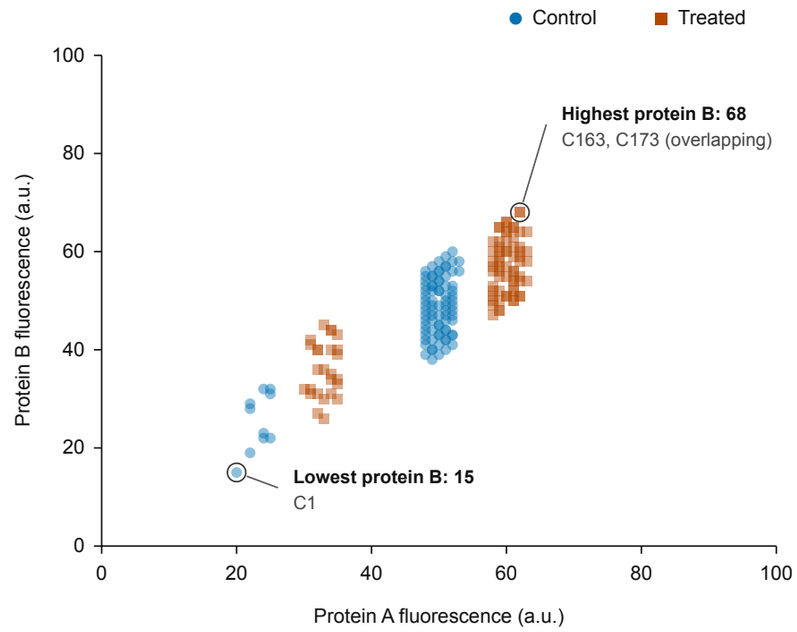
Refine annotation placement

Move labels into clear space and attach leader lines beside the text, ending at the highlight boundary. Combine duplicate leaders for overlapping observations and explicitly identify the overlap.

Starting figure



✓ Final improvement



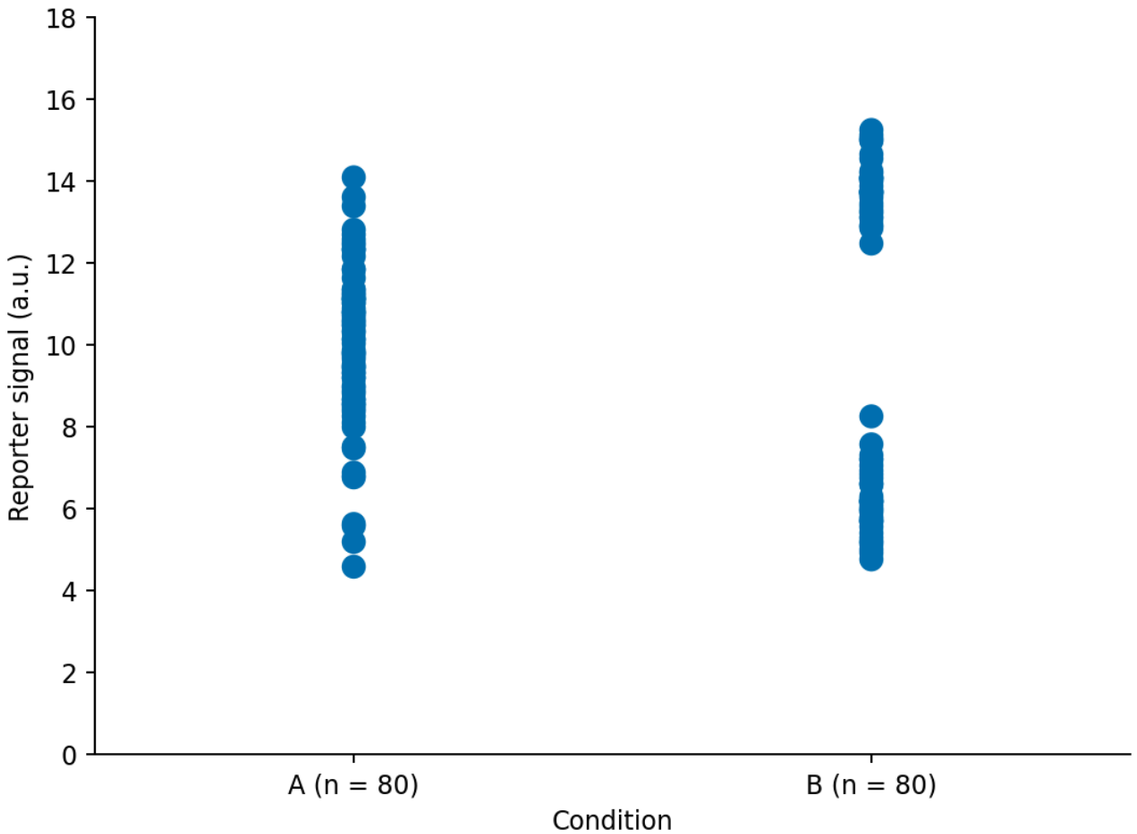
> **Show code changes**

> **Show full Python code for these plots**

> **About the data**

Overlapping points by condition

Each dot is a simulated cell. Large opaque markers hide how many cells occupy the same region. Can you see how the distributions differ?



Current view

Next reveals a clearer view of the same cells.

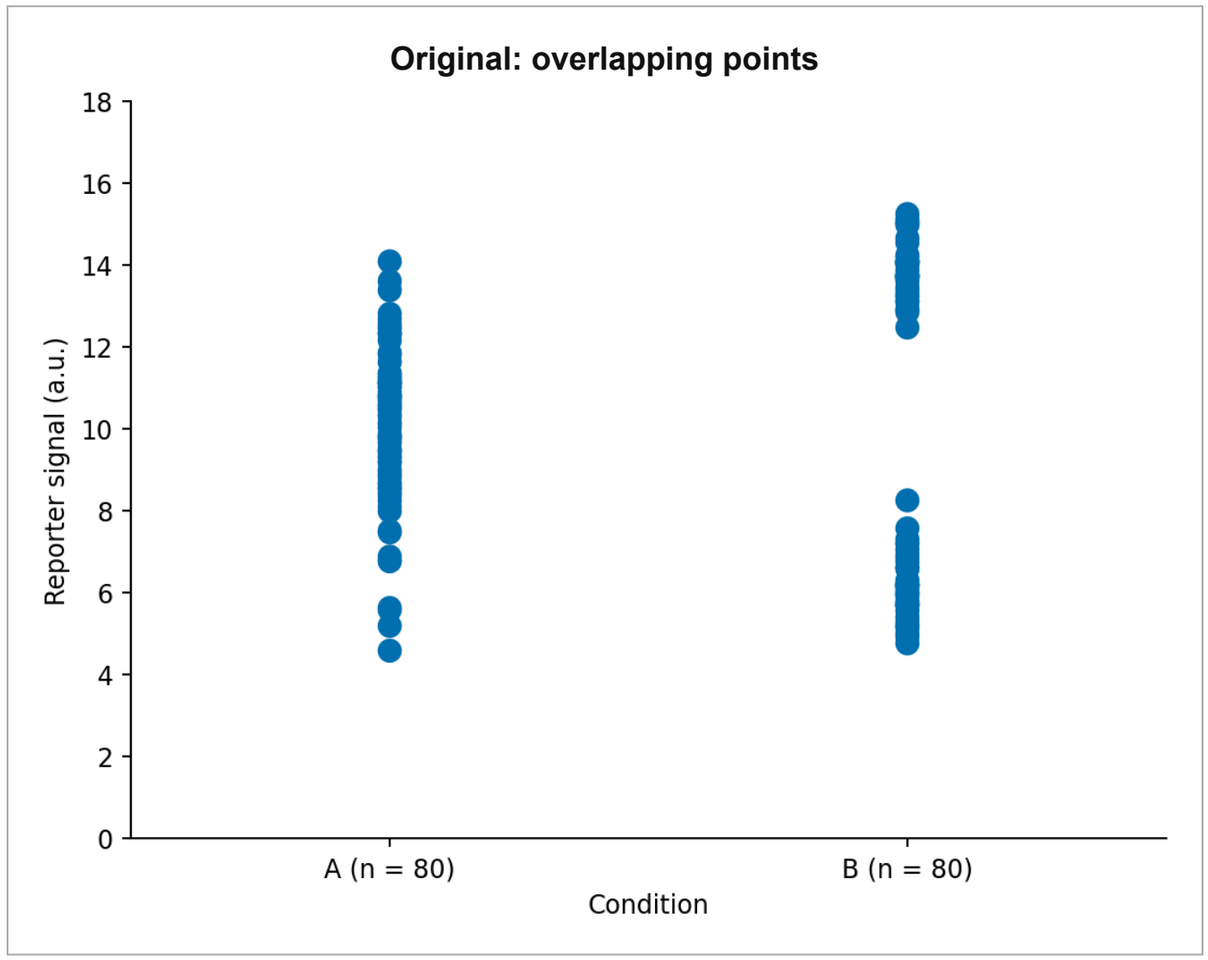
Show code changes

Data & interpretation

Show full Python code

Separate overlapping observations

Smaller, transparent points with horizontal jitter expose both distributions. Horizontal offsets have no measured meaning; all signal values stay unchanged.



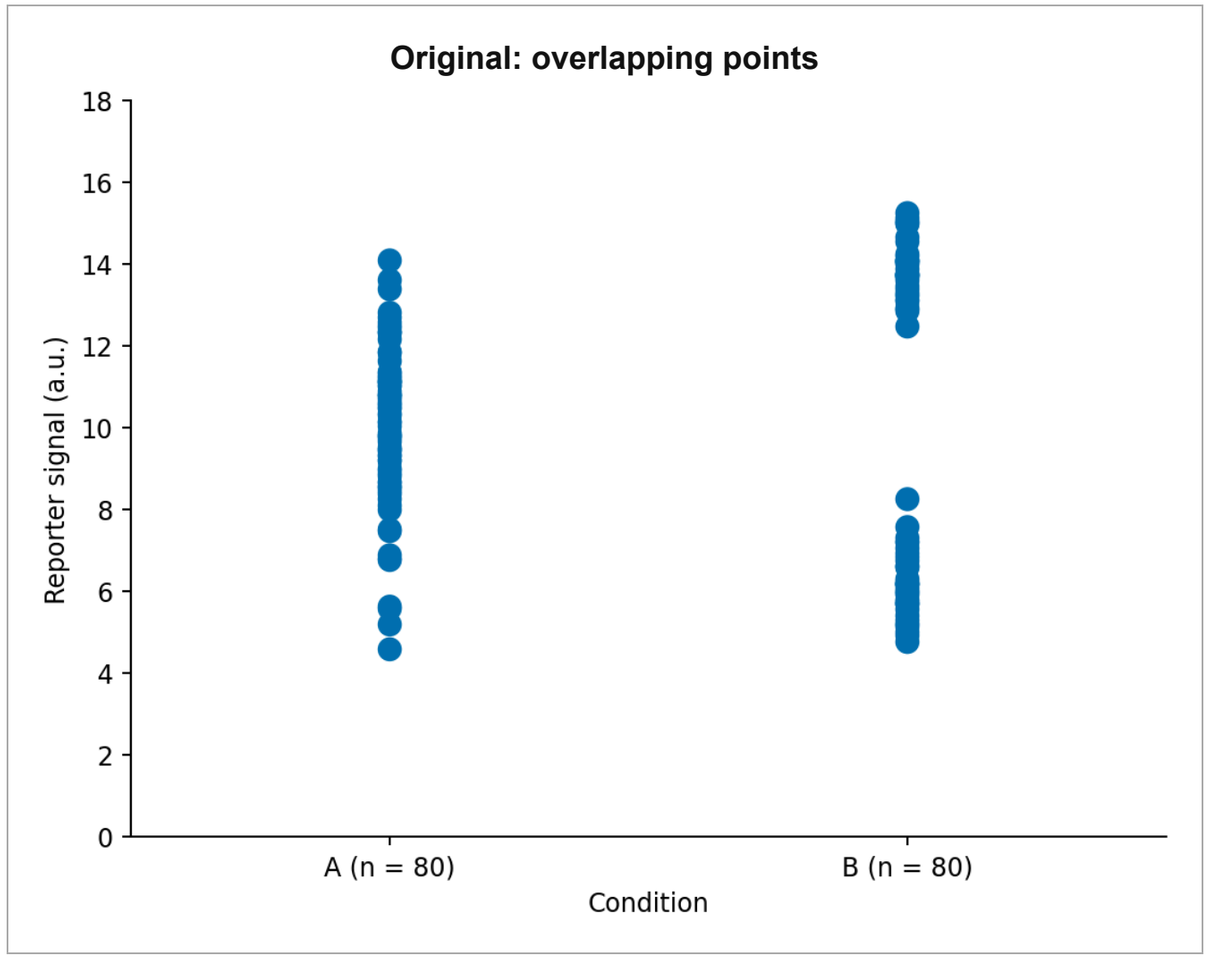
Show code changes

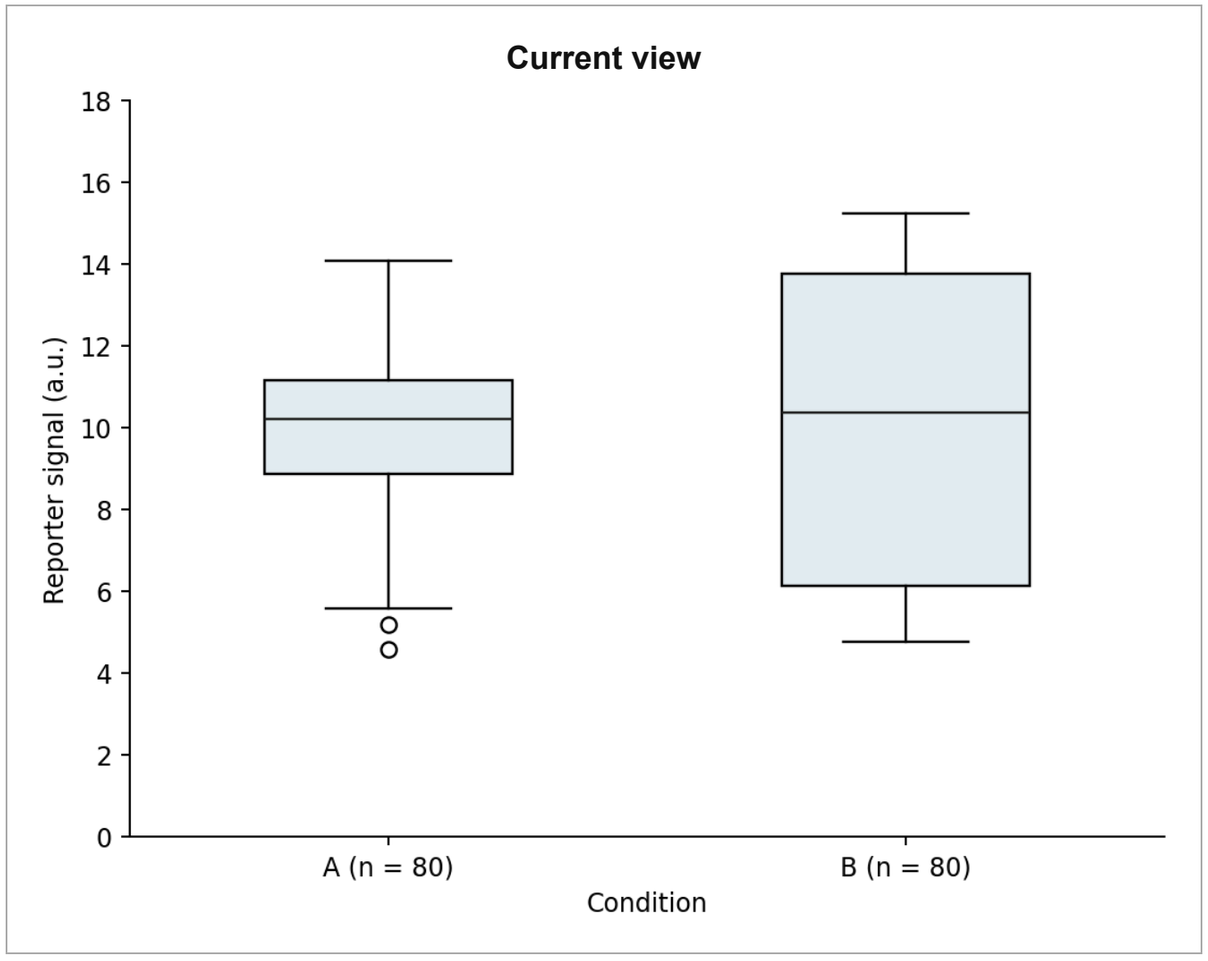
Data & interpretation

Show full Python code

Try a box plot: is this better?

A box plot gives a compact median and middle 50%, with whiskers extending to 1.5 × IQR and any outliers shown separately. But the two concentrations in B are hidden by the summary.





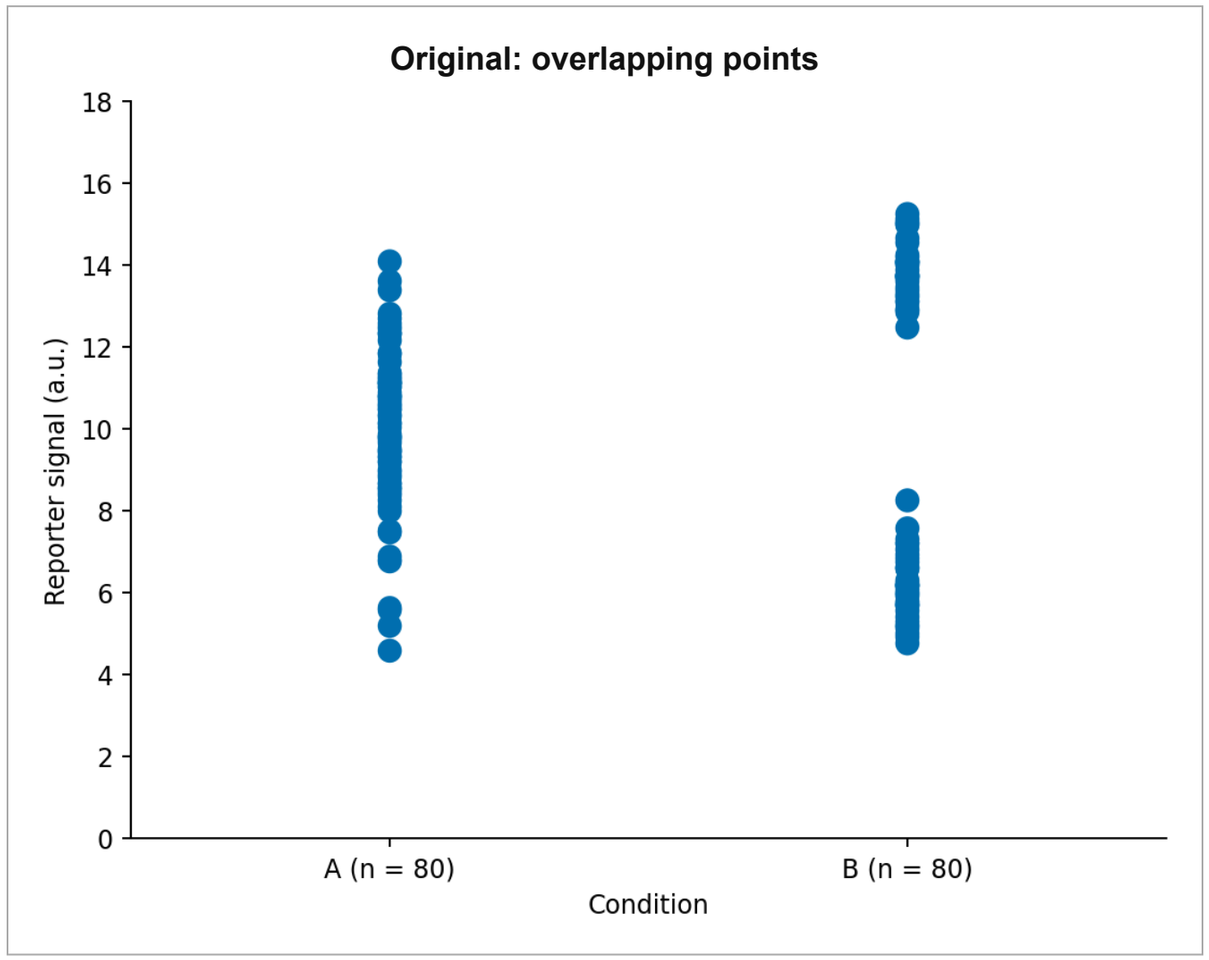
Show code changes

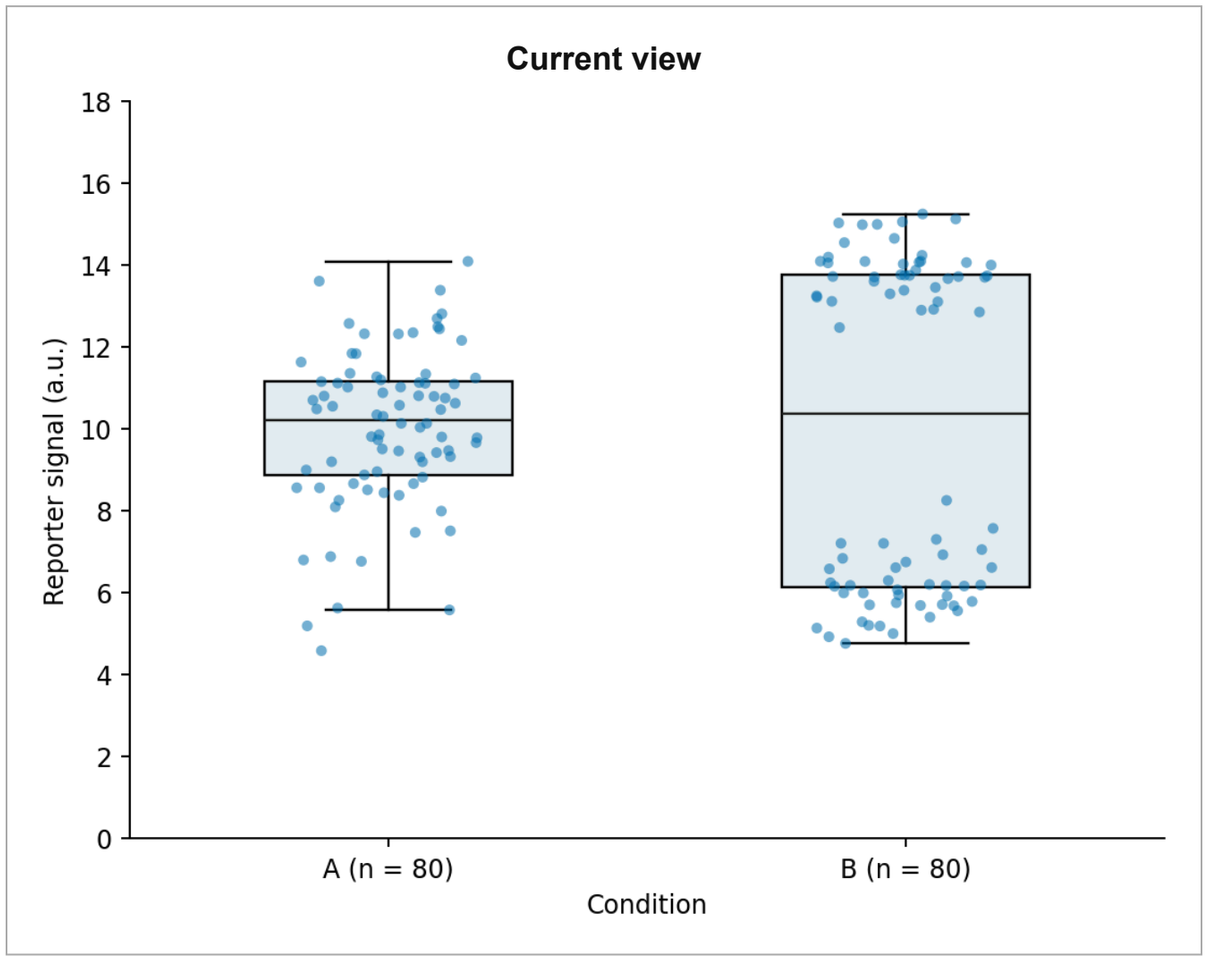
Data & interpretation

Show full Python code

Keep the box, bring back the cells

Individual points recover the distribution detail while the box supplies a summary. B has two concentrations that its median alone cannot describe.





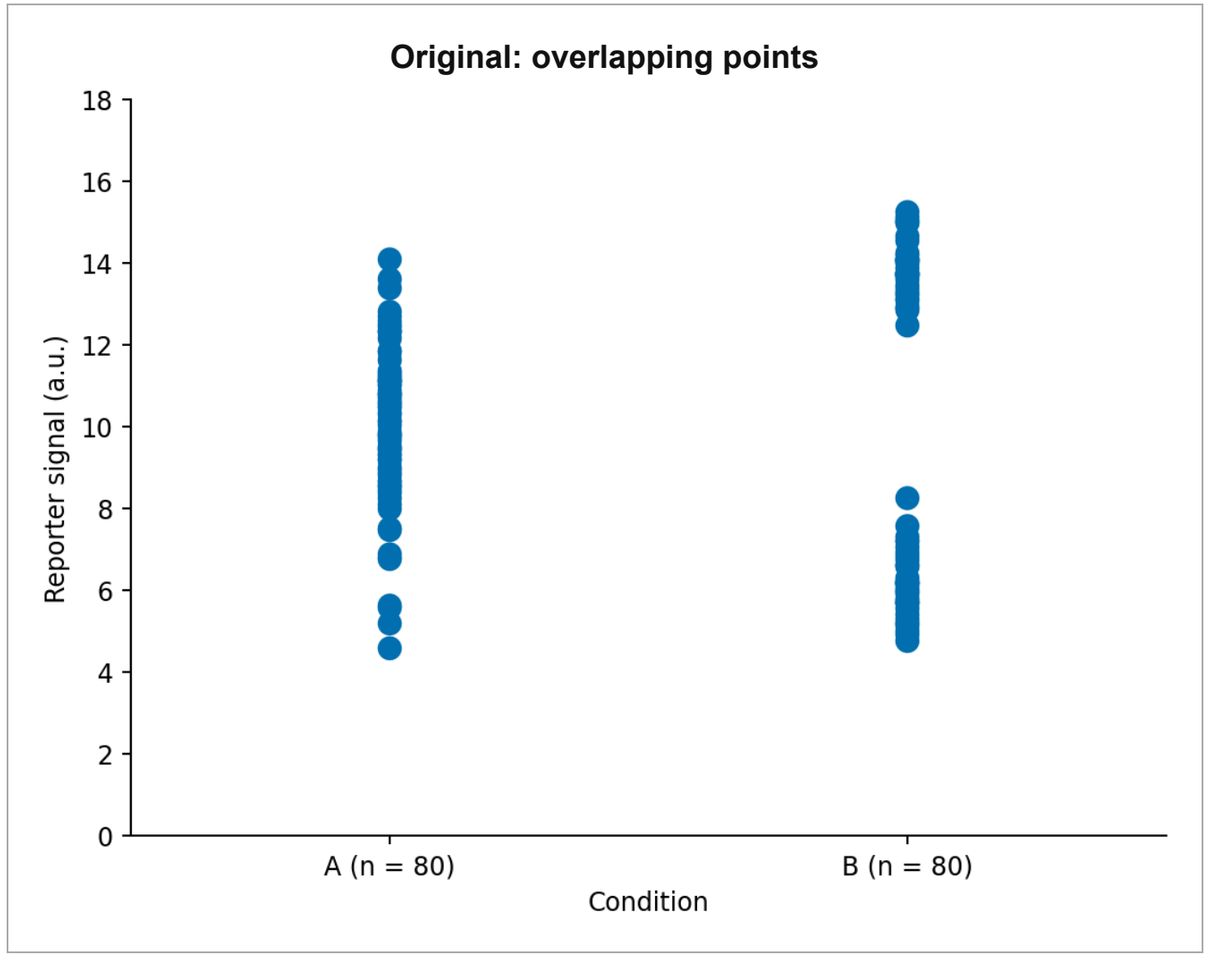
Show code changes

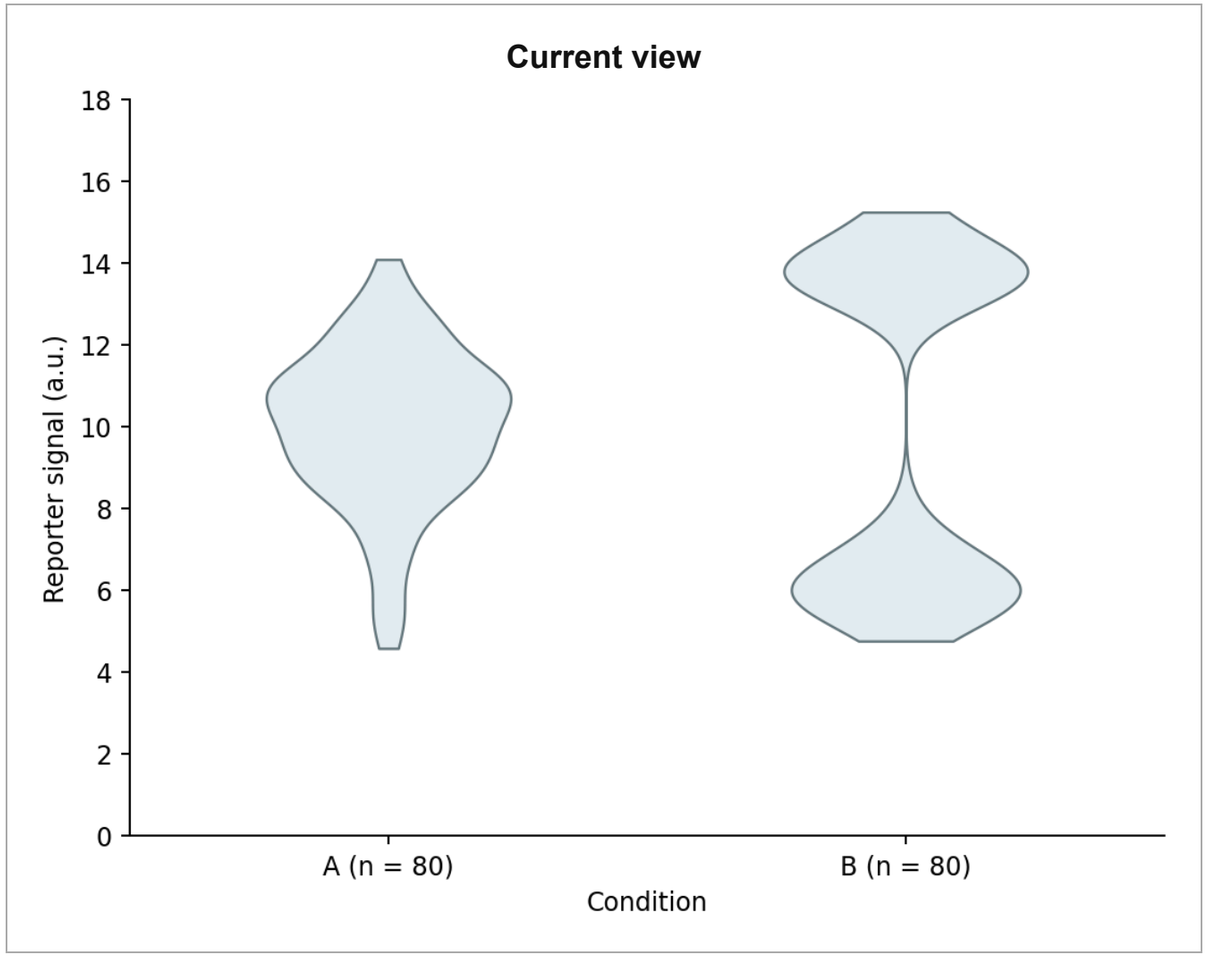
Data & interpretation

Show full Python code

Try a violin: what does it add?

A smoothed density makes B’s two concentrations easier to see, but hides the individual observations. Both violins use the same bandwidth (0.65 signal units) and width rule; smoothing still affects the shape.





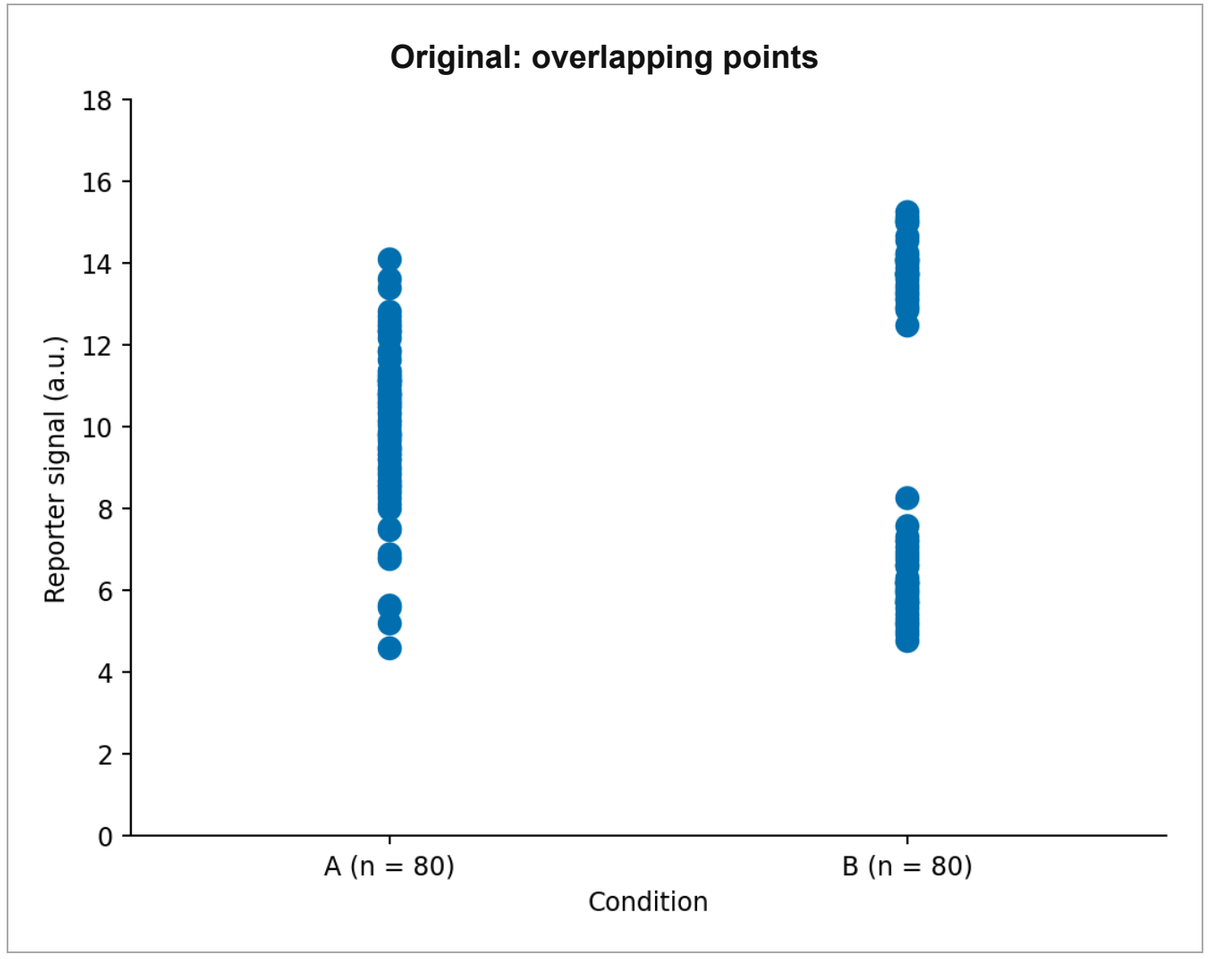
Show code changes

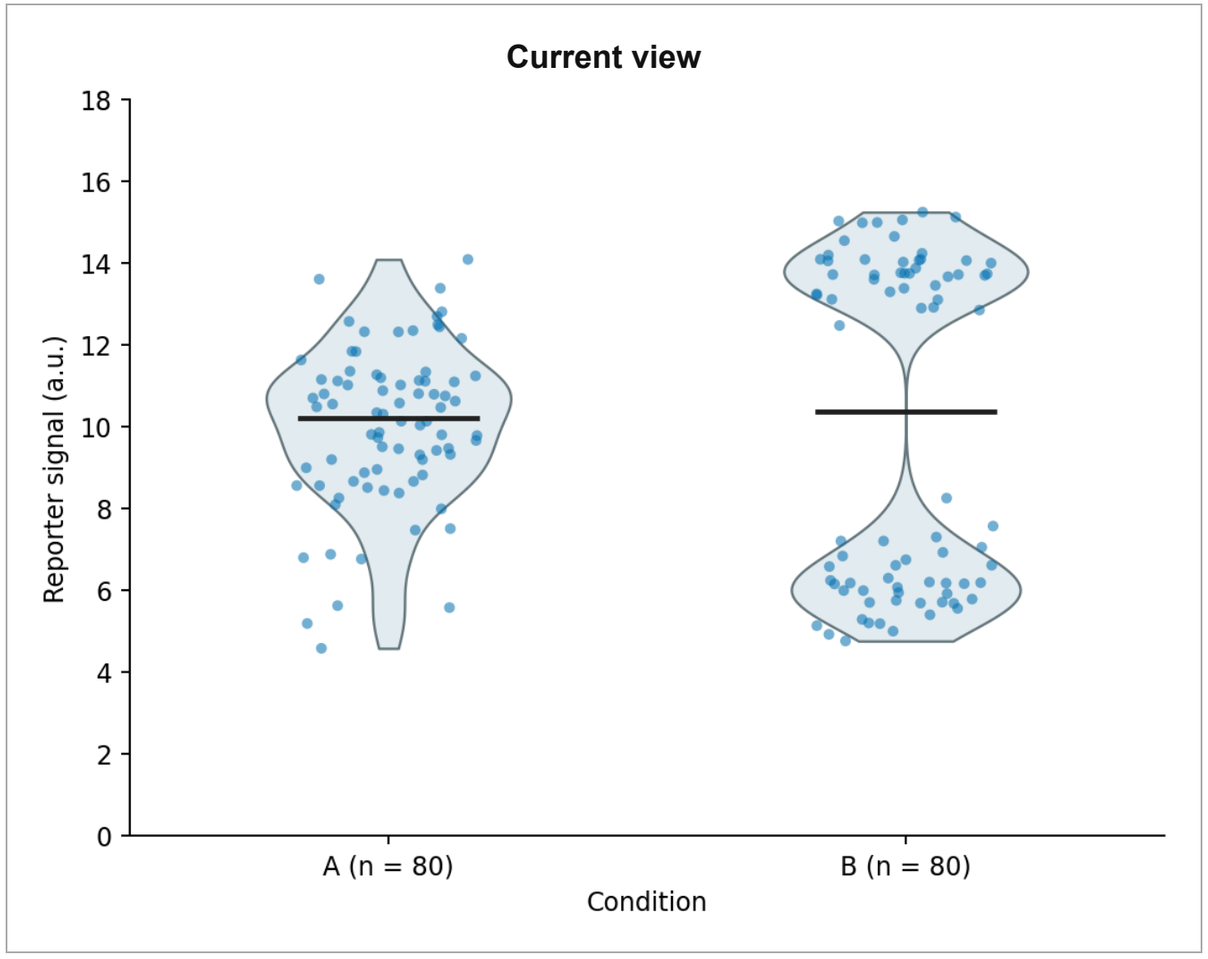
Data & interpretation

Show full Python code

Combine shape, observations and median

The violin shows shape, dots show all 80 cells per condition, and a short line marks the median. This is useful for this question; a simpler point plot may already be sufficient.





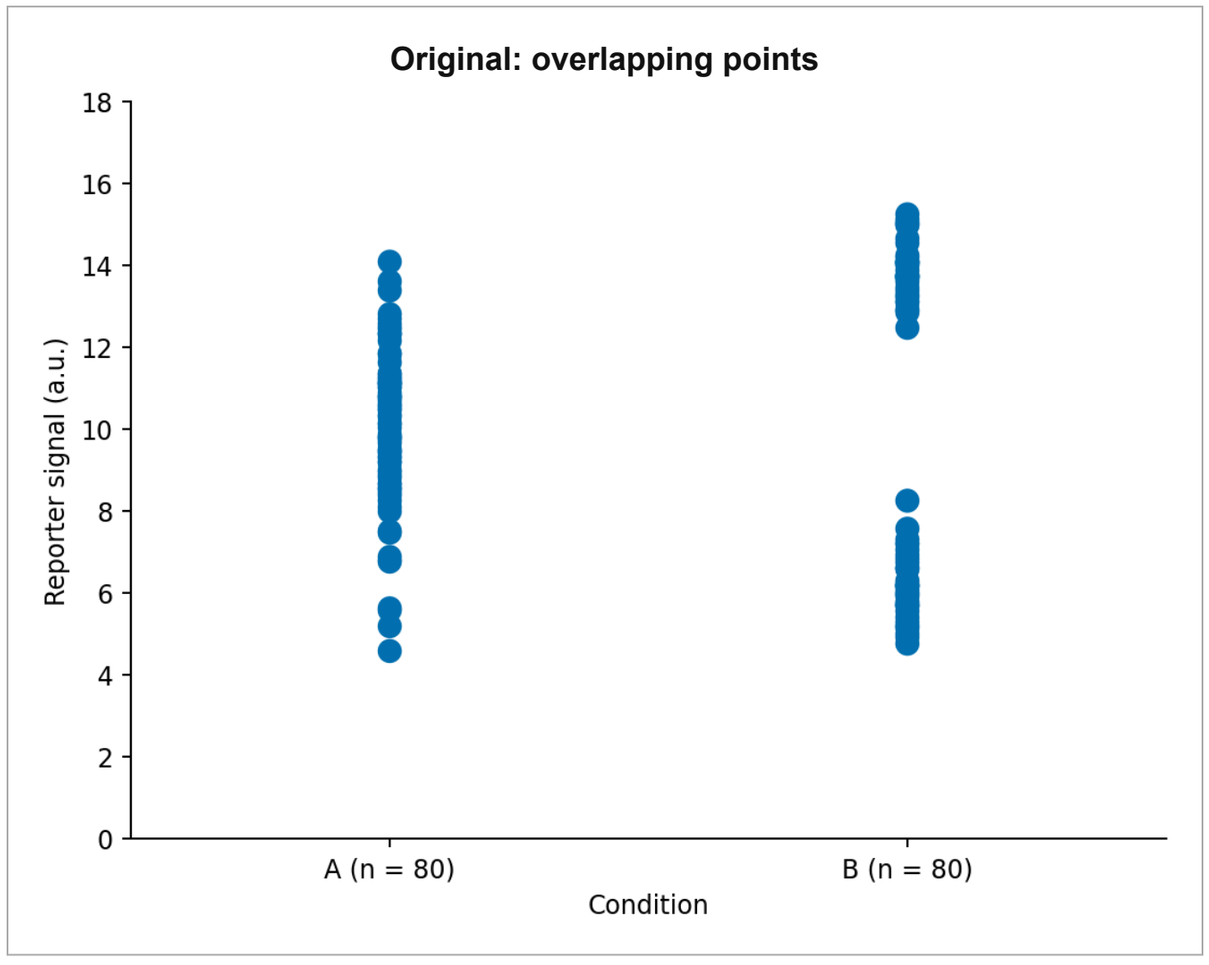
Show code changes

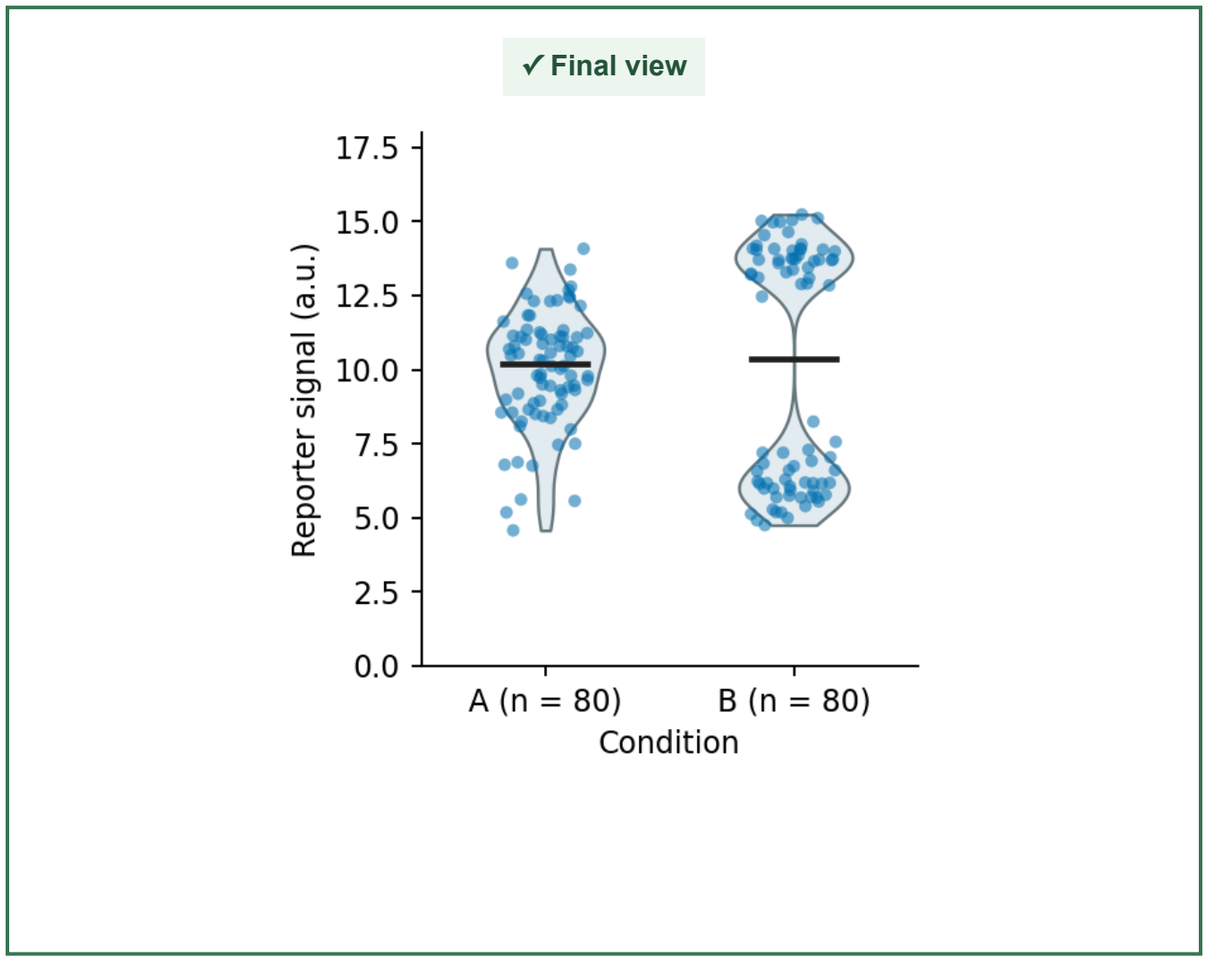
Data & interpretation

Show full Python code

Compact publication layout

Reduce the canvas from 6.2 × 4.6 to 3 × 3 inches. All cells, density shapes, medians, sample sizes and axes remain; text stays at its original point size. Check the exported figure at the intended publication size.





Download publication PDF · Download SVG

Show code changes

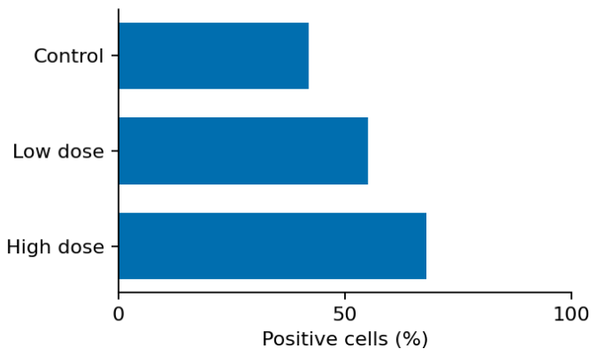
Data & interpretation

Show full Python code

Too much empty canvas

The plot occupies only the middle of an 8 × 5 inch canvas. Empty margins consume space without helping the comparison.

Original · 8 × 5 in



Current · 8 × 5 in

Next reveals a tighter layout.

Both previews use the same size ratio. Screen inches are not physical inches; inspect the exported figure at its intended size.

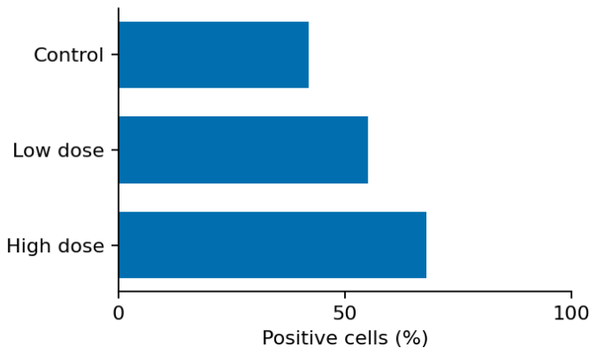
How to make this figure compact

Show full Python code

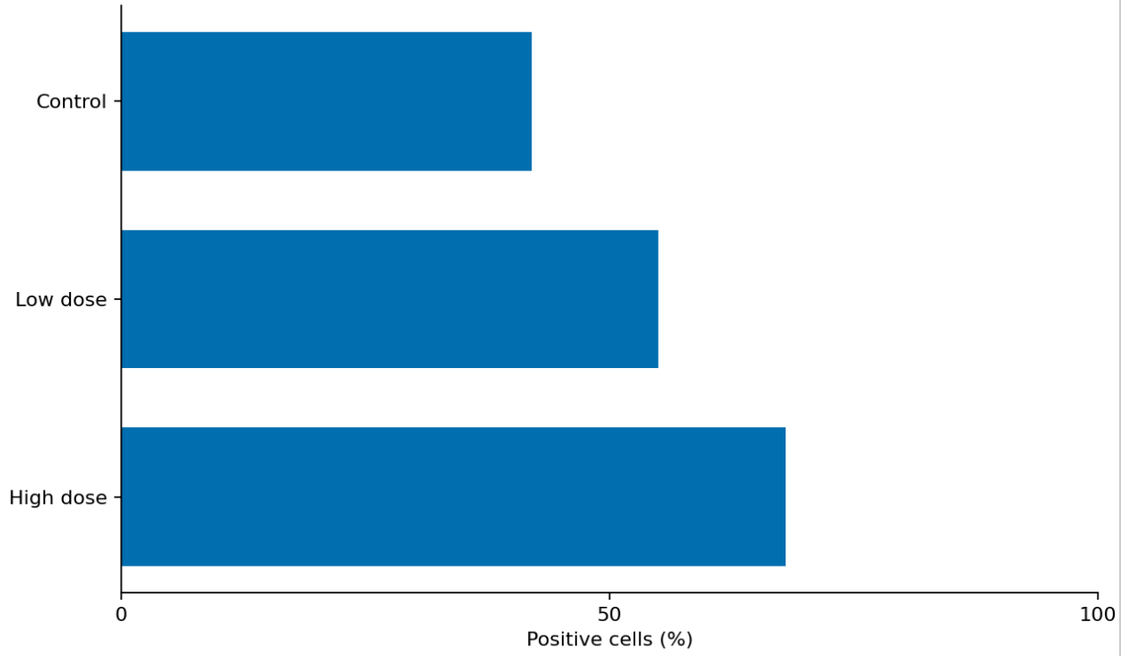
Trim the outer margins

Keep the canvas size and data unchanged, but let the plotting area use the available space. Preserve room for labels.

Original · 8 × 5 in



Current · 8 × 5 in



Both previews use the same size ratio. Screen inches are not physical inches; inspect the exported figure at its intended size.

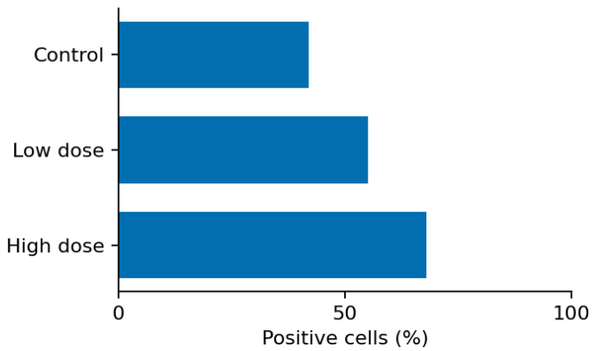
How to make this figure compact

Show full Python code

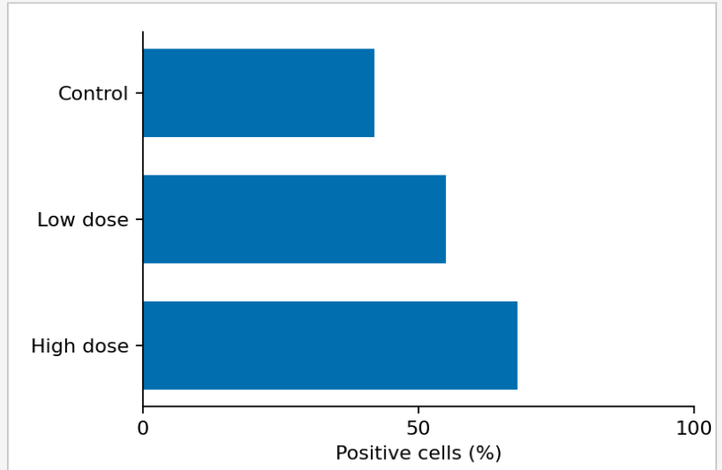
Try a smaller canvas

Reduce to 5 × 3.5 inches while keeping labels at 10 pt. Fewer pixels are not the goal: the printed or displayed size must remain readable.

Original · 8 × 5 in



Current · 5 × 3.5 in



Both previews use the same size ratio. Screen inches are not physical inches; inspect the exported figure at its intended size.

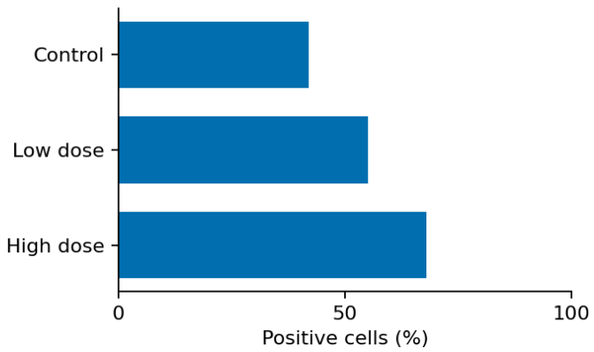
How to make this figure compact

Show full Python code

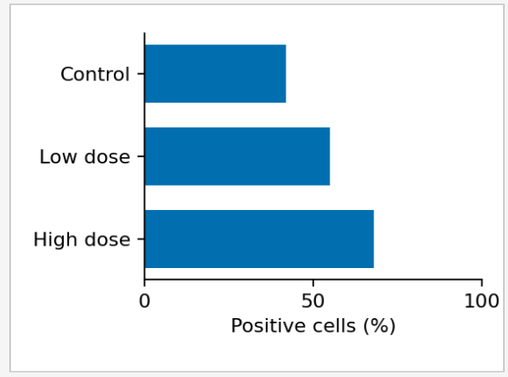
Try a compact figure

At 3.5 × 2.6 inches, all three categories, the zero baseline and the percentage scale still fit. This is a candidate for a single-column figure.

Original · 8 × 5 in



Current · 3.5 × 2.6 in



Both previews use the same size ratio. Screen inches are not physical inches; inspect the exported figure at its intended size.

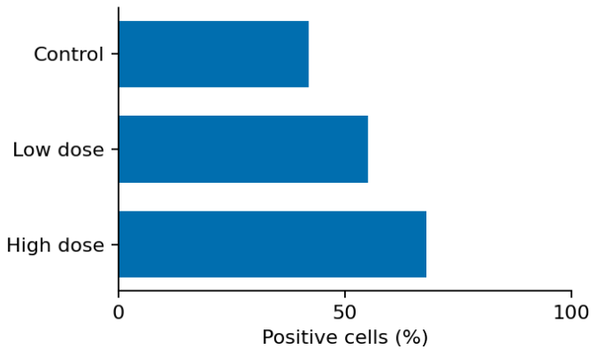
How to make this figure compact

Show full Python code

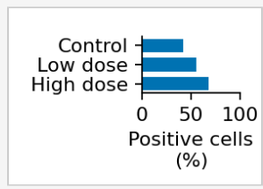
Find the limit: too small

At 1.8 × 1.25 inches, the plot becomes cramped even with a wrapped axis title. Keeping 10 pt text exposes the limit; shrinking text would hide the problem rather than solve it.

Original · 8 × 5 in



Current · 1.8 × 1.25 in



Both previews use the same size ratio. Screen inches are not physical inches; inspect the exported figure at its intended size.

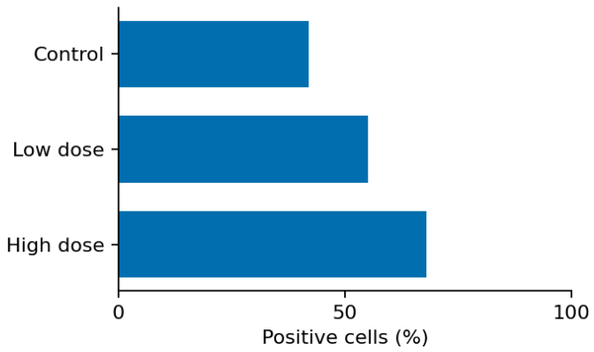
How to make this figure compact

Show full Python code

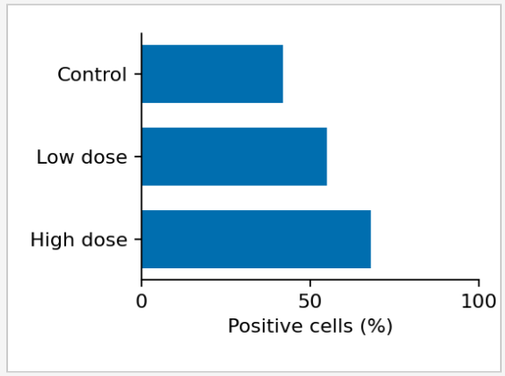
Choose the smallest readable candidate

Return to 3.5 × 2.6 inches: the smallest readable size among the versions tested here. There is no universal minimum; check the actual output size, audience and publication requirements.

Original · 8 × 5 in



✓ Smallest readable candidate



Both previews use the same size ratio. Screen inches are not physical inches; inspect the exported figure at its intended size.

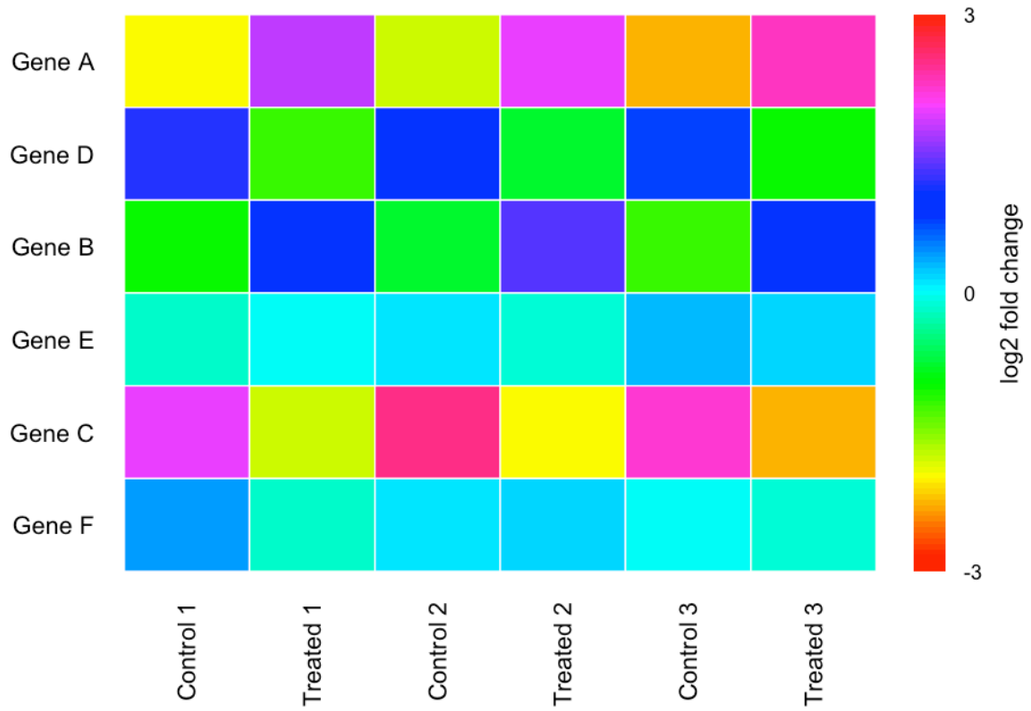
How to make this figure compact

Show full Python code

Start: rainbow colours, mixed order

All measurements and a colour key are present. But it takes effort to relate the colours to direction and compare the interleaved groups.

Starting figure



Next version

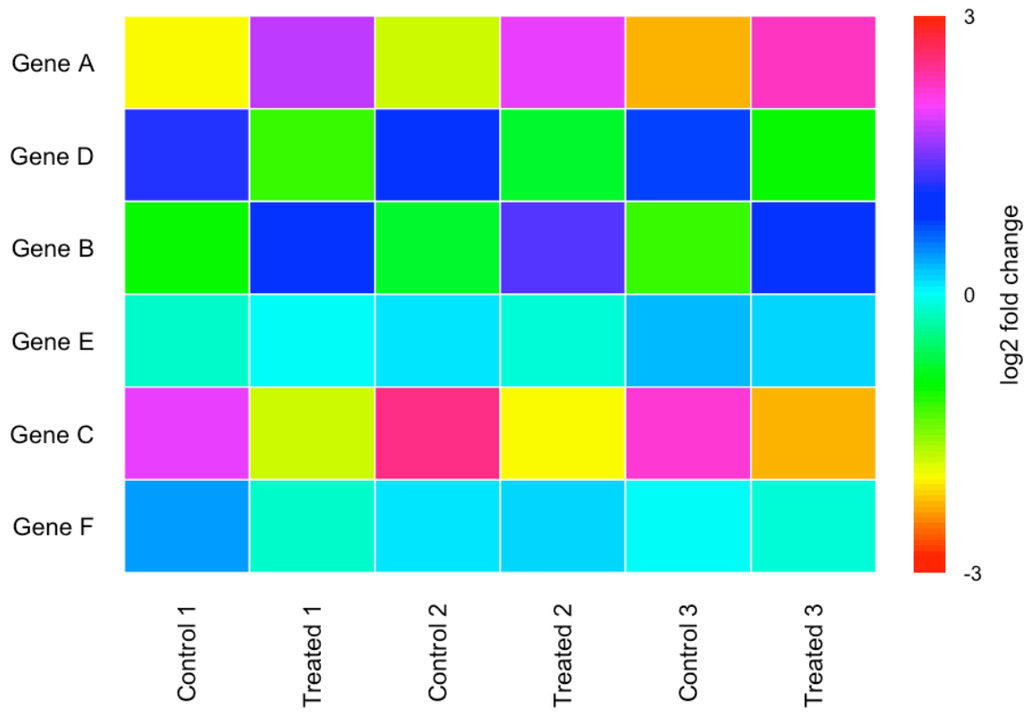
What would make the response patterns easier to compare?

Data & interpretation

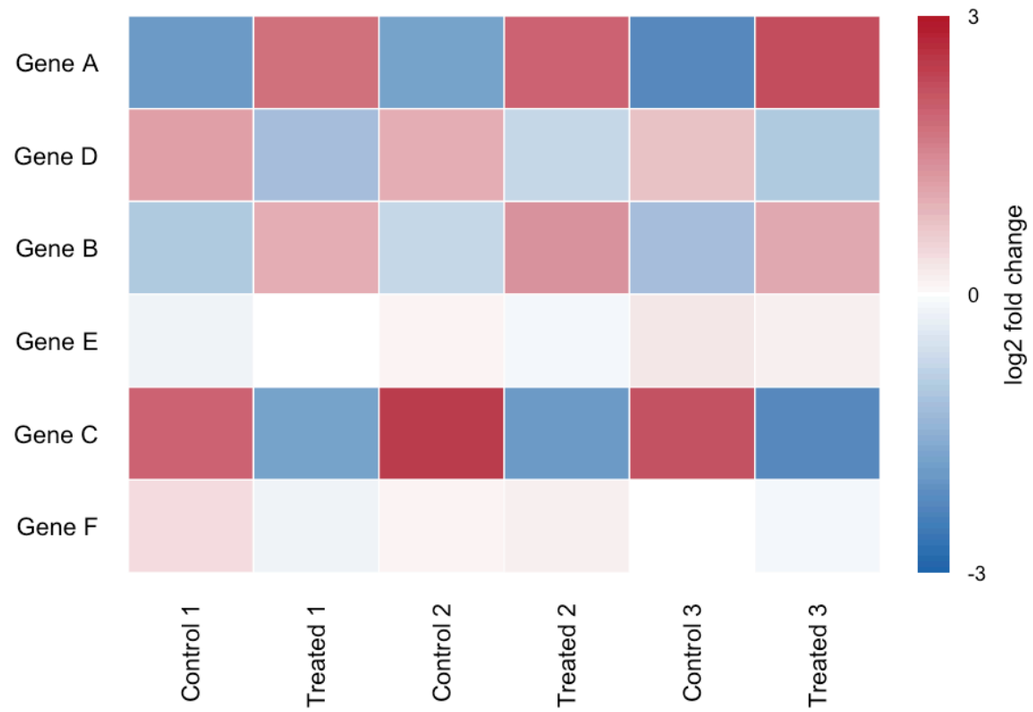
Give zero a clear meaning

Use white at zero, blue for negative values, and red for positive values. The symmetric limits are unchanged. A diverging scale suits this meaningful reference point; it is not the default for every heatmap.

Starting figure



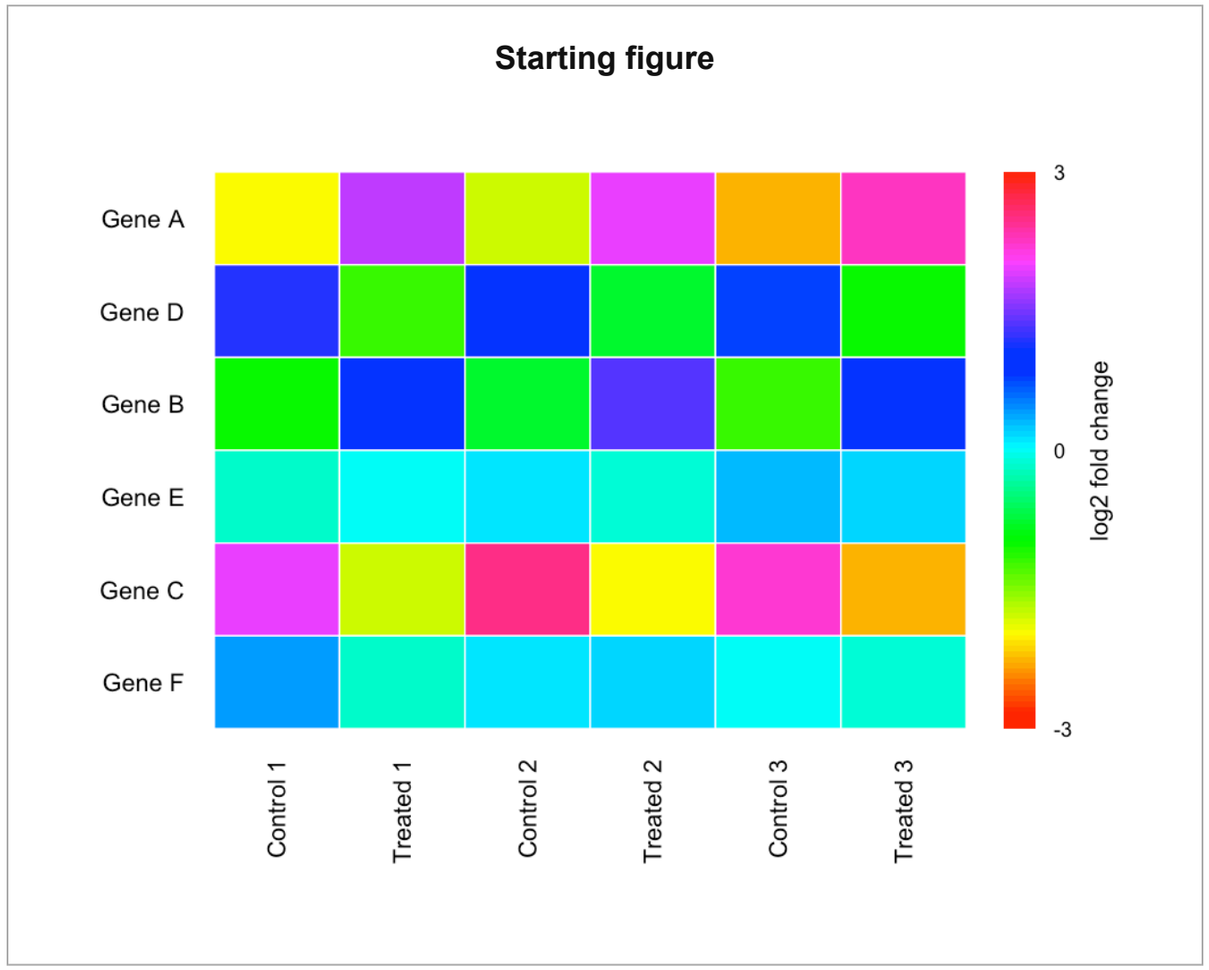
Next version

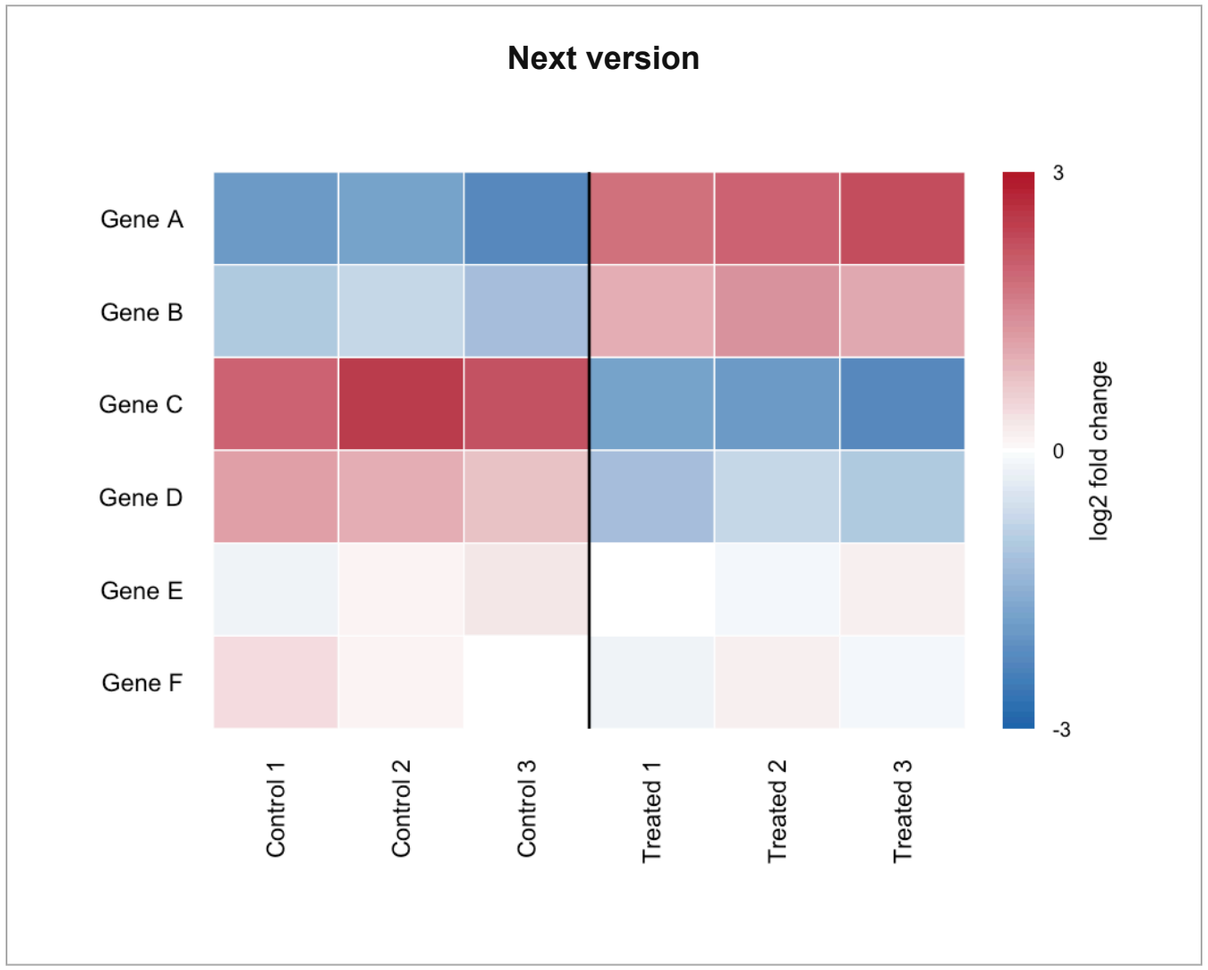


Data & interpretation

Arrange samples and genes

Group control and treated samples, then place genes with similar responses together. Only the order changes: no values are rescaled. The divider marks the two sample groups. This is a teaching order, not a statistical clustering result.

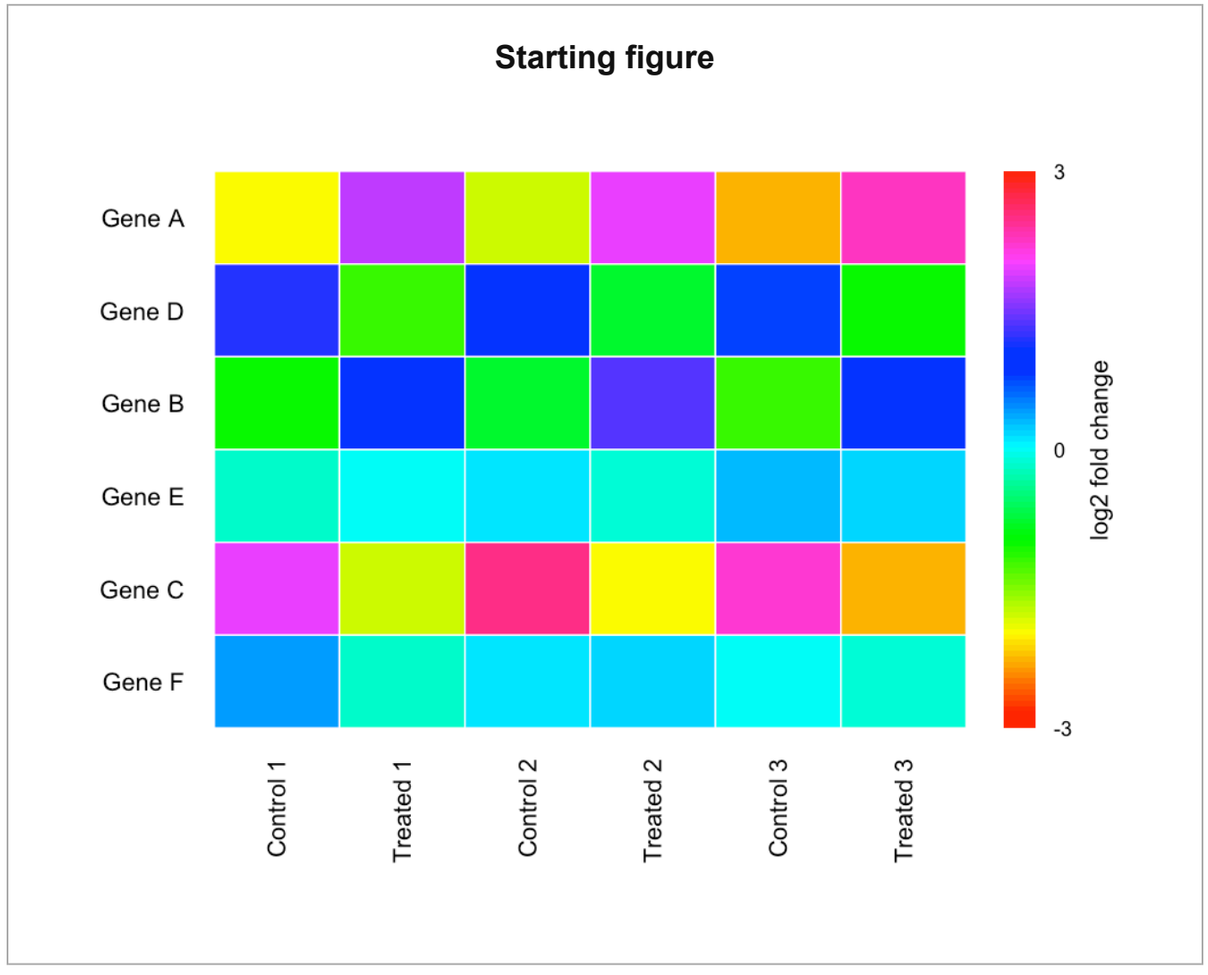


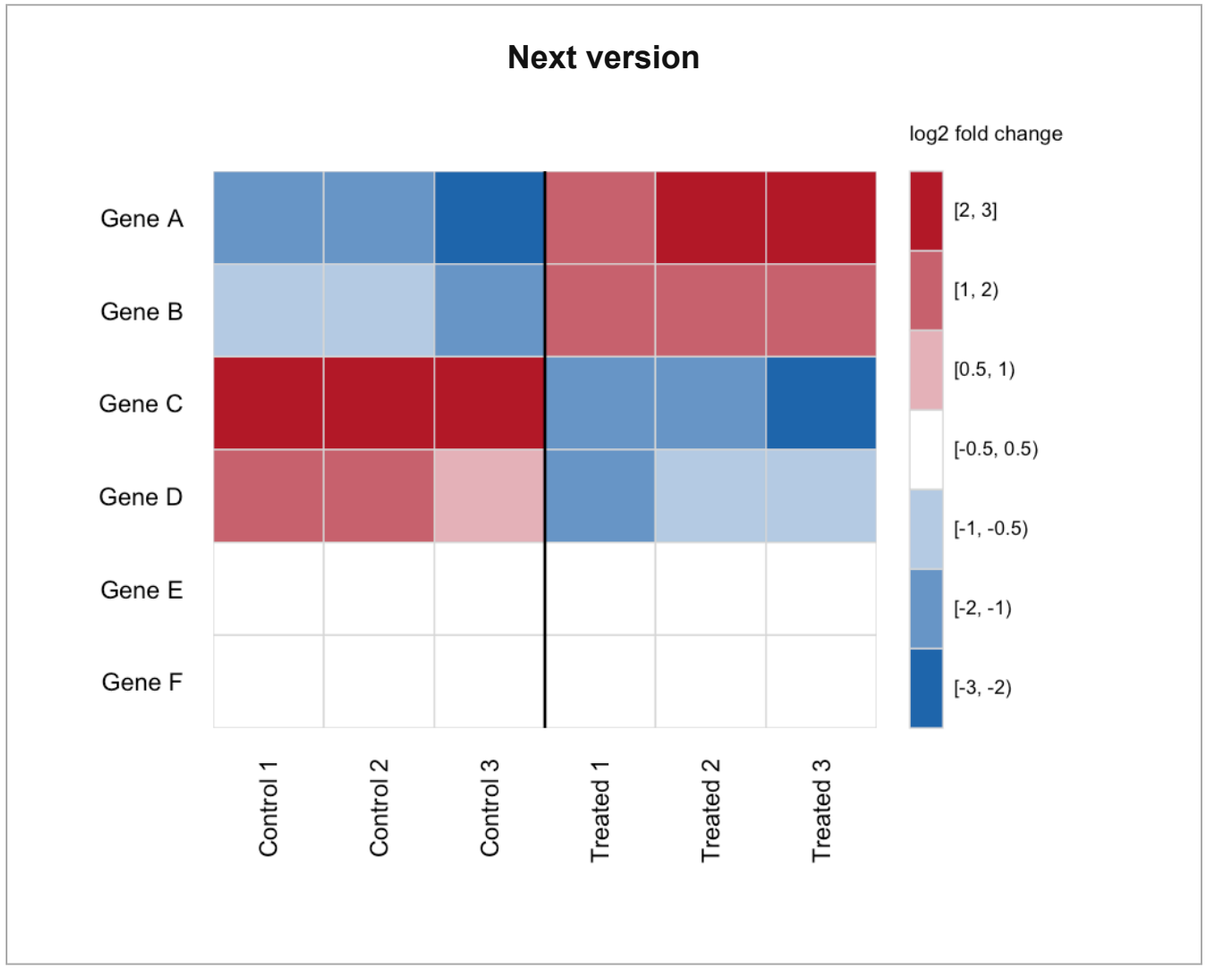


Data & interpretation

An alternative: seven colour bins

Group values into seven labelled colour ranges. This makes broad categories easier to distinguish, but hides differences within each bin and creates sharp visual boundaries. White now means −0.5 to below +0.5, not exactly zero. In the legend, [includes a boundary and) excludes it. These cutoffs are teaching choices, not biological thresholds; use bins when ranges matter more than fine differences.



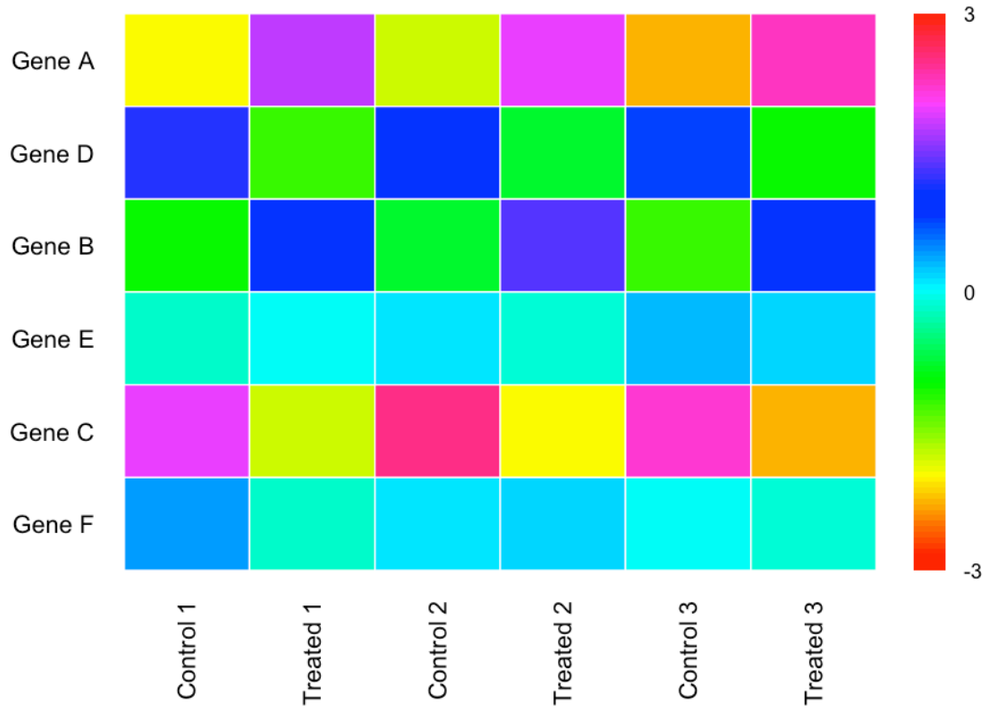


Data & interpretation

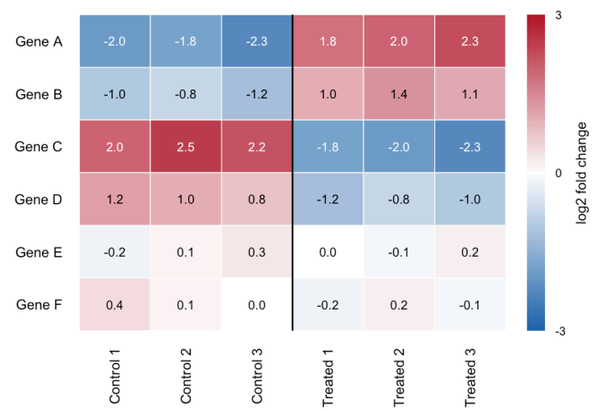
Compare start and finish

The same measurements, with meaningful colour, organised groups and readable values.

Starting figure



✓ Final improvement



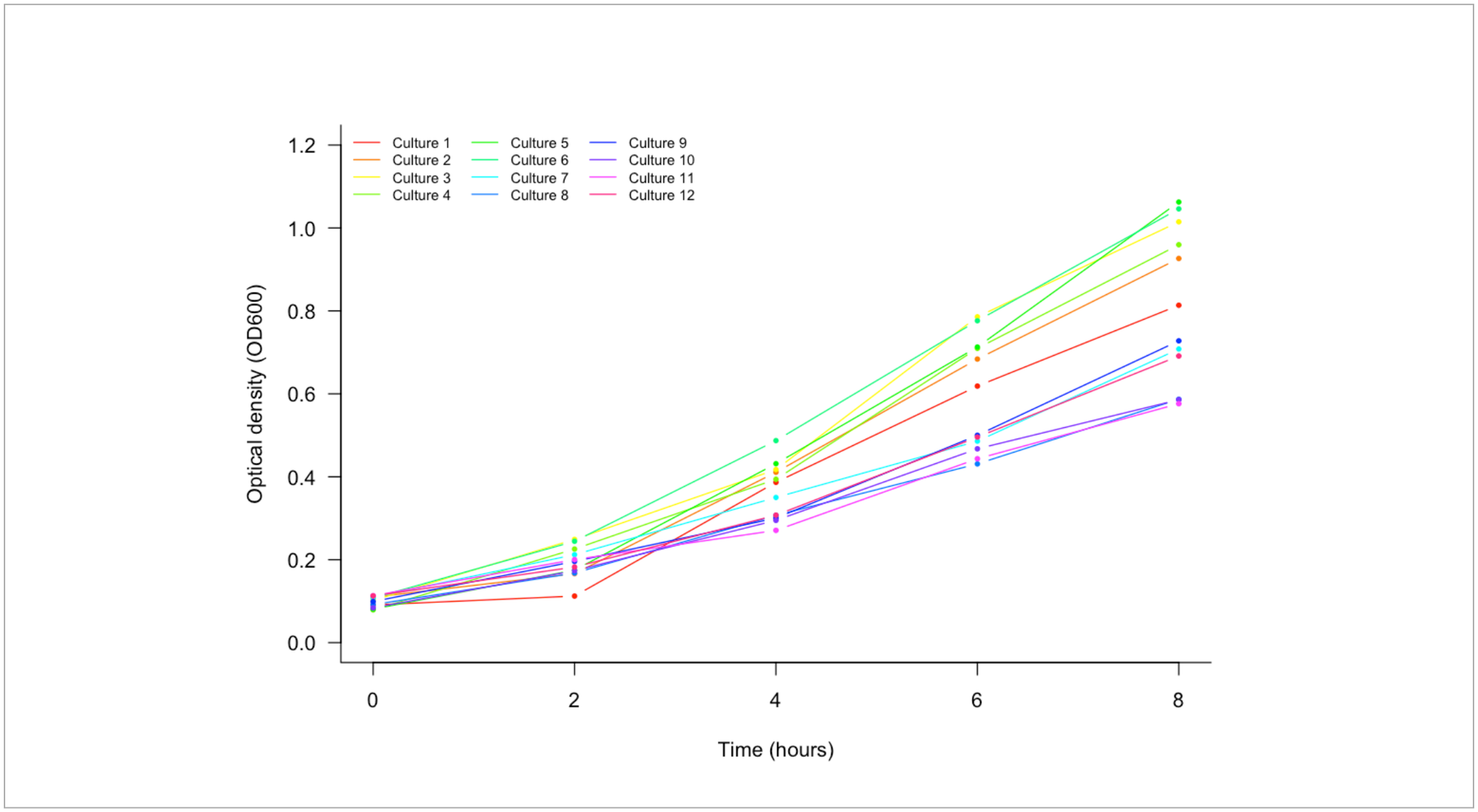
Data & interpretation

From crowded time courses to a clear comparison

Six independent bacterial cultures per condition were followed at 0, 2, 4, 6, and 8 hours. The vertical axis is optical density at 600 nm (OD600), a proxy for culture turbidity. Each line follows the same culture.

Source: Course example.

How could we compare control and treatment without losing the individual cultures?



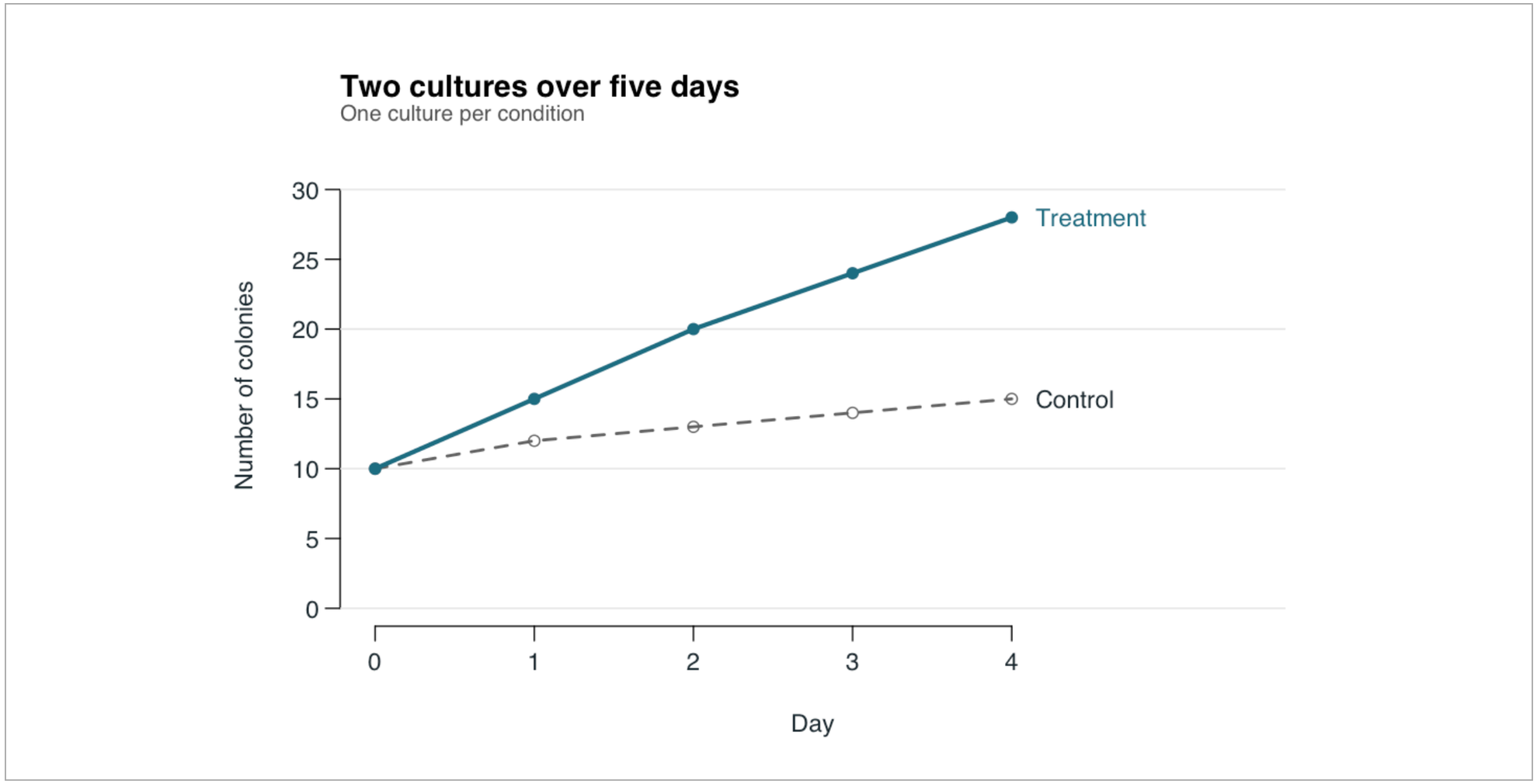
Show the individual cultures and make the comparison clearer.

Replace all measurements with one overall average.

Further reading & data

Follow a line without a lookup

Why are end labels useful for these two lines?

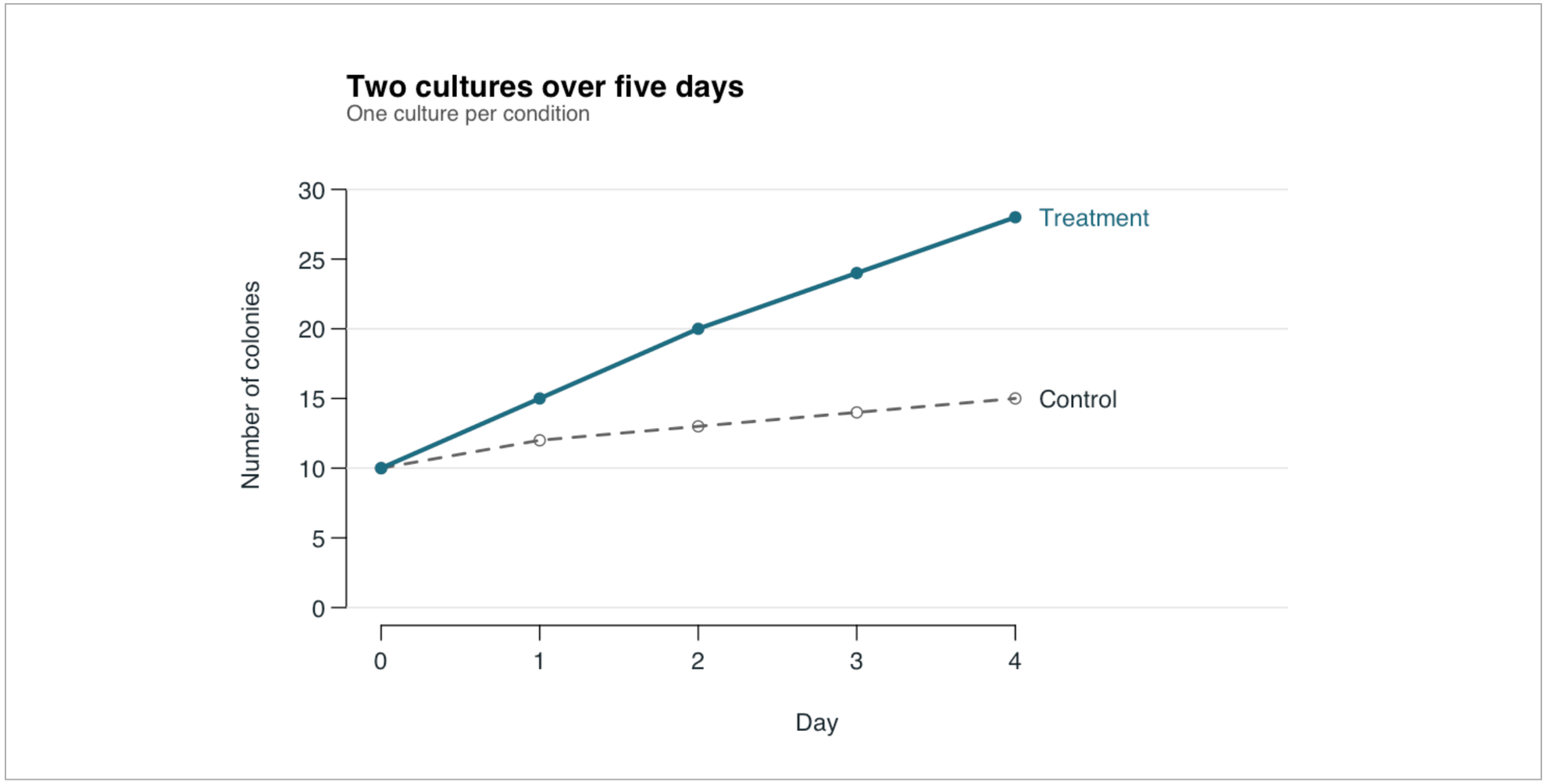


Reveal what works

Context & source

Follow a line without a lookup

Why are end labels useful for these two lines?

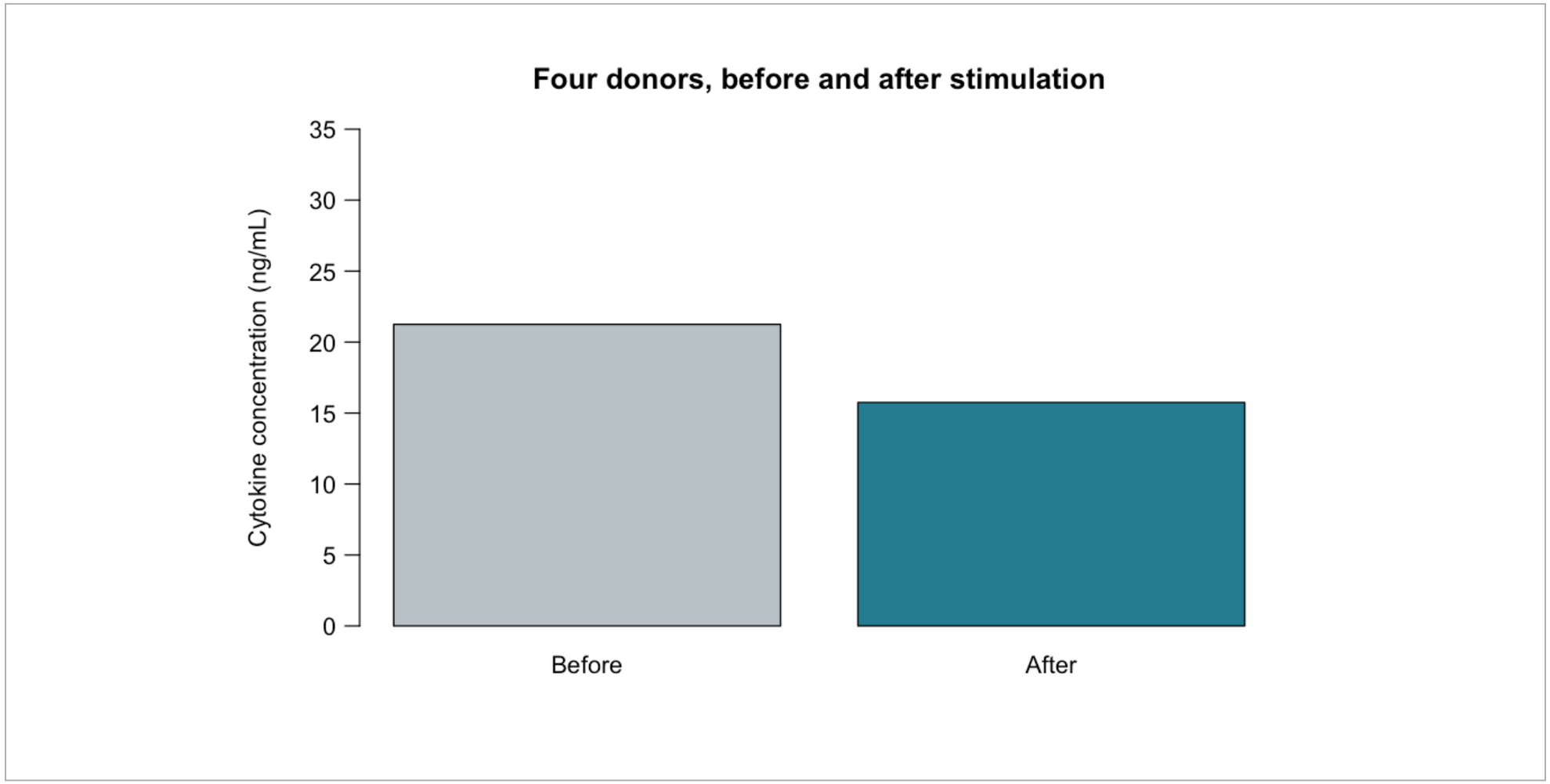


Two well-separated line ends leave room for labels and avoid looking back and forth to a legend. With many overlapping lines, this approach may become crowded.

Context & source

Compare changes within each donor

Which view lets you compare the change for each donor?

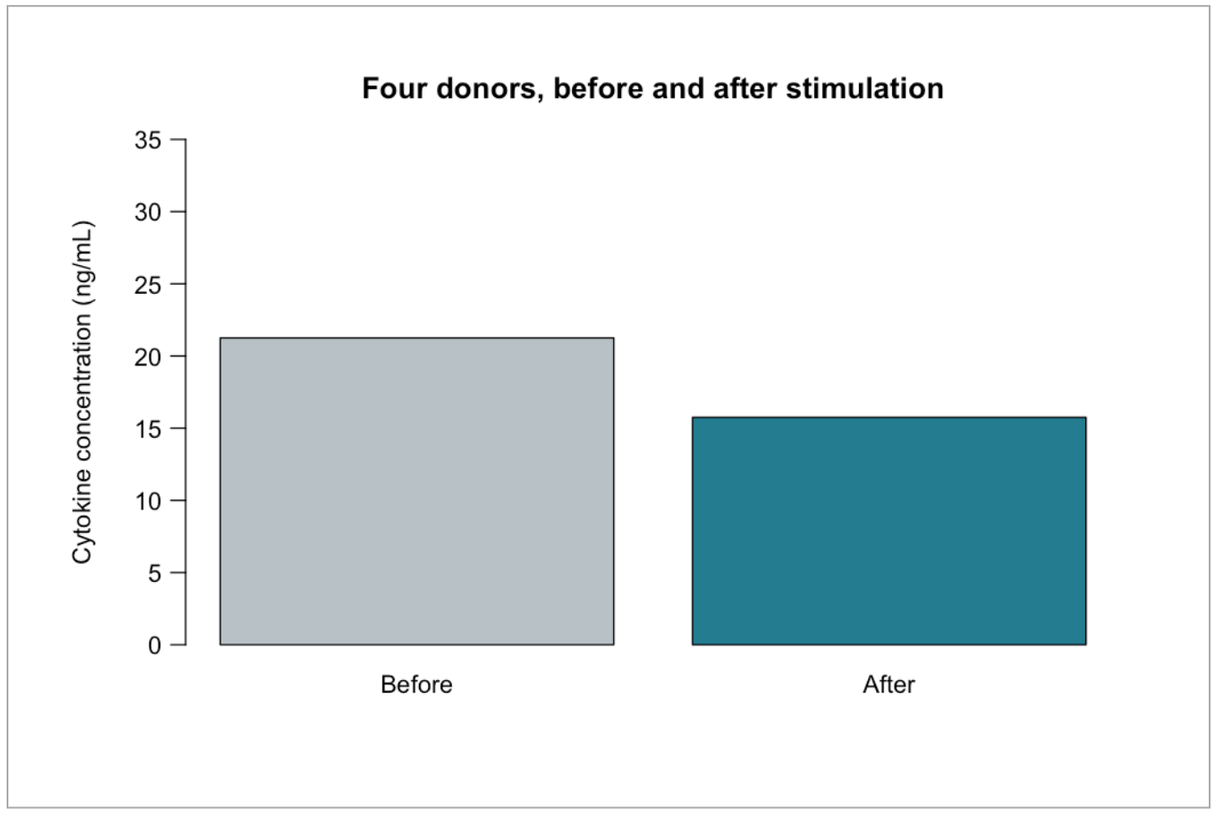


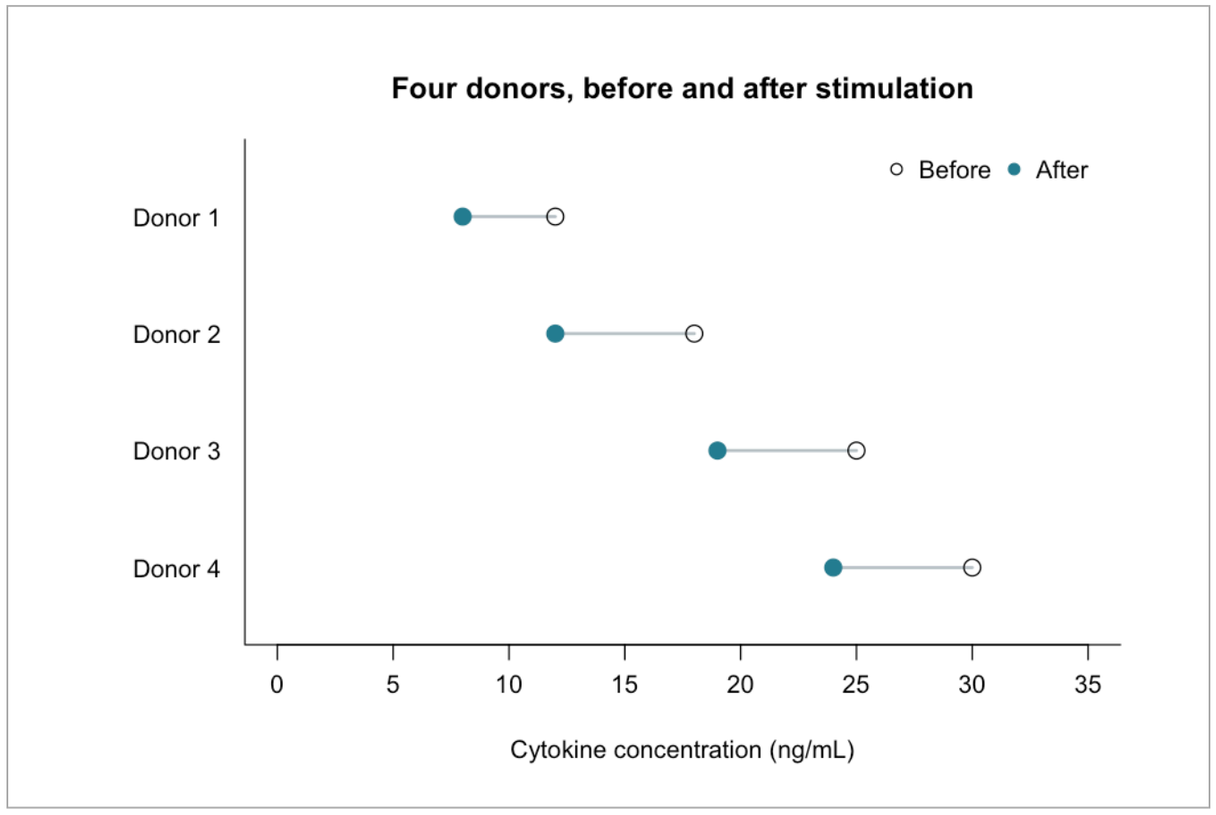
Reveal alternative figure

Context & source

Compare changes within each donor

Which view lets you compare the change for each donor?



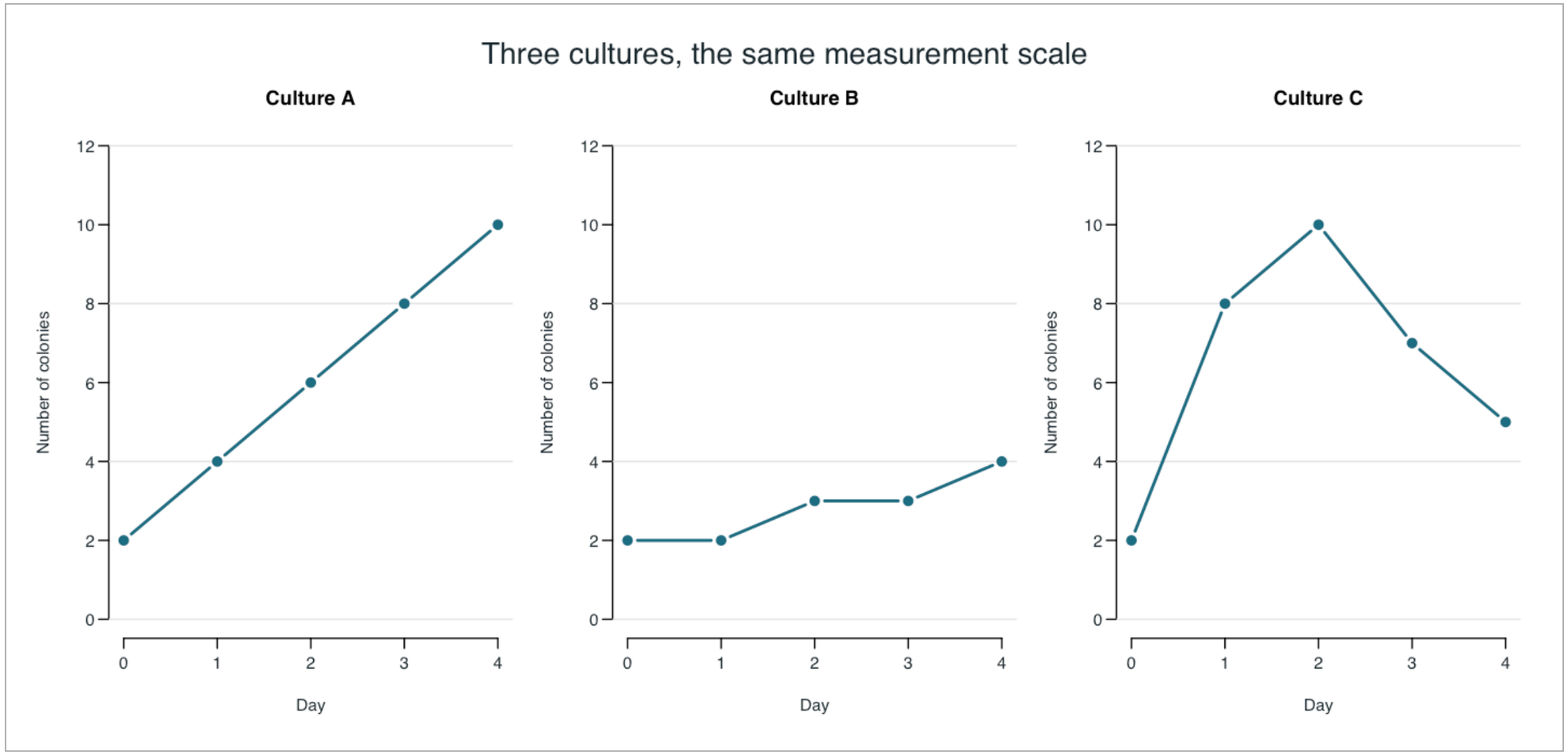


B keeps each donor’s pair together. You can compare individual changes while still seeing the original concentrations. This display alone does not establish what caused the change.

> **Context & source**

A small layout that stays readable

Why keep the same vertical scale in all three panels?

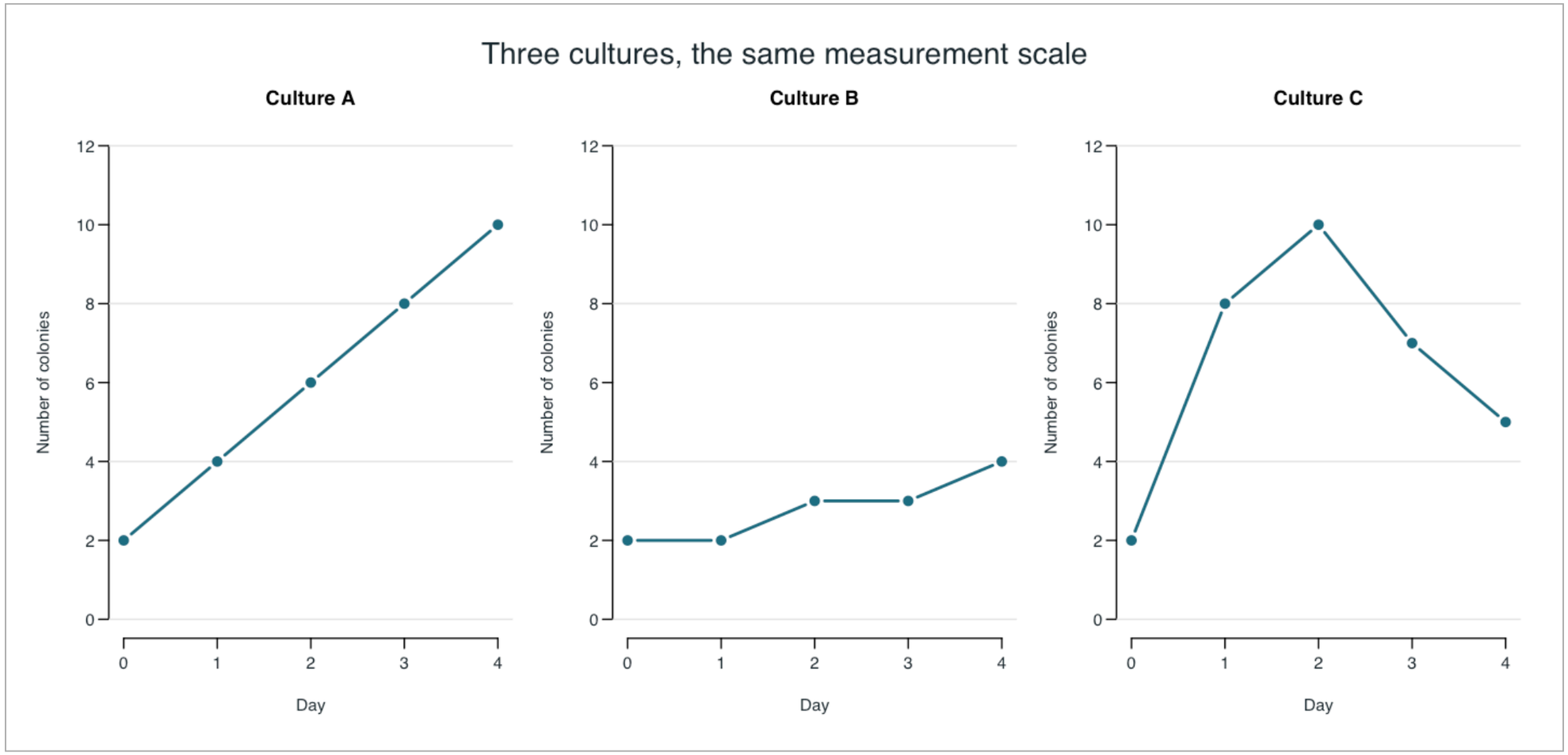


Reveal what works

Context & source

A small layout that stays readable

Why keep the same vertical scale in all three panels?



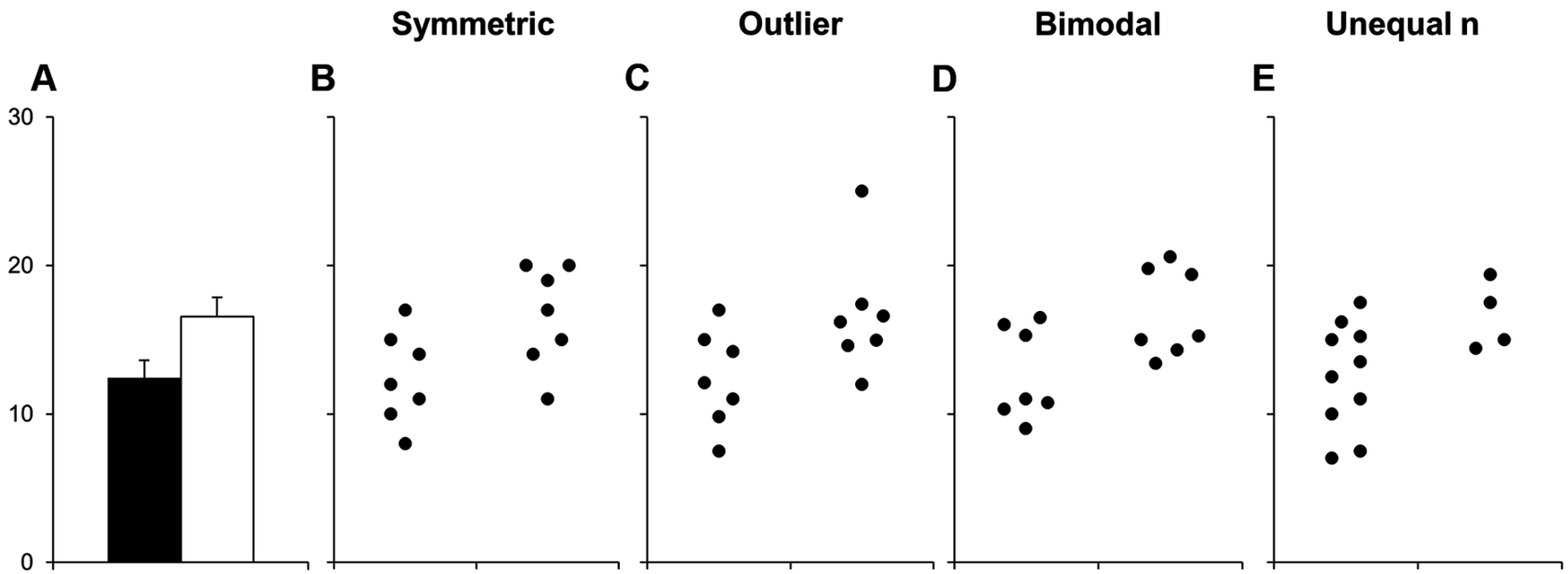
The same height means the same count in every panel. This supports comparison across cultures.

Context & source

One summary, many possible stories

Teaching datasets compare two groups, placed side by side in each panel. The vertical axis shows numerical values without specified units. Panel A displays means with standard-error bars; each dot in B–E is an observation. The table below reports statistical test results.

Source: Weissgerber et al. · PLOS Biology · 2015 · Figure 1



Test				
	p value			
T-test: Equal var.	0.035	0.050	0.026	0.063
T-test: Unequal var.	0.035	0.050	0.026	0.035
Wilcoxon	0.054	0.073	0.128	0.103

Reveal the discussion question

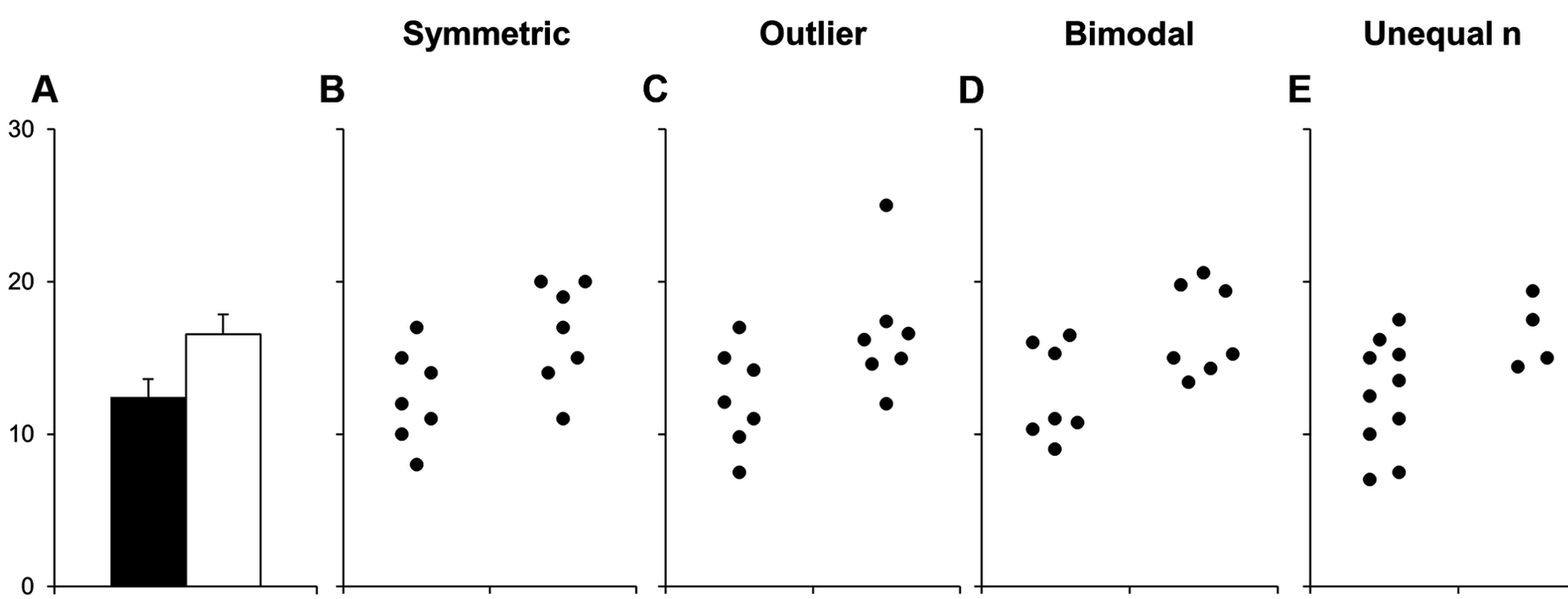
Credits

One summary, many possible stories

Teaching datasets compare two groups, placed side by side in each panel. The vertical axis shows numerical values without specified units. Panel A displays means with standard-error bars; each dot in B–E is an observation. The table below reports statistical test results.

Source: Weissgerber et al. · PLOS Biology · 2015 · Figure 1

Look at all five panels. What additional evidence do the points provide?



Test				
	p value			
T-test: Equal var.	0.035	0.050	0.026	0.063
T-test: Unequal var.	0.035	0.050	0.026	0.035
Wilcoxon	0.054	0.073	0.128	0.103

Proof that bars are always wrong.

Exact causes of differences between groups.

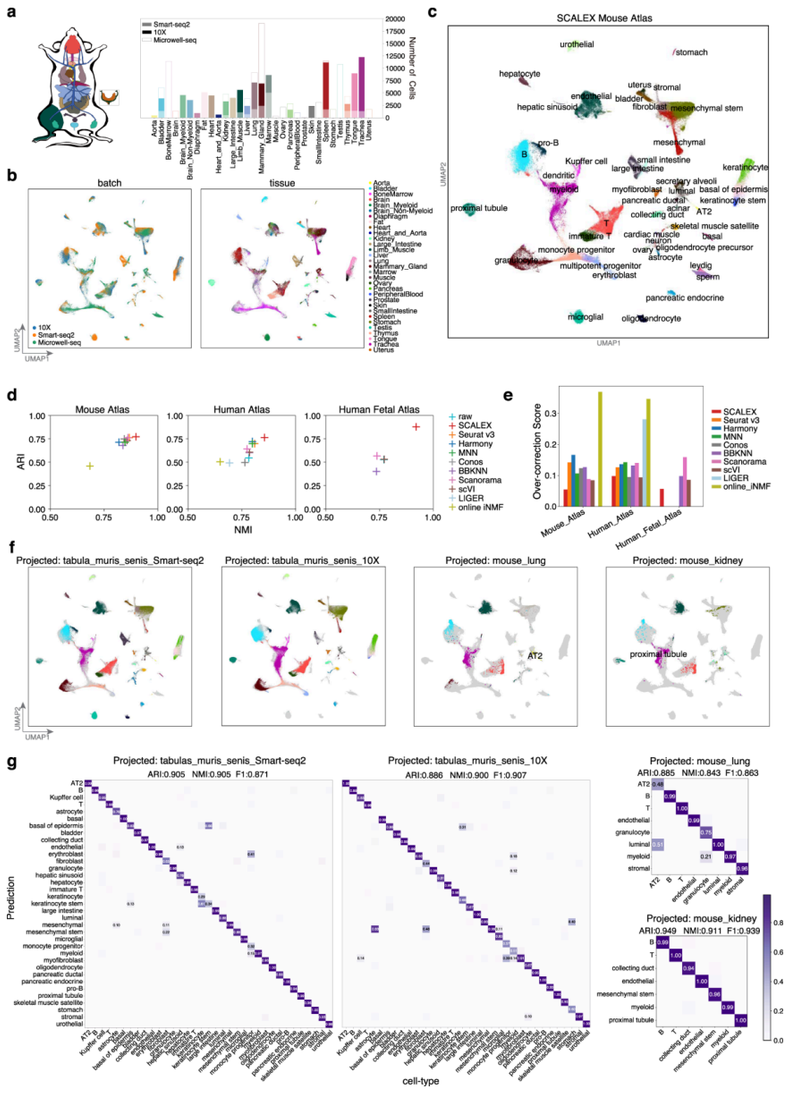
Variation, outliers, and the number of observations.

> **Credits**

A paper figure is a reading journey

A published single-cell atlas figure. Panel a shows tissues and cell counts. In b, c, and f, dots are cells positioned by two unitless embedding coordinates (UMAP1/2). Panel d compares two agreement scores: normalized mutual information horizontally and adjusted Rand index vertically. Panel e plots over-correction scores by atlas and method; g compares cell-type labels on both axes, with colour showing proportions.

Source: Xiong et al. · Nature Communications · 2022 · Figure 4



Reveal the discussion question

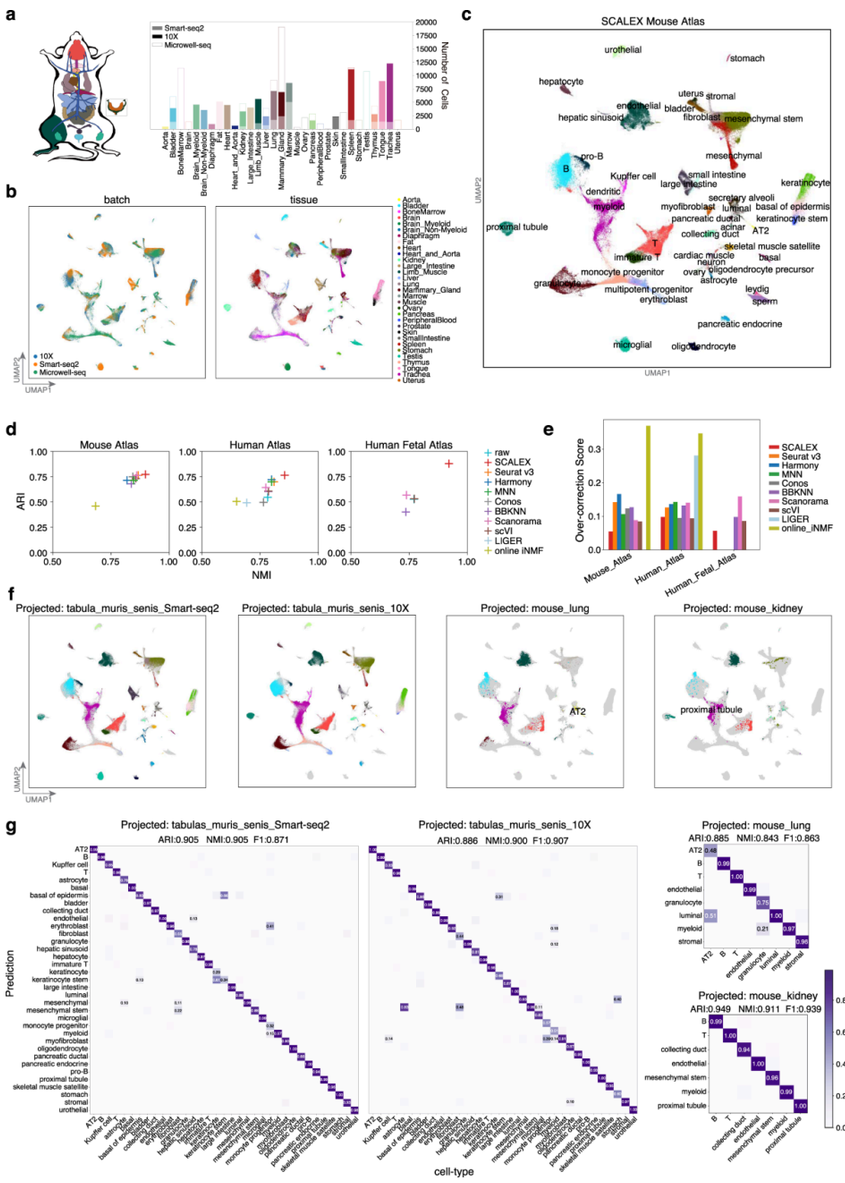
> **Credits**

A paper figure is a reading journey

A published single-cell atlas figure. Panel a shows tissues and cell counts. In b, c, and f, dots are cells positioned by two unitless embedding coordinates (UMAP1/2). Panel d compares two agreement scores: normalized mutual information horizontally and adjusted Rand index vertically. Panel e plots over-correction scores by atlas and method; g compares cell-type labels on both axes, with colour showing proportions.

Source: Xiong et al. · Nature Communications · 2022 · Figure 4

This Nature Communications figure combines several views of a cell atlas. Which habit is worth borrowing?



Include as many panels as possible.	Give each panel a role and connect them with labels and consistent visual conventions.	Assume a prestigious journal guarantees a good figure.

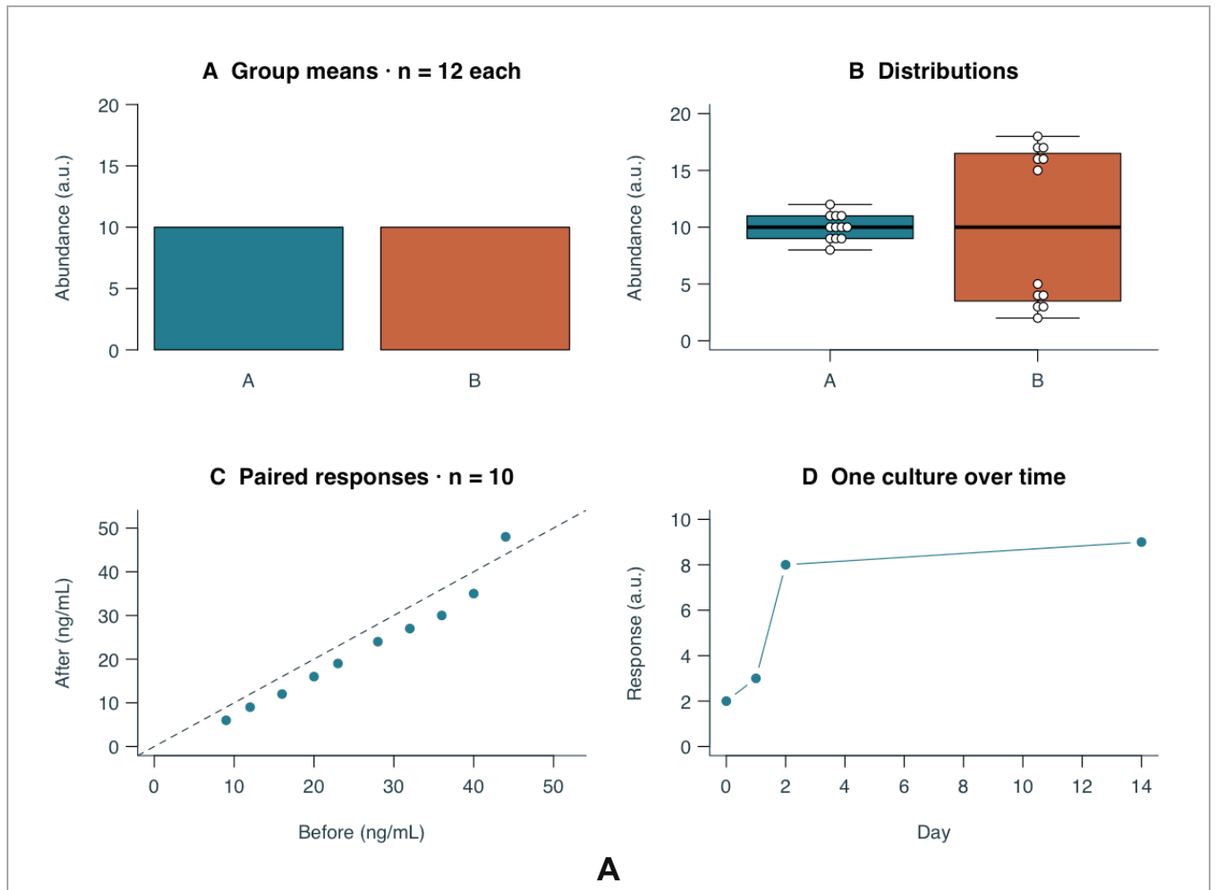
> **Credits**

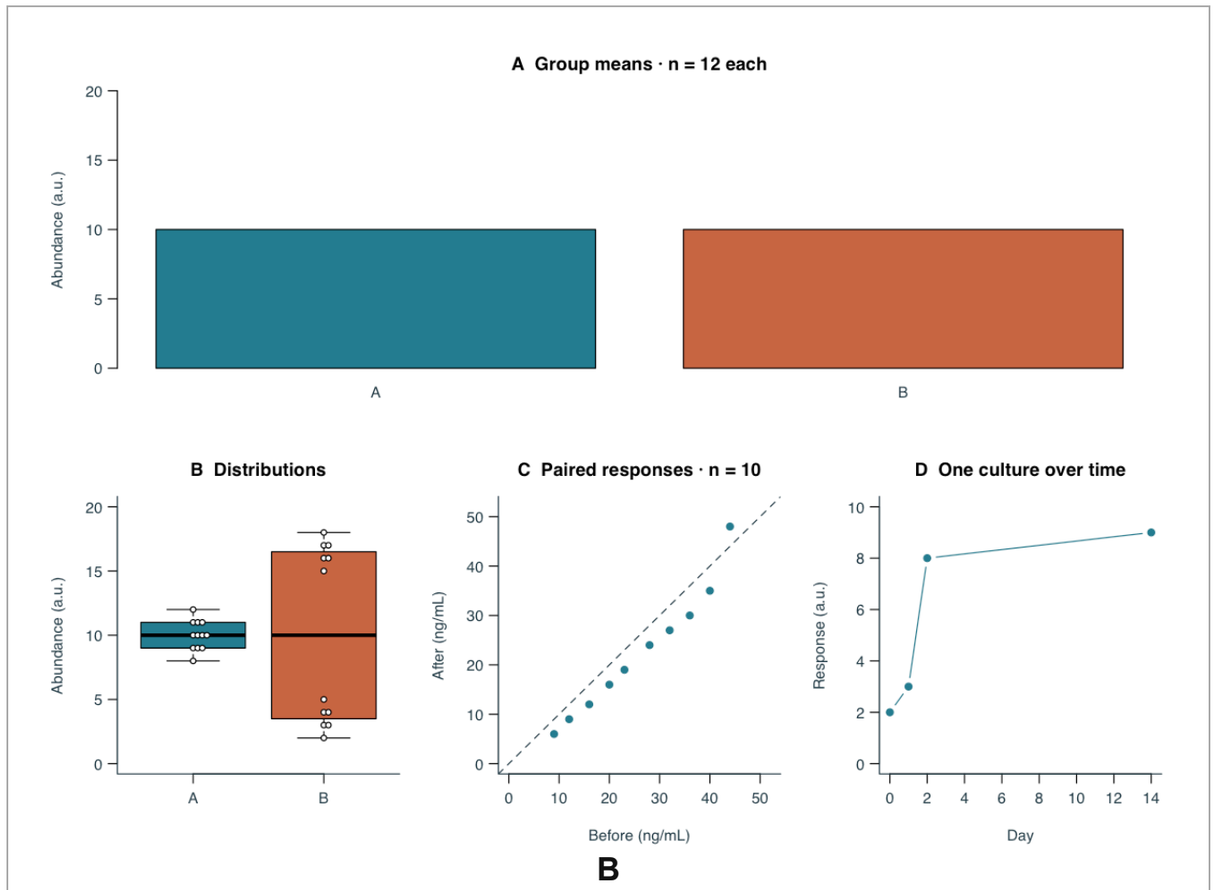
Arrange the evidence

Two layouts show the same four panels of data. Panels A–B compare groups horizontally and protein abundance vertically (arbitrary units). Panel C plots before-treatment concentration against after-treatment concentration (ng/mL), with a dashed equality line. Panel D plots days horizontally and response vertically (arbitrary units). These are separate teaching datasets.

Source: Course example.

You need readers to inspect all four panels at paper-reading size. Which layout gives the evidence more balanced space?





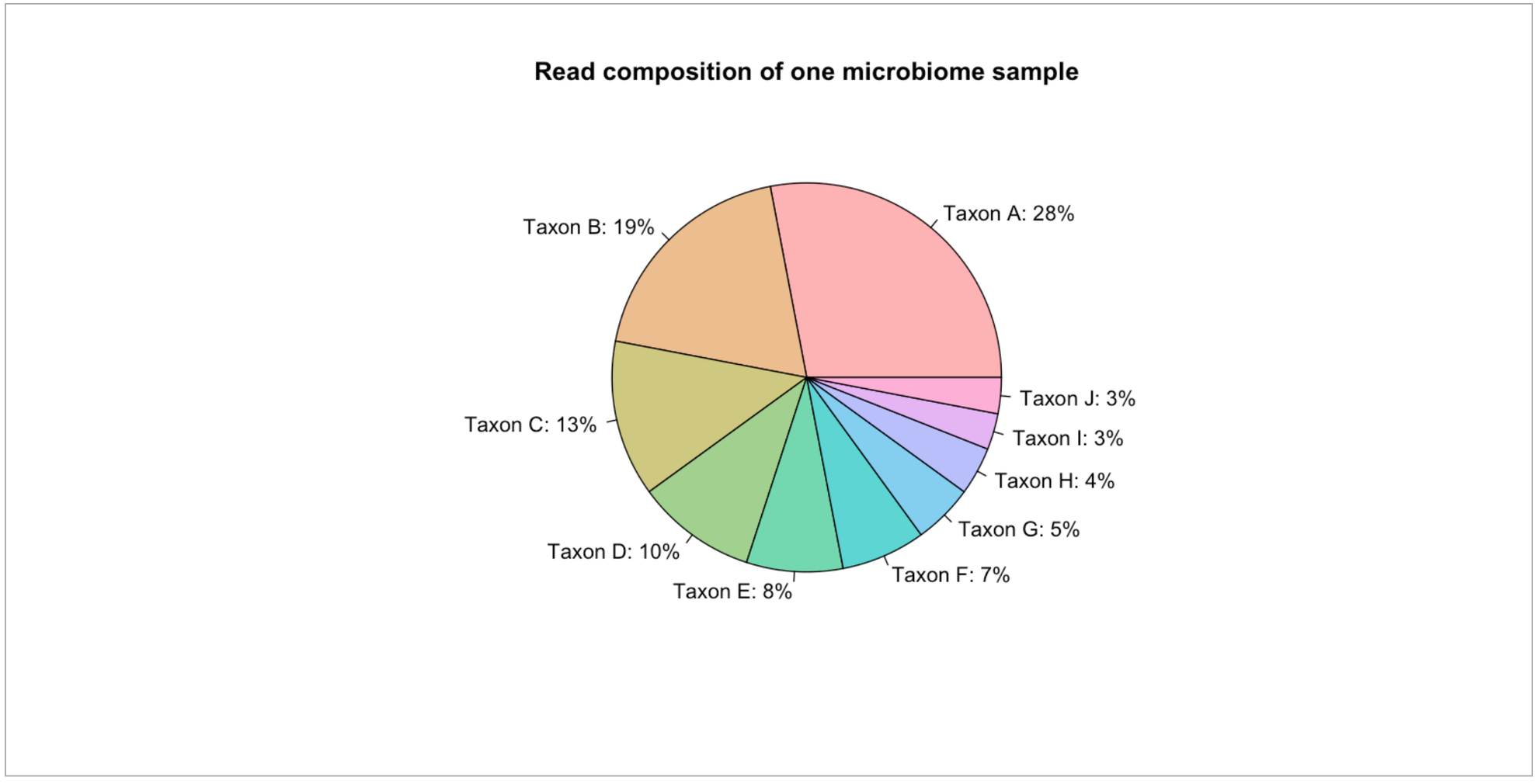
Choose A

Choose B

> **Further reading**

Compare the smaller microbial groups

Which view makes it easier to compare taxa D, E, and F?



Reveal alternative figure

Context & source

Compare the smaller microbial groups

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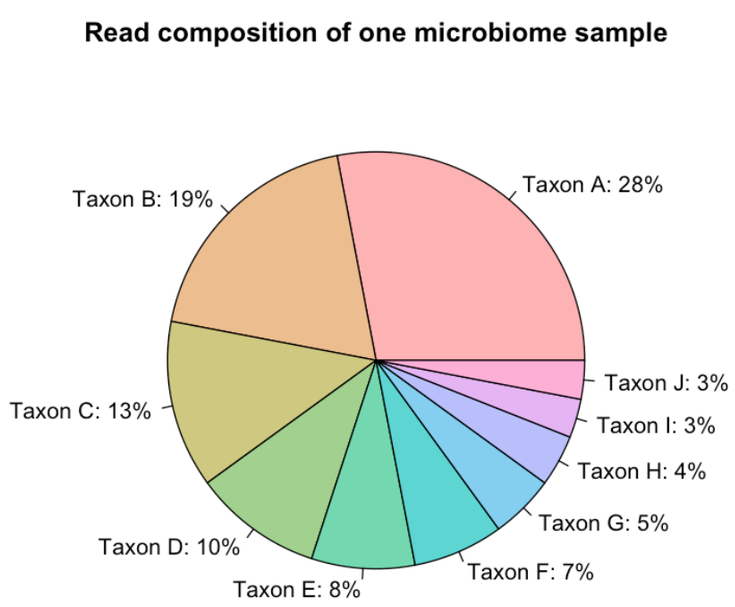
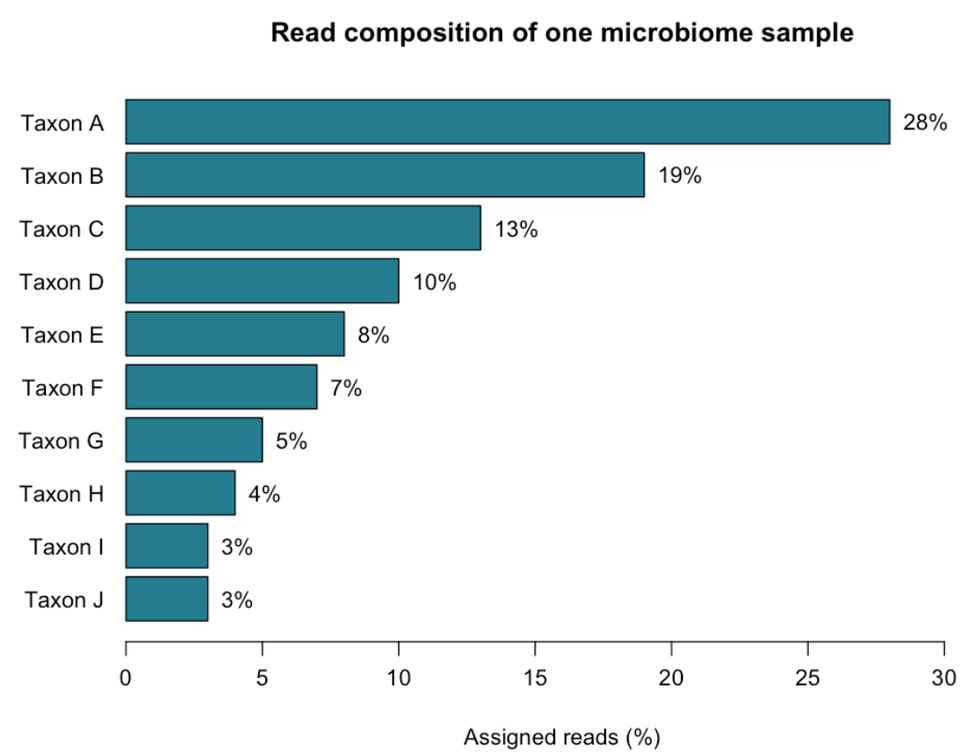




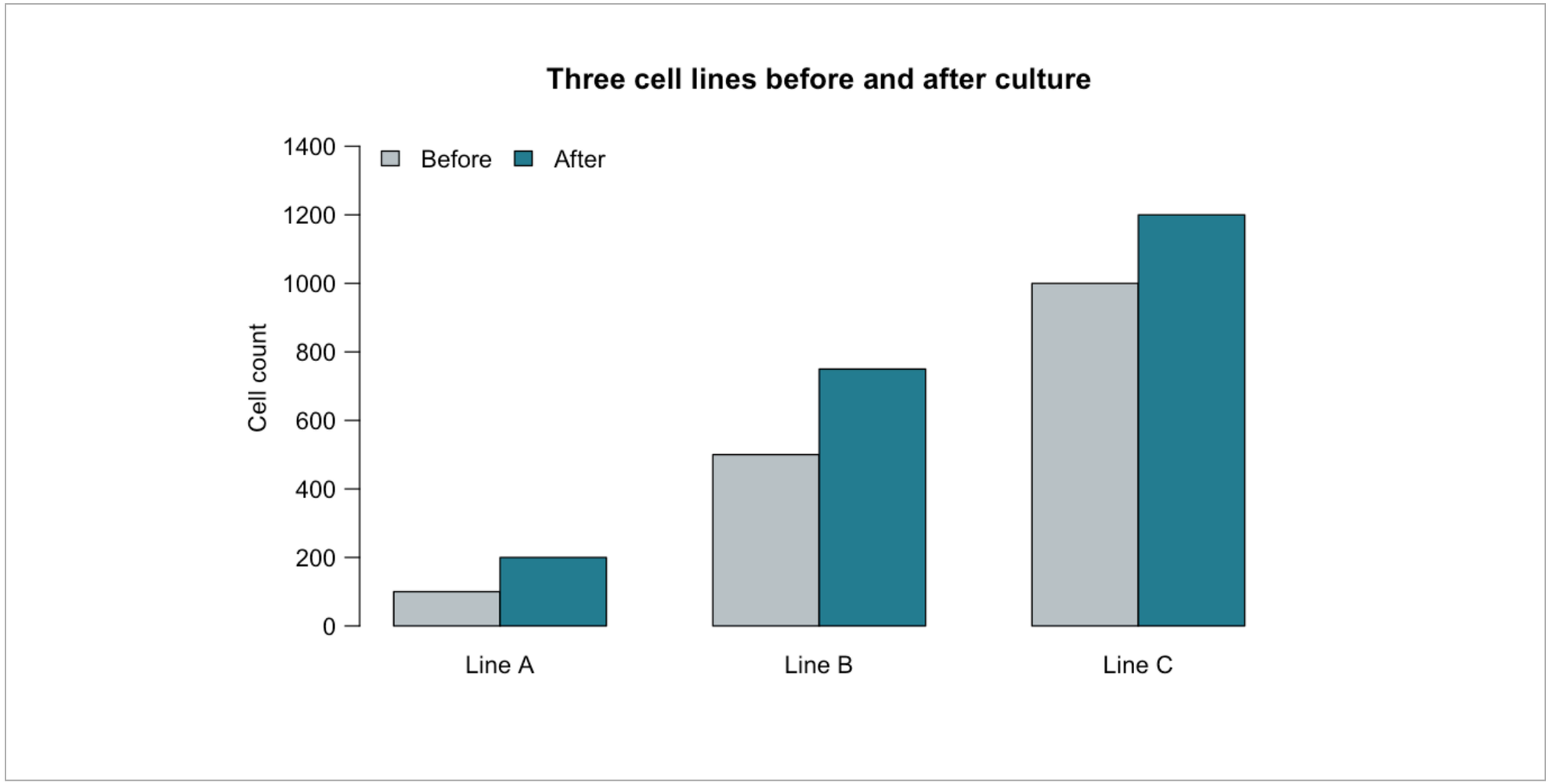
The bars share a baseline and are ordered by abundance, making 10%, 8%, and 7% easy to compare. These are read proportions, not necessarily cell proportions.

Also compare a 100% stacked bar

Context & source

Which cell line grew fastest relative to its start?

Which view answers the relative-growth question most directly?



Reveal alternative figure

Context & source

Which cell line grew fastest relative to its start?

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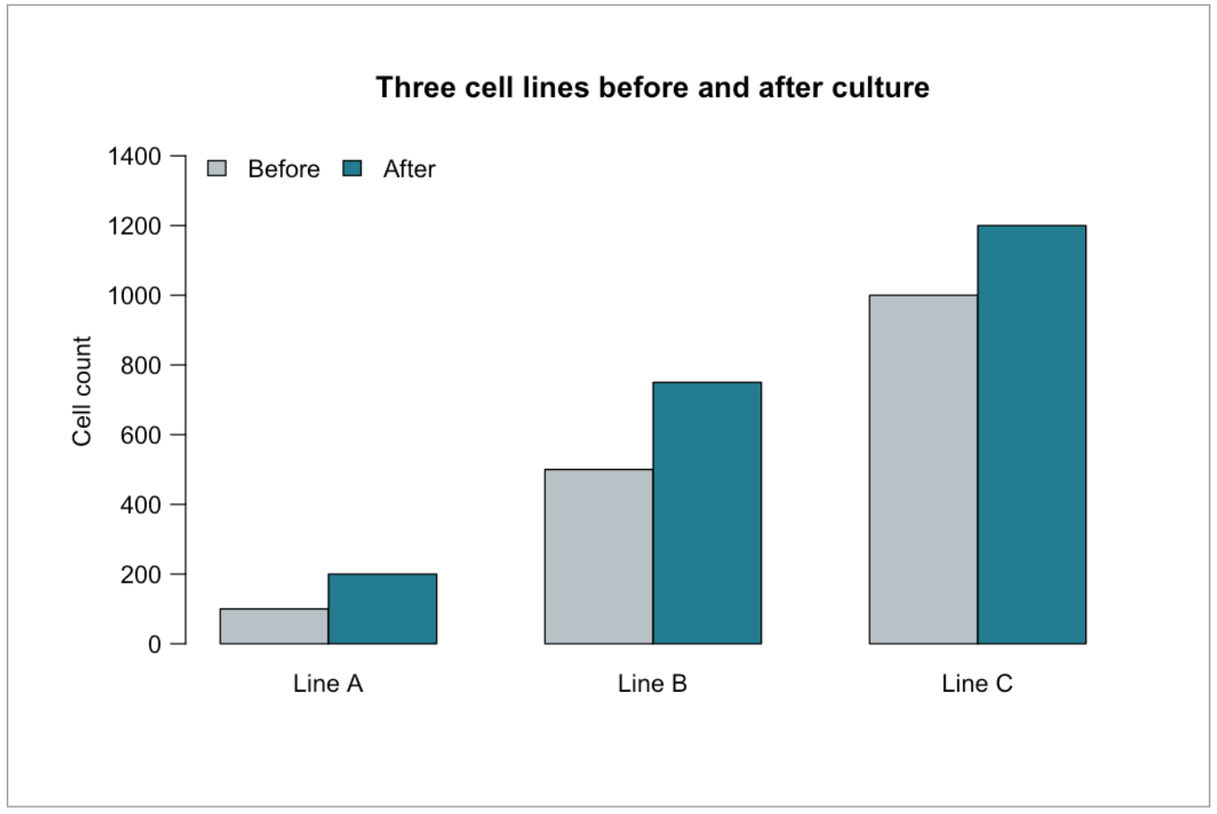
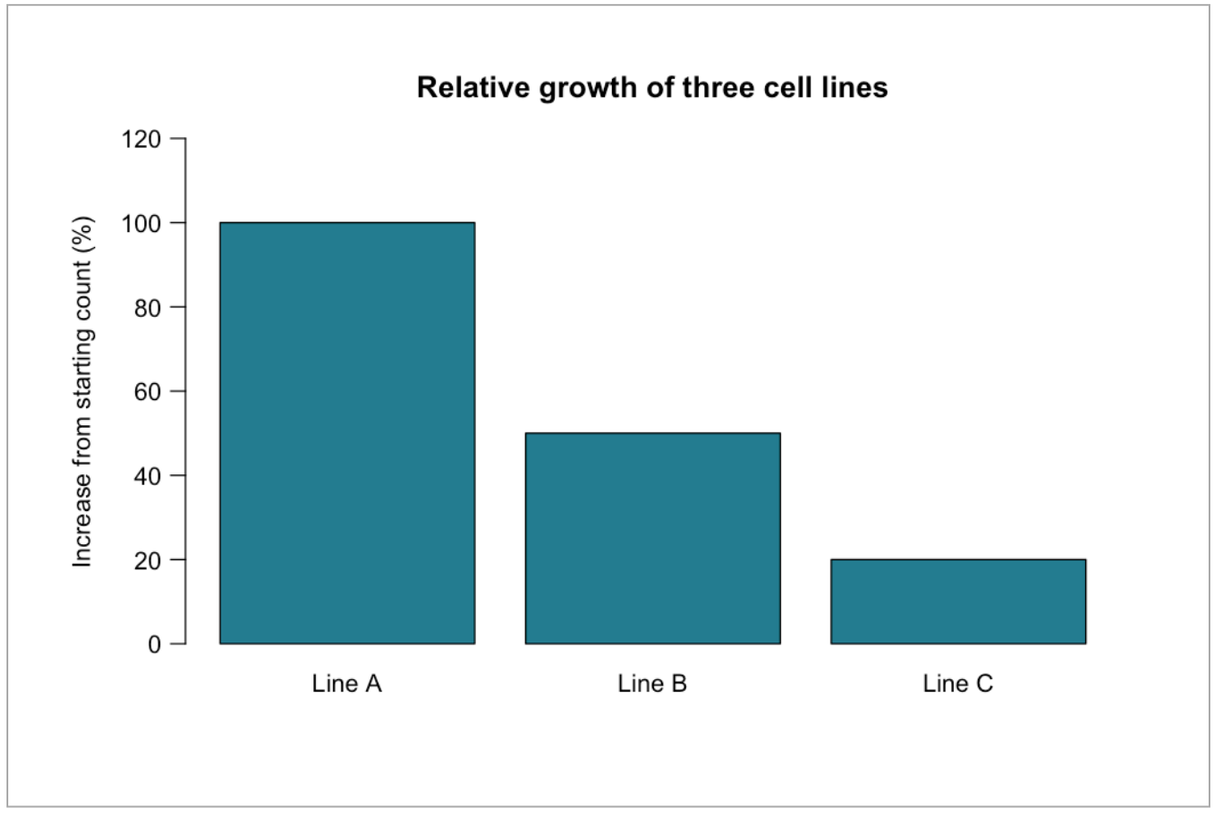




B shows proportional growth directly: A increases 100%, B 50%, and C 20%. It is better for this question; the original counts are still needed to understand the scale of each culture.

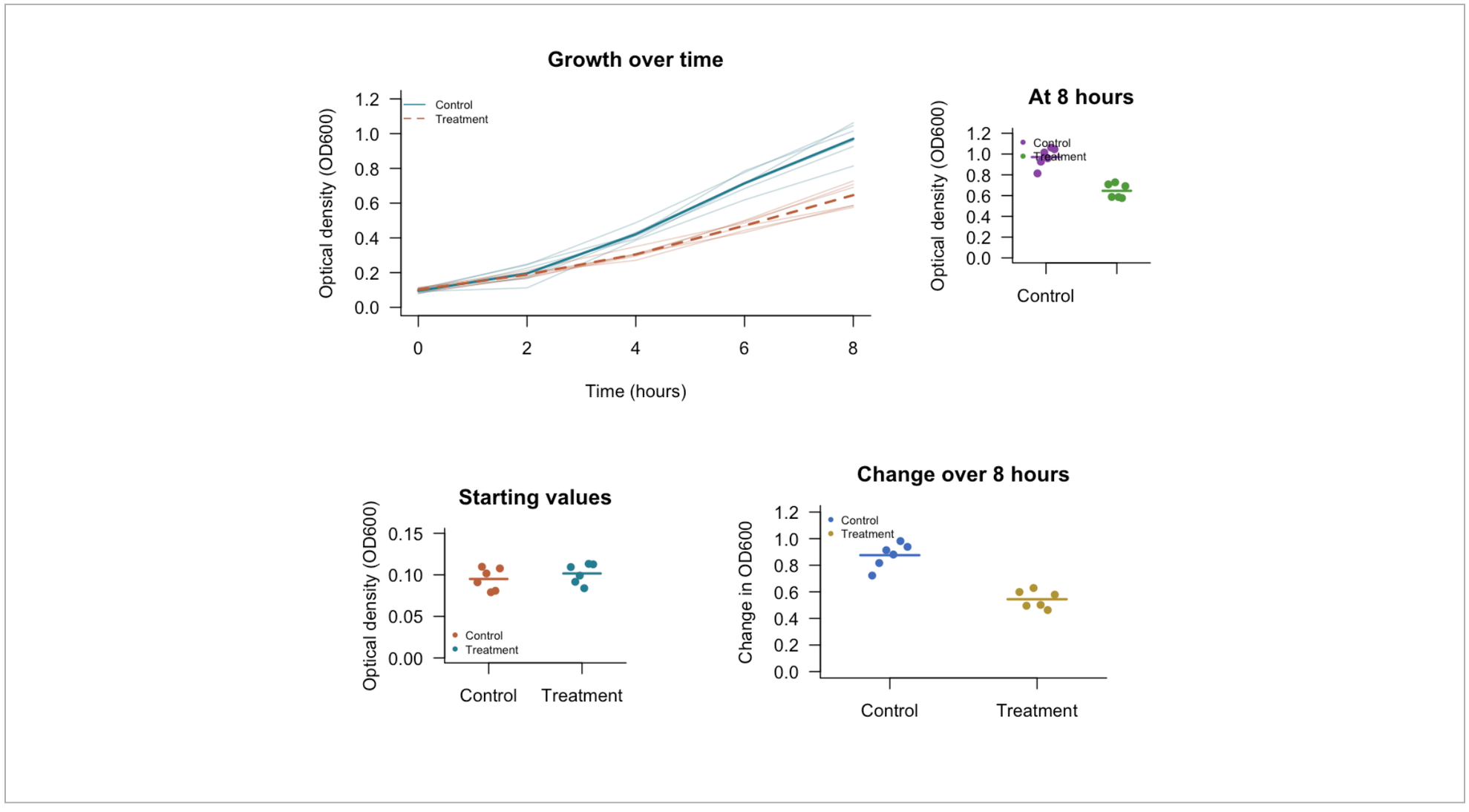
> **Context & source**

Build a publication figure from four plots

The same growth experiment is shown four ways: trajectories, values at 8 hours, starting values, and changes from 0 to 8 hours.
There are six independent cultures per condition. OD600 measures culture turbidity.

Source: Course example.

How can these four views tell one clear story?



Show the individual cultures and make the comparison clearer.

Replace all measurements with one overall average.

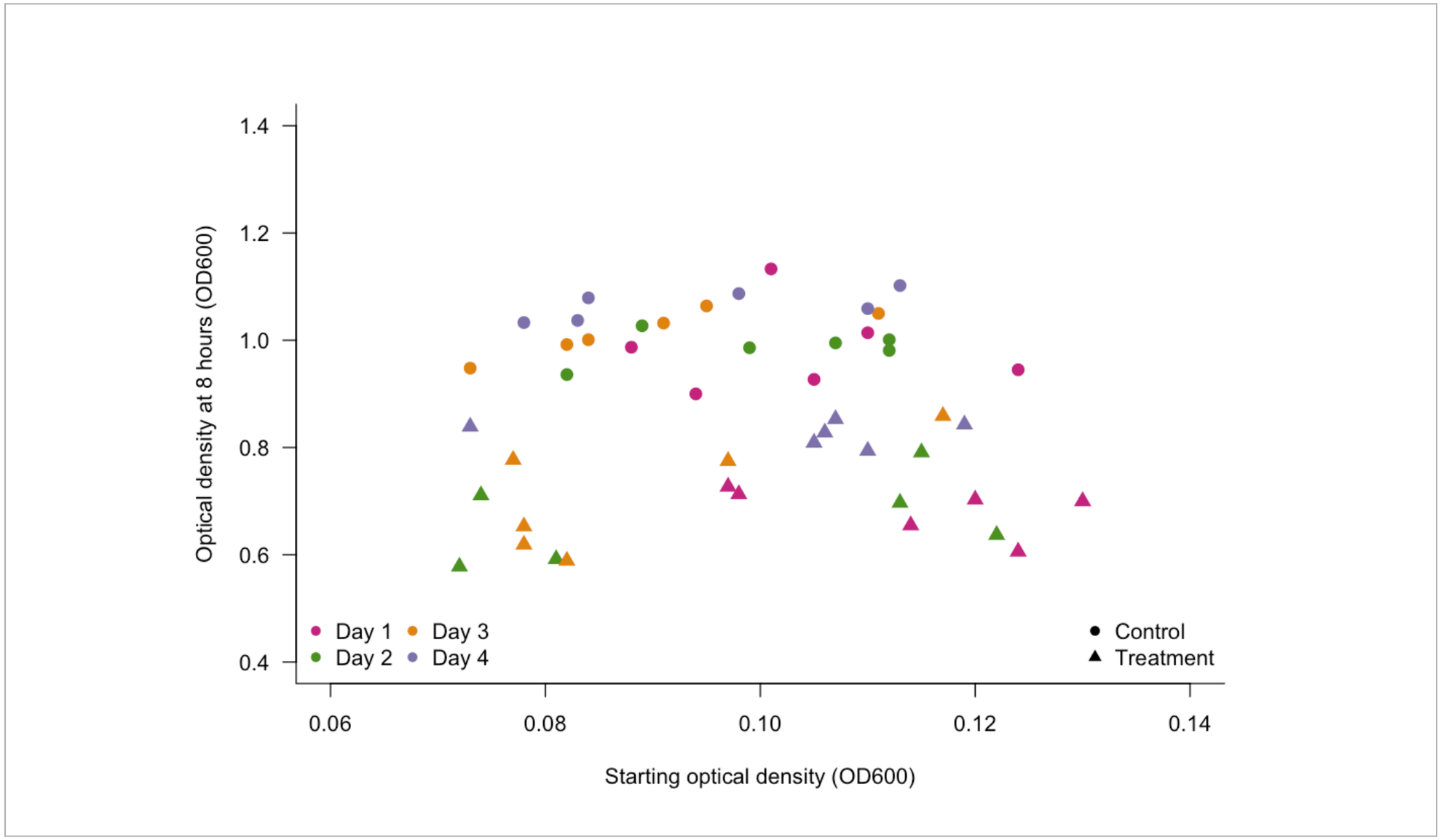
Further reading & data

What should stand out first?

Forty-eight independent bacterial cultures were assayed on four consecutive experimental days: six per condition per day. Each assay lasts eight hours; different cultures are used on each day. The horizontal axis shows starting OD600; the vertical axis shows OD600 at 8 hours. The main question is how control and treatment compare. Every version uses the same measurements.

Source: Course example.

Does the initial colour choice support the main comparison?



Yes: the brightest grouping must be the most important one.

Not well: colour emphasises experimental day rather than condition.

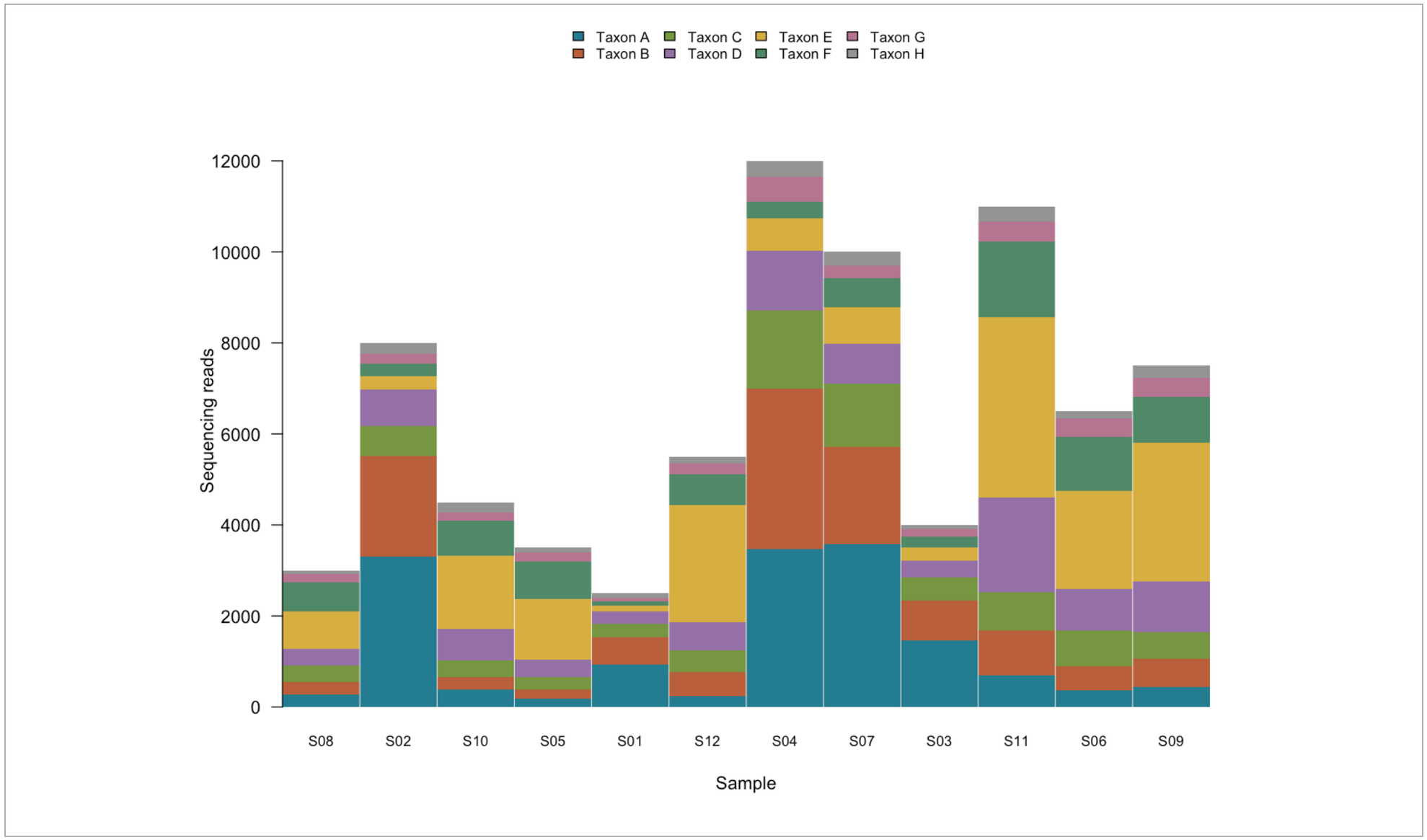
Further reading & data

Order samples to reveal taxon profiles

Twelve soil samples contain sequencing reads assigned to eight taxa. Six samples are bulk soil and six are root-associated. Stacked colours identify taxa; each bar is one sample. Sequencing depth varies. All versions use the same count table.

Source: Course example.

How could we compare sample profiles without confusing sequencing depth with composition?



Compare proportions, then choose an ordering that fits the biological question.

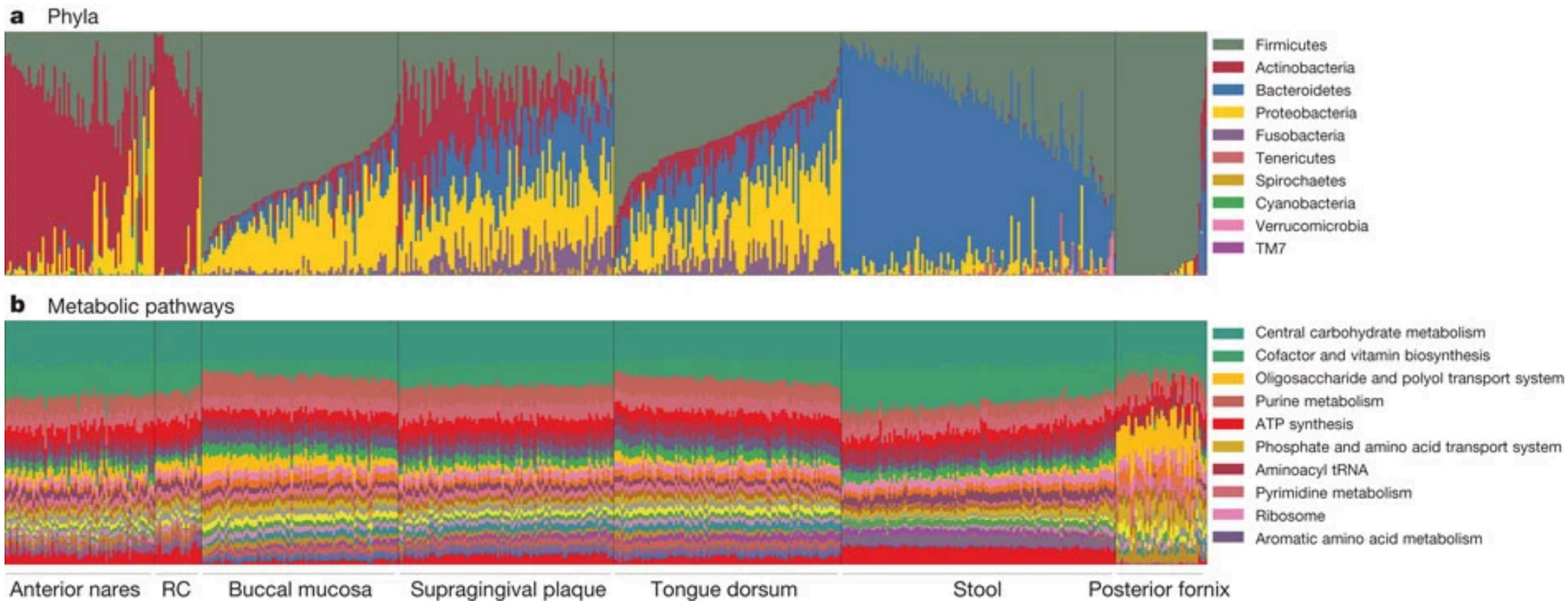
Treat the tallest bar as the sample with the most bacteria.

Further reading & data

Published example: the Human Microbiome Project

Each narrow bar is a microbiome sample, grouped horizontally by seven body sites. Stacked heights represent relative abundance: panel a shows microbial phyla; panel b shows metabolic pathways. Colours have separate meanings in the two panels. RC means retroauricular crease, the skin behind the ear. This is a comparison of samples, not a time series.

Source: Human Microbiome Project Consortium · Nature · 2012 · Figure 2



Reveal the discussion question

Credits

Published example: the Human Microbiome Project

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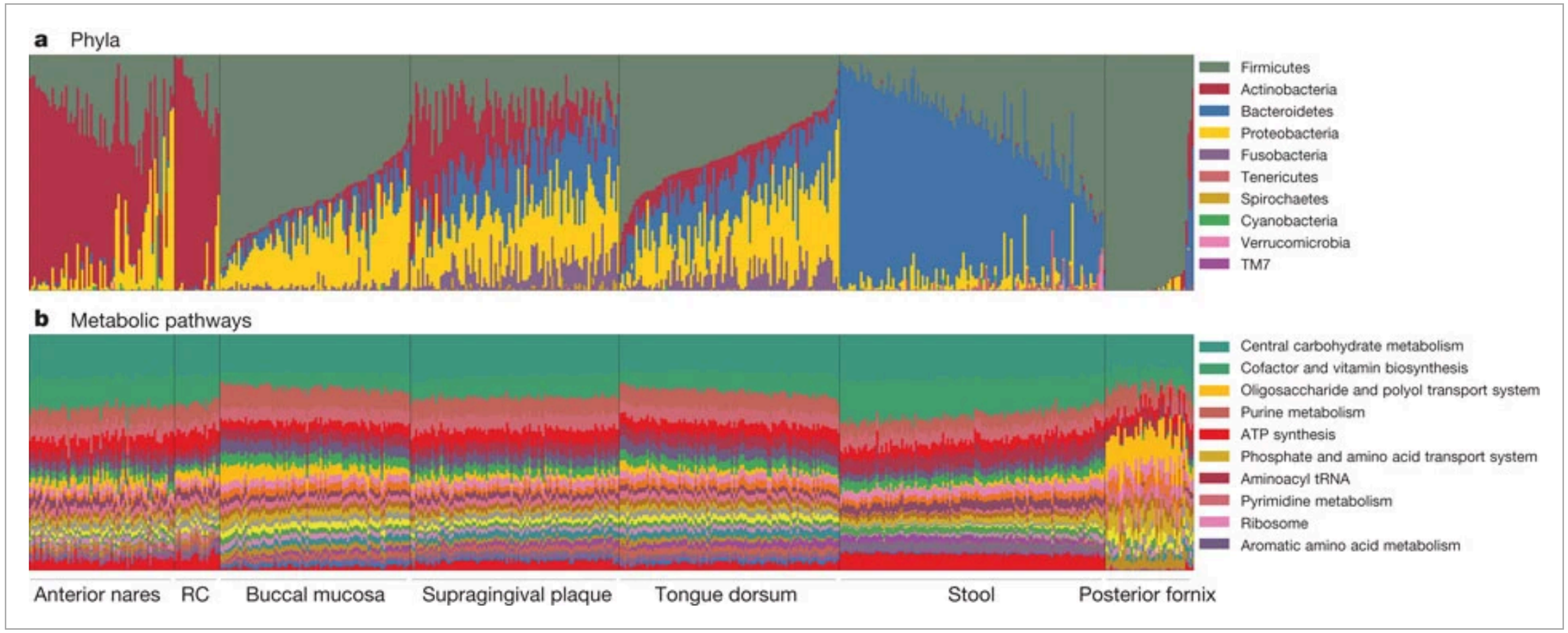
Source: Human Microbiome Project Consortium · Nature · 2012 · Figure 2

What does grouping by body site and aligning the panels make easier to compare?



Taxonomic composition and functional profiles across the same body-site groups.	How one person’s microbiome changed over time.	Absolute numbers of microbes in each sample.

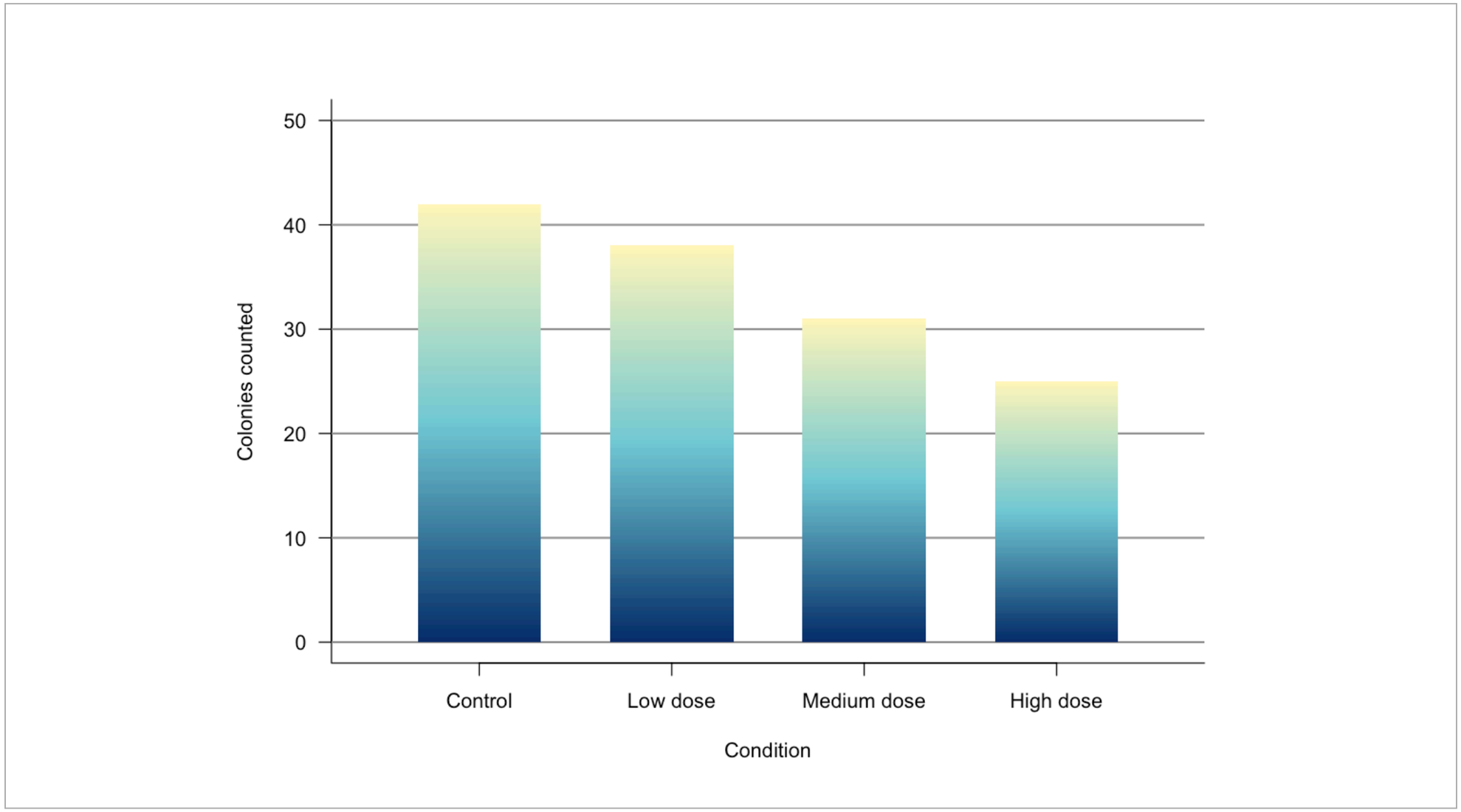
> **Credits**

Function over decoration: does the gradient help?

Colony counts from one plate per condition. Bar height gives the count; the gradient inside every bar encodes no additional measurement. This example asks how easily the counts can be compared.

Source: Course example.

Does the colour gradient help answer the question?



Yes: darker areas must represent a second measurement.

No: height already encodes the count; the gradient adds no data.

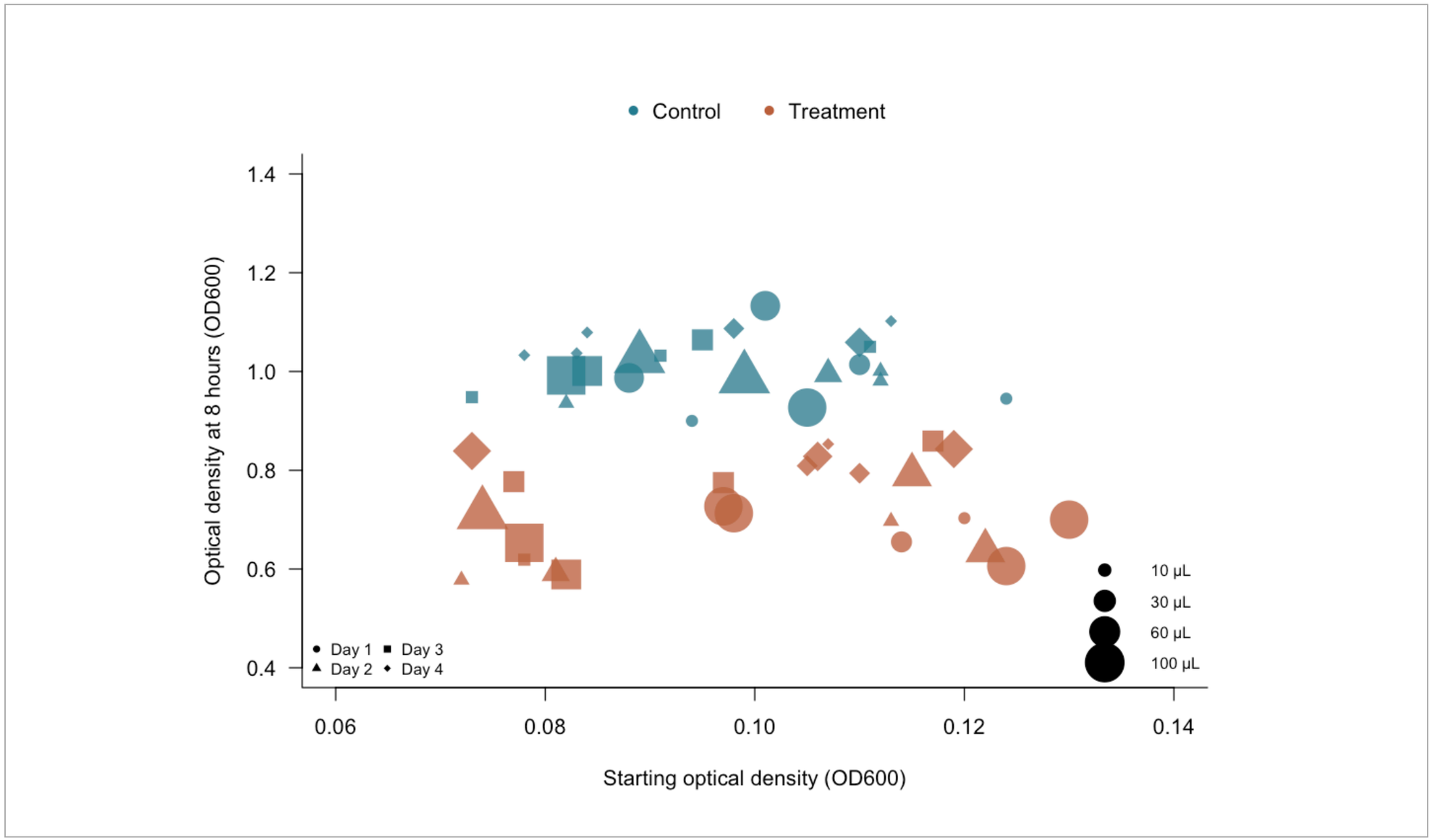
Further reading & data

Function over decoration: too many encodings?

Forty-eight independent cultures have starting and final OD600, condition, experimental day, and inoculum volume recorded. The initial question is how the two conditions compare at similar starting OD600. Point area encodes inoculum volume, colour condition, and shape day.

Source: Course example.

Does showing every recorded variable make this comparison easier?



Not necessarily: extra encodings compete with the main question.

Yes: every available variable should always be visible.

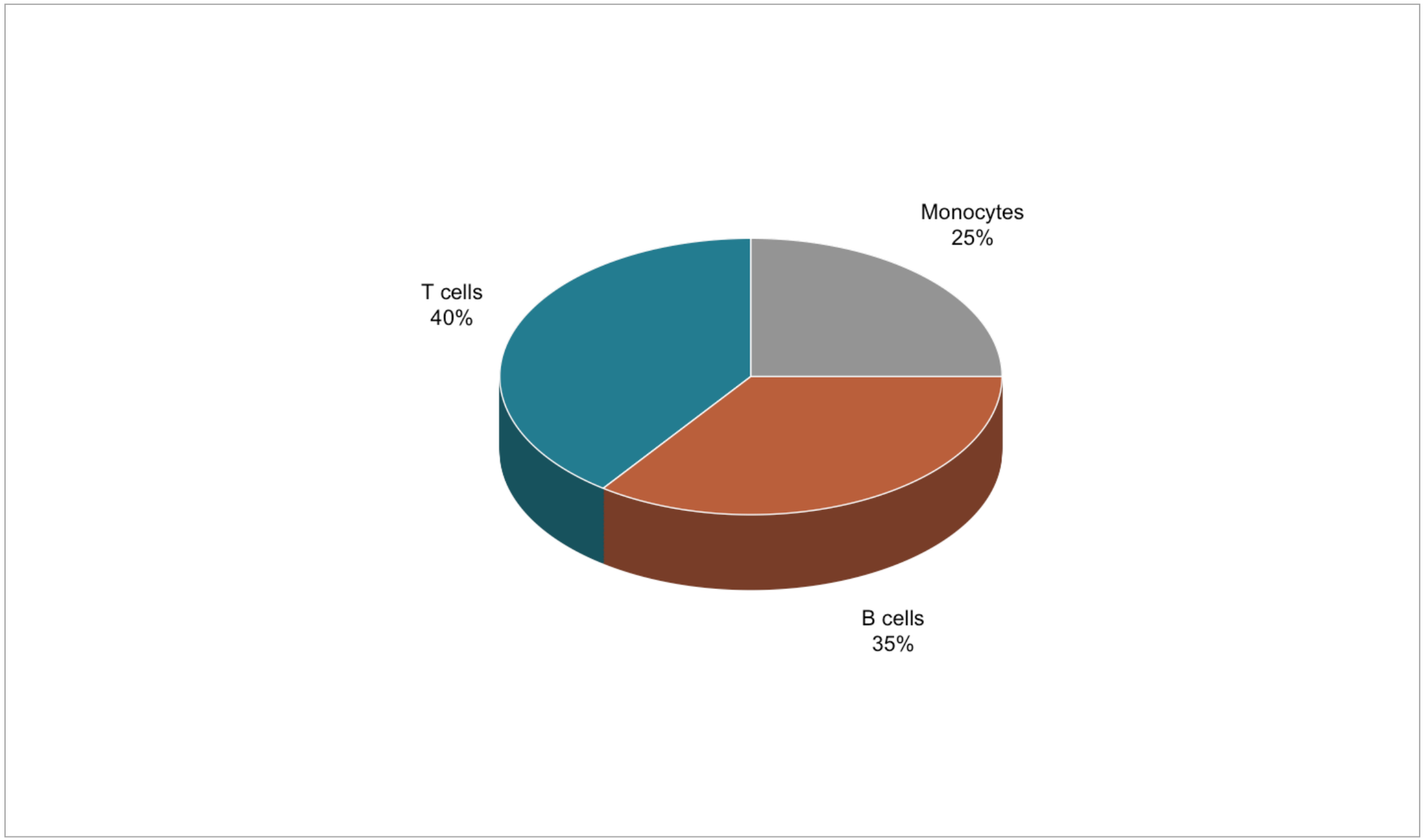
Further reading & data

Function over decoration: does 3D help?

One sample contains 100 classified cells: 40 T cells, 35 B cells, and 25 monocytes. Each cell belongs to exactly one category. All three versions show these same counts.

Source: Course example.

Does the added depth make the proportions easier to compare?



Yes: the visible front surface represents additional cells.

No: depth adds visual area without adding data.

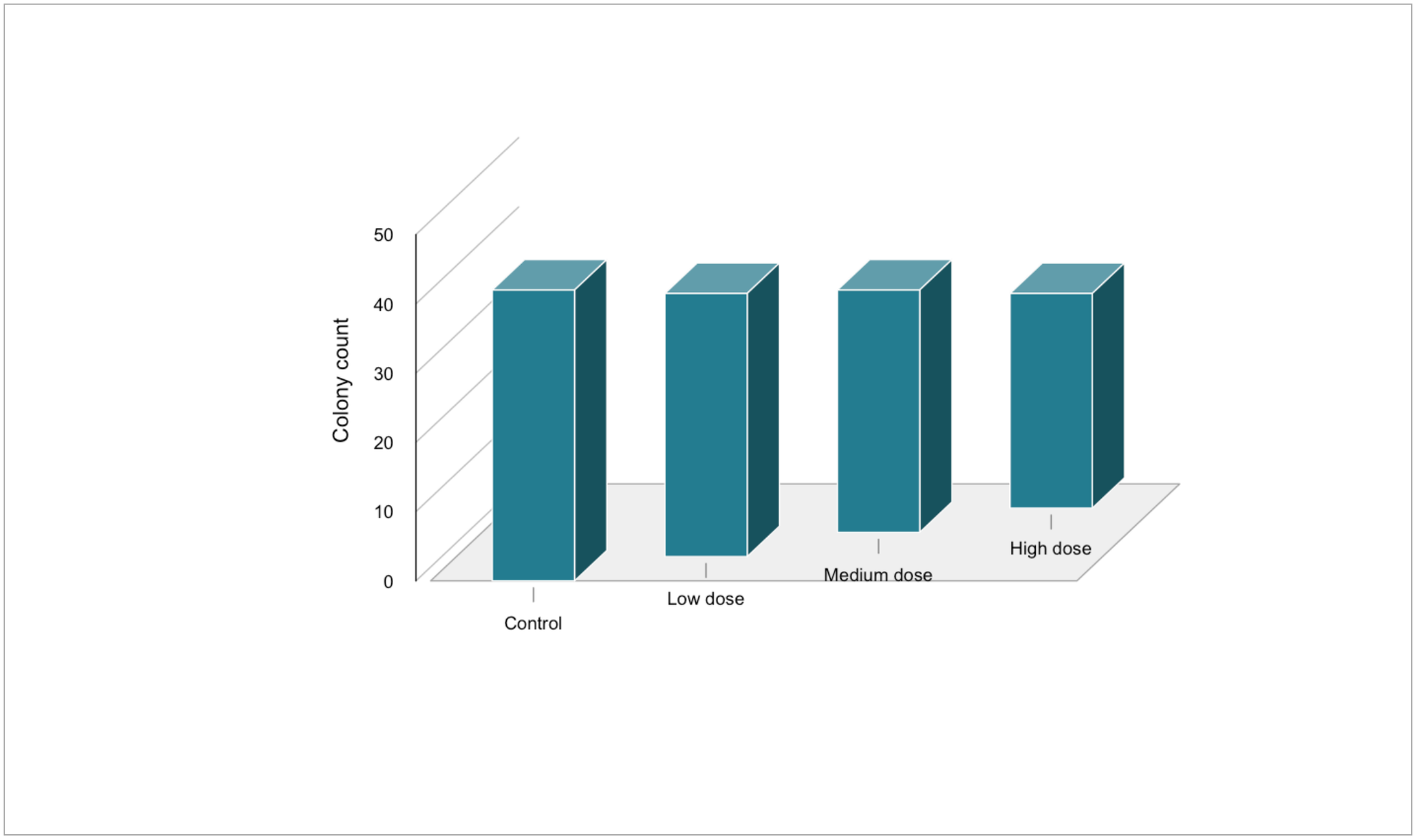
Further reading & data

Function over decoration: 3D bars

Colony counts from one plate per condition: control, low dose, medium dose, and high dose. Bar height encodes count. In the first version, bars are staggered in decorative depth; depth represents no measured variable.

Source: Course example.

Can the screen positions of the bar tops be compared directly?



Yes: nearly level tops mean nearly equal counts.

No: depth shifts both the baselines and the bar tops.

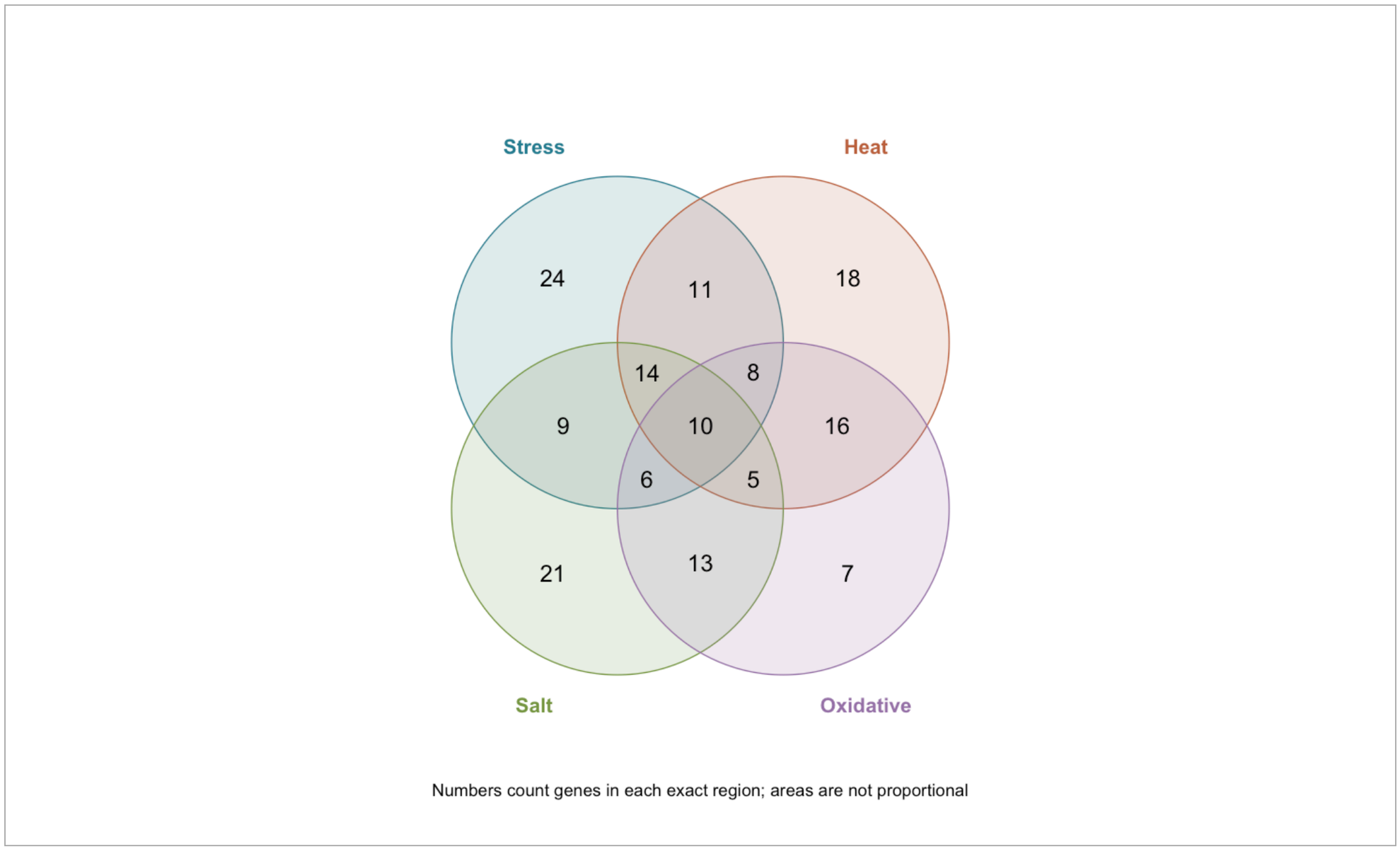
Further reading & data

When overlapping circles stop helping

Four gene lists come from stress, heat, salt, and oxidative-response comparisons. Numbers count exact membership combinations: a gene in three lists belongs only to that three-list region. Only genes in at least one list are shown. Both displays use the same memberships and counts.

Source: Course example.

How easy is it to find the largest intersection involving more than one list?



The circle areas let us rank the counts directly.

We must trace the regions and compare their printed counts.

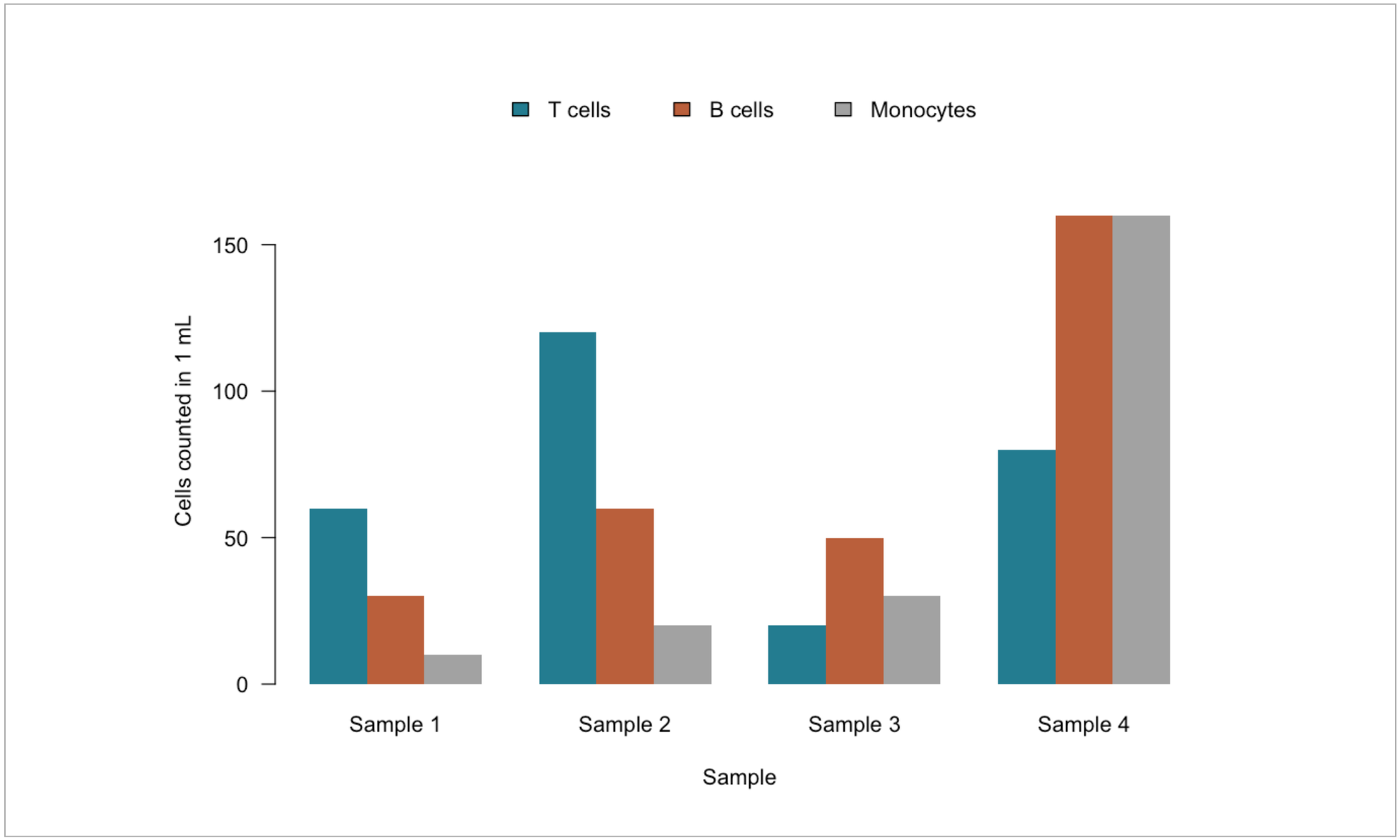
Further reading & data

Grouped, stacked, or 100% stacked?

Four samples were counted using the same volume: 1 mL each. Cells are classified as T cells, B cells, or monocytes. Every version uses the same counts and category colours; only the layout or conversion to percentages changes.

Source: Course example.

Which design is best for these measurements?



It depends: compare individual cell types, total counts, or composition.

100% stacked bars are always best because all samples have equal height.

> **Further reading & data**